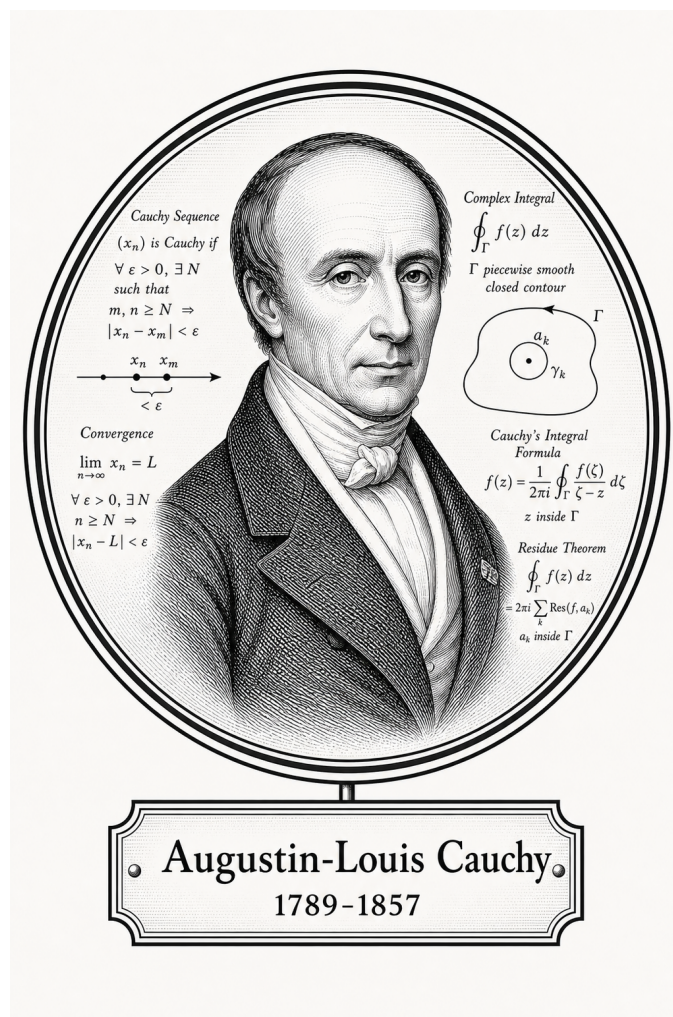


FROM CANTOR TO ITO

# Classical Analysis



Augustin-Louis Cauchy

1789-1857

*Self-Study Notes*

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July 1, 2026

*For every reader learning to make mathematics their own.*

# Contents

|          |                                                               |            |
|----------|---------------------------------------------------------------|------------|
| <b>I</b> | <b>Classical Analysis</b>                                     | <b>1</b>   |
| <b>1</b> | <b>Real Analysis</b>                                          | <b>2</b>   |
| 1.1      | Foundational Analysis Properties . . . . .                    | 3          |
| 1.1.1    | Foundational Analysis Properties . . . . .                    | 3          |
| 1.1.2    | Manipulating Inequalities and the Logic of Bounding . . . . . | 10         |
| 1.1.3    | Completeness as a Constructive Engine . . . . .               | 15         |
| 1.1.4    | Walking a Sequence with a Predicate . . . . .                 | 19         |
| 1.1.5    | Interval Bisection and Nested Intervals . . . . .             | 24         |
| 1.1.6    | Subsequence Partition via Residue Classes . . . . .           | 28         |
| 1.2      | Asymptotic Notation . . . . .                                 | 33         |
| 1.3      | Inequality . . . . .                                          | 35         |
| 1.4      | Modulus . . . . .                                             | 41         |
| <b>2</b> | <b>Structure of the Real Line</b>                             | <b>87</b>  |
| 2.1      | Real Line One Dimensional . . . . .                           | 87         |
| 2.2      | Order Distance Length . . . . .                               | 87         |
| 2.3      | Absolute Value as Distance . . . . .                          | 89         |
| 2.4      | Intervals as Subsets . . . . .                                | 89         |
| 2.5      | Neighborhoods and Balls . . . . .                             | 92         |
| 2.6      | Open Sets . . . . .                                           | 93         |
| 2.7      | Closed Sets . . . . .                                         | 94         |
| 2.8      | Interior Exterior Boundary . . . . .                          | 95         |
| 2.9      | Limit and Isolated Points . . . . .                           | 97         |
| 2.10     | Unions and Intersections . . . . .                            | 98         |
| 2.11     | Covers and Subcovers . . . . .                                | 98         |
| 2.12     | Compactness on $\mathbb{R}$ . . . . .                         | 99         |
| 2.13     | Heine Borel . . . . .                                         | 101        |
| <b>3</b> | <b>Bounds</b>                                                 | <b>112</b> |
| 3.1      | Completeness Axiom . . . . .                                  | 112        |
| 3.1.1    | Nested Intervals . . . . .                                    | 114        |
| 3.1.2    | Archimedean Property . . . . .                                | 116        |
| 3.1.3    | Density . . . . .                                             | 118        |
| 3.1.4    | Completeness Summary . . . . .                                | 124        |
| 3.2      | Bounds and Extremal Values . . . . .                          | 125        |

|          |                                            |            |
|----------|--------------------------------------------|------------|
| 3.2.1    | Upper and Lower Bounds                     | 125        |
| 3.2.2    | Suprema and Infima                         | 128        |
| 3.2.3    | Maxima and Minima                          | 137        |
| 3.2.4    | Epsilon Characterization                   | 141        |
| 3.2.5    | Extremal Bounds Summary                    | 144        |
| 3.3      | Algebra of Bounds                          | 144        |
| 3.3.1    | Set Operations and Bounds                  | 144        |
| 3.3.2    | Lattice Operations and Bounds              | 150        |
| 3.3.3    | Images, Inverse Images, and Extrema        | 154        |
| 3.3.4    | Dilation and Displacement                  | 162        |
| 3.3.5    | Algebra of Suprema and Infima              | 168        |
| <b>4</b> | <b>Functions</b>                           | <b>277</b> |
| 4.1      | Algebra of Functions                       | 277        |
| 4.2      | Real-Valued Functions                      | 281        |
| 4.3      | Subsets of $\mathbb{R}$                    | 293        |
| <b>5</b> | <b>Discrete Calculus</b>                   | <b>312</b> |
| 5.1      | Motivation                                 | 312        |
| 5.2      | Foundations                                | 315        |
| 5.2.1    | Foundations                                | 315        |
| 5.2.2    | Summation Notation                         | 315        |
| 5.2.3    | The Binomial Theorem                       | 317        |
| 5.3      | Algebra of Summations                      | 319        |
| 5.4      | Difference Operators                       | 322        |
| 5.4.1    | Difference Operators                       | 322        |
| 5.4.2    | Higher Differences                         | 323        |
| 5.4.3    | Properties of the Difference Operator      | 324        |
| 5.4.4    | Difference Tables and Polynomial Detection | 325        |
| 5.5      | Telescoping Sums                           | 327        |
| 5.5.1    | Telescoping Sums                           | 327        |
| 5.6      | Discrete Antiderivatives                   | 329        |
| 5.6.1    | Discrete Antiderivatives                   | 329        |
| 5.7      | Polynomial Basis                           | 332        |
| 5.7.1    | Polynomial Basis: Falling Factorials       | 332        |
| 5.8      | Calculus Rules                             | 335        |
| 5.8.1    | Calculus Rules                             | 335        |
| 5.9      | Expansions                                 | 337        |
| 5.9.1    | Newton's Forward Difference Expansion      | 337        |
| 5.10     | Special Functions                          | 339        |
| 5.10.1   | Special Functions in Discrete Calculus     | 339        |
| 5.10.2   | The Function $2^n$ in Discrete Calculus    | 340        |

|           |                                               |            |
|-----------|-----------------------------------------------|------------|
| <b>6</b>  | <b>Sequences</b>                              | <b>355</b> |
| 6.1       | Sequence Definitions . . . . .                | 355        |
| 6.1.1     | Definitions . . . . .                         | 355        |
| 6.1.2     | Null and Constant Sequences . . . . .         | 357        |
| 6.2       | Examples and Counterexamples . . . . .        | 369        |
| 6.3       | Convergence . . . . .                         | 371        |
| 6.3.1     | Limits . . . . .                              | 374        |
| 6.3.2     | Algebra of Limits . . . . .                   | 379        |
| 6.4       | Tails . . . . .                               | 391        |
| 6.5       | Monotonicity . . . . .                        | 395        |
| 6.6       | Subsequences . . . . .                        | 406        |
| 6.7       | Divergence . . . . .                          | 416        |
| 6.8       | Liminf and Limsup . . . . .                   | 422        |
| 6.9       | Order-Theoretic Limit Constructions . . . . . | 430        |
| 6.10      | Cluster Values of Sequences . . . . .         | 433        |
| 6.11      | Cauchy . . . . .                              | 436        |
| 6.12      | Applications of Sequences . . . . .           | 445        |
| <b>7</b>  | <b>Series</b>                                 | <b>556</b> |
| 7.1       | Finite Series . . . . .                       | 556        |
| 7.2       | Infinite Series . . . . .                     | 565        |
| <b>8</b>  | <b>Function Sequences</b>                     | <b>585</b> |
| <b>9</b>  | <b>Elementary Functions</b>                   | <b>586</b> |
| 9.1       | Hyperbolic Functions . . . . .                | 586        |
| 9.2       | The Logarithm . . . . .                       | 590        |
| 9.3       | Power Functions and the Exponential . . . . . | 593        |
| 9.4       | Trigonometric Functions . . . . .             | 598        |
| <b>10</b> | <b>Continuity</b>                             | <b>616</b> |
| 10.1      | Limits . . . . .                              | 616        |
| 10.2      | Point Continuity . . . . .                    | 633        |
| 10.3      | Global Theorems . . . . .                     | 646        |
| 10.4      | Uniform Continuity . . . . .                  | 655        |
| 10.5      | Monotone Functions . . . . .                  | 666        |
| 10.6      | Limits at Infinity . . . . .                  | 676        |
| 10.7      | Approximation . . . . .                       | 678        |
| 10.8      | Gauge . . . . .                               | 683        |
| <b>11</b> | <b>Differentiation</b>                        | <b>746</b> |
| 11.1      | The Tangent Problem . . . . .                 | 746        |
| 11.2      | The Derivative . . . . .                      | 748        |
| 11.2.1    | First Consequences . . . . .                  | 754        |
| 11.3      | One-Sided Derivatives . . . . .               | 757        |

|           |                                                   |            |
|-----------|---------------------------------------------------|------------|
| 11.4      | Carathéodory and the Chain Rule . . . . .         | 760        |
| 11.4.1    | Carathéodory Characterization . . . . .           | 760        |
| 11.5      | The Shape Questions . . . . .                     | 765        |
| 11.5.1    | Interior Extrema . . . . .                        | 766        |
| 11.6      | The Mean Value Theorems . . . . .                 | 770        |
| 11.7      | Reading the Graph from $f'$ and $f''$ . . . . .   | 776        |
| 11.7.1    | First-Order Shape . . . . .                       | 776        |
| 11.7.2    | Second-Order Shape . . . . .                      | 782        |
| 11.7.3    | Higher-Order Derivatives . . . . .                | 782        |
| 11.7.4    | Second-Order Shape . . . . .                      | 783        |
| 11.7.5    | Darboux's Theorem . . . . .                       | 786        |
| 11.7.6    | Smoothness Classes . . . . .                      | 786        |
| 11.8      | The Algebra of Derivatives . . . . .              | 794        |
| 11.8.1    | From Characterization to Computation . . . . .    | 794        |
| 11.8.2    | Closure . . . . .                                 | 797        |
| 11.8.3    | Global Forms on an Interval . . . . .             | 799        |
| 11.8.4    | Inverting the Operation . . . . .                 | 803        |
| 11.8.5    | Indeterminate Forms: L'Hôpital's Rule . . . . .   | 805        |
| 11.9      | Taylor: Construction and Its Error . . . . .      | 806        |
| 11.9.1    | The Differential . . . . .                        | 811        |
| <b>12</b> | <b>Integration</b>                                | <b>877</b> |
| 12.1      | Partitions and Tags . . . . .                     | 877        |
| 12.2      | The Cauchy Integral . . . . .                     | 879        |
| 12.3      | The Riemann Integral . . . . .                    | 882        |
| 12.4      | The Darboux Integral . . . . .                    | 884        |
| 12.5      | The Riemann–Stieltjes Integral . . . . .          | 889        |
| 12.6      | The Henstock–Kurzweil Integral . . . . .          | 892        |
| 12.7      | The McShane Integral . . . . .                    | 896        |
| 12.8      | Measure Zero and the Lebesgue Criterion . . . . . | 900        |

# Part I

## Classical Analysis



# Chapter 1

## Real Analysis



# 1.1 Foundational Analysis Properties

## 1.1.1 Foundational Analysis Properties

### Foundational Analysis Properties — Quick Reference

| Concept                  | Goal                         | Proof mechanism / logical form                                                            |
|--------------------------|------------------------------|-------------------------------------------------------------------------------------------|
| Well-definedness         | Consistency                  | $x \sim y \Rightarrow F(x) = F(y)$                                                        |
| Existence                | Feasibility                  | $\exists x P(x)$                                                                          |
| Uniqueness               | Singularity                  | $P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2$                                              |
| Linearity                | Decomposition                | $L(f + g) = L(f) + L(g)$ and $L(cf) = cL(f)$                                              |
| Uniformity               | One choice works globally    | $\forall \varepsilon > 0, \exists \delta > 0, \forall x, y, Q(\varepsilon, \delta, x, y)$ |
| Equivalence              | Interchangeable descriptions | $A \Rightarrow B$ and $B \Rightarrow A$                                                   |
| Characterisation         | Usable test                  | $P(x) \iff C(x)$                                                                          |
| Estimate                 | Control                      | $A \leq B,  A  < \varepsilon$ , or $d(A, B) < \varepsilon$                                |
| Boundedness              | Two-sided control            | $\exists l, u \in S \forall x \in A (l \leq x \leq u)$                                    |
| Upper/lower control      | One-sided domination         | $\forall x \in A (x \leq u)$ or $\forall x \in A (l \leq x)$                              |
| Extremality              | Realized optimality          | $m \in A \wedge \text{Bound}(m, A)$                                                       |
| Least/greatest character | Optimal bound                | $\text{Bound}(b, A) \wedge \forall c (\text{Bound}(c, A) \Rightarrow b \preceq c)$        |
| Monotonicity             | Order preservation           | Start with $x \leq y$ and prove $f(x) \leq f(y)$ , or the reverse                         |
| Closure                  | Staying inside a class       | $P(x) \wedge P(y) \Rightarrow P(x \star y)$                                               |
| Approximation            | Arbitrary precision          | $\forall \varepsilon > 0 \exists x d(x, a) < \varepsilon$                                 |

Many proofs in analysis are not isolated inventions. They prove recurring structural properties: a definition is independent of choices, an object exists, there is only one possible object, an operation respects addition and scalars, a single parameter works everywhere, or two definitions describe the same phenomenon. Naming the property first tells us what the proof must do.

## Well-Definedness

Well-definedness appears whenever an object is defined using choices: representatives of equivalence classes, selected sequences, selected partitions, or selected approximations. The goal is to prove that different allowed choices give the same output.

$$x \sim x' \Rightarrow F(x) = F(x').$$

*Remark 1.1.1* (Proof sketch). 1. Identify the choice built into the definition.

2. Take two allowed choices, usually  $x$  and  $x'$  with  $x \sim x'$ .

3. Compute or compare the two proposed outputs.

4. Show the outputs agree in the codomain.

*Remark 1.1.2* (Crucial logic). The output must be independent of the chosen representative. If changing the representative changes the output, the proposed definition is not a function on the quotient or chosen domain.

## Existence

Existence proves that at least one object satisfying a property is available:

$$\exists x P(x).$$

In analysis the object might be a limit, a supremum, a point in an interval, a subsequence, a bound, an index  $N$ , a radius  $\delta$ , or a partition.

*Remark 1.1.3* (Proof sketch). 1. Determine every requirement in the property  $P$ .

2. Produce a candidate explicitly, or invoke a theorem that guarantees one.

3. Verify the candidate belongs to the required domain.

4. Verify the candidate satisfies every part of  $P$ .

*Remark 1.1.4* (Constructive and nonconstructive existence). A constructive proof names the object directly. A nonconstructive proof uses a theorem such as completeness, the intermediate value theorem, or Bolzano–Weierstrass. In both cases, the final burden is the same: the object must satisfy the stated property.

## Uniqueness

Uniqueness proves that at most one object can satisfy a property:

$$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2.$$

It does not by itself prove existence. It only proves that two successful candidates cannot be distinct.

*Remark 1.1.5* (Proof sketch). 1. Assume  $x_1$  and  $x_2$  both satisfy  $P$ .

2. Use the defining property of  $x_1$  and the defining property of  $x_2$ .

3. Derive two-sided comparison, zero distance, or identical defining data.

4. Conclude  $x_1 = x_2$ .

*Remark 1.1.6* (Common analysis forms). For suprema, show  $s \leq t$  and  $t \leq s$ . For limits, show  $|L - M| < \varepsilon$  for every  $\varepsilon > 0$ , hence  $L = M$ . In metric spaces, show  $d(x_1, x_2) = 0$ .

## Linearity

Linearity proves that an operation decomposes over addition and scalar multiplication:

$$L(f + g) = L(f) + L(g), \quad L(cf) = cL(f).$$

The goal is not merely algebraic neatness. Linearity says that a complicated input can be analysed by splitting it into simpler pieces.

*Remark 1.1.7* (Proof sketch). 1. Take arbitrary admissible inputs  $f, g$  and scalar  $c$ .

2. Expand  $L(f + g)$  using the definition of  $L$  and the definition of addition.

3. Rearrange the expression until  $L(f) + L(g)$  appears.

4. Repeat with  $L(cf)$  and show it equals  $cL(f)$ .

*Remark 1.1.8* (Typical analysis examples). Limits, derivatives, integrals, and many summation operators have linearity theorems. Each proof follows the same shape: expand the left side, split the expression, and identify the two pieces.

## Uniformity

Uniformity means that one choice works for all relevant points at once. The prototype is not just uniform continuity, but any statement where a witness is chosen before the point is allowed to vary:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y, Q(\varepsilon, \delta, x, y).$$

The key phrase is “one choice.” The same  $N$ ,  $\delta$ , bound, or partition must work globally over the specified domain.

*Remark 1.1.9* (Proof sketch). 1. Fix the tolerance or target requirement.

2. Choose a single witness that depends only on the allowed global data.

3. Let the point, pair of points, function, or index vary arbitrarily.

4. Verify that the same witness works in every permitted case.

*Remark 1.1.10* (Contrast with pointwise statements). Pointwise statements allow the witness to depend on the point. Uniform statements do not. The order of quantifiers is the whole issue:

$$\forall x \exists \delta \text{ is pointwise, while } \exists \delta \forall x \text{ is uniform.}$$

## Equivalence

Equivalence proves that two statements are logically interchangeable:

$$A \iff B.$$

Such theorems are common because analysis often has several descriptions of the same phenomenon: epsilon-delta, sequential, order-theoretic, topological, or integral.

*Remark 1.1.11* (Proof sketch). 1. Prove  $A \Rightarrow B$ .

2. Prove  $B \Rightarrow A$ .

3. Keep the assumptions for the two directions separate.

4. After both directions are proved, either description may be used.

*Remark 1.1.12* (Typical analysis examples). Sequential and epsilon-delta definitions of continuity are equivalent. Riemann and Darboux integrability are equivalent under the appropriate hypotheses. A set is closed exactly when it contains the limits of all convergent sequences from the set.

## Characterisation

A characterisation is an equivalence whose purpose is to turn a definition into a usable test:

$$P(x) \iff C(x).$$

The theorem does not merely say two statements agree. It supplies a criterion that can be used in later proofs.

*Remark 1.1.13* (Proof sketch). 1. Identify the original definition.

2. Identify the proposed criterion.

3. Prove definition  $\Rightarrow$  criterion.

4. Prove criterion  $\Rightarrow$  definition.

*Remark 1.1.14* (Typical analysis examples). The epsilon characterisation of the supremum turns least-upper-bound language into an approximation rule. Cauchy criteria turn convergence questions into internal distance estimates. Sequential criteria turn topological conditions into statements about sequences.

## Estimates and Bounds

An estimate proves control:

$$A \leq B, \quad |A| < \varepsilon, \quad d(A, B) < \varepsilon.$$

The aim is not to compute the exact value. The aim is to put the target quantity inside a range that is strong enough for the theorem.

*Remark 1.1.15* (Proof sketch). 1. Identify the quantity that must be controlled.

2. Replace it by a larger or simpler quantity using valid inequalities.

3. Use hypotheses to replace variable terms by constants or small errors.

4. End with the required bound.

*Remark 1.1.16* (Typical analysis examples). Showing a sequence is bounded, proving an error term tends to zero, bounding a Riemann-sum error, or proving  $|f(x) - L| < \varepsilon$  are all estimate problems. The proof is a chain of control.

## Boundedness

Boundedness proves two-sided control. In an ordered setting, the usual form is that a set or sequence is bounded above and bounded below:

$$\exists l, u \in S \forall x \in A (l \leq x \leq u).$$

*Remark 1.1.17* (Proof sketch). 1. Prove bounded above by finding a candidate upper bound.

2. Prove bounded below by finding a candidate lower bound.

3. Verify both candidates work for every element under consideration.

4. Conclude that the object is bounded.

## Upper and Lower Control

Upper and lower control prove one-sided domination. The proof focuses on a candidate bound and checks that every element lies on the required side of it:

$$\forall x \in A (x \leq u), \quad \forall x \in A (l \leq x).$$

*Remark 1.1.18* (Proof sketch). 1. Name the candidate upper or lower bound.

2. Let an arbitrary element of the set be given.

3. Use the hypotheses or definition of the set to compare that element with the candidate.

4. Since the element was arbitrary, conclude the one-sided bound.

## Extremality

Extremality proves that an optimal value is actually realized by the object being studied. A maximum is not merely an upper bound; it is an upper bound that belongs to the set. A minimum is the corresponding realized lower bound.

$$\text{Maximum}(m, A) \iff (m \in A) \wedge \text{UpperBound}(m, A),$$

$$\text{Minimum}(m, A) \iff (m \in A) \wedge \text{LowerBound}(m, A).$$

*Remark 1.1.19* (Proof sketch). 1. Prove the candidate belongs to the set.

2. Prove the candidate is the appropriate one-sided bound.

3. Combine membership with the bound condition.

4. Conclude maximum or minimum.

## Least and Greatest Character

Least and greatest character proves optimality among competitors. For a supremum, the candidate must first be an upper bound, and then it must be no larger than every other upper bound. Infimum proofs reverse the direction.

$$\text{Supremum}(s, A) \iff \text{UpperBound}(s, A) \wedge \forall u \in S (\text{UpperBound}(u, A) \Rightarrow s \leq u),$$

$$\text{Infimum}(i, A) \iff \text{LowerBound}(i, A) \wedge \forall l \in S (\text{LowerBound}(l, A) \Rightarrow l \leq i).$$

*Remark 1.1.20* (Proof sketch). 1. Prove the candidate is a bound of the required kind.

2. Let an arbitrary competing bound be given.
3. Compare the candidate with that competing bound.
4. Conclude the candidate is the least upper bound or greatest lower bound.

## Monotonicity

Monotonicity proves order preservation or order reversal:

$$x \leq y \Rightarrow f(x) \leq f(y), \quad x \leq y \Rightarrow f(y) \leq f(x).$$

The proof begins with an arbitrary ordered pair and carries that order through the definitions.

*Remark 1.1.21* (Proof sketch). 1. Take arbitrary points  $x$  and  $y$  with  $x \leq y$ .

2. Expand the definitions of the function or operation.
3. Use order rules and hypotheses to compare the outputs.
4. Conclude that the order is preserved or reversed.

## Closure

Closure proves that a class is stable under an operation. One starts with objects already known to belong to the class, combines them, and then verifies that the result still satisfies the defining property.

$$P(x) \wedge P(y) \Rightarrow P(x \star y).$$

For closure under scalar operations, the same pattern becomes

$$P(x) \Rightarrow P(c \star x)$$

for every admissible scalar or parameter  $c$ .

*Remark 1.1.22* (Proof sketch). 1. Take arbitrary objects from the class.

2. Form the object produced by the operation.
3. Check every condition in the definition of the class.
4. Conclude the result remains inside the class.

## Approximation

Approximation proves arbitrary precision. The proof usually begins with a tolerance and constructs an object close enough to a target:

$$\forall \varepsilon > 0 \exists x d(x, a) < \varepsilon.$$

*Remark 1.1.23* (Proof sketch). 1. Fix an arbitrary tolerance  $\varepsilon > 0$ .

2. Choose or construct an approximating object.
3. Estimate the error between the approximation and the target.
4. Show the error is below  $\varepsilon$ .

## A Classification Pass

Before proving a theorem, classify its job:

- Does the statement depend on choices? Check well-definedness.
- Does it ask for an object? Prove existence.
- Does it say only one object can work? Prove uniqueness.
- Does it say an operation splits across sums and scalars? Prove linearity.
- Does one witness have to work for every point? Prove uniformity.
- Does it say two descriptions agree? Prove equivalence.
- Does it give a test for a definition? Prove a characterisation.
- Does it ask for smallness or control? Prove an estimate.
- Does it ask for two-sided order control? Prove boundedness.
- Does it ask for one-sided order control? Prove an upper or lower bound.
- Does it ask for a realized best element? Prove extremality.
- Does it ask for the best bound among all bounds? Prove least or greatest character.
- Does it preserve or reverse order? Prove monotonicity.
- Does it stay inside a class after an operation? Prove closure.
- Does it ask for arbitrary precision? Prove approximation.

*Remark 1.1.24* (The point). The classification does not solve the proof, but it tells us what a completed proof must contain. For example, an existence proof must exhibit or obtain an object, while a uniqueness proof must compare two candidates. An equivalence proof must have two directions, while a uniformity proof must choose one witness before the arbitrary point is fixed.



## 1.1.2 Manipulating Inequalities and the Logic of Bounding

### Inequality Toolkit — Quick Reference

| Tool                        | Statement                                                                                         | Typical use                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Triangle inequality         | $ a + b  \leq  a  +  b $                                                                          | Split a distance at an intermediate point       |
| Distance form               | $ a - b  \leq  a - c  +  c - b $                                                                  | Introduce pivot $c$ to separate two estimates   |
| Reverse triangle ineq.      | $  a  -  b   \leq  a - b $                                                                        | Bound magnitudes using closeness                |
| Absolute value interval     | $ x  \leq y \iff -y \leq x \leq y$                                                                | Convert $ \cdot $ bound to two-sided inequality |
| Boundedness replacement     | $ b_n  \leq M$ (constant): replace $ b_n $ by $M$                                                 | Eliminate variable factors in products          |
| $\varepsilon$ -budget split | $ A  < \frac{\varepsilon}{2}$ and $ B  < \frac{\varepsilon}{2} \Rightarrow  A + B  < \varepsilon$ | Combine two independent estimates               |
| +1 trick                    | Replace $ L $ by $ L  + 1$ in denominators                                                        | Avoid division by zero when $L$ may vanish      |

## The Fundamental Asymmetry Between Equality and Inequality

Equality is bilateral:  $a = b$  licenses substitution in either direction and carries no information about magnitude. An inequality is directional:  $a \leq b$  locates  $a$  relative to  $b$  on the line, says nothing about the gap, and does not support backwards substitution.

This asymmetry is the source of a strategic shift. In an equality argument the goal is to show that two expressions are the same object. In a bounding argument the goal is to show that one expression is *controlled* by another. Control is inherently one-sided: a ceiling above  $|f(x)|$  is useful regardless of how far below it  $|f(x)|$  actually falls.

*Remark 1.1.25* (Consequence for proof strategy). The question in analysis is rarely “what is  $|a_n - L|$ ?” It is “is  $|a_n - L|$  smaller than  $\varepsilon$ ?” A valid crude bound that answers the second question is more useful than an exact answer to the first. Deliberate over-estimation — absent from algebra — is a required habit.

*Remark 1.1.26* (Common error). Treating an inequality as if it were an equality by reversing direction. Given  $a \leq b$ , negation gives  $-a \geq -b$ , not  $-a \leq -b$ . Multiplication by a negative constant reverses direction. Failing to track sign changes when multiplying or negating is the most frequent algebraic error in  $\varepsilon$ - $\delta$  computations.

## The Basic Rules of Inequality Arithmetic

**Proposition 1.1.27** (Order arithmetic). *Let  $a, b, c, d \in \mathbb{R}$ .*

- (i) *If  $a \leq b$  and  $c \leq d$ , then  $a + c \leq b + d$ .*
- (ii) *If  $a \leq b$  and  $c > 0$ , then  $ac \leq bc$ .*
- (iii) *If  $a \leq b$  and  $c < 0$ , then  $ac \geq bc$ .*
- (iv)  *$|x| \leq y$  (with  $y \geq 0$ ) if and only if  $-y \leq x \leq y$ .*

*Go to proof.*

*Remark.*

*Remark 1.1.28* (Fully quantified form). (i)  $\forall a, b, c, d \in \mathbb{R}, (a \leq b) \wedge (c \leq d) \Rightarrow a + c \leq b + d$ .  
(iv)  $\forall x \in \mathbb{R}, \forall y \geq 0, |x| \leq y \Leftrightarrow -y \leq x \leq y$ .

*Remark 1.1.29* (Inequalities add but do not subtract). Rule (i) licenses adding inequalities in the same direction. There is no subtraction rule:  $a \leq b$  and  $c \leq d$  yields no conclusion about  $a - c$  versus  $b - d$  in general. The direction of  $a - c$  versus  $b - d$  depends on the relative sizes of  $(b - a)$  and  $(d - c)$ , which are not controlled.

*Remark 1.1.30* (Consequence for absolute values). Rule (iv) is the bridge between  $|\cdot|$  bounds and two-sided linear inequalities. It is invoked silently every time an  $\varepsilon$ - $N$  argument converts  $|a_n - L| < \varepsilon$  into positional information about  $a_n$ .

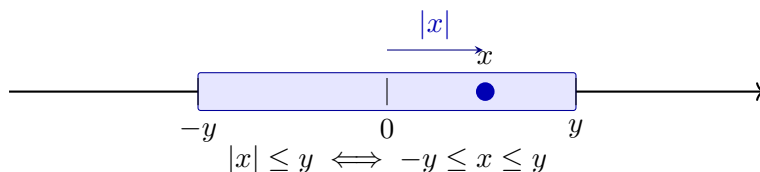


Figure 1.1: The absolute value characterisation as a symmetric interval. The shaded region is  $[-y, y]$ ; the point  $x$  lies inside with  $|x| \leq y$ . The equivalence  $|x| \leq y \Leftrightarrow -y \leq x \leq y$  (Proposition 1.1.27(iv)) converts an absolute value bound into a two-sided linear constraint.

## The Triangle Inequality and the Pivot Move

### Theorem (Triangle Inequality)

For all  $a, b \in \mathbb{R}$ ,  $|a + b| \leq |a| + |b|$ . Equivalently, for all  $a, b, c \in \mathbb{R}$ ,  $|a - b| \leq |a - c| + |c - b|$ .

*Remark 1.1.31* (Fully quantified form).  $\forall a, b \in \mathbb{R}, |a + b| \leq |a| + |b|$ .

*Remark 1.1.32* (The pivot move). The distance form  $|a - b| \leq |a - c| + |c - b|$  is the operative version in analysis. The element  $c$  is a *pivot*: an intermediate point that decomposes a hard distance into two pieces, each controllable by hypothesis. In the proof that  $a_n + b_n \rightarrow L + M$ , the pivot  $c = L + b_n$  gives

$$|(a_n + b_n) - (L + M)| = |(a_n - L) + (b_n - M)| \leq |a_n - L| + |b_n - M|,$$

and the two pieces are controlled by the individual convergences.

*Remark 1.1.33* (Reverse triangle inequality).  $||a| - |b|| \leq |a - b|$ . Proof: apply the standard inequality to  $a = (a - b) + b$  to get  $|a| - |b| \leq |a - b|$ ; symmetry gives the other side. Consequence:  $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$  is Lipschitz with constant 1, so closeness of inputs implies closeness of magnitudes.

*Remark 1.1.34* (Common error). Writing  $|a| + |b| \leq |a + b|$ . The triangle inequality always produces an *upper* bound on a sum, never a lower bound. The reverse ( $|1| + |-1| = 2 > |1 + (-1)| = 0$ ) is a one-line counterexample.

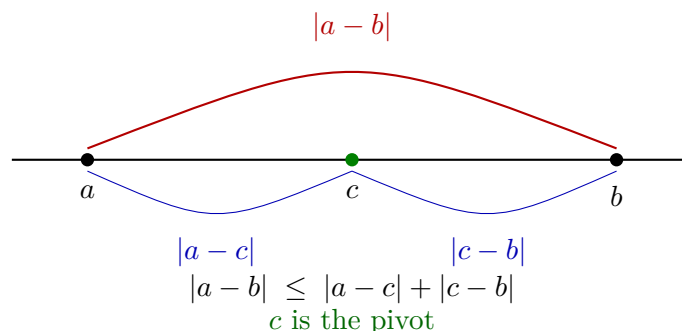


Figure 1.2: The triangle inequality in distance form. The direct distance  $|a - b|$  (red) is bounded above by the sum of indirect distances  $|a - c|$  and  $|c - b|$  (blue) through pivot  $c$ . In analysis,  $c$  is chosen to split  $|a - b|$  into two pieces each controlled by a separate hypothesis.

## Bounding Is Not Solving

In algebra, a computation terminates when the exact value is found. In analysis, a bounding argument terminates when the expression is shown to fall within an acceptable range. These are structurally different goals.

*Remark 1.1.35* (Deliberate crudeness is valid). To bound  $|x^2 - 9x + 1|$  on  $[-1, 5]$ :

$$|x^2 - 9x + 1| \leq |x|^2 + 9|x| + 1 \leq 25 + 45 + 1 = 71.$$

The actual supremum is far smaller than 71. That is irrelevant. The bound is valid, explicit, and was obtained cheaply. The test for a bound is not “is this sharp?” but “is this true and sufficient for the argument’s purpose?”

*Remark 1.1.36* (Consequence for proof reading). A reader who checks that each inequality is valid can trust the conclusion without caring whether the bounds are optimal. Optimality is irrelevant to logical correctness.

## The $\varepsilon$ -Budget Split

The prototype of all two-piece bounding arguments:

$$|A + B| \leq |A| + |B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The total tolerance  $\varepsilon$  is a budget allocated among the summands. With  $k$  summands, allocate  $\varepsilon/k$  each; with summands of unequal difficulty, split unevenly provided the shares sum to at most  $\varepsilon$ .

*Remark 1.1.37* (The four-step procedure). Standard structure of an  $\varepsilon$ - $N$  or  $\varepsilon$ - $\delta$  proof:

1. Apply the triangle inequality: express  $|f - L|$  as a sum  $|A_1| + \cdots + |A_k|$ .
2. Assign budgets  $\varepsilon_i > 0$  with  $\sum_i \varepsilon_i = \varepsilon$ .

3. For each piece, find  $N_i$  (or  $\delta_i$ ) so that  $|A_i| < \varepsilon_i$  for all  $n \geq N_i$  (or  $|x - a| < \delta_i$ ).
4. Set  $N = \max_i N_i$  (or  $\delta = \min_i \delta_i$ ) to enforce all constraints simultaneously.

Step 4 reflects the logical structure of the conclusion: all conditions must hold at once, so the threshold is the most restrictive one.

*Remark 1.1.38* (Common error). Taking  $N = N_1$  and ignoring  $N_2$ . For  $n \geq N_1$  the first piece is controlled; the second is not. Both pieces must be within budget simultaneously, which requires  $n \geq \max\{N_1, N_2\}$ .

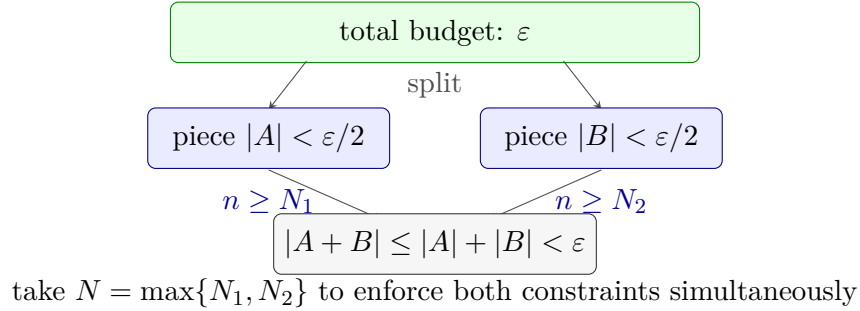


Figure 1.3: The  $\varepsilon$ -budget split. The total tolerance  $\varepsilon$  is distributed as shares  $\varepsilon/2$  between two pieces. Each piece is controlled independently for  $n \geq N_i$ . Taking  $N = \max\{N_1, N_2\}$  enforces both bounds simultaneously, giving  $|A + B| < \varepsilon$  for all  $n \geq N$ . With  $k$  pieces, assign  $\varepsilon/k$  each and take  $N = \max_i N_i$ .

## The Boundedness Replacement

**Proposition 1.1.39** (Convergent sequences are bounded). *If  $(b_n) \rightarrow b$ , then there exists  $M > 0$  such that  $|b_n| \leq M$  for all  $n \in \mathbb{N}$ . Go to proof.*

*Remark.*

*Remark 1.1.40* (Fully quantified form).  $\forall (b_n) \subset \mathbb{R}, \lim_{n \rightarrow \infty} b_n = b \Rightarrow \exists M > 0, \forall n \in \mathbb{N}, |b_n| \leq M$ .

*Remark 1.1.41* (The replacement move). Given  $|b_n| \leq M$ :

$$|b_n| \cdot |a_n - L| \leq M \cdot |a_n - L|.$$

The variable factor is gone. Choose  $N$  so that  $|a_n - L| < \varepsilon/M$  for  $n \geq N$ ; the whole product is then less than  $\varepsilon$ . The constant  $M$  need not be tight.

*Remark 1.1.42* (Common error). Choosing  $|a_n - L| < \varepsilon/|b_n|$ . This is circular:  $N$  would depend on  $n$ , since  $|b_n|$  changes with  $n$ . The threshold  $N$  must be a fixed number, independent of  $n$ . Replacing  $|b_n|$  by the constant  $M$  breaks the circularity.

## The Work-Backwards Principle

$\varepsilon$ - $N$  proofs are *stated* forwards but *discovered* backwards. The proof asserts: “Let  $\varepsilon > 0$ . Choose  $N = \lceil C/\varepsilon \rceil$ .” The value  $N$  was found by scratchwork: “the expression simplifies to  $C/n$ ; this is less than  $\varepsilon$  when  $n > C/\varepsilon$ ; so set  $N = \lceil C/\varepsilon \rceil$ .”

The scratchwork is a private computation that produces the *witness*. The proof states the witness and verifies it, working forwards.

**Scratchwork (private):** Work backwards from  $|a_n - L| < \varepsilon$  to find what  $N$  must satisfy.

**Proof (public):** State  $N$  explicitly. Verify  $n \geq N \Rightarrow |a_n - L| < \varepsilon$ .

*Remark 1.1.43* (Common error). Allowing scratchwork to appear in the proof: “we need  $n > C/\varepsilon$ , so choose  $n$  this large.” The proof must assert a fixed  $N$  and verify the conclusion holds. A threshold stated as “ $n$  large enough” is an incomplete scratchwork note, not a proof.

*Remark 1.1.44* (Connection to the geometric phase). The work-backwards process is the analytic counterpart of the geometric phase in limit proofs. The geometric phase identifies structure and chooses a pivot; the backwards computation determines the explicit threshold. The proof states both and verifies the chain forwards.

## Chaining Inequalities

A standard proof move builds a chain  $|f(x)| \leq A(x) \leq B \leq \varepsilon$ , where each step uses a different tool: the triangle inequality, a domain bound, and the choice of  $\delta$ .

*Remark 1.1.45* (Replacing by a larger quantity). A quantity on the right-hand side of  $\leq$  may be freely replaced by anything larger, since this only weakens the inequality. If  $|x| \leq |x-1|+1$  and  $|x-1| < \delta$ , then  $|x| < \delta+1$ . The replacement of  $|x-1|$  by  $\delta$  is valid because  $|x-1|$  appears on the right. The same replacement on the *left* would be invalid.

*Remark 1.1.46* (Common error). Including a step where the inequality points the wrong direction, or where a step holds only generically rather than pointwise. A chain is only as strong as its weakest link; every  $\leq$  must be unconditionally valid.

## The +1 Trick for Denominators

When  $\varepsilon/|L|$  appears in an estimate and  $L$  may be zero, replace  $|L|$  by  $|L| + 1 \geq 1$ :

$$\frac{\varepsilon}{|L| + 1} \leq \varepsilon.$$

*Remark 1.1.47* (Fully quantified form).  $\forall L \in \mathbb{R}$ ,  $|L| + 1 \geq 1 > 0$ . Hence  $\varepsilon/(|L| + 1)$  is well-defined and positive for all  $\varepsilon > 0$  and all  $L \in \mathbb{R}$ .

*Remark 1.1.48* (Generalisation). The +1 is an instance of a wider principle: any constant intended for a denominator must be guaranteed strictly positive. If the constant is a given quantity  $C$  that might be zero, replace it by  $C+1$ . The choice of 1 is conventional; any fixed positive number serves the same purpose.

*Remark 1.1.49* (Common error). Using  $|L|$  as a denominator without first verifying  $L \neq 0$ . The product limit proof  $a_n b_n \rightarrow LM$  is the canonical location for this error: the bound  $\varepsilon/(2|L|)$  fails when  $L = 0$ , and the  $L = 0$  case requires separate treatment or the use of  $|L| + 1$ .

## Summary: The Bounding Mindset

| Algebra                                   | Analysis                                      |
|-------------------------------------------|-----------------------------------------------|
| Find the exact value of $x$               | Show $ x - L  < \varepsilon$                  |
| Equality: both sides are the same object  | Inequality: one side controls the other       |
| Work backwards and forwards symmetrically | Discover witness backwards; verify forwards   |
| Exact answer is the goal                  | Any valid bound is sufficient                 |
| Fractions must be simplified              | Crude over-estimates are acceptable           |
| Multiply both sides freely                | Track sign changes; direction is load-bearing |

*Remark 1.1.50* (The structure of a bounding proof). A bounding proof is a *chain of control*: inequalities connecting the target expression to  $\varepsilon$ , where each link uses one of the toolkit tools above. The proof is complete when the chain reaches  $\varepsilon$ . Every link must be unconditionally valid; every variable factor must be replaced by a constant before the chain can close.

### 1.1.3 Completeness as a Constructive Engine

#### Completeness Construction Toolkit — Quick Reference

| Tool                                                             | What it supplies                                                     |
|------------------------------------------------------------------|----------------------------------------------------------------------|
| <a href="#"><math>\varepsilon</math>-characterisation of sup</a> | $\forall \varepsilon > 0, \exists x \in S, x > \sup S - \varepsilon$ |
| <a href="#">Inductive selection</a>                              | Monotone $(x_n) \subset S$ with $x_n \rightarrow \sup S$             |
| <a href="#">Monotone Approximation</a>                           | Such a sequence exists for any nonempty bounded $S$                  |
| <a href="#">LUB <math>\Leftrightarrow</math> MCT</a>             | The two completeness statements are equivalent                       |

## The $\varepsilon$ -Characterisation of Suprema and Infima

**Proposition 1.1.51** ( $\varepsilon$ -characterisation of sup). *Let  $S \subset \mathbb{R}$  be nonempty and bounded above, and let  $s = \sup S$ . Then for every  $\varepsilon > 0$  there exists  $x \in S$  with  $x > s - \varepsilon$ . [Go to proof.](#)*

*Remark.*

*Remark 1.1.52* (Fully quantified form).  $\forall \varepsilon > 0, \exists x \in S, s - \varepsilon < x \leq s$ . Equivalently: any  $M < s$  is not an upper bound for  $S$ .

*Remark 1.1.53* (Proof strategy). If no such  $x$  existed for some  $\varepsilon_0 > 0$ , then  $s - \varepsilon_0$  would be an upper bound for  $S$  smaller than  $s$ , contradicting  $s = \sup S$ .

*Remark 1.1.54* (Operative role). The  $\varepsilon$ -characterisation is the *existential engine* of constructive completeness arguments. At each inductive step it is invoked with a specific  $\varepsilon$  value, discharging the universal quantifier and yielding a concrete existence claim. Existential instantiation then names a witness — call it  $x_{n+1}$  — and the rest of the argument uses only the single property  $x_{n+1} > s - \varepsilon$ . The witness is underspecified; the argument makes no commitment about which element of  $S$  is chosen, and no further properties of  $x_{n+1}$  are used.

*Remark 1.1.55* (Symmetric form for inf). For  $t = \inf S$ : for every  $\varepsilon > 0$  there exists  $y \in S$  with  $y < t + \varepsilon$ . The proof is symmetric: any  $M > t$  is not a lower bound.

## Inductive Selection at Bounds

### Theorem (Inductive Selection at sup)

Let  $S \subset \mathbb{R}$  be nonempty and bounded above, and let  $s = \sup S$ . Then there exists a strictly increasing sequence  $(x_n)_{n=1}^\infty \subset S$  with  $x_n \rightarrow s$ .

*Remark 1.1.56* (Fully quantified form).  $\exists(x_n) \subset S: \forall n, x_n < x_{n+1}; \forall n, s - \frac{1}{n} < x_n \leq s; \lim_{n \rightarrow \infty} x_n = s$ .

*Remark 1.1.57* (Why no algebraic formula is possible in general). The set  $S$  may have large gaps; any fixed formula might land outside  $S$  entirely. The construction must be inductive, selecting from  $S$  at each stage using the  $\varepsilon$ -characterisation rather than evaluating a closed expression.

*Remark 1.1.58* (Proof strategy — the max device). The inductive step must simultaneously satisfy two lower bounds:  $x_{n+1} > x_n$  (monotonicity) and  $x_{n+1} > s - \frac{1}{n+1}$  (proximity). A single invocation of the existential can impose only one threshold. Setting

$$M_n := \max\left(x_n, s - \frac{1}{n+1}\right)$$

collapses both requirements into one:  $x_{n+1} > M_n$  delivers both  $x_{n+1} > x_n$  and  $x_{n+1} > s - \frac{1}{n+1}$  simultaneously. Since  $M_n < s$  (both entries are strictly below  $s$  whenever  $s \notin S$ , or the constant sequence  $x_n = s$  handles the case  $s \in S$ ), the  $\varepsilon$ -characterisation with  $\varepsilon = s - M_n > 0$  supplies the required witness.

**Base case.** Set  $\varepsilon = 1$ . The  $\varepsilon$ -characterisation gives  $x_1 \in S$  with  $x_1 > s - 1$ ; since  $x_1 \in S$ , also  $x_1 \leq s$ .

**Inductive step.** Given  $x_n$  with  $s - \frac{1}{n} < x_n \leq s$ , form  $M_n$  as above and instantiate the  $\varepsilon$ -characterisation with  $\varepsilon = s - M_n > 0$  to obtain  $x_{n+1} \in S$  with  $x_{n+1} > M_n$ . Then:

$$\begin{aligned} x_{n+1} > M_n \geq x_n &\Rightarrow x_{n+1} > x_n, \\ x_{n+1} > M_n \geq s - \frac{1}{n+1} &\Rightarrow s - \frac{1}{n+1} < x_{n+1} \leq s. \end{aligned}$$

**Convergence.**  $0 \leq s - x_n < \frac{1}{n}$ ; the squeeze theorem and the Archimedean Property give  $x_n \rightarrow s$ .

*Remark 1.1.59* (What the construction requires). Three ingredients: (i) completeness of  $\mathbb{R}$  (sup  $S$  exists); (ii) the  $\varepsilon$ -characterisation ([Proposition 1.1.51](#)), supplying the witness at each step; (iii) the Archimedean Property ([Theorem 3.1.4](#)), forcing  $|s - x_n| \leq \frac{1}{n} \rightarrow 0$ . The internal structure of  $S$  — its cardinality, openness, density — plays no role.



*Remark 1.1.60* (Common error). Instantiating the existential with only the proximity threshold  $s - \frac{1}{n+1}$  and ignoring  $x_n$ . The resulting witness satisfies proximity but not monotonicity: one might obtain  $x_1 = 0.9$ ,  $x_2 = 0.4$ ,  $x_3 = 0.85$ , each close to  $s$  but not increasing. The max is not a convenience; it is the minimal device that encodes both requirements into one threshold.

*Remark 1.1.61* (Infimum case). Replace max by min,  $s - \frac{1}{n}$  by  $t + \frac{1}{n}$ , and reverse all strict inequalities. The resulting sequence  $(y_n) \subset S$  is strictly decreasing with  $y_n \rightarrow t = \inf S$ .

## Monotone Approximation of Bounds

**Proposition 1.1.62** (Monotone Approximation of Bounds). *Let  $S \subset \mathbb{R}$  be nonempty and bounded. Then there exist monotone sequences  $(x_n), (y_n) \subset S$  with  $x_n \rightarrow \sup S$  and  $y_n \rightarrow \inf S$ . [Go to proof.](#)*

*Remark.*

*Remark 1.1.63* (Fully quantified form).  $\forall S \neq \emptyset$  bounded:  $\exists (x_n) \subset S, x_n \uparrow \sup S$ ;  $\exists (y_n) \subset S, y_n \downarrow \inf S$ .

*Remark 1.1.64* (Proof strategy). The strictly increasing sequence from [Section 1.1.3](#) is already monotone and converges to  $\sup S$ . Taking  $X_n = \max\{x_1, \dots, x_n\}$  makes it explicitly non-decreasing (useful when the inductive step does not guarantee strict increase). The infimum case is symmetric.

*Remark 1.1.65* (Consequence). Every supremum is the limit of a sequence from the set. The sequential characterisation of sup and inf — that they are realised as limits of internal sequences — follows from this construction, making [Proposition 1.1.62](#) the canonical proof of that characterisation.

*Remark 1.1.66* (Forward connections). The inductive selection pattern reappears in:

- **Monotone Convergence Theorem** — every increasing bounded sequence converges to the supremum of its range; this construction exhibits that the supremum is always realised as such a limit.
- **Nested Interval Property** — the left and right endpoints of nested intervals form monotone sequences converging to a common point by exactly this mechanism.
- **Existence of square roots** — the standard proof sets  $s = \sup\{x \in \mathbb{R} : x^2 < 2\}$  and extracts an approximating sequence from that set by inductive selection.

## LUB $\iff$ MCT: The Completeness Equivalence

### Definition (Least Upper Bound Property)

$\mathbb{R}$  has the **least upper bound property** if every nonempty set  $A \subset \mathbb{R}$  that is bounded above has a supremum in  $\mathbb{R}$ .



### Theorem (LUB $\iff$ MCT)

The following are equivalent over the Archimedean ordered field axioms:

- (i)  $\mathbb{R}$  has the least upper bound property (LUB).
- (ii) Every bounded monotone sequence of real numbers converges (MCT).

*Remark 1.1.67* (Fully quantified form). (i)  $\forall A \subset \mathbb{R}, (A \neq \emptyset \wedge \exists M, \forall a \in A, a \leq M) \Rightarrow \exists \sup A \in \mathbb{R}$ .

(ii)  $\forall (x_n) \subset \mathbb{R}, (\forall n, x_n \leq x_{n+1}) \wedge (\exists M, \forall n, x_n \leq M) \Rightarrow \exists L \in \mathbb{R}, \lim_{n \rightarrow \infty} x_n = L$ .

*Remark 1.1.68* (Proof strategy — LUB  $\Rightarrow$  MCT). Let  $(x_n)$  be increasing and bounded. The image set  $A = \{x_n : n \in \mathbb{N}\}$  is bounded above, so LUB gives  $\alpha = \sup A$ . The  $\varepsilon$ -characterisation produces  $N$  with  $x_N > \alpha - \varepsilon$ . Monotonicity promotes this to a tail statement:  $\alpha - \varepsilon < x_n \leq \alpha$  for all  $n \geq N$ . Hence  $x_n \rightarrow \alpha$ .

The quantifier structure is:

$$\alpha = \sup A \Rightarrow (\forall \varepsilon > 0) \exists N (x_N > \alpha - \varepsilon) \xrightarrow{\text{monotone}} (\forall \varepsilon > 0) \exists N (\forall n \geq N) (|x_n - \alpha| < \varepsilon).$$

*Remark 1.1.69* (Proof strategy — MCT  $\Rightarrow$  LUB). Given  $A \neq \emptyset$  and bounded above by  $u_1$ , pick  $l_1 \in A$ . Build a bisection bracket: at each stage, let  $m_n = (l_n + u_n)/2$ ; if  $m_n$  is an upper bound for  $A$  set  $u_{n+1} = m_n$ , otherwise set  $l_{n+1} = m_n$  (and pick a witnessing  $a \in A$  with  $a > l_{n+1}$ ). The sequences  $(l_n)$  and  $(u_n)$  are monotone and bounded, so MCT gives limits  $l_n \rightarrow s$  and  $u_n \rightarrow s$  (since  $u_n - l_n = (u_1 - l_1)/2^n \rightarrow 0$ ). The invariants — “ $u_n$  is an upper bound for  $A$ ” and “ $l_n$  is not” — force  $s = \sup A$ .

The construction maintains:

- a *universal* invariant:  $(\forall a \in A)(a \leq u_n)$ ,
- a *failure* invariant:  $(\exists a \in A)(a > l_n)$ .

These two invariants, together with  $u_n - l_n \rightarrow 0$ , pin  $s$  as the least upper bound.

*Remark 1.1.70* (Why this matters structurally). The chain LUB  $\iff$  MCT  $\iff$  NIP  $\iff$  BW  $\iff$  CC (Cauchy Criterion) is the backbone of the equivalence cluster at the heart of real analysis. All five statements assert that  $\mathbb{R}$  has no gaps, each in its own language. The inductive selection technique — building sequences from sets using the  $\varepsilon$ -characterisation — is the constructive face of all five equivalences.

*Remark 1.1.71* (Downstream reuse). The quantifier pattern

$$(\forall \varepsilon > 0)(\exists \text{witness}) \Rightarrow (\exists \text{sequence with controlled tail})$$

reappears in Bolzano–Weierstrass (extracting convergent subsequences),  $\limsup/\liminf$  (tail suprema as a decreasing sequence), compactness (finite subcovers as finite witnesses), and approximation theorems throughout analysis.

## 1.1.4 Walking a Sequence with a Predicate

### Predicate-Walking Toolkit — Quick Reference

| Theorem                             | Predicate $P(n)$                 | Tool                    |
|-------------------------------------|----------------------------------|-------------------------|
| <a href="#">Monotone Subseq.</a>    | $n$ is a peak of $(x_n)$         | Dichotomy on peak count |
| <a href="#">Bolzano–Weierstrass</a> | interval has $\infty$ many terms | Bisection + pigeonhole  |
| <a href="#">Arzelà–Ascoli</a>       | converges at $d_1, \dots, d_m$   | Diagonal extraction     |

## The Infinitely-Often / Eventually Dichotomy

### Definition (Infinitely Often / Eventually)

Let  $(x_n)$  be a sequence and  $P$  a predicate on  $\mathbb{N}$ .  $P$  holds **infinitely often** if  $\{n \in \mathbb{N} : P(n)\}$  is infinite;  $P$  holds **eventually** if there exists  $N \in \mathbb{N}$  such that  $P(n)$  holds for all  $n \geq N$ .

*Remark 1.1.72* (Fully quantified form).  $P$  infinitely often:  $\forall N \in \mathbb{N}, \exists n \geq N, P(n)$ .  
 $P$  eventually:  $\exists N \in \mathbb{N}, \forall n \geq N, P(n)$ .

**Proposition 1.1.73** (The dichotomy). *For any predicate  $P$  on  $\mathbb{N}$ , exactly one of the following holds:*

- (i)  $P$  holds infinitely often.
- (ii)  $\neg P$  holds eventually.

*Go to proof.*

*Remark.*

*Remark 1.1.74* (Fully quantified form).  $(\forall N, \exists n \geq N, P(n)) \vee (\exists N, \forall n \geq N, \neg P(n))$ . The two cases are exhaustive because  $\mathbb{N}$  is infinite: if  $P$  fails for all but finitely many  $n$ , those indices have a maximum  $N$ , and  $\neg P(n)$  holds for all  $n > N$ .

*Remark 1.1.75* (Relation to Infinite Pigeonhole). The index set  $\mathbb{N}$  is partitioned into  $\{n : P(n)\}$  and  $\{n : \neg P(n)\}$ ; an infinite set cannot have both parts finite. The “infinitely often” case provides an inexhaustible reservoir of indices from which to select subsequence terms inductively.

## The Abstract Extraction Template

*Remark 1.1.76* (Four-step pattern). Every predicate-walking argument follows the same skeleton:

1. *Define the predicate.* Identify the property  $P(n)$  to propagate: peak, infinite-occupancy, pointwise convergence, etc.

2. *Apply the dichotomy.* Either  $P$  holds infinitely often, or  $\neg P$  holds eventually. Handle each case.
3. *Extract the subsequence.* In the “infinitely often” case, enumerate indices satisfying  $P$  in increasing order:  $n_1 < n_2 < \dots$ . In the “eventually” case, work in the  $\neg P$ -tail and build inductively, using  $\neg P$  as the fuel.
4. *Verify the property.* Show  $(x_{n_k})$  has the desired property, using the predicate and any additional hypotheses (boundedness, MCT, NIP, equicontinuity).

The critical structural requirement: at each inductive stage, one must find  $n_{k+1} > n_k$  satisfying the relevant condition. The “infinitely often” guarantee ensures the reservoir is never exhausted.

## The Peak Argument: Monotone Subsequence Theorem

### Definition (Peak Index)

An index  $m \in \mathbb{N}$  is a **peak** of  $(x_n)$  if  $x_m \geq x_n$  for all  $n \geq m$ .

*Remark 1.1.77* (Fully quantified form).  $m$  is a peak:  $\forall n \in \mathbb{N}, n \geq m \Rightarrow x_m \geq x_n$ .

### Theorem (Monotone Subsequence)

Every sequence  $(x_n)$  in  $\mathbb{R}$  has a monotone subsequence.

*Remark 1.1.78* (Fully quantified form).  $\forall (x_n) \subset \mathbb{R}, \exists (n_k)$  strictly increasing,  $(x_{n_k})$  is monotone

*Remark 1.1.79* (Proof strategy — peak dichotomy). Apply the dichotomy to  $P(m) = “m$  is a peak.”

**Case 1: infinitely many peaks.** Let  $n_1 < n_2 < \dots$  enumerate the peaks. Since  $n_k$  is a peak and  $n_{k+1} > n_k$ :  $x_{n_k} \geq x_{n_{k+1}}$ . The subsequence is decreasing.

**Case 2: finitely many peaks.** Let  $N$  be an index beyond all peaks. No  $n \geq N$  is a peak, so for each such  $n$  there exists  $m > n$  with  $x_m > x_n$ . Set  $n_1 = N$ . Since  $n_1$  is not a peak,  $\exists n_2 > n_1$  with  $x_{n_2} > x_{n_1}$ . Inductively, this produces a strictly increasing subsequence.

The no-peak condition  $\neg P(n_k)$  is the fuel for each inductive step: it is precisely the statement “there exists a later term exceeding  $x_{n_k}$ ,” which supplies  $n_{k+1}$ .

*Remark 1.1.80* (Consequence: Bolzano–Weierstrass). Every bounded sequence has a bounded monotone subsequence. By MCT, that subsequence converges. Bolzano–Weierstrass follows immediately.

*Remark 1.1.81* (Common error). Assuming Case 2 produces an increasing subsequence “by default.” It does, but only because  $\neg P(n)$  supplies a strictly larger term. If the predicate were defined differently,  $\neg P$  might provide no useful structure. The conclusion depends on the specific content of  $\neg P$ , not the abstract case structure.

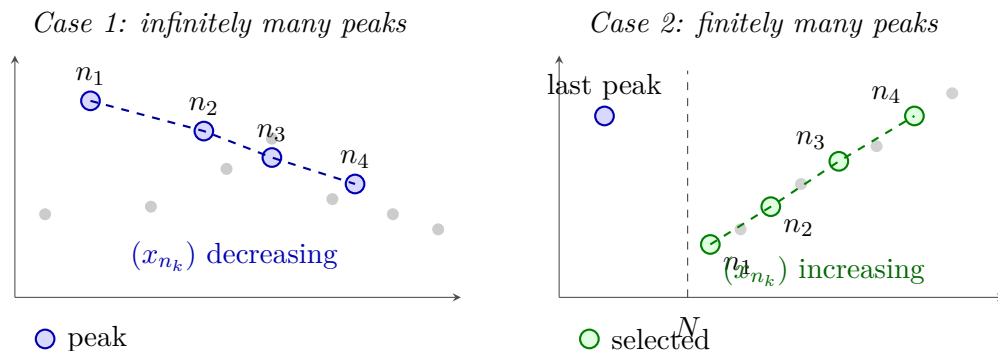


Figure 1.4: The two cases of the Monotone Subsequence Theorem (Section 1.1.4). *Left:* Infinitely many peaks (blue) yield a decreasing subsequence (dashed blue). *Right:* After the last peak, every  $n \geq N$  is a non-peak, so a later term always exceeds the current one; this fuels the inductive construction of an increasing subsequence (dashed green).

## The Infinite-Occupancy Predicate: Bolzano–Weierstrass via Bisection

**Proposition 1.1.82** (Bolzano–Weierstrass via bisection). *Every bounded sequence in  $\mathbb{R}$  has a convergent subsequence. [Go to proof](#).*

*Remark.*

*Remark 1.1.83* (Proof strategy — infinite-occupancy predicate). The predicate at each stage:  $P(I) = \text{“interval } I \text{ contains infinitely many terms of } (x_n)\text{.”}$  Since  $(x_n) \subseteq [a_0, b_0]$  for some bounded interval,  $P([a_0, b_0])$  holds.

**Inductive step.** Given  $I_k = [a_k, b_k]$  with  $P(I_k)$ , bisect at  $m_k = (a_k + b_k)/2$ . The two halves cover  $I_k$ ; at least one contains infinitely many terms. Set  $I_{k+1}$  to be that half. Pick  $x_{n_{k+1}} \in I_{k+1}$  with  $n_{k+1} > n_k$ . The predicate  $P(I_{k+1})$  holds by construction.

**Convergence.** The intervals have  $|I_k| = (b_0 - a_0)/2^k \rightarrow 0$ . By NIP there exists a unique  $L \in \bigcap I_k$ . Since  $a_k \leq x_{n_k} \leq b_k$  and both endpoints converge to  $L$ , the squeeze theorem gives  $x_{n_k} \rightarrow L$ .

*Remark 1.1.84* (Comparison with the peak proof). The peak proof is combinatorial: it operates on the values of the sequence directly and produces a monotone subsequence without requiring boundedness. Convergence follows from MCT.

The bisection proof is geometric: it operates on intervals containing the sequence’s range and produces a subsequence squeezed to a limit directly. Boundedness is required from the outset to produce the initial interval.

Both proofs use the same dichotomy, but in the peak proof it is applied to an index predicate; in the bisection proof, to an interval predicate.

## Diagonal Extraction: Arzelà–Ascoli

*Remark 1.1.85* (Diagonal extraction pattern). The diagonal argument generalises predicate-walking to sequences of functions. Let  $(f_n)$  be bounded and equicontinuous in  $C^0([a, b], \mathbb{R})$ , and let  $D = \{d_1, d_2, \dots\} = \mathbb{Q} \cap [a, b]$ .

**Stage 1.**  $(f_n(d_1))$  is bounded; extract by BW a subsequence  $(f_{1,k})$  converging at  $d_1$ .

**Stage 2.**  $(f_{1,k}(d_2))$  is bounded; extract a sub-subsequence  $(f_{2,k})$  converging at  $d_2$ . As a subsequence of  $(f_{1,k})$ , it still converges at  $d_1$ .

**Stage  $m$ .** Extract  $(f_{m,k})$  as a subsequence of  $(f_{m-1,k})$  converging at  $d_m$ . By induction,  $(f_{m,k})$  converges at  $d_1, \dots, d_m$ .

**Diagonal step.** Define  $g_m = f_{m,m}$ . For any fixed  $j$  and  $m \geq j$ , the term  $g_m = f_{m,m}$  is a term of  $(f_{j,k})$  with index  $k = m \geq j$ . Hence  $g_m(d_j) \rightarrow y_j$ . The sequence  $(g_m)$  converges pointwise on all of  $D$ .

Equicontinuity then propagates pointwise convergence on  $D$  to uniform convergence on  $[a, b]$  via an  $\varepsilon/3$  argument.

*Remark 1.1.86* (Why the diagonal works). At stage  $m$ , the predicate “converges at  $d_1, \dots, d_m$ ” holds for all terms of  $(f_{m,k})$ . The diagonal term  $g_m = f_{m,m}$  is a tail term of every earlier row  $f_{j,k}$  for  $j \leq m$ , so it satisfies the predicate for all  $j \leq m$  simultaneously. Sending  $m \rightarrow \infty$  gives convergence at every  $d_j$ .

|                      | $d_1$ | $d_2$ | $d_3$ | $d_4$ | $\dots$ |
|----------------------|-------|-------|-------|-------|---------|
| Stage 1: $(f_{1,k})$ | conv. | ?     | ?     | ?     |         |
| Stage 2: $(f_{2,k})$ | conv. | conv. | ?     | ?     |         |
| Stage 3: $(f_{3,k})$ | conv. | conv. | conv. | ?     |         |
| Stage 4: $(f_{4,k})$ | conv. | conv. | conv. | conv. |         |

(g<sub>m</sub>) converges pointwise on all of  $D = \{d_1, d_2, \dots\}$

Figure 1.5: Diagonal extraction in the Arzelà–Ascoli argument. Each row  $(f_{m,k})$  converges at  $d_1, \dots, d_m$  (green) and is a subsequence of the row above. The diagonal  $g_m = f_{m,m}$  (red) is a tail of every row, so it converges at every  $d_j$ . Equicontinuity then extends pointwise convergence on the dense set  $D$  to uniform convergence on  $[a, b]$ .

## Limitations of the Technique

*Remark 1.1.87* (The predicate must be decidable at each stage). At each inductive step, one must determine which branch of the dichotomy applies. If the predicate requires knowledge of the limit — a quantity not yet known — the construction cannot proceed. Predicate-walking requires a finitary, combinatorial, or topological condition on the terms already seen.

*Remark 1.1.88* (Multiple simultaneous predicates may exhaust the reservoir). The conjunction  $P \wedge Q$  can hold only finitely often even when  $P$  and  $Q$  each hold infinitely often. The

diagonal argument avoids this by handling one predicate at a time, composing via the diagonal.

*Remark 1.1.89* (Subsequences, not sequences). Predicate-walking yields existence of a subsequence with the desired property, not a conclusion about the full sequence. The Monotone Subsequence Theorem does not assert  $(x_n)$  is monotone, or that the extracted subsequence is unique.

*Remark 1.1.90* (Convergence requires a separate argument). The extracted subsequence has structural properties but not yet a limit. Converting structure into convergence requires a separate theorem: monotone subsequence  $\rightarrow$  MCT (requires boundedness); bisection subsequence  $\rightarrow$  NIP and squeeze; pointwise convergence on  $D \rightarrow$  equicontinuity and  $\varepsilon/3$ .

*Remark 1.1.91* (Quantitative control is absent). Predicate-walking proofs are pure existence arguments. They specify neither the rate of convergence nor which indices are selected. Explicit rates or constructive witnesses require a different approach.

*Remark 1.1.92* (Countability is essential for the diagonal). The Arzelà–Ascoli diagonal requires a countable dense  $D$  so that points can be enumerated and handled one at a time. In non-separable metric spaces (those without a countable dense subset) the argument breaks down.

## The Unifying Principle

*Remark 1.1.93* (Infinite Pigeonhole as the common core). Every predicate-walking argument rests on the fact that an infinite set cannot be partitioned into two finite parts. Applied to  $\mathbb{N}$ : if  $P$  holds only finitely often, the  $P$ -indices have a maximum, and  $\neg P$  holds from that point on. The method reduces a complex existence question to a sequence of binary choices, each justified by the Infinite Pigeonhole Principle. The construction is inductive, automatic, and never runs dry, provided the predicate is well-chosen and the reservoir is genuinely infinite at each stage.

*Remark 1.1.94* (Dependency summary). 1. Monotone Subsequence Theorem — peak dichotomy alone; no completeness.

2. Bolzano–Weierstrass via MCT — Monotone Subseq. + MCT (requires completeness via AoC).

3. Bolzano–Weierstrass via bisection — NIP + squeeze (requires completeness via NIP  $\equiv$  AoC).

4. Arzelà–Ascoli — BW at each diagonal stage + equicontinuity.

The Monotone Subsequence Theorem is the only result in the cluster requiring no completeness. All others depend on completeness via MCT or NIP.

## 1.1.5 Interval Bisection and Nested Intervals

### Bisection and NIP Toolkit — Quick Reference

| Label | Statement                                                                       | Proof method                    |
|-------|---------------------------------------------------------------------------------|---------------------------------|
| NIP   | $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ for nested closed bounded intervals | Supremum of left endpoints      |
| IVT   | $f$ continuous, $f(a) < L < f(b) \Rightarrow \exists c \in (a, b), f(c) = L$    | Bisection + NIP                 |
| UNC   | $\mathbb{R}$ is uncountable                                                     | Nested interval diagonalisation |

## The Nested Interval Property

### Theorem (Nested Interval Property)

For each  $n \in \mathbb{N}$ , let  $I_n = [a_n, b_n]$  be a closed bounded interval with  $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ .  
Then  $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ .

*Remark 1.1.95* (Fully quantified form).  $\forall n \in \mathbb{N}, I_n = [a_n, b_n] \neq \emptyset, [a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \Rightarrow \exists x \in \mathbb{R}, \forall n, x \in I_n$ .

*Remark 1.1.96* (Proof strategy). Let  $A = \{a_n : n \in \mathbb{N}\}$ . Nestedness implies every  $b_n$  is an upper bound for  $A$ . By AoC,  $x = \sup A$  exists. Then  $a_n \leq x \leq b_n$  for all  $n$ , so  $x \in I_n$  for all  $n$ .

*Remark 1.1.97* (Closedness is essential). The open intervals  $(0, 1/n)$  are nested but  $\bigcap (0, 1/n) = \emptyset$ . Closedness ensures the endpoints can serve as witnesses in the AoC argument.

*Remark 1.1.98* (Boundedness is essential). The unbounded closed intervals  $[n, \infty)$  are nested but their intersection is empty. Both hypotheses are necessary.

*Remark 1.1.99* (Singleton intersection). If additionally  $b_n - a_n \rightarrow 0$ , the intersection is a singleton  $\{x\}$ , since  $a_n \leq x \leq b_n$  and the squeeze theorem forces  $a_n \rightarrow x$  and  $b_n \rightarrow x$ . This strengthened form is used in most applications.

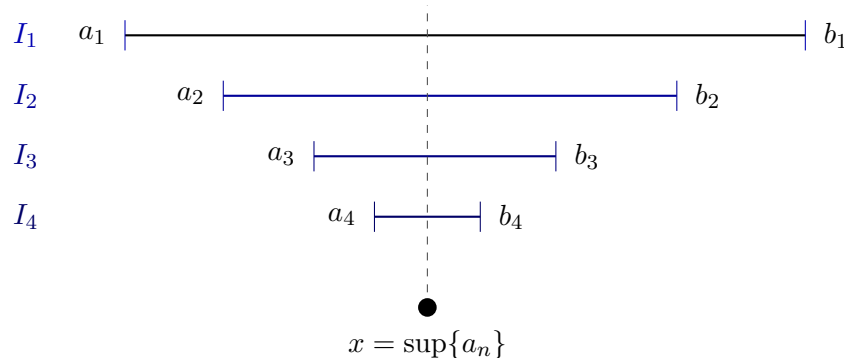


Figure 1.6: The Nested Interval Property (Section 1.1.5). Left endpoints  $a_n$  increase; right endpoints  $b_n$  decrease; each  $b_n$  is an upper bound for  $\{a_n\}$ . By the Axiom of Completeness,  $x = \sup\{a_n\}$  exists and satisfies  $a_n \leq x \leq b_n$  for all  $n$ , so  $x \in \bigcap I_n$ .

*Remark 1.1.100* (Equivalence with completeness). NIP is equivalent to AoC over the Archimedean ordered field axioms. In particular, NIP fails over  $\mathbb{Q}$ : the intervals  $[1, \sqrt{2}] \cap \mathbb{Q}$  shrink to a gap that  $\mathbb{Q}$  cannot fill. ■

## The Bisection Technique

### Definition (Bisection Sequence)

Let  $I_0 = [a_0, b_0]$  carry a property  $P$ . The **bisection sequence**  $(I_n)$  is defined inductively: given  $I_n = [a_n, b_n]$ , let  $m_n = (a_n + b_n)/2$  and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } [a_n, m_n] \text{ inherits } P, \\ [m_n, b_n] & \text{otherwise.} \end{cases}$$

*Remark 1.1.101* (Fully quantified form).  $I_0 = [a_0, b_0]$ ;  $\forall n, I_{n+1} \subseteq I_n$ ;  $|I_n| = (b_0 - a_0)/2^n$ .

*Remark 1.1.102* (Length control). Bisection gives automatic length control:  $|I_n| = (b_0 - a_0)/2^n \rightarrow 0$ . No separate argument for the length condition of NIP is required.

*Remark 1.1.103* (Abstract template). Every bisection proof has the same five-step shape:

1. *Initialize.* Identify  $P$  and starting interval  $I_0$ .
2. *Bisect.* Check which half inherits  $P$ .
3. *Extract.* Apply NIP to obtain  $x \in \bigcap I_n$ ; note  $a_n \rightarrow x$  and  $b_n \rightarrow x$ .
4. *Conclude.* Pass  $P$  through the limit, using continuity, MCT, or squeeze as appropriate.
5. *Verify hypotheses.* Confirm intervals are closed and nested; length control is free from bisection.

*Remark 1.1.104* (Connection to predicate-walking). The bisection technique is a special case of predicate-walking: the predicate  $P(I) = \text{“interval } I \text{ inherits the relevant property”}$  is propagated at each stage by bisecting and keeping the half that satisfies  $P$ . The infinite-pigeonhole dichotomy appears in the Bolzano–Weierstrass proof ([Proposition 1.1.82](#)) but not in the IVT proof, where neither half is discarded based on cardinality — instead the sign-test determines the choice deterministically.



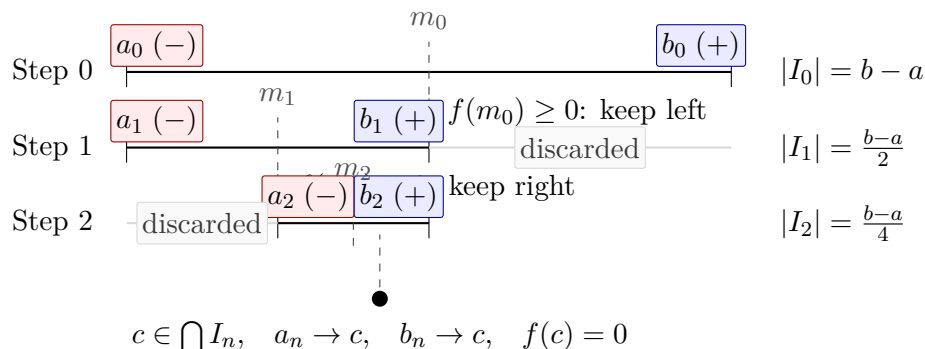


Figure 1.7: Three steps of the bisection proof of the Intermediate Value Theorem ([Proposition 1.1.105](#)) for  $f(a) < 0 < f(b)$ . At each step the half preserving a sign change is retained and the other discarded; lengths halve as  $(b-a)/2^n \rightarrow 0$ . By NIP,  $c \in \bigcap I_n$  exists uniquely; continuity forces  $f(c) = 0$ .

## The Intermediate Value Theorem via Bisection

**Proposition 1.1.105** (Intermediate Value Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. If  $L \in \mathbb{R}$  satisfies  $f(a) < L < f(b)$  or  $f(a) > L > f(b)$ , then there exists  $c \in (a, b)$  with  $f(c) = L$ . [Go to proof](#).*

*Remark.*

*Remark 1.1.106* (Proof strategy — bisection approach). Reduce to  $L = 0$ ,  $f(a) < 0 < f(b)$  (replace  $f$  by  $f - L$ ).

**Property to propagate.**  $f(a_n) < 0$  and  $f(b_n) \geq 0$  — a sign change across the interval.

**Branching rule.** Let  $m_n = (a_n + b_n)/2$ : if  $f(m_n) \geq 0$  set  $[a_{n+1}, b_{n+1}] = [a_n, m_n]$ ; otherwise set  $[a_{n+1}, b_{n+1}] = [m_n, b_n]$ . The sign-change property is preserved in both cases.

**Closing argument.** NIP yields  $c \in \bigcap I_n$  with  $a_n \rightarrow c$  and  $b_n \rightarrow c$ . Since  $f(a_n) < 0$  for all  $n$ : continuity gives  $f(c) \leq 0$ . Since  $f(b_n) \geq 0$  for all  $n$ : continuity gives  $f(c) \geq 0$ . Therefore  $f(c) = 0$ .

*Remark 1.1.107* (Two completeness paths). IVT admits two proofs from completeness:

1. *AoC path.* Set  $K = \{x \in [a, b] : f(x) \leq 0\}$ ,  $c = \sup K$ ; rule out  $f(c) > 0$  and  $f(c) < 0$  using the LUB property.
2. *NIP path.* Bisection as above.

The two paths are equivalent over the Archimedean ordered field axioms.

*Remark 1.1.108* (Applied interpretation). The bisection construction is directly algorithmic: it locates a root to any desired precision in finitely many steps. After  $n$  bisections the root is known to within  $(b-a)/2^n$ . This is the bisection method of numerical analysis.

## Uncountability of $\mathbb{R}$ via Nested Intervals

**Proposition 1.1.109.**  $\mathbb{R}$  is uncountable. [Go to proof.](#)

*Remark.*

*Remark 1.1.110* (Proof strategy — nested interval diagonalisation). Suppose for contradiction  $\mathbb{R} = \{x_1, x_2, \dots\}$ . Build nested intervals  $(I_n)$  inductively with  $I_{n+1} \subseteq I_n$  and  $x_n \notin I_{n+1}$ : at each step  $I_n$  contains two disjoint closed subintervals;  $x_{n+1}$  lies in at most one, so choose the other.

By NIP,  $\bigcap I_n \neq \emptyset$ : pick  $y \in \bigcap I_n$ . Then  $y \neq x_n$  for every  $n$  (since  $y \in I_n$  but  $x_n \notin I_n$ ), contradicting the assumption that  $\{x_n\}$  enumerates  $\mathbb{R}$ .

*Remark 1.1.111* (Relation to Cantor’s diagonal argument). This is the geometric form of Cantor’s diagonalisation: instead of altering a decimal digit, the  $n$ th real is excluded from the  $n$ th nested interval. The witness  $y$  is the interval analogue of the “diagonal” real.

## Summary: When to Use Each Technique

|                                                                 | Feature            | Bisection                    |
|-----------------------------------------------------------------|--------------------|------------------------------|
| <i>Remark 1.1.112</i> (Bisection vs. general nested intervals). | How built          | Halving at midpoints         |
|                                                                 | Length control     | Automatic: $(b_0 - a_0)/2^n$ |
|                                                                 | Typical use        | Property propagation via sig |
|                                                                 | Canonical examples | IVT, BW                      |
|                                                                 | Guarantor          | NIP $\equiv$ AoC             |

*Remark 1.1.113* (Completeness is indispensable). Neither technique works over  $\mathbb{Q}$ . The bisection procedure for  $f(x) = x^2 - 2$  on  $[1, 2]$  runs mechanically over  $\mathbb{Q}$ , but the limiting point  $\sqrt{2}$  is not in  $\mathbb{Q}$ . Every bisection proof is, at bottom, an appeal to the completeness of  $\mathbb{R}$ .

*Remark 1.1.114* (Cross-section of the chapter). The four sections of this chapter are not independent: each builds on the last. Inequality bounding (§1.1.2) requires only the ordered field axioms. Completeness as a constructive engine (§1.1.3) requires AoC and the Archimedean Property. Predicate-walking (§1.1.4) requires completeness via MCT or NIP, depending on the variant. Bisection and NIP (§1.1.5) is the geometric synthesis: it encapsulates completeness in a form that drives the IVT, BW, and uncountability proofs simultaneously, connecting back to both the inductive-selection constructions of §1.1.3 and the predicate-walking patterns of §1.1.4.

## 1.1.6 Subsequence Partition via Residue Classes

### Residue Partition Toolkit — Quick Reference

| Tool                                 | Statement                                                                 | Use                                  |
|--------------------------------------|---------------------------------------------------------------------------|--------------------------------------|
| $k$ -Periodicity Criterion           | $(a_n) \rightarrow L \Leftrightarrow \forall r, (a_{kn+r}) \rightarrow L$ | Prove convergence by analysing $k$ c |
| Forward direction                    | Subseqs. of convergent seqs. converge to $L$                              | Reduce to subsequence analysis       |
| <a href="#">Divergence corollary</a> | Two classes $\rightarrow$ different limits $\Rightarrow (a_n)$ diverges   | Detect divergence by comparing res   |
| Index translation                    | $m = kn + r \Rightarrow n \geq N_r$ once $m \geq k \max_r N_r$            | Core of the backward direction       |

### The Residue-Class Partition

For any sequence  $(a_n)$  and any integer  $k \geq 2$ , the division algorithm partitions  $\mathbb{N}$  into  $k$  disjoint residue classes:

$$\mathbb{N} = \bigsqcup_{r=0}^{k-1} \{kn + r : n \geq 0\}.$$

Each class defines a subsequence:  $(a_{kn+r})_{n \geq 0}$  selects every  $k$ th term of  $(a_n)$  beginning at position  $r$ . These  $k$  subsequences partition  $(a_n)$  exactly — every term appears in precisely one class, order within each class is preserved, and their union recovers  $(a_n)$ .

*Remark 1.1.115* (Fully quantified form of the partition).  $\forall n \in \mathbb{N}, \exists! r \in \{0, \dots, k-1\}, \exists! q \in \mathbb{N}, n = kq + r$ . Hence  $\{(a_{kn+r})_{n \geq 0} : r = 0, \dots, k-1\}$  is a partition of  $(a_n)$  into  $k$  subsequences.

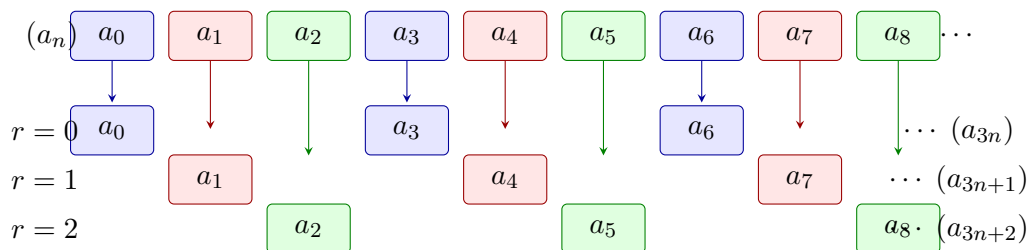


Figure 1.8: The residue-class partition of  $(a_n)$  with modulus  $k = 3$ . Each term is coloured by residue class: blue ( $r = 0$ ), red ( $r = 1$ ), green ( $r = 2$ ). The partition is exact; the  $k$ -Periodicity Criterion ([Section 1.1.6](#)) states that  $(a_n) \rightarrow L$  if and only if each coloured subsequence converges to the same  $L$ .

### The $k$ -Periodicity Convergence Criterion

#### Theorem ( $k$ -Periodicity Convergence Criterion)

Let  $(a_n)$  be a sequence,  $k \geq 2$  an integer, and  $L \in \mathbb{R}$ . The following are equivalent:

- (i)  $(a_n) \rightarrow L$ .
- (ii) For every  $r \in \{0, 1, \dots, k-1\}$ , the subsequence  $(a_{kn+r}) \rightarrow L$ .

*Remark 1.1.116* (Fully quantified form).  $(i) \Leftrightarrow (ii)$ , where:

$(i)$ :  $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m \geq N, |a_m - L| < \varepsilon$ .

$(ii)$ :  $\forall r \in \{0, \dots, k-1\}, \forall \varepsilon > 0, \exists N_r \in \mathbb{N}, \forall n \geq N_r, |a_{kn+r} - L| < \varepsilon$ .

*Remark 1.1.117* (Proof strategy). **Forward direction**  $(i) \Rightarrow (ii)$ . Each  $(a_{kn+r})$  is a subsequence of  $(a_n)$ . Subsequences of a convergent sequence converge to the same limit ([Proposition 1.1.39](#)).

**Backward direction**  $(ii) \Rightarrow (i)$ . The operative direction. Given  $\varepsilon > 0$ , for each residue  $r$  let  $N_r$  satisfy  $|a_{kn+r} - L| < \varepsilon$  for all  $n \geq N_r$ . Set  $N^* = k \cdot \max_r N_r$ . For any  $m \geq N^*$ , write  $m = kn + r$  via the division algorithm. Then:

$$n = \frac{m - r}{k} \geq \frac{N^* - (k - 1)}{k} \geq \frac{k \max_r N_r - (k - 1)}{k} \geq N_r,$$

so  $|a_m - L| = |a_{kn+r} - L| < \varepsilon$ .

*Remark 1.1.118* (The index translation). The key step converts a global index  $m$  to a local index  $n$  within its residue class via  $m = kn + r$ . The factor  $k$  in the definition of  $N^*$  absorbs the offset  $r \leq k-1$  from every class simultaneously, ensuring that  $n \geq N_r$  holds for whichever class  $m$  lands in.

## The Divergence Corollary

**Proposition 1.1.119** (Residue divergence criterion). *If there exist residues  $r, s \in \{0, \dots, k-1\}$  such that  $(a_{kn+r}) \rightarrow L$  and  $(a_{kn+s}) \rightarrow M$  with  $L \neq M$ , then  $(a_n)$  diverges.* [Go to proof](#).

*Remark.*

*Remark 1.1.120* (Fully quantified form).  $(a_{kn+r}) \rightarrow L \wedge (a_{kn+s}) \rightarrow M \wedge L \neq M \Rightarrow (a_n)$  diverges.

*Remark 1.1.121* (Proof strategy). Contrapositively: if  $(a_n) \rightarrow L$ , then every subsequence converges to  $L$  (forward direction of [Section 1.1.6](#)). If two residue-class subsequences converge to different limits,  $(a_n)$  cannot converge.

*Remark 1.1.122* (Canonical example). The sequence  $((-1)^n)$  has  $a_{2n} = 1 \rightarrow 1$  and  $a_{2n+1} = -1 \rightarrow -1$ . Since  $1 \neq -1$ , the sequence diverges — established immediately without any  $\varepsilon$ - $N$  computation.

## The Alternating Series Test via Even/Odd Partition

The  $k = 2$  case of [Section 1.1.6](#) drives the cleanest proof of the Alternating Series Test.

### Theorem (Alternating Series Test)

Let  $(a_n)$  be a positive decreasing sequence with  $a_n \rightarrow 0$ . Then the alternating series  $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$  converges.

*Remark 1.1.123* (Fully quantified form).  $(\forall n, a_n > 0) \wedge (\forall n, a_{n+1} \leq a_n) \wedge (a_n \rightarrow 0) \Rightarrow \exists L, \sum_{n=1}^{\infty} (-1)^{n+1} a_n = L$ .

*Remark 1.1.124* (Proof strategy — even/odd partition). Let  $s_n$  denote the  $n$ th partial sum. Apply [Section 1.1.6](#) with  $k = 2$ : it suffices to show  $(s_{2k}) \rightarrow L$  and  $(s_{2k+1}) \rightarrow L$  for the same  $L$ .

**Even subsequence  $(s_{2k})$ : decreasing and bounded.** Group consecutive pairs:

$$s_{2k} = \sum_{j=1}^k (-a_{2j-1} + a_{2j}).$$

Since  $(a_n)$  is decreasing, each summand  $-a_{2j-1} + a_{2j} \leq 0$ , so  $(s_{2k})$  is decreasing. Also:

$$s_{2k} = -a_1 + \sum_{j=1}^{k-1} (a_{2j} - a_{2j+1}) + a_{2k} \geq -a_1,$$

since each term in the sum is non-negative. Thus  $(s_{2k})$  is decreasing and bounded below by  $-a_1$ . By MCT,  $s_{2k} \rightarrow L$  for some  $L \geq -a_1$ .

**Odd subsequence  $(s_{2k+1})$ : linked to the even.** Observe  $s_{2k+1} = s_{2k} + a_{2k+1}$ . Since  $s_{2k} \rightarrow L$  and  $a_{2k+1} \rightarrow 0$ , the Algebraic Limit Theorem gives  $s_{2k+1} \rightarrow L$ .

**Applying the criterion.** Both residue-class subsequences converge to the same  $L$ . By [Section 1.1.6](#) with  $k = 2$ ,  $(s_n) \rightarrow L$ .

*Remark 1.1.125* (Why the partition is the right tool). The even and odd partial sums have structurally different characters: the even sums are monotone (enabling MCT), the odd sums are then controlled via the algebraic relationship  $s_{2k+1} = s_{2k} + a_{2k+1}$ . No single uniform argument handles both. The partition separates the two cases into independent analyses, then the criterion reassembles the conclusion.

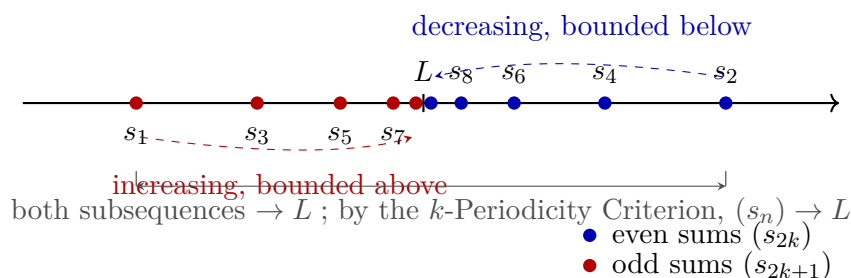


Figure 1.9: Even and odd partial sums of an alternating series  $\sum (-1)^{n+1} a_n$  with  $(a_n)$  positive, decreasing, and tending to zero. Even sums  $(s_{2k})$  (blue) decrease and are bounded below; by MCT they converge to  $L$ . Odd sums  $(s_{2k+1})$  (red) satisfy  $s_{2k+1} = s_{2k} + a_{2k+1} \rightarrow L$ . Since both residue classes (mod  $k = 2$ ) converge to  $L$ , the  $k$ -Periodicity Criterion ([Section 1.1.6](#)) gives  $(s_n) \rightarrow L$ .

*Remark 1.1.126* (Alternative proofs). The Alternating Series Test admits three proofs from different completeness avatars:

1. *Cauchy criterion* — show  $(s_n)$  is eventually  $\varepsilon$ -steady directly.

2. *NIP* — the nested intervals  $[s_{2k}, s_{2k-1}]$  shrink to a point by the Nested Interval Property.
3. *MCT + residue partition* — as above.

The partition proof is the most transparent structurally: the monotone behaviour and the limit identity are isolated into the two residue classes rather than handled simultaneously.

## Higher $k$ and Choosing the Right Period

For sequences with period  $k > 2$ , the criterion applies directly. The sequence  $a_n = \cos(2\pi n/3)$  takes values  $1, -\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}, \dots$ , cycling with period 3. Partitioning modulo 3:

$$\begin{aligned} a_{3n} &= 1 \rightarrow 1, \\ a_{3n+1} &= -\frac{1}{2} \rightarrow -\frac{1}{2}, \\ a_{3n+2} &= -\frac{1}{2} \rightarrow -\frac{1}{2}. \end{aligned}$$

The three limits differ ( $1 \neq -\frac{1}{2}$ ), so the sequence diverges by [Proposition 1.1.119](#).

*Remark 1.1.127* (Choosing  $k$  to match the period). The right choice of  $k$  is the natural period of the sequence’s structure. Using  $k = 2$  for a period-3 sequence produces residue classes that are themselves period-3 subsequences; convergence of each class then requires a further partition with  $k = 3$ . Choosing  $k$  to match the period compresses the argument to a single application of the criterion.

*Remark 1.1.128* (Applying the criterion to prove convergence). To prove  $(a_n) \rightarrow L$  via the criterion, one must:

1. Select  $k$  matching the period of the sequence’s behaviour.
2. Prove  $(a_{kn+r}) \rightarrow L$  for each  $r \in \{0, \dots, k-1\}$ , using whatever tool fits that residue class.
3. Invoke [Section 1.1.6](#) to conclude  $(a_n) \rightarrow L$ .

The method does not produce  $L$  — it confirms a candidate. Identifying  $L$  requires separate work (direct computation, MCT applied to a class, etc.) before the criterion is invoked.

## Contrast with Predicate-Walking

*Remark 1.1.129* (Dual logical directions). The residue partition technique and predicate-walking (§1.1.4) operate in opposite logical directions.

| Feature           | Predicate-walking             | Residue partition                   |
|-------------------|-------------------------------|-------------------------------------|
| Flow              | Whole $\rightarrow$ one part  | All parts $\rightarrow$ whole       |
| Goal              | Property of extracted subseq. | Convergence of full $(a_n)$         |
| Coverage          | One “good” subsequence        | All $k$ residue classes             |
| Divergence use    | Two classes, different limits | Same: two classes, different limits |
| Completeness role | Via MCT or NIP                | Via MCT or algebraic limit theorem  |

Predicate-walking extracts what is needed; the residue partition reassembles from all pieces. The divergence criterion is available in both frameworks: two subsequences with different limits immediately imply divergence.

## Limitations

*Remark 1.1.130* (The same-limit condition is non-negotiable). The backward direction of [Section 1.1.6](#) requires all  $k$  residue-class subsequences to converge to the *same*  $L$ . There is no weakening: a convergent sequence has all its subsequences converging to the identical limit, so any two classes converging to different limits already certify divergence. The technique cannot produce convergence from mixed limits.

*Remark 1.1.131* (The limit must be known in advance). The criterion confirms a candidate limit  $L$ ; it does not produce  $L$  from scratch. In practice,  $L$  is identified by direct computation on one residue class (e.g., the even partial sums converge to  $L$  by MCT), then the criterion assembles global convergence.

*Remark 1.1.132* (Exact periodicity is required). The technique applies when the sequence's qualitative behaviour repeats with exact period  $k$  in the index. For sequences whose oscillation decays without exact periodicity, the residue classes are not individually tractable, and a direct  $\varepsilon$ - $N$  argument or a squeeze is more appropriate.

*Remark 1.1.133* (Common error). Applying the criterion with  $k$  smaller than the true period, then concluding convergence after verifying only the even residue class. If the odd class is not verified, the conclusion does not follow. Every residue class must be checked.

*Remark 1.1.134* (Forward connections). The residue partition technique reappears in:

- **Cesàro means** — the Cesàro average of a sequence  $(a_n)$  can be analysed by partitioning the partial sums of the average into residue classes.
- **Fourier series** — the convergence of partial sums of a Fourier series is often established by analysing even and odd partial sums (or partial sums along specific subsequences of the approximation order).
- **Recursively defined sequences** — when a recurrence has period  $k$ , the residue-class subsequences each satisfy a simpler recurrence, enabling separate analysis.

## 1.2 Asymptotic Notation

### Asymptotic Notation — Quick Reference

| Concept/Statement       | Meaning/Role                                                  |
|-------------------------|---------------------------------------------------------------|
| Little- $o$ at a point  | One function is negligible relative to another near a point.  |
| Little- $o$ at infinity | Negligibility along an unbounded input range.                 |
| Increment little- $o$   | The local error notation $r(h) = o(h)$ as $h \rightarrow 0$ . |
| Quotient criterion      | Little- $o$ is detected by a quotient tending to 0.           |
| Sum rule                | Negligible errors add.                                        |
| Bounded factor rule     | Bounded multipliers preserve negligible errors.               |

### Little- $o$ Notation

Asymptotic notation separates the main term of an expression from the part that becomes negligible in a limiting process. The notation  $f = o(g)$  means that  $f$  is smaller than every fixed multiple of  $g$  near the limiting point. It does not say that  $f$  is zero. It says that  $f$  is negligible relative to the chosen scale  $g$ .

**Definition 1.2.1** (Little- $o$  at a Point). Let  $a \in \mathbb{R}$ , let  $f$  and  $g$  be real-valued functions defined on a punctured neighbourhood of  $a$ , and suppose  $g(x) \neq 0$  for all  $x$  sufficiently close to  $a$  with  $x \neq a$ . We write

$$f(x) = o(g(x)) \quad (x \rightarrow a)$$

if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$0 < |x - a| < \delta \implies |f(x)| \leq \varepsilon |g(x)|.$$

*Remark* (Standard quantified statement).

$$f(x) = o(g(x)) \ (x \rightarrow a) \iff \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \\ (0 < |x - a| < \delta \implies |f(x)| \leq \varepsilon |g(x)|).$$

*Remark* (Interpretation). The scale  $g$  is the unit of comparison. The assertion  $f = o(g)$  says that no matter how small a tolerance multiple  $\varepsilon g$  is chosen,  $f$  eventually fits inside that multiple.

**Definition 1.2.2** (Little- $o$  at Infinity). Let  $f$  and  $g$  be real-valued functions defined on some interval  $(M, \infty)$ , and suppose  $g(x) \neq 0$  for all sufficiently large  $x$ . We write

$$f(x) = o(g(x)) \quad (x \rightarrow \infty)$$

if and only if for every  $\varepsilon > 0$  there exists  $R > 0$  such that

$$x > R \implies |f(x)| \leq \varepsilon |g(x)|.$$

*Remark* (Standard quantified statement).

$$f(x) = o(g(x)) \ (x \rightarrow \infty) \iff \forall \varepsilon > 0 \ \exists R > 0 \ \forall x \\ (x > R \implies |f(x)| \leq \varepsilon |g(x)|).$$



*Remark* (Interpretation). The point version and the infinity version have the same logical shape. Only the neighbourhood changes: small punctured intervals around  $a$  are replaced by tails  $(R, \infty)$ .

**Definition 1.2.3** (Increment Little-o). Let  $r$  be a real-valued function defined for all sufficiently small nonzero  $h$ . We write

$$r(h) = o(h) \quad (h \rightarrow 0)$$

if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$0 < |h| < \delta \implies |r(h)| \leq \varepsilon|h|.$$

*Remark* (Standard quantified statement).

$$r(h) = o(h) \ (h \rightarrow 0) \iff \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall h \ (0 < |h| < \delta \implies |r(h)| \leq \varepsilon|h|).$$

*Remark* (Interpretation). This is the form used in first-order approximation. A remainder  $r(h)$  is smaller than the increment  $h$  itself, so the quotient  $r(h)/h$  tends to zero.

*Remark* (Dependencies).

- [Little-o at a Point](#)

**Proposition 1.2.4** (Little-o Quotient Characterisation). *Let  $a \in \mathbb{R}$ , and let  $f$  and  $g$  be real-valued functions defined on a punctured neighbourhood of  $a$ , with  $g(x) \neq 0$  for  $x$  sufficiently close to  $a$  and  $x \neq a$ . Then*

$$f(x) = o(g(x)) \ (x \rightarrow a) \iff \frac{f(x)}{g(x)} \rightarrow 0 \ (x \rightarrow a).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$[\forall \varepsilon > 0 \ \exists \delta > 0 \ 0 < |x - a| < \delta \implies |f(x)| \leq \varepsilon|g(x)|] \iff \frac{f(x)}{g(x)} \rightarrow 0.$$

*Remark* (Interpretation). The quotient form is often the fastest test for little-o. The epsilon form is often the safest proof form because it avoids manipulating a quotient until the nonvanishing of the scale has been checked.

*Remark* (Dependencies).

- [Little-o at a Point](#)

**Proposition 1.2.5** (Sum Rule for Little-o). *If  $f_1(x) = o(g(x))$  and  $f_2(x) = o(g(x))$  as  $x \rightarrow a$ , then*

$$f_1(x) + f_2(x) = o(g(x)) \quad (x \rightarrow a).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$f_1 = o(g) \wedge f_2 = o(g) \implies f_1 + f_2 = o(g).$$

*Remark* (Interpretation). Negligible errors can be added. The proof is the usual epsilon-budget split: make each error smaller than half the final tolerance.

*Remark* (Dependencies).

- [Little-o at a Point](#)

**Proposition 1.2.6** (Bounded Factor Rule for Little-o). *If  $f(x) = o(g(x))$  as  $x \rightarrow a$  and  $m$  is bounded near  $a$ , then*

$$m(x)f(x) = o(g(x)) \quad (x \rightarrow a).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$f = o(g) \wedge \exists M > 0 \exists \eta > 0 \forall x (0 < |x - a| < \eta \implies |m(x)| \leq M) \implies mf = o(g).$$

*Remark* (Interpretation). A bounded multiplier cannot turn a negligible error into a main-order term. It only changes the constant in the epsilon estimate.

*Remark* (Dependencies).

- [Little-o at a Point](#)

## 1.3 Inequality

### Inequality — Quick Reference

| Concept/Statement                        | Meaning/Role                                         |
|------------------------------------------|------------------------------------------------------|
| <a href="#">Add both sides</a>           | Adding the same term preserves strict inequalities.  |
| <a href="#">Nonstrict add both sides</a> | Addition also preserves nonstrict inequalities.      |
| <a href="#">Add inequalities</a>         | Pairwise strict inequalities add.                    |
| <a href="#">Mixed addition</a>           | Strict and nonstrict hypotheses combine predictably. |
| <a href="#">Multiply by positive</a>     | Positive factors preserve order.                     |
| <a href="#">Multiply by negative</a>     | Negative factors reverse order.                      |
| <a href="#">Squeeze theorem</a>          | Double bounding by one value forces equality.        |
| <a href="#">Reciprocal order</a>         | Reciprocals reverse positive inequalities.           |
| <a href="#">Square monotonicity</a>      | Squaring preserves order on nonnegative inputs.      |
| <a href="#">AM–GM</a>                    | Arithmetic mean dominates geometric mean.            |
| <a href="#">Bernoulli inequality</a>     | Powers of $1 + x$ dominate the tangent line.         |

## Inequality

**Theorem 1.3.1** (Addition Preserves Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \Rightarrow a + c < b + c.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \Rightarrow a + c < b + c.$$

*Remark* (Dependencies).

[Ordered field](#), [ordered field axioms](#), [additive cancellation](#).

**Theorem 1.3.2** (Addition Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \Rightarrow a + c \leq b + c.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \Rightarrow a + c \leq b + c.$$

*Remark* (Dependencies).

[Ordered field](#), [addition preserves strict inequality](#).

**Theorem 1.3.3** (Addition of Strict Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a < b \wedge c < d \Rightarrow a + c < b + d.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a < b \wedge c < d \Rightarrow a + c < b + d.$$

*Remark* (Dependencies).

[addition preserves strict inequality](#), [transitivity of strict inequality](#).

**Theorem 1.3.4** (Addition of Nonstrict Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a \leq b \wedge c \leq d \Rightarrow a + c \leq b + d.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \wedge c \leq d \Rightarrow a + c \leq b + d.$$

*Remark* (Dependencies).

[addition preserves nonstrict inequality](#).

**Theorem 1.3.5** (Mixed Addition of Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a \leq b \wedge c < d \Rightarrow a + c < b + d.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \wedge c < d \Rightarrow a + c < b + d.$$

*Remark* (Dependencies).

[addition preserves nonstrict inequality](#), [addition preserves strict inequality](#), [mixed transitivity](#).

**Theorem 1.3.6** (Multiplication by a Positive Scalar Preserves Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge 0 < c \Rightarrow ac < bc.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \wedge 0 < c \Rightarrow ac < bc.$$

*Remark* (Dependencies).

[Ordered field](#), multiply positive, [positive cone](#).

**Theorem 1.3.7** (Multiplication by a Negative Scalar Reverses Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge c < 0 \Rightarrow ac > bc.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \wedge c < 0 \Rightarrow ac > bc.$$

*Remark* (Dependencies).

[Ordered field](#), multiply negative, [positive cone](#).

**Theorem 1.3.8** (Multiplication by a Positive Scalar Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge 0 < c \Rightarrow ac \leq bc.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge 0 < c \Rightarrow ac \leq bc.$$

*Remark* (Dependencies).

[multiplication by positive preserves strict inequality](#).

**Theorem 1.3.9** (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge 0 \leq c \Rightarrow ac \leq bc.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge 0 \leq c \Rightarrow ac \leq bc.$$

*Remark* (Interpretation). This extends [the positive-scalar result](#) to include  $c = 0$ , where both sides become 0 and the inequality holds trivially.

*Remark* (Dependencies).

[multiplication by positive preserves nonstrict inequality](#), [ordered field](#).

#### Theorem (Squeeze)

**Theorem 1.3.10** (Squeeze). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \leq c \wedge a = c \Rightarrow b = a.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \leq c \wedge a = c \Rightarrow b = a.$$

*Remark* (Interpretation). The squeeze principle forces equality when a value is simultaneously bounded above and below by equal quantities. The continuous analogue (if  $f(x) \leq g(x) \leq h(x)$  and  $f(x) \rightarrow L$  and  $h(x) \rightarrow L$  then  $g(x) \rightarrow L$ ) is the central tool for computing limits. The algebraic version stated here is its foundation.

*Remark* (Dependencies).

[Ordered field](#), [additive cancellation](#).

**Theorem 1.3.11** (Transitivity of Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge b < c \Rightarrow a < c.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \wedge b < c \Rightarrow a < c.$$

*Remark* (Dependencies).

[Ordered field](#), [positive cone](#).

**Theorem 1.3.12** (Mixed Transitivity). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge b < c \Rightarrow a < c.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge b < c \Rightarrow a < c.$$

*Remark* (Dependencies).

[Ordered field](#), [transitivity of strict inequality](#).

**Theorem 1.3.13** (Reciprocal Reverses Strict Inequality for Positives). *For all  $a, b \in \mathbb{R}$ ,*

$$0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

*Remark* (Dependencies).

[Ordered field](#), [inverse positive](#), [reciprocal order for positive elements](#), [multiplication by positive preserves strict inequality](#).

**Theorem 1.3.14** (Reciprocal Equivalence for Positives). *For all  $a, b \in \mathbb{R}$  with  $a > 0$  and  $b > 0$ ,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad a > 0 \wedge b > 0 \Rightarrow (a < b \iff \frac{1}{b} < \frac{1}{a}).$$

*Remark* (Dependencies).

[reciprocal reverses strict inequality for positives](#), [reciprocal order for positive elements](#).

**Theorem 1.3.15** (Square Is Strictly Monotone on Nonneg Reals). *For all  $a, b \in \mathbb{R}$ ,*

$$0 \leq a < b \Rightarrow a^2 < b^2.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a < b \Rightarrow a^2 < b^2.$$

*Remark* (Dependencies).

[Ordered field](#), [addition of strict inequalities](#), [multiplication by positive preserves strict inequality](#), [squares are nonnegative](#).

**Theorem 1.3.16** (Square Root Is Monotone on Nonneg Reals). *For all  $a, b \in \mathbb{R}$ ,*

$$0 \leq a \leq b \Rightarrow \sqrt{a} \leq \sqrt{b}.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a \leq b \Rightarrow \sqrt{a} \leq \sqrt{b}.$$

*Remark* (Audit note). This theorem depends on `def:square-root`, which does not yet appear in the repository label inventory. That definition must be created and registered before this result can be fully formalised.

*Remark* (Dependencies).

`def:square-root` (not yet registered — see audit note above), [square is strictly monotone on nonneg reals](#).

**Theorem 1.3.17** (AM-GM Inequality for Two Terms). *For all  $a, b \geq 0$ ,*

$$\sqrt{ab} \leq \frac{a+b}{2}.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad a \geq 0 \wedge b \geq 0 \Rightarrow \sqrt{ab} \leq \frac{a+b}{2}.$$

*Remark* (Interpretation). The arithmetic–geometric mean inequality for two terms is the simplest case of the AM-GM family. The standard proof proceeds by squaring both sides (using [square monotonicity](#)) and reducing to  $(\sqrt{a} - \sqrt{b})^2 \geq 0$ , which follows from [squares being nonnegative](#).

*Remark* (Dependencies).

[squares are nonnegative](#), [square is strictly monotone on nonneg reals](#), `def:square-root` (not yet registered — see audit note in [square root monotonicity](#)).

#### Theorem (Bernoulli's Inequality)

**Theorem 1.3.18** (Bernoulli's Inequality). *For all  $x \in \mathbb{R}$  with  $x \geq -1$  and all  $n \in \mathbb{N}$ ,*

$$(1+x)^n \geq 1+nx.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R}, \quad \forall n \in \mathbb{N}, \quad x \geq -1 \Rightarrow (1+x)^n \geq 1+nx.$$

*Remark* (Interpretation). Bernoulli's inequality is proved by induction and is a key estimate in analysis: it provides a linear lower bound for exponential growth and is used to establish the divergence of the harmonic series, bounds on compound interest, and properties of the exponential function.

*Remark* (Dependencies).

Ordered field, multiplication by nonneg preserves nonstrict inequality, addition preserves nonstrict inequality.

## 1.4 Modulus

### Modulus — Quick Reference

| Concept/Statement            | Meaning/Role                                               |
|------------------------------|------------------------------------------------------------|
| Absolute value               | Magnitude stripped of sign.                                |
| Nonnegativity                | Modulus is always nonnegative.                             |
| Zero characterisation        | Only 0 has modulus 0.                                      |
| Self or negation             | $ a $ is either $a$ or $-a$ .                              |
| Symmetry                     | Negation does not change magnitude.                        |
| Product rule                 | Modulus is multiplicative.                                 |
| Quotient rule                | Division respects modulus when the denominator is nonzero. |
| Bounds by modulus            | $a$ lies between $- a $ and $ a $ .                        |
| Nonstrict modulus inequality | $ a  \leq r$ is equivalent to a two-sided bound.           |
| Strict modulus inequality    | $ a  < r$ is equivalent to a strict two-sided bound.       |
| Reverse triangle inequality  | Magnitudes vary no faster than distance.                   |
| Finite triangle inequality   | The modulus of a finite sum is bounded by summed moduli.   |

## Modulus

### Definition (Absolute Value)

**Definition 1.4.1** (Absolute Value). Let  $a \in \mathbb{R}$ . The *absolute value* (or *modulus*) of  $a$  is

$$|a| := \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Equivalently,  $|a| := \max(a, -a)$ .

*Remark* (Standard quantified statement).

$$|a| = \max(a, -a) := \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

*Remark* (Interpretation). The absolute value measures magnitude independent of sign. On the real line it equals the distance from  $a$  to 0:  $|a| = d(a, 0)$  under the standard metric. The two-branch definition partitions  $\mathbb{R}$  at 0 and reflects negative values to positive ones.



*Remark* (Dependencies).

Ordered field,  $\mathbb{R}$  is an ordered field.

**Theorem 1.4.2** (Absolute Value Is Nonnegative). *For all  $a \in \mathbb{R}$ ,*

$$|a| \geq 0.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| \geq 0.$$

*Remark* (Dependencies).

Absolute value, squares are nonnegative, ordered field.

**Theorem 1.4.3** (Absolute Value Zero Iff Zero). *For all  $a \in \mathbb{R}$ ,*

$$|a| = 0 \iff a = 0.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| = 0 \iff a = 0.$$

*Remark* (Dependencies).

Absolute value, absolute value is nonnegative, ordered field.

**Theorem 1.4.4** (Absolute Value Equals Self or Negation). *For all  $a \in \mathbb{R}$ , either  $|a| = a$  or  $|a| = -a$ .*

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| = a \vee |a| = -a.$$

*Remark* (Dependencies).

Absolute value.

**Theorem 1.4.5** (Absolute Value Is Symmetric). *For all  $a \in \mathbb{R}$ ,*

$$|-a| = |a|.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |-a| = |a|.$$

*Remark* (Dependencies).

[Absolute value](#), double negation.

**Theorem 1.4.6** (Absolute Value of a Product). *For all  $a, b \in \mathbb{R}$ ,*

$$|ab| = |a| |b|.$$

*Remark* (Proof). [Go to proof](#).

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |ab| = |a| |b|.$$

*Remark* (Dependencies).

[Absolute value](#), negation of product, product of negatives, [squares are nonnegative](#).

**Theorem 1.4.7** (Absolute Value of a Quotient). *For all  $a \in \mathbb{R}$  and  $b \in \mathbb{R}$  with  $b \neq 0$ ,*

$$\left| \frac{a}{b} \right| = \frac{|a|}{|b|}.$$

*Remark* (Proof). [Go to proof](#).

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \forall b \in \mathbb{R} \setminus \{0\}, \quad \left| \frac{a}{b} \right| = \frac{|a|}{|b|}.$$

*Remark* (Dependencies).

[Absolute value](#), [absolute value of a product](#), [absolute value zero iff zero](#), [uniqueness of multiplicative inverse](#).

**Theorem 1.4.8** (Bound by Modulus). *For all  $a \in \mathbb{R}$ ,*

$$-|a| \leq a \leq |a|.$$

*Remark* (Proof). [Go to proof](#).

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad -|a| \leq a \leq |a|.$$

*Remark* (Dependencies).

[Absolute value](#), [absolute value is nonnegative](#), [ordered field](#).

#### Theorem (Nonstrict Modulus Inequality)

**Theorem 1.4.9** (Nonstrict Modulus Inequality). *For all  $a \in \mathbb{R}$  and  $r \geq 0$ ,*

$$|a| \leq r \iff -r \leq a \leq r.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \forall r \in \mathbb{R}, r \geq 0 \Rightarrow (|a| \leq r \iff -r \leq a \leq r).$$

*Remark* (Interpretation). This biconditional is the primary tool for replacing a modulus inequality with a pair of simultaneous linear inequalities. It underlies the  $\varepsilon$ - $\delta$  manipulations throughout analysis: the condition  $|x - a| \leq \varepsilon$  is equivalent to  $a - \varepsilon \leq x \leq a + \varepsilon$ .

*Remark* (Dependencies).

[Absolute value](#), [bound by modulus](#), [ordered field](#).

**Theorem 1.4.10** (Strict Modulus Inequality). *For all  $a \in \mathbb{R}$  and  $r > 0$ ,*

$$|a| < r \iff -r < a < r.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R}, \forall r \in \mathbb{R}, r > 0 \Rightarrow (|a| < r \iff -r < a < r).$$

*Remark* (Dependencies).

[Absolute value](#), [nonstrict modulus inequality](#), [ordered field](#).

#### Theorem (Triangle Inequality)

**Theorem** (Triangle Inequality). *For all  $a, b \in \mathbb{R}$ ,*

$$|a + b| \leq |a| + |b|.$$

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |a + b| \leq |a| + |b|.$$

*Remark* (Interpretation). The triangle inequality is the foundational estimate of analysis. Every convergence argument, continuity proof, and metric-space result that requires bounding a sum reduces, at some step, to this inequality. The name reflects its geometric meaning: the length of one side of a triangle does not exceed the sum of the other two.

**Theorem 1.4.11** (Reverse Triangle Inequality). *For all  $a, b \in \mathbb{R}$ ,*

$$||a| - |b|| \leq |a - b|.$$

*Remark* (Proof). [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad ||a| - |b|| \leq |a - b|.$$

*Remark* (Interpretation). The reverse triangle inequality provides a lower bound on  $|a - b|$  in terms of the difference of moduli. It is used to bound the modulus of a difference from below, particularly when proving that a function is not uniformly continuous or that a sequence does not converge.

*Remark* (Dependencies).

Absolute value, [triangle inequality](#), absolute value is symmetric.

**Theorem 1.4.12** (Generalised Triangle Inequality). *For all  $n \in \mathbb{N}$  and  $a_1, \dots, a_n \in \mathbb{R}$ ,*

$$\left| \sum_{k=1}^n a_k \right| \leq \sum_{k=1}^n |a_k|.$$

*Remark* (Proof). [Go to proof](#).

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N}, \forall a_1, \dots, a_n \in \mathbb{R}, \quad \left| \sum_{k=1}^n a_k \right| \leq \sum_{k=1}^n |a_k|.$$

*Remark* (Dependencies).

Absolute value, [triangle inequality](#).

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Order arithmetic). *Let  $a, b, c, d \in \mathbb{R}$ .*

- (i) If  $a \leq b$  and  $c \leq d$ , then  $a + c \leq b + d$ .*
- (ii) If  $a \leq b$  and  $c > 0$ , then  $ac \leq bc$ .*
- (iii) If  $a \leq b$  and  $c < 0$ , then  $ac \geq bc$ .*
- (iv)  $|x| \leq y$  (with  $y \geq 0$ ) if and only if  $-y \leq x \leq y$ .*

*Professional Standard Proof.*

TODO: professional standard proof for order-arithmetic. □

*Detailed Learning Proof.*

TODO: detailed learning proof for order-arithmetic. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Convergent sequences are bounded). *If  $(b_n) \rightarrow b$ , then there exists  $M > 0$  such that  $|b_n| \leq M$  for all  $n \in \mathbb{N}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for convergent-bounded. □

*Detailed Learning Proof.*

TODO: detailed learning proof for convergent-bounded. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (-characterisation of ). *Let  $S \subset \mathbb{R}$  be nonempty and bounded above, and let  $s = \sup S$ . Then for every  $\varepsilon > 0$  there exists  $x \in S$  with  $x > s - \varepsilon$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eps-char-sup. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eps-char-sup. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Monotone Approximation of Bounds). *Let  $S \subset \mathbb{R}$  be nonempty and bounded. Then there exist monotone sequences  $(x_n), (y_n) \subset S$  with  $x_n \rightarrow \sup S$  and  $y_n \rightarrow \inf S$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-approx-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-approx-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (The dichotomy). *For any predicate  $P$  on  $\mathbb{N}$ , exactly one of the following holds:*

(i)  $P$  holds infinitely often.

(ii)  $\neg P$  holds eventually.

*Professional Standard Proof.*

TODO: professional standard proof for io-ev-dichotomy. □

*Detailed Learning Proof.*

TODO: detailed learning proof for io-ev-dichotomy. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Bolzano–Weierstrass via bisection). *Every bounded sequence in  $\mathbb{R}$  has a convergent subsequence.*

*Professional Standard Proof.*

TODO: professional standard proof for bw-bisection. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bw-bisection. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Intermediate Value Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. If  $L \in \mathbb{R}$  satisfies  $f(a) < L < f(b)$  or  $f(a) > L > f(b)$ , then there exists  $c \in (a, b)$  with  $f(c) = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for ivt-bisection. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ivt-bisection. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (r-uncountable).  $\mathbb{R}$  is uncountable.

*Professional Standard Proof.*

TODO: professional standard proof for r-uncountable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for r-uncountable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Residue divergence criterion). *If there exist residues  $r, s \in \{0, \dots, k-1\}$  such that  $(a_{kn+r}) \rightarrow L$  and  $(a_{kn+s}) \rightarrow M$  with  $L \neq M$ , then  $(a_n)$  diverges.*

*Professional Standard Proof.*

TODO: professional standard proof for residue-divergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for residue-divergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Little-o Quotient Characterisation). *Let  $a \in \mathbb{R}$ , and let  $f$  and  $g$  be real-valued functions defined on a punctured neighbourhood of  $a$ , with  $g(x) \neq 0$  for  $x$  sufficiently close to  $a$  and  $x \neq a$ . Then*

$$f(x) = o(g(x)) \ (x \rightarrow a) \iff \frac{f(x)}{g(x)} \rightarrow 0 \ (x \rightarrow a).$$

*Professional Standard Proof.*

TODO: professional standard proof for little-o-quotient-characterization. □

*Detailed Learning Proof.*

TODO: detailed learning proof for little-o-quotient-characterization. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Sum Rule for Little-o). *If  $f_1(x) = o(g(x))$  and  $f_2(x) = o(g(x))$  as  $x \rightarrow a$ , then*

$$f_1(x) + f_2(x) = o(g(x)) \quad (x \rightarrow a).$$

*Professional Standard Proof.*

TODO: professional standard proof for little-o-sum-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for little-o-sum-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Bounded Factor Rule for Little-o). *If  $f(x) = o(g(x))$  as  $x \rightarrow a$  and  $m$  is bounded near  $a$ , then*

$$m(x)f(x) = o(g(x)) \quad (x \rightarrow a).$$

*Professional Standard Proof.*

TODO: professional standard proof for little-o-bounded-factor-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for little-o-bounded-factor-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Addition Preserves Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \Rightarrow a + c < b + c.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-add-both-sides. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-add-both-sides. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Addition Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \Rightarrow a + c \leq b + c.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-nonstrict-add-both-sides. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-nonstrict-add-both-sides. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Addition of Strict Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a < b \wedge c < d \Rightarrow a + c < b + d.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-add-inequalities. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-add-inequalities. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Addition of Nonstrict Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a \leq b \wedge c \leq d \Rightarrow a + c \leq b + d.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-nonstrict-add-inequalities. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-nonstrict-add-inequalities. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Mixed Addition of Inequalities). *For all  $a, b, c, d \in \mathbb{R}$ ,*

$$a \leq b \wedge c < d \Rightarrow a + c < b + d.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-mixed-add. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-mixed-add. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Multiplication by a Positive Scalar Preserves Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge 0 < c \Rightarrow ac < bc.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-multiply-positive. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-multiply-positive. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Multiplication by a Negative Scalar Reverses Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge c < 0 \Rightarrow ac > bc.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-multiply-negative. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-multiply-negative. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Multiplication by a Positive Scalar Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge 0 < c \Rightarrow ac \leq bc.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-nonstrict-multiply-positive. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-nonstrict-multiply-positive. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge 0 \leq c \Rightarrow ac \leq bc.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-nonstrict-multiply-nonneg. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-nonstrict-multiply-nonneg. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Squeeze). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \leq c \wedge a = c \Rightarrow b = a.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-squeeze. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-squeeze. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Transitivity of Strict Inequality). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a < b \wedge b < c \Rightarrow a < c.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-transitivity-strict. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-transitivity-strict. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Mixed Transitivity). *For all  $a, b, c \in \mathbb{R}$ ,*

$$a \leq b \wedge b < c \Rightarrow a < c.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-transitivity-mixed. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-transitivity-mixed. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Reciprocal Reverses Strict Inequality for Positives). *For all  $a, b \in \mathbb{R}$ ,*

$$0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-reciprocal-positive. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-reciprocal-positive. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Reciprocal Equivalence for Positives). *For all  $a, b \in \mathbb{R}$  with  $a > 0$  and  $b > 0$ ,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-reciprocal-flip. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-reciprocal-flip. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Square Is Strictly Monotone on Nonneg Reals). *For all  $a, b \in \mathbb{R}$ ,*

$$0 \leq a < b \Rightarrow a^2 < b^2.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-square-monotone. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-square-monotone. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Square Root Is Monotone on Nonneg Reals). *For all  $a, b \in \mathbb{R}$ ,*

$$0 \leq a \leq b \Rightarrow \sqrt{a} \leq \sqrt{b}.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-square-root-monotone. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-square-root-monotone. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (AM-GM Inequality for Two Terms). *For all  $a, b \geq 0$ ,*

$$\sqrt{ab} \leq \frac{a+b}{2}.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-am-gm-two. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-am-gm-two. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bernoulli's Inequality). *For all  $x \in \mathbb{R}$  with  $x \geq -1$  and all  $n \in \mathbb{N}$ ,*

$$(1 + x)^n \geq 1 + nx.$$

*Professional Standard Proof.*

TODO: professional standard proof for ineq-bernoulli. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ineq-bernoulli. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value Is Nonnegative). *For all  $a \in \mathbb{R}$ ,*

$$|a| \geq 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-nonneg. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-nonneg. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value Zero Iff Zero). *For all  $a \in \mathbb{R}$ ,*

$$|a| = 0 \iff a = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-zero-iff-zero. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-zero-iff-zero. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value Equals Self or Negation). *For all  $a \in \mathbb{R}$ , either  $|a| = a$  or  $|a| = -a$ .*

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-self-or-neg. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-self-or-neg. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value Is Symmetric). *For all  $a \in \mathbb{R}$ ,*

$$|-a| = |a|.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-symmetric. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-symmetric. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value of a Product). *For all  $a, b \in \mathbb{R}$ ,*

$$|ab| = |a| |b|.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-product. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-product. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Value of a Quotient). *For all  $a \in \mathbb{R}$  and  $b \in \mathbb{R}$  with  $b \neq 0$ ,*

$$\left| \frac{a}{b} \right| = \frac{|a|}{|b|}.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-quotient.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-quotient.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bound by Modulus). *For all  $a \in \mathbb{R}$ ,*

$$-|a| \leq a \leq |a|.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Nonstrict Modulus Inequality). *For all  $a \in \mathbb{R}$  and  $r \geq 0$ ,*

$$|a| \leq r \iff -r \leq a \leq r.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-le-iff. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-le-iff. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Strict Modulus Inequality). *For all  $a \in \mathbb{R}$  and  $r > 0$ ,*

$$|a| < r \iff -r < a < r.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-lt-iff. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-lt-iff. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Reverse Triangle Inequality). *For all  $a, b \in \mathbb{R}$ ,*

$$||a| - |b|| \leq |a - b|.$$

*Professional Standard Proof.*

TODO: professional standard proof for reverse-triangle-inequality. □

*Detailed Learning Proof.*

TODO: detailed learning proof for reverse-triangle-inequality. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Generalised Triangle Inequality). *For all  $n \in \mathbb{N}$  and  $a_1, \dots, a_n \in \mathbb{R}$ ,*

$$\left| \sum_{k=1}^n a_k \right| \leq \sum_{k=1}^n |a_k|.$$

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-sum-bound. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-sum-bound. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

# Chapter 2

## Structure of the Real Line



### 2.1 Real Line One Dimensional

#### The Line as Order Plus Distance

The real numbers carry two compatible structures at once: a linear order  $<$  and a notion of distance  $d(x, y) = |x - y|$ . Together these make  $\mathbb{R}$  a one-dimensional space — a line. This chapter develops only the metric topology of  $\mathbb{R}$ , not topology in general spaces. Every open set is built from intervals, and compactness is introduced only on  $\mathbb{R}$ , with Heine–Borel as the capstone.

The intuition is geometric: points are positions on a line, order is “left of,” and distance is the length of the segment between two points. The caution to keep throughout is that  $\mathbb{R}$  is special. Order and metric agree here in a way that fails on a general space, so definitions are stated in the form that survives to metric spaces and manifolds, with  $\mathbb{R}$  as the first instance rather than the only one.

### 2.2 Order Distance Length

#### Distance on the Line

The absolute value of a difference turns  $\mathbb{R}$  into a metric space. Length of an interval is the distance between its endpoints. These are the quantitative tools underneath every later definition of nearness.

### Definition (Distance on the Real Line)

**Definition 2.2.1** (Distance on the Real Line). The *distance* on  $\mathbb{R}$  is the map  $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  given by

$$d(x, y) = |x - y|.$$

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} \ (d(x, y) = |x - y|).$$

*Remark* (Interpretation). The distance between two points is the length of the segment joining them. It is the metric that makes “close” precise and is the basis for balls, neighborhoods, and open sets.

*Remark* (Dependencies).

- [Triangle Inequality](#)

### Definition (Length of a Bounded Interval)

**Definition 2.2.2** (Length of a Bounded Interval). Let  $a, b \in \mathbb{R}$  with  $a \leq b$ . The *length* of any bounded interval with endpoints  $a$  and  $b$  is

$$\ell = b - a.$$

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \ (a \leq b \implies \ell([a, b]) = b - a = d(a, b)).$$

*Remark* (Interpretation). Length depends only on the endpoints, not on whether they are included. It is the distance  $d(a, b)$  and measures the size of the basic pieces of the line.

*Remark* (Dependencies).

- [Distance on the Real Line](#)

### Theorem (The Distance Function Is a Metric)

**Theorem 2.2.3** (The Distance Function Is a Metric). The map  $d(x, y) = |x - y|$  satisfies, for all  $x, y, z \in \mathbb{R}$ : (i)  $d(x, y) \geq 0$  with  $d(x, y) = 0$  iff  $x = y$ ; (ii)  $d(x, y) = d(y, x)$ ; and (iii)  $d(x, z) \leq d(x, y) + d(y, z)$ . Hence  $(\mathbb{R}, d)$  is a metric space. *Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y, z \in \mathbb{R} \ (d(x, y) \geq 0 \wedge (d(x, y) = 0 \iff x = y) \wedge d(x, y) = d(y, x) \wedge d(x, z) \leq d(x, y) + d(y, z)). \blacksquare$$

*Remark* (Interpretation). The three metric axioms are exactly the properties of  $|\cdot|$ : nonnegativity with the zero-distance characterization, symmetry, and the triangle inequality. This is the precise sense in which  $\mathbb{R}$  is a one-dimensional metric space.

*Remark* (Dependencies).

- [Distance on the Real Line](#)
- [Triangle Inequality](#)

## 2.3 Absolute Value as Distance

### Absolute Value Is Distance to Zero

The metric  $d$  specializes to absolute value when one point is the origin. This is the geometric reading of  $|a|$ : how far  $a$  sits from 0 on the line.

#### Proposition (Absolute Value Is Distance to the Origin)

**Proposition 2.3.1** (Absolute Value Is Distance to the Origin). *For every  $a \in \mathbb{R}$ ,  $|a| = d(a, 0)$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall a \in \mathbb{R} (|a| = d(a, 0)).$$

*Remark* (Interpretation). Absolute value is not a separate idea from distance; it is the distance to the origin. The two-sided bound  $|x| \leq r \Leftrightarrow -r \leq x \leq r$  is then exactly the statement that  $x$  lies within distance  $r$  of 0, i.e. in the interval  $[-r, r]$ .

*Remark* (Dependencies).

- [Distance on the Real Line](#)

## 2.4 Intervals as Subsets

### Intervals as the Basic Pieces

The interval shapes were introduced in §12.2. Here they are recast as subsets of the metric line and supplemented with boundedness, which is the size condition compactness will need.

#### Definition (Bounded Subset of the Real Line)

**Definition 2.4.1** (Bounded Subset of the Real Line). A set  $E \subseteq \mathbb{R}$  is *bounded* exactly when there exists  $M \geq 0$  such that  $|x| \leq M$  for all  $x \in E$ ; equivalently,  $E$  is contained in some closed interval  $[-M, M]$ .

*Remark* (Standard quantified statement).

$$E \text{ is bounded} \iff \exists M \geq 0 \forall x \in E (|x| \leq M).$$

*Remark* (Negated quantified statement).

$$E \text{ is unbounded} \iff \forall M \geq 0 \exists x \in E (|x| > M).$$

*Remark* (Interpretation). Boundedness says  $E$  fits inside a single closed interval — it does not run off to infinity. Together with closedness it is one half of the Heine–Borel characterization of compact subsets of  $\mathbb{R}$ .



*Remark* (Dependencies).

- [Closed Interval](#)
- [Triangle Inequality](#)

## Set Operations on Intervals

### Intervals as Sets

Once intervals are regarded as subsets of  $\mathbb{R}$ , the usual set operations apply to them. This is the first place where intervals behave like the building blocks of topology rather than merely as pieces of notation on the number line. Intersections often remain intervals, while unions, complements, and differences may split into several interval pieces.

#### Definition (Union of Intervals)

**Definition 2.4.2** (Union of Intervals). Let  $I, J \subseteq \mathbb{R}$  be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

*Remark* (Dependencies).

- [Open Interval](#)
- [Closed Interval](#)
- [Union](#)

#### Definition (Intersection of Intervals)

**Definition 2.4.3** (Intersection of Intervals). Let  $I, J \subseteq \mathbb{R}$  be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

*Remark* (Dependencies).

- [Open Interval](#)
- [Closed Interval](#)
- [Intersection](#)

#### Definition (Complement of an Interval)

**Definition 2.4.4** (Complement of an Interval). Let  $I \subseteq \mathbb{R}$  be an interval. Its *complement* in  $\mathbb{R}$  is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

*Remark* (Dependencies).

- [Open Interval](#)
- [Closed Interval](#)
- [Complement](#)

#### Definition (Difference of Intervals)

**Definition 2.4.5** (Difference of Intervals). Let  $I, J \subseteq \mathbb{R}$  be intervals. The *difference* of  $I$  by  $J$  is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

*Remark* (Predicate readings). For any  $x \in \mathbb{R}$ ,

$$x \in I \cup J \iff (x \in I) \vee (x \in J),$$

$$x \in I \cap J \iff (x \in I) \wedge (x \in J),$$

$$x \in I^c \iff x \notin I, \quad x \in I \setminus J \iff (x \in I) \wedge (x \notin J).$$

These are set-theoretic operations; no arithmetic operation is being performed on the endpoints.

*Remark* (Dependencies).

- [Open Interval](#)
- [Closed Interval](#)
- [Left-Half-Open Interval](#)
- [Right-Half-Open Interval](#)
- [Union of Intervals](#)
- [Intersection of Intervals](#)
- [Complement of an Interval](#)

*Remark* (Intersections of intervals). The intersection of two intervals is either empty or another interval. For example, if  $a \leq b$  and  $c \leq d$ , then

$$[a, b] \cap [c, d] = \begin{cases} [\max\{a, c\}, \min\{b, d\}], & \max\{a, c\} \leq \min\{b, d\}, \\ \emptyset, & \max\{a, c\} > \min\{b, d\}. \end{cases}$$

Endpoint inclusion follows the endpoint rules of the original intervals.

*Remark* (Unions of intervals). The union of two intervals need not be an interval:

$$(0, 1) \cup (2, 3)$$

has a gap. If two intervals overlap or touch without a missing endpoint, their union is again an interval, for instance

$$(0, 2] \cup [2, 5) = (0, 5).$$

The missing or included endpoints determine whether the joined set is truly one interval.

*Remark* (Complements of intervals). The complement of a bounded interval usually splits into two rays:

$$(a, b)^c = (-\infty, a] \cup [b, \infty), \quad [a, b]^c = (-\infty, a) \cup (b, \infty).$$

Complements reverse endpoint inclusion: an endpoint excluded from the interval is included in the complement, and an endpoint included in the interval is excluded from the complement.

*Remark* (Differences of intervals). A difference is an intersection with a complement, so it can also split:

$$(0, 5) \setminus [2, 3] = (0, 2) \cup (3, 5).$$

If the removed interval only cuts off one side, the result may remain a single interval:

$$(0, 5) \setminus (3, \infty) = (0, 3].$$

The bracket at 3 appears because  $3 \notin (3, \infty)$ , so it is not removed.

*Remark* (Topological motivation). Open sets in  $\mathbb{R}$  are built from unions of open intervals. Closed sets are understood through complements of open sets. Boundaries arise when a point is not cleanly inside a set or its complement. Thus interval set operations are the grammar behind the metric topology of the real line.

## 2.5 Neighborhoods and Balls

### Balls Are Open Intervals

On the line, the ball of radius  $r$  about a point is an open interval centered at that point. Neighborhoods are sets that contain such a ball. These are the local objects from which open sets are assembled.

#### Definition (Open Ball on the Real Line)

**Definition 2.5.1** (Open Ball on the Real Line). Let  $x \in \mathbb{R}$  and  $r > 0$ . The *open ball* of radius  $r$  about  $x$  is

$$B_r(x) = \{y \in \mathbb{R} : |y - x| < r\} = (x - r, x + r).$$

*Remark* (Standard quantified statement).

$$B_r(x) = \{y \in \mathbb{R} : d(y, x) < r\} = (x - r, x + r).$$

*Remark* (Interpretation). A ball on the line is just a symmetric open interval: all points strictly within distance  $r$  of  $x$ . Writing balls as intervals keeps the chapter inside the metric topology of  $\mathbb{R}$ .

*Remark* (Dependencies).

- [Distance on the Real Line](#)

- [Open Interval](#)

### Definition (Neighborhood of a Point)

**Definition 2.5.2** (Neighborhood of a Point). A set  $N \subseteq \mathbb{R}$  is a *neighborhood* of  $x \in \mathbb{R}$  exactly when there exists  $r > 0$  with  $B_r(x) \subseteq N$ .

*Remark* (Standard quantified statement).

$$N \text{ is a neighborhood of } x \iff \exists r > 0 (B_r(x) \subseteq N).$$

*Remark* (Interpretation). A neighborhood of  $x$  is any set with room to spare around  $x$  — it contains a whole ball, not just  $x$  itself. Neighborhoods are the language of “near  $x$ ” used to define interior points and limits.

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

## 2.6 Open Sets

### Sets Built from Balls

An open set is one that contains a ball around each of its points. On the line these are exactly the unions of open intervals. The closure properties recorded here are the metric-topology core.

### Definition (Open Set in $\mathbb{R}$ )

**Definition 2.6.1** (Open Set in  $\mathbb{R}$ ). A set  $U \subseteq \mathbb{R}$  is *open* exactly when

$$\forall x \in U \exists r > 0 (B_r(x) \subseteq U).$$

*Remark* (Standard quantified statement).

$$U \text{ is open} \iff \forall x \in U \exists r > 0 (B_r(x) \subseteq U).$$

*Remark* (Negated quantified statement).

$$U \text{ is not open} \iff \exists x \in U \forall r > 0 (B_r(x) \not\subseteq U).$$

*Remark* (Failure modes). Openness fails exactly when some point of  $U$  has no ball contained in  $U$  — a boundary point trapped inside  $U$ , every ball around which pokes outside.

*Remark* (Interpretation). An open set has interior room around each of its points: a small step in either direction remains inside the set. The empty set and  $\mathbb{R}$  are open, and open intervals are the model examples.

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

### Theorem (Open Intervals Are Open)

**Theorem 2.6.2** (Open Intervals Are Open). *Every open interval  $(a, b)$ , every open ray  $(a, \infty)$  and  $(-\infty, b)$ , and  $\mathbb{R}$  itself are open sets. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall a, b \in \mathbb{R} \ (a < b \implies (a, b) \text{ is open}), \\ \forall a \in \mathbb{R} \ ((a, \infty) \text{ is open} \wedge (-\infty, a) \text{ is open}), \quad \mathbb{R} \text{ is open.} \end{aligned}$$

*Remark* (Interpretation). The name “open interval” is justified: each such interval satisfies the open-set condition, with the radius at  $x$  taken as the distance to the nearer endpoint.

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)
- [Open Interval](#)

### Theorem (Unions and Finite Intersections of Open Sets)

**Theorem 2.6.3** (Unions and Finite Intersections of Open Sets). *An arbitrary union of open subsets of  $\mathbb{R}$  is open, and a finite intersection of open subsets of  $\mathbb{R}$  is open. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\left( \forall i \in I \ U_i \text{ open} \right) \implies \left( \bigcup_{i \in I} U_i \right) \text{ open}; \quad \left( \bigwedge_{k=1}^n U_k \text{ open} \right) \implies \left( \bigcap_{k=1}^n U_k \right) \text{ open}.$$

*Remark* (Interpretation). Openness is preserved by arbitrary unions but only finite intersections. The infinite intersection  $\bigcap_n (-1/n, 1/n) = \{0\}$  shows why “finite” cannot be dropped: a shrinking family of open intervals can close down onto a single point.

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)

## 2.7 Closed Sets

### Complements of Open Sets

Closed sets are the complements of open sets. They can equivalently be described as the sets that contain all of their limit points, the description used when reasoning about convergence.

### Definition (Closed Set in $\mathbb{R}$ )

**Definition 2.7.1** (Closed Set in  $\mathbb{R}$ ). A set  $F \subseteq \mathbb{R}$  is *closed* exactly when its complement  $\mathbb{R} \setminus F$  is open.

*Remark* (Standard quantified statement).

$$F \text{ is closed} \iff (\mathbb{R} \setminus F) \text{ is open.}$$

*Remark* (Interpretation). Closed and open are complementary, not opposite: a set can be both (e.g.  $\emptyset, \mathbb{R}$ ) or neither (e.g.  $[0, 1)$ ). Closed intervals  $[a, b]$  are closed, as are finite sets and  $\mathbb{R}$  itself.

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)

### Theorem (Closed Sets Contain Their Limit Points)

**Theorem 2.7.2** (Closed Sets Contain Their Limit Points). A set  $F \subseteq \mathbb{R}$  is closed if and only if it contains every one of its limit points. [Go to proof](#).

*Remark* (Standard quantified statement).

$$F \text{ closed} \iff \forall x \in \mathbb{R} (x \text{ is a limit point of } F \Rightarrow x \in F).$$

*Remark* (Interpretation). This is the convergence-flavored description of closedness: a closed set cannot be approached from outside without already containing the approached point. It is the form used when showing limits of sequences in  $F$  stay in  $F$ .

*Remark* (Dependencies).

- [Closed Set in  \$\mathbb{R}\$](#)
- [Limit Point](#)

## 2.8 Interior Exterior Boundary

### Inside, Outside, and Edge

Relative to a set  $E$ , each point of the line is interior to  $E$ , exterior to  $E$ , or on its boundary. These three regions partition  $\mathbb{R}$  and refine the open/closed distinction.

### Definition (Interior Point)

**Definition 2.8.1** (Interior Point). Let  $E \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$ . Then  $x$  is an interior point of  $E$  exactly when

$$\exists r > 0 (B_r(x) \subseteq E).$$

*Remark* (Standard quantified statement).

$$x \text{ is an interior point of } E \iff \exists r > 0 (B_r(x) \subseteq E).$$

*Remark* (Interpretation). An interior point of  $E$  sits inside  $E$  with room to spare: a whole ball around it lies in  $E$ .

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

#### Definition (Exterior Point)

**Definition 2.8.2** (Exterior Point). Let  $E \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$ . Then  $x$  is an exterior point of  $E$  exactly when

$$\exists r > 0 (B_r(x) \subseteq \mathbb{R} \setminus E).$$

*Remark* (Standard quantified statement).

$$x \text{ is an exterior point of } E \iff \exists r > 0 (B_r(x) \subseteq \mathbb{R} \setminus E).$$

*Remark* (Interpretation). An exterior point of  $E$  is interior to the complement: a whole ball around it misses  $E$ .

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

#### Definition (Boundary Point)

**Definition 2.8.3** (Boundary Point). Let  $E \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$ . Then  $x$  is a boundary point of  $E$  exactly when

$$\forall r > 0 (B_r(x) \cap E \neq \emptyset \wedge B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

*Remark* (Standard quantified statement).

$$x \text{ is a boundary point of } E \iff \forall r > 0 (B_r(x) \cap E \neq \emptyset \wedge B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

*Remark* (Interpretation). A boundary point of  $E$  is approached from both sides: every ball around it meets both  $E$  and its complement.

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

#### Definition (Interior of a Set)

**Definition 2.8.4** (Interior of a Set). The *interior* of  $E \subseteq \mathbb{R}$ , written  $\text{int}(E)$ , is the set of all interior points of  $E$ .

*Remark* (Standard quantified statement).

$$\text{int}(E) = \{ x \in \mathbb{R} : \exists r > 0 (B_r(x) \subseteq E) \}.$$

*Remark* (Interpretation). The interior is the largest open set inside  $E$ ;  $E$  is open exactly when  $E = \text{int}(E)$ . The boundary is what  $E$  and its complement share, and a set is closed exactly when it contains its boundary.

*Remark* (Dependencies).

- [Interior Point](#)

## 2.9 Limit and Isolated Points

### Approachable and Detached Points

A limit point of  $E$  is one that  $E$  crowds arbitrarily closely, even with the point itself removed. An isolated point of  $E$  belongs to  $E$  but has a ball meeting  $E$  only at itself. These notions drive the convergence description of closed sets.

#### Definition (Limit Point)

**Definition 2.9.1** (Limit Point). Let  $E \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$ . Then  $x$  is a *limit point* of  $E$  exactly when

$$\forall r > 0 \exists y \in E (0 < |y - x| < r).$$

*Remark* (Standard quantified statement).

$$x \text{ is a limit point of } E \iff \forall r > 0 \exists y \in E (0 < |y - x| < r).$$

*Remark* (Negated quantified statement).

$$x \text{ is not a limit point of } E \iff \exists r > 0 \forall y \in E (y = x \vee |y - x| \geq r).$$

*Remark* (Interpretation). Every punctured ball around a limit point meets  $E$ : the set accumulates at  $x$ . The strict inequality  $0 < |y - x|$  excludes  $x$  itself, so  $x$  need not belong to  $E$  to be a limit point. This is the exact condition under which a sequence in  $E$  can converge to  $x$ .

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

#### Definition (Isolated Point)

**Definition 2.9.2** (Isolated Point). Let  $E \subseteq \mathbb{R}$  and  $x \in E$ . Then  $x$  is an *isolated point* of  $E$  exactly when

$$\exists r > 0 \forall y \in E (|y - x| < r \Rightarrow y = x).$$



*Remark* (Standard quantified statement).

$$x \text{ is an isolated point of } E \iff x \in E \wedge \exists r > 0 \forall y \in E (|y - x| < r \Rightarrow y = x).$$

*Remark* (Interpretation). An isolated point sits in  $E$  but stands apart: some ball around it contains no other point of  $E$ . A point of  $E$  is either isolated or a limit point of  $E$ , never both.

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

## 2.10 Unions and Intersections

### Closure Laws for Open and Closed Sets

The open closure laws of §2.6 dualize to closed sets by complementation. Collecting them gives the bookkeeping rules used throughout the rest of the development.

#### Theorem (Closure Laws for Closed Sets)

**Theorem 2.10.1** (Closure Laws for Closed Sets). *An arbitrary intersection of closed subsets of  $\mathbb{R}$  is closed, and a finite union of closed subsets of  $\mathbb{R}$  is closed. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\left( \forall i \in I \ F_i \text{ closed} \right) \implies \left( \bigcap_{i \in I} F_i \right) \text{ closed}; \quad \left( \bigwedge_{k=1}^n F_k \text{ closed} \right) \implies \left( \bigcup_{k=1}^n F_k \right) \text{ closed}.$$

*Remark* (Interpretation). The roles of union and intersection swap relative to open sets: closed sets are stable under arbitrary intersection but only finite union. The asymmetry mirrors the open case under De Morgan's laws.

*Remark* (Dependencies).

- [Closed Set in  \$\mathbb{R}\$](#)
- [Unions and Finite Intersections of Open Sets](#)

## 2.11 Covers and Subcovers

### Covering a Set with Open Pieces

A cover of  $E$  is a family of sets whose union contains  $E$ ; an open cover uses open sets. A subcover keeps enough of the family to still cover, and a finite subcover keeps only finitely many. These are the exact ingredients of compactness, defined so as to survive beyond the

line.

### Definition (Open Cover)

**Definition 2.11.1** (Open Cover). Let  $E \subseteq \mathbb{R}$ . An *open cover* of  $E$  is a family  $\{U_i\}_{i \in I}$  of open subsets of  $\mathbb{R}$  with

$$E \subseteq \bigcup_{i \in I} U_i.$$

*Remark* (Standard quantified statement).

$$\{U_i\}_{i \in I} \text{ is an open cover of } E \iff (\forall i \in I \ U_i \text{ open}) \wedge E \subseteq \bigcup_{i \in I} U_i.$$

*Remark* (Interpretation). An open cover is a way of surrounding every point of  $E$  with an open piece of the family. The index set  $I$  may be infinite; the question compactness asks is whether finitely many pieces already suffice.

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)

### Definition (Subcover and Finite Subcover)

**Definition 2.11.2** (Subcover and Finite Subcover). Let  $\{U_i\}_{i \in I}$  be an open cover of  $E$ . A *subcover* is a subfamily  $\{U_i\}_{i \in J}$  with  $J \subseteq I$  that still covers  $E$ . It is a *finite subcover* when  $J$  is finite.

*Remark* (Standard quantified statement).

$$\{U_i\}_{i \in J} \text{ is a finite subcover of } E \iff (J \subseteq I \wedge J \text{ is finite} \wedge E \subseteq \bigcup_{i \in J} U_i).$$

*Remark* (Interpretation). A subcover discards some pieces while still covering  $E$ ; a finite subcover discards all but finitely many. The existence of a finite subcover for *every* open cover is the defining property of compactness.

*Remark* (Dependencies).

- [Open Cover](#)

## 2.12 Compactness on $\mathbb{R}$

### The Open-Cover Definition

Compactness is defined by the open-cover property, deliberately and not as “closed and bounded.” The open-cover formulation is the one that survives to general metric spaces and to manifolds such as the torus, where “bounded” has no intrinsic meaning. On  $\mathbb{R}$

the two descriptions coincide, and that coincidence is the Heine–Borel theorem of the next section, proved rather than assumed.

### Definition (Compact Set in $\mathbb{R}$ )

**Definition 2.12.1** (Compact Set in  $\mathbb{R}$ ). A set  $K \subseteq \mathbb{R}$  is *compact* exactly when every open cover of  $K$  admits a finite subcover:

$$\forall \{U_i\}_{i \in I} \text{ open cover of } K \exists \text{finite } J \subseteq I \left( K \subseteq \bigcup_{i \in J} U_i \right).$$

*Remark* (Standard quantified statement).

$$K \text{ compact} \iff \forall \{U_i\}_{i \in I} \left( \left( \forall i \in I \ U_i \text{ open} \right) \wedge K \subseteq \bigcup_{i \in I} U_i \Rightarrow \exists \text{finite } J \subseteq I \ K \subseteq \bigcup_{i \in J} U_i \right).$$

*Remark* (Negated quantified statement).

$$K \text{ not compact} \iff \exists \{U_i\}_{i \in I} \left( \left( \forall i \in I \ U_i \text{ open} \right) \wedge K \subseteq \bigcup_{i \in I} U_i \wedge \forall \text{finite } J \subseteq I \ K \not\subseteq \bigcup_{i \in J} U_i \right).$$

*Remark* (Failure modes). Compactness fails exactly when some open cover has no finite subcover. Two standard witnesses:  $(0, 1]$  fails via the cover  $\{(1/n, 2)\}_n$  (not closed), and  $[0, \infty)$  fails via  $\{(-1, n)\}_n$  (not bounded).

*Remark* (Interpretation). Compactness is a finiteness property phrased entirely in open sets: no matter how a set is covered by open pieces, finitely many already do the job. This is what later guarantees, on a compact domain, that continuous functions attain extrema and are uniformly continuous — the facts Analysis I relies on. Crucially, this definition does not mention order or boundedness, so it transfers unchanged to the torus.

*Remark* (Dependencies).

- [Open Cover](#)
- [Subcover and Finite Subcover](#)

### Theorem (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)

**Theorem 2.12.2** (Compact Subsets of  $\mathbb{R}$  Are Closed and Bounded). *If  $K \subseteq \mathbb{R}$  is compact, then  $K$  is closed and bounded. [Go to proof](#).*

*Remark* (Standard quantified statement).

$$K \text{ compact} \implies (K \text{ closed} \wedge K \text{ bounded}).$$

*Remark* (Interpretation). This is the easy half of Heine–Borel and holds the same way in every metric space. Boundedness comes from covering  $K$  by growing balls about a fixed point; closedness comes from separating an outside point from  $K$  by disjoint balls. The converse — closed and bounded implies compact — is special to  $\mathbb{R}^n$  and is the next theorem.

*Remark* (Dependencies).

- [Compact Set in  \$\mathbb{R}\$](#)
- [Closed Set in  \$\mathbb{R}\$](#)
- [Bounded Subset of the Real Line](#)

## 2.13 Heine Borel

### The Capstone of the Real Line

Heine–Borel closes the chapter by proving the converse direction: on  $\mathbb{R}$ , closed and bounded is enough for compactness. The keystone is that every closed bounded interval  $[a, b]$  is compact; the general statement follows because a closed subset of a compact set is compact.

#### Theorem (Closed Bounded Intervals Are Compact)

**Theorem 2.13.1** (Closed Bounded Intervals Are Compact). *Every closed bounded interval  $[a, b] \subseteq \mathbb{R}$  is compact. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \ (a \leq b \implies [a, b] \text{ is compact}).$$

*Remark* (Interpretation). This is the hard half. The standard proof uses completeness directly: let  $S$  be the set of  $x \in [a, b]$  such that  $[a, x]$  has a finite subcover, and show  $\sup S = b$  is attained. Bisection (the nested-interval method) gives the same result. Either route shows completeness is what makes the line compact in the large.

*Remark* (Dependencies).

- [Compact Set in  \$\mathbb{R}\$](#)
- [Closed Interval](#)
- [Least-Upper-Bound Property](#)

#### Theorem (Heine–Borel Theorem for $\mathbb{R}$ )

**Theorem 2.13.2** (Heine–Borel Theorem for  $\mathbb{R}$ ). *A set  $K \subseteq \mathbb{R}$  is compact if and only if  $K$  is closed and bounded. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$K \text{ compact} \iff (K \text{ closed} \wedge K \text{ bounded}).$$

*Remark* (Interpretation). Heine–Borel is the bridge into Analysis I: it certifies that the closed bounded sets — the ones on which the Extreme Value and uniform-continuity theorems live — are exactly the compact ones. It is stated here as the equivalence *on*  $\mathbb{R}$ ; the open-cover side is the definition that later carries to the torus, while “closed and bounded” does not.

*Remark* (Exposition). With Heine–Borel in hand the chain is complete: number construction  $\rightarrow$  embedded number systems  $\rightarrow$  the real line as an ordered metric space  $\rightarrow$  Analysis I. Completeness, introduced as the last rung of the embedding ladder, is exactly the property that makes the line compact in the large, and compactness is the hypothesis Analysis I leans on.

*Remark* (Dependencies).

- [Compact Subsets of  \$\mathbb{R}\$  Are Closed and Bounded](#)
- [Closed Bounded Intervals Are Compact](#)

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Theorem** (The Distance Function Is a Metric). *The map  $d(x, y) = |x - y|$  satisfies, for all  $x, y, z \in \mathbb{R}$ : (i)  $d(x, y) \geq 0$  with  $d(x, y) = 0$  iff  $x = y$ ; (ii)  $d(x, y) = d(y, x)$ ; and (iii)  $d(x, z) \leq d(x, y) + d(y, z)$ . Hence  $(\mathbb{R}, d)$  is a metric space.*

*Professional Standard Proof.*

TODO: professional standard proof for distance-is-a-metric. □

*Detailed Learning Proof.*

TODO: detailed learning proof for distance-is-a-metric. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Distance on the Real Line](#)
- TODO: link to the triangle inequality for absolute value.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Absolute Value Is Distance to the Origin). *For every  $a \in \mathbb{R}$ ,  $|a| = d(a, 0)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for abs-value-is-distance-to-zero. □

*Detailed Learning Proof.*

TODO: detailed learning proof for abs-value-is-distance-to-zero. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Distance on the Real Line](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Open Intervals Are Open). *Every open interval  $(a, b)$ , every open ray  $(a, \infty)$  and  $(-\infty, b)$ , and  $\mathbb{R}$  itself are open sets.*

*Professional Standard Proof.*

TODO: professional standard proof for open-interval-is-open. □

*Detailed Learning Proof.*

TODO: detailed learning proof for open-interval-is-open. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)
- [Open Interval](#)



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Unions and Finite Intersections of Open Sets). *An arbitrary union of open subsets of  $\mathbb{R}$  is open, and a finite intersection of open subsets of  $\mathbb{R}$  is open.*

*Professional Standard Proof.*

TODO: professional standard proof for open-set-closure-operations. □

*Detailed Learning Proof.*

TODO: detailed learning proof for open-set-closure-operations. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Open Set in  \$\mathbb{R}\$](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Closed Sets Contain Their Limit Points). *A set  $F \subseteq \mathbb{R}$  is closed if and only if it contains every one of its limit points.*

*Professional Standard Proof.*

TODO: professional standard proof for closed-iff-contains-limit-points. □

*Detailed Learning Proof.*

TODO: detailed learning proof for closed-iff-contains-limit-points. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Closed Set in  \$\mathbb{R}\$](#)
- [Limit Point](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Closure Laws for Closed Sets). *An arbitrary intersection of closed subsets of  $\mathbb{R}$  is closed, and a finite union of closed subsets of  $\mathbb{R}$  is closed.*

*Professional Standard Proof.*

TODO: professional standard proof for closed-set-closure-operations. □

*Detailed Learning Proof.*

TODO: detailed learning proof for closed-set-closure-operations. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Closed Set in  \$\mathbb{R}\$](#)
- [Unions and Finite Intersections of Open Sets](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Compact Subsets of  $\mathbb{R}$  Are Closed and Bounded). *If  $K \subseteq \mathbb{R}$  is compact, then  $K$  is closed and bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for compact-implies-closed-bounded. □

*Detailed Learning Proof.*

TODO: detailed learning proof for compact-implies-closed-bounded. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Compact Set in  \$\mathbb{R}\$](#)
- [Closed Set in  \$\mathbb{R}\$](#)
- [Bounded Subset of the Real Line](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Closed Bounded Intervals Are Compact). *Every closed bounded interval  $[a, b] \subseteq \mathbb{R}$  is compact.*

*Professional Standard Proof.*

TODO: professional standard proof for closed-bounded-interval-compact. □

*Detailed Learning Proof.*

TODO: detailed learning proof for closed-bounded-interval-compact. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Compact Set in  \$\mathbb{R}\$](#)
- [Closed Interval](#)
- TODO: link to the Least Upper Bound Property / Nested Interval Property.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Heine–Borel Theorem for  $\mathbb{R}$ ). *A set  $K \subseteq \mathbb{R}$  is compact if and only if  $K$  is closed and bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for heine-borel. □

*Detailed Learning Proof.*

TODO: detailed learning proof for heine-borel. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Compact Subsets of  \$\mathbb{R}\$  Are Closed and Bounded](#)
- [Closed Bounded Intervals Are Compact](#)

# Chapter 3

## Bounds



### 3.1 Completeness Axiom

#### Axiom of Completeness

*Remark* (Why completeness is needed). The ordered field axioms alone do not prevent “holes” in the number line. The rationals  $\mathbb{Q}$  satisfy all field and order axioms, yet fail completeness. Consider the set

$$S = \{x \in \mathbb{Q} : x^2 < 2\}.$$

This set is nonempty (e.g.  $1 \in S$ ) and bounded above in  $\mathbb{Q}$  (e.g. 2 is an upper bound). Yet  $\sup S$  does not exist as a rational number: the candidate  $\sqrt{2}$  is irrational. The set  $S$  has no least upper bound in  $\mathbb{Q}$  — a hole sits exactly where  $\sqrt{2}$  should be.

The Completeness Axiom asserts that  $\mathbb{R}$  has no such holes: every nonempty bounded-above set has a supremum *inside*  $\mathbb{R}$ . This single axiom is what distinguishes  $\mathbb{R}$  from  $\mathbb{Q}$ , and it underlies every major theorem in analysis.

#### Definition (Least Upper Bound Property)

**Definition 3.1.1** (Least Upper Bound Property). Let  $(S, \leq)$  be an ordered set. The set  $S$  has the *Least Upper Bound Property* if and only if every nonempty subset of  $S$  that is bounded above in  $S$  has a supremum in  $S$ .

*Remark* (Standard quantified statement).

Let  $(S, \leq)$  be an ordered set.

$S$  has the Least Upper Bound Property

$$\iff \forall A \subseteq S \left[ (A \neq \emptyset \wedge \exists u \in S \forall x \in A (x \leq u)) \rightarrow \exists s \in S \text{ Supremum}(s, A) \right].$$

*Remark* (Definition predicate reading).

$$\text{LeastUpperBoundProperty}(S) := \forall A \subseteq S \left[ (A \neq \emptyset \wedge \exists u \in S \forall x \in A (x \leq u)) \rightarrow \exists s \in S \text{ Supremum}(s, A) \right]$$

*Remark* (Negated quantified statement).

Let  $(S, \leq)$  be an ordered set.

$S$  does not have the Least Upper Bound Property

$$\iff \exists A \subseteq S \left[ A \neq \emptyset \wedge \exists u \in S \forall x \in A (x \leq u) \wedge \forall s \in S \neg \text{Supremum}(s, A) \right].$$

*Remark* (Negation predicate reading).

$$\neg \text{LeastUpperBoundProperty}(S) \iff \exists A \subseteq S \left[ A \neq \emptyset \wedge \exists u \in S \forall x \in A (x \leq u) \wedge \forall s \in S \neg \text{Supremum}(s, A) \right]$$

*Remark* (Failure modes). The property fails when there is a nonempty subset of the ambient ordered set that has at least one upper bound but has no least upper bound inside that same ambient set.

*Remark* (Failure mode decomposition).

$$\neg \text{LeastUpperBoundProperty}(S) = \underbrace{\exists A \subseteq S (A \neq \emptyset)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in S \forall x \in A (x \leq u)}_{\text{bounded above in } S} \wedge \underbrace{\forall s \in S \neg \text{Supremum}(s, A)}_{\text{no supremum in } S}$$

*Remark* (Interpretation). The Least Upper Bound Property says that upper-bounded nonempty subsets do not point toward missing ceiling elements. Whenever the order supplies at least one upper bound, it also supplies a smallest upper bound in the same ambient set. Failure therefore indicates a hole in the ordered structure rather than a defect in the subset alone.

*Remark* (Dependencies).

- [Ordered Set](#)
- [Subset](#)
- [Upper Bound](#)
- [Supremum](#)

#### Axiom (Completeness of $\mathbb{R}$ )

**Axiom 3.1.2** (Completeness of  $\mathbb{R}$ ). The ordered field  $\mathbb{R}$  has the Least Upper Bound Property.

*Remark* (Standard quantified statement).

$$\text{LeastUpperBoundProperty}(\mathbb{R}).$$

Equivalently,

$$\forall A \subseteq \mathbb{R} \left[ (A \neq \emptyset \wedge \exists u \in \mathbb{R} \forall x \in A (x \leq u)) \rightarrow \exists s \in \mathbb{R} \text{ Supremum}(s, A) \right].$$



*Remark* (Axiom predicate reading).

$$\text{LeastUpperBoundProperty}(\mathbb{R}).$$

*Remark* (Negated quantified statement).

$$\neg \text{LeastUpperBoundProperty}(\mathbb{R}) \\ \iff \exists A \subseteq \mathbb{R} \left[ A \neq \emptyset \wedge \exists u \in \mathbb{R} \forall x \in A (x \leq u) \wedge \forall s \in \mathbb{R} \neg \text{Supremum}(s, A) \right].$$

*Remark* (Negation predicate reading).

$$\neg \text{LeastUpperBoundProperty}(\mathbb{R}) \iff \exists A \subseteq \mathbb{R} \left[ A \neq \emptyset \wedge \exists u \in \mathbb{R} \forall x \in A (x \leq u) \wedge \forall s \in \mathbb{R} \neg \text{Supremum}(s, A) \right].$$

*Remark* (Failure modes). Completeness would fail if some nonempty subset of  $\mathbb{R}$  were bounded above in  $\mathbb{R}$  but had no supremum in  $\mathbb{R}$ .

*Remark* (Failure mode decomposition).

$$\neg \text{LeastUpperBoundProperty}(\mathbb{R}) = \underbrace{\exists A \subseteq \mathbb{R} (A \neq \emptyset)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in \mathbb{R} \forall x \in A (x \leq u)}_{\text{bounded above in } \mathbb{R}} \wedge \underbrace{\forall s \in \mathbb{R} \neg \text{Supremum}(s, A)}_{\text{missing real supremum}}$$

*Remark* (Interpretation). The completeness axiom asserts that the real line has no order gaps of the least-upper-bound kind. Nonempty sets of real numbers that are bounded above always determine a real number sitting exactly at the least possible ceiling. In Volume II,  $\mathbb{R}$  is constructed as the complete ordered field arising from Dedekind cuts; in this volume, completeness is used as the standing least-upper-bound principle for real analysis. This is the structural property that distinguishes  $\mathbb{R}$  from ordered fields such as  $\mathbb{Q}$ .

*Remark* (Dependencies).

- [The Real Numbers](#)
- [Ordered Field](#)
- [Least Upper Bound Property](#)

### 3.1.1 Nested Intervals

#### Nested Interval Property

##### Theorem (Nested Interval Property)

**Theorem 3.1.3** (Nested Interval Property). *Let  $(I_n)$  be a sequence of nonempty closed intervals in  $\mathbb{R}$ . If  $I_{n+1} \subseteq I_n$  for every  $n \in \mathbb{N}$ , then*

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (a_n) \subseteq \mathbb{R} \, \forall (b_n) \subseteq \mathbb{R} \\ & \left[ \left( \forall n \in \mathbb{N} (a_n \leq b_n) \right) \wedge \left( \forall n \in \mathbb{N} ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]) \right) \right. \\ & \quad \left. \rightarrow \exists x \in \mathbb{R} \, \forall n \in \mathbb{N} (x \in [a_n, b_n]) \right]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{NestedClosedIntervals}(I_n) \implies \text{NonemptyIntersection} \left( \bigcap_{n=1}^{\infty} I_n \right).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} & \exists (a_n) \subseteq \mathbb{R} \, \exists (b_n) \subseteq \mathbb{R} \\ & \left[ \left( \forall n \in \mathbb{N} (a_n \leq b_n) \right) \wedge \left( \forall n \in \mathbb{N} ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]) \right) \right. \\ & \quad \left. \wedge \forall x \in \mathbb{R} \, \exists n \in \mathbb{N} (x \notin [a_n, b_n]) \right]. \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{NestedClosedIntervals}(I_n) \wedge \neg \text{NonemptyIntersection} \left( \bigcap_{n=1}^{\infty} I_n \right).$$

*Remark* (Failure modes). The theorem fails when a nested sequence of nonempty closed real intervals has no common real point.

*Remark* (Failure mode decomposition).

$$\underbrace{\text{NestedClosedIntervals}(I_n)}_{\text{closed intervals remain nested}} \wedge \underbrace{\forall x \in \mathbb{R} \, \exists n \in \mathbb{N} (x \notin I_n)}_{\text{no real point lies in every interval}}.$$

*Remark* (Interpretation). Nested closed intervals cannot keep shrinking or relocating in a way that loses every real point. Nesting keeps the intervals aligned, nonemptiness prevents trivial collapse at each stage, and completeness supplies a real point that survives all stages at once.

*Remark* (Dependencies).

[Axiom 3.1.2.](#)

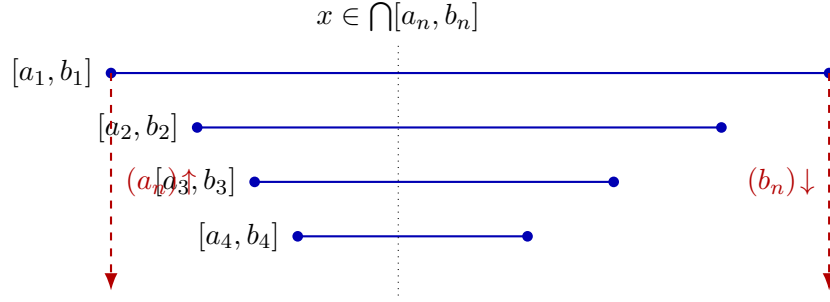


Figure 3.1: Nested intervals  $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$  pinch toward a common point. The sequence of left endpoints is increasing and bounded above; its supremum  $x$  lies in every interval.

### 3.1.2 Archimedean Property

#### Archimedean Property

*Remark* (Why this requires proof). The Archimedean Property states that the natural numbers are unbounded in  $\mathbb{R}$ . That conclusion is not a formal consequence of the ordered-field axioms alone. There exist ordered fields satisfying all ordered-field axioms in which the natural numbers are bounded above, so the statement must be justified from stronger structure.

The needed structure is completeness. If  $\mathbb{N}$  were bounded above in  $\mathbb{R}$ , then the Least Upper Bound Property would produce a supremum  $\alpha \in \mathbb{R}$  for  $\mathbb{N}$ . But then  $\alpha - 1$  cannot be an upper bound, so there exists  $n \in \mathbb{N}$  with  $n > \alpha - 1$ . Hence  $n + 1 > \alpha$ , contradicting the fact that  $\alpha$  is an upper bound of  $\mathbb{N}$ .

*Go to proof.*

#### Theorem (Archimedean Property)

**Theorem 3.1.4** (Archimedean Property). *For every real number  $x$ , there exists a natural number  $n$  such that  $n > x$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} \exists n \in \mathbb{N} (n > x).$$

*Remark* (Predicate reading).

$$\text{ArchimedeanProperty}(\mathbb{R}) \iff \forall x \in \mathbb{R} \exists n \in \mathbb{N} (n > x).$$

*Remark* (Negated quantified statement).

$$\exists x \in \mathbb{R} \forall n \in \mathbb{N} (n \leq x).$$

*Remark* (Negation predicate reading).

$$\neg \text{ArchimedeanProperty}(\mathbb{R}) \iff \exists x \in \mathbb{R} \forall n \in \mathbb{N} (n \leq x).$$

*Remark* (Failure modes). The theorem fails exactly when the natural numbers are bounded above by some real number.

*Remark* (Failure mode decomposition).

$$\neg \text{ArchimedeanProperty}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}}_{\text{real upper bound}} \quad \underbrace{\forall n \in \mathbb{N} (n \leq x)}_{\text{all natural numbers remain below it}} .$$

*Remark* (Interpretation). The Archimedean Property says that the natural numbers do not stop below any real ceiling. No real number is so large that every natural number remains below it. In this chapter, the result is a consequence of completeness: a bounded copy of  $\mathbb{N}$  would have a supremum, and that supremum is contradicted by shifting one natural number upward.

*Remark* (Dependencies).

[Axiom 3.1.2](#), [Definition 3.2.7](#).

#### Corollary (Archimedean Reciprocal Corollary)

**Corollary 3.1.5** (Archimedean Reciprocal Corollary). *For every  $x > 0$  and every  $y \in \mathbb{R}$ , there exists  $n \in \mathbb{N}$  such that  $nx > y$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} (x > 0 \rightarrow \exists n \in \mathbb{N} (nx > y)).$$

*Remark* (Predicate reading).

$$\text{ArchimedeanReciprocal}(\mathbb{R}) \iff \forall x, y \in \mathbb{R} (x > 0 \rightarrow \exists n \in \mathbb{N} (nx > y)).$$

*Remark* (Negated quantified statement).

$$\exists x, y \in \mathbb{R} (x > 0 \wedge \forall n \in \mathbb{N} (nx \leq y)).$$

*Remark* (Negation predicate reading).

$$\neg \text{ArchimedeanReciprocal}(\mathbb{R}) \iff \exists x, y \in \mathbb{R} (x > 0 \wedge \forall n \in \mathbb{N} (nx \leq y)).$$

*Remark* (Failure modes). The corollary fails when a positive real step size has all of its natural multiples bounded above by some real threshold.

*Remark* (Failure mode decomposition).

$$\neg \text{ArchimedeanReciprocal}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R} (x > 0)}_{\text{positive step size}} \wedge \underbrace{\exists y \in \mathbb{R} \forall n \in \mathbb{N} (nx \leq y)}_{\text{all natural multiples stay bounded}} .$$

*Remark* (Interpretation). Any fixed positive real step eventually passes any real threshold when repeated enough times. The statement is the Archimedean Property applied after rescaling by the positive number  $x$ : growth by units becomes growth by steps of size  $x$ .

*Remark* (Dependencies).

[Theorem 3.1.4](#).

### 3.1.3 Density

#### Density

##### Definition (Dense Subset)

**Definition 3.1.6** (Dense Subset). Let  $D \subseteq S \subseteq \mathbb{R}$ . The set  $D$  is *dense in*  $S$  if and only if for every  $x \in S$  and every  $\varepsilon > 0$  there exists  $d \in D$  such that  $|x - d| < \varepsilon$ .

*Remark* (Standard quantified statement).

Let  $D \subseteq S \subseteq \mathbb{R}$ .

$D$  is dense in  $S \iff \forall x \in S \forall \varepsilon > 0 \exists d \in D (|x - d| < \varepsilon)$ .

*Remark* (Definition predicate reading).

$\text{DenseSubset}(D, S) := \forall x \in S \forall \varepsilon > 0 \exists d \in D (|x - d| < \varepsilon)$ .

*Remark* (Negated quantified statement).

Let  $D \subseteq S \subseteq \mathbb{R}$ .

$D$  is not dense in  $S \iff \exists x \in S \exists \varepsilon_0 > 0 \forall d \in D (|x - d| \geq \varepsilon_0)$ .

*Remark* (Negation predicate reading).

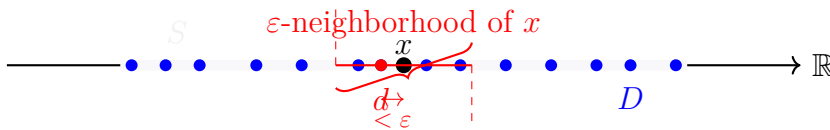
$\neg \text{DenseSubset}(D, S) \iff \exists x \in S \exists \varepsilon_0 > 0 \forall d \in D (|x - d| \geq \varepsilon_0)$ .

*Remark* (Failure modes). Density fails when some point of  $S$  has a positive-radius neighborhood that contains no point of  $D$ .

*Remark* (Failure mode decomposition).

$$\neg \text{DenseSubset}(D, S) = \underbrace{\exists x \in S}_{\text{point not approximated}} \underbrace{\exists \varepsilon_0 > 0}_{\text{positive exclusion radius}} \underbrace{\forall d \in D (|x - d| \geq \varepsilon_0)}_{\text{no point of } D \text{ enters the neighborhood}}.$$

*Remark* (Interpretation). Density is an approximation property inside  $S$ . It does not say that every point of  $S$  already belongs to  $D$ ; it says that points of  $D$  occur arbitrarily close to every point of  $S$ . Thus  $D$  can be a proper subset of  $S$  while still leaving no positive-width gap inside  $S$ .



**Figure.** Illustration of a Dense Subset  $D \subseteq S$ .

Given any  $x \in S$  (black dot) and any  $\varepsilon > 0$  (red interval width), there exists  $d \in D$  (red-filled blue dot) such that  $|x - d| < \varepsilon$ .

### Definition (Order Dense Subset)

**Definition 3.1.7** (Order Dense Subset). Let  $X$  be a linearly ordered set and let  $A \subseteq X$ . The subset  $A$  is *order dense in  $X$*  if and only if for every  $a, b \in X$  with  $a < b$  there exists  $c \in A$  such that  $a < c < b$ .

*Remark* (Standard quantified statement).

Let  $X$  be a linearly ordered set and  $A \subseteq X$ .

$A$  is order dense in  $X \iff \forall a, b \in X (a < b \rightarrow \exists c \in A (a < c < b))$ .

*Remark* (Definition predicate reading).

$\text{OrderDenseSubset}(A, X) := \forall a, b \in X (a < b \rightarrow \exists c \in A (a < c < b))$ .

*Remark* (Negated quantified statement).

Let  $X$  be a linearly ordered set and  $A \subseteq X$ .

$A$  is not order dense in  $X \iff \exists a, b \in X (a < b \wedge \forall c \in A (c \leq a \vee b \leq c))$ .

*Remark* (Negation predicate reading).

$\neg \text{OrderDenseSubset}(A, X) \iff \exists a, b \in X (a < b \wedge \forall c \in A (c \leq a \vee b \leq c))$ .

*Remark* (Failure modes). Order density fails when there is a nonempty open interval of  $X$  that contains no point of  $A$ .

*Remark* (Failure mode decomposition).

$$\neg \text{OrderDenseSubset}(A, X) = \underbrace{\exists a, b \in X (a < b)}_{\text{nonempty interval}} \wedge \underbrace{\forall c \in A (c \leq a \vee b \leq c)}_{\text{no point of } A \text{ lies between } a \text{ and } b}.$$

*Remark* (Interpretation). Order density is an interval-penetration property. It ignores metric distance and asks only whether every ordered gap contains a point of the subset. A subset can therefore be order dense in  $X$  precisely when no nonempty open interval of  $X$  is missed entirely by that subset.

*Remark* (Dependencies).

- [Total Order](#)
- [Subset](#)

### Theorem (Density of the Rational Numbers in $\mathbb{R}$ )

**Theorem 3.1.8** (Density of the Rational Numbers in  $\mathbb{R}$ ). *For every  $a, b \in \mathbb{R}$  with  $a < b$ , there exists  $q \in \mathbb{Q}$  such that  $a < q < b$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} (a < b \rightarrow \exists q \in \mathbb{Q} (a < q < b)).$$

*Remark* (Predicate reading).

$$\text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R} (a < b \rightarrow \exists q \in \mathbb{Q} (a < q < b)).$$

*Remark* (Negated quantified statement).

$$\exists a, b \in \mathbb{R} (a < b \wedge \forall q \in \mathbb{Q} (q \leq a \vee b \leq q)).$$

*Remark* (Negation predicate reading).

$$\neg \text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R} (a < b \wedge \forall q \in \mathbb{Q} (q \leq a \vee b \leq q)).$$

*Remark* (Failure modes). The theorem would fail precisely if some nonempty real interval contained no rational number.

*Remark* (Failure mode decomposition).

$$\neg \text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R} (a < b)}_{\text{nonempty real interval}} \wedge \underbrace{\forall q \in \mathbb{Q} (q \leq a \vee b \leq q)}_{\text{no rational lies inside}}.$$

*Remark* (Interpretation). Rational numbers penetrate every real interval. The theorem does not say that every real number is rational; it says that rational numbers are never separated from a real interval by a positive order gap. This is the interval form of rational density in the real line.

*Remark* (Dependencies).

Definition 3.1.7, Theorem 3.1.4.

#### Theorem (Density of the Irrational Numbers in $\mathbb{R}$ )

**Theorem 3.1.9** (Density of the Irrational Numbers in  $\mathbb{R}$ ). *For every  $a, b \in \mathbb{R}$  with  $a < b$ , there exists  $s \in \mathbb{R} \setminus \mathbb{Q}$  such that  $a < s < b$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} (a < b \rightarrow \exists s \in \mathbb{R} \setminus \mathbb{Q} (a < s < b)).$$

*Remark* (Predicate reading).

$$\text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R} (a < b \rightarrow \exists s \in \mathbb{R} \setminus \mathbb{Q} (a < s < b)).$$

*Remark* (Negated quantified statement).

$$\exists a, b \in \mathbb{R} (a < b \wedge \forall s \in \mathbb{R} \setminus \mathbb{Q} (s \leq a \vee b \leq s)).$$

*Remark* (Negation predicate reading).

$$\neg \text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R} (a < b \wedge \forall s \in \mathbb{R} \setminus \mathbb{Q} (s \leq a \vee b \leq s)).$$

*Remark* (Failure modes). The theorem would fail precisely if some nonempty real interval contained no irrational number.

*Remark* (Failure mode decomposition).

$$\neg \text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R} (a < b)}_{\text{nonempty real interval}} \wedge \underbrace{\forall s \in \mathbb{R} \setminus \mathbb{Q} (s \leq a \vee b \leq s)}_{\text{no irrational lies inside}}.$$

*Remark* (Interpretation). Irrational numbers also penetrate every real interval. One way to see the result is to insert a rational point into a translated and rescaled interval, then shift it by an irrational amount. The outcome is that the irrational numbers, like the rational numbers, leave no interval gap in  $\mathbb{R}$ .

*Remark* (Dependencies).

Definition 3.1.7, Theorem 3.1.8.

#### Corollary (Rational and Irrational Points in Every Open Interval)

**Corollary 3.1.10** (Rational and Irrational Points in Every Open Interval). *Every open interval in  $\mathbb{R}$  contains both a rational number and an irrational number. Go to proof.*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \left[ a < b \rightarrow (\exists q \in \mathbb{Q} (a < q < b) \wedge \exists s \in \mathbb{R} \setminus \mathbb{Q} (a < s < b)) \right].$$

*Remark* (Predicate reading).

$$\text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \wedge \text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}).$$

*Remark* (Negated quantified statement).

$$\exists a, b \in \mathbb{R} \left[ a < b \wedge (\forall q \in \mathbb{Q} (q \leq a \vee b \leq q) \vee \forall s \in \mathbb{R} \setminus \mathbb{Q} (s \leq a \vee b \leq s)) \right].$$

*Remark* (Negation predicate reading).

$$\neg \left[ \text{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \wedge \text{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \right].$$

*Remark* (Failure modes). The corollary fails if a nonempty real interval misses all rational points or misses all irrational points.

*Remark* (Failure mode decomposition).

$$\exists a, b \in \mathbb{R} (a < b) \wedge \underbrace{\left[ \forall q \in \mathbb{Q} (q \leq a \vee b \leq q) \right]}_{\text{no rational point}} \vee \underbrace{\left[ \forall s \in \mathbb{R} \setminus \mathbb{Q} (s \leq a \vee b \leq s) \right]}_{\text{no irrational point}}.$$

*Remark* (Interpretation). The corollary combines the two density theorems. Every real interval, no matter how small, contains representatives of both arithmetic types. Thus neither the rational numbers nor the irrational numbers occupy isolated regions of the real line.



*Remark* (Dependencies).

Theorem 3.1.8, Theorem 3.1.9.

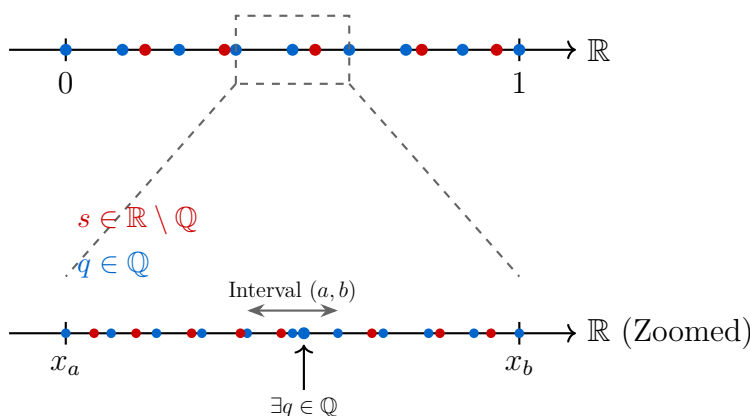


Figure 3.2: Visualization of the Density of  $\mathbb{Q}$  and  $\mathbb{R} \setminus \mathbb{Q}$  in  $\mathbb{R}$ . Zooming into any interval reveals that both sets are always present.

*Remark* (Zoom invariance and structural consequences).

Density in  $\mathbb{R}$  has a scale-invariant consequence: no matter how small a nonempty open interval becomes, it still contains both rational and irrational points. The interleaving of  $\mathbb{Q}$  and  $\mathbb{R} \setminus \mathbb{Q}$  is therefore not a large-scale phenomenon. It persists at every interval scale.

This has three important consequences. First, the continuum does not become locally discrete under magnification. For every nonempty open interval  $(a, b)$ , both

$$\mathbb{Q} \cap (a, b) \quad \text{and} \quad (\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$$

remain nonempty, and in fact infinite. Density is therefore stable under repeated restriction to smaller intervals.

Second, this behavior sharply distinguishes the real line from discrete numerical models. Floating-point systems form a discrete set of representable values, so repeated magnification eventually reaches a scale at which adjacent representable numbers leave no further point strictly between them. At that stage, interval penetration fails in the discrete model even though it continues to hold in  $\mathbb{R}$ .

Third, the Archimedean Property supplies the scale mechanism behind this behavior. Given any positive scale  $\varepsilon$ , there exists  $n \in \mathbb{N}$  with

$$\frac{1}{n} < \varepsilon.$$

This makes it possible to construct rational approximations at arbitrarily small interval scales, which is one of the basic engines behind density arguments.

The resulting contrast is structural. Smooth mathematical models are built in a setting where interval penetration persists at every scale, whereas finite computational models eventually encounter a smallest representable separation. The transition from continuum behavior to discrete behavior occurs exactly where nonempty intervals in the numerical model cease to contain further representable points.

### Density Dependency Map

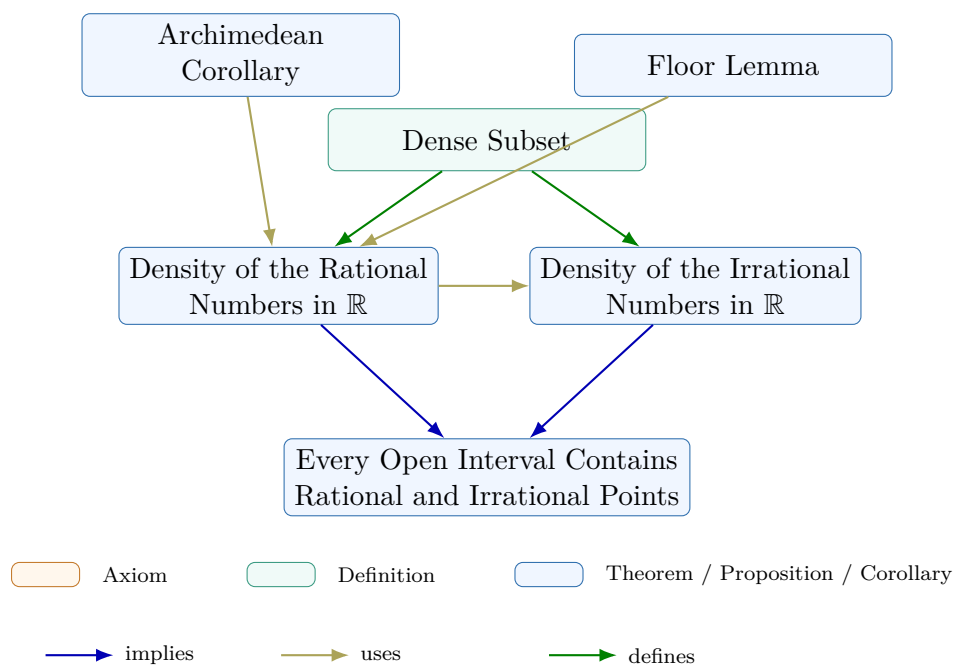


Figure 3.3: Dependency map for the density subchapter. The Floor Lemma and the Archimedean Corollary support the proof of the density of the rational numbers in  $\mathbb{R}$ . The irrational density theorem is then derived from the rational density result, and the final corollary combines both density theorems into a single interval statement. The definition records the local vocabulary of the subchapter.

### 3.1.4 Completeness Summary

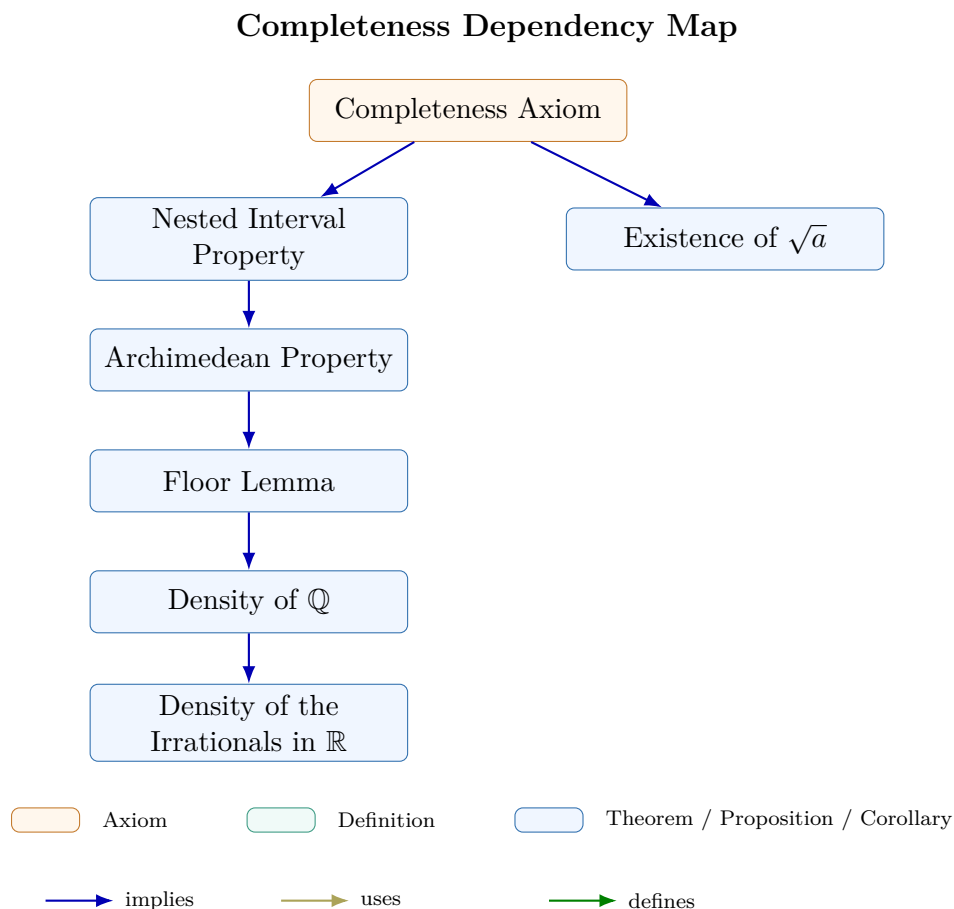


Figure 3.4: Dependency map for the completeness subchapter. The Completeness Axiom yields the Nested Interval Property and the existence of  $\sqrt{a}$ ; the Nested Interval Property then supports the Archimedean Property, the Floor Lemma, and the density results for  $\mathbb{Q}$  and for the irrational numbers.

*Remark* (Connections forward). Completeness appears in every major limit theorem in the notes. The Monotone Convergence Theorem (convergence chapter) constructs the limit of a bounded monotone sequence as  $\sup\{a_n : n \in \mathbb{N}\}$ ; existence of this supremum requires completeness. The Bolzano–Weierstrass Theorem applies the Nested Interval Property directly to extract a convergent subsequence from any bounded sequence. The Cauchy Criterion identifies convergent sequences purely in terms of their own spacing, without naming the limit in advance; its proof constructs the limit as a supremum and again requires completeness. In metric spaces (Distance chapter), completeness in the sense of Cauchy sequences is the natural generalization of the same idea to abstract settings. Every  $\varepsilon$ - $\delta$  proof that locates a limit or extremum is drawing on this chapter’s infrastructure.

## 3.2 Bounds and Extremal Values

### 3.2.1 Upper and Lower Bounds

Bounds are the first level of extremal information. They compare one candidate element with every element of a nonempty set, but they do not require the candidate to be sharp or to belong to the set.

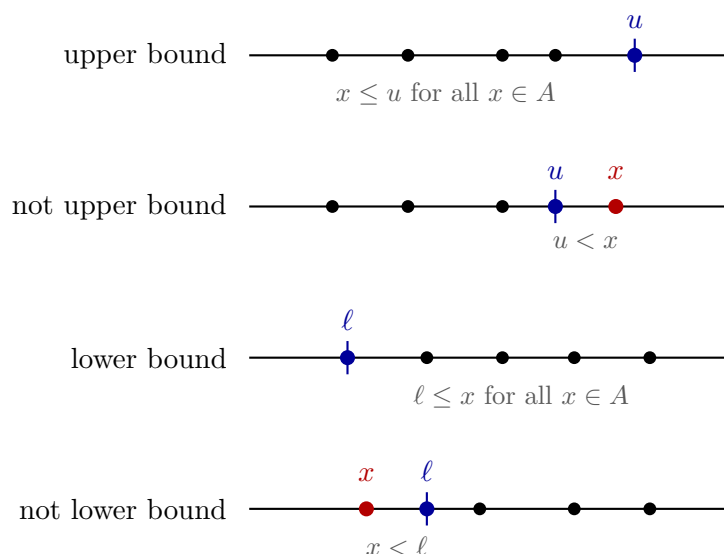


Figure 3.5: Upper and lower bound conditions on a nonempty set.

#### Bound

**Definition 3.2.1** (Bound). Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and let  $b \in S$ . A *bound* of  $A$  means a one-sided comparison between  $b$  and every element of  $A$ : either  $x \leq b$  for all  $x \in A$ , or  $b \leq x$  for all  $x \in A$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad [(\forall x \in A)(x \leq b) \text{ or } (\forall x \in A)(b \leq x)].$$

*Remark* (Definition predicate reading).

$$\text{Bound}_{S, \leq}(b, A) := \text{UB}_{S, \leq}(b, A) \vee \text{LB}_{S, \leq}(b, A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad [(\exists x \in A)(b < x) \text{ and } (\exists y \in A)(y < b)].$$

*Remark* (Negation predicate reading).

$$\neg \text{Bound}_{S, \leq}(b, A) \iff A = \emptyset \vee (\neg \text{UB}_{S, \leq}(b, A) \wedge \neg \text{LB}_{S, \leq}(b, A)).$$

*Remark* (Failure modes). The statement fails if the set is empty, or if  $b$  lies strictly between two elements of  $A$ .

*Remark* (Failure mode decomposition). Failure of the upper-bound side means some element of  $A$  is strictly larger than  $b$ . Failure of the lower-bound side means some element of  $A$  is strictly smaller than  $b$ . Failure to be any bound requires both witnesses.

### Upper Bound

**Definition 3.2.2** (Upper Bound). Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and let  $u \in S$ . The element  $u$  is an *upper bound* of  $A$  if every element of  $A$  is less than or equal to  $u$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\forall x \in A)(x \leq u).$$

*Remark* (Definition predicate reading).

$$\text{UB}_{S, \leq}(u, A) := A \neq \emptyset \wedge (\forall x \in A)(x \leq u).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\exists x \in A)(u < x).$$

*Remark* (Negation predicate reading).

$$\neg \text{UB}_{S, \leq}(u, A) \iff A = \emptyset \vee (\exists x \in A)(u < x).$$

*Remark* (Failure modes). The set may be empty, or a counterexample element of  $A$  may sit strictly above the proposed bound.

*Remark* (Failure mode decomposition). To disprove that  $u$  is an upper bound, produce either the structural failure  $A = \emptyset$  or a witness  $x \in A$  with  $u < x$ .

### Lower Bound

**Definition 3.2.3** (Lower Bound). Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and let  $\ell \in S$ . The element  $\ell$  is a *lower bound* of  $A$  if every element of  $A$  is greater than or equal to  $\ell$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\forall x \in A)(\ell \leq x).$$

*Remark* (Definition predicate reading).

$$\text{LB}_{S, \leq}(\ell, A) := A \neq \emptyset \wedge (\forall x \in A)(\ell \leq x).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\exists x \in A)(x < \ell).$$

*Remark* (Negation predicate reading).

$$\neg \text{LB}_{S,\leq}(\ell, A) \iff A = \emptyset \vee (\exists x \in A)(x < \ell).$$

*Remark* (Failure modes). The set may be empty, or a counterexample element of  $A$  may sit strictly below the proposed bound.

*Remark* (Failure mode decomposition). To disprove that  $\ell$  is a lower bound, produce either the structural failure  $A = \emptyset$  or a witness  $x \in A$  with  $x < \ell$ .

### Bounded Above

**Definition 3.2.4** (Bounded Above). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. The set  $A$  is *bounded above* if it has at least one upper bound in  $S$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\exists u \in S)(\forall x \in A)(x \leq u).$$

*Remark* (Definition predicate reading).

$$\text{BddAbove}_{S,\leq}(A) := (\exists u \in S) \text{UB}_{S,\leq}(u, A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\forall u \in S)(\exists x \in A)(u < x).$$

*Remark* (Negation predicate reading).

$$\neg \text{BddAbove}_{S,\leq}(A) \iff A = \emptyset \vee (\forall u \in S) \neg \text{UB}_{S,\leq}(u, A).$$

*Remark* (Failure modes). The set may be empty, or every proposed upper bound may be defeated by some larger element of  $A$ .

*Remark* (Failure mode decomposition). Unboundedness above is not one failed candidate; it is a uniform procedure that defeats every candidate  $u \in S$ .

### Bounded Below

**Definition 3.2.5** (Bounded Below). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. The set  $A$  is *bounded below* if it has at least one lower bound in  $S$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\exists \ell \in S)(\forall x \in A)(\ell \leq x).$$

*Remark* (Definition predicate reading).

$$\text{BddBelow}_{S,\leq}(A) := (\exists \ell \in S) \text{LB}_{S,\leq}(\ell, A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\forall \ell \in S)(\exists x \in A)(x < \ell).$$

*Remark* (Negation predicate reading).

$$\neg \text{BddBelow}_{S,\leq}(A) \iff A = \emptyset \vee (\forall \ell \in S) \neg \text{LB}_{S,\leq}(\ell, A).$$

*Remark* (Failure modes). The set may be empty, or every proposed lower bound may be defeated by some smaller element of  $A$ .

*Remark* (Failure mode decomposition). Unboundedness below is a uniform defeat of all lower-bound candidates.

### Bounded Set

**Definition 3.2.6** (Bounded Set). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. The set  $A$  is *bounded* if it is bounded above and bounded below.

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\exists \ell, u \in S)(\forall x \in A)(\ell \leq x \leq u).$$

*Remark* (Definition predicate reading).

$$\text{Bdd}_{S,\leq}(A) := \text{BddBelow}_{S,\leq}(A) \wedge \text{BddAbove}_{S,\leq}(A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\forall \ell, u \in S)(\exists x \in A)(x < \ell \text{ or } u < x).$$

*Remark* (Negation predicate reading).

$$\neg \text{Bdd}_{S,\leq}(A) \iff A = \emptyset \vee \neg \text{BddBelow}_{S,\leq}(A) \vee \neg \text{BddAbove}_{S,\leq}(A).$$

*Remark* (Failure modes). The set may be empty, unbounded below, unbounded above, or both.

*Remark* (Failure mode decomposition). Boundedness has two independent one-sided requirements. A failure on either side is enough to destroy boundedness.

## 3.2.2 Suprema and Infima

Suprema and infima are sharp bounds. A supremum is the least upper bound; an infimum is the greatest lower bound. The set remains required to be nonempty throughout this section.

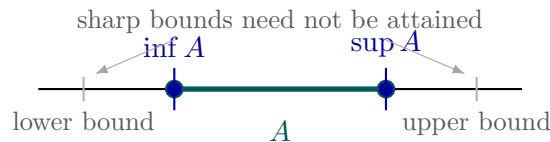


Figure 3.6: Suprema and infima are sharp bounds, whether or not they are elements of the set.

## Supremum

**Definition 3.2.7** (Supremum). Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and let  $s \in S$ . The element  $s$  is the *supremum* of  $A$  if  $s$  is an upper bound of  $A$  and no smaller element of  $S$  is an upper bound of  $A$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\forall x \in A)(x \leq s) \quad \text{and} \quad (\forall u \in S)[(\forall x \in A)(x \leq u) \Rightarrow s \leq u].$$

*Remark* (Definition predicate reading).

$$\text{Sup}_{S,\leq}(s, A) := \text{UB}_{S,\leq}(s, A) \wedge (\forall u \in S)(\text{UB}_{S,\leq}(u, A) \Rightarrow s \leq u).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\exists x \in A)(s < x) \quad \text{or} \quad (\exists u \in S)[(\forall x \in A)(x \leq u) \text{ and } u < s].$$

*Remark* (Negation predicate reading).

$$\neg \text{Sup}_{S,\leq}(s, A) \iff A = \emptyset \vee \neg \text{UB}_{S,\leq}(s, A) \vee (\exists u \in S)(\text{UB}_{S,\leq}(u, A) \wedge u < s).$$

*Remark* (Failure modes). The set may be empty;  $s$  may fail to be an upper bound; or  $s$  may be an upper bound that is not least among upper bounds.

*Remark* (Failure mode decomposition). The first clause is a domain failure. The second is a bound failure. The third is a sharpness failure witnessed by a strictly smaller upper bound.

## Infimum

**Definition 3.2.8** (Infimum). Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and let  $i \in S$ . The element  $i$  is the *infimum* of  $A$  if  $i$  is a lower bound of  $A$  and no larger element of  $S$  is a lower bound of  $A$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad (\forall x \in A)(i \leq x) \quad \text{and} \quad (\forall \ell \in S)[(\forall x \in A)(\ell \leq x) \Rightarrow \ell \leq i].$$

*Remark* (Definition predicate reading).

$$\text{Inf}_{S,\leq}(i, A) := \text{LB}_{S,\leq}(i, A) \wedge (\forall \ell \in S)(\text{LB}_{S,\leq}(\ell, A) \Rightarrow \ell \leq i).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad (\exists x \in A)(x < i) \quad \text{or} \quad (\exists \ell \in S)[(\forall x \in A)(\ell \leq x) \text{ and } i < \ell].$$

*Remark* (Negation predicate reading).

$$\neg \text{Inf}_{S,\leq}(i, A) \iff A = \emptyset \vee \neg \text{LB}_{S,\leq}(i, A) \vee (\exists \ell \in S)(\text{LB}_{S,\leq}(\ell, A) \wedge i < \ell).$$

*Remark* (Failure modes). The set may be empty;  $i$  may fail to be a lower bound; or  $i$  may be a lower bound that is not greatest among lower bounds.



*Remark* (Failure mode decomposition). The first clause is a domain failure. The second is a bound failure. The third is a sharpness failure witnessed by a strictly larger lower bound.

### Uniqueness of the Supremum

**Proposition 3.2.9** (Uniqueness of the Supremum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $s, t \in S$  are both suprema of  $A$ , then  $s = t$ .*  
*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Sup}_{S, \leq}(s, A) \quad \text{and} \quad \text{Sup}_{S, \leq}(t, A) \quad \implies \quad s = t.$$

*Remark* (Predicate reading).

$$\text{Sup}_{S, \leq}(s, A) \wedge \text{Sup}_{S, \leq}(t, A) \implies s = t.$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Sup}_{S, \leq}(s, A) \quad \text{and} \quad \text{Sup}_{S, \leq}(t, A) \quad \text{and} \quad s \neq t.$$

*Remark* (Negation predicate reading).

$$\text{Sup}_{S, \leq}(s, A) \wedge \text{Sup}_{S, \leq}(t, A) \wedge s \neq t.$$

*Remark* (Failure modes). Failure would require two distinct least upper bounds for the same nonempty set.

*Remark* (Failure mode decomposition). Each candidate is an upper bound and each lies below every upper bound; applying these minimality clauses to the other candidate gives both  $s \leq t$  and  $t \leq s$ .

### Uniqueness of the Infimum

**Proposition 3.2.10** (Uniqueness of the Infimum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $i, j \in S$  are both infima of  $A$ , then  $i = j$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Inf}_{S, \leq}(i, A) \quad \text{and} \quad \text{Inf}_{S, \leq}(j, A) \quad \implies \quad i = j.$$

*Remark* (Predicate reading).

$$\text{Inf}_{S, \leq}(i, A) \wedge \text{Inf}_{S, \leq}(j, A) \implies i = j.$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Inf}_{S, \leq}(i, A) \quad \text{and} \quad \text{Inf}_{S, \leq}(j, A) \quad \text{and} \quad i \neq j.$$

*Remark* (Negation predicate reading).

$$\text{Inf}_{S, \leq}(i, A) \wedge \text{Inf}_{S, \leq}(j, A) \wedge i \neq j.$$

*Remark* (Failure modes). Failure would require two distinct greatest lower bounds for the same nonempty set.

*Remark* (Failure mode decomposition). Each candidate is a lower bound and each lies above every lower bound; applying these maximality clauses to the other candidate gives both  $i \leq j$  and  $j \leq i$ .

### Upper Bound Property of the Supremum

**Proposition 3.2.11** (Upper Bound Property of the Supremum). *Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and suppose that  $s = \sup A$  exists in  $S$ . Then every element of  $A$  lies below  $s$ :*

$$(\forall x \in A)(x \leq s).$$

*Go to proof.*

*Remark* (Predicate reading).

$$\text{Sup}_{S, \leq}(s, A) \implies \text{UB}_{S, \leq}(s, A).$$

*Remark* (Negated quantified statement).

$$\text{Sup}_{S, \leq}(s, A) \quad \text{and} \quad (\exists x \in A)(s < x).$$

*Remark* (Failure modes). Failure would mean that a supremum is not an upper bound.

*Remark* (Failure mode decomposition). This is exactly the first clause in the definition of supremum.

### Lower Bound Property of the Infimum

**Proposition 3.2.12** (Lower Bound Property of the Infimum). *Let  $(S, \leq)$  be an ordered set, let  $A \subseteq S$  be nonempty, and suppose that  $i = \inf A$  exists in  $S$ . Then every element of  $A$  lies above  $i$ :*

$$(\forall x \in A)(i \leq x).$$

*Remark* (Predicate reading).

$$\text{Inf}_{S, \leq}(i, A) \implies \text{LB}_{S, \leq}(i, A).$$

*Remark* (Negated quantified statement).

$$\text{Inf}_{S, \leq}(i, A) \quad \text{and} \quad (\exists x \in A)(x < i).$$

*Remark* (Failure modes). Failure would mean that an infimum is not a lower bound.

*Remark* (Failure mode decomposition). This is exactly the first clause in the definition of infimum.

### Subset Inclusion Preserves Upper Bounds

**Lemma 3.2.13** (Subset Inclusion Preserves Upper Bounds). *Let  $A \subseteq B \subseteq S$  be nonempty subsets of an ordered set  $(S, \leq)$ . If  $u$  is an upper bound of  $B$ , then  $u$  is an upper bound of  $A$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad A \subseteq B, \quad (\forall y \in B)(y \leq u) \implies (\forall x \in A)(x \leq u).$$

*Remark* (Predicate reading).

$$A \subseteq B \wedge \text{UB}_{S,\leq}(u, B) \implies \text{UB}_{S,\leq}(u, A).$$

*Remark* (Negated quantified statement).

$$A \subseteq B, \quad \text{UB}_{S,\leq}(u, B), \quad \text{and} \quad (\exists x \in A)(u < x).$$

*Remark* (Failure modes). Failure would require a counterexample in the smaller set even though the larger set is already bounded by  $u$ .

*Remark* (Failure mode decomposition). Any counterexample  $x \in A$  is also an element of  $B$ , contradicting the upper-bound hypothesis for  $B$ .

#### Subset Inclusion Preserves Lower Bounds

**Lemma 3.2.14** (Subset Inclusion Preserves Lower Bounds). *Let  $A \subseteq B \subseteq S$  be nonempty subsets of an ordered set  $(S, \leq)$ . If  $\ell$  is a lower bound of  $B$ , then  $\ell$  is a lower bound of  $A$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad A \subseteq B, \quad (\forall y \in B)(\ell \leq y) \implies (\forall x \in A)(\ell \leq x).$$

*Remark* (Predicate reading).

$$A \subseteq B \wedge \text{LB}_{S,\leq}(\ell, B) \implies \text{LB}_{S,\leq}(\ell, A).$$

*Remark* (Negated quantified statement).

$$A \subseteq B, \quad \text{LB}_{S,\leq}(\ell, B), \quad \text{and} \quad (\exists x \in A)(x < \ell).$$

*Remark* (Failure modes). Failure would require a smaller-set counterexample to a lower bound that already works on the larger set.

*Remark* (Failure mode decomposition). Any counterexample  $x \in A$  lies in  $B$ , contradicting the lower-bound hypothesis for  $B$ .

#### Supremum Is Monotone Under Inclusion

**Theorem 3.2.15** (Supremum Is Monotone Under Inclusion). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded above. If  $A \subseteq B$ , then  $\sup A \leq \sup B$ .*  
*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset, \quad A \subseteq B, \quad \text{Sup}_{\mathbb{R},\leq}(s_A, A), \quad \text{Sup}_{\mathbb{R},\leq}(s_B, B) \implies s_A \leq s_B.$$

*Remark* (Predicate reading).

$$A \subseteq B \wedge \text{Sup}(s_A, A) \wedge \text{Sup}(s_B, B) \implies s_A \leq s_B.$$

*Remark* (Negated quantified statement).

$$A \subseteq B, \quad \text{Sup}(s_A, A), \quad \text{Sup}(s_B, B), \quad s_B < s_A.$$

*Remark* (Failure modes). Failure would mean enlarging a nonempty set lowers its least upper bound.

*Remark* (Failure mode decomposition). The value  $\sup B$  bounds  $B$ , hence bounds  $A$ ; since  $\sup A$  is least among upper bounds of  $A$ ,  $\sup A \leq \sup B$ .

### Infimum Is Monotone Under Inclusion

**Theorem 3.2.16** (Infimum Is Monotone Under Inclusion). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded below. If  $A \subseteq B$ , then  $\inf B \leq \inf A$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset, \quad A \subseteq B, \quad \inf_{\mathbb{R}, \leq}(i_A, A), \quad \inf_{\mathbb{R}, \leq}(i_B, B) \implies i_B \leq i_A.$$

*Remark* (Predicate reading).

$$A \subseteq B \wedge \inf(i_A, A) \wedge \inf(i_B, B) \implies i_B \leq i_A.$$

*Remark* (Negated quantified statement).

$$A \subseteq B, \quad \inf(i_A, A), \quad \inf(i_B, B), \quad i_A < i_B.$$

*Remark* (Failure modes). Failure would mean enlarging a nonempty set raises its greatest lower bound.

*Remark* (Failure mode decomposition). The value  $\inf B$  bounds  $B$  below, hence bounds  $A$  below; since  $\inf A$  is greatest among lower bounds of  $A$ ,  $\inf B \leq \inf A$ .

### Upper Bound Comparison with the Supremum

**Proposition 3.2.17** (Upper Bound Comparison with the Supremum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $s = \sup A$ . For  $u \in \mathbb{R}$ , the number  $u$  is an upper bound of  $A$  if and only if  $s \leq u$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad \text{Sup}(s, A) \implies (\forall u \in \mathbb{R})[\text{UB}(u, A) \iff s \leq u].$$

*Remark* (Predicate reading).

$$\text{Sup}(s, A) \implies \forall u (\text{UB}(u, A) \leftrightarrow s \leq u).$$

*Remark* (Negated quantified statement).

$$\text{Sup}(s, A) \quad \text{and} \quad (\exists u)[(\text{UB}(u, A) \wedge u < s) \vee (s \leq u \wedge \neg \text{UB}(u, A))].$$

*Remark* (Failure modes). Failure would mean an upper bound below the supremum, or a number above the supremum that fails to bound the set.

*Remark* (Failure mode decomposition). Leastness gives the forward implication. The fact that  $s$  bounds  $A$  gives the reverse implication by transitivity.

#### Lower Bound Comparison with the Infimum

**Proposition 3.2.18** (Lower Bound Comparison with the Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $i = \inf A$ . For  $\ell \in \mathbb{R}$ , the number  $\ell$  is a lower bound of  $A$  if and only if  $\ell \leq i$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad \text{Inf}(i, A) \implies (\forall \ell \in \mathbb{R})[\text{LB}(\ell, A) \iff \ell \leq i].$$

*Remark* (Predicate reading).

$$\text{Inf}(i, A) \implies \forall \ell (\text{LB}(\ell, A) \leftrightarrow \ell \leq i).$$

*Remark* (Negated quantified statement).

$$\text{Inf}(i, A) \quad \text{and} \quad (\exists \ell)[(\text{LB}(\ell, A) \wedge i < \ell) \vee (\ell \leq i \wedge \neg \text{LB}(\ell, A))].$$

*Remark* (Failure modes). Failure would mean a lower bound above the infimum, or a number below the infimum that fails to bound the set below.

*Remark* (Failure mode decomposition). Greatestness gives the forward implication. The fact that  $i$  bounds  $A$  below gives the reverse implication by transitivity.

#### Every Element Lies Below the Supremum

**Proposition 3.2.19** (Every Element Lies Below the Supremum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $s = \sup A$ . Then  $x \leq s$  for every  $x \in A$ .  
Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad \text{Sup}(s, A) \implies (\forall x \in A)(x \leq s).$$

*Remark* (Predicate reading).

$$\text{Sup}(s, A) \implies \text{UB}(s, A).$$

*Remark* (Negated quantified statement).

$$\text{Sup}(s, A) \quad \text{and} \quad (\exists x \in A)(s < x).$$

*Remark* (Failure modes). Failure would mean the supremum is not an upper bound.

*Remark* (Failure mode decomposition). This is the upper-bound clause in the definition of supremum.

### Every Element Lies Above the Infimum

**Proposition 3.2.20** (Every Element Lies Above the Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $i = \inf A$ . Then  $i \leq x$  for every  $x \in A$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad \text{Inf}(i, A) \implies (\forall x \in A)(i \leq x).$$

*Remark* (Predicate reading).

$$\text{Inf}(i, A) \implies \text{LB}(i, A).$$

*Remark* (Negated quantified statement).

$$\text{Inf}(i, A) \quad \text{and} \quad (\exists x \in A)(x < i).$$

*Remark* (Failure modes). Failure would mean the infimum is not a lower bound.

*Remark* (Failure mode decomposition). This is the lower-bound clause in the definition of infimum.

### Infimum Is Less Than or Equal to the Supremum

**Theorem 3.2.21** (Infimum Is Less Than or Equal to the Supremum). *Let  $A \subseteq \mathbb{R}$  be nonempty, bounded below, and bounded above. Then  $\inf A \leq \sup A$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad \text{Inf}(i, A), \quad \text{Sup}(s, A) \implies i \leq s.$$

*Remark* (Predicate reading).

$$\text{Inf}(i, A) \wedge \text{Sup}(s, A) \implies i \leq s.$$

*Remark* (Negated quantified statement).

$$\text{Inf}(i, A), \quad \text{Sup}(s, A), \quad \text{and} \quad s < i.$$

*Remark* (Failure modes). Failure would put the greatest lower bound strictly above the least upper bound of the same nonempty set.

*Remark* (Failure mode decomposition). Choose  $a \in A$ . Then  $i \leq a$  because  $i$  is a lower bound, and  $a \leq s$  because  $s$  is an upper bound.

### Order Comparison of Suprema

**Proposition 3.2.22** (Order Comparison of Suprema). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded above. If every  $x \in A$  is below some  $y \in B$ , then  $\sup A \leq \sup B$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset, \quad (\forall x \in A)(\exists y \in B)(x \leq y) \implies \sup A \leq \sup B.$$

*Remark* (Predicate reading).

$$[\forall x \in A \exists y \in B (x \leq y)] \wedge \text{Sup}(s_A, A) \wedge \text{Sup}(s_B, B) \implies s_A \leq s_B.$$

*Remark* (Negated quantified statement).

$$[\forall x \in A \exists y \in B (x \leq y)] \quad \text{and} \quad \sup B < \sup A.$$

*Remark* (Failure modes). Failure would mean  $A$  reaches higher in supremum than  $B$  even though each element of  $A$  is dominated by an element of  $B$ .

*Remark* (Failure mode decomposition). Every element of  $B$  lies below  $\sup B$ , so every element of  $A$  also lies below  $\sup B$ . Thus  $\sup B$  is an upper bound for  $A$ .

### Least Upper Bound Property Implies Existence of Suprema

**Theorem 3.2.23** (Least Upper Bound Property Implies Existence of Suprema). *Let  $S \subseteq \mathbb{R}$  be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number  $s$  such that  $s = \sup S$ .*

*Remark* (Standard quantified statement).

$$S \neq \emptyset \quad \text{and} \quad (\exists u \in \mathbb{R})(\forall x \in S)(x \leq u) \implies (\exists s \in \mathbb{R}) \text{Sup}_{\mathbb{R}, \leq}(s, S).$$

*Remark* (Predicate reading).

$$S \neq \emptyset \wedge \text{BddAbove}_{\mathbb{R}, \leq}(S) \implies (\exists s \in \mathbb{R}) \text{Sup}_{\mathbb{R}, \leq}(s, S).$$

*Remark* (Negated quantified statement).

$$S \neq \emptyset, \quad \text{BddAbove}_{\mathbb{R}, \leq}(S), \quad \text{and} \quad (\forall s \in \mathbb{R}) \neg \text{Sup}_{\mathbb{R}, \leq}(s, S).$$

*Remark* (Failure modes). In  $\mathbb{R}$  the theorem cannot fail once the set is nonempty and bounded above; outside complete ordered settings, the existence clause can fail.

*Remark* (Failure mode decomposition). The hypotheses separate domain nonemptiness from one-sided boundedness. The conclusion asserts existence of the sharp upper bound.

### Greatest Lower Bound Property Implies Existence of Infima

**Theorem 3.2.24** (Greatest Lower Bound Property Implies Existence of Infima). *Let  $S \subseteq \mathbb{R}$  be nonempty and bounded below. Then there exists a real number  $i$  such that  $i = \inf S$ .*

*Remark* (Standard quantified statement).

$$S \neq \emptyset \quad \text{and} \quad (\exists \ell \in \mathbb{R})(\forall x \in S)(\ell \leq x) \implies (\exists i \in \mathbb{R}) \text{Inf}_{\mathbb{R}, \leq}(i, S).$$

*Remark* (Predicate reading).

$$S \neq \emptyset \wedge \text{BddBelow}_{\mathbb{R}, \leq}(S) \implies (\exists i \in \mathbb{R}) \text{Inf}_{\mathbb{R}, \leq}(i, S).$$

*Remark* (Negated quantified statement).

$$S \neq \emptyset, \quad \text{BddBelow}_{\mathbb{R}, \leq}(S), \quad \text{and} \quad (\forall i \in \mathbb{R}) \neg \text{Inf}_{\mathbb{R}, \leq}(i, S).$$

*Remark* (Failure modes). In  $\mathbb{R}$  the theorem cannot fail once the set is nonempty and bounded below.

*Remark* (Failure mode decomposition). Apply the least upper bound property to  $-S$  or argue by the order-dual form of completeness.

### 3.2.3 Maxima and Minima

Maxima and minima are attained sharp bounds. They are stronger than suprema and infima because the extremal element must belong to the nonempty set itself.

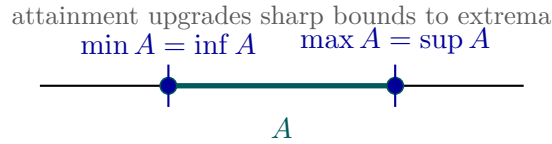


Figure 3.7: Maxima and minima are suprema and infima that are attained by the set.

#### Maximum

**Definition 3.2.25** (Maximum). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. An element  $m \in S$  is the *maximum* of  $A$  if  $m \in A$  and every element of  $A$  is less than or equal to  $m$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad m \in A \quad \text{and} \quad (\forall x \in A)(x \leq m).$$

*Remark* (Definition predicate reading).

$$\text{Max}_{S, \leq}(m, A) := m \in A \wedge \text{UB}_{S, \leq}(m, A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad m \notin A \quad \text{or} \quad (\exists x \in A)(m < x).$$



*Remark* (Negation predicate reading).

$$\neg \text{Max}_{S,\leq}(m, A) \iff A = \emptyset \vee m \notin A \vee \neg \text{UB}_{S,\leq}(m, A).$$

*Remark* (Failure modes). The set may be empty; the candidate may not be a member of the set; or a larger element of the set may exist.

*Remark* (Failure mode decomposition). Maximum combines membership and an upper-bound condition. Failure of either component prevents attainment.

### Minimum

**Definition 3.2.26** (Minimum). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. An element  $m \in S$  is the *minimum* of  $A$  if  $m \in A$  and every element of  $A$  is greater than or equal to  $m$ .

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad m \in A \quad \text{and} \quad (\forall x \in A)(m \leq x).$$

*Remark* (Definition predicate reading).

$$\text{Min}_{S,\leq}(m, A) := m \in A \wedge \text{LB}_{S,\leq}(m, A).$$

*Remark* (Negated quantified statement).

$$A = \emptyset \quad \text{or} \quad m \notin A \quad \text{or} \quad (\exists x \in A)(x < m).$$

*Remark* (Negation predicate reading).

$$\neg \text{Min}_{S,\leq}(m, A) \iff A = \emptyset \vee m \notin A \vee \neg \text{LB}_{S,\leq}(m, A).$$

*Remark* (Failure modes). The set may be empty; the candidate may not be a member of the set; or a smaller element of the set may exist.

*Remark* (Failure mode decomposition). Minimum combines membership and a lower-bound condition. Failure of either component prevents attainment.

### Uniqueness of the Maximum

**Proposition 3.2.27** (Uniqueness of the Maximum). Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $m, n \in S$  are both maxima of  $A$ , then  $m = n$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Max}_{S,\leq}(m, A) \quad \text{and} \quad \text{Max}_{S,\leq}(n, A) \implies m = n.$$

*Remark* (Predicate reading).

$$\text{Max}_{S,\leq}(m, A) \wedge \text{Max}_{S,\leq}(n, A) \implies m = n.$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Max}_{S,\leq}(m, A) \quad \text{and} \quad \text{Max}_{S,\leq}(n, A) \quad \text{and} \quad m \neq n.$$

*Remark* (Negation predicate reading).

$$\text{Max}_{S,\leq}(m, A) \wedge \text{Max}_{S,\leq}(n, A) \wedge m \neq n.$$

*Remark* (Failure modes). Failure would require two distinct elements of  $A$  each to dominate the entire set.

*Remark* (Failure mode decomposition). Since  $m$  is maximal over all elements and  $n \in A$ , we get  $n \leq m$ . Since  $n$  is maximal over all elements and  $m \in A$ , we get  $m \leq n$ .

### Uniqueness of the Minimum

**Proposition 3.2.28** (Uniqueness of the Minimum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $m, n \in S$  are both minima of  $A$ , then  $m = n$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Min}_{S,\leq}(m, A) \quad \text{and} \quad \text{Min}_{S,\leq}(n, A) \quad \implies \quad m = n.$$

*Remark* (Predicate reading).

$$\text{Min}_{S,\leq}(m, A) \wedge \text{Min}_{S,\leq}(n, A) \implies m = n.$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Min}_{S,\leq}(m, A) \quad \text{and} \quad \text{Min}_{S,\leq}(n, A) \quad \text{and} \quad m \neq n.$$

*Remark* (Negation predicate reading).

$$\text{Min}_{S,\leq}(m, A) \wedge \text{Min}_{S,\leq}(n, A) \wedge m \neq n.$$

*Remark* (Failure modes). Failure would require two distinct elements of  $A$  each to lie below the entire set.

*Remark* (Failure mode decomposition). Since  $m$  is minimal and  $n \in A$ , we get  $m \leq n$ . Since  $n$  is minimal and  $m \in A$ , we get  $n \leq m$ .

### Maximum Implies Supremum

**Proposition 3.2.29** (Maximum Implies Supremum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $m = \max A$ , then  $m = \sup A$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Max}_{S,\leq}(m, A) \quad \implies \quad \text{Sup}_{S,\leq}(m, A).$$

*Remark* (Predicate reading).

$$\text{Max}_{S,\leq}(m, A) \implies \text{Sup}_{S,\leq}(m, A).$$

*Remark* (Negated quantified statement).

$$\text{Max}_{S,\leq}(m, A) \quad \text{and} \quad \neg \text{Sup}_{S,\leq}(m, A).$$

*Remark* (Failure modes). Failure would require an attained upper bound that is not least among upper bounds.

*Remark* (Failure mode decomposition). Membership gives  $m \in A$ . Any upper bound of  $A$  must lie above every element of  $A$ , hence above  $m$  itself.

#### Minimum Implies Infimum

**Proposition 3.2.30** (Minimum Implies Infimum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $m = \min A$ , then  $m = \inf A$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Min}_{S,\leq}(m, A) \implies \text{Inf}_{S,\leq}(m, A).$$

*Remark* (Predicate reading).

$$\text{Min}_{S,\leq}(m, A) \implies \text{Inf}_{S,\leq}(m, A).$$

*Remark* (Negated quantified statement).

$$\text{Min}_{S,\leq}(m, A) \quad \text{and} \quad \neg \text{Inf}_{S,\leq}(m, A).$$

*Remark* (Failure modes). Failure would require an attained lower bound that is not greatest among lower bounds.

*Remark* (Failure mode decomposition). Membership gives  $m \in A$ . Any lower bound of  $A$  must lie below every element of  $A$ , hence below  $m$  itself.

#### Supremum in the Set is the Maximum

**Proposition 3.2.31** (Supremum in the Set is the Maximum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $s = \sup A$  and  $s \in A$ , then  $s = \max A$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Sup}_{S,\leq}(s, A) \quad \text{and} \quad s \in A \implies \text{Max}_{S,\leq}(s, A).$$

*Remark* (Predicate reading).

$$\text{Sup}_{S,\leq}(s, A) \wedge s \in A \implies \text{Max}_{S,\leq}(s, A).$$

*Remark* (Negated quantified statement).

$$\text{Sup}_{S,\leq}(s, A) \quad \text{and} \quad s \in A \quad \text{and} \quad \neg \text{Max}_{S,\leq}(s, A).$$

*Remark* (Failure modes). Failure would require a supremum that belongs to the set but is not its maximum.

*Remark* (Failure mode decomposition). The supremum condition supplies the upper-bound clause. Membership supplies attainment.

#### Infimum in the Set is the Minimum

**Proposition 3.2.32** (Infimum in the Set is the Minimum). *Let  $(S, \leq)$  be an ordered set and let  $A \subseteq S$  be nonempty. If  $i = \inf A$  and  $i \in A$ , then  $i = \min A$ .*

*Remark* (Standard quantified statement).

$$A \neq \emptyset \quad \text{and} \quad \text{Inf}_{S,\leq}(i, A) \quad \text{and} \quad i \in A \quad \implies \quad \text{Min}_{S,\leq}(i, A).$$

*Remark* (Predicate reading).

$$\text{Inf}_{S,\leq}(i, A) \wedge i \in A \implies \text{Min}_{S,\leq}(i, A).$$

*Remark* (Negated quantified statement).

$$\text{Inf}_{S,\leq}(i, A) \quad \text{and} \quad i \in A \quad \text{and} \quad \neg \text{Min}_{S,\leq}(i, A).$$

*Remark* (Failure modes). Failure would require an infimum that belongs to the set but is not its minimum.

*Remark* (Failure mode decomposition). The infimum condition supplies the lower-bound clause. Membership supplies attainment.

### 3.2.4 Epsilon Characterization

#### $\varepsilon$ -Characterization of Supremum

##### Definition (Epsilon Characterization of Supremum)

**Definition 3.2.33** (Epsilon Characterization of Supremum). Let  $A \subseteq \mathbb{R}$  and let  $s \in \mathbb{R}$ . The element  $s$  satisfies the *epsilon characterization of the supremum* of  $A$  if and only if  $s$  is an upper bound of  $A$  and for every  $\varepsilon > 0$  there exists  $a \in A$  such that  $s - \varepsilon < a$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}. \\ &s \text{ satisfies the epsilon characterization of } \sup A \\ &\iff \left[ \forall x \in A (x \leq s) \right] \wedge \left[ \forall \varepsilon > 0 \exists a \in A (s - \varepsilon < a) \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{EpsilonSupremum}(s, A) := \text{UpperBound}(s, A) \wedge \forall \varepsilon > 0 \exists a \in A (s - \varepsilon < a).$$

*Remark* (Negated quantified statement).

Let  $A \subseteq \mathbb{R}$  and  $s \in \mathbb{R}$ .

$s$  does not satisfy the epsilon characterization of  $\sup A$

$$\iff \left[ \exists x \in A (s < x) \right] \vee \left[ \exists \varepsilon_0 > 0 \forall a \in A (a \leq s - \varepsilon_0) \right].$$

*Remark* (Negation predicate reading).

$$\neg \text{EpsilonSupremum}(s, A) \iff \left[ \exists x \in A (s < x) \right] \vee \left[ \exists \varepsilon_0 > 0 \forall a \in A (a \leq s - \varepsilon_0) \right].$$

*Remark* (Failure modes). The characterization fails either because  $s$  is not an upper bound of  $A$ , or because  $A$  remains at least some positive distance below  $s$ .

*Remark* (Failure mode decomposition).

$$\neg \text{EpsilonSupremum}(s, A) = \underbrace{\exists x \in A (s < x)}_{\text{upper-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0 \forall a \in A (a \leq s - \varepsilon_0)}_{\text{positive gap below } s}.$$

*Remark* (Interpretation). This condition says that  $s$  caps the set from above while points of  $A$  approach  $s$  from below at every positive scale. The upper-bound clause prevents  $A$  from crossing above  $s$ , and the epsilon clause prevents  $s$  from floating strictly above the set. Geometrically,  $s$  is a ceiling that the set presses against arbitrarily closely.

*Remark* (Dependencies).

[Definition 10.1.7.](#)

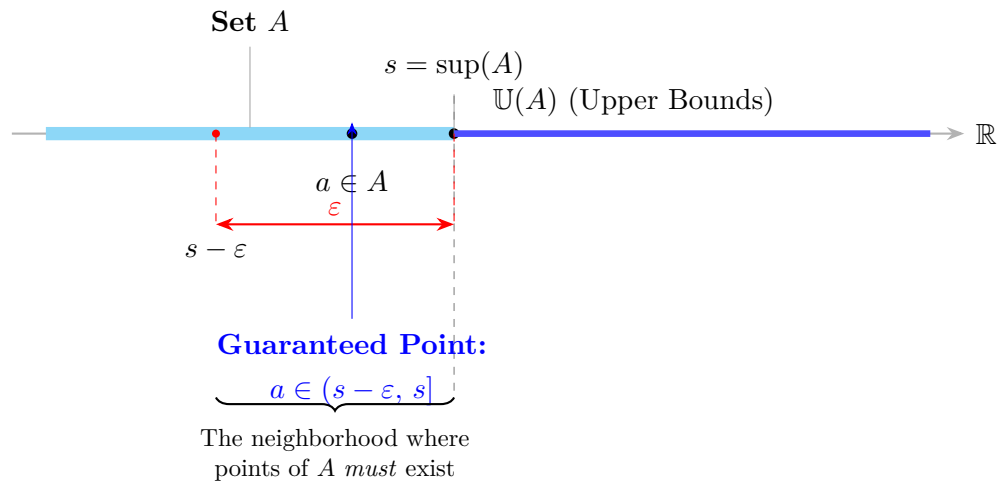


Figure 3.8: Refined visualization of the  $\varepsilon$ -characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

### Definition (Epsilon Characterization of Infimum)

**Definition 3.2.34** (Epsilon Characterization of Infimum). Let  $A \subseteq \mathbb{R}$  and let  $t \in \mathbb{R}$ . The element  $t$  satisfies the *epsilon characterization of the infimum* of  $A$  if and only if  $t$  is a lower bound of  $A$  and for every  $\varepsilon > 0$  there exists  $a \in A$  such that  $a < t + \varepsilon$ .

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$  and  $t \in \mathbb{R}$ .

$t$  satisfies the epsilon characterization of  $\inf A$

$$\iff \left[ \forall x \in A (t \leq x) \right] \wedge \left[ \forall \varepsilon > 0 \exists a \in A (a < t + \varepsilon) \right].$$

*Remark* (Definition predicate reading).

$$\text{EpsilonInfimum}(t, A) := \text{LowerBound}(t, A) \wedge \forall \varepsilon > 0 \exists a \in A (a < t + \varepsilon).$$

*Remark* (Negated quantified statement).

Let  $A \subseteq \mathbb{R}$  and  $t \in \mathbb{R}$ .

$t$  does not satisfy the epsilon characterization of  $\inf A$

$$\iff \left[ \exists x \in A (x < t) \right] \vee \left[ \exists \varepsilon_0 > 0 \forall a \in A (t + \varepsilon_0 \leq a) \right].$$

*Remark* (Negation predicate reading).

$$\neg \text{EpsilonInfimum}(t, A) \iff \left[ \exists x \in A (x < t) \right] \vee \left[ \exists \varepsilon_0 > 0 \forall a \in A (t + \varepsilon_0 \leq a) \right].$$

*Remark* (Failure modes). The characterization fails either because  $t$  is not a lower bound of  $A$ , or because  $A$  remains at least some positive distance above  $t$ .

*Remark* (Failure mode decomposition).

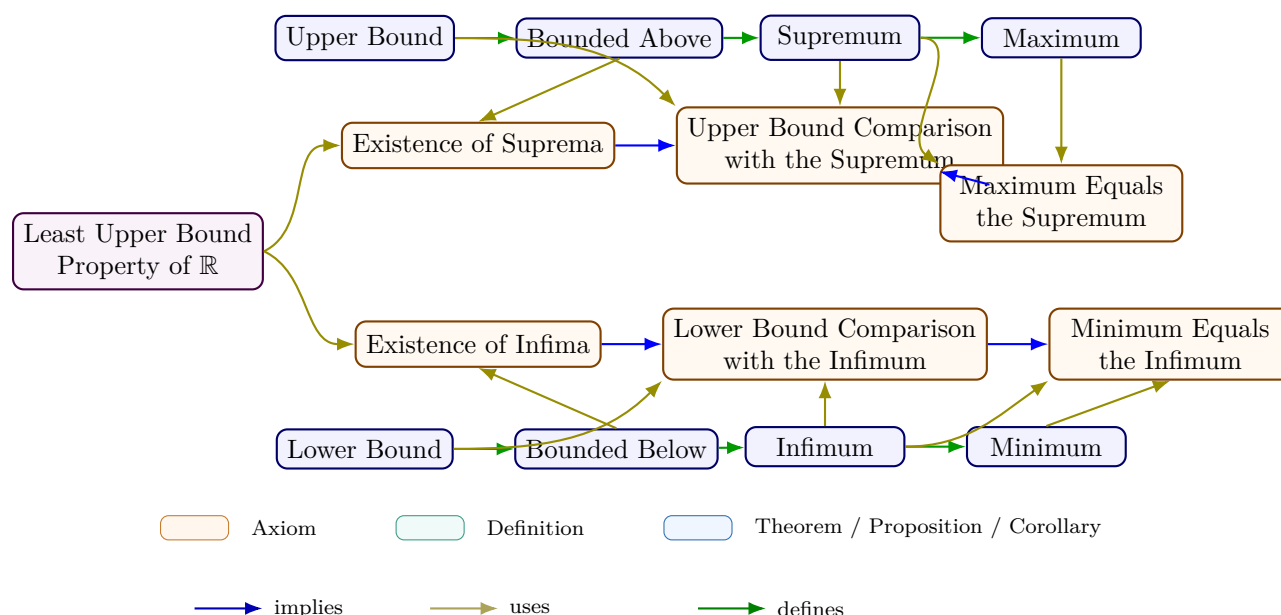
$$\neg \text{EpsilonInfimum}(t, A) = \underbrace{\exists x \in A (x < t)}_{\text{lower-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0 \forall a \in A (t + \varepsilon_0 \leq a)}_{\text{positive gap above } t}.$$

*Remark* (Interpretation). This condition says that  $t$  sits below the set while points of  $A$  approach  $t$  from above at every positive scale. The lower-bound clause prevents  $A$  from crossing below  $t$ , and the epsilon clause prevents  $t$  from sitting strictly below the set with a positive gap. Geometrically,  $t$  is a floor that the set presses against arbitrarily closely from above.

*Remark* (Dependencies).

[Definition 10.1.7.](#)

### 3.2.5 Extremal Bounds Summary



*Remark* (Equivalent formulations).

Over the ordered-field axioms, completeness is equivalent to each of the following:

- Every nonempty set bounded below has an infimum.
- Every Cauchy sequence in  $\mathbb{R}$  converges.
- (*Nested Interval Property*) Every nested sequence of nonempty closed bounded intervals has nonempty intersection.

## 3.3 Algebra of Bounds

### 3.3.1 Set Operations and Bounds

*Remark* (Orientation). Set operations change the collection of possible counterexamples to a proposed bound. Each result below isolates one operation and one direction of bound transport.

#### Lemma (Union Preserves Upper Bounds)

**Lemma 3.3.1** (Union Preserves Upper Bounds). *Let  $A, B \subseteq \mathbb{R}$ . If  $u_A$  is an upper bound of  $A$  and  $u_B$  is an upper bound of  $B$ , then  $\max\{u_A, u_B\}$  is an upper bound of  $A \cup B$ . Go to proof.*

*Remark* (Standard quantified statement).

$$(\forall a \in A)(a \leq u_A) \wedge (\forall b \in B)(b \leq u_B) \implies (\forall x \in A \cup B)(x \leq \max\{u_A, u_B\}).$$

*Remark* (Predicate reading).

$$\text{UB}(u_A, A) \wedge \text{UB}(u_B, B) \implies \text{UB}(\max\{u_A, u_B\}, A \cup B).$$

*Remark* (Negated quantified statement).

$$(\forall a \in A)(a \leq u_A), \quad (\forall b \in B)(b \leq u_B), \quad (\exists x \in A \cup B)(\max\{u_A, u_B\} < x).$$

*Remark* (Negation predicate reading).

$$\text{UB}(u_A, A) \wedge \text{UB}(u_B, B) \wedge \neg \text{UB}(\max\{u_A, u_B\}, A \cup B).$$

*Remark* (Failure modes). A failure would require a union element larger than both proposed upper bounds.

*Remark* (Failure mode decomposition). Every  $x \in A \cup B$  lies in  $A$  or in  $B$ . In the first case  $x \leq u_A \leq \max\{u_A, u_B\}$ ; in the second case  $x \leq u_B \leq \max\{u_A, u_B\}$ .

*Remark* (Dependencies).

Upper Bound.

#### Lemma (Union Preserves Lower Bounds)

**Lemma 3.3.2** (Union Preserves Lower Bounds). *Let  $A, B \subseteq \mathbb{R}$ . If  $\ell_A$  is a lower bound of  $A$  and  $\ell_B$  is a lower bound of  $B$ , then  $\min\{\ell_A, \ell_B\}$  is a lower bound of  $A \cup B$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$(\forall a \in A)(\ell_A \leq a) \wedge (\forall b \in B)(\ell_B \leq b) \implies (\forall x \in A \cup B)(\min\{\ell_A, \ell_B\} \leq x).$$

*Remark* (Predicate reading).

$$\text{LB}(\ell_A, A) \wedge \text{LB}(\ell_B, B) \implies \text{LB}(\min\{\ell_A, \ell_B\}, A \cup B).$$

*Remark* (Negated quantified statement).

$$(\forall a \in A)(\ell_A \leq a), \quad (\forall b \in B)(\ell_B \leq b), \quad (\exists x \in A \cup B)(x < \min\{\ell_A, \ell_B\}).$$

*Remark* (Negation predicate reading).

$$\text{LB}(\ell_A, A) \wedge \text{LB}(\ell_B, B) \wedge \neg \text{LB}(\min\{\ell_A, \ell_B\}, A \cup B).$$

*Remark* (Failure modes). A failure would require a union element smaller than both proposed lower bounds.

*Remark* (Failure mode decomposition). Every  $x \in A \cup B$  lies in  $A$  or in  $B$ . In either case the appropriate lower bound lies below  $x$ , and  $\min\{\ell_A, \ell_B\}$  lies below that lower bound.



Remark (Dependencies).

Lower Bound.

**Proposition (Union Is Bounded Above Exactly When Both Pieces Are)**

**Proposition 3.3.3** (Union Is Bounded Above Exactly When Both Pieces Are). *Let  $A, B \subseteq \mathbb{R}$  be nonempty. Then  $A \cup B$  is bounded above if and only if  $A$  is bounded above and  $B$  is bounded above. Go to proof.*

Remark (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset \quad \implies \quad \text{BddAbove}(A \cup B) \iff \text{BddAbove}(A) \wedge \text{BddAbove}(B).$$

Remark (Predicate reading).

$$\text{BddAbove}(A \cup B) \iff \text{BddAbove}(A) \wedge \text{BddAbove}(B).$$

Remark (Negated quantified statement). The equivalence fails precisely if one side holds and the other does not.

Remark (Negation predicate reading).

$$\text{BddAbove}(A \cup B) \oplus (\text{BddAbove}(A) \wedge \text{BddAbove}(B)).$$

Remark (Failure modes). The forward direction could fail only if a bound for the union did not bound a piece. The reverse direction could fail only if bounds for both pieces did not combine into a bound for the union.

Remark (Failure mode decomposition). Use subset inheritance for the forward direction and [Union Preserves Upper Bounds](#) for the reverse direction.

Remark (Dependencies).

Bounded Above; Union Preserves Upper Bounds; Subsets Preserve Upper Bounds.

**Proposition (Union Is Bounded Below Exactly When Both Pieces Are)**

**Proposition 3.3.4** (Union Is Bounded Below Exactly When Both Pieces Are). *Let  $A, B \subseteq \mathbb{R}$  be nonempty. Then  $A \cup B$  is bounded below if and only if  $A$  is bounded below and  $B$  is bounded below. Go to proof.*

Remark (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset \quad \implies \quad \text{BddBelow}(A \cup B) \iff \text{BddBelow}(A) \wedge \text{BddBelow}(B).$$

Remark (Predicate reading).

$$\text{BddBelow}(A \cup B) \iff \text{BddBelow}(A) \wedge \text{BddBelow}(B).$$

Remark (Negated quantified statement). The equivalence fails precisely if one side holds and the other does not.

*Remark* (Negation predicate reading).

$$\text{BddBelow}(A \cup B) \oplus (\text{BddBelow}(A) \wedge \text{BddBelow}(B)).$$

*Remark* (Failure modes). The forward direction could fail only if a lower bound for the union did not bound a piece. The reverse direction could fail only if lower bounds for both pieces did not combine into a lower bound for the union.

*Remark* (Failure mode decomposition). Use subset inheritance for the forward direction and [Union Preserves Lower Bounds](#) for the reverse direction.

*Remark* (Dependencies).

[Bounded Below](#); [Union Preserves Lower Bounds](#); [Subsets Preserve Lower Bounds](#).

#### Corollary (Union Is Bounded Exactly When Both Pieces Are)

**Corollary 3.3.5** (Union Is Bounded Exactly When Both Pieces Are). *Let  $A, B \subseteq \mathbb{R}$  be nonempty. Then  $A \cup B$  is bounded if and only if  $A$  is bounded and  $B$  is bounded. [Go to proof](#).*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad B \neq \emptyset \quad \implies \quad \text{Bdd}(A \cup B) \iff \text{Bdd}(A) \wedge \text{Bdd}(B).$$

*Remark* (Predicate reading).

$$\text{Bdd}(A \cup B) \iff \text{Bdd}(A) \wedge \text{Bdd}(B).$$

*Remark* (Negated quantified statement). The two-sided boundedness equivalence fails if either the upper or lower one-sided equivalence fails.

*Remark* (Negation predicate reading).

$$\text{Bdd}(A \cup B) \oplus (\text{Bdd}(A) \wedge \text{Bdd}(B)).$$

*Remark* (Failure modes). Two-sided boundedness fails exactly through an upper-bound failure or a lower-bound failure.

*Remark* (Failure mode decomposition). Apply the bounded-above and bounded-below union equivalences separately.

*Remark* (Dependencies).

[Bounded Set](#); [Union Is Bounded Above Exactly When Both Pieces Are](#); [Union Is Bounded Below Exactly When Both Pieces Are](#).

#### Lemma (Subsets Preserve Upper Bounds)

**Lemma 3.3.6** (Subsets Preserve Upper Bounds). *Let  $C \subseteq A \subseteq \mathbb{R}$ . If  $u$  is an upper bound of  $A$ , then  $u$  is an upper bound of  $C$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$C \subseteq A \wedge (\forall a \in A)(a \leq u) \implies (\forall c \in C)(c \leq u).$$

*Remark* (Predicate reading).

$$C \subseteq A \wedge \text{UB}(u, A) \implies \text{UB}(u, C).$$

*Remark* (Negated quantified statement).

$$C \subseteq A, \quad (\forall a \in A)(a \leq u), \quad (\exists c \in C)(u < c).$$

*Remark* (Negation predicate reading).

$$C \subseteq A \wedge \text{UB}(u, A) \wedge \neg \text{UB}(u, C).$$

*Remark* (Failure modes). A failure would require a counterexample in  $C$  that is not already a counterexample in  $A$ .

*Remark* (Failure mode decomposition). Since  $C \subseteq A$ , every  $c \in C$  is an element of  $A$ . The upper-bound condition on  $A$  therefore applies directly to  $c$ .

*Remark* (Dependencies).

Upper Bound.

#### Lemma (Subsets Preserve Lower Bounds)

**Lemma 3.3.7** (Subsets Preserve Lower Bounds). *Let  $C \subseteq A \subseteq \mathbb{R}$ . If  $\ell$  is a lower bound of  $A$ , then  $\ell$  is a lower bound of  $C$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$C \subseteq A \wedge (\forall a \in A)(\ell \leq a) \implies (\forall c \in C)(\ell \leq c).$$

*Remark* (Predicate reading).

$$C \subseteq A \wedge \text{LB}(\ell, A) \implies \text{LB}(\ell, C).$$

*Remark* (Negated quantified statement).

$$C \subseteq A, \quad (\forall a \in A)(\ell \leq a), \quad (\exists c \in C)(c < \ell).$$

*Remark* (Negation predicate reading).

$$C \subseteq A \wedge \text{LB}(\ell, A) \wedge \neg \text{LB}(\ell, C).$$

*Remark* (Failure modes). A failure would require a point in  $C$  below  $\ell$ , hence a point in  $A$  below  $\ell$ .

*Remark* (Failure mode decomposition). Subset inheritance removes possible counterexamples; it cannot create a new one.

*Remark* (Dependencies).

Lower Bound.

#### Corollary (Intersections Inherit Bounds)

**Corollary 3.3.8** (Intersections Inherit Bounds). *Let  $A, B \subseteq \mathbb{R}$ . Every upper bound of  $A$  is an upper bound of  $A \cap B$ , and every lower bound of  $A$  is a lower bound of  $A \cap B$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$A \cap B \subseteq A.$$

*Remark* (Predicate reading).

$$\text{UB}(u, A) \implies \text{UB}(u, A \cap B),$$

and similarly for lower bounds.

*Remark* (Negated quantified statement). A failure would be a failure of subset inheritance for  $A \cap B \subseteq A$ .

*Remark* (Negation predicate reading).

$$\text{UB}(u, A) \wedge \neg \text{UB}(u, A \cap B)$$

is contradictory, and the same holds for lower bounds.

*Remark* (Failure modes). Intersection can lose elements, but it cannot introduce an element outside the parent set  $A$ .

*Remark* (Failure mode decomposition). Apply subset inheritance to  $C = A \cap B$ .

*Remark* (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

#### Corollary (Differences Inherit Bounds)

**Corollary 3.3.9** (Differences Inherit Bounds). *Let  $A, B \subseteq \mathbb{R}$ . Every upper bound of  $A$  is an upper bound of  $A \setminus B$ , and every lower bound of  $A$  is a lower bound of  $A \setminus B$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$A \setminus B \subseteq A.$$

*Remark* (Predicate reading).

$$\text{UB}(u, A) \implies \text{UB}(u, A \setminus B),$$

and similarly for lower bounds.

*Remark* (Negated quantified statement). A failure would be a failure of subset inheritance for  $A \setminus B \subseteq A$ .

*Remark* (Negation predicate reading).

$$\text{UB}(u, A) \wedge \neg \text{UB}(u, A \setminus B)$$

is contradictory, and the same holds for lower bounds.

*Remark* (Failure modes). Difference can remove elements from  $A$ , but it cannot create a new element outside  $A$ .

*Remark* (Failure mode decomposition). Apply subset inheritance to  $C = A \setminus B$ .

*Remark* (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

#### Corollary (Complements Inherit Ambient Bounds)

**Corollary 3.3.10** (Complements Inherit Ambient Bounds). *Let  $A \subseteq E \subseteq \mathbb{R}$ , and take complements relative to  $E$ . Every upper bound of  $E$  is an upper bound of  $E \setminus A$ , and every lower bound of  $E$  is a lower bound of  $E \setminus A$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$E \setminus A \subseteq E.$$

*Remark* (Predicate reading).

$$\text{UB}(u, E) \implies \text{UB}(u, E \setminus A), \quad \text{LB}(\ell, E) \implies \text{LB}(\ell, E \setminus A).$$

*Remark* (Negated quantified statement).

$$\text{UB}(u, E) \wedge \neg \text{UB}(u, E \setminus A)$$

or

$$\text{LB}(\ell, E) \wedge \neg \text{LB}(\ell, E \setminus A).$$

*Remark* (Negation predicate reading). Both negated configurations contradict subset inheritance because  $E \setminus A \subseteq E$ .

*Remark* (Failure modes). The complement does not inherit bounds from  $A$ ; it inherits them from the ambient set  $E$ .

*Remark* (Failure mode decomposition). Every point of  $E \setminus A$  is a point of  $E$ , so every ambient bound of  $E$  applies to the complement.

*Remark* (Dependencies).

Subsets Preserve Upper Bounds; Subsets Preserve Lower Bounds.

### 3.3.2 Lattice Operations and Bounds

*Remark* (Orientation). In this section, *lattice* refers to the max–min structure already present in the usual order on  $\mathbb{R}$ . For two real numbers, their join is their maximum and their meet is their minimum:

$$x \vee y = \max\{x, y\}, \quad x \wedge y = \min\{x, y\}.$$

Thus lattice operations are not a new order relation; they are the operations created by the order. The set-level constructions below apply those pairwise operations to all pairs of points from two nonempty sets.

### Definition (Pointwise Maximum and Minimum Sets)

**Definition 3.3.11** (Pointwise Maximum and Minimum Sets). Let  $A, B \subseteq \mathbb{R}$  be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A, b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A, b \in B\}.$$

*Remark* (Standard quantified statement).

$$x \in \max(A, B) \iff (\exists a \in A)(\exists b \in B)(x = \max\{a, b\}),$$

and similarly for  $\min(A, B)$  with  $\min\{a, b\}$ .

*Remark* (Definition predicate reading).

$$\text{PointwiseMaxSet}(A, B) = \max(A, B), \quad \text{PointwiseMinSet}(A, B) = \min(A, B).$$

*Remark* (Negated quantified statement).

$$x \notin \max(A, B) \iff (\forall a \in A)(\forall b \in B)(x \neq \max\{a, b\}),$$

and similarly for  $\min(A, B)$ .

*Remark* (Negation predicate reading).

$$\neg \text{PointwiseMaxSet}(A, B) = C \quad \text{means} \quad (\exists x)[x \in C \triangle \max(A, B)],$$

and similarly for  $\text{PointwiseMinSet}$ .

*Remark* (Failure modes). The definition fails only if the image set omits a pairwise maximum or minimum, or if it includes a value not produced by one of the pairwise rules.

*Remark* (Failure mode decomposition). Every membership claim decomposes into two input witnesses and one order-based operation: either the larger input or the smaller input is selected.

*Remark* (Dependencies).

Maximum; Minimum.

### Theorem (Supremum of the Pointwise Maximum Set)

**Theorem 3.3.12** (Supremum of the Pointwise Maximum Set). Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded above. Then

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded above} \implies \sup\{\max\{a, b\} : a \in A, b \in B\} = \max\{\sup A, \sup B\}. \blacksquare$$

*Remark* (Predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies \text{Supremum}(\max\{s_A, s_B\}, \max(A, B)).$$

*Remark* (Negated quantified statement).

$$\sup \max(A, B) \neq \max\{\sup A, \sup B\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg \text{Supremum}(\max\{s_A, s_B\}, \max(A, B)).$$

*Remark* (Failure modes). The formula could fail only if the maximum of the two suprema failed to bound all pairwise maxima, or if it were not the least such bound.

*Remark* (Failure mode decomposition). Every pairwise maximum is bounded by  $\max\{\sup A, \sup B\}$  ■ Sharpness comes from holding one input fixed in the nonempty other set and letting the relevant set approach its own supremum.

*Remark* (Dependencies).

Supremum; Pointwise Maximum and Minimum Sets.

#### Theorem (Infimum of the Pointwise Maximum Set)

**Theorem 3.3.13** (Infimum of the Pointwise Maximum Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded below. Then*

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded below} \implies \inf\{\max\{a, b\} : a \in A, b \in B\} = \max\{\inf A, \inf B\}. \blacksquare$$

*Remark* (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \implies \text{Infimum}(\max\{t_A, t_B\}, \max(A, B)).$$

*Remark* (Negated quantified statement).

$$\inf \max(A, B) \neq \max\{\inf A, \inf B\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \neg \text{Infimum}(\max\{t_A, t_B\}, \max(A, B)).$$

*Remark* (Failure modes). The formula could fail only if the larger of the two lower thresholds were not a lower bound for all pairwise maxima, or if pairwise maxima could not approach that threshold from above.

*Remark* (Failure mode decomposition). Every pairwise maximum is at least both lower thresholds, hence at least their maximum. Approximation of the two infima gives pairwise maxima arbitrarily close to the larger lower threshold.

*Remark* (Dependencies).

Infimum; Pointwise Maximum and Minimum Sets.

#### Theorem (Supremum of the Pointwise Minimum Set)

**Theorem 3.3.14** (Supremum of the Pointwise Minimum Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded above. Then*

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded above} \implies \sup\{\min\{a, b\} : a \in A, b \in B\} = \min\{\sup A, \sup B\}. \blacksquare$$

*Remark* (Predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies \text{Supremum}(\min\{s_A, s_B\}, \min(A, B)).$$

*Remark* (Negated quantified statement).

$$\sup \min(A, B) \neq \min\{\sup A, \sup B\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg \text{Supremum}(\min\{s_A, s_B\}, \min(A, B)).$$

*Remark* (Failure modes). The formula could fail only if the smaller supremum failed to bound all pairwise minima, or if that upper bound were not sharp.

*Remark* (Failure mode decomposition). Every pairwise minimum is below both suprema, hence below their minimum. Sharpness comes from taking both inputs near their own suprema.

*Remark* (Dependencies).

Supremum; Pointwise Maximum and Minimum Sets.



### Theorem (Infimum of the Pointwise Minimum Set)

**Theorem 3.3.15** (Infimum of the Pointwise Minimum Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded below. Then*

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded below} \implies \inf\{\min\{a, b\} : a \in A, b \in B\} = \min\{\inf A, \inf B\}. \blacksquare$$

*Remark* (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \implies \text{Infimum}(\min\{t_A, t_B\}, \min(A, B)).$$

*Remark* (Negated quantified statement).

$$\inf \min(A, B) \neq \min\{\inf A, \inf B\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \neg \text{Infimum}(\min\{t_A, t_B\}, \min(A, B)).$$

*Remark* (Failure modes). The formula could fail only if the smaller infimum failed to bound all pairwise minima from below, or if that lower bound were not sharp.

*Remark* (Failure mode decomposition). Every pairwise minimum is at least the smaller of the two infima. Sharpness comes from holding one set nonempty and approaching the smaller lower threshold.

*Remark* (Dependencies).

[Infimum; Pointwise Maximum and Minimum Sets.](#)

### 3.3.3 Images, Inverse Images, and Extrema

*Remark* (Orientation). One cannot usually move a supremum or infimum through an arbitrary function:

$$\sup f(A) \neq f(\sup A)$$

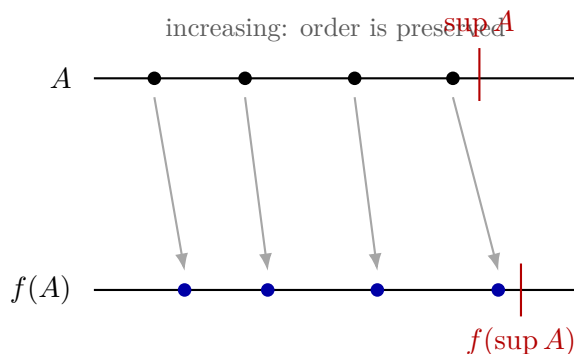
in general. Equality is an order-and-continuity statement. Monotone functions preserve or reverse the relevant inequalities, and continuity at the endpoint turns the endpoint bound into a sharp bound for the image. The inverse function statements are the same principle applied to  $f^{-1}$ .

*Remark* (Geometric meaning of the hypotheses). Think of a nonempty bounded set  $A \subseteq \mathbb{R}$  as a cloud of points on the real line. The number  $\sup A$  is the right-hand wall of that cloud: it may or may not belong to  $A$ , but points of  $A$  press arbitrarily close to it from the left. Applying a function  $f$  moves every point of the cloud and produces the image cloud

$$f(A) = \{f(a) : a \in A\}.$$

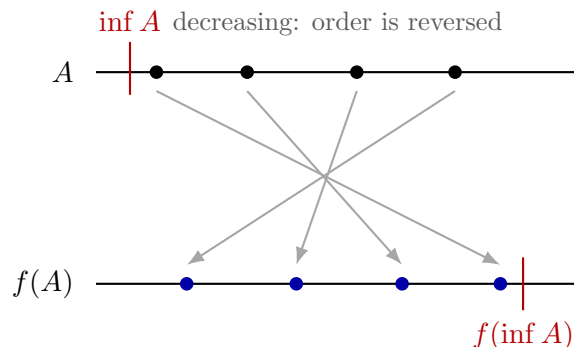
The question  $\sup f(A) = f(\sup A)$  asks whether the function carries the right-hand wall of the original cloud to the right-hand wall of the image cloud.

*Example* (Increasing images keep the right edge on the right).



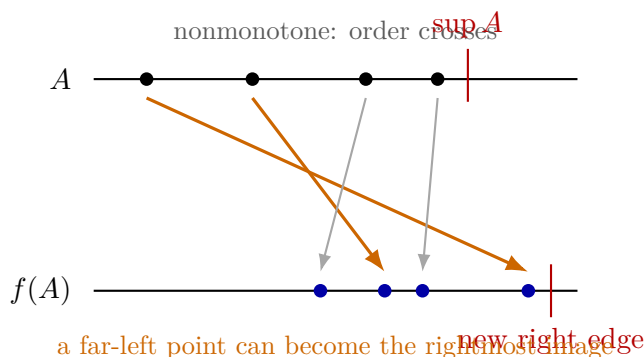
When  $f$  is increasing, order is preserved. A point that starts farther to the right cannot land to the left of a point that starts farther to the left. Thus the right edge of the image cloud is controlled by the right edge of the original cloud, provided continuity prevents a jump at the endpoint.

*Example* (Decreasing images swap the endpoint roles).



When  $f$  is decreasing, order is reversed. The left edge of the original cloud becomes the right edge of the image cloud, and the right edge of the original cloud becomes the left edge of the image cloud. This is why decreasing image theorems exchange suprema and infima.

*Example* (Nonmonotone images can cross).



If  $f$  is not monotone on the relevant domain, the arrows can cross. Then the right edge of  $f(A)$  need not come from  $\sup A$  or from  $\inf A$  alone in a uniform way. The image edge may be produced by an interior point, by the left branch of a piecewise monotone function, or by the right branch. In that situation the set must be split into pieces where  $f$  is monotone, and the image pieces must be recombined using the algebra of suprema over unions.

*Example* (Why  $x \mapsto x^2$  does not fit either global theorem). The function  $f(x) = x^2$  is not increasing on all of  $\mathbb{R}$ . Indeed,  $-2 \leq 1$ , but

$$(-2)^2 = 4 > 1 = 1^2.$$

It is also not decreasing on all of  $\mathbb{R}$ . Indeed,  $1 \leq 2$ , but

$$1^2 = 1 < 4 = 2^2.$$

Thus neither the increasing image theorem nor the decreasing image theorem can be applied globally to  $x \mapsto x^2$  on  $\mathbb{R}$ .

Let

$$A = (-1, 2).$$

Then  $\inf A = -1$  and  $\sup A = 2$ . The image under the square map is

$$f(A) = \{x^2 : -1 < x < 2\} = [0, 4),$$

so

$$\sup f(A) = 4.$$

The increasing-theorem expression gives

$$f(\sup A) = f(2) = 4,$$

which happens to agree in this example. But the agreement is not justified by global increasingness, because  $x^2$  is not globally increasing.

The decreasing-theorem expression gives

$$f(\inf A) = f(-1) = 1,$$

and this does not agree:

$$\sup f(A) = 4 \neq 1 = f(\inf A).$$

The failure occurs because the square map folds the real line at 0. It is decreasing on  $(-\infty, 0]$  and increasing on  $[0, \infty)$ , so a set that straddles 0 must be analyzed branch by branch. For a nonempty bounded set  $A \subseteq \mathbb{R}$  with finite endpoints  $\alpha = \inf A$  and  $\beta = \sup A$ , the largest square is controlled by whichever endpoint is farther from 0:

$$\sup\{x^2 : x \in A\} = \max\{\alpha^2, \beta^2\}.$$

### Theorem (Increasing Image Preserves Suprema)

**Theorem 3.3.16** (Increasing Image Preserves Suprema). *Let  $I \subseteq \mathbb{R}$ , let  $A \subseteq I$  be nonempty and bounded above, and let  $f : I \rightarrow \mathbb{R}$  be increasing. Suppose  $s = \sup A$  belongs to  $I$  and  $f$  is continuous at  $s$ . Then*

$$\sup f(A) = f(\sup A), \quad f(A) = \{f(a) : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad s = \sup A \in I, \quad f \text{ increasing and continuous at } s \implies \sup\{f(a) : a \in A\} = f(s).$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, s) \implies \text{Supremum}(f(s), f(A)).$$

*Remark* (Negated quantified statement).

$$\sup f(A) \neq f(\sup A)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, s) \wedge \neg \text{Supremum}(f(s), f(A)).$$

*Remark* (Failure modes). The equality can fail if  $f$  is not increasing, if  $\sup A$  is not in the domain, or if the endpoint value of  $f$  is not approached by values  $f(a)$  with  $a \in A$ .

*Remark* (Failure mode decomposition). Increasingness gives  $f(a) \leq f(s)$  for every  $a \in A$ . The defining approximation property of the supremum gives points of  $A$  arbitrarily close to  $s$  from below, and continuity at  $s$  makes their images approach  $f(s)$ .

*Remark* (Dependencies).

Supremum; Increasing Function; Continuity at a Point.

### Theorem (Increasing Image Preserves Infima)

**Theorem 3.3.17** (Increasing Image Preserves Infima). *Let  $I \subseteq \mathbb{R}$ , let  $A \subseteq I$  be nonempty and bounded below, and let  $f : I \rightarrow \mathbb{R}$  be increasing. Suppose  $t = \inf A$  belongs to  $I$  and  $f$  is continuous at  $t$ . Then*

$$\inf f(A) = f(\inf A).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad t = \inf A \in I, \quad f \text{ increasing and continuous at } t \implies \inf\{f(a) : a \in A\} = f(t).$$

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, t) \implies \text{Infimum}(f(t), f(A)).$$

*Remark* (Negated quantified statement).

$$\inf f(A) \neq f(\inf A)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Increasing}(f, I) \wedge \text{ContinuousAt}(f, t) \wedge \neg \text{Infimum}(f(t), f(A)).$$

*Remark* (Failure modes). The equality can fail if  $f$  is not increasing, if  $\inf A$  is not in the domain, or if the endpoint value is not reached as a limit of image values.

*Remark* (Failure mode decomposition). Increasingness gives  $f(t) \leq f(a)$  for every  $a \in A$ . Approximation of the infimum from above and continuity at  $t$  make this lower bound sharp.

*Remark* (Dependencies).

[Infimum](#); [Increasing Function](#); [Continuity at a Point](#).

#### Theorem (Decreasing Image Sends Infimum to Supremum)

**Theorem 3.3.18** (Decreasing Image Sends Infimum to Supremum). *Let  $I \subseteq \mathbb{R}$ , let  $A \subseteq I$  be nonempty and bounded below, and let  $f : I \rightarrow \mathbb{R}$  be decreasing. Suppose  $t = \inf A$  belongs to  $I$  and  $f$  is continuous at  $t$ . Then*

$$\sup f(A) = f(\inf A).$$

[Go to proof](#).

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad t = \inf A \in I, \quad f \text{ decreasing and continuous at } t \implies \sup\{f(a) : a \in A\} = f(t).$$

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, t) \implies \text{Supremum}(f(t), f(A)).$$

*Remark* (Negated quantified statement).

$$\sup f(A) \neq f(\inf A)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, t) \wedge \neg \text{Supremum}(f(t), f(A)).$$

*Remark* (Failure modes). The equality can fail if  $f$  is not decreasing, if  $\inf A$  is outside the domain, or if  $f$  is not continuous at the lower endpoint.

*Remark* (Failure mode decomposition). Decreasingness gives  $f(a) \leq f(t)$  for every  $a \in A$ . Approximation of  $t$  from above and continuity at  $t$  make this upper bound sharp.

*Remark* (Dependencies).

Infimum; Supremum; Decreasing Function; Continuity at a Point.

#### Theorem (Decreasing Image Sends Supremum to Infimum)

**Theorem 3.3.19** (Decreasing Image Sends Supremum to Infimum). *Let  $I \subseteq \mathbb{R}$ , let  $A \subseteq I$  be nonempty and bounded above, and let  $f : I \rightarrow \mathbb{R}$  be decreasing. Suppose  $s = \sup A$  belongs to  $I$  and  $f$  is continuous at  $s$ . Then*

$$\inf f(A) = f(\sup A).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad s = \sup A \in I, \quad f \text{ decreasing and continuous at } s \implies \inf\{f(a) : a \in A\} = f(s).$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, s) \implies \text{Infimum}(f(s), f(A)).$$

*Remark* (Negated quantified statement).

$$\inf f(A) \neq f(\sup A)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Decreasing}(f, I) \wedge \text{ContinuousAt}(f, s) \wedge \neg \text{Infimum}(f(s), f(A)).$$

*Remark* (Failure modes). The equality can fail if  $f$  is not decreasing, if  $\sup A$  is outside the domain, or if  $f$  is not continuous at the upper endpoint.

*Remark* (Failure mode decomposition). Decreasingness gives  $f(s) \leq f(a)$  for every  $a \in A$ . Approximation of  $s$  from below and continuity at  $s$  make this lower bound sharp.

*Remark* (Dependencies).

Supremum; Infimum; Decreasing Function; Continuity at a Point.

### Theorem (Increasing Inverse Preserves Suprema)

**Theorem 3.3.20** (Increasing Inverse Preserves Suprema). *Let  $I, J \subseteq \mathbb{R}$ , and let  $f : I \rightarrow J$  be bijective. Suppose  $f^{-1} : J \rightarrow I$  is increasing and continuous. If  $B \subseteq J$  is nonempty and bounded above, and if  $\sup B \in J$ , then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$B \neq \emptyset, \quad \sup B \in J, \quad f^{-1} \text{ increasing and continuous at } \sup B \implies \sup f^{-1}(B) = f^{-1}(\sup B). \blacksquare$$

*Remark* (Predicate reading).

$$\text{Supremum}(u, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \implies \text{Supremum}(f^{-1}(u), f^{-1}(B)). \blacksquare$$

*Remark* (Negated quantified statement).

$$\sup f^{-1}(B) \neq f^{-1}(\sup B)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(u, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \wedge \neg \text{Supremum}(f^{-1}(u), f^{-1}(B)).$$

*Remark* (Failure modes). The formula can fail if the inverse is not increasing, if  $\sup B$  is outside the codomain set  $J$ , or if the inverse is not continuous at  $\sup B$ .

*Remark* (Failure mode decomposition). Apply the increasing image supremum theorem to  $g = f^{-1}$  and the set  $B$ .

*Remark* (Dependencies).

[Increasing Image Preserves Suprema.](#)

### Theorem (Increasing Inverse Preserves Infima)

**Theorem 3.3.21** (Increasing Inverse Preserves Infima). *Let  $I, J \subseteq \mathbb{R}$ , and let  $f : I \rightarrow J$  be bijective. Suppose  $f^{-1} : J \rightarrow I$  is increasing and continuous. If  $B \subseteq J$  is nonempty and bounded below, and if  $\inf B \in J$ , then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$B \neq \emptyset, \quad \inf B \in J, \quad f^{-1} \text{ increasing and continuous at } \inf B \implies \inf f^{-1}(B) = f^{-1}(\inf B).$$

*Remark* (Predicate reading).

$$\text{Infimum}(v, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, v) \implies \text{Infimum}(f^{-1}(v), f^{-1}(B)).$$

*Remark* (Negated quantified statement).

$$\inf f^{-1}(B) \neq f^{-1}(\inf B)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(v, B) \wedge \text{Increasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, v) \wedge \neg \text{Infimum}(f^{-1}(v), f^{-1}(B)).$$

*Remark* (Failure modes). The formula can fail if the inverse is not increasing, if  $\inf B$  is outside the codomain set  $J$ , or if the inverse is not continuous at  $\inf B$ .

*Remark* (Failure mode decomposition). Apply the increasing image infimum theorem to  $g = f^{-1}$  and the set  $B$ .

*Remark* (Dependencies).

[Increasing Image Preserves Infima.](#)

#### Theorem (Decreasing Inverse Sends Infimum to Supremum)

**Theorem 3.3.22** (Decreasing Inverse Sends Infimum to Supremum). *Let  $I, J \subseteq \mathbb{R}$ , and let  $f : I \rightarrow J$  be bijective. Suppose  $f^{-1} : J \rightarrow I$  is decreasing and continuous. If  $B \subseteq J$  is nonempty and bounded below, and if  $\inf B \in J$ , then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$B \neq \emptyset, \quad \inf B \in J, \quad f^{-1} \text{ decreasing and continuous at } \inf B \implies \sup f^{-1}(B) = f^{-1}(\inf B). \blacksquare$$

*Remark* (Predicate reading).

$$\text{Infimum}(v, B) \wedge \text{Decreasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, v) \implies \text{Supremum}(f^{-1}(v), f^{-1}(B)).$$

*Remark* (Negated quantified statement).

$$\sup f^{-1}(B) \neq f^{-1}(\inf B)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(v, B) \wedge \text{Decreasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, v) \wedge \neg \text{Supremum}(f^{-1}(v), f^{-1}(B)).$$



*Remark* (Failure modes). The formula can fail if the inverse is not decreasing, if  $\inf B$  is outside  $J$ , or if the inverse is not continuous at  $\inf B$ .

*Remark* (Failure mode decomposition). Apply the decreasing image infimum-to-supremum theorem to  $g = f^{-1}$  and  $B$ .

*Remark* (Dependencies).

Decreasing Image Sends Infimum to Supremum.

#### Theorem (Decreasing Inverse Sends Supremum to Infimum)

**Theorem 3.3.23** (Decreasing Inverse Sends Supremum to Infimum). *Let  $I, J \subseteq \mathbb{R}$ , and let  $f : I \rightarrow J$  be bijective. Suppose  $f^{-1} : J \rightarrow I$  is decreasing and continuous. If  $B \subseteq J$  is nonempty and bounded above, and if  $\sup B \in J$ , then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$B \neq \emptyset, \quad \sup B \in J, \quad f^{-1} \text{ decreasing and continuous at } \sup B \implies \inf f^{-1}(B) = f^{-1}(\sup B).$  ■

*Remark* (Predicate reading).

$\text{Supremum}(u, B) \wedge \text{Decreasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \implies \text{Infimum}(f^{-1}(u), f^{-1}(B)).$

*Remark* (Negated quantified statement).

$$\inf f^{-1}(B) \neq f^{-1}(\sup B)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$\text{Supremum}(u, B) \wedge \text{Decreasing}(f^{-1}, J) \wedge \text{ContinuousAt}(f^{-1}, u) \wedge \neg \text{Infimum}(f^{-1}(u), f^{-1}(B)).$

*Remark* (Failure modes). The formula can fail if the inverse is not decreasing, if  $\sup B$  is outside  $J$ , or if the inverse is not continuous at  $\sup B$ .

*Remark* (Failure mode decomposition). Apply the decreasing image supremum-to-infimum theorem to  $g = f^{-1}$  and  $B$ .

*Remark* (Dependencies).

Decreasing Image Sends Supremum to Infimum.

### 3.3.4 Dilation and Displacement

*Remark* (Orientation). Dilation and displacement are affine operations on sets. Translation preserves the order of all comparisons, positive dilation preserves order, and negative dilation reverses order.

### Definition (Displacement, Dilation, and Reflection)

**Definition 3.3.24** (Displacement, Dilation, and Reflection). Let  $A \subseteq \mathbb{R}$  and let  $c, \lambda \in \mathbb{R}$ . Define

$$A + c := \{a + c : a \in A\}, \quad A - c := \{a - c : a \in A\}, \quad \lambda A := \{\lambda a : a \in A\}.$$

The set  $-A := (-1)A$  is the *reflection* of  $A$  through the origin.

*Remark* (Standard quantified statement).

$$x \in A + c \iff (\exists a \in A)(x = a + c), \quad x \in \lambda A \iff (\exists a \in A)(x = \lambda a).$$

*Remark* (Definition predicate reading).

$$\text{Translate}(A, c) = A + c, \quad \text{Dilate}(A, \lambda) = \lambda A, \quad \text{Reflect}(A) = -A.$$

*Remark* (Negated quantified statement).

$$x \notin A + c \iff (\forall a \in A)(x \neq a + c), \quad x \notin \lambda A \iff (\forall a \in A)(x \neq \lambda a).$$

*Remark* (Negation predicate reading).

$$\neg \text{Translate}(A, c) = B \quad \text{means} \quad (\exists x)[x \in B \Delta (A + c)],$$

and similarly for dilation and reflection.

*Remark* (Failure modes). The definition fails only if the image set omits an actual transformed point or contains a point that is not produced by transforming an element of  $A$ .

*Remark* (Failure mode decomposition). Every membership claim in a transformed set decomposes into an original element and the transformation rule that produced the new point.

*Remark* (Interpretation). Displacement slides the set, dilation stretches or collapses it, and reflection is the special dilation that reverses the real line.

*Remark* (Dependencies).

[Subset](#); [Multiplication on  \$\mathbb{R}\$](#) ; [Addition on  \$\mathbb{R}\$](#) .

### Proposition (Translation Preserves Upper Bounds)

**Proposition 3.3.25** (Translation Preserves Upper Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty. If  $u$  is an upper bound of  $A$ , then  $u + c$  is an upper bound of  $A + c$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$(\forall a \in A)(a \leq u) \implies (\forall y \in A + c)(y \leq u + c).$$

*Remark* (Predicate reading).

$$\text{UB}(u, A) \implies \text{UB}(u + c, A + c).$$

*Remark* (Negated quantified statement).

$$(\forall a \in A)(a \leq u), \quad (\exists y \in A + c)(u + c < y).$$

*Remark* (Negation predicate reading).

$$\text{UB}(u, A) \wedge \neg \text{UB}(u + c, A + c).$$

*Remark* (Failure modes). A failure would require a translated point above the translated upper bound.

*Remark* (Failure mode decomposition). Write  $y = a + c$  with  $a \in A$ . Then  $a \leq u$  gives  $a + c \leq u + c$ .

*Remark* (Dependencies).

[Upper Bound](#); [Displacement, Dilation, and Reflection](#).

#### Proposition (Translation Preserves Lower Bounds)

**Proposition 3.3.26** (Translation Preserves Lower Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty. If  $\ell$  is a lower bound of  $A$ , then  $\ell + c$  is a lower bound of  $A + c$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$(\forall a \in A)(\ell \leq a) \implies (\forall y \in A + c)(\ell + c \leq y).$$

*Remark* (Predicate reading).

$$\text{LB}(\ell, A) \implies \text{LB}(\ell + c, A + c).$$

*Remark* (Negated quantified statement).

$$(\forall a \in A)(\ell \leq a), \quad (\exists y \in A + c)(y < \ell + c).$$

*Remark* (Negation predicate reading).

$$\text{LB}(\ell, A) \wedge \neg \text{LB}(\ell + c, A + c).$$

*Remark* (Failure modes). A failure would require a translated point below the translated lower bound.

*Remark* (Failure mode decomposition). Write  $y = a + c$  with  $a \in A$ . Then  $\ell \leq a$  gives  $\ell + c \leq a + c$ .

*Remark* (Dependencies).

[Lower Bound](#); [Displacement, Dilation, and Reflection](#).

#### Proposition (Positive Dilation Preserves Upper Bounds)

**Proposition 3.3.27** (Positive Dilation Preserves Upper Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty and let  $\lambda > 0$ . If  $u$  is an upper bound of  $A$ , then  $\lambda u$  is an upper bound of  $\lambda A$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\lambda > 0 \wedge (\forall a \in A)(a \leq u) \implies (\forall y \in \lambda A)(y \leq \lambda u).$$

*Remark* (Predicate reading).

$$\lambda > 0 \wedge \text{UB}(u, A) \implies \text{UB}(\lambda u, \lambda A).$$

*Remark* (Negated quantified statement).

$$\lambda > 0, \quad (\forall a \in A)(a \leq u), \quad (\exists y \in \lambda A)(\lambda u < y).$$

*Remark* (Negation predicate reading).

$$\lambda > 0 \wedge \text{UB}(u, A) \wedge \neg \text{UB}(\lambda u, \lambda A).$$

*Remark* (Failure modes). A failure would require positive multiplication to reverse an inequality.

*Remark* (Failure mode decomposition). Write  $y = \lambda a$  with  $a \in A$ . Since  $a \leq u$  and  $\lambda > 0$ ,  $\lambda a \leq \lambda u$ .

*Remark* (Dependencies).

[Upper Bound](#); [Displacement](#), [Dilation](#), and [Reflection](#).

#### Proposition (Positive Dilation Preserves Lower Bounds)

**Proposition 3.3.28** (Positive Dilation Preserves Lower Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty and let  $\lambda > 0$ . If  $\ell$  is a lower bound of  $A$ , then  $\lambda \ell$  is a lower bound of  $\lambda A$ . Go to proof.*

*Remark* (Standard quantified statement).

$$\lambda > 0 \wedge (\forall a \in A)(\ell \leq a) \implies (\forall y \in \lambda A)(\lambda \ell \leq y).$$

*Remark* (Predicate reading).

$$\lambda > 0 \wedge \text{LB}(\ell, A) \implies \text{LB}(\lambda \ell, \lambda A).$$

*Remark* (Negated quantified statement).

$$\lambda > 0, \quad (\forall a \in A)(\ell \leq a), \quad (\exists y \in \lambda A)(y < \lambda \ell).$$

*Remark* (Negation predicate reading).

$$\lambda > 0 \wedge \text{LB}(\ell, A) \wedge \neg \text{LB}(\lambda \ell, \lambda A).$$

*Remark* (Failure modes). A failure would require a positively dilated point below the positively dilated lower bound.

*Remark* (Failure mode decomposition). Write  $y = \lambda a$  with  $a \in A$ . Since  $\ell \leq a$  and  $\lambda > 0$ ,  $\lambda\ell \leq \lambda a$ .

*Remark* (Dependencies).

[Lower Bound](#); [Displacement](#), [Dilation](#), and [Reflection](#).

**Proposition (Negative Dilation Sends Lower Bounds to Upper Bounds)**

**Proposition 3.3.29** (Negative Dilation Sends Lower Bounds to Upper Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty and let  $\lambda < 0$ . If  $\ell$  is a lower bound of  $A$ , then  $\lambda\ell$  is an upper bound of  $\lambda A$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\lambda < 0 \wedge (\forall a \in A)(\ell \leq a) \implies (\forall y \in \lambda A)(y \leq \lambda\ell).$$

*Remark* (Predicate reading).

$$\lambda < 0 \wedge \text{LB}(\ell, A) \implies \text{UB}(\lambda\ell, \lambda A).$$

*Remark* (Negated quantified statement).

$$\lambda < 0, \quad (\forall a \in A)(\ell \leq a), \quad (\exists y \in \lambda A)(\lambda\ell < y).$$

*Remark* (Negation predicate reading).

$$\lambda < 0 \wedge \text{LB}(\ell, A) \wedge \neg \text{UB}(\lambda\ell, \lambda A).$$

*Remark* (Failure modes). A failure would require negative multiplication not to reverse the lower-bound inequality.

*Remark* (Failure mode decomposition). Write  $y = \lambda a$  with  $a \in A$ . Since  $\ell \leq a$  and  $\lambda < 0$ ,  $\lambda a \leq \lambda\ell$ .

*Remark* (Dependencies).

[Lower Bound](#); [Upper Bound](#); [Displacement](#), [Dilation](#), and [Reflection](#).

**Proposition (Negative Dilation Sends Upper Bounds to Lower Bounds)**

**Proposition 3.3.30** (Negative Dilation Sends Upper Bounds to Lower Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty and let  $\lambda < 0$ . If  $u$  is an upper bound of  $A$ , then  $\lambda u$  is a lower bound of  $\lambda A$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\lambda < 0 \wedge (\forall a \in A)(a \leq u) \implies (\forall y \in \lambda A)(\lambda u \leq y).$$

*Remark* (Predicate reading).

$$\lambda < 0 \wedge \text{UB}(u, A) \implies \text{LB}(\lambda u, \lambda A).$$

*Remark* (Negated quantified statement).

$$\lambda < 0, \quad (\forall a \in A)(a \leq u), \quad (\exists y \in \lambda A)(y < \lambda u).$$

*Remark* (Negation predicate reading).

$$\lambda < 0 \wedge \text{UB}(u, A) \wedge \neg \text{LB}(\lambda u, \lambda A).$$

*Remark* (Failure modes). A failure would require negative multiplication not to reverse the upper-bound inequality.

*Remark* (Failure mode decomposition). Write  $y = \lambda a$  with  $a \in A$ . Since  $a \leq u$  and  $\lambda < 0$ ,  $\lambda u \leq \lambda a$ .

*Remark* (Dependencies).

[Upper Bound](#); [Lower Bound](#); [Displacement](#), [Dilation](#), and [Reflection](#).

#### Corollary (Reflection Swaps Upper and Lower Bounds)

**Corollary 3.3.31** (Reflection Swaps Upper and Lower Bounds). *Let  $A \subseteq \mathbb{R}$  be nonempty. If  $\ell$  is a lower bound of  $A$ , then  $-\ell$  is an upper bound of  $-A$ . If  $u$  is an upper bound of  $A$ , then  $-u$  is a lower bound of  $-A$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$(\forall a \in A)(\ell \leq a) \implies (\forall y \in -A)(y \leq -\ell),$$

and

$$(\forall a \in A)(a \leq u) \implies (\forall y \in -A)(-u \leq y).$$

*Remark* (Predicate reading).

$$\text{LB}(\ell, A) \implies \text{UB}(-\ell, -A), \quad \text{UB}(u, A) \implies \text{LB}(-u, -A).$$

*Remark* (Negated quantified statement). A failure is a failure of negative dilation at the scalar  $\lambda = -1$ .

*Remark* (Negation predicate reading).

$$\text{LB}(\ell, A) \wedge \neg \text{UB}(-\ell, -A)$$

or

$$\text{UB}(u, A) \wedge \neg \text{LB}(-u, -A).$$

*Remark* (Failure modes). Reflection reverses the real line, so lower information becomes upper information and upper information becomes lower information.

*Remark* (Failure mode decomposition). Apply negative dilation with  $\lambda = -1$ .

*Remark* (Dependencies).

[Negative Dilation Sends Lower Bounds to Upper Bounds](#); [Negative Dilation Sends Upper Bounds to Lower Bounds](#).

### 3.3.5 Algebra of Suprema and Infima

#### Current Algebraic Properties of Supremum and Infimum

*Remark* (Section boundary). Basic bound, supremum, infimum, maximum, and minimum facts now live in [Upper and Lower Bounds](#), [Suprema and Infima](#), and [Maxima and Minima](#). This section begins with algebraic transformations of those extremal values.

#### Definition (Set Arithmetic Images)

**Definition 3.3.32** (Set Arithmetic Images). Let  $A, B \subseteq \mathbb{R}$ . Define

$$\begin{aligned} A + B &:= \{a + b : a \in A, b \in B\}, & A - B &:= \{a - b : a \in A, b \in B\}, \\ AB &:= \{ab : a \in A, b \in B\}, & A/B &:= \{a/b : a \in A, b \in B, b \neq 0\}. \end{aligned}$$

*Remark* (Standard quantified statement).

$$x \in A + B \iff (\exists a \in A)(\exists b \in B)(x = a + b),$$

and similarly for difference, product, and quotient with the quotient condition  $b \neq 0$ .

*Remark* (Definition predicate reading).

$$\text{SetSum}(A, B) = A + B, \quad \text{SetDifference}(A, B) = A - B, \quad \text{SetProduct}(A, B) = AB, \quad \text{SetQuotient}(A, B) = A/B$$

*Remark* (Negated quantified statement).

$$x \notin A + B \iff (\forall a \in A)(\forall b \in B)(x \neq a + b),$$

with the analogous readings for the other arithmetic images.

*Remark* (Negation predicate reading).

$$\neg \text{SetProduct}(A, B) = C \quad \text{means} \quad (\exists x)[x \in C \triangle AB],$$

and similarly for sum, difference, and quotient.

*Remark* (Failure modes). An arithmetic-image definition fails only if it omits a value produced by the rule or includes a value not produced by the rule.

*Remark* (Failure mode decomposition). Every membership claim decomposes into witnesses from the input sets and the arithmetic operation that produced the image point.

*Remark* (Dependencies).

[Subset](#); [Addition on  \$\mathbb{R}\$](#) ; [Multiplication on  \$\mathbb{R}\$](#) .

#### Definition (Reciprocal Image)

**Definition 3.3.33** (Reciprocal Image). Let  $A \subseteq \mathbb{R}$  and suppose  $0 \notin A$ . The reciprocal image of  $A$  is

$$A^{-1} := \{1/a : a \in A\}.$$

*Remark* (Standard quantified statement).

$$x \in A^{-1} \iff (\exists a \in A)(x = 1/a).$$

*Remark* (Definition predicate reading).

$$\text{ReciprocalImage}(A) = A^{-1}.$$

*Remark* (Negated quantified statement).

$$x \notin A^{-1} \iff (\forall a \in A)(x \neq 1/a).$$

*Remark* (Negation predicate reading).

$$\neg \text{ReciprocalImage}(A) = B \quad \text{means} \quad (\exists x)[x \in B \Delta A^{-1}].$$

*Remark* (Failure modes). The reciprocal image is undefined if zero is allowed as an input value. Once zero is excluded, the only definition failures are omitted or spurious reciprocal values.

*Remark* (Failure mode decomposition). Every element of  $A^{-1}$  has the form  $1/a$  for a nonzero  $a \in A$ .

*Remark* (Dependencies).

[Subset](#); [Multiplication on  \$\mathbb{R}\$](#) .

#### Theorem (Translation Invariance of the Supremum)

**Theorem 3.3.34** (Translation Invariance of the Supremum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $c \in \mathbb{R}$ . Then*

$$\sup(A + c) = \sup A + c, \quad A + c = \{a + c : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } c \in \mathbb{R}. \\ &\sup\{a + c : a \in A\} = \sup A + c. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \implies \text{Supremum}(s + c, A + c).$$

*Remark* (Negated quantified statement).

$$\sup(A + c) \neq \sup A + c$$

is impossible under the stated hypotheses.



*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \neg \text{Supremum}(s + c, A + c)$$

is a contradictory configuration.

*Remark* (Failure modes). The formula could fail only if translation either failed to preserve upper bounds or failed to preserve leastness. Adding the same real number to every comparison preserves the order, so neither failure mode can occur.

*Remark* (Failure mode decomposition).

$$\begin{aligned} \forall a \in A (a \leq s) &\iff \forall a \in A (a + c \leq s + c), \\ \forall u [\text{UpperBound}(u, A) \implies s \leq u] &\iff \forall v [\text{UpperBound}(v, A + c) \implies s + c \leq v]. \end{aligned}$$

*Remark* (Interpretation). Translation shifts the entire set and all of its ceilings by the same amount. The least ceiling therefore shifts by exactly that amount.

*Remark* (Dependencies).

[Upper Bound](#); [Supremum](#).

#### Theorem (Translation Invariance of the Infimum)

**Theorem 3.3.35** (Translation Invariance of the Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $c \in \mathbb{R}$ . Then*

$$\inf(A + c) = \inf A + c.$$

[Go to proof](#).

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad A \text{ bounded below} \implies \inf\{a + c : a \in A\} = \inf A + c.$$

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \implies \text{Infimum}(t + c, A + c).$$

*Remark* (Negated quantified statement).

$$\inf(A + c) \neq \inf A + c$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \neg \text{Infimum}(t + c, A + c).$$

*Remark* (Failure modes). The formula could fail only if translation failed to preserve lower bounds or failed to preserve greatestness among lower bounds.

*Remark* (Failure mode decomposition).

$$(\forall a \in A)(t \leq a) \iff (\forall a \in A)(t + c \leq a + c).$$

*Remark* (Interpretation). Translation shifts every floor by the same amount, so the greatest floor shifts by that amount.

*Remark* (Dependencies).

[Lower Bound](#); [Infimum](#); [Set Arithmetic Images](#).

#### Theorem (Scalar Multiplication by a Positive Scalar)

**Theorem 3.3.36** (Scalar Multiplication by a Positive Scalar). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $k > 0$ . Then*

$$\sup(kA) = k \sup A, \quad kA = \{ka : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded above, and let } k > 0. \\ &\sup\{ka : a \in A\} = k \sup A. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \implies \text{Supremum}(ks, kA).$$

*Remark* (Negated quantified statement).

$$k > 0, \quad \sup(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \wedge \neg \text{Supremum}(ks, kA)$$

is a contradictory configuration.

*Remark* (Failure modes). The formula could fail only if multiplying by a positive scalar failed to preserve upper bounds or failed to preserve leastness. Positive multiplication preserves the order, so both properties scale exactly.

*Remark* (Failure mode decomposition).

$$\begin{aligned} \forall a \in A (a \leq s) &\iff \forall a \in A (ka \leq ks), \\ \text{UpperBound}(u, A) &\iff \text{UpperBound}(ku, kA). \end{aligned}$$

*Remark* (Interpretation). Positive scaling stretches the set without reversing order. The least ceiling stretches by the same positive factor.

Remark (Dependencies).

Upper Bound; Supremum.

#### Theorem (Positive Scalar Multiplication of the Infimum)

**Theorem 3.3.37** (Positive Scalar Multiplication of the Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $k > 0$ . Then*

$$\inf(kA) = k \inf A.$$

*Go to proof.*

Remark (Standard quantified statement).

$$k > 0, \quad A \neq \emptyset, \quad A \text{ bounded below} \implies \inf\{ka : a \in A\} = k \inf A.$$

Remark (Predicate reading).

$$\text{Infimum}(t, A) \wedge k > 0 \implies \text{Infimum}(kt, kA).$$

Remark (Negated quantified statement).

$$k > 0, \quad \inf(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge k > 0 \wedge \neg \text{Infimum}(kt, kA).$$

Remark (Failure modes). The formula could fail only if positive multiplication failed to preserve lower bounds or greatestness among lower bounds.

Remark (Failure mode decomposition).

$$(\forall a \in A)(t \leq a) \iff (\forall a \in A)(kt \leq ka) \quad (k > 0).$$

Remark (Interpretation). Positive scaling stretches the floor of the set by the same factor.

Remark (Dependencies).

Lower Bound; Infimum; Displacement, Dilation, and Reflection.

#### Theorem (Scalar Multiplication by a Negative Scalar)

**Theorem 3.3.38** (Scalar Multiplication by a Negative Scalar). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $k < 0$ . Then*

$$\sup(kA) = k \inf A, \quad kA = \{ka : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $k < 0$ .  
 $\sup\{ka : a \in A\} = k \inf A$ .

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \wedge k < 0 \implies \text{Supremum}(kt, kA).$$

*Remark* (Negated quantified statement).

$$k < 0, \quad \sup(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge k < 0 \wedge \neg \text{Supremum}(kt, kA)$$

is a contradictory configuration.

*Remark* (Failure modes). The formula could fail only if negative multiplication did not reverse lower bounds into upper bounds, or if the extremal value lost its leastness after reversal. Multiplication by a negative scalar reverses order exactly, so the infimum of  $A$  becomes the supremum of  $kA$ .

*Remark* (Failure mode decomposition).

$$\begin{aligned} \forall a \in A (t \leq a) &\iff \forall a \in A (ka \leq kt), \\ \text{LowerBound}(v, A) &\iff \text{UpperBound}(kv, kA) \quad (k < 0). \end{aligned}$$

*Remark* (Interpretation). Negative scaling reflects the set and changes floors into ceilings. The lowest point threshold of  $A$  becomes the highest point threshold of the reflected and scaled set.

*Remark* (Dependencies).

[Lower Bound](#); [Upper Bound](#); [Infimum](#); [Supremum](#).

#### Theorem (Negative Scalar Multiplication of the Infimum)

**Theorem 3.3.39** (Negative Scalar Multiplication of the Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $k < 0$ . Then*

$$\inf(kA) = k \sup A.$$

[Go to proof](#).

*Remark* (Standard quantified statement).

$$k < 0, \quad A \neq \emptyset, \quad A \text{ bounded above} \implies \inf\{ka : a \in A\} = k \sup A.$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge k < 0 \implies \text{Infimum}(ks, kA).$$

*Remark* (Negated quantified statement).

$$k < 0, \quad \inf(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge k < 0 \wedge \neg \text{Infimum}(ks, kA).$$

*Remark* (Failure modes). The formula could fail only if negative multiplication failed to reverse upper bounds into lower bounds or failed to preserve the extremal threshold after reversal.

*Remark* (Failure mode decomposition).

$$(\forall a \in A)(a \leq s) \iff (\forall a \in A)(ks \leq ka) \quad (k < 0).$$

*Remark* (Interpretation). Negative scaling reflects the set, so the ceiling of  $A$  becomes the floor of  $kA$ .

*Remark* (Dependencies).

[Upper Bound](#); [Lower Bound](#); [Supremum](#); [Infimum](#).

#### Theorem (Negation Exchanges Supremum and Infimum)

**Theorem 3.3.40** (Negation Exchanges Supremum and Infimum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below. Then  $-A = \{-a : a \in A\}$  is bounded above and*

$$\sup(-A) = -\inf A.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ be nonempty and bounded below.} \\ &\sup\{-a : a \in A\} = -\inf A. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \implies \text{Supremum}(-t, -A).$$

*Remark* (Negated quantified statement).

$$\sup(-A) \neq -\inf A$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \neg \text{Supremum}(-t, -A)$$

is a contradictory configuration.

*Remark* (Failure modes). The theorem could fail only if negation did not convert lower bounds into upper bounds or did not preserve the extremal threshold after reversing order. The map  $x \mapsto -x$  reverses all inequalities, so the exchange is exact.

*Remark* (Failure mode decomposition).

$$\begin{aligned} \forall a \in A (t \leq a) &\iff \forall a \in A (-a \leq -t), \\ \text{LowerBound}(v, A) &\iff \text{UpperBound}(-v, -A). \end{aligned}$$

*Remark* (Interpretation). Negation reflects the real line through the origin. Floors become ceilings, so the infimum of the original set becomes the negative of the supremum of the reflected set.

*Remark* (Dependencies).

[Lower Bound](#); [Upper Bound](#); [Infimum](#); [Supremum](#).

#### Theorem (Supremum of a Sum Set)

**Theorem 3.3.41** (Supremum of a Sum Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \quad A + B = \{a + b : a \in A, b \in B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.} \\ &\sup\{a + b : a \in A, b \in B\} = \sup A + \sup B. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies \text{Supremum}(s_A + s_B, A + B).$$

*Remark* (Negated quantified statement).

$$\sup(A + B) \neq \sup A + \sup B$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge \neg \text{Supremum}(s_A + s_B, A + B)$$

is a contradictory configuration.

*Remark* (Failure modes). The equality could fail only if the sum of the two suprema did not bound the sum set, or if it were not the least such bound. The first part follows from adding the two upper-bound inequalities; the second follows by approximating each supremum from within its set.

*Remark* (Failure mode decomposition).

$$\begin{aligned} a \leq \sup A, \quad b \leq \sup B &\implies a + b \leq \sup A + \sup B, \\ \varepsilon > 0 &\implies \exists a \in A, \quad b \in B : \sup A + \sup B - \varepsilon < a + b. \end{aligned}$$

*Remark* (Interpretation). The highest limiting value of all pairwise sums is obtained by combining values that approach the separate ceilings of  $A$  and  $B$ .

*Remark* (Dependencies).

Supremum; Epsilon Characterization of Supremum.

#### Theorem (Infimum of a Sum Set)

**Theorem 3.3.42** (Infimum of a Sum Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded below. Then*

$$\inf(A + B) = \inf A + \inf B.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded below} \implies \inf\{a + b : a \in A, \quad b \in B\} = \inf A + \inf B.$$

*Remark* (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \implies \text{Infimum}(t_A + t_B, A + B).$$

*Remark* (Negated quantified statement).

$$\inf(A + B) \neq \inf A + \inf B$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \neg \text{Infimum}(t_A + t_B, A + B).$$

*Remark* (Failure modes). The equality could fail only if the sum of the two infima failed to be a lower bound of the sum set or failed to be the greatest such lower bound.

*Remark* (Failure mode decomposition).

$$t_A \leq a, \quad t_B \leq b \implies t_A + t_B \leq a + b,$$

and sharpness follows by choosing elements near each infimum.

*Remark* (Interpretation). The lowest limiting value of pairwise sums is obtained by combining values near the separate floors of the two sets.

*Remark* (Dependencies).

[Infimum](#); [Set Arithmetic Images](#).

#### Theorem (Supremum of a Difference Set)

**Theorem 3.3.43** (Supremum of a Difference Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty, with  $A$  bounded above and  $B$  bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \quad A - B = \{a - b : a \in A, b \in B\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A, B \subseteq \mathbb{R}$  be nonempty,  $A$  bounded above, and  $B$  bounded below.

$$\sup\{a - b : a \in A, b \in B\} = \sup A - \inf B.$$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \implies \text{Supremum}(s - t, A - B).$$

*Remark* (Negated quantified statement).

$$\sup(A - B) \neq \sup A - \inf B$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \wedge \neg \text{Supremum}(s - t, A - B)$$

is a contradictory configuration.

*Remark* (Failure modes). The formula could fail only if subtracting elements of  $B$  did not convert the lower extremal behavior of  $B$  into an upper contribution. Since  $a - b = a + (-b)$ , this is the sum theorem combined with the negation exchange.

*Remark* (Failure mode decomposition).

$$A - B = A + (-B), \quad \sup(-B) = -\inf B, \quad \sup(A + (-B)) = \sup A + \sup(-B).$$

*Remark* (Interpretation). The largest possible difference comes from making the first term as large as possible and the subtracted term as small as possible.

*Remark* (Dependencies).

[Supremum](#); [Infimum](#); [Negation Exchanges Supremum and Infimum](#); [Supremum of a Sum Set](#).



### Theorem (Infimum of a Difference Set)

**Theorem 3.3.44** (Infimum of a Difference Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty, with  $A$  bounded below and  $B$  bounded above. Then*

$$\inf(A - B) = \inf A - \sup B.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A \text{ bounded below}, \quad B \text{ bounded above} \implies \inf\{a-b : a \in A, b \in B\} = \inf A - \sup B. \blacksquare$$

*Remark* (Predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, B) \implies \text{Infimum}(t - s, A - B).$$

*Remark* (Negated quantified statement).

$$\inf(A - B) \neq \inf A - \sup B$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Infimum}(t, A) \wedge \text{Supremum}(s, B) \wedge \neg \text{Infimum}(t - s, A - B).$$

*Remark* (Failure modes). The formula could fail only if subtracting  $B$  failed to convert its upper extremal behavior into a lower contribution.

*Remark* (Failure mode decomposition).

$$A - B = A + (-B), \quad \inf(-B) = -\sup B, \quad \inf(A + (-B)) = \inf A + \inf(-B).$$

*Remark* (Interpretation). The smallest possible difference comes from making the first term as small as possible and the subtracted term as large as possible.

*Remark* (Dependencies).

[Infimum](#); [Supremum](#); [Infimum of a Sum Set](#); [Negative Scalar Multiplication of the Infimum](#).

### Theorem (Supremum of a Dilation)

**Theorem 3.3.45** (Supremum of a Dilation). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded, and let  $\lambda \in \mathbb{R}$ . Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \quad \lambda A = \{\lambda a : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$  be nonempty and bounded, and let  $\lambda \in \mathbb{R}$ .  
 $\sup\{\lambda a : a \in A\} = \max\{\lambda \sup A, \lambda \inf A\}.$

*Remark* (Predicate reading).

$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \implies \text{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A).$

*Remark* (Negated quantified statement).

$$\sup(\lambda A) \neq \max\{\lambda \sup A, \lambda \inf A\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \wedge \neg \text{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A)$

is a contradictory configuration.

*Remark* (Failure modes). The formula separates into the cases  $\lambda > 0$ ,  $\lambda = 0$ , and  $\lambda < 0$ . Positive dilation scales the supremum, zero dilation collapses the set to  $\{0\}$ , and negative dilation turns the infimum into the supremum.

*Remark* (Failure mode decomposition).

$$\sup(\lambda A) = \begin{cases} \lambda \sup A, & \lambda > 0, \\ 0, & \lambda = 0, \\ \lambda \inf A, & \lambda < 0. \end{cases}$$

*Remark* (Interpretation). A dilation either preserves the order, collapses all points to zero, or reverses the order. The displayed maximum packages those three cases into one formula.

*Remark* (Dependencies).

[Supremum](#); [Infimum](#); [Maximum](#); [Scalar Multiplication by a Positive Scalar](#); [Scalar Multiplication by a Negative Scalar](#).

#### Theorem (Supremum of the Absolute-Value Image)

**Theorem 3.3.46** (Supremum of the Absolute-Value Image). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \quad |A| = \{|a| : a \in A\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$  be nonempty and bounded.  
 $\sup\{|a| : a \in A\} = \max\{|\sup A|, |\inf A|\}.$

*Remark* (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \implies \text{Supremum}(\max\{|s|, |t|\}, |A|).$$

*Remark* (Negated quantified statement).

$$\sup |A| \neq \max\{|\sup A|, |\inf A|\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \wedge \neg \text{Supremum}(\max\{|s|, |t|\}, |A|)$$

is a contradictory configuration.

*Remark* (Failure modes). The formula could fail only if the farthest absolute-value size were not achieved at one of the two extremal thresholds. Since every  $a \in A$  lies between  $\inf A$  and  $\sup A$ , its distance from zero is bounded by the larger endpoint magnitude.

*Remark* (Failure mode decomposition).

$$\inf A \leq a \leq \sup A \implies |a| \leq \max\{|\sup A|, |\inf A|\}.$$

*Remark* (Interpretation). The absolute-value image measures distance from the origin. For a bounded subset of the real line, the largest possible distance is controlled by whichever extremal endpoint lies farther from zero.

*Remark* (Dependencies).

Supremum; Infimum; Maximum; Every Element Lies Below the Supremum.

#### Theorem (Supremum of a Reciprocal Set)

**Theorem 3.3.47** (Supremum of a Reciprocal Set). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded, and suppose that  $0 < \inf A$  or  $\sup A < 0$ . Define*

$$A^{-1} := \{1/a : a \in A\}.$$

*Then*

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A] \implies \sup\{1/a : a \in A\} = (\inf A)^{-1}.$$

*Remark* (Predicate reading).

$$\text{Infimum}(m, A) \wedge \text{Supremum}(M, A) \wedge 0 \notin [m, M] \implies \text{Supremum}(m^{-1}, A^{-1}).$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A], \quad \sup(A^{-1}) \neq (\inf A)^{-1}.$$

*Remark* (Negation predicate reading).

$$\text{Infimum}(m, A) \wedge 0 \notin [m, \sup A] \wedge \neg \text{Supremum}(m^{-1}, A^{-1})$$

is contradictory under the stated hypotheses.

*Remark* (Failure modes). The formula fails without a nonzero gap from zero: reciprocal images may be unbounded or undefined. With the gap, reciprocal is order-reversing on the whole interval containing  $A$ , so the floor of  $A$  becomes the ceiling of  $A^{-1}$ .

*Remark* (Failure mode decomposition). Every  $a \in A$  lies between  $\inf A$  and  $\sup A$ , and the interval does not cross zero. The reciprocal map reverses comparisons there, so  $1/a \leq 1/\inf A$ . Leastness follows by applying the same reversal to values of  $A$  approaching  $\inf A$ .

*Remark* (Interpretation). Taking reciprocals turns small nonzero magnitudes into large magnitudes. The least value threshold of the original set therefore controls the largest value threshold of the reciprocal set.

*Remark* (Dependencies).

Bounded Set; Infimum; Supremum.

#### Theorem (Infimum of a Reciprocal Set)

**Theorem 3.3.48** (Infimum of a Reciprocal Set). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded, and suppose that  $0 < \inf A$  or  $\sup A < 0$ . Define*

$$A^{-1} := \{1/a : a \in A\}.$$

*Then*

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A \neq \emptyset, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A] \implies \inf\{1/a : a \in A\} = (\sup A)^{-1}.$$

*Remark* (Predicate reading).

$$\text{Infimum}(m, A) \wedge \text{Supremum}(M, A) \wedge 0 \notin [m, M] \implies \text{Infimum}(M^{-1}, A^{-1}).$$

*Remark* (Negated quantified statement).

$$A \neq \emptyset, \quad A \text{ bounded}, \quad 0 \notin [\inf A, \sup A], \quad \inf(A^{-1}) \neq (\sup A)^{-1}.$$

*Remark* (Negation predicate reading).

$$\text{Supremum}(M, A) \wedge 0 \notin [\inf A, M] \wedge \neg \text{Infimum}(M^{-1}, A^{-1})$$

is contradictory under the stated hypotheses.

*Remark* (Failure modes). The formula fails without excluding zero from the closed extremal interval. Once zero is excluded, reciprocal reverses order and the ceiling of  $A$  becomes the floor of  $A^{-1}$ .

*Remark* (Failure mode decomposition). Every reciprocal  $1/a$  lies above  $1/\sup A$  because reciprocal reverses order on the nonzero interval containing  $A$ . Greatestness of  $\sup A$  gives elements of  $A$  arbitrarily close to that ceiling, which makes the lower bound sharp.

*Remark* (Interpretation). For nonzero bounded sets that stay on one side of zero, reciprocal swaps the two extremal thresholds.

*Remark* (Dependencies).

Bounded Set; Infimum; Supremum; Supremum of a Reciprocal Set.

### Theorem (Supremum of a Product Set)

**Theorem 3.3.49** (Supremum of a Product Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded} \quad \implies \quad \sup\{ab : a \in A, b \in B\}$$

equals the maximum of the four endpoint products.

*Remark* (Predicate reading).

$$\text{Infimum}(\alpha, A) \wedge \text{Supremum}(\beta, A) \wedge \text{Infimum}(\gamma, B) \wedge \text{Supremum}(\delta, B) \implies \text{Supremum}(\max\{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}, AB)$$

*Remark* (Negated quantified statement).

$$\sup(AB) \neq \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading). The product endpoint maximum is not the supremum of  $AB$ .

*Remark* (Failure modes). A failure would require a product above all four endpoint products or would require the endpoint maximum not to be least.

*Remark* (Failure mode decomposition). Every pair  $(a, b)$  lies in the rectangle  $[\inf A, \sup A] \times [\inf B, \sup B]$ . A bilinear product on a rectangle has its extreme values at corners, and elements can approximate the relevant corners from within the two sets.

*Remark* (Dependencies).

Set Arithmetic Images; Infimum; Supremum; Maximum.

### Theorem (Infimum of a Product Set)

**Theorem 3.3.50** (Infimum of a Product Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded} \implies \inf\{ab : a \in A, b \in B\}$$

equals the minimum of the four endpoint products.

*Remark* (Predicate reading).

$$\text{Infimum}(\alpha, A) \wedge \text{Supremum}(\beta, A) \wedge \text{Infimum}(\gamma, B) \wedge \text{Supremum}(\delta, B) \implies \text{Infimum}(\min\{\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta\}, AB)$$

*Remark* (Negated quantified statement).

$$\inf(AB) \neq \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading). The product endpoint minimum is not the infimum of  $AB$ .

*Remark* (Failure modes). A failure would require a product below all four endpoint products or would require the endpoint minimum not to be greatest.

*Remark* (Failure mode decomposition). Reduce to the same endpoint analysis as the supremum product formula, replacing the corner maximum by the corner minimum.

*Remark* (Dependencies).

Set Arithmetic Images; Infimum; Supremum; Minimum; Supremum of a Product Set.

### Theorem (Supremum of a Quotient Set)

**Theorem 3.3.51** (Supremum of a Quotient Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded, and suppose  $0 < \inf B$  or  $\sup B < 0$ . Then*

$$\sup(A/B) = \max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded}, \quad 0 \notin [\inf B, \sup B]$$

implies that the supremum of  $\{a/b : a \in A, b \in B\}$  is the maximum of the four endpoint quotients.

*Remark* (Predicate reading).

$$\text{Supremum}\left(\max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}, A/B\right).$$

*Remark* (Negated quantified statement). The displayed endpoint maximum is not the supremum of  $A/B$ .

*Remark* (Negation predicate reading).

$$\neg \text{Supremum}(\text{endpoint quotient maximum}, A/B).$$

*Remark* (Failure modes). A failure would require the quotient set to see values not controlled by the reciprocal endpoint interval for  $B$ .

*Remark* (Failure mode decomposition). Use  $A/B = AB^{-1}$ , the reciprocal endpoint formulas for  $B^{-1}$ , and the product endpoint formula.

*Remark* (Dependencies).

Set Arithmetic Images; Reciprocal Image; Supremum of a Reciprocal Set; Infimum of a Reciprocal Set; Supremum of a Product Set.

#### Theorem (Infimum of a Quotient Set)

**Theorem 3.3.52** (Infimum of a Quotient Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded, and suppose  $0 < \inf B$  or  $\sup B < 0$ . Then*

$$\inf(A/B) = \min\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$A, B \neq \emptyset, \quad A, B \text{ bounded}, \quad 0 \notin [\inf B, \sup B]$$

implies that the infimum of  $\{a/b : a \in A, b \in B\}$  is the minimum of the four endpoint quotients.

*Remark* (Predicate reading).

$$\text{Infimum}\left(\min\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}, A/B\right).$$

*Remark* (Negated quantified statement). The displayed endpoint minimum is not the infimum of  $A/B$ .

*Remark* (Negation predicate reading).

$$\neg \text{Infimum}(\text{endpoint quotient minimum}, A/B).$$

*Remark* (Failure modes). A failure would require a quotient below all endpoint quotients or would require the endpoint minimum not to be greatest among lower bounds.

*Remark* (Failure mode decomposition). Use  $A/B = AB^{-1}$  and combine the reciprocal endpoint formulas with the product endpoint infimum formula.

*Remark* (Dependencies).

Set Arithmetic Images; Reciprocal Image; Supremum of a Reciprocal Set; Infimum of a Reciprocal Set; Infimum of a Product Set.

**Proposition (Bounds for Suprema and Infima Under Set Addition)**

**Proposition 3.3.53** (Bounds for Suprema and Infima Under Set Addition). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \quad \text{and} \quad \sup(A + B) \leq \sup A + \sup B.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded.

$$\inf A + \inf B \leq \inf(A + B) \quad \wedge \quad \sup(A + B) \leq \sup A + \sup B.$$

*Remark* (Predicate reading).

$$\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies t_A + t_B \leq \inf(A + B) \leq \sup(A + B) \leq s_A + s_B$$

*Remark* (Negated quantified statement).

$$\inf(A + B) < \inf A + \inf B \quad \text{or} \quad \sup A + \sup B < \sup(A + B)$$

is impossible under the stated hypotheses.

*Remark* (Negation predicate reading).

$$[\text{Infimum}(t_A, A) \wedge \text{Infimum}(t_B, B) \wedge \inf(A + B) < t_A + t_B] \vee [\text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge s_A + s_B < \sup(A + B)]$$

is a contradictory configuration.

*Remark* (Failure modes). The lower inequality could fail only if a sum fell below the sum of the two lower thresholds. The upper inequality could fail only if a sum rose above the sum of the two upper thresholds.

*Remark* (Failure mode decomposition).

$$\inf A \leq a \leq \sup A, \quad \inf B \leq b \leq \sup B \implies \inf A + \inf B \leq a + b \leq \sup A + \sup B.$$

*Remark* (Interpretation). Every element of the sum set is built from one element of  $A$  and one element of  $B$ . Adding the individual lower and upper controls gives immediate lower and upper controls for every such sum.



Remark (Dependencies).

Infimum; Supremum; Supremum of a Sum Set.

**Proposition (Upper Bounds Depend on the Ambient Order)**

**Proposition 3.3.54** (Upper Bounds Depend on the Ambient Order). *Let  $(S, \leq_S)$  and  $(T, \leq_T)$  be ordered sets, and let  $A \subseteq S \cap T$ . Whether  $u$  is an upper bound of  $A$  must be interpreted relative to the chosen ambient order. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} u \text{ is an upper bound of } A \text{ in } S &\iff u \in S \text{ and } \forall x \in A (x \leq_S u), \\ u \text{ is an upper bound of } A \text{ in } T &\iff u \in T \text{ and } \forall x \in A (x \leq_T u). \end{aligned}$$

Remark (Predicate reading).

$$\text{UpperBound}_S(u, A) \iff u \in S \wedge \forall x \in A (x \leq_S u).$$

Remark (Negated quantified statement).

The phrase “ $u$  is an upper bound of  $A$ ”

is not well-determined until the ambient ordered set and its order relation are fixed.

Remark (Negation predicate reading).

$$\neg \text{WellDetermined}(\text{UpperBound}(u, A))$$

when the ambient order is omitted.

Remark (Failure modes). Ambiguity occurs when the same set  $A$  is considered inside different ordered ambient sets or under different order relations. The candidate bound may belong to one ambient set and not another, or the relevant comparison relation may change.

Remark (Failure mode decomposition).

$$\text{UpperBound}_S(u, A) \text{ and } \text{UpperBound}_T(u, A)$$

are separate assertions unless  $S = T$  and  $\leq_S = \leq_T$  on the elements being compared.

Remark (Interpretation). An upper bound is not only about the subset; it is also about where the subset lives. Changing the ambient order changes the universe of possible bounds and the comparison used to test them.

Remark (Dependencies).

Upper Bound.

**Proposition (Suprema Depend on the Ambient Set)**

**Proposition 3.3.55** (Suprema Depend on the Ambient Set). *Let  $A \subseteq S \subseteq S'$  be subsets of an ordered set. The supremum of  $A$  relative to  $S$ , if it exists, need not agree with the supremum of  $A$  relative to  $S'$ . Go to proof.*

*Remark* (Standard quantified statement).

$$\sup_S A \text{ and } \sup_{S'} A$$

are ambient-relative objects and may differ in existence or value.

*Remark* (Predicate reading).

$$\text{Supremum}_S(s, A) \text{ and } \text{Supremum}_{S'}(s, A)$$

are different assertions unless the ambient upper-bound sets agree.

*Remark* (Negated quantified statement).

$$\sup_S A = \sup_{S'} A$$

is not guaranteed by  $A \subseteq S \subseteq S'$  alone.

*Remark* (Negation predicate reading).

$$\text{Supremum}_{S'}(s, A) \wedge \neg \text{Supremum}_S(s, A)$$

is possible when the larger ambient set contains the least upper bound but the smaller one does not.

*Remark* (Failure modes). The ambient-relative supremum can fail to transfer because the candidate least upper bound may not belong to the smaller ambient set, or because the smaller ambient set may have a different collection of upper bounds.

*Remark* (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S \implies s \text{ cannot be the supremum of } A \text{ relative to } S.$$

*Remark* (Interpretation). Suprema are least upper bounds inside a chosen universe. Enlarging or shrinking that universe can create or remove the point that should serve as the least ceiling.

*Remark* (Dependencies).

Upper Bound; Supremum.

#### Theorem (Ambient Existence of Supremum)

**Theorem 3.3.56** (Ambient Existence of Supremum). *There are ordered sets  $S \subseteq S'$  and a subset  $A \subseteq S$  such that  $A$  has a supremum relative to  $S'$  but has no supremum relative to  $S$ . Go to proof.*

*Remark* (Standard quantified statement).

$$\exists S \subseteq S' \exists A \subseteq S : \sup_{S'} A \text{ exists and } \sup_S A \text{ does not exist.}$$

*Remark* (Predicate reading).

$$\exists S \subseteq S' \exists A \subseteq S \exists s \in S' : \text{Supremum}_{S'}(s, A) \wedge \neg \exists t \in S \text{Supremum}_S(t, A).$$

*Remark* (Negated quantified statement).

Every ambient enlargement preserves and reflects existence of suprema.

This assertion is false.

*Remark* (Negation predicate reading).

$$\forall S \subseteq S' \forall A \subseteq S : [\exists s \in S' \text{Supremum}_{S'}(s, A) \implies \exists t \in S \text{Supremum}_S(t, A)]$$

is false.

*Remark* (Failure modes). The transfer from  $S'$  to  $S$  fails when the least upper bound exists in the larger ambient set but is missing from the smaller one. The smaller ambient set may contain upper bounds without containing a least upper bound.

*Remark* (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S, \quad \text{and no } t \in S \text{ is least among the upper bounds of } A \text{ in } S.$$

*Remark* (Interpretation). Completeness is an ambient property. A larger ordered universe can fill a gap that remains absent in a smaller ordered universe.

*Remark* (Dependencies).

Upper Bound; Supremum; Suprema Depend on the Ambient Set.

Lemma (Rational Gap Example for Suprema)

**Lemma 3.3.57** (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

*Then  $A$  is nonempty and bounded above in  $\mathbb{Q}$ , but  $A$  has no supremum relative to  $\mathbb{Q}$ . Go to proof.*

*Remark* (Standard quantified statement).

$$A = \{q \in \mathbb{Q} : q^2 < 2\} \implies \sup_{\mathbb{Q}} A \text{ does not exist.}$$

*Remark* (Predicate reading).

$$\neg \exists s \in \mathbb{Q} \text{Supremum}_{\mathbb{Q}}(s, A), \quad A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

*Remark* (Negated quantified statement).

$$\exists s \in \mathbb{Q} (s = \sup_{\mathbb{Q}} A)$$

is false for this set.

*Remark* (Negation predicate reading).

$$\exists s \in \mathbb{Q} \text{ Supremum}_{\mathbb{Q}}(s, A)$$

is false.

*Remark* (Failure modes). A rational candidate below  $\sqrt{2}$  fails to be an upper bound. A rational candidate above  $\sqrt{2}$  is an upper bound but is not least, because another rational upper bound lies between it and  $\sqrt{2}$ .

*Remark* (Failure mode decomposition).

$$\begin{cases} s^2 < 2 & \implies s \text{ is not an upper bound of } A, \\ s^2 > 2 & \implies s \text{ is not the least upper bound of } A. \end{cases} \quad (s \in \mathbb{Q})$$

*Remark* (Interpretation). The set wants  $\sqrt{2}$  as its least ceiling, but that number is not rational. The rational ambient set therefore has a visible gap where the supremum should be.

*Remark* (Dependencies).

Upper Bound; Supremum; Ambient Existence of Supremum.

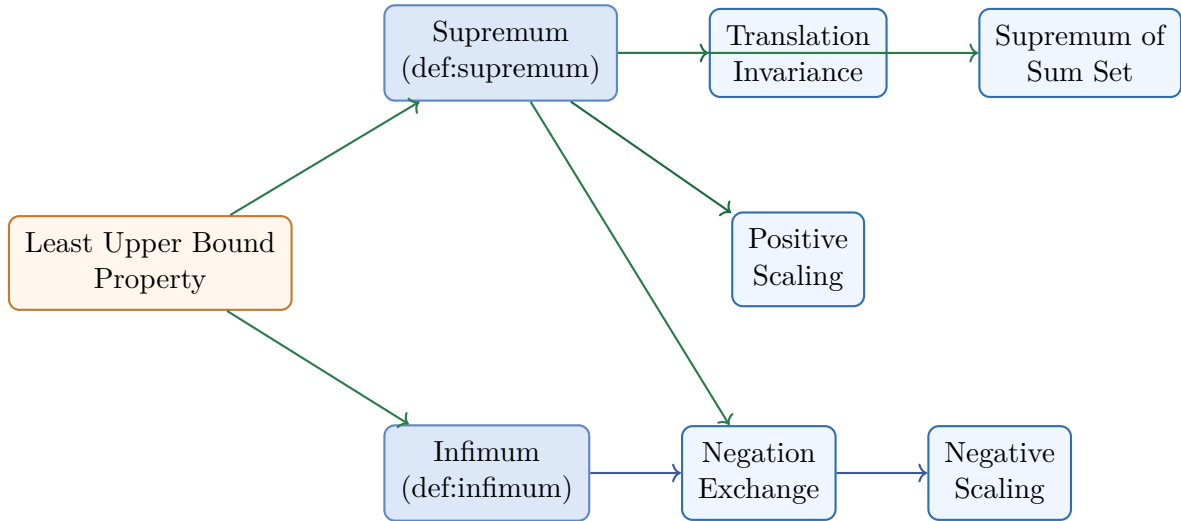


Figure 3.9: Bound Algebra Cluster Dependency Map. The map originates from the ‘Least Upper Bound Property’ (Axiom) and branches into the dual properties of the continuum. The Negation Exchange serves as the structural bridge between supremum and infimum algebra.



## Proofs

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Archimedean Property). *For every  $x \in \mathbb{R}$ , there exists  $n \in \mathbb{N}$  such that  $n > x$ .*

*Professional Standard Proof.*

TODO: professional standard proof for archimedean-property. □

*Detailed Learning Proof.*

TODO: detailed learning proof for archimedean-property. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Archimedean Corollary). *For every  $\varepsilon \in \mathbb{R}$  such that  $\varepsilon > 0$ , there exists  $n \in \mathbb{N}$  such that  $\frac{1}{n} < \varepsilon$ .*

*Professional Standard Proof.*

TODO: professional standard proof for archimedean-reciprocal. □

*Detailed Learning Proof.*

TODO: detailed learning proof for archimedean-reciprocal. □

*Remark* (Strategy: The Choice of Leastness Method). Because this corollary is a direct algebraic consequence of the Archimedean Property, we prove it through simple variable transformation ( $x = 1/\varepsilon$ ) rather than using the supremum's  $\varepsilon$ -characterization directly. In the hierarchy of your repository, the Archimedean Property (which relies on "leastness") does the heavy lifting, allowing the corollary to be a "Side A" algebraic derivation.

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Density of the Rational Numbers in  $\mathbb{R}$ ). *For every  $a, b \in \mathbb{R}$  with  $a < b$ , there exists  $q \in \mathbb{Q}$  such that  $a < q < b$ .*

*Professional Standard Proof.*

TODO: professional standard proof for density-of-rationals-in-reals. □

*Detailed Learning Proof.*

TODO: detailed learning proof for density-of-rationals-in-reals. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Density of the Irrational Numbers in  $\mathbb{R}$ ). *For every  $a, b \in \mathbb{R}$  with  $a < b$ , there exists  $s \in \mathbb{R} \setminus \mathbb{Q}$  such that  $a < s < b$ .*

*Professional Standard Proof.*

TODO: professional standard proof for density-of-irrationals-in-reals. □

*Detailed Learning Proof.*

TODO: detailed learning proof for density-of-irrationals-in-reals. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Corollary** (Rational and Irrational Points in Every Open Interval). *Every open interval in  $\mathbb{R}$  contains both a rational number and an irrational number.*

*Professional Standard Proof.*

TODO: professional standard proof for every-open-interval-contains-rational-and-irrational. □

*Detailed Learning Proof.*

TODO: detailed learning proof for every-open-interval-contains-rational-and-irrational. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Nested Interval Theorem). *Let  $I_n = [a_n, b_n]$  be a sequence of non-empty closed bounded intervals such that  $I_{n+1} \subseteq I_n$ . Then  $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ .*

*Professional Standard Proof.*

TODO: professional standard proof for nested-interval-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for nested-interval-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Least Upper Bound Property Implies Existence of Suprema). *Let  $S$  be a partially ordered set. If  $S$  has the Least Upper Bound Property, then every nonempty subset of  $S$  that is bounded above in  $S$  has a supremum in  $S$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lub-property-implies-existence-of-suprema. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lub-property-implies-existence-of-suprema. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Upper Bound Property of Supremum). *Let  $A \subseteq \mathbb{R}$  and let  $s \in \mathbb{R}$ . If  $s = \sup A$ , then*

$$x \leq s \quad \text{for every } x \in A.$$

*Professional Standard Proof.*

TODO: professional standard proof for upper-bound-property-of-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for upper-bound-property-of-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Uniqueness of the Minimum). *If a set has a minimum, then that minimum is unique.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-minimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-minimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Minimum Implies Infimum). *If a nonempty ordered set has a minimum, then that minimum is its infimum.*

*Professional Standard Proof.*

TODO: professional standard proof for minimum-implies-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for minimum-implies-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Infimum in the Set is the Minimum). *If the infimum of a nonempty set belongs to the set, then it is the minimum.*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-in-the-set-is-the-minimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-in-the-set-is-the-minimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Uniqueness of the Infimum). *If a set has an infimum, then that infimum is unique.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Lower Bound Property of the Infimum). *If  $i$  is the infimum of  $A$ , then  $i$  is a lower bound of  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lower-bound-property-of-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lower-bound-property-of-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Lemma** (Subset Inclusion Preserves Lower Bounds). *If  $A$  is a subset of  $B$  and  $ell$  is a lower bound of  $B$ , then  $ell$  is a lower bound of  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subset-inclusion-preserves-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subset-inclusion-preserves-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum Is Monotone Under Inclusion). *If  $A$  is a subset of  $B$  and both infima exist, then  $\inf B$  is less than or equal to  $\inf A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-is-monotone-under-inclusion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-is-monotone-under-inclusion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Lower Bound Comparison with the Infimum). *For a nonempty set with infimum  $i$ , lower bounds are exactly the numbers less than or equal to  $i$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lower-bound-comparison-with-the-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lower-bound-comparison-with-the-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Every Element Lies Above the Infimum). *If  $i$  is the infimum of  $A$ , then every element of  $A$  lies above  $i$ .*

*Professional Standard Proof.*

TODO: professional standard proof for every-element-lies-above-the-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for every-element-lies-above-the-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Greatest Lower Bound Property Implies Existence of Infima). *Every nonempty subset of  $R$  that is bounded below has an infimum in  $R$ .*

*Professional Standard Proof.*

TODO: professional standard proof for glb-property-implies-existence-of-infima. □

*Detailed Learning Proof.*

TODO: detailed learning proof for glb-property-implies-existence-of-infima. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Union Preserves Upper Bounds). *If  $u_A$  is an upper bound of  $A$  and  $u_B$  is an upper bound of  $B$ , then  $\max\{u_A, u_B\}$  is an upper bound of  $A \cup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for union-preserves-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for union-preserves-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Union Preserves Lower Bounds). *If  $\ell_A$  is a lower bound of  $A$  and  $\ell_B$  is a lower bound of  $B$ , then  $\min\{\ell_A, \ell_B\}$  is a lower bound of  $A \cup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for union-preserves-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for union-preserves-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Proposition](#)

**Proposition** (Union Is Bounded Above Exactly When Both Pieces Are). *For nonempty  $A, B \subseteq \mathbb{R}$ ,  $A \cup B$  is bounded above if and only if  $A$  and  $B$  are bounded above.*

*Professional Standard Proof.*

TODO: professional standard proof for union-bounded-above-iff-pieces-bounded-above.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for union-bounded-above-iff-pieces-bounded-above.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Union Is Bounded Below Exactly When Both Pieces Are). *For nonempty  $A, B \subseteq \mathbb{R}$ ,  $A \cup B$  is bounded below if and only if  $A$  and  $B$  are bounded below.*

*Professional Standard Proof.*

TODO: professional standard proof for union-bounded-below-iff-pieces-bounded-below.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for union-bounded-below-iff-pieces-bounded-below.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Union Is Bounded Exactly When Both Pieces Are). *For nonempty  $A, B \subseteq \mathbb{R}$ ,  $A \cup B$  is bounded if and only if  $A$  and  $B$  are bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for union-bounded-iff-pieces-bounded. □

*Detailed Learning Proof.*

TODO: detailed learning proof for union-bounded-iff-pieces-bounded. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Subsets Preserve Upper Bounds). *If  $C \subseteq A \subseteq \mathbb{R}$  and  $u$  is an upper bound of  $A$ , then  $u$  is an upper bound of  $C$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsets-preserve-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsets-preserve-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Subsets Preserve Lower Bounds). *If  $C \subseteq A \subseteq \mathbb{R}$  and  $\ell$  is a lower bound of  $A$ , then  $\ell$  is a lower bound of  $C$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsets-preserve-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsets-preserve-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Intersections Inherit Bounds). *Every upper bound of  $A$  is an upper bound of  $A \cap B$ , and every lower bound of  $A$  is a lower bound of  $A \cap B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for intersections-inherit-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for intersections-inherit-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Differences Inherit Bounds). *Every upper bound of  $A$  is an upper bound of  $A \setminus B$ , and every lower bound of  $A$  is a lower bound of  $A \setminus B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for differences-inherit-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for differences-inherit-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Complements Inherit Ambient Bounds). *For  $A \subseteq E \subseteq \mathbb{R}$ , every upper bound of  $E$  is an upper bound of  $E \setminus A$ , and every lower bound of  $E$  is a lower bound of  $E \setminus A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for complements-inherit-ambient-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for complements-inherit-ambient-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of the Pointwise Maximum Set). *For nonempty bounded-above  $A, B \subseteq \mathbb{R}$ ,  $\sup \max(A, B) = \max\{\sup A, \sup B\}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-pointwise-maximum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-pointwise-maximum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of the Pointwise Maximum Set). *For nonempty bounded-below  $A, B \subseteq \mathbb{R}$ ,  $\inf \max(A, B) = \max\{\inf A, \inf B\}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-pointwise-maximum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-pointwise-maximum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of the Pointwise Minimum Set). *For nonempty bounded-above  $A, B \subseteq \mathbb{R}$ ,  $\sup \min(A, B) = \min\{\sup A, \sup B\}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-pointwise-minimum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-pointwise-minimum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of the Pointwise Minimum Set). *For nonempty bounded-below  $A, B \subseteq \mathbb{R}$ ,  $\inf \min(A, B) = \min\{\inf A, \inf B\}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-pointwise-minimum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-pointwise-minimum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Increasing Image Preserves Suprema). *If  $f$  is increasing and continuous at  $\sup A$ , then  $\sup f(A) = f(\sup A)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for increasing-image-preserves-suprema. □

*Detailed Learning Proof.*

TODO: detailed learning proof for increasing-image-preserves-suprema. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Increasing Image Preserves Infima). *If  $f$  is increasing and continuous at  $\inf A$ , then  $\inf f(A) = f(\inf A)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for increasing-image-preserves-infima. □

*Detailed Learning Proof.*

TODO: detailed learning proof for increasing-image-preserves-infima. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decreasing Image Sends Infimum to Supremum). *If  $f$  is decreasing and continuous at  $\inf A$ , then  $\sup f(A) = f(\inf A)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for decreasing-image-infimum-to-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decreasing-image-infimum-to-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decreasing Image Sends Supremum to Infimum). *If  $f$  is decreasing and continuous at  $\sup A$ , then  $\inf f(A) = f(\sup A)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for decreasing-image-supremum-to-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decreasing-image-supremum-to-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Increasing Inverse Preserves Suprema). *If  $f^{-1}$  is increasing and continuous at  $\sup B$ , then  $\sup f^{-1}(B) = f^{-1}(\sup B)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for increasing-inverse-preserves-suprema. □

*Detailed Learning Proof.*

TODO: detailed learning proof for increasing-inverse-preserves-suprema. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Increasing Inverse Preserves Infima). *If  $f^{-1}$  is increasing and continuous at  $\inf B$ , then  $\inf f^{-1}(B) = f^{-1}(\inf B)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for increasing-inverse-preserves-infima. □

*Detailed Learning Proof.*

TODO: detailed learning proof for increasing-inverse-preserves-infima. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decreasing Inverse Sends Infimum to Supremum). *If  $f^{-1}$  is decreasing and continuous at  $\inf B$ , then  $\sup f^{-1}(B) = f^{-1}(\inf B)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for decreasing-inverse-infimum-to-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decreasing-inverse-infimum-to-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decreasing Inverse Sends Supremum to Infimum). *If  $f^{-1}$  is decreasing and continuous at  $\sup B$ , then  $\inf f^{-1}(B) = f^{-1}(\sup B)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for decreasing-inverse-supremum-to-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decreasing-inverse-supremum-to-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Translation Preserves Upper Bounds). *If  $u$  is an upper bound of  $A$ , then  $u + c$  is an upper bound of  $A + c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for translation-preserves-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for translation-preserves-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Translation Preserves Lower Bounds). *If  $\ell$  is a lower bound of  $A$ , then  $\ell + c$  is a lower bound of  $A + c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for translation-preserves-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for translation-preserves-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Positive Dilation Preserves Upper Bounds). *If  $\lambda > 0$  and  $u$  is an upper bound of  $A$ , then  $\lambda u$  is an upper bound of  $\lambda A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for positive-dilation-preserves-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for positive-dilation-preserves-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Positive Dilation Preserves Lower Bounds). *If  $\lambda > 0$  and  $\ell$  is a lower bound of  $A$ , then  $\lambda\ell$  is a lower bound of  $\lambda A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for positive-dilation-preserves-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for positive-dilation-preserves-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Proposition](#)

**Proposition** (Negative Dilation Sends Lower Bounds to Upper Bounds). *If  $\lambda < 0$  and  $\ell$  is a lower bound of  $A$ , then  $\lambda\ell$  is an upper bound of  $\lambda A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for negative-dilation-sends-lower-to-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for negative-dilation-sends-lower-to-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Negative Dilation Sends Upper Bounds to Lower Bounds). *If  $\lambda < 0$  and  $u$  is an upper bound of  $A$ , then  $\lambda u$  is a lower bound of  $\lambda A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for negative-dilation-sends-upper-to-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for negative-dilation-sends-upper-to-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Reflection Swaps Upper and Lower Bounds). *If  $\ell$  is a lower bound of  $A$ , then  $-\ell$  is an upper bound of  $-A$ . If  $u$  is an upper bound of  $A$ , then  $-u$  is a lower bound of  $-A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for reflection-swaps-upper-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for reflection-swaps-upper-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Uniqueness of the Maximum). *If a set  $A \subseteq \mathbb{R}$  has a maximum, then that maximum is unique.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-maximum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-maximum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Uniqueness of the Supremum). *If a set  $A \subseteq \mathbb{R}$  has a supremum, then that supremum is unique.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Proposition** (Maximum Implies Supremum). *If a set  $A \subseteq \mathbb{R}$  has a maximum  $m$ , then  $m$  is also the supremum of  $A$ .*

*Professional Standard Proof.*

Suppose  $m$  is a maximum of  $A$ . Then  $m \in A$  and

$$x \leq m$$

for every  $x \in A$ . Hence  $m$  is an upper bound of  $A$ .

Now let  $u$  be any upper bound of  $A$ . Since  $m \in A$ , the upper-bound property of  $u$  gives  $m \leq u$ . Thus every upper bound  $u$  of  $A$  satisfies  $m \leq u$ . Therefore  $m$  is the least upper bound of  $A$ , so

$$m = \sup A.$$

□

*Detailed Learning Proof.*

We prove that  $m$  satisfies the two defining properties of the supremum of  $A$ .

First, since  $m$  is a maximum of  $A$ , the definition of maximum gives

$$m \in A \quad \text{and} \quad x \leq m \quad \text{for every } x \in A.$$

The second condition says exactly that  $m$  is an upper bound of  $A$ .

Second, we show that  $m$  is the least upper bound. Let  $u$  be an arbitrary upper bound of  $A$ . Because  $u$  is an upper bound, every element of  $A$  is less than or equal to  $u$ . Since  $m \in A$ , we may apply this to the element  $m$  itself and obtain

$$m \leq u.$$

Thus every upper bound  $u$  of  $A$  lies above  $m$ .

So  $m$  is an upper bound of  $A$ , and every upper bound of  $A$  is at least  $m$ . By the definition of supremum,

$$m = \sup A.$$

□

*Remark* (Proof structure). A maximum is already an upper bound because it dominates every element of the set. It is least among upper bounds because it is itself an element of the set: any upper bound must bound  $m$  too.

*Remark* (Dependencies).

- [Maximum](#)
- [Upper Bound](#)
- [Supremum](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Supremum in the Set Is the Maximum). *Let  $s = \sup A$ . If  $s \in A$ , then  $s$  is the maximum of  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-in-the-set-is-the-maximum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-in-the-set-is-the-maximum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Subset Inclusion Preserves Upper Bounds). *Let  $A \subseteq B \subseteq \mathbb{R}$ . If  $u$  is an upper bound of  $B$ , then  $u$  is an upper bound of  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subset-inclusion-preserves-upper-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subset-inclusion-preserves-upper-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum Is Monotone Under Inclusion). *Let  $A$  and  $B$  be non-empty subsets of  $\mathbb{R}$  that are bounded above. If  $A \subseteq B$ , then  $\sup A \leq \sup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-is-monotone-under-inclusion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-is-monotone-under-inclusion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Upper Bound Comparison with the Supremum). *Let  $A \subseteq \mathbb{R}$  be non-empty and bounded above, and let  $s = \sup A$ . For any  $u \in \mathbb{R}$ ,  $u$  is an upper bound of  $A$  if and only if  $s \leq u$ .*

*Professional Standard Proof.*

TODO: professional standard proof for upper-bound-comparison-with-the-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for upper-bound-comparison-with-the-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Proposition** (Every Element Lies Below the Supremum). *If  $s = \sup A$ , then for every  $x \in A$ ,  $x \leq s$ .*

*Professional Standard Proof.*

Since  $s = \sup A$ , the definition of supremum says, in particular, that  $s$  is an upper bound of  $A$ . Hence every element of  $A$  lies below  $s$ : for every  $x \in A$ ,

$$x \leq s.$$

□

*Detailed Learning Proof.*

Let  $x \in A$  be arbitrary. We need to prove that  $x \leq s$ .

The hypothesis  $s = \sup A$  means that  $s$  is the least upper bound of  $A$ . In particular, before using the “least” part, we know that  $s$  is an upper bound of  $A$ . By the definition of upper bound, this means that every element of  $A$  is less than or equal to  $s$ . Since  $x \in A$ , it follows that

$$x \leq s.$$

Because  $x \in A$  was arbitrary,  $x \leq s$  for every  $x \in A$ .

□

*Remark* (Proof structure). This proof only unpacks the definition of supremum. The conclusion uses the upper-bound part of  $s = \sup A$ ; the minimality of  $s$  among upper bounds is not needed.

*Remark* (Dependencies).

- [Upper Bound](#)
- [Supremum](#)

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Infimum-Supremum Comparison). *For any non-empty bounded set  $A \subseteq \mathbb{R}$ , the infimum of  $A$  is less than or equal to the supremum of  $A$ .*

*Professional Standard Proof.*

Because  $A$  is nonempty, choose  $x \in A$ . Since  $\inf A$  is a lower bound of  $A$ , we have

$$\inf A \leq x.$$

Since  $\sup A$  is an upper bound of  $A$ , we have

$$x \leq \sup A.$$

By transitivity of the order on  $\mathbb{R}$ ,

$$\inf A \leq \sup A.$$

□

*Detailed Learning Proof.*

The set  $A$  is nonempty, so there exists an element  $x \in A$ .

By definition,  $\inf A$  is the greatest lower bound of  $A$ . In particular, it is a lower bound of  $A$ . Thus every element of  $A$  lies above  $\inf A$ , so for our chosen  $x \in A$ ,

$$\inf A \leq x.$$

Similarly,  $\sup A$  is the least upper bound of  $A$ . In particular, it is an upper bound of  $A$ . Thus every element of  $A$  lies below  $\sup A$ , so for the same  $x \in A$ ,

$$x \leq \sup A.$$

Combining the two inequalities gives

$$\inf A \leq x \leq \sup A.$$

Therefore

$$\inf A \leq \sup A.$$

□

*Remark* (Proof structure). Choose one element of the nonempty set. The lower-bound part of the infimum definition places  $\inf A$  below that element, and the upper-bound part of the supremum definition places the element below  $\sup A$ . Transitivity finishes the comparison.

*Remark* (Dependencies).

- [Lower Bound](#)
- [Upper Bound](#)
- [Infimum](#)
- [Supremum](#)
- [Real Ordered Field](#)

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Order Comparison of Supremum). *Let  $A, B \subseteq \mathbb{R}$  be non-empty sets that are bounded above. If for every  $x \in A$ , there exists  $y \in B$  such that  $x \leq y$ , then  $\sup A \leq \sup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for order-comparison-of-suprema. □

*Detailed Learning Proof.*

TODO: detailed learning proof for order-comparison-of-suprema. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Translation Invariance of the Supremum). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $c \in \mathbb{R}$ . Then*

$$\sup(A + c) = \sup A + c, \quad A + c = \{a + c : a \in A\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for translation-invariance-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for translation-invariance-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Translation Invariance of the Infimum). *For nonempty  $A$  bounded below,  $\inf(A + c) = \inf A + c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for translation-invariance-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for translation-invariance-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Scalar Multiplication by a Positive Scalar). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded above, and let  $k > 0$ . Then*

$$\sup(kA) = k \sup A, \quad kA = \{ka : a \in A\}.$$

*Professional Standard Proof.*

Let  $s = \sup A$ . Since  $s$  is an upper bound of  $A$ , we have  $a \leq s$  for every  $a \in A$ . Because  $k > 0$ , multiplication by  $k$  preserves order, so  $ka \leq ks$  for every  $a \in A$ . Thus  $ks$  is an upper bound of  $kA$ .

Let  $t = \sup(kA)$ . Since  $ks$  is an upper bound of  $kA$ , the least-upper-bound property gives  $t \leq ks$ . Conversely, since  $t$  is an upper bound of  $kA$ , we have  $ka \leq t$  for every  $a \in A$ . Dividing by  $k > 0$  gives  $a \leq t/k$  for every  $a \in A$ , so  $t/k$  is an upper bound of  $A$ . Since  $s = \sup A$ , we get  $s \leq t/k$ , and multiplying by  $k > 0$  gives  $ks \leq t$ . Hence  $t = ks$ , so  $\sup(kA) = k \sup A$ .  $\square$

*Detailed Learning Proof.*

Set  $s = \sup A$ . We prove that  $ks$  is exactly the least upper bound of  $kA$ .

First we show that  $ks$  is an upper bound of  $kA$ . Let  $y \in kA$ . By the definition of  $kA$ , there is some  $a \in A$  such that  $y = ka$ . Since  $s = \sup A$ , every element of  $A$  is at most  $s$ , so  $a \leq s$ . Because  $k > 0$ , multiplication by  $k$  preserves the inequality, and therefore  $ka \leq ks$ . Hence  $y \leq ks$ . Since  $y \in kA$  was arbitrary,  $ks$  is an upper bound of  $kA$ .

Now let  $t = \sup(kA)$ . From the preceding paragraph,  $ks$  is an upper bound of  $kA$ , so the defining leastness of  $t$  gives

$$t \leq ks.$$

It remains to prove the reverse inequality. Since  $t$  is an upper bound of  $kA$ , for each  $a \in A$  the element  $ka$  belongs to  $kA$  and therefore  $ka \leq t$ . Because  $k > 0$ , dividing by  $k$  preserves order, so

$$a \leq \frac{t}{k}$$

for every  $a \in A$ . Thus  $t/k$  is an upper bound of  $A$ . Since  $s = \sup A$ ,  $s$  is less than or equal to every upper bound of  $A$ , so

$$s \leq \frac{t}{k}.$$

Multiplying by  $k > 0$  gives  $ks \leq t$ .

Together,  $t \leq ks$  and  $ks \leq t$  imply  $t = ks$ . Since  $t = \sup(kA)$  and  $s = \sup A$ , this is precisely

$$\sup(kA) = k \sup A.$$

$\square$

*Remark* (Proof structure). Show that the proposed value  $k \sup A$  is an upper bound of  $kA$ , then use the least-upper-bound property in the reverse direction by dividing an arbitrary upper bound of  $kA$  by the positive scalar  $k$ .



*Remark* (Dependencies).

- Upper bound.
- Supremum.
- Upper Bound Comparison with the Supremum.
- Multiplication by a positive scalar preserves non-strict order.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Positive Scalar Multiplication of the Infimum). *If  $k > 0$  and  $A$  is nonempty and bounded below, then  $\inf(kA) = k \inf A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for positive-scalar-mult-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for positive-scalar-mult-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Scalar Multiplication by a Negative Scalar). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded below, and let  $k < 0$ . Then*

$$\sup(kA) = k \inf A, \quad kA = \{ka : a \in A\}.$$

*Professional Standard Proof.*

Let  $N = -k$ . Since  $k < 0$ , we have  $N > 0$ . Also

$$kA = \{ka : a \in A\} = \{N(-a) : a \in A\} = N(-A).$$

Because  $A$  is bounded below, the set  $-A$  is bounded above. By scalar multiplication by a positive scalar,

$$\sup(N(-A)) = N \sup(-A).$$

By the theorem that negation exchanges supremum and infimum,

$$\sup(-A) = -\inf A.$$

Therefore

$$\sup(kA) = \sup(N(-A)) = N \sup(-A) = N(-\inf A) = -N \inf A.$$

Since  $N = -k$ , we have  $-N = k$ , and hence  $\sup(kA) = k \inf A$ . □

*Detailed Learning Proof.*

We reduce the negative-scalar case to two results that are already known: positive scalar multiplication of suprema, and the exchange of supremum and infimum under negation.

Let  $N = -k$ . The hypothesis  $k < 0$  implies  $N > 0$ . For every  $a \in A$ ,

$$ka = (-N)a = N(-a).$$

Therefore the image of  $A$  under multiplication by  $k$  is the same set as the image of  $-A$  under multiplication by the positive scalar  $N$ :

$$kA = N(-A).$$

Since  $A$  is bounded below, negating all of its elements produces a set  $-A = \{-a : a \in A\}$  that is bounded above. The positive-scalar theorem applies to the nonempty bounded-above set  $-A$  and the positive scalar  $N$ , so

$$\sup(N(-A)) = N \sup(-A).$$

The negation theorem gives

$$\sup(-A) = -\inf A.$$

Substituting this into the preceding equality yields

$$\sup(N(-A)) = N(-\inf A).$$

Finally, because  $kA = N(-A)$ , the two sets have the same supremum. Hence

$$\sup(kA) = \sup(N(-A)) = N(-\inf A) = -N \inf A.$$

Since  $N = -k$ , we have  $-N = k$ . Thus

$$\sup(kA) = k \inf A,$$

as required. □

*Remark* (Proof structure). Convert the negative multiplier  $k$  into a positive multiplier  $N = -k$  applied to the negated set  $-A$ . Then combine the positive-scalar supremum formula with the theorem that negation exchanges supremum and infimum.

*Remark* (Dependencies).

- [Scalar Multiplication by a Positive Scalar](#).
- [Negation Exchanges Supremum and Infimum](#).
- [Supremum](#).
- [Infimum](#).

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Negative Scalar Multiplication of the Infimum). *If  $k < 0$  and  $A$  is nonempty and bounded above, then  $\inf(kA) = k \sup A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for negative-scalar-mult-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for negative-scalar-mult-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

*Remark* (Proof vault). [Handwritten proof record](#)

**Theorem** (Negation Exchanges Supremum and Infimum). *Let  $A \subseteq \mathbb{R}$  be a non-empty set. If  $A$  is bounded below, then  $\sup(-A) = -\inf A$ .*

*Professional Standard Proof.*

Let  $i = \inf A$ . We prove that  $-i$  is the supremum of  $-A$ .

First let  $y \in -A$ . Then  $y = -x$  for some  $x \in A$ . Since  $i$  is a lower bound of  $A$ , we have  $i \leq x$ . Multiplying by  $-1$  reverses the inequality, so

$$y = -x \leq -i.$$

Thus  $-i$  is an upper bound of  $-A$ . In particular,  $-A$  is bounded above.

Now let  $u$  be any upper bound of  $-A$ . For every  $x \in A$ , the element  $-x$  belongs to  $-A$ , so

$$-x \leq u.$$

Multiplying by  $-1$  gives

$$-u \leq x$$

for every  $x \in A$ . Hence  $-u$  is a lower bound of  $A$ . Since  $i = \inf A$  is the greatest lower bound of  $A$ , we have  $-u \leq i$ . Multiplying by  $-1$  again gives

$$-i \leq u.$$

Therefore every upper bound  $u$  of  $-A$  lies above  $-i$ . Since  $-i$  is itself an upper bound of  $-A$ , it is the least upper bound of  $-A$ . Hence

$$\sup(-A) = -i = -\inf A.$$

□

*Detailed Learning Proof.*

Set

$$i = \inf A.$$

We want to show that  $-i$  is the least upper bound of the reflected set

$$-A = \{-a : a \in A\}.$$

First we show that  $-i$  is an upper bound of  $-A$ . Take an arbitrary element  $y \in -A$ . By the definition of  $-A$ , there is some  $x \in A$  such that

$$y = -x.$$

Because  $i = \inf A$ , the number  $i$  is a lower bound of  $A$ . Thus

$$i \leq x.$$

After multiplying by  $-1$ , the inequality reverses:

$$-x \leq -i.$$

Since  $y = -x$ , this gives  $y \leq -i$ . Therefore every element of  $-A$  is less than or equal to  $-i$ , so  $-i$  is an upper bound of  $-A$ . This also shows that  $-A$  is bounded above.

It remains to show that  $-i$  is the least upper bound. Let  $u$  be any upper bound of  $-A$ . Then every element of  $-A$  is less than or equal to  $u$ . In particular, for each  $x \in A$ , the element  $-x \in -A$ , so

$$-x \leq u.$$

Multiplying by  $-1$  reverses the inequality:

$$-u \leq x.$$

This holds for every  $x \in A$ , so  $-u$  is a lower bound of  $A$ . Since  $i = \inf A$  is the greatest lower bound of  $A$ , every lower bound of  $A$  is less than or equal to  $i$ . Hence

$$-u \leq i.$$

Multiplying by  $-1$  one more time gives

$$-i \leq u.$$

Thus every upper bound  $u$  of  $-A$  lies above  $-i$ .

We have shown that  $-i$  is an upper bound of  $-A$  and that every upper bound of  $-A$  is at least  $-i$ . By the definition of supremum,

$$\sup(-A) = -i = -\inf A.$$

□

*Remark* (Strategy: The Choice of Leastness Method). In this proof, we establish the "leastness" property by comparing  $-i$  against an arbitrary upper bound  $u$  (the "Side A" method) rather than using the  $\varepsilon$ -characterization ("Side B"). While the  $\varepsilon$ -characterization is mathematically equivalent, it often requires "double negation" logic here—showing that  $a < i + \varepsilon$  implies  $-a > -i - \varepsilon$ . When a proof already involves a set transformation like negation, it is often procedurally cleaner to use the direct definition of the supremum to avoid the mental overhead of tracking flipped inequalities across a tolerance variable.

*Remark* (Proof structure). Negation reverses order. Lower bounds of  $A$  become upper bounds of  $-A$ , and the greatest lower bound becomes the least upper bound after multiplying all inequalities by  $-1$ .

*Remark* (Dependencies).

- [Lower Bound](#)
- [Upper Bound](#)
- [Infimum](#)
- [Supremum](#)
- [Real Ordered Field](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Sum Set). *Let  $A, B \subseteq \mathbb{R}$  be non-empty sets bounded above. Define  $A + B = \{a + b : a \in A, b \in B\}$ . Then  $\sup(A + B) = \sup A + \sup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-sum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-sum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of a Sum Set). *For nonempty bounded-below sets  $A, B$ ,  $\inf(A + B) = \inf A + \inf B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-sum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-sum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Difference Set). *Let  $A, B \subseteq \mathbb{R}$  be nonempty, with  $A$  bounded above and  $B$  bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \quad A - B = \{a - b : a \in A, b \in B\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for supremum-difference-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-difference-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of a Difference Set). *For nonempty  $A, B$ , with  $A$  bounded below and  $B$  bounded above,  $\inf(A - B) = \inf A - \sup B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-difference-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-difference-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Dilation). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded, and let  $\lambda \in \mathbb{R}$ . Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \quad \lambda A = \{\lambda a : a \in A\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for supremum-dilation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-dilation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of the Absolute-Value Image). *Let  $A \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \quad |A| = \{|a| : a \in A\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for supremum-absolute-value-image. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-absolute-value-image. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Reciprocal Set). *For a nonempty bounded set separated from zero,*

$$\sup(A^{-1}) = (\inf A)^{-1}.$$

*Professional Standard Proof.*

TODO: professional standard proof for supremum-reciprocal-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-reciprocal-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of a Reciprocal Set). *For a nonempty bounded set separated from zero,*

$$\inf(A^{-1}) = (\sup A)^{-1}.$$

*Professional Standard Proof.*

TODO: professional standard proof for infimum-reciprocal-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-reciprocal-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Product Set). *For nonempty bounded  $A, B$ ,  $\sup(AB)$  is the maximum of the four endpoint products.*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-product-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-product-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of a Product Set). *For nonempty bounded  $A, B$ ,  $\inf(AB)$  is the minimum of the four endpoint products.*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-product-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-product-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Supremum of a Quotient Set). *If  $A, B$  are nonempty and bounded and  $B$  is separated from zero, then  $\sup(A/B)$  is the maximum of the four endpoint quotients.*

*Professional Standard Proof.*

TODO: professional standard proof for supremum-quotient-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for supremum-quotient-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Infimum of a Quotient Set). *If  $A, B$  are nonempty and bounded and  $B$  is separated from zero, then  $\inf(A/B)$  is the minimum of the four endpoint quotients.*

*Professional Standard Proof.*

TODO: professional standard proof for infimum-quotient-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for infimum-quotient-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Bounds for Suprema and Infima Under Set Addition). *Let  $A, B \subseteq \mathbb{R}$  be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \quad \text{and} \quad \sup(A + B) \leq \sup A + \sup B.$$

*Professional Standard Proof.*

TODO: professional standard proof for bounds-sum-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounds-sum-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Upper Bounds Depend on the Ambient Order). *Let  $(S, \leq_S)$  and  $(T, \leq_T)$  be ordered sets, and let  $A \subseteq S \cap T$ . Whether  $u$  is an upper bound of  $A$  must be interpreted relative to the chosen ambient order.*

*Professional Standard Proof.*

TODO: professional standard proof for upper-bounds-ambient-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for upper-bounds-ambient-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Suprema Depend on the Ambient Set). *Let  $A \subseteq S \subseteq S'$  be subsets of an ordered set. The supremum of  $A$  relative to  $S$ , if it exists, need not agree with the supremum of  $A$  relative to  $S'$ .*

*Professional Standard Proof.*

TODO: professional standard proof for suprema-ambient-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for suprema-ambient-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Ambient Existence of Supremum). *There are ordered sets  $S \subseteq S'$  and a subset  $A \subseteq S$  such that  $A$  has a supremum relative to  $S'$  but has no supremum relative to  $S$ .*

*Professional Standard Proof.*

TODO: professional standard proof for ambient-existence-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ambient-existence-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Lemma** (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

*Then  $A$  is nonempty and bounded above in  $\mathbb{Q}$ , but  $A$  has no supremum relative to  $\mathbb{Q}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for rational-gap-suprema. □

*Detailed Learning Proof.*

TODO: detailed learning proof for rational-gap-suprema. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



# Capstone

## Capstone Theorem: The Minimax Inequality

**Theorem.** If  $X$  and  $Y$  are nonempty sets and  $\Phi : X \times Y \rightarrow \mathbb{R}$  is bounded, then

$$\sup_{x \in X} \inf_{y \in Y} \Phi(x, y) \leq \inf_{y \in Y} \sup_{x \in X} \Phi(x, y).$$

*Remark (Abstract).* This capstone proves that a player who must commit first cannot guarantee more than the opposing player can force by committing second. No topology, continuity, compactness, or convexity is involved. The proof uses only the order meaning of supremum and infimum, plus boundedness to ensure that all displayed extrema are real numbers.

*Remark (Notation).* For each  $x \in X$ , call

$$r(x) := \inf_{y \in Y} \Phi(x, y)$$

the floor of the row indexed by  $x$ . For each  $y \in Y$ , call

$$c(y) := \sup_{x \in X} \Phi(x, y)$$

the ceiling of the column indexed by  $y$ . The lower and upper values are

$$\underline{V} := \sup_{x \in X} r(x), \quad \overline{V} := \inf_{y \in Y} c(y).$$

The theorem is the comparison  $\underline{V} \leq \overline{V}$ .

*Remark (Orientation).* The theorem should be read against a simpler warning about coupled and independent indices. If  $f, g : X \rightarrow \mathbb{R}$  are bounded, then

$$\sup_{x \in X} (f(x) + g(x)) \leq \sup_{x \in X} f(x) + \sup_{x \in X} g(x),$$

and the inequality can be strict when the two suprema occur at different points. By contrast, the independent grid identity

$$\sup_{(x,y) \in X \times Y} (f(x) + g(y)) = \sup_{x \in X} f(x) + \sup_{y \in Y} g(y)$$

is exactly the result proved in [Supremum of a Sum Set](#). The minimax inequality is another place where the order and independence of indices matter.

*Remark (Lemma 1: existence of the displayed extrema).* The row floors  $r(x)$ , column ceilings  $c(y)$ , lower value  $\underline{V}$ , and upper value  $\overline{V}$  all exist as real numbers.

*Proof task.* Use the bounds  $m \leq \Phi(x, y) \leq M$ . Each row image  $\{\Phi(x_0, y) : y \in Y\}$  and each column image  $\{\Phi(x, y_0) : x \in X\}$  is nonempty and bounded because  $X$  and  $Y$  are nonempty. Therefore each row has an infimum and each column has a supremum. The family of row floors is bounded by  $m$  and  $M$ , and the family of column ceilings is also bounded by  $m$  and  $M$ . Apply the chapter's existence machinery for suprema and infima.

*Remark* (Lemma 2: the bridge). For every  $x_0 \in X$  and  $y_0 \in Y$ ,

$$r(x_0) = \inf_{y \in Y} \Phi(x_0, y) \leq \Phi(x_0, y_0) \leq \sup_{x \in X} \Phi(x, y_0) = c(y_0).$$

*Proof task.* The left inequality is the lower-bound property of the row infimum. The right inequality is the upper-bound property of the column supremum. The single point-height  $\Phi(x_0, y_0)$  is the bridge between the row floor and the column ceiling.

*Remark* (Proposition 1: every column ceiling dominates the lower value). For every fixed  $y_0 \in Y$ ,

$$\underline{V} \leq c(y_0).$$

*Proof task.* By Lemma 2,  $r(x) \leq c(y_0)$  for every  $x \in X$ . Thus  $c(y_0)$  is an upper bound for the set of row floors  $\{r(x) : x \in X\}$ . Since  $\underline{V}$  is the supremum of that set, the upper-bound comparison property gives  $\underline{V} \leq c(y_0)$ .

*Remark* (Proof of the theorem). Proposition 1 holds for every  $y_0 \in Y$ . Hence  $\underline{V}$  is a lower bound for the set of column ceilings  $\{c(y) : y \in Y\}$ . Since  $\bar{V}$  is the infimum of that set,

$$\underline{V} \leq \inf_{y \in Y} c(y) = \bar{V}.$$

Expanding the definitions of  $r$ ,  $c$ ,  $\underline{V}$ , and  $\bar{V}$  gives

$$\sup_{x \in X} \inf_{y \in Y} \Phi(x, y) \leq \inf_{y \in Y} \sup_{x \in X} \Phi(x, y).$$

*Remark* (Proposition 2: a saddle point forces equality). Call  $(x^*, y^*)$  a saddle point of  $\Phi$  if

$$\Phi(x, y^*) \leq \Phi(x^*, y^*) \leq \Phi(x^*, y) \quad \text{for all } x \in X, y \in Y.$$

If a saddle point exists, then

$$\underline{V} = \bar{V} = \Phi(x^*, y^*).$$

*Proof task.* The left saddle inequality makes  $\Phi(x^*, y^*)$  the supremum of the column at  $y^*$ . The right saddle inequality makes it the infimum of the row at  $x^*$ . Thus

$$\bar{V} \leq \Phi(x^*, y^*) \leq \underline{V}.$$

Combine this with the minimax inequality  $\underline{V} \leq \bar{V}$ .

*Remark* (Application: one-step pursuit on the circle). Let  $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$ , with wrap-around metric

$$d_{\mathbb{S}^1}(a, b) = \min_{k \in \mathbb{Z}} |a - b + k| \in \left[0, \frac{1}{2}\right].$$

Place a pursuer at 0 and an evader at  $d_0 \in (0, \frac{1}{2}]$ . In one step the pursuer chooses  $u \in [-s_P, s_P]$ , and the evader chooses  $v \in [-s_E, s_E]$ , where  $s_P, s_E > 0$ . Define

$$\Psi(v, u) := d_{\mathbb{S}^1}(u, d_0 + v).$$

The evader maximizes distance and the pursuer minimizes distance, so the one-step lower and upper values are

$$\sup_{v \in [-s_E, s_E]} \inf_{u \in [-s_P, s_P]} \Psi(v, u) \leq \inf_{u \in [-s_P, s_P]} \sup_{v \in [-s_E, s_E]} \Psi(v, u).$$

As a final task, determine whether this circle model has a saddle point, and compare the result with the corresponding one-step model on the line.

*Remark* (Limits of the theorem). Equality need not hold without extra structure. In particular,  $\underline{V} = \bar{V}$  does not by itself imply that a saddle point exists. Existence of saddle points or equality of values usually requires additional hypotheses such as compactness, convexity in one variable, and concavity in the other. Those hypotheses belong to later minimax theorems, not to this bounds-only result.

*Remark* (Dependencies to state). • [Supremum](#)

- [Infimum](#)

*Remark* (Dependencies to prove). • [Supremum](#)

- [Infimum](#)
- [Upper-Bound Comparison with the Supremum](#)
- [Infimum Less Than Supremum](#)
- [Supremum of a Sum Set](#)

*Remark* (Dependency ceiling). This capstone may use only the order, supremum, infimum, and boundedness results routed at or before the Bounds chapter.

*Remark* (Dependencies).

- [Supremum](#)
- [Infimum](#)
- [Upper-Bound Comparison with the Supremum](#)
- [Infimum Less Than Supremum](#)
- [Supremum of a Sum Set](#)

# Chapter 4

## Functions



### 4.1 Algebra of Functions

#### Algebra of Functions — Quick Reference

| Statement                              | Role                                                |
|----------------------------------------|-----------------------------------------------------|
| <a href="#">Injective composition</a>  | Injectivity is preserved by composition             |
| <a href="#">Surjective composition</a> | Surjectivity is preserved by composition            |
| <a href="#">Bijective composition</a>  | Bijections compose to bijections                    |
| <a href="#">Inverse bijection</a>      | Bijections have bijective inverses                  |
| <a href="#">Preimage algebra</a>       | Preimages preserve unions and intersections exactly |

### Algebra of Functions

*Remark* (Intervals on  $\mathbb{R}$ ). Every open interval  $(a, b)$  consists entirely of cluster points;  $[a, b]$  is closed and bounded. Natural domains for functions are typically intervals or unions of intervals.

#### Proposition

**Proposition 4.1.1** (Composition of Injections Is Injective). *Let  $f : A \rightarrow B$  and let  $g : B \rightarrow C$ . If  $f$  is injective and  $g$  is injective, then  $g \circ f : A \rightarrow C$  is injective.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\left( \forall a_1, a_2 \in A (f(a_1) = f(a_2) \Rightarrow a_1 = a_2) \right) \wedge \left( \forall b_1, b_2 \in B (g(b_1) = g(b_2) \Rightarrow b_1 = b_2) \right) \\ \Rightarrow \forall a_1, a_2 \in A (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2).$$

*Remark* (Definition predicate reading).

$$\text{Injective}(f) \wedge \text{Injective}(g) \implies \text{Injective}(g \circ f).$$

*Remark* (Negated quantified statement).

$$\exists a_1, a_2 \in A (a_1 \neq a_2 \wedge g(f(a_1)) = g(f(a_2))).$$

*Remark* (Negation predicate reading).

$$\neg \text{Injective}(g \circ f).$$

*Remark* (Failure modes). The composite fails to be injective exactly when two distinct inputs of  $A$  are sent to the same output in  $C$  after applying both functions.

*Remark* (Failure mode decomposition).

$$\exists a_1, a_2 \in A (a_1 \neq a_2 \wedge g(f(a_1)) = g(f(a_2))).$$

*Remark* (Interpretation). Injectivity survives composition: equality after  $g \circ f$  can be pulled back first through  $g$  and then through  $f$ .

*Remark* (Dependencies).

- [Composition](#)
- [Function](#)

#### Proposition

**Proposition 4.1.2** (Composition of Surjections Is Surjective).  $\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \text{Surjective}(g \circ f)$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \forall c \in C, \exists a \in A : g(f(a)) = c.$$

*Remark* (Definition predicate reading).

$$\text{Surjective}(g \circ f, C) \equiv \forall c \in C, \exists a \in A : (g \circ f)(a) = c.$$

*Remark* (Dependencies).

- [Composition](#)

- [Function](#)

#### Corollary

**Corollary 4.1.3** (Composition of Bijections Is Bijective). *Let  $f : A \rightarrow B$  and let  $g : B \rightarrow C$ . If  $f$  is bijective and  $g$  is bijective, then  $g \circ f : A \rightarrow C$  is bijective.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ (\forall a_1, a_2 \in A (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)) \wedge (\forall b \in B \exists a \in A (f(a) = b)) \right] \\ & \wedge \left[ (\forall b_1, b_2 \in B (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)) \wedge (\forall c \in C \exists b \in B (g(b) = c)) \right] \\ & \Rightarrow \left[ (\forall a_1, a_2 \in A (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2)) \wedge (\forall c \in C \exists a \in A (g(f(a)) = c)) \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{Bijective}(f) \wedge \text{Bijective}(g) \implies \text{Bijective}(g \circ f).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} & \exists a_1, a_2 \in A (a_1 \neq a_2 \wedge g(f(a_1)) = g(f(a_2))) \\ & \vee \exists c \in C \forall a \in A (g(f(a)) \neq c). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{Bijective}(g \circ f).$$

*Remark* (Failure modes). A composite bijection can fail only by losing injectivity or by missing some target value in  $C$ .

*Remark* (Failure mode decomposition).

$$\neg \text{Bijective}(g \circ f) \iff \neg \text{Injective}(g \circ f) \vee \neg \text{Surjective}(g \circ f).$$

*Remark* (Interpretation). A bijection is an injective and surjective function. Since both of those properties are preserved by composition, bijectivity is preserved too.

*Remark* (Dependencies).

- [Composition](#)
- [Function](#)

#### Proposition

**Proposition 4.1.4** (Inverse of a Bijection Is a Bijection).  $\text{Bijective}(f) \implies \text{Bijective}(f^{-1})$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Bijective}(f) \Rightarrow \text{InverseFunction}(f, f^{-1}), f^{-1} \circ f = \text{id}_A, f \circ f^{-1} = \text{id}_B.$$

*Remark* (Definition predicate reading).

$$\text{InverseFunction}(f, f^{-1}) \equiv f^{-1} \circ f = \text{id}_A \wedge f \circ f^{-1} = \text{id}_B.$$

*Remark* (Dependencies).

- [Composition](#)
- [Function](#)

### Proposition

**Proposition 4.1.5** (Preimage Distributes over Union and Intersection). *Let  $f : A \rightarrow B$ ,  $S, T \subseteq B$ . Then:*

- (i)  $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$ .
- (ii)  $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$ .
- (iii)  $f^{-1}(S^c) = (f^{-1}(S))^c$ .

[Go to proof.](#)

*Remark* (Standard quantified statement).

- (i)  $x \in f^{-1}(S \cup T) \iff f(x) \in S \vee f(x) \in T \iff x \in f^{-1}(S) \cup f^{-1}(T)$ .
- (ii)  $x \in f^{-1}(S \cap T) \iff f(x) \in S \wedge f(x) \in T \iff x \in f^{-1}(S) \cap f^{-1}(T)$ .

*Remark* (Negated quantified statement).

$$x \notin f^{-1}(S) \iff f(x) \notin S \iff x \in f^{-1}(S^c).$$

*Remark* (Interpretation). Preimage is a complete lattice homomorphism: it preserves all set operations including complement. Contrast with image:  $f(A \cap B) \subseteq f(A) \cap f(B)$  with strict containment possible when  $f$  is not injective. Preimage identities are load-bearing in topology (preimages of open sets are open) and measure theory (preimages of measurable sets are measurable).

*Remark* (Dependencies).

- [Function](#)

## 4.2 Real-Valued Functions

### Real-Valued Functions — Quick Reference

| Concept            | Meaning                                    |
|--------------------|--------------------------------------------|
| Pointwise sum      | Algebraic addition of function values      |
| Pointwise product  | Product $fg$ , distinct from composition   |
| Pointwise quotient | Quotient where denominator is nonzero      |
| Bounded function   | A single bound controls all values on $A$  |
| Supremum on a set  | Least upper bound of $f(A)$                |
| Maximum point      | Point where the supremum is attained       |
| Monotone function  | Increasing or decreasing behavior on a set |
| Constant function  | Same value at every input                  |

## Real-Valued Functions

### Definition (Pointwise Sum of Functions)

**Definition 4.2.1** (Pointwise Sum of Functions). Let  $X \subseteq \mathbb{R}$  and let  $f, g : X \rightarrow \mathbb{R}$ . The pointwise sum  $f + g : X \rightarrow \mathbb{R}$  is the function defined by

$$(f + g)(x) = f(x) + g(x) \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X ((f + g)(x) = f(x) + g(x)).$$

*Remark* (Interpretation). The sum is computed output by output; at each input, add the two function values at that same input.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Addition on  \$\mathbb{R}\$](#)

### Definition (Pointwise Difference of Functions)

**Definition 4.2.2** (Pointwise Difference of Functions). Let  $X \subseteq \mathbb{R}$  and let  $f, g : X \rightarrow \mathbb{R}$ . The pointwise difference  $f - g : X \rightarrow \mathbb{R}$  is the function defined by

$$(f - g)(x) = f(x) - g(x) \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X ((f - g)(x) = f(x) - g(x)).$$



*Remark* (Interpretation). The difference subtracts the output of  $g$  from the output of  $f$  at each input separately.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Subtraction on  \$\mathbb{R}\$](#)

#### Definition (Pointwise Product of Functions)

**Definition 4.2.3** (Pointwise Product of Functions). Let  $X \subseteq \mathbb{R}$  and let  $f, g : X \rightarrow \mathbb{R}$ . The pointwise product  $fg : X \rightarrow \mathbb{R}$  is the function defined by

$$(fg)(x) = f(x)g(x) \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X \ ((fg)(x) = f(x)g(x)).$$

*Remark* (Interpretation). The product is not composition: it multiplies the two real outputs at the same input.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Multiplication on  \$\mathbb{R}\$](#)

#### Definition (Pointwise Scalar Multiple of a Function)

**Definition 4.2.4** (Pointwise Scalar Multiple of a Function). Let  $X \subseteq \mathbb{R}$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $c \in \mathbb{R}$ . The pointwise scalar multiple  $cf : X \rightarrow \mathbb{R}$  is the function defined by

$$(cf)(x) = cf(x) \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X \ ((cf)(x) = cf(x)).$$

*Remark* (Interpretation). Scalar multiplication stretches every output of the function by the same real scalar.

*Remark* (Dependencies).

- [Function](#)

- [Subset](#)
- [Multiplication on  \$\mathbb{R}\$](#)

#### Definition (Pointwise Absolute Value of a Function)

**Definition 4.2.5** (Pointwise Absolute Value of a Function). Let  $X \subseteq \mathbb{R}$  and let  $f : X \rightarrow \mathbb{R}$ . The pointwise absolute value  $|f| : X \rightarrow \mathbb{R}$  is the function defined by

$$|f|(x) = |f(x)| \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X \ (|f|(x) = |f(x)|).$$

*Remark* (Interpretation). The absolute value operation applies the ordinary real absolute value to each output.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Absolute Value on  \$\mathbb{R}\$](#)

#### Definition (Pointwise Maximum of Two Functions)

**Definition 4.2.6** (Pointwise Maximum of Two Functions). Let  $X \subseteq \mathbb{R}$  and let  $f, g : X \rightarrow \mathbb{R}$ . The pointwise maximum  $\max(f, g) : X \rightarrow \mathbb{R}$  is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X \ (\max(f, g)(x) = \max\{f(x), g(x)\}).$$

*Remark* (Interpretation). At each input, the new function keeps the larger of the two output values.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Maximum](#)

### Definition (Pointwise Quotient of Functions)

**Definition 4.2.7** (Pointwise Quotient of Functions). Let  $X \subseteq \mathbb{R}$  and let  $f, g : X \rightarrow \mathbb{R}$ . Suppose  $g(x) \neq 0$  for every  $x \in X$ . The pointwise quotient  $f/g : X \rightarrow \mathbb{R}$  is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$(\forall x \in X (g(x) \neq 0)) \implies \forall x \in X ((f/g)(x) = f(x)/g(x)).$$

*Remark* (Interpretation). The quotient is defined pointwise only when the denominator function never vanishes on the domain.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Division on  \$\mathbb{R}\$](#)

### Definition (Linear Combination of Real-Valued Functions)

**Definition 4.2.8** (Linear Combination of Real-Valued Functions). Let  $X \subseteq \mathbb{R}$ , let  $f, g : X \rightarrow \mathbb{R}$ , and let  $\alpha, \beta \in \mathbb{R}$ . The real linear combination  $\alpha f + \beta g : X \rightarrow \mathbb{R}$  is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \quad (x \in X).$$

*Remark* (Standard quantified statement).

$$\forall x \in X ((\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x)).$$

*Remark* (Interpretation). A linear combination is built only from scalar multiples and sums. It is the shared algebraic pattern behind sum rules, constant-multiple rules, and finite linear-combination rules for functions.

*Remark* (Dependencies).

- [Pointwise Scalar Multiple of a Function](#)
- [Pointwise Sum of Functions](#)

### Definition (Real-Linear Rule)

**Definition 4.2.9** (Real-Linear Rule). Let  $\mathcal{C}$  be a class of real-valued functions closed under real linear combinations. A rule  $T$  on  $\mathcal{C}$  is real-linear if, whenever  $f, g \in \mathcal{C}$  and  $\alpha, \beta \in \mathbb{R}$ ,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

*Remark* (Standard quantified statement).

$$f, g \in \mathcal{C} \implies T(\alpha f + \beta g) = \alpha T(f) + \beta T(g).$$

*Remark* (Interpretation). Linearity records exactly the behavior of preserving addition and scalar multiplication. Nonlinear operations such as products, quotients, absolute values, and order comparisons require their own dependency information.

*Remark* (Dependencies).

- [Function](#)
- [Linear Combination of Real-Valued Functions](#)

#### Definition (Bounded Above Function)

**Definition 4.2.10** (Bounded Above Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is bounded above on  $A$  if there exists  $M \in \mathbb{R}$  such that

$$f(x) \leq M \quad (x \in A).$$

*Remark* (Standard quantified statement).

$$\exists M \in \mathbb{R} \forall x \in A (f(x) \leq M).$$

*Remark* (Negated quantified statement).

$$\forall M \in \mathbb{R} \exists x \in A (f(x) > M).$$

*Remark* (Failure modes). No proposed real ceiling works: whenever a candidate upper bound is chosen, some input has an output larger than that candidate.

*Remark* (Failure mode decomposition).

$$\forall M \in \mathbb{R} \exists x \in A (f(x) > M).$$

*Remark* (Interpretation). Bounded above means all output values lie below one common real ceiling.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)
- [Bounded Above](#)

### Definition (Bounded Below Function)

**Definition 4.2.11** (Bounded Below Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is bounded below on  $A$  if there exists  $m \in \mathbb{R}$  such that

$$m \leq f(x) \quad (x \in A).$$

*Remark* (Standard quantified statement).

$$\exists m \in \mathbb{R} \forall x \in A (m \leq f(x)).$$

*Remark* (Negated quantified statement).

$$\forall m \in \mathbb{R} \exists x \in A (f(x) < m).$$

*Remark* (Failure modes). No proposed real floor works: whenever a candidate lower bound is chosen, some input has an output smaller than that candidate.

*Remark* (Failure mode decomposition).

$$\forall m \in \mathbb{R} \exists x \in A (f(x) < m).$$

*Remark* (Interpretation). Bounded below means all output values lie above one common real floor.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)
- [Bounded Below](#)

### Definition (Bounded Function)

**Definition 4.2.12** (Bounded Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is bounded on  $A$  if there exists  $B > 0$  such that

$$|f(x)| \leq B \quad (x \in A).$$

*Remark* (Standard quantified statement).

$$\exists B > 0 \forall x \in A (|f(x)| \leq B).$$

*Remark* (Definition predicate reading).

$$\text{BoundedOn}(f, A) := \exists B > 0 \forall x \in A (|f(x)| \leq B).$$

*Remark* (Negated quantified statement).

$$\forall B > 0 \exists x \in A (|f(x)| > B).$$

*Remark* (Negation predicate reading).

$$\neg \text{BoundedOn}(f, A) \iff \forall B > 0 \exists x \in A (|f(x)| > B).$$

*Remark* (Failure modes). Every symmetric interval around 0 fails to contain all output values of  $f$  on  $A$ .

*Remark* (Failure mode decomposition).

$$\forall B > 0 \exists x \in A (f(x) > B \vee f(x) < -B).$$

*Remark* (Interpretation). A bounded function has all of its values trapped inside some symmetric interval  $[-B, B]$ . Being bounded above alone is not enough; the function  $f(x) = -x$  on  $\mathbb{R}$  is bounded above by 0 but is not bounded.

*Remark* (Dependencies).

- [Bounded Above Function](#)
- [Bounded Below Function](#)

#### Definition (Supremum of a Function on a Set)

**Definition 4.2.13** (Supremum of a Function on a Set). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The supremum of  $f$  on  $A$  is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of  $f(A)$  exists.

*Remark* (Standard quantified statement).

$$s = \sup_{x \in A} f(x) \iff (\forall x \in A (f(x) \leq s)) \wedge (\forall u \in \mathbb{R} ((\forall x \in A (f(x) \leq u)) \Rightarrow s \leq u)).$$

*Remark* (Interpretation). The supremum is the least real ceiling for the output set  $f(A)$ ; it need not be an output value of  $f$ .

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Image Set](#)
- [Supremum](#)

### Definition (Infimum of a Function on a Set)

**Definition 4.2.14** (Infimum of a Function on a Set). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The infimum of  $f$  on  $A$  is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of  $f(A)$  exists.

*Remark* (Standard quantified statement).

$$t = \inf_{x \in A} f(x) \iff (\forall x \in A (t \leq f(x))) \wedge (\forall \ell \in \mathbb{R} ((\forall x \in A (\ell \leq f(x))) \Rightarrow \ell \leq t)).$$

*Remark* (Interpretation). The infimum is the greatest real floor for the output set  $f(A)$ ; it need not be an output value of  $f$ .

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Image Set](#)
- [Infimum](#)

### Definition (Maximum Point of a Function)

**Definition 4.2.15** (Maximum Point of a Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . A point  $x^* \in A$  is a maximum point of  $f$  on  $A$  if

$$f(x) \leq f(x^*) \quad (x \in A).$$

*Remark* (Standard quantified statement).

$$x^* \in A \wedge \forall x \in A (f(x) \leq f(x^*)).$$

*Remark* (Negated quantified statement).

$$x^* \notin A \vee \exists x \in A (f(x) > f(x^*)).$$

*Remark* (Failure modes). A proposed maximum point fails either by not belonging to the domain or by being beaten by another output value.

*Remark* (Failure mode decomposition).

$$x^* \notin A \vee \exists x \in A (f(x) > f(x^*)).$$

*Remark* (Interpretation). A maximum point is an input where the largest output value is attained.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)
- [Maximum](#)

### Definition (Minimum Point of a Function)

**Definition 4.2.16** (Minimum Point of a Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . A point  $x_* \in A$  is a minimum point of  $f$  on  $A$  if

$$f(x_*) \leq f(x) \quad (x \in A).$$

*Remark* (Standard quantified statement).

$$x_* \in A \wedge \forall x \in A (f(x_*) \leq f(x)).$$

*Remark* (Negated quantified statement).

$$x_* \notin A \vee \exists x \in A (f(x) < f(x_*)).$$

*Remark* (Failure modes). A proposed minimum point fails either by not belonging to the domain or by being undercut by another output value.

*Remark* (Failure mode decomposition).

$$x_* \notin A \vee \exists x \in A (f(x) < f(x_*)).$$

*Remark* (Interpretation). A minimum point is an input where the smallest output value is attained.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)
- [Minimum](#)

### Definition (Increasing Function)

**Definition 4.2.17** (Increasing Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is increasing on  $A$  if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \quad (x_1, x_2 \in A).$$



*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)).$$

*Remark* (Negated quantified statement).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2)).$$

*Remark* (Failure modes). The function fails to be increasing if some later input has a smaller output than an earlier input.

*Remark* (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2)).$$

*Remark* (Interpretation). As the input moves to the right in  $A$ , the output is not allowed to go down.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Strictly Increasing Function)

**Definition 4.2.18** (Strictly Increasing Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is strictly increasing on  $A$  if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \quad (x_1, x_2 \in A).$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) < f(x_2)).$$

*Remark* (Negated quantified statement).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) \geq f(x_2)).$$

*Remark* (Failure modes). Strict increase fails if some later input has an output that is no larger than the earlier output.

*Remark* (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) \geq f(x_2)).$$

*Remark* (Interpretation). Strict increase requires every genuine move to the right in the domain to produce a genuine increase in output.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Strict Order on  \$\mathbb{R}\$](#)

#### Definition (Decreasing Function)

**Definition 4.2.19** (Decreasing Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is decreasing on  $A$  if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \quad (x_1, x_2 \in A).$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 < x_2 \implies f(x_1) \geq f(x_2)).$$

*Remark* (Negated quantified statement).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) < f(x_2)).$$

*Remark* (Failure modes). The function fails to be decreasing if some later input has a larger output than an earlier input.

*Remark* (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) < f(x_2)).$$

*Remark* (Interpretation). As the input moves to the right in  $A$ , the output is not allowed to go up.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Strictly Decreasing Function)

**Definition 4.2.20** (Strictly Decreasing Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is strictly decreasing on  $A$  if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \quad (x_1, x_2 \in A).$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 < x_2 \implies f(x_1) > f(x_2)).$$

*Remark* (Negated quantified statement).

$$\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) \leq f(x_2)).$$

*Remark* (Failure modes). Strict decrease fails if some later input has an output that is no smaller than the earlier output.

*Remark* (Failure mode decomposition).

$$\exists x_1, x_2 \in A (x_1 < x_2 \wedge f(x_1) \leq f(x_2)).$$

*Remark* (Interpretation). Strict decrease requires every genuine move to the right in the domain to produce a genuine decrease in output.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Strict Order on  \$\mathbb{R}\$](#)

#### Definition (Monotone Function)

**Definition 4.2.21** (Monotone Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is monotone on  $A$  if  $f$  is increasing on  $A$  or decreasing on  $A$ .

*Remark* (Standard quantified statement).

$$(\forall x_1, x_2 \in A (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))) \vee (\forall x_1, x_2 \in A (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2))).$$

*Remark* (Definition predicate reading).

$$\text{MonotoneFunction}(f, A) := (\forall x_1, x_2 \in A (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))) \vee (\forall x_1, x_2 \in A (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2))).$$

*Remark* (Negated quantified statement).

$$(\exists x_1, x_2 \in A (x_1 < x_2 \wedge f(x_1) > f(x_2))) \wedge (\exists y_1, y_2 \in A (y_1 < y_2 \wedge f(y_1) < f(y_2))).$$

*Remark* (Negation predicate reading).

$$\neg \text{MonotoneFunction}(f, A)$$

*Remark* (Failure modes). A nonmonotone function has evidence of both kinds of reversal: one pair where the outputs go down as inputs increase, and another pair where the outputs go up as inputs increase.

*Remark* (Failure mode decomposition).

$$\begin{aligned} &\exists x_1, x_2 \in A (x_1 < x_2 \wedge f(x_1) > f(x_2)) \\ &\wedge \exists y_1, y_2 \in A (y_1 < y_2 \wedge f(y_1) < f(y_2)). \end{aligned}$$

*Remark* (Interpretation). A monotone function moves consistently in one direction: nondecreasing or nonincreasing.

*Remark* (Dependencies).

- [Increasing Function](#)
- [Decreasing Function](#)

### Definition (Constant Function)

**Definition 4.2.22** (Constant Function). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is constant on  $A$  if

$$f(x_1) = f(x_2) \quad (x_1, x_2 \in A).$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (f(x_1) = f(x_2)).$$

*Remark* (Negated quantified statement).

$$\exists x_1, x_2 \in A \ (f(x_1) \neq f(x_2)).$$

*Remark* (Failure modes). Constancy fails as soon as two inputs in the domain have different outputs.

*Remark* (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ (f(x_1) \neq f(x_2)).$$

*Remark* (Interpretation). A constant function assigns the same real value to every input in its domain.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)

## 4.3 Subsets of $\mathbb{R}$

### Subsets of $\mathbb{R}$ — Quick Reference

| Concept                               | Meaning                                                     |
|---------------------------------------|-------------------------------------------------------------|
| <a href="#">Epsilon neighbourhood</a> | Unpunctured condition $ x - c  < r$                         |
| <a href="#">Deleted neighbourhood</a> | Punctured condition $0 <  x - c  < \delta$                  |
| <a href="#">Cluster point</a>         | Every neighbourhood contains a distinct point of $X$        |
| <a href="#">Adherent point</a>        | Every neighbourhood contains some point of $X$              |
| <a href="#">Closure</a>               | Points adherent to $X$                                      |
| <a href="#">Closed set</a>            | Set containing its closure or sequential limits             |
| <a href="#">Bounded set</a>           | Set contained in a large interval                           |
| <a href="#">Heine–Borel</a>           | Closed and bounded iff sequentially compact in $\mathbb{R}$ |

## Subsets of $\mathbb{R}$

*Remark* (Terminological note). Cluster point = limit point = accumulation point. Bartle and Lebl use *cluster point*; Rudin uses *limit point*; Tao uses both. Adherent point (Tao) is the weaker notion that includes isolated points.

### Definition ( $\varepsilon$ -Neighbourhood)

**Definition 4.3.1** ( $\varepsilon$ -Neighbourhood). Let  $x \in \mathbb{R}$  and let  $\varepsilon > 0$ . The  $\varepsilon$ -neighbourhood of  $x$  is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

*Remark* (Standard quantified statement).

$$y \in V_\varepsilon(x) \iff |y - x| < \varepsilon.$$

*Remark* (Interpretation). The  $\varepsilon$ -neighbourhood is the open interval of radius  $\varepsilon$  centered at  $x$ .

*Remark* (Dependencies).

- [Open Ball on the Real Line](#)

### Definition (Deleted $\varepsilon$ -Neighbourhood)

**Definition 4.3.2** (Deleted  $\varepsilon$ -Neighbourhood). Let  $x \in \mathbb{R}$  and let  $\varepsilon > 0$ . The **deleted  $\varepsilon$ -neighbourhood** of  $x$  is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

*Remark* (Standard quantified statement).

$$y \in V_\varepsilon^*(x) \iff 0 < |y - x| < \varepsilon.$$

*Remark* (Interpretation). The deleted neighbourhood removes the base point. It is the punctured input region used in limit definitions.

*Remark* (Dependencies).

- [ε-Neighbourhood](#)

### Definition (Cluster Point)

**Definition 4.3.3** (Cluster Point). Let  $X \subseteq \mathbb{R}$  and  $x \in \mathbb{R}$ . The point  $x$  is a **cluster point** of  $X$  if

$$\forall \varepsilon > 0, \exists y \in X \setminus \{x\} : |y - x| < \varepsilon.$$

*Remark* (Standard quantified statement).

$$x \text{ is a cluster point of } X \iff \forall \varepsilon > 0, \exists y \in X \setminus \{x\} : |y - x| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$x \text{ is a cluster point of } X \equiv \forall \varepsilon > 0, V_\varepsilon^*(x) \cap X \neq \emptyset.$$

*Remark* (Negated quantified statement).

$$\neg x \text{ is a cluster point of } X \iff \exists \varepsilon > 0, \forall y \in X \setminus \{x\} : \neg |y - x| < \varepsilon.$$

*Remark* (Negation predicate reading).

$$\neg x \text{ is a cluster point of } X \equiv \exists \varepsilon > 0 : V_\varepsilon^*(x) \cap X = \emptyset.$$

*Remark* (Interpretation). A cluster point is approached by  $X$  via distinct witnesses arbitrarily close. A cluster point need not belong to  $X$ :  $0$  is a cluster point of  $(0, 1)$  but  $0 \notin (0, 1)$ . The condition  $\lim_{x \rightarrow c} f(x) = L$  requires  $\text{ClusterPoint}(c, \text{dom}(f))$ ; without it the punctured ball condition is vacuous.

*Remark* (Dependencies).

- Deleted  $\varepsilon$ -Neighbourhood

#### Theorem

**Theorem 4.3.4** (Sequential Characterization of Cluster Points).  $c$  is a cluster point of  $A$  if and only if there exists  $(a_n) \subseteq A \setminus \{c\}$  with  $a_n \rightarrow c$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$c \text{ is a cluster point of } A \iff \exists (a_n) \subseteq A \setminus \{c\} : a_n \rightarrow c.$$

*Remark* (Interpretation). The forward direction: use  $\varepsilon = 1/n$  to build the witness sequence. The reverse direction is immediate since the sequence witnesses the ball condition for every  $\varepsilon$ . This bridges the  $\varepsilon$ - $\delta$  definition with the sequential language.

*Remark* (Dependencies).

- Cluster Point

#### Definition (Adherent Point)

**Definition 4.3.5** (Adherent Point). Let  $X \subseteq \mathbb{R}$  and let  $x \in \mathbb{R}$ . The point  $x$  is an **adherent point** of  $X$  if

$$\forall \varepsilon > 0 \exists y \in X (|y - x| < \varepsilon).$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 (V_\varepsilon(x) \cap X \neq \emptyset).$$

*Remark* (Definition predicate reading).

$$\text{AdherentPoint}(x, X) := \forall \varepsilon > 0 \exists y \in X (|y - x| < \varepsilon).$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon > 0 \forall y \in X (|y - x| \geq \varepsilon).$$

*Remark* (Interpretation). An adherent point can be witnessed by points of  $X$  arbitrarily close to  $x$ , including possibly  $x$  itself.

*Remark* (Dependencies).

- [ε-Neighbourhood](#)

#### Definition (Isolated Point)

**Definition 4.3.6** (Isolated Point). Let  $X \subseteq \mathbb{R}$  and let  $x \in X$ . The point  $x$  is an **isolated point** of  $X$  if

$$\exists \varepsilon > 0 (V_\varepsilon(x) \cap X = \{x\}).$$

*Remark* (Standard quantified statement).

$$\exists \varepsilon > 0 \forall y \in X (|y - x| < \varepsilon \Rightarrow y = x).$$

*Remark* (Definition predicate reading).

$$\text{IsolatedPoint}(x, X) := x \in X \wedge \exists \varepsilon > 0 (V_\varepsilon(x) \cap X = \{x\}).$$

*Remark* (Negated quantified statement).

$$\forall \varepsilon > 0 \exists y \in X \setminus \{x\} (|y - x| < \varepsilon).$$

*Remark* (Interpretation). An isolated point belongs to the set but has a neighbourhood containing no other points of the set.

*Remark* (Dependencies).

- [Adherent Point](#)
- [Cluster Point](#)

#### Definition (Interior Point)

**Definition 4.3.7** (Interior Point). Let  $X \subseteq \mathbb{R}$  and let  $x \in \mathbb{R}$ . The point  $x$  is an **interior point** of  $X$  if

$$\exists \varepsilon > 0 (V_\varepsilon(x) \subseteq X).$$

*Remark* (Standard quantified statement).

$$\exists \varepsilon > 0 \forall y \in \mathbb{R} (|y - x| < \varepsilon \Rightarrow y \in X).$$

*Remark* (Definition predicate reading).

$$\text{InteriorPoint}(x, X) := \exists \varepsilon > 0 \ (V_\varepsilon(x) \subseteq X).$$

*Remark* (Interpretation). An interior point has a full open neighbourhood contained in the set.

*Remark* (Dependencies).

- $\varepsilon$ -Neighbourhood

#### Definition (Boundary Point)

**Definition 4.3.8** (Boundary Point). Let  $X \subseteq \mathbb{R}$  and let  $x \in \mathbb{R}$ . The point  $x$  is a **boundary point** of  $X$  if

$$\forall \varepsilon > 0 \ (V_\varepsilon(x) \cap X \neq \emptyset) \wedge (V_\varepsilon(x) \cap X^c \neq \emptyset).$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \ (V_\varepsilon(x) \cap X \neq \emptyset) \wedge (V_\varepsilon(x) \cap X^c \neq \emptyset).$$

*Remark* (Definition predicate reading).

$$\text{BoundaryPoint}(x, X) := \forall \varepsilon > 0 \ (V_\varepsilon(x) \cap X \neq \emptyset) \wedge (V_\varepsilon(x) \cap X^c \neq \emptyset).$$

*Remark* (Interpretation). A boundary point sees both the set and its complement in every neighbourhood.

*Remark* (Dependencies).

- $\varepsilon$ -Neighbourhood

#### Definition (Interior)

**Definition 4.3.9** (Interior). Let  $X \subseteq \mathbb{R}$ . The **interior** of  $X$  is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 \ V_\varepsilon(x) \subseteq X\}.$$

*Remark* (Standard quantified statement).

$$x \in X^\circ \iff \exists \varepsilon > 0 \ V_\varepsilon(x) \subseteq X.$$

*Remark* (Interpretation). The interior collects exactly the points around which the set contains a whole neighbourhood.

*Remark* (Dependencies).

- Interior Point



### Definition (Boundary)

**Definition 4.3.10** (Boundary). Let  $X \subseteq \mathbb{R}$ . The **boundary** of  $X$  is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0 (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

*Remark* (Standard quantified statement).

$$x \in \partial X \iff \forall \varepsilon > 0 (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset).$$

*Remark* (Interpretation). The boundary records the points where every neighbourhood touches both inside and outside of the set.

*Remark* (Dependencies).

- [Boundary Point](#)

### Definition (Closure)

**Definition 4.3.11** (Closure).

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0 V_\varepsilon(x) \cap X \neq \emptyset\}.$$

*Remark* (Standard quantified statement).

$$x \in \overline{X} \iff x \text{ is an adherent point of } X \iff \forall \varepsilon > 0 : V_\varepsilon(x) \cap X \neq \emptyset.$$

*Remark* (Interpretation). The closure adds all adherent points of  $X$ , so it fills in the limiting points that can be reached by points of  $X$ .

*Remark* (Canonical examples).  $\overline{(a, b)} = [a, b]$ ;  $\overline{\mathbb{Q}} = \mathbb{R}$ ;  $\overline{\mathbb{Z}} = \mathbb{Z}$ ;  $\overline{\{1/n\}} = \{1/n\} \cup \{0\}$ .

*Remark* (Dependencies).

- [Adherent Point](#)

### Lemma

**Lemma 4.3.12** (Elementary Properties of Closures). Let  $X, Y \subseteq \mathbb{R}$ . Then: (i)  $X \subseteq \overline{X}$ ; (ii)  $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$ ; (iii)  $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$ ; (iv)  $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$\overline{X \cup Y} = \overline{X} \cup \overline{Y}; \quad \overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}.$$

*Remark* (Failure of equality in (iii)). Take  $X = (0, 1)$ ,  $Y = (1, 2)$ :  $\overline{X \cap Y} = \emptyset$  but  $\overline{X} \cap \overline{Y} = \{1\}$ .

*Remark* (Interpretation). Closure preserves inclusions and finite unions, but intersection can lose limit points that are approached separately by the two sets.

*Remark* (Dependencies).

- Closure

**Definition (Closed Subset of  $\mathbb{R}$ )**

**Definition 4.3.13** (Closed Subset of  $\mathbb{R}$ ).  $E$  is closed  $\iff \overline{E} = E$ .

*Remark* (Standard quantified statement).

$$E \text{ is closed} \iff \forall x \in \mathbb{R} : x \text{ is an adherent point of } E \Rightarrow x \in E.$$

*Remark* (Definition predicate reading).

$$E \text{ is closed} \equiv \forall x : x \text{ is an adherent point of } E \Rightarrow x \in E.$$

*Remark* (Negated quantified statement).

$$\neg E \text{ is closed} \iff \exists x \notin E : x \text{ is an adherent point of } E.$$

*Remark* (Negation predicate reading).

$$\neg E \text{ is closed} \equiv \exists x \notin E : \forall \varepsilon > 0 : V_\varepsilon(x) \cap E \neq \emptyset.$$

*Remark* (Interpretation).  $E$  is closed iff it contains its own boundary. Examples:  $[a, b]$  closed;  $(a, b)$  not closed;  $\mathbb{R}$  and  $\emptyset$  both closed;  $\mathbb{Q}$  not closed.

*Remark* (Dependencies).

- Closure

**Corollary**

**Corollary 4.3.14** (Closed iff Sequentially Closed).  $X$  is closed  $\iff \forall (a_n) \subseteq X, a_n \rightarrow x \Rightarrow x \in X$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$$X \text{ is closed} \iff \forall (a_n) \subseteq X, a_n \rightarrow x \Rightarrow x \in X.$$

*Remark* (Interpretation). A closed set cannot be escaped by convergent sequences. Every adherent point has a witness sequence; closedness forces the limit into the set.

*Remark* (Dependencies).

- Closed Subset of  $\mathbb{R}$
- Sequential Characterization of Cluster Points

### Corollary

**Corollary 4.3.15** (Every Point of an Interval Is a Cluster Point). *If  $I$  is an interval, then every  $x \in I$  is a cluster point of  $I$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall x \in I \forall \varepsilon > 0 \exists y \in I \setminus \{x\} (|y - x| < \varepsilon).$$

*Remark* (Interpretation). This justifies intervals as natural domains for function limits.  $\lim_{x \rightarrow c} f(x) = L$  requires  $\text{ClusterPoint}(c, \text{dom}(f))$ ; on any interval this is free. ■

*Remark* (Dependencies).

- [Cluster Point](#)

### Definition (Bounded Subset of $\mathbb{R}$ )

**Definition 4.3.16** (Bounded Subset of  $\mathbb{R}$ ).  $X$  is bounded  $\iff \exists M > 0, \forall x \in X : |x| \leq M$ .

*Remark* (Standard quantified statement).

$$\exists M > 0 \forall x \in X (|x| \leq M).$$

*Remark* (Negated quantified statement).

$$\forall M > 0 \exists x \in X (|x| > M).$$

*Remark* (Interpretation). A bounded subset of  $\mathbb{R}$  fits inside some symmetric interval  $[-M, M]$ .

*Remark* (Dependencies).

- [Subset](#)
- [Absolute Value on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

### Theorem

**Theorem 4.3.17** (Heine–Borel Theorem for  $\mathbb{R}$ ).  $X$  is closed  $\wedge X$  is bounded  $\iff$  every  $(a_n) \subseteq X$  has a subsequence converging to a limit in  $X$ .

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$X \text{ is closed} \wedge X \text{ is bounded} \iff \forall (a_n) \subseteq X, \exists (a_{n_k}), \exists L \in X : a_{n_k} \rightarrow L.$$

*Remark* (Definition predicate reading).

$$\text{SequentiallyCompact}(X) :\equiv X \text{ is closed} \wedge X \text{ is bounded.}$$

*Remark* (Negated quantified statement).

$$\neg(X \text{ is closed} \wedge X \text{ is bounded}) \iff \exists(a_n) \subseteq X : \text{no subsequence converges to a limit in } X.$$

*Remark* (Failure modes). The compactness conclusion fails when either closedness or boundedness fails.

*Remark* (Failure mode decomposition). Either  $X$  is bounded fails — a sequence escapes to  $\pm\infty$  — or  $X$  is closed fails — a convergent subsequence exists but its limit lies outside  $X$ .

*Remark* (Interpretation). Closed prevents limits from escaping through boundary gaps; bounded prevents escape to infinity. Both are required. Forward: bounded gives Bolzano–Weierstrass; closed keeps the limit in  $X$ . Reverse: negate either hypothesis to construct a sequence with no convergent subsequence in  $X$ .

*Remark* (Dependencies).

- Closed Subset of  $\mathbb{R}$
- Bounded Subset of  $\mathbb{R}$

#### Definition (True Near a Point)

**Definition 4.3.18** (True Near a Point). Property  $P(f(x))$  is **true near**  $x_0$  if  $\exists\delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x))$ .

*Remark* (Standard quantified statement).

$$P(f(x)) \text{ true near } x_0 \iff \exists\delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x)).$$

*Remark* (Interpretation).  $\lim_{x \rightarrow x_0} f(x) = L$  states: for every  $\varepsilon > 0$ , the property  $|f(x) - L| < \varepsilon$  is true near  $x_0$ .

*Remark* (Dependencies).

- $\varepsilon$ -Neighbourhood

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Composition of Injections Is Injective). *Let  $f : A \rightarrow B$  and let  $g : B \rightarrow C$ . If  $f$  is injective and  $g$  is injective, then  $g \circ f : A \rightarrow C$  is injective.*

*Professional Standard Proof.*

TODO: professional standard proof for composition-injective. □

*Detailed Learning Proof.*

TODO: detailed learning proof for composition-injective. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Composition of Surjections Is Surjective).  $\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \text{Surjective}(g \circ f)$ .

*Professional Standard Proof.*

TODO: professional standard proof for composition-surjective. □

*Detailed Learning Proof.*

TODO: detailed learning proof for composition-surjective. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Composition of Bijections Is Bijective). *Let  $f : A \rightarrow B$  and let  $g : B \rightarrow C$ . If  $f$  is bijective and  $g$  is bijective, then  $g \circ f : A \rightarrow C$  is bijective.*

*Professional Standard Proof.*

TODO: professional standard proof for composition-bijective.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for composition-bijective.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Inverse of a Bijection Is a Bijection).  $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$ .

*Professional Standard Proof.*

TODO: professional standard proof for inverse-bijection. □

*Detailed Learning Proof.*

TODO: detailed learning proof for inverse-bijection. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Preimage Distributes over Union and Intersection). *Let  $f : A \rightarrow B$ ,  $S, T \subseteq B$ . Then:*

$$(i) \quad f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T).$$

$$(ii) \quad f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

$$(iii) \quad f^{-1}(S^c) = (f^{-1}(S))^c.$$

*Professional Standard Proof.*

TODO: professional standard proof for preimage-union-intersection. □

*Detailed Learning Proof.*

TODO: detailed learning proof for preimage-union-intersection. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sequential Characterization of Cluster Points).  *$c$  is a cluster point of  $A$  if and only if there exists  $(a_n) \subseteq A \setminus \{c\}$  with  $a_n \rightarrow c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cluster-point-sequential. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cluster-point-sequential. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Lemma** (Elementary Properties of Closures). *Let  $X, Y \subseteq \mathbb{R}$ . Then: (i)  $X \subseteq \overline{X}$ ; (ii)  $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$ ; (iii)  $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$ ; (iv)  $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for closure-elementary. □

*Detailed Learning Proof.*

TODO: detailed learning proof for closure-elementary. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Closed iff Sequentially Closed).  $X$  is closed  $\iff \forall (a_n) \subseteq X, a_n \rightarrow x \Rightarrow x \in X$ .

*Professional Standard Proof.*

TODO: professional standard proof for closed-iff-seq-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for closed-iff-seq-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Every Point of an Interval Is a Cluster Point). *If  $I$  is an interval, then every  $x \in I$  is a cluster point of  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for interval-all-limit-points. □

*Detailed Learning Proof.*

TODO: detailed learning proof for interval-all-limit-points. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Heine–Borel Theorem for  $\mathbb{R}$ ).  $X$  is closed  $\wedge$   $X$  is bounded  $\iff$  every  $(a_n) \subseteq X$  has a subsequence converging to a limit in  $X$ .

*Professional Standard Proof.*

TODO: professional standard proof for heine-borel-subsets-real-line. □

*Detailed Learning Proof.*

TODO: detailed learning proof for heine-borel-subsets-real-line. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

# Chapter 5

## Discrete Calculus



### 5.1 Motivation

#### Motivation

Classical calculus studies functions defined on continuous domains, typically functions  $f : \mathbb{R} \rightarrow \mathbb{R}$ . Its central tools are the derivative and the integral:

- The derivative measures *instantaneous change*.
- The integral accumulates *total change*.

However, many important mathematical objects are inherently *discrete*. Examples include

- sequences  $a_n$ ,
- recurrence relations,
- sums of integers or combinatorial quantities,
- discrete probability distributions,
- algorithms and computational processes.

In these contexts the domain is typically the natural numbers

$$f : \mathbb{Z} \rightarrow \mathbb{R}$$

rather than the real line.

To study change in such settings, calculus is replaced by its discrete counterpart.

## Discrete Analogs of Calculus

The central idea of *discrete calculus* is that the roles of derivative and integral are played by two simple operations.

| Continuous vs. Discrete Calculus |                                                  |
|----------------------------------|--------------------------------------------------|
| Continuous Calculus              | Discrete Calculus                                |
| Derivative $\frac{d}{dx}f(x)$    | Forward difference $\Delta f(n) = f(n+1) - f(n)$ |
| Integral $\int f(x) dx$          | Summation $\sum f(n)$                            |

The forward difference operator measures change between consecutive values of a function.

$$\Delta f(n) = f(n+1) - f(n).$$

This quantity plays the same role for sequences that the derivative plays for functions of a real variable.

Likewise, summation serves as the discrete analogue of integration.

$$\sum_{k=a}^b f(k)$$

accumulates the values of a function across a finite discrete interval.

## Telescoping Phenomena

One of the most important identities in discrete mathematics arises when differences are summed.

$$\sum_{k=a}^{b-1} (f(k+1) - f(k)) = f(b) - f(a).$$

The intermediate terms cancel, leaving only the endpoints. Such sums are called *telescoping sums*.

This identity is the discrete counterpart of the *Fundamental Theorem of Calculus*.

## Why Discrete Calculus Matters

Discrete calculus appears throughout mathematics:

- in combinatorics and enumerative formulas,
- in recurrence relations and generating functions,
- in probability theory and stochastic processes,



- in numerical analysis and finite difference methods,
- in computer science and algorithm analysis.

For students of analysis, discrete calculus is particularly valuable because it reveals the structural skeleton behind many identities involving sums and sequences.

Many techniques used to estimate limits of sequences or series ultimately rely on discrete analogues of differential identities.

## A Structural View

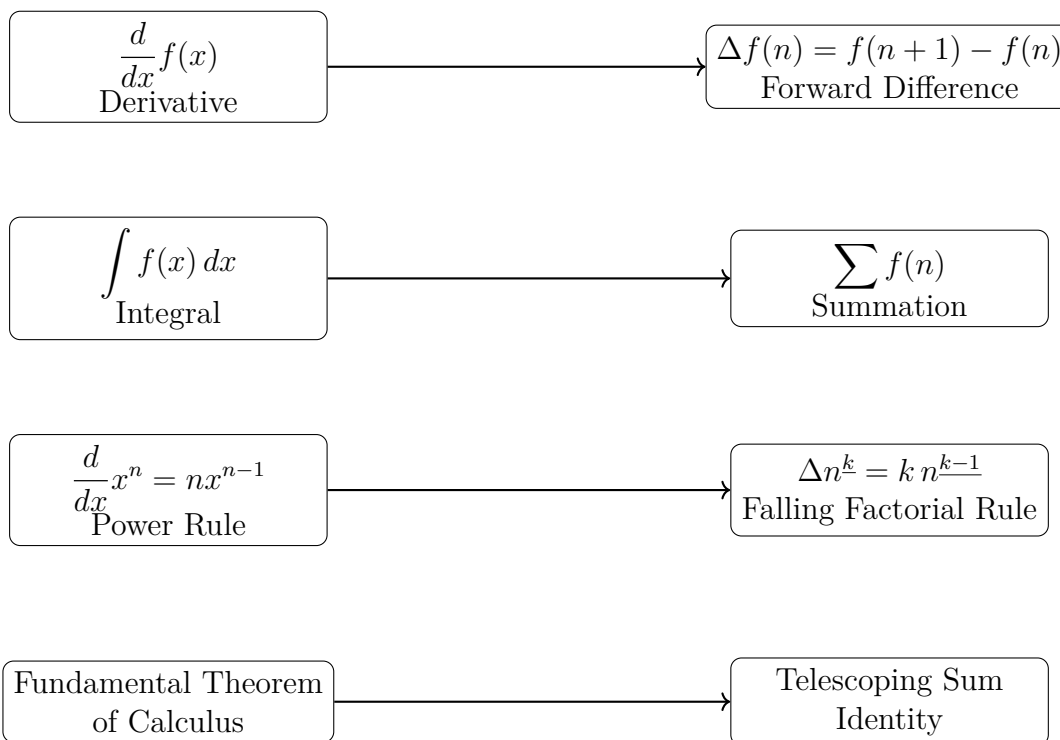
Continuous calculus studies how functions change along the real line.

Discrete calculus studies how functions change along the integers.

Despite the difference in setting, the two theories mirror each other remarkably closely. Understanding this parallel provides powerful intuition and useful techniques for manipulating sums, sequences, and recurrence relations.

In the sections that follow we develop the basic tools of discrete calculus, beginning with the forward difference operator and the algebra of finite differences.

### Structural Correspondence Between Continuous and Discrete Calculus



## 5.2 Foundations

### 5.2.1 Foundations

#### Definition (Discrete Function)

A *discrete function* is a function

$$f : \mathbb{Z} \rightarrow \mathbb{R}$$

(or more generally  $f : \mathbb{N} \rightarrow \mathbb{R}$ ). It assigns a real value to each integer input, producing a sequence  $f(0), f(1), f(2), \dots$

*Remark* (Fully quantified form).  $\forall n \in \mathbb{Z}, f(n) \in \mathbb{R}$ . The domain is the integers (or natural numbers); the codomain is the real line.

*Remark* (Discrete vs. continuous). In discrete calculus the independent variable moves in *integer steps* rather than continuously. Change is therefore measured between successive integer inputs rather than through infinitesimal limits. The fundamental operator of discrete calculus is the *difference operator*

$$\Delta f(n) = f(n+1) - f(n),$$

which plays a role analogous to the derivative in continuous calculus.

*Remark* (Sequences as discrete functions). A discrete function is simply another way of viewing a *sequence*. A sequence  $a_0, a_1, a_2, \dots$  is identified with

$$f : \mathbb{N} \rightarrow \mathbb{R}, \quad f(n) = a_n.$$

Several equivalent notations appear in the literature:  $f(n)$ ,  $a_n$ ,  $(a_n)_{n \in \mathbb{N}}$ ,  $\{a_n\}_{n=0}^{\infty}$ . All describe the same object. The functional notation  $f(n)$  is preferred in discrete calculus because the difference operator acts on it much as a derivative acts on functions in continuous calculus.

### 5.2.2 Summation Notation

#### Definition (Summation)

Let  $f$  be a function defined on the integers. For integers  $a \leq b$ , the expression

$$\sum_{k=a}^b f(k)$$

denotes the sum  $f(a) + f(a+1) + f(a+2) + \dots + f(b)$ . The symbol  $\sum$  (Greek capital sigma) represents summation;  $k$  is the *index of summation*;  $a$  and  $b$  are the *lower* and *upper limits*;  $f(k)$  is the *summand*.

*Remark* (Standard quantified statement).  $\sum_{k=a}^b f(k) := f(a) + f(a+1) + \dots + f(b)$ , where  $a, b \in \mathbb{Z}$ ,  $a \leq b$ , and  $f : \mathbb{Z} \rightarrow \mathbb{R}$ .

*Remark* (Interpretation). Sigma notation compresses a finite list of additions into one indexed symbol.

*Remark* (Dummy variable). The index of summation is a *dummy variable*:  $\sum_{k=a}^b f(k) = \sum_{i=a}^b f(i)$ . Its name carries no information about the value of the sum.

## Basic Properties of Summation

*Remark* (Basic properties of summation). For functions  $f$  and  $g$  and constants  $c$ ,

$$\sum_{k=m}^n (f(k) + g(k)) = \sum_{k=m}^n f(k) + \sum_{k=m}^n g(k),$$

$$\sum_{k=m}^n c f(k) = c \sum_{k=m}^n f(k).$$

These express the *linearity* of summation.

## Splitting Sums

*Remark* (Splitting sums). A sum over an interval may be divided into smaller pieces:

$$\sum_{k=a}^c f(k) = \sum_{k=a}^b f(k) + \sum_{k=b+1}^c f(k) \quad \text{whenever } a \leq b < c.$$

## Changing the Index

*Remark* (Changing the index). It is often useful to shift the summation index. For example,

$$\sum_{k=0}^n f(k) = \sum_{j=1}^{n+1} f(j-1).$$

Such substitutions frequently simplify expressions or align sums with known formulas.

## Empty Sums

*Remark* (Empty sums). If the upper bound is smaller than the lower bound, the sum contains no terms. By convention,

$$\sum_{k=a}^b f(k) = 0 \quad \text{when } a > b.$$

This convention simplifies many algebraic identities.

## Nested Summations

*Remark* (Nested summations). Summations may be nested:

$$\sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{i=1}^n (a_{i1} + a_{i2} + \cdots + a_{im}).$$

The inner sum is evaluated first. When the limits are independent, the order of summation may be exchanged:

$$\sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{j=1}^m \sum_{i=1}^n a_{ij}.$$

*Remark* (Analogy with integration). Nested sums behave like iterated integrals: exchanging summation order corresponds to Fubini's theorem in the discrete setting.

## Common Examples

*Remark* (Common examples).

$$\sum_{k=1}^n 1 = n, \quad \sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

These formulas appear throughout discrete mathematics and analysis.

### 5.2.3 The Binomial Theorem

**Theorem 5.2.1** (Binomial Theorem). *For any integer  $n \geq 0$  and real numbers  $x, y$ ,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k,$$

where the binomial coefficients are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

*Remark* (Standard quantified statement).  $\forall n \in \mathbb{N}, \forall x, y \in \mathbb{R} : (x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$ .

*Remark* (Interpretation). The theorem expands a power of a sum by counting how many terms choose each possible number of  $y$ -factors.

*Remark* (Combinatorial meaning). The binomial coefficients count the number of ways to choose  $k$  factors of  $y$  from the  $n$  factors in the product  $(x + y)(x + y) \cdots (x + y)$ . Thus  $\binom{n}{k}$  is the number of ways to select  $k$  objects from a set of  $n$ , and the theorem describes how a power of a sum expands into a sum of products.

## Basic Properties of Binomial Coefficients

*Remark* (Basic properties of binomial coefficients).

$$\binom{n}{0} = 1, \quad \binom{n}{n} = 1, \quad \binom{n}{k} = \binom{n}{n-k}.$$

They also satisfy the *Pascal recursion*:

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}.$$

This recursion produces Pascal's triangle, in which each entry is the sum of the two entries directly above it.

$$\begin{array}{ccccccc} & & & & \binom{0}{0} & & \\ & & & & & & \\ & & \binom{1}{0} & & \binom{1}{1} & & \\ & & & & & & \\ & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & \\ & & & & & & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} \\ & & & & & & \\ & \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} \\ & & & & & & \\ & \binom{5}{0} & & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} \end{array}$$

*Remark* (Pascal's triangle). Each entry  $\binom{n}{k}$  is obtained from the sum of the two entries above it:  $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ . These coefficients appear throughout discrete calculus, especially in formulas for higher finite differences and in Newton's forward difference expansion.

## Connection with Discrete Calculus

*Remark* (Connection with discrete calculus). The binomial theorem plays a central role in discrete calculus because difference operators naturally produce binomial coefficients.

**Proposition 5.2.2** (Higher difference formula). *For any function  $f$ ,*

$$\Delta^n f(x) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x+k).$$

*Go to proof.*

*Remark* (Standard quantified statement).  $\forall n \in \mathbb{N}, \forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall x \in \mathbb{Z} : \Delta^n f(x) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x+k).$

*Remark* (Interpretation). The  $n$ -th forward difference is a signed binomial combination of  $n + 1$  successive function values.

*Remark* (Proof idea). Applying the difference operator  $\Delta = E - I$  (where  $E$  is the shift operator) repeatedly gives  $\Delta^n = (E - I)^n$ . Expanding by the binomial theorem yields the coefficients  $(-1)^{n-k} \binom{n}{k}$ , and  $E^k f(x) = f(x + k)$ . This derivation is carried out in full in the Difference Operators section.

*Remark* (Dependencies).

- [Binomial Theorem](#).
- [Forward Difference Operator](#).
- [Higher Differences](#).

## Explicit Examples

*Remark* (Explicit examples). Applying the difference operator repeatedly to a function  $f$  yields

$$\begin{aligned}\Delta f(x) &= f(x + 1) - f(x), \\ \Delta^2 f(x) &= f(x + 2) - 2f(x + 1) + f(x), \\ \Delta^3 f(x) &= f(x + 3) - 3f(x + 2) + 3f(x + 1) - f(x).\end{aligned}$$

The coefficients 1, 2, 1 and 1, 3, 3, 1 are precisely the rows of Pascal's triangle, confirming that binomial coefficients determine the pattern of coefficients in higher-order finite differences.

## 5.3 Algebra of Summations

### Summation Algebra — Quick Reference

|                     |                                                                    |
|---------------------|--------------------------------------------------------------------|
| <b>Linearity</b>    | $\sum (af + bg) = a \sum f + b \sum g$                             |
| <b>Splitting</b>    | $\sum_a^c = \sum_a^b + \sum_{b+1}^c$                               |
| <b>Index shift</b>  | $\sum_{k=a}^b f(k) = \sum_{k=a+1}^{b+1} f(k - 1)$                  |
| <b>Constant sum</b> | $\sum_{k=a}^b c = (b - a + 1)c$                                    |
| <b>Empty sum</b>    | $\sum_{k=a}^b f(k) = 0$ when $a > b$                               |
| <b>Nested sums</b>  | $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ (independent limits) |

*Remark* (Orientation). The difference operator and telescoping identity provide the background for the basic algebraic rules used to manipulate summations. These identities allow sums to be combined, split, and reindexed while preserving their value.

## Linearity

*Remark* (Linearity). Summation is a linear operator. For functions  $f$  and  $g$  and constants  $a$  and  $b$ ,

$$\sum_{k=m}^n (a f(k) + b g(k)) = a \sum_{k=m}^n f(k) + b \sum_{k=m}^n g(k).$$

This allows constants to be factored out of a sum and sums of expressions to be separated.

*Example.*  $\sum_{k=1}^n (2k + 3) = 2 \sum_{k=1}^n k + 3 \sum_{k=1}^n 1.$

## Splitting a Sum

*Remark* (Splitting a sum). A sum over an interval may be divided into pieces:

$$\sum_{k=a}^c f(k) = \sum_{k=a}^b f(k) + \sum_{k=b+1}^c f(k) \quad \text{whenever } a \leq b < c.$$

*Example.*  $\sum_{k=1}^{10} f(k) = \sum_{k=1}^5 f(k) + \sum_{k=6}^{10} f(k).$

## Index Renaming

*Remark* (Index renaming). The index of summation is a *dummy variable*:

$$\sum_{k=a}^b f(k) = \sum_{i=a}^b f(i).$$

This is useful when manipulating expressions containing several sums, since the index variable can be renamed to avoid conflicts.

## Index Shifting

*Remark* (Index shifting). It is often convenient to shift the summation index:

$$\sum_{k=a}^b f(k) = \sum_{k=a+1}^{b+1} f(k-1).$$

Index shifting allows sums to be aligned so that terms cancel or combine properly. It appears frequently in proofs involving recurrence relations and discrete calculus identities.

*Example.*  $\sum_{k=0}^n f(k+1) = \sum_{k=1}^{n+1} f(k).$

## Constant Sum

*Remark* (Constant sum). If the summand does not depend on the index, the sum counts terms:

$$\sum_{k=a}^b c = (b - a + 1) c.$$

The quantity  $b - a + 1$  is the number of integers between  $a$  and  $b$ , inclusive.

*Example.*  $\sum_{k=3}^7 5 = 5 \cdot 5 = 25.$

## Empty Sum

*Remark* (Empty sum). If the upper limit is smaller than the lower limit, the sum contains no terms. By convention,

$$\sum_{k=a}^b f(k) = 0 \quad \text{whenever } a > b.$$

This convention simplifies many algebraic identities and allows formulas to remain valid without special cases.

## Nested Sums

*Remark* (Nested sums). Summations may be nested:

$$\sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{i=1}^n \left( a_{i1} + a_{i2} + \cdots + a_{im} \right).$$

When the limits are independent, the order may be exchanged:

$$\sum_{i=1}^n \sum_{j=1}^m a_{ij} = \sum_{j=1}^m \sum_{i=1}^n a_{ij}.$$

*Remark* (Analogy with double integrals). Exchanging summation order is the discrete analogue of Fubini's theorem for iterated integrals.

## Common Summation Identities

*Remark* (Common summation identities). Many sums that appear in discrete mathematics and discrete calculus have closed-form expressions. The following identities are among the most frequently encountered.



### Common Summation Identities

$$\begin{aligned} \sum_{k=1}^n 1 &= n & \sum_{k=1}^n k &= \frac{n(n+1)}{2} \\ \sum_{k=1}^n k^2 &= \frac{n(n+1)(2n+1)}{6} & \sum_{k=1}^n k^3 &= \left(\frac{n(n+1)}{2}\right)^2 \\ \sum_{k=0}^n r^k &= \frac{1-r^{n+1}}{1-r} \quad (r \neq 1) & \sum_{k=0}^n \binom{n}{k} &= 2^n \end{aligned}$$

*Remark* (Role in discrete calculus). Polynomial summation formulas play an important role when working with finite differences and Newton series, while geometric sums arise naturally in recurrence relations and generating functions. Many of these identities can be derived systematically using the falling factorial basis and the Fundamental Theorem of Finite Calculus introduced later in this chapter.

## 5.4 Difference Operators

### 5.4.1 Difference Operators

Discrete calculus studies functions defined on the integers and the way their values change from one integer to the next. The central tool for measuring this change is the *difference operator*, which plays a role analogous to the derivative in continuous calculus.

### Motivation

In ordinary calculus the derivative measures the rate of change of a function:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

In the discrete setting the independent variable changes in steps of size 1. Instead of taking a limit, we measure the difference between successive values of the function directly.

### Forward Difference

**Definition 5.4.1** (Forward Difference Operator). Let  $f : \mathbb{Z} \rightarrow \mathbb{R}$ . The *forward difference operator*  $\Delta$  is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function  $\Delta f$  measures the change in  $f$  when the input increases by one unit.

*Remark* (Fully quantified form).  $\forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall n \in \mathbb{Z} : \Delta f(n) := f(n+1) - f(n)$ .

*Remark* (Interpretation). The forward difference records the one-step change of a discrete function.

*Remark* (Analogy with differentiation). The operator  $\Delta$  is the discrete analogue of the derivative. While the derivative measures infinitesimal change through a limit, the forward difference measures change between successive integer inputs exactly, with no limit required.

## Example

Consider the function  $f(n) = n^2$ . Applying the difference operator gives

$$\begin{aligned}\Delta f(n) &= f(n+1) - f(n) \\ &= (n+1)^2 - n^2 \\ &= 2n+1.\end{aligned}$$

Thus the forward difference of the quadratic function  $n^2$  is the linear function  $2n+1$ .

*Remark* (Degree reduction). This mirrors a familiar fact from ordinary calculus:  $\frac{d}{dx}x^2 = 2x$ . Just as derivatives reduce the degree of a polynomial by one, finite differences also lower the degree of polynomial sequences. Note, however, that  $\Delta(n^2) = 2n+1$  rather than  $2n$ ; ordinary powers do not behave as cleanly under  $\Delta$  as they do under differentiation. This motivates introducing the *falling factorial* basis later in this chapter.

## 5.4.2 Higher Differences

**Definition 5.4.2** (Higher Differences). The *second difference* of  $f$  is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the  $k$ -th difference  $\Delta^k f(n)$  denotes the  $k$ -fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \quad \Delta^k f = \Delta(\Delta^{k-1} f).$$

*Remark* (Fully quantified form).  $\forall k \in \mathbb{N}, \forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall n \in \mathbb{Z} : \Delta^k f(n) := \underbrace{(\Delta \circ \cdots \circ \Delta)}_k f(n)$

*Remark* (Interpretation). Higher differences measure repeated one-step change, just as higher derivatives measure repeated infinitesimal change.

*Remark* (Dependencies).

- [Forward Difference Operator](#).

## Example

Let  $f(n) = n^2$ . Then

$$\begin{aligned}\Delta f(n) &= 2n+1, \\ \Delta^2 f(n) &= \Delta(2n+1) = 2(n+1) + 1 - (2n+1) = 2, \\ \Delta^3 f(n) &= \Delta(2) = 0.\end{aligned}$$

*Remark* (Analogy with higher derivatives). Higher differences behave analogously to higher derivatives. For polynomial functions of degree  $d$ , the  $d$ -th difference is constant and all higher differences are zero — just as the  $d$ -th derivative of a degree- $d$  polynomial is constant and the next derivative vanishes.

### 5.4.3 Properties of the Difference Operator

The forward difference operator satisfies algebraic properties that closely mirror the familiar rules of differentiation.

#### Linearity

**Proposition 5.4.3** (Linearity of  $\Delta$ ). *For functions  $f, g : \mathbb{Z} \rightarrow \mathbb{R}$  and constants  $a, b \in \mathbb{R}$ ,*

$$\Delta(af + bg) = a \Delta f + b \Delta g.$$

*Remark* (Fully quantified form).  $\forall f, g : \mathbb{Z} \rightarrow \mathbb{R}, \forall a, b \in \mathbb{R}, \forall n \in \mathbb{Z} : \Delta(af + bg)(n) = a \Delta f(n) + b \Delta g(n)$ .

*Remark* (Interpretation). Taking a forward difference respects pointwise linear combinations.

*Remark* (Dependencies).

- [Forward Difference Operator](#).

*Proof.* By definition of the difference operator,

$$\begin{aligned} \Delta(af + bg)(n) &= (af + bg)(n + 1) - (af + bg)(n) \\ &= af(n + 1) + bg(n + 1) - af(n) - bg(n) \\ &= a(f(n + 1) - f(n)) + b(g(n + 1) - g(n)) \\ &= a \Delta f(n) + b \Delta g(n). \quad \square \end{aligned}$$

#### Constant Rule

**Proposition 5.4.4** (Constant rule). *If  $c$  is a constant function, then  $\Delta c = 0$ .*

*Remark* (Fully quantified form).  $\forall c \in \mathbb{R}, \forall n \in \mathbb{Z} : \Delta c(n) = 0$ .

*Remark* (Interpretation). A constant discrete function has no one-step change.

*Remark* (Dependencies).

- [Forward Difference Operator](#).

*Proof.*  $\Delta c(n) = c - c = 0$ .  $\square$

*Remark* (Analogy). This mirrors  $\frac{d}{dx}c = 0$  in continuous calculus.

### The Shift Operator and Operator Algebra

**Definition 5.4.5** (Shift Operator). The *shift operator*  $E$  and the *identity operator*  $I$  are defined by

$$Ef(n) = f(n + 1), \quad If(n) = f(n).$$

*Remark* (Fully quantified form).  $\forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall n \in \mathbb{Z} : Ef(n) = f(n + 1)$  and  $If(n) = f(n)$ .

*Remark* (Interpretation). The shift operator advances the input by one step, while the identity operator leaves the input unchanged.

**Theorem 5.4.6** ( $\Delta = E - I$ ).

$$\Delta = E - I.$$

*Remark* (Fully quantified form).  $\forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall n \in \mathbb{Z} : \Delta f(n) = (E - I)f(n) = f(n + 1) - f(n)$ .

*Remark* (Interpretation). The forward difference is the shifted value minus the original value.

*Remark* (Dependencies).

- [Forward Difference Operator](#).
- [Shift Operator](#).

*Proof*. Applying the operator  $E - I$  to a function  $f$  gives

$$(E - I)f(n) = Ef(n) - If(n) = f(n + 1) - f(n) = \Delta f(n). \quad \square$$

*Remark* (Higher differences via the binomial theorem). Expressing  $\Delta = E - I$  reveals that higher differences can be computed using ordinary algebra. Applying the binomial theorem to  $\Delta^n = (E - I)^n$  gives

$$\Delta^n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} E^k.$$

Since  $E^k f(n) = f(n + k)$ , this yields

$$\Delta^n f(n) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(n + k).$$

This explains why binomial coefficients and Pascal's triangle appear naturally throughout discrete calculus: they are a direct consequence of the algebraic identity  $\Delta = E - I$ .

### 5.4.4 Difference Tables and Polynomial Detection

One of the most useful tools in discrete calculus is the *difference table*. By repeatedly applying the difference operator to a sequence, one can detect whether the sequence arises from a polynomial function and determine its degree.

**Definition 5.4.7** (Difference Table). Given a sequence  $f(0), f(1), f(2), \dots$ , the *difference table* is constructed by listing the successive values of the function and then computing their forward differences  $\Delta f(n) = f(n + 1) - f(n)$ , then  $\Delta^2 f$ ,  $\Delta^3 f$ , and so on.

*Remark* (Interpretation). A difference table organizes repeated forward differences into rows so that eventual constant or zero rows become visible.

*Remark* (Dependencies).

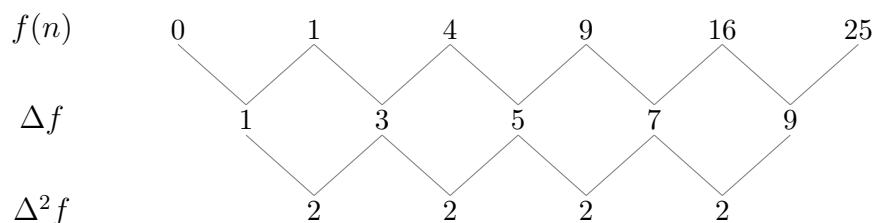
- [Forward Difference Operator](#).
- [Higher Differences](#).

## Example

For  $f(n) = n^2$ :

| $n$ | $f(n)$ | $\Delta f(n)$ | $\Delta^2 f(n)$ |
|-----|--------|---------------|-----------------|
| 0   | 0      | 1             |                 |
| 1   | 1      | 3             | 2               |
| 2   | 4      | 5             | 2               |
| 3   | 9      | 7             | 2               |
| 4   | 16     | 9             | 2               |
| 5   | 25     |               |                 |

The second differences are constant (all equal to 2), and the third differences are zero.



**Theorem 5.4.8** (Polynomial detection). *Let  $f(n)$  be a sequence generated by a polynomial of degree  $d$ . Then  $\Delta^d f(n)$  is constant, and  $\Delta^{d+1} f(n) = 0$ .*

*Proof.* Write  $f(n) = a_d n^d + a_{d-1} n^{d-1} + \cdots + a_0$  with  $a_d \neq 0$ . We show by induction on  $d$  that  $\Delta f$  is a polynomial of degree  $d - 1$  with leading coefficient  $d a_d$ .

*Base case* ( $d = 0$ ): a constant polynomial has  $\Delta f = 0$ . ✓

*Inductive step:* Assume the result holds for all polynomials of degree less than  $d$ . Then

$$\Delta f(n) = f(n+1) - f(n) = a_d((n+1)^d - n^d) + (\text{lower-degree terms}).$$

Expanding  $(n+1)^d - n^d = d n^{d-1} + \binom{d}{2} n^{d-2} + \cdots$ , the leading term is  $d a_d n^{d-1}$ . Thus  $\Delta f$  is a polynomial of degree  $d - 1$  with leading coefficient  $d a_d \neq 0$ .

By induction, applying  $\Delta$  a further  $d - 1$  times yields the constant  $\Delta^d f(n) = d! a_d$ , and one further application gives 0.  $\square$

*Remark* (Fully quantified form).  $\forall d \in \mathbb{N}, \forall f : \mathbb{Z} \rightarrow \mathbb{R}$  polynomial of degree  $d$ :  
 $\exists c \in \mathbb{R} : \forall n \in \mathbb{Z}, \Delta^d f(n) = c$ , and  $\forall n \in \mathbb{Z}, \Delta^{d+1} f(n) = 0$ .

*Remark* (Interpretation). Repeated differences reveal polynomial degree: a degree  $d$  polynomial has constant  $d$ -th differences and vanishing next differences.

*Remark* (Consequence). Difference tables provide a practical method for identifying polynomial sequences and determining their degree. This observation leads naturally to the *falling factorial basis*: a family of polynomials that interact particularly cleanly with the difference operator, forming the foundation for Newton's discrete analogue of the Taylor expansion.

*Remark* (Dependencies).

- [Forward Difference Operator](#).
- [Higher Differences](#).
- [Difference Table](#).
- [Binomial Theorem](#).

## 5.5 Telescoping Sums

### 5.5.1 Telescoping Sums

One of the most useful patterns in summation is the phenomenon of *telescoping*. In a telescoping sum, consecutive terms cancel with one another, leaving only a small number of terms from the beginning and end of the expression.

#### A Simple Example

Consider the sum

$$(2 - 1) + (3 - 2) + (4 - 3) + (5 - 4).$$

Expanding gives

$$2 - 1 + 3 - 2 + 4 - 3 + 5 - 4.$$

Most terms cancel in adjacent pairs:

$$\cancel{2} - 1 + \cancel{3} - \cancel{2} + \cancel{4} - \cancel{3} + 5 - \cancel{4}.$$

After cancellation, only the first and last terms remain:  $5 - 1$ . Thus

$$(2 - 1) + (3 - 2) + (4 - 3) + (5 - 4) = 5 - 1.$$

#### Definition (Telescoping Sum)

A sum is called a *telescoping sum* if consecutive terms cancel when the sum is expanded, leaving only a few remaining terms — typically the first and last.

#### General Form

The general telescoping structure is

$$\sum_{k=a}^{b-1} (f(k+1) - f(k)).$$

Writing out the terms explicitly:

$$\begin{aligned} & (f(a+1) - f(a)) \\ & + (f(a+2) - f(a+1)) \\ & + (f(a+3) - f(a+2)) \\ & \vdots \\ & + (f(b) - f(b-1)). \end{aligned}$$

All intermediate terms cancel in pairs, leaving only  $f(b) - f(a)$ .

*Remark* (Why “telescoping”). The name comes from an old collapsible telescope: each section slides into the next. The intermediate terms in the sum “slide together” and vanish, collapsing the entire sum down to its endpoints.

## The Telescoping Identity

### Theorem (Telescoping Identity)

For any function  $f : \mathbb{Z} \rightarrow \mathbb{R}$  and integers  $a \leq b$ ,

$$\sum_{k=a}^{b-1} (f(k+1) - f(k)) = f(b) - f(a).$$

*Remark* (Fully quantified form).  $\forall f : \mathbb{Z} \rightarrow \mathbb{R}, \forall a, b \in \mathbb{Z}, a \leq b : \sum_{k=a}^{b-1} (f(k+1) - f(k)) = f(b) - f(a)$ .

*Proof.* Expanding the sum gives

$$\begin{aligned} & (f(a+1) - f(a)) + (f(a+2) - f(a+1)) \\ & \quad + \cdots + (f(b) - f(b-1)). \end{aligned}$$

Every intermediate term  $f(a+1), f(a+2), \dots, f(b-1)$  appears once with a positive sign and once with a negative sign, so they all cancel. The remaining terms are  $f(b)$  and  $-f(a)$ , giving  $f(b) - f(a)$ .  $\square$

## Cancellation Pattern

The cancellation can be seen explicitly:

$$\begin{array}{ccc}
f(a+1) - f(a) & & -f(a) \\
\\
f(a+2) - f(a+1) & & \cancel{f(a+1)} - \cancel{f(a+1)} \\
\\
f(a+3) - f(a+2) & & \cancel{f(a+2)} - \cancel{f(a+2)} \\
\\
\vdots & & \vdots \\
\\
f(b) - f(b-1) & & f(b)
\end{array}$$

*Remark* (Intuition). Each intermediate term appears twice in the expanded sum: once with a positive sign and once with a negative sign. These pairs cancel, leaving only the first and last terms.

## Operator Form

Using the forward difference operator  $\Delta f(n) = f(n+1) - f(n)$ , the telescoping identity takes the compact form

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a).$$

*Remark* (Discrete Fundamental Theorem of Calculus). This identity is the discrete analogue of the Fundamental Theorem of Calculus:

$$\int_a^b f'(x) dx = f(b) - f(a).$$

It reveals a deep relationship between the difference operator and summation: summation is the inverse of differencing, just as integration is the inverse of differentiation. This connection is made precise in the next section through the concept of the *discrete antiderivative*.

## 5.6 Discrete Antiderivatives

### 5.6.1 Discrete Antiderivatives

The telescoping identity established in the previous section,

$$\sum_{k=a}^{b-1} \Delta f(k) = f(b) - f(a),$$



reveals that summation acts as an inverse operation to the difference operator, much like integration acts as an inverse to differentiation in ordinary calculus. This motivates the following definition.

**Definition 5.6.1** (Discrete Antiderivative). Let  $f : \mathbb{Z} \rightarrow \mathbb{R}$  be a discrete function. A function  $F : \mathbb{Z} \rightarrow \mathbb{R}$  is called a *discrete antiderivative* (or *discrete indefinite sum*) of  $f$  if

$$\Delta F(n) = f(n),$$

that is,  $F(n+1) - F(n) = f(n)$  for all  $n \in \mathbb{Z}$ .

*Remark* (Standard quantified statement).  $\forall f, F : \mathbb{Z} \rightarrow \mathbb{R} : F \text{ is a discrete antiderivative of } f \iff \forall n \in \mathbb{Z}, F(n+1) - F(n) = f(n).$  ■

*Remark* (Interpretation). A discrete antiderivative is a function whose one-step increments reproduce the given summand.

*Remark* (Analogy). A discrete antiderivative plays the same role in discrete calculus that an antiderivative plays in ordinary calculus: it reverses the fundamental differentiation-like operator of the theory. Instead of reversing differentiation, it reverses the forward difference  $\Delta$ .

*Remark* (Dependencies).

- [Forward Difference Operator](#).

## Examples

**Example 1.** Let  $f(n) = 1$ . Take  $F(n) = n$ . Then

$$\Delta F(n) = (n+1) - n = 1 = f(n). \checkmark$$

**Example 2.** Let  $f(n) = n$ . Take  $F(n) = \frac{n(n-1)}{2}$ . Then

$$\Delta F(n) = \frac{(n+1)n}{2} - \frac{n(n-1)}{2} = \frac{n(n+1-n+1)}{2} = n = f(n). \checkmark$$

*Remark* (Connection to falling factorials). Both examples above are instances of a general pattern: the discrete antiderivative of the falling factorial  $n^{\underline{k}}$  is  $\frac{1}{k+1}n^{\underline{k+1}}$ , exactly mirroring the antiderivative rule for ordinary powers. This is developed in the Polynomial Basis section.

## Constant of Integration

Discrete antiderivatives are not unique.

**Proposition 5.6.2** (Constant of integration). *If  $F$  is a discrete antiderivative of  $f$ , then so is  $F + C$  for any constant  $C \in \mathbb{R}$ .*

*Remark* (Fully quantified form).  $\forall C \in \mathbb{R} : \Delta F = f \Rightarrow \Delta(F + C) = f.$

*Remark* (Interpretation). Adding a constant changes the level of a discrete antiderivative but not its one-step increments.

*Remark* (Dependencies).

- [Discrete Antiderivative](#).
- [Forward Difference Operator](#).

*Proof*.

$$\Delta(F + C)(n) = (F + C)(n + 1) - (F + C)(n) = F(n + 1) - F(n) = \Delta F(n) = f(n). \quad \square$$

## Fundamental Theorem of Finite Calculus

### Theorem (Fundamental Theorem of Finite Calculus)

Let  $F$  be a discrete antiderivative of  $f$ , so that  $\Delta F(n) = f(n)$  for all  $n$ . Then

$$\sum_{k=a}^{b-1} f(k) = F(b) - F(a).$$

*Remark* (Fully quantified form).  $\forall f, F : \mathbb{Z} \rightarrow \mathbb{R}, \forall a, b \in \mathbb{Z}, a \leq b : (\forall n, \Delta F(n) = f(n)) \Rightarrow \sum_{k=a}^{b-1} f(k) = F(b) - F(a).$

*Proof*. Since  $f(k) = \Delta F(k)$  for all  $k$ , substituting gives

$$\sum_{k=a}^{b-1} f(k) = \sum_{k=a}^{b-1} \Delta F(k).$$

By the Telescoping Identity (Theorem [5.5.1](#)),

$$\sum_{k=a}^{b-1} \Delta F(k) = F(b) - F(a). \quad \square$$

*Remark* (Analogy with continuous calculus). This theorem is the discrete analogue of the Fundamental Theorem of Calculus,

$$\int_a^b f'(x) dx = f(b) - f(a).$$

In continuous calculus, differentiation and integration are inverse operations. In discrete calculus, the analogous inverse pair is the difference operator  $\Delta$  and summation. Any sum can be evaluated by finding a discrete antiderivative of the summand — just as any definite integral can be evaluated by finding an antiderivative.

*Remark* (Practical consequence). The theorem reduces the problem of computing  $\sum_{k=a}^{b-1} f(k)$  to the algebraic problem of finding  $F$  with  $\Delta F = f$ . Combined with the falling factorial rules developed in the next section, this gives a systematic method for evaluating a wide family of sums in closed form.

## 5.7 Polynomial Basis

### 5.7.1 Polynomial Basis: Falling Factorials

In ordinary calculus, polynomial functions are naturally expressed using the monomial powers  $1, x, x^2, x^3, \dots$  because the derivative acts simply on them:  $\frac{d}{dx}x^n = nx^{n-1}$ .

In discrete calculus, however, the difference operator does not behave as cleanly on ordinary powers. For example,

$$\Delta(n^2) = (n+1)^2 - n^2 = 2n + 1.$$

Although the degree decreases by one, the result contains an extra constant term. For this reason it is more natural to work with a different family of polynomials — the *falling factorials* — which interact with  $\Delta$  exactly as monomials interact with  $\frac{d}{dx}$ .

#### Definition (Falling Factorial)

For a nonnegative integer  $k$ , the *falling factorial* (or *falling power*) is defined by

$$x^{\underline{k}} = x(x-1)(x-2)\cdots(x-k+1) \quad (k \text{ factors}).$$

By convention,  $x^{\underline{0}} = 1$ .

*Remark* (Fully quantified form).  $\forall k \in \mathbb{N}, \forall x \in \mathbb{R} : x^{\underline{k}} := \prod_{j=0}^{k-1} (x-j)$ , with  $x^{\underline{0}} := 1$ .

*Remark* (Relation to factorials). When  $x = n \in \mathbb{N}$  and  $k \leq n$ , the falling factorial equals  $\frac{n!}{(n-k)!}$ . In particular,  $n^{\underline{n}} = n!$ . Thus falling factorials generalise factorials to arbitrary inputs.

The first few falling factorials are:

$$\begin{aligned} x^{\underline{0}} &= 1, \\ x^{\underline{1}} &= x, \\ x^{\underline{2}} &= x(x-1), \\ x^{\underline{3}} &= x(x-1)(x-2). \end{aligned}$$

### The Difference Rule for Falling Factorials

#### Theorem (Difference Rule for Falling Factorials)

For  $k \geq 1$ ,

$$\Delta x^{\underline{k}} = k x^{\underline{k-1}}.$$

*Remark* (Fully quantified form).  $\forall k \in \mathbb{N}, k \geq 1, \forall x \in \mathbb{R} : (x+1)^{\underline{k}} - x^{\underline{k}} = k x^{\underline{k-1}}$ .

*Proof.* Observe that

$$(x+1)^{\underline{k}} = (x+1)x(x-1)\cdots(x-k+2).$$

Therefore

$$\Delta x^{\underline{k}} = (x+1)^{\underline{k}} - x^{\underline{k}} = x(x-1)\cdots(x-k+2)[(x+1) - (x-k+1)].$$

The bracket simplifies:

$$(x+1) - (x-k+1) = k.$$

The remaining product  $x(x-1)\cdots(x-k+2)$  is exactly  $x^{\underline{k-1}}$ . Thus

$$\Delta x^{\underline{k}} = k x^{\underline{k-1}}. \quad \square$$

*Remark* (Mirror of the power rule). This formula exactly mirrors the continuous power rule  $\frac{d}{dx}x^k = kx^{k-1}$ . Because of this clean behaviour under  $\Delta$ , falling factorials form the natural polynomial basis for discrete calculus.

*Remark* (Antiderivative rule). The theorem also yields the discrete antiderivative rule:

$$\Delta\left(\frac{x^{\underline{k+1}}}{k+1}\right) = x^{\underline{k}}.$$

Thus  $\frac{x^{\underline{k+1}}}{k+1}$  is a discrete antiderivative of  $x^{\underline{k}}$ , mirroring the continuous rule  $\int x^k dx = \frac{x^{k+1}}{k+1}$ .

## Summary Table

| $f(x)$                                 | $\Delta f(x)$  |
|----------------------------------------|----------------|
| $x^{\underline{1}} = x$                | 1              |
| $x^{\underline{2}} = x(x-1)$           | $2x$           |
| $x^{\underline{3}} = x(x-1)(x-2)$      | $3x(x-1)$      |
| $x^{\underline{4}} = x(x-1)(x-2)(x-3)$ | $4x(x-1)(x-2)$ |

## Polynomial Representation in the Falling Factorial Basis

Every polynomial can be written as a linear combination of falling factorials:

$$f(x) = c_0 + c_1 x + c_2 x^{\underline{2}} + c_3 x^{\underline{3}} + \cdots + c_n x^{\underline{n}}.$$

*Remark* (Role in Newton's expansion). This representation plays the same role in discrete calculus that the Taylor expansion plays in continuous calculus. The coefficients  $c_k$  are determined by the successive finite differences of  $f$  evaluated at 0:  $c_k = \Delta^k f(0)/k!$ . This leads directly to Newton's forward difference expansion, developed in the Expansions section.

## Consequences of the Difference Rule

The difference rule  $\Delta x^{\underline{k}} = k x^{\underline{k-1}}$  (Theorem 5.7.1) has several important consequences.

## Higher Differences of Falling Factorials

Applying the rule repeatedly gives:

$$\Delta^m x^{\underline{k}} = \begin{cases} \frac{k!}{(k-m)!} x^{\underline{k-m}} & \text{if } m \leq k, \\ 0 & \text{if } m > k. \end{cases}$$

In particular, evaluating at  $x = 0$ :

$$\Delta^m x^{\underline{k}}|_{x=0} = \begin{cases} k! & \text{if } m = k, \\ 0 & \text{if } m \neq k. \end{cases}$$

*Remark* (Basis orthogonality). This vanishing property means the falling factorials behave like a “triangular” basis:  $\Delta^m$  kills all  $x^{\underline{k}}$  with  $k < m$ , and returns a scalar multiple for  $k = m$ . This is what makes the coefficient extraction in Newton’s expansion work cleanly.

## Connection to Binomial Coefficients

The higher difference formula from the Binomial Theorem section (Proposition 5.2.2),

$$\Delta^n f(x) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x+k),$$

together with the difference rule, gives an explicit formula for the coefficients of the falling factorial expansion:

$$c_k = \frac{\Delta^k f(0)}{k!}.$$

*Remark* (Pascal’s triangle revisited). The coefficients appearing in higher finite differences follow exactly the pattern of Pascal’s triangle — a direct consequence of the algebraic identity  $\Delta = E - I$  and the binomial theorem.

## Worked Example: Expanding $n^2$

We find the falling factorial expansion of  $f(n) = n^2$ .

Computing the difference table at  $n = 0$ :

$$f(0) = 0, \quad \Delta f(0) = 1, \quad \Delta^2 f(0) = 2, \quad \Delta^3 f(0) = 0.$$

The coefficients are  $c_k = \Delta^k f(0)/k!$ :

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{2}{2!} = 1, \quad c_3 = 0.$$

Therefore:

$$n^2 = 0 + n^{\underline{1}} + n^{\underline{2}} = n + n(n-1) = n^2 - n + n = n^2. \checkmark$$

*Remark* (Practical value). Once a function is expressed in the falling factorial basis, both differencing and antidifferencing become purely mechanical, following the rule  $\Delta x^{\underline{k}} = k x^{\underline{k-1}}$  and its inverse. This makes the Fundamental Theorem of Finite Calculus immediately applicable to compute sums like  $\sum_{k=0}^{n-1} k^2$  without clever tricks.

## 5.8 Calculus Rules

### 5.8.1 Calculus Rules

With the difference operator and the polynomial basis in place, we can now develop the full suite of differentiation-like rules for discrete calculus. The linearity of  $\Delta$  was established in the Difference Operators section (Proposition 5.4.3). Here we add the *product rule* — the discrete counterpart of the Leibniz rule  $(fg)' = f'g + fg'$ .

#### Product Rule

##### Theorem (Discrete Product Rule)

For functions  $f, g : \mathbb{Z} \rightarrow \mathbb{R}$ ,

$$\Delta(fg)(n) = f(n) \Delta g(n) + g(n+1) \Delta f(n).$$

*Remark* (Fully quantified form).  $\forall f, g : \mathbb{Z} \rightarrow \mathbb{R}, \forall n \in \mathbb{Z} : (f(n+1)g(n+1)) - (f(n)g(n)) = f(n)(g(n+1) - g(n)) + g(n+1)(f(n+1) - f(n)).$

*Proof.* By definition of  $\Delta$ ,

$$\Delta(fg)(n) = f(n+1)g(n+1) - f(n)g(n).$$

Add and subtract  $f(n)g(n+1)$ :

$$= f(n+1)g(n+1) - f(n)g(n+1) + f(n)g(n+1) - f(n)g(n).$$

Grouping the first two and last two terms:

$$= g(n+1)(f(n+1) - f(n)) + f(n)(g(n+1) - g(n)) = g(n+1) \Delta f(n) + f(n) \Delta g(n). \quad \square$$

*Remark* (Comparison with the continuous product rule). In continuous calculus,  $(fg)' = f'g + fg'$  is symmetric in  $f$  and  $g$ . In discrete calculus the rule is slightly asymmetric: one factor is evaluated at  $n$  and the other at  $n+1$ . This asymmetry is unavoidable and is a characteristic feature of the discrete setting.

*Remark* (Consequence — summation by parts). The product rule leads directly to the summation by parts formula: summing both sides of  $\Delta(fg) = f \Delta g + g(E) \Delta f$  over  $k = a$  to  $b-1$  and applying the Fundamental Theorem gives the formula in the next section.

#### Summation by Parts

Summation by parts is the discrete analogue of integration by parts. It is derived from the product rule exactly as the continuous version is derived from the Leibniz rule.

### Theorem (Summation by Parts)

For functions  $f, g : \mathbb{Z} \rightarrow \mathbb{R}$  and integers  $a \leq b$ ,

$$\sum_{k=a}^{b-1} f(k) \Delta g(k) = \left[ f(k)g(k) \right]_{k=a}^{k=b} - \sum_{k=a}^{b-1} g(k+1) \Delta f(k).$$

That is,

$$\sum_{k=a}^{b-1} f(k) \Delta g(k) = f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1) \Delta f(k).$$

*Remark* (Fully quantified form).  $\forall f, g : \mathbb{Z} \rightarrow \mathbb{R}, \forall a, b \in \mathbb{Z}, a \leq b : \sum_{k=a}^{b-1} f(k) \Delta g(k) =$

$$f(b)g(b) - f(a)g(a) - \sum_{k=a}^{b-1} g(k+1) \Delta f(k).$$

*Proof.* By the Discrete Product Rule (Theorem 5.8.1),

$$\Delta(fg)(k) = f(k) \Delta g(k) + g(k+1) \Delta f(k).$$

Rearranging,

$$f(k) \Delta g(k) = \Delta(fg)(k) - g(k+1) \Delta f(k).$$

Sum both sides from  $k = a$  to  $k = b - 1$ :

$$\sum_{k=a}^{b-1} f(k) \Delta g(k) = \sum_{k=a}^{b-1} \Delta(fg)(k) - \sum_{k=a}^{b-1} g(k+1) \Delta f(k).$$

Applying the Fundamental Theorem of Finite Calculus (Theorem 5.6.1) to the first sum on the right gives

$$\sum_{k=a}^{b-1} \Delta(fg)(k) = f(b)g(b) - f(a)g(a),$$

which completes the proof.  $\square$

*Remark* (Analogy with integration by parts). The continuous integration by parts formula,

$$\int_a^b f dg = [fg]_a^b - \int_a^b g df,$$

is recovered from summation by parts by replacing  $\sum$  with  $\int$  and  $\Delta$  with  $d$ . The asymmetry in the discrete formula —  $g(k+1)$  rather than  $g(k)$  in the remaining sum — reflects the shift inherent in the product rule.

*Remark* (Applications). Summation by parts is useful for evaluating sums involving products, particularly when one factor is easy to sum (as a falling factorial) and the other is easy to difference. Classic applications include evaluating  $\sum kr^k$  (where  $f(k) = k$ ,  $\Delta g(k) = r^k$  leads to a geometric-type sum) and establishing Abel's summation formula.

## 5.9 Expansions

### 5.9.1 Newton's Forward Difference Expansion

In continuous calculus, smooth functions can be expanded as power series called Taylor series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$$

The coefficients are determined by successive derivatives evaluated at 0, and the basis functions are ordinary powers  $x^k$ .

Discrete calculus has a precise analogue: the *Newton forward difference expansion*, in which derivatives are replaced by finite differences and ordinary powers are replaced by falling factorials.

#### Theorem (Newton Forward Difference Expansion)

Let  $f : \mathbb{Z} \rightarrow \mathbb{R}$ . Then  $f$  can be expressed as

$$f(x) = \sum_{k=0}^n \frac{\Delta^k f(0)}{k!} x^{\underline{k}} = f(0) + \Delta f(0)x + \frac{\Delta^2 f(0)}{2!}x^2 + \frac{\Delta^3 f(0)}{3!}x^3 + \dots,$$

where  $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$  is the falling factorial. For a polynomial of degree  $n$ , the expansion is exact (the series terminates at  $k = n$ ).

*Remark* (Fully quantified form).  $\forall f : \mathbb{Z} \rightarrow \mathbb{R}$  polynomial of degree  $\leq n$ ,  $\forall x \in \mathbb{Z} : f(x) = \sum_{k=0}^n \binom{x}{k} \Delta^k f(0)$ , where  $\binom{x}{k} := x^{\underline{k}}/k!$ .

### Structural Parallel with the Taylor Series

The Newton expansion mirrors the Taylor series at every point. The correspondence is exact:

| Taylor Series (continuous)      | Newton Expansion (discrete)         |
|---------------------------------|-------------------------------------|
| Coefficients $f^{(k)}(0)/k!$    | Coefficients $\Delta^k f(0)/k!$     |
| Basis functions $x^k$           | Basis functions $x^{\underline{k}}$ |
| Derivative $f^{(k)}$            | Difference $\Delta^k$               |
| Convergence required in general | Exact for polynomials               |

*Remark* (Why falling factorials play the role of powers). In the Taylor series, powers  $x^k$  are the natural basis because the derivative satisfies  $\frac{d}{dx}x^k = kx^{k-1}$  and  $\frac{d^m}{dx^m}x^k|_{x=0} = k!$  when  $m = k$ , and 0 otherwise. The falling factorials satisfy the same two properties under  $\Delta$ :  $\Delta x^{\underline{k}} = kx^{\underline{k-1}}$  and  $\Delta^m x^{\underline{k}}|_{x=0} = k!$  when  $m = k$ , else 0. This “triangular” property is exactly what allows the coefficients  $c_k = \Delta^k f(0)/k!$  to be read off independently, level by level.



## Proof Sketch

We construct the expansion term by term, matching the values of  $f$  at successive integers.

The constant term  $f(0)$  matches  $f$  at  $x = 0$ .

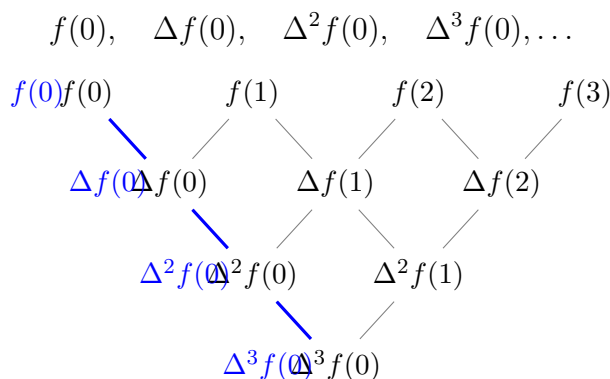
Adding the linear term  $\Delta f(0) \cdot x$  matches  $f$  at  $x = 0$  and  $x = 1$ .

Adding the quadratic term  $\frac{\Delta^2 f(0)}{2!} x^2$  corrects the value at  $x = 2$  while leaving earlier matches intact, because  $x^2 = x(x-1)$  vanishes at  $x = 0$  and  $x = 1$ .

At each stage,  $x^k$  vanishes at  $x = 0, 1, \dots, k-1$ , so the  $k$ -th term corrects only the value at  $x = k$ . The coefficient  $\Delta^k f(0)/k!$  achieves this correction by the triangular property noted above.

## Reading Coefficients from the Difference Table

The coefficients in the Newton expansion are read from the *left edge* of the difference table — the “Newton diagonal”:



Each level of the difference table contributes exactly one term to the falling-factorial expansion.

## Worked Example: Expanding $f(n) = n^3$

We compute the difference table at  $n = 0$ :

| $n$ | $f(n)$ | $\Delta f(n)$ | $\Delta^2 f(n)$ | $\Delta^3 f(n)$ |
|-----|--------|---------------|-----------------|-----------------|
| 0   | 0      | 1             | 6               | 6               |
| 1   | 1      | 7             | 12              |                 |
| 2   | 8      | 19            |                 |                 |
| 3   | 27     |               |                 |                 |

The coefficients are  $c_k = \Delta^k f(0)/k!$ :

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = \frac{6}{2} = 3, \quad c_3 = \frac{6}{6} = 1.$$

Therefore the Newton expansion is:

$$n^3 = n + 3n^2 + n^3 = n + 3n(n-1) + n(n-1)(n-2).$$

Verification:  $n + 3n^2 - 3n + n^3 - 3n^2 + 2n = n^3$ . ✓

## Application: Computing $\sum_{k=0}^{n-1} k^2$

Using the expansion  $k^2 = k^1 + k^2$  and the Fundamental Theorem of Finite Calculus (Theorem 5.6.1),

$$\sum_{k=0}^{n-1} k^2 = \sum_{k=0}^{n-1} k^1 + \sum_{k=0}^{n-1} k^2.$$

Since  $\Delta\left(\frac{k^2}{2}\right) = k^1$  and  $\Delta\left(\frac{k^3}{3}\right) = k^2$ , the Fundamental Theorem gives:

$$\sum_{k=0}^{n-1} k^1 = \frac{n^2}{2} - \frac{0^2}{2} = \frac{n(n-1)}{2},$$

$$\sum_{k=0}^{n-1} k^2 = \frac{n^3}{3} - \frac{0^3}{3} = \frac{n(n-1)(n-2)}{3}.$$

Adding:

$$\sum_{k=0}^{n-1} k^2 = \frac{n(n-1)}{2} + \frac{n(n-1)(n-2)}{3} = \frac{n(n-1)}{6} [3 + 2(n-2)] = \frac{n(n-1)(2n-1)}{6}.$$

*Remark* (Check against the standard formula). The standard formula is  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ .

The computed result  $\sum_{k=0}^{n-1} k^2 = \frac{(n-1)n(2n-1)}{6}$  matches: substitute  $n \mapsto n-1$  in the standard formula to get  $\frac{(n-1)n(2n-1)}{6}$ . ✓

This example illustrates the power of the method: the sum was evaluated by pure algebraic antidifferentiation, with no need for induction or clever manipulation.

## 5.10 Special Functions

### 5.10.1 Special Functions in Discrete Calculus

Just as continuous calculus has its distinguished special functions — the exponential  $e^{ax}$ , the trigonometric functions, and the logarithm — discrete calculus has analogous functions that behave simply under the difference operator.

### The Discrete Exponential

In continuous calculus, the exponential function  $e^{ax}$  is its own derivative (up to a constant):

$$\frac{d}{dx} e^{ax} = a e^{ax}.$$

The analogous function in discrete calculus is the *discrete exponential*  $(1+a)^n$ .

**Definition 5.10.1** (Discrete Exponential). For a constant  $a \neq -1$  and  $n \in \mathbb{Z}$ , the *discrete exponential* with base  $(1 + a)$  is the function

$$E_a(n) = (1 + a)^n.$$

*Remark* (Fully quantified form).  $\forall a \in \mathbb{R} \setminus \{-1\}, \forall n \in \mathbb{Z} : E_a(n) := (1 + a)^n$ .

*Remark* (Interpretation). The discrete exponential is the geometric function whose ratio between successive inputs is  $1 + a$ .

**Proposition 5.10.2** (Discrete exponential rule).

$$\Delta(1 + a)^n = a(1 + a)^n.$$

*Remark* (Fully quantified form).  $\forall a \in \mathbb{R} \setminus \{-1\}, \forall n \in \mathbb{Z} : (1 + a)^{n+1} - (1 + a)^n = a(1 + a)^n$ .

*Remark* (Interpretation). The forward difference of a discrete exponential is a scalar multiple of the same discrete exponential.

*Proof.*

$$\Delta(1 + a)^n = (1 + a)^{n+1} - (1 + a)^n = (1 + a)^n[(1 + a) - 1] = a(1 + a)^n. \quad \square$$

*Remark* (Analogy). The continuous exponential satisfies  $\frac{d}{dx}e^{ax} = ae^{ax}$ . The discrete exponential satisfies  $\Delta(1 + a)^n = a(1 + a)^n$ . The base  $e$  in continuous calculus is replaced by  $(1 + a)$  in discrete calculus. When  $a$  is small,  $(1 + a) \approx e^a$ , recovering the continuous limit.

*Remark* (Antiderivative). From the proposition, a discrete antiderivative of  $a(1 + a)^n$  is  $(1 + a)^n$  itself. Equivalently, a discrete antiderivative of  $(1 + a)^n$  is  $\frac{(1+a)^n}{a}$ . Combined with the Fundamental Theorem of Finite Calculus, this gives the geometric series formula:

$$\sum_{k=0}^{n-1} (1 + a)^k = \frac{(1 + a)^n - 1}{a} \quad (a \neq 0).$$

*Remark* (Dependencies).

- [Discrete Exponential](#).
- [Forward Difference Operator](#).

## 5.10.2 The Function $2^n$ in Discrete Calculus

Among all discrete exponentials, the case  $a = 1$  — giving  $(1 + 1)^n = 2^n$  — is perhaps the most important. It appears in combinatorics as the count of subsets, in probability as the sample space of fair coin sequences, in computer science as the natural measure of binary information, and in discrete calculus as a canonical test function. This section develops its properties systematically.

## Differencing $2^n$

Setting  $a = 1$  in the discrete exponential rule (Proposition 5.10.2) gives immediately:

**Proposition** ( $\Delta 2^n = 2^n$ )

$$\Delta 2^n = 2^n.$$

*Remark* (Standard quantified statement).  $\forall n \in \mathbb{Z} : 2^{n+1} - 2^n = 2^n$ .

*Proof.*

$$\Delta 2^n = 2^{n+1} - 2^n = 2 \cdot 2^n - 2^n = (2 - 1) \cdot 2^n = 2^n. \quad \square$$

*Remark* (Self-similarity). This says that  $2^n$  is its own forward difference. In continuous calculus, the analogous property is  $\frac{d}{dx}e^x = e^x$ : the exponential is its own derivative. The function  $2^n$  is the discrete analogue of  $e^x$  in the sense that  $\Delta(2^n) = 2^n$ . More precisely,  $2^n = e^{n \ln 2}$ , and the relationship  $\Delta(2^n) = 2^n$  corresponds to the special case  $a = 1$  of  $\Delta(1 + a)^n = a(1 + a)^n$ .

## Antiderivative and Geometric Sum

Since  $\Delta(2^n) = 2^n$ , a discrete antiderivative of  $2^n$  is  $2^n$  itself (up to a constant).

**Proposition** (Antiderivative of  $2^n$ )

A discrete antiderivative of  $2^n$  is  $2^n$  itself:

$$\Delta(2^n) = 2^n \quad \text{so} \quad \sum_{k=0}^{n-1} 2^k = 2^n - 1.$$

*Remark* (Standard quantified statement).  $\forall n \in \mathbb{N} : \sum_{k=0}^{n-1} 2^k = 2^n - 1$ .

*Proof.* Applying the Fundamental Theorem of Finite Calculus (Theorem 5.6.1) with  $F(k) = 2^k$  and  $f(k) = \Delta F(k) = 2^k$ :

$$\sum_{k=0}^{n-1} 2^k = F(n) - F(0) = 2^n - 2^0 = 2^n - 1. \quad \square$$

*Remark* (Verification by expansion). Expanding:  $1 + 2 + 4 + 8 + \cdots + 2^{n-1}$ . Multiply the partial sum  $S = \sum_{k=0}^{n-1} 2^k$  by 2 to get  $2S = \sum_{k=1}^n 2^k$ . Subtracting:  $2S - S = 2^n - 1$ , so  $S = 2^n - 1$ . This is the classical derivation of the geometric series formula; the discrete calculus approach via antidifferentiation gives the same result mechanically.

## Higher Differences of $2^n$

**Proposition 5.10.3** (Higher differences of  $2^n$ ). *For all  $k \geq 0$ ,*

$$\Delta^k 2^n = 2^n.$$

*Remark* (Standard quantified statement).  $\forall k \in \mathbb{N}, \forall n \in \mathbb{Z} : \Delta^k(2^n) = 2^n$ .

*Remark* (Interpretation). Every forward-difference row for  $2^n$  reproduces the same function.

*Proof.* By induction. The base case  $k = 0$  is  $\Delta^0(2^n) = 2^n$  by definition. For the inductive step, assume  $\Delta^k(2^n) = 2^n$ . Then

$$\Delta^{k+1}(2^n) = \Delta(\Delta^k(2^n)) = \Delta(2^n) = 2^n. \quad \square$$

*Remark* (Consequence for difference tables). Since every finite difference of  $2^n$  is  $2^n$  again, the difference table of  $2^n$  consists entirely of the sequence  $2^n$  at every level. This is the discrete analogue of the fact that the Taylor series of  $e^x$  has the same coefficient pattern at every derivative level. In particular,  $2^n$  is *not* a polynomial — a polynomial of degree  $d$  would reach 0 at the  $(d + 1)$ -th difference row. The non-termination of the difference table is the signature of a genuinely transcendental discrete function.

*Remark* (Dependencies).

- [Higher Differences](#).
- [Discrete Exponential Rule](#).

## $2^n$ in Combinatorics and the Newton Expansion

The Binomial Theorem (Theorem 5.2.1) gives a direct connection: setting  $x = y = 1$  yields

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

This counts the total number of subsets of an  $n$ -element set: there are  $\binom{n}{k}$  subsets of size  $k$ , and the sum over all  $k$  gives  $2^n$ .

More deeply, the Newton expansion (Theorem 5.9.1) of  $2^n$  in the falling factorial basis is:

$$2^n = \sum_{k=0}^n \binom{n}{k} = \sum_{k=0}^n \frac{n^{\underline{k}}}{k!},$$

since  $\Delta^k(2^n)|_{n=0} = 2^0 = 1$  for every  $k$ . This expresses  $2^n$  as the sum of all normalised falling factorials up to degree  $n$ , and reveals why binomial coefficients count subsets: they are the coefficients of  $2^n$  in its own Newton expansion.

*Remark* (Analogy with  $e^x$ ). In continuous calculus,  $e^x = \sum_{k=0}^{\infty} x^k/k!$ . The Newton expansion gives  $2^n = \sum_{k=0}^n n^{\underline{k}}/k!$ . Both express a canonical exponential as a sum of normalised basis functions, with all coefficients equal to 1. The analogy is exact: ordinary powers replace falling factorials, and the infinite series terminates at degree  $n$  in the discrete case because  $2^n$  has degree exactly  $n$  in the falling factorial basis.

## Discrete Differential Equation

In continuous calculus, the equation  $f' = f$  (with  $f(0) = 1$ ) has the unique solution  $f(x) = e^x$ . In discrete calculus, the analogous *difference equation* is:

$$\Delta F(n) = F(n), \quad F(0) = 1.$$

That is,  $F(n+1) - F(n) = F(n)$ , so  $F(n+1) = 2F(n)$ . The unique solution to this initial value problem is  $F(n) = 2^n$ .

**Proposition 5.10.4** (Discrete IVP). *The unique function  $F : \mathbb{N} \rightarrow \mathbb{R}$  satisfying*

$$F(n+1) = 2F(n), \quad F(0) = 1$$

*is  $F(n) = 2^n$ .*

*Remark* (Standard quantified statement).  $\forall n \in \mathbb{N} : (F(n+1) = 2F(n) \text{ and } F(0) = 1) \Rightarrow F(n) = 2^n$ .

*Remark* (Interpretation). The recurrence  $F(n+1) = 2F(n)$  with initial value 1 determines the discrete exponential  $2^n$  uniquely.

*Proof.* Uniqueness: if  $G$  is another solution, then  $G(0) = 1 = F(0)$ , and inductively  $G(n+1) = 2G(n) = 2F(n) = F(n+1)$ , so  $G = F$ . Existence: verify directly that  $2^{n+1} = 2 \cdot 2^n$  and  $2^0 = 1$ .  $\square$

*Remark* (Significance). This characterises  $2^n$  as the unique fixed point of “multiply by 2” starting from 1. The analogous characterisation of  $e^x$  is: unique solution to  $f' = f$ ,  $f(0) = 1$ . Both are initial-value problems in their respective calculi, and both have closed-form solutions that serve as the canonical exponential of that calculus.

*Remark* (Dependencies).

- [Discrete Exponential](#).

## Difference Formulas for Special Functions

The table below collects the difference formulas for the principal special functions of discrete calculus.

| Function                                                                   | Forward Difference      |
|----------------------------------------------------------------------------|-------------------------|
| $x^{\underline{k}}$ (falling factorial)                                    | $k x^{\underline{k-1}}$ |
| $(1+a)^n$ (discrete exponential)                                           | $a(1+a)^n$              |
| <i>Remark</i> (Difference formulas). $\binom{x}{k} = x^{\underline{k}}/k!$ |                         |
| $c$ (constant)                                                             | 0                       |
| $n$                                                                        | 1                       |
| $n^2 = n(n-1)$                                                             | $2n$                    |

*Remark* (Binomial coefficient as a special case). The formula  $\Delta \binom{x}{k} = \binom{x}{k-1}$  follows immediately from the falling factorial rule  $\Delta x^{\underline{k}} = k x^{\underline{k-1}}$  by dividing through by  $k!$ . It says that the binomial coefficient  $\binom{x}{k}$  is its own natural “unit” in discrete calculus, analogous to the monomial  $x^k/k!$  in the Taylor expansion.

## Antiderivative Table

By reversing the difference formulas, we obtain the corresponding discrete antiderivatives:

| Function $f(n)$                                                    | Antiderivative $F(n)$ with $\Delta F = f$ |
|--------------------------------------------------------------------|-------------------------------------------|
| $x^{\underline{k}}$                                                | $\frac{x^{\underline{k+1}}}{k+1}$         |
| <i>Remark</i> (Discrete antiderivatives). $(1+a)^n$ ( $a \neq 0$ ) | $\frac{(1+a)^n}{a}$                       |
| $\binom{x}{k}$                                                     | $\binom{x}{k+1}$                          |
| $c$ (constant)                                                     | $cn$                                      |

*Remark* (Using the Fundamental Theorem). Each antiderivative in the table, combined with the Fundamental Theorem of Finite Calculus (Theorem 5.6.1), yields an immediate summation formula. For instance, from  $\Delta \binom{x^{\underline{k+1}}}{k+1} = x^{\underline{k}}$ :

$$\sum_{j=0}^{n-1} j^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}.$$

This is the discrete analogue of  $\int_0^n x^k dx = \frac{n^{k+1}}{k+1}$ .

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Binomial Theorem). *For any integer  $n \geq 0$  and real numbers  $x, y$ ,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

*Professional Standard Proof.*

TODO: professional standard proof for binomial. □

*Detailed Learning Proof.*

TODO: detailed learning proof for binomial. □

*Remark* (Proof structure).



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Higher difference formula). *For any function  $f$ ,*

$$\Delta^n f(x) = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f(x+k).$$

*Professional Standard Proof.*

TODO: professional standard proof for higher-diff-formula. □

*Detailed Learning Proof.*

TODO: detailed learning proof for higher-diff-formula. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Binomial Theorem](#).
- [Forward Difference Operator](#).
- [Higher Differences](#).

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Linearity of  $\Delta$ ). *The forward difference operator is linear.*

*Professional Standard Proof.*

TODO: professional standard proof for delta-linearity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for delta-linearity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Forward Difference Operator](#).

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Constant rule). *The forward difference of a constant sequence is zero.*

*Professional Standard Proof.*

TODO: professional standard proof for delta-constant. □

*Detailed Learning Proof.*

TODO: detailed learning proof for delta-constant. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Forward Difference Operator](#).

*Remark* (Return). [Return to Theorem](#)

**Theorem** ( $\Delta = E - I$ ). *The forward difference operator is the shift operator minus the identity operator.*

*Professional Standard Proof.*

TODO: professional standard proof for delta-shift-minus-identity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for delta-shift-minus-identity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Forward Difference Operator](#).
- [Shift Operator](#).

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Polynomial detection). *Finite differences detect polynomial degree.*

*Professional Standard Proof.*

TODO: professional standard proof for polynomial-detection. □

*Detailed Learning Proof.*

TODO: detailed learning proof for polynomial-detection. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Forward Difference Operator](#).
- [Higher Differences](#).
- [Difference Table](#).
- [Binomial Theorem](#).

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Constant of integration). *Discrete antiderivatives are unique up to an additive constant.*

*Professional Standard Proof.*

TODO: professional standard proof for constant-of-integration. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-of-integration. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Discrete Antiderivative.](#)
- [Forward Difference Operator.](#)

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Discrete exponential rule). *The discrete exponential has geometric forward differences.*

*Professional Standard Proof.*

TODO: professional standard proof for discrete-exponential-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for discrete-exponential-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Discrete Exponential](#).
- [Forward Difference Operator](#).

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Higher differences of  $2^n$ ). *All higher forward differences of  $2^n$  are again  $2^n$ .*

*Professional Standard Proof.*

TODO: professional standard proof for higher-diff-two-n. □

*Detailed Learning Proof.*

TODO: detailed learning proof for higher-diff-two-n. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Higher Differences](#).
- [Discrete Exponential Rule](#).



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Discrete IVP). *A first-order discrete initial value problem has the stated solution.*

*Professional Standard Proof.*

TODO: professional standard proof for discrete-ivp. □

*Detailed Learning Proof.*

TODO: detailed learning proof for discrete-ivp. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Discrete Exponential](#).

# Chapter 6

## Sequences



### 6.1 Sequence Definitions

#### 6.1.1 Definitions

##### Definition (Sequence)

**Definition 6.1.1** (Sequence). Let  $X$  be a nonempty set. A **sequence** in  $X$  is a function

$$x : \mathbb{N} \rightarrow X.$$

The value  $x(n)$  is written  $x_n$  and called the  **$n$ -th term** of the sequence. The sequence itself is denoted  $(x_n)_{n \in \mathbb{N}}$ , or simply  $(x_n)$  when the index set is clear.

*Remark* (Standard quantified statement).

$$\forall X \neq \emptyset \forall x : \mathbb{N} \rightarrow X \left( \text{Sequence}(x, X) \iff x : \mathbb{N} \rightarrow X \right).$$

*Remark* (Definition predicate reading).

$$\text{Sequence}(x_n, X) := x : \mathbb{N} \rightarrow X.$$

*Remark* (Interpretation). A sequence is not a set of terms but a function: the index carries independent weight. Two sequences  $(x_n)$  and  $(y_n)$  in  $X$  are equal precisely when  $x_n = y_n$  for every  $n \in \mathbb{N}$ , so order and repetition are intrinsic to the object. The notational collapse from  $x(n)$  to  $x_n$  is conventional suppression of function-application syntax; the underlying object remains a function  $\mathbb{N} \rightarrow X$ . When  $X = \mathbb{R}$ , the sequence is called a **real sequence**, which is the default ambient context for this chapter.

*Remark* (Dependencies).

- [Function](#)
- [Natural Numbers](#)

#### Definition (Real Sequence)

**Definition 6.1.2** (Real Sequence). A **real sequence** is a sequence whose image is contained in  $\mathbb{R}$ . Equivalently, a real sequence is a function

$$x : \mathbb{N} \rightarrow \mathbb{R}.$$

*Remark* (Standard quantified statement).

$$\text{RealSequence}(x_n) \iff x : \mathbb{N} \rightarrow \mathbb{R}.$$

*Remark* (Definition predicate reading).

$$\text{RealSequence}(x_n) := \text{Sequence}(x_n, \mathbb{R}).$$

*Remark* (Interpretation). A real sequence is not a new kind of indexing object; it is the specialization of a sequence to the ambient space  $\mathbb{R}$ . Thus the generic function structure is inherited from sequences, while order, absolute value, and arithmetic enter through the real codomain.

*Remark* (Dependencies).

- [Sequence](#)
- [Real Numbers](#)

*Example* (Constant Sequence). Let  $c \in \mathbb{R}$ . The **constant sequence** is defined by

$$x_n := c \quad \forall n \in \mathbb{N}.$$

The constant sequence converges to  $c$ : given any  $\varepsilon > 0$ , take  $N = 1$ , and observe  $|x_n - c| = 0 < \varepsilon$  for all  $n \geq 1$ . It is the degenerate case of convergence in which the tail is not merely eventually close to  $L$  but identically equal to it from the first term.

*Example* (The Sequence  $(1/n)$ ). The sequence  $(x_n)$  defined by

$$x_n := \frac{1}{n} \quad \forall n \in \mathbb{N}$$

converges to 0. Given  $\varepsilon > 0$ , the Archimedean property supplies  $N \in \mathbb{N}$  with  $N > 1/\varepsilon$ , and then for all  $n \geq N$ ,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This is the canonical example of a sequence that approaches its limit monotonically from above, never reaching it. It also serves as the standard witness in squeeze arguments and in the proof that  $\inf\{1/n : n \in \mathbb{N}\} = 0$ .

*Example* (The Sequence  $(n)$ ). The sequence  $(x_n)$  defined by

$$x_n := n \quad \forall n \in \mathbb{N}$$

is divergent. For any proposed limit  $L \in \mathbb{R}$ , take  $\varepsilon = 1$ ; the Archimedean property supplies  $n \in \mathbb{N}$  with  $n > L + 1$ , so  $|x_n - L| = |n - L| > 1 = \varepsilon$  for infinitely many  $n$ . The sequence is unbounded, and every convergent sequence is bounded, so divergence also follows from that theorem once it is established. This is the canonical example of divergence to  $+\infty$ : the terms are not oscillating outside a neighborhood of  $L$  but growing without bound in a single direction.

## 6.1.2 Null and Constant Sequences

### Null and Constant Sequences

#### Definition (Constant Sequence)

**Definition 6.1.3** (Constant Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **constant** exactly when there exists  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \in \mathbb{N}$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &(x_n) \text{ is constant} \iff \exists c \in \mathbb{R} \, \forall n \in \mathbb{N} \, (x_n = c). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{ConstantSequence}(x_n, c) := \forall n \in \mathbb{N} \, (x_n = c).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &(x_n) \text{ is not constant} \iff \forall c \in \mathbb{R} \, \exists n \in \mathbb{N} \, (x_n \neq c). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \exists c \in \mathbb{R} \, \text{ConstantSequence}(x_n, c).$$

*Remark* (Failure modes). Every proposed constant value fails at some index.

*Remark* (Failure mode decomposition).

$$\forall c \in \mathbb{R} \quad \underbrace{\exists n \in \mathbb{N} \, (x_n \neq c)}_{\text{a term differs from the proposed constant value}}.$$

*Remark* (Interpretation). A constant sequence has no genuine dependence on the index. The indexing is still part of the object, but every index is assigned the same real value.

*Remark* (Dependencies).

- Real Sequence

### Definition (Null Sequence)

**Definition 6.1.4** (Null Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is a **null sequence** exactly when for every  $\varepsilon > 0$  there exists  $N \in \mathbb{N}$  such that  $|x_n| < \varepsilon$  for every  $n \geq N$ .

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is a null sequence  $\iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon)$ .

*Remark* (Definition predicate reading).

$\text{NullSequence}(x_n) := \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon)$ .

*Remark* (Negated quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is not a null sequence  $\iff \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n| \geq \varepsilon)$ .

*Remark* (Negation predicate reading).

$\neg \text{NullSequence}(x_n)$ .

*Remark* (Failure modes). Some positive tolerance is missed infinitely far out in the sequence.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon > 0 \forall N \in \mathbb{N} \underbrace{\exists n \geq N (|x_n| \geq \varepsilon)}_{\text{the tail is not trapped inside the } \varepsilon\text{-band}}.$$

*Remark* (Interpretation). A null sequence is a sequence whose tail is eventually contained in every symmetric neighborhood of zero. The first finitely many terms are irrelevant; only the eventual behavior matters.

*Remark* (Dependencies).

- Real Sequence

### Theorem (Constant Sequence Convergence)

**Theorem 6.1.5** (Constant Sequence Convergence). Let  $(x_n)$  be a real sequence, and let  $c \in \mathbb{R}$ . If  $x_n = c$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  converges to  $c$ .

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and  $c \in \mathbb{R}$ .

$(\forall n \in \mathbb{N} (x_n = c)) \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - c| < \varepsilon)$ .

*Remark* (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{ConvergentSequence}(x_n, c).$$

*Remark* (Interpretation). A constant sequence is already equal to its proposed limit at every index. Every positive tolerance is therefore satisfied by any tail of the sequence.

*Remark* (Dependencies).

- [Constant Sequence](#)
- [Convergent Sequence](#)

#### Theorem (Zero Sequence is Null)

**Theorem 6.1.6** (Zero Sequence is Null). *Let  $(x_n)$  be a real sequence. If  $x_n = 0$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  is a null sequence.*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$(\forall n \in \mathbb{N} (x_n = 0)) \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon).$$

*Remark* (Predicate reading).

$$\text{ConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

*Remark* (Interpretation). The zero sequence satisfies every null-sequence tolerance immediately because the distance from each term to zero is exactly zero.

*Remark* (Dependencies).

- [Null Sequence](#)
- [Constant Sequence](#)

#### Theorem (Constant Null Sequence)

**Theorem 6.1.7** (Constant Null Sequence). *Let  $(x_n)$  be a real sequence, and let  $c \in \mathbb{R}$ . If  $x_n = c$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  is a null sequence if and only if  $c = 0$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and  $c \in \mathbb{R}$ .

$$(\forall n \in \mathbb{N} (x_n = c)) \implies \left( [\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon)] \iff c = 0 \right).$$

*Remark* (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies (\text{NullSequence}(x_n) \iff c = 0).$$

*Remark* (Interpretation). A sequence that never changes can approach zero only when its fixed value is already zero. Conversely, the fixed value zero gives the zero sequence.

*Remark* (Dependencies).

- [Constant Sequence](#)
- [Null Sequence](#)

#### Definition (Ultimately Constant Sequence)

**Definition 6.1.8** (Ultimately Constant Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **ultimately constant** exactly when there exist  $N \in \mathbb{N}$  and  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \geq N$ .

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$(x_n) \text{ is ultimately constant} \iff \exists N \in \mathbb{N} \exists c \in \mathbb{R} \forall n \geq N (x_n = c).$$

*Remark* (Definition predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) := \exists N \in \mathbb{N} \forall n \geq N (x_n = c).$$

*Remark* (Negated quantified statement).

Let  $(x_n)$  be a real sequence.

$$(x_n) \text{ is not ultimately constant} \iff \forall N \in \mathbb{N} \forall c \in \mathbb{R} \exists n \geq N (x_n \neq c).$$

*Remark* (Negation predicate reading).

$$\neg \exists c \in \mathbb{R} \text{ UltimatelyConstantSequence}(x_n, c).$$

*Remark* (Failure modes). No tail of the sequence becomes permanently equal to a single real value.

*Remark* (Failure mode decomposition).

$$\forall N \in \mathbb{N} \forall c \in \mathbb{R} \underbrace{\exists n \geq N (x_n \neq c)}_{\text{the proposed tail value fails later in the sequence}}.$$

*Remark* (Interpretation). An ultimately constant sequence may vary on an initial finite segment, but its tail becomes fixed. The concept isolates eventual behavior from finite initial behavior.

*Remark* (Dependencies).

- Real Sequence

### Theorem (Difference from the Limit is Null)

**Theorem 6.1.9** (Difference from the Limit is Null). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . The sequence  $(x_n)$  converges to  $L$  if and only if the sequence  $(x_n - L)$  is a null sequence.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence and } L \in \mathbb{R}. \\ &[\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon)] \\ &\iff [\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon)]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \iff \text{NullSequence}(x_n - L).$$

*Remark* (Interpretation). Subtracting the proposed limit translates the convergence problem to the special case of convergence to zero. Null sequences are therefore the local normal form for sequence convergence.

*Remark* (Dependencies).

- Convergent Sequence
- Null Sequence
- Pointwise Difference of Sequences

### Theorem (Ultimately Constant Sequence Convergence)

**Theorem 6.1.10** (Ultimately Constant Sequence Convergence). *Let  $(x_n)$  be a real sequence. If there exist  $N_0 \in \mathbb{N}$  and  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \geq N_0$ , then  $(x_n)$  converges to  $c$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\forall c \in \mathbb{R} \left( [\exists N_0 \in \mathbb{N} \forall n \geq N_0 (x_n = c)] \right. \\ &\quad \implies [\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - c| < \varepsilon)] \Big). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies \text{ConvergentSequence}(x_n, c).$$



*Remark* (Interpretation). Once the tail is equal to a fixed value, all sufficiently late terms are already exactly at the proposed limit.

*Remark* (Dependencies).

- [Ultimately Constant Sequence](#)
- [Convergent Sequence](#)

### Theorem (Constant Implies Ultimately Constant)

**Theorem 6.1.11** (Constant Implies Ultimately Constant). *Every constant real sequence is ultimately constant.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &[\exists c \in \mathbb{R} \forall n \in \mathbb{N} (x_n = c)] \\ &\implies [\exists N \in \mathbb{N} \exists c \in \mathbb{R} \forall n \geq N (x_n = c)]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{UltimatelyConstantSequence}(x_n, c).$$

*Remark* (Interpretation). A constant sequence has already stabilized from the first index, so it is automatically stabilized on some tail.

*Remark* (Dependencies).

- [Constant Sequence](#)
- [Ultimately Constant Sequence](#)

### Theorem (Ultimately Zero Sequence is Null)

**Theorem 6.1.12** (Ultimately Zero Sequence is Null). *Let  $(x_n)$  be a real sequence. If there exists  $N_0 \in \mathbb{N}$  such that  $x_n = 0$  for every  $n \geq N_0$ , then  $(x_n)$  is a null sequence.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &[\exists N_0 \in \mathbb{N} \forall n \geq N_0 (x_n = 0)] \\ &\implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

*Remark* (Interpretation). If the tail is identically zero, then every sufficiently late term lies inside every positive tolerance around zero.

*Remark* (Dependencies).

- [Ultimately Constant Sequence](#)
- [Null Sequence](#)

#### Theorem (Ultimately Constant Null Sequence)

**Theorem 6.1.13** (Ultimately Constant Null Sequence). *Let  $(x_n)$  be a real sequence. Suppose there exist  $N_0 \in \mathbb{N}$  and  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \geq N_0$ . Then  $(x_n)$  is a null sequence if and only if  $c = 0$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\forall c \in \mathbb{R} \left( \left[ \exists N_0 \in \mathbb{N} \forall n \geq N_0 (x_n = c) \right] \right. \\ &\quad \left. \implies \left( \left[ \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n| < \varepsilon) \right] \iff c = 0 \right) \right). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies (\text{NullSequence}(x_n) \iff c = 0).$$

*Remark* (Interpretation). Only the eventual value controls whether an ultimately constant sequence is null. A finite initial segment cannot prevent or create null behavior.

*Remark* (Dependencies).

- [Ultimately Constant Sequence](#)
- [Null Sequence](#)

#### Theorem (Tail Equality Preserves Convergence)

**Theorem 6.1.14** (Tail Equality Preserves Convergence). *Let  $(x_n)$  and  $(y_n)$  be real sequences. If there exists  $N_0 \in \mathbb{N}$  such that  $x_n = y_n$  for every  $n \geq N_0$ , then for every  $L \in \mathbb{R}$ ,  $(x_n)$  converges to  $L$  if and only if  $(y_n)$  converges to  $L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.} \\ &\left[ \exists N_0 \in \mathbb{N} \forall n \geq N_0 (x_n = y_n) \right] \\ &\implies \forall L \in \mathbb{R} \left( \left[ \forall \varepsilon > 0 \exists N_x \in \mathbb{N} \forall n \geq N_x (|x_n - L| < \varepsilon) \right] \right. \\ &\quad \left. \iff \left[ \forall \varepsilon > 0 \exists N_y \in \mathbb{N} \forall n \geq N_y (|y_n - L| < \varepsilon) \right] \right). \end{aligned}$$

*Remark* (Predicate reading).

$\text{EventuallyEqual}(x_n, y_n) \implies \forall L \in \mathbb{R} \left( \text{ConvergentSequence}(x_n, L) \iff \text{ConvergentSequence}(y_n, L) \right).$  ■

*Remark* (Interpretation). Convergence is a tail property. Changing finitely many initial terms cannot change whether a sequence converges to a specified real number.

*Remark* (Dependencies).

- [Convergent Sequence](#)

#### Definition (Sequence Bounded Above)

**Definition 6.1.15** (Sequence Bounded Above). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **bounded above** exactly when there exists  $M \in \mathbb{R}$  such that  $x_n \leq M$  for every  $n \in \mathbb{N}$ .

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is bounded above  $\iff \exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)$ .

*Remark* (Definition predicate reading).

$\text{BoundedAboveSequence}(x_n) := \exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)$ .

*Remark* (Negated quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is not bounded above  $\iff \forall M \in \mathbb{R} \exists n \in \mathbb{N} (M < x_n)$ .

*Remark* (Negation predicate reading).

$\neg \text{BoundedAboveSequence}(x_n)$ .

*Remark* (Failure modes). Every proposed upper bound is exceeded by some term of the sequence.

*Remark* (Failure mode decomposition).

$$\forall M \in \mathbb{R} \quad \underbrace{\exists n \in \mathbb{N} (M < x_n)}_{\text{a term rises above the proposed ceiling}}.$$

*Remark* (Interpretation). A bounded-above sequence has a finite ceiling. Failure means no real number serves as a ceiling for all terms.

*Remark* (Dependencies).

- [Real Sequence](#)

### Definition (Sequence Bounded Below)

**Definition 6.1.16** (Sequence Bounded Below). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **bounded below** exactly when there exists  $m \in \mathbb{R}$  such that  $m \leq x_n$  for every  $n \in \mathbb{N}$ .

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is bounded below  $\iff \exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n)$ .

*Remark* (Definition predicate reading).

$\text{BoundedBelowSequence}(x_n) := \exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n)$ .

*Remark* (Negated quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is not bounded below  $\iff \forall m \in \mathbb{R} \exists n \in \mathbb{N} (x_n < m)$ .

*Remark* (Negation predicate reading).

$\neg \text{BoundedBelowSequence}(x_n)$ .

*Remark* (Failure modes). Every proposed lower bound is undercut by some term of the sequence.

*Remark* (Failure mode decomposition).

$$\forall m \in \mathbb{R} \quad \underbrace{\exists n \in \mathbb{N} (x_n < m)}_{\text{a term falls below the proposed floor}}.$$

*Remark* (Interpretation). A bounded-below sequence has a finite floor. Failure means no real number serves as a floor for all terms.

*Remark* (Dependencies).

- [Real Sequence](#)

### Definition (Bounded Sequence)

**Definition 6.1.17** (Bounded Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **bounded** exactly when there exists  $M > 0$  such that  $|x_n| \leq M$  for every  $n \in \mathbb{N}$ .

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$(x_n)$  is bounded  $\iff \exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M)$ .

*Remark* (Definition predicate reading).

$$\text{BoundedSequence}(x_n) := \exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M).$$

*Remark* (Negated quantified statement).

Let  $(x_n)$  be a real sequence.

$$(x_n) \text{ is not bounded} \iff \forall M > 0 \exists n \in \mathbb{N} (|x_n| > M).$$

*Remark* (Negation predicate reading).

$$\neg \text{BoundedSequence}(x_n).$$

*Remark* (Failure modes). Every symmetric bound around zero is escaped by some term of the sequence.

*Remark* (Failure mode decomposition).

$$\forall M > 0 \quad \underbrace{\exists n \in \mathbb{N} (|x_n| > M)}_{\text{a term escapes the symmetric band of radius } M}.$$

*Remark* (Interpretation). A bounded sequence is trapped in a finite symmetric band around zero. This is equivalent to having both a finite ceiling and a finite floor.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Eventually Bounded Above Tail Formulation)

**Theorem 6.1.18** (Eventually Bounded Above Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded above if and only if there exist  $N_0 \in \mathbb{N}$  and  $M \in \mathbb{R}$  such that  $x_n \leq M$  for every  $n \geq N_0$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$\begin{aligned} & [\exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)] \\ \iff & [\exists N_0 \in \mathbb{N} \exists M \in \mathbb{R} \forall n \geq N_0 (x_n \leq M)]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{BoundedAboveSequence}(x_n) \iff \exists M \in \mathbb{R} \text{ Eventually}(x_n, x_n \leq M).$$

*Remark* (Interpretation). One-sided boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a larger upper bound.

Remark (Dependencies).

- Sequence Bounded Above

#### Theorem (Eventually Bounded Below Tail Formulation)

**Theorem 6.1.19** (Eventually Bounded Below Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded below if and only if there exist  $N_0 \in \mathbb{N}$  and  $m \in \mathbb{R}$  such that  $m \leq x_n$  for every  $n \geq N_0$ .*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &[\exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n)] \\ &\iff [\exists N_0 \in \mathbb{N} \exists m \in \mathbb{R} \forall n \geq N_0 (m \leq x_n)]. \end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedBelowSequence}(x_n) \iff \exists m \in \mathbb{R} \text{ Eventually}(x_n, m \leq x_n).$$

Remark (Interpretation). One-sided lower boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a lower bound.

Remark (Dependencies).

- Sequence Bounded Below

#### Theorem (Eventually Bounded Tail Formulation)

**Theorem 6.1.20** (Eventually Bounded Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded if and only if there exist  $N_0 \in \mathbb{N}$  and  $M > 0$  such that  $|x_n| \leq M$  for every  $n \geq N_0$ .*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &[\exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M)] \\ &\iff [\exists N_0 \in \mathbb{N} \exists M > 0 \forall n \geq N_0 (|x_n| \leq M)]. \end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff \exists M > 0 \text{ Eventually}(x_n, |x_n| \leq M).$$

Remark (Interpretation). Boundedness is a tail property up to finitely many exceptions. A finite initial segment can be enclosed by increasing the bound.

Remark (Dependencies).

- Bounded Sequence

**Theorem (Bounded Sequence If and Only If Bounded Above and Below)**

**Theorem 6.1.21** (Bounded Sequence If and Only If Bounded Above and Below). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded if and only if it is both bounded above and bounded below.*

*Go to proof.*

Remark (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$\begin{aligned} & [\exists K > 0 \forall n \in \mathbb{N} (|x_n| \leq K)] \\ & \iff [\exists m \in \mathbb{R} \exists M \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n \wedge x_n \leq M)]. \end{aligned}$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff (\text{BoundedBelowSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n)).$$

Remark (Interpretation). An absolute-value bound gives a symmetric interval around zero. Having both a floor and a ceiling gives an interval that may not be symmetric, but it can be enlarged to a symmetric one.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

**Theorem (Absolute Bound Gives Upper and Lower Bounds)**

**Theorem 6.1.22** (Absolute Bound Gives Upper and Lower Bounds). *Let  $(x_n)$  be a real sequence, and let  $K > 0$ . If  $|x_n| \leq K$  for every  $n \in \mathbb{N}$ , then  $-K \leq x_n \leq K$  for every  $n \in \mathbb{N}$ .*

*Go to proof.*

Remark (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$\begin{aligned} & \forall K > 0 [\forall n \in \mathbb{N} (|x_n| \leq K)] \\ & \implies \forall n \in \mathbb{N} (-K \leq x_n \wedge x_n \leq K). \end{aligned}$$

Remark (Interpretation). An absolute-value inequality simultaneously controls the positive and negative directions. The upper barrier is  $K$ , and the lower barrier is  $-K$ .

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

### Theorem (Upper and Lower Bounds Give an Absolute Bound)

**Theorem 6.1.23** (Upper and Lower Bounds Give an Absolute Bound). *Let  $(x_n)$  be a real sequence. If there exist  $m, M \in \mathbb{R}$  such that  $m \leq x_n \leq M$  for every  $n \in \mathbb{N}$ , then there exists  $K > 0$  such that  $|x_n| \leq K$  for every  $n \in \mathbb{N}$ .*

*Go to proof.*

Remark (Standard quantified statement).

Let  $(x_n)$  be a real sequence.

$$\begin{aligned} & [\exists m \in \mathbb{R} \exists M \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n \wedge x_n \leq M)] \\ & \implies [\exists K > 0 \forall n \in \mathbb{N} (|x_n| \leq K)]. \end{aligned}$$

Remark (Interpretation). Two one-sided barriers can always be replaced by one symmetric barrier around zero. Any positive  $K$  at least as large as both  $|m|$  and  $|M|$  works.

Remark (Dependencies).

- Sequence Bounded Above
- Sequence Bounded Below
- Bounded Sequence

## 6.2 Examples and Counterexamples

### Sequence Examples — Quick Reference

| Example                      | Role                                      |
|------------------------------|-------------------------------------------|
| Constant sequence            | Basic convergent model                    |
| Reciprocal sequence          | Canonical null sequence                   |
| Alternating null sequence    | Sign oscillation with convergence to 0    |
| Oscillating sequence         | Bounded nonconvergent model               |
| Geometric sequence           | Ratio-driven convergence and divergence   |
| Bounded not convergent       | Boundedness alone is insufficient         |
| Small differences not Cauchy | Vanishing increments do not ensure Cauchy |



## Examples and Counterexamples

Examples provide test objects for the definitions. A good sequence example usually isolates one behavior: settling to a limit, oscillating between several values, escaping to infinity, or satisfying a tempting condition that is still too weak to force convergence.

*Example* (Constant Sequence). Let  $c \in \mathbb{R}$  and define  $x_n = c$  for every  $n \in \mathbb{N}$ . Then  $(x_n)$  converges to  $c$ .

*Remark* (Interpretation). A constant sequence is the baseline example of convergence: every term is already equal to the proposed limit.

*Example* (Reciprocal Sequence). Define  $x_n = 1/n$  for every  $n \in \mathbb{N}$ . Then  $(x_n)$  is decreasing, bounded below by 0, and converges to 0.

*Remark* (Interpretation). The reciprocal sequence is the standard positive null sequence. It is also the model for estimates of the form “eventually smaller than any fixed positive tolerance.”

*Example* (Alternating Null Sequence). Define

$$x_n = \frac{(-1)^n}{n} \quad \text{for every } n \in \mathbb{N}.$$

Then  $(x_n)$  is not monotone, but it converges to 0.

*Remark* (Interpretation). Oscillation in sign does not by itself prevent convergence. The amplitudes must be watched: here the oscillation collapses to zero.

*Example* (Oscillating Sequence). Define

$$x_n = (-1)^n \quad \text{for every } n \in \mathbb{N}.$$

Then  $(x_n)$  is bounded and divergent. Its even-indexed subsequence converges to 1, and its odd-indexed subsequence converges to  $-1$ .

*Remark* (Interpretation). This is the canonical oscillator. The sequence never settles because two subsequences settle to different limits.

*Example* (Geometric Sequence). Let  $r \in \mathbb{R}$  and define  $x_n = r^n$  for every  $n \in \mathbb{N}$ . Then:

- if  $|r| < 1$ , then  $x_n \rightarrow 0$ ;
- if  $r = 1$ , then  $x_n \rightarrow 1$ ;
- if  $r = -1$ , then  $(x_n)$  oscillates and diverges;
- if  $|r| > 1$ , then  $(x_n)$  is unbounded.

*Remark* (Interpretation). The geometric sequence is the standard laboratory for rate, sign, and magnitude. The threshold  $|r| = 1$  separates decay from growth.

*Example* (Bounded Does Not Imply Convergent). The sequence  $x_n = (-1)^n$  is bounded, since  $|x_n| \leq 1$  for every  $n \in \mathbb{N}$ , but it does not converge.

*Remark* (Interpretation). Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

*Example* (Vanishing Successive Differences Do Not Imply Cauchy). Let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} \quad \text{for every } n \in \mathbb{N}.$$

Then

$$x_{n+1} - x_n = \frac{1}{n+1} \rightarrow 0,$$

but  $(x_n)$  is not Cauchy.

*Remark* (Interpretation). Cauchy control requires all sufficiently late terms to be mutually close. Controlling only adjacent jumps is weaker.

## 6.3 Convergence

### Definition (Convergence of a Sequence)

**Definition 6.3.1** (Convergence of a Sequence). Let  $(x_n) \subseteq \mathbb{R}$ . The sequence  $(x_n)$  **converges** to  $L \in \mathbb{R}$  if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon).$$

The value  $L$  is called the **limit** of  $(x_n)$ , written  $x_n \rightarrow L$  or  $\lim_{n \rightarrow \infty} x_n = L$ . A sequence that converges to some  $L \in \mathbb{R}$  is called **convergent**; otherwise it is called **divergent**.

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \\ & \left( x_n \rightarrow L \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) := \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N \text{OutputTolerance}(x_n, L, \varepsilon).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} x_n \not\rightarrow L & \iff \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N \\ & (|x_n - L| \geq \varepsilon). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) := \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| \geq \varepsilon).$$

*Remark* (Failure modes). Failure of convergence to a fixed  $L$  means that some positive tolerance cannot be controlled on any tail. The escaped terms may oscillate, drift monotonically away, or grow without bound, but all such behavior is captured by the same persistent-tolerance obstruction.

*Remark* (Failure mode decomposition). The negation decomposes into a single structural obstruction:

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed tolerance}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{|x_n - L| \geq \varepsilon}_{\text{outside the target interval}}.$$

The sequence fails to converge to  $L$  not by drifting away once, but by returning outside the  $\varepsilon$ -interval infinitely often regardless of how far along the tail one looks.

*Remark* (Interpretation). The quantifier order  $\forall \varepsilon \exists N \forall n \geq N$  encodes a strict dependency:  $N$  is permitted to depend on  $\varepsilon$ , but not on  $n$ . This asymmetry is load-bearing. Reversing the outer quantifiers to  $\exists N \forall \varepsilon$  would assert that a single index  $N$  controls all tolerances simultaneously, which is a strictly stronger and generally false condition. The condition  $|x_n - L| < \varepsilon$  is the membership assertion  $x_n \in (L - \varepsilon, L + \varepsilon)$ ; convergence is therefore the statement that the tail of the sequence is eventually contained in every  $\varepsilon$ -neighborhood of  $L$ .

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Convergence — Neighborhood Formulation)

**Definition 6.3.2** (Convergence — Neighborhood Formulation). Let  $(x_n) \subseteq \mathbb{R}$  and let  $L \in \mathbb{R}$ . The sequence  $(x_n)$  **converges** to  $L$  if for every  $\varepsilon > 0$  the  $\varepsilon$ -neighborhood

$$V_\varepsilon(L) := (L - \varepsilon, L + \varepsilon)$$

contains all but finitely many terms of  $(x_n)$ ; that is,

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (x_n \in V_\varepsilon(L)).$$

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \\ & \left( x_n \rightarrow L \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (x_n \in V_\varepsilon(L)) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) := \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (x_n \in V_\varepsilon(L)).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} x_n \not\rightarrow L & \iff \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N \\ & (x_n \notin V_\varepsilon(L)). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) := \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (x_n \notin V_\varepsilon(L)).$$

*Remark* (Failure modes). Failure of the neighborhood formulation is exactly failure to trap a tail in every neighborhood of  $L$ . Different examples may escape by oscillation, by monotone drift, or by unbounded growth, but the formal failure is always the existence of one neighborhood that every tail misses at least once.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed neighborhood}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{x_n \notin V_\varepsilon(L)}_{\text{outside every candidate neighborhood}} .$$

The sequence fails to converge to  $L$  precisely when some neighborhood of  $L$  contains only finitely many terms: infinitely many terms lie permanently outside it.

*Remark* (Interpretation). The neighborhood formulation reframes convergence as a set-containment condition rather than an inequality condition. The assertion  $x_n \in V_\varepsilon(L)$  is identical to  $|x_n - L| < \varepsilon$ ; the two definitions are logically equivalent and the choice between them is one of language, not content. The neighborhood language is the natural bridge to the topological formulation: in a general topological space, open sets replace  $\varepsilon$ -neighborhoods and the definition reads identically with  $V_\varepsilon(L)$  replaced by any open set  $U$  containing  $L$ . The phrase *all but finitely many terms* is the canonical prose compression of the quantifier block  $\exists N \in \mathbb{N} \forall n \geq N$ , and is standard in the literature.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [ε-neighbourhood](#)

### Theorem (Equivalence of Convergence Formulations)

**Theorem 6.3.3** (Equivalence of Convergence Formulations). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $L \in \mathbb{R}$ . The following statements are equivalent.*

- (a)  $x_n \rightarrow L$ .
- (b)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (|x_n - L| < \varepsilon)$ .
- (c)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (L - \varepsilon < x_n < L + \varepsilon)$ .
- (d)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (x_n \in V_\varepsilon(L))$ .

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \left( (a) \iff (b) \iff (c) \iff (d) \right).$$

*Remark* (Theorem predicate reading).

$\text{EquivalentConvergenceForms}(x_n, L) := \text{ConvergentSequence}(x_n, L) \iff \text{NeighborhoodConvergentSequence}(x_n, L)$

*Remark* (Interpretation). The four conditions are the same geometric fact written in four notational registers. Condition (a) is the informal assertion. Condition (b) is the canonical  $\varepsilon$ - $K$  definition: the absolute value formulation. Condition (c) makes the interval structure explicit by expanding  $|x_n - L| < \varepsilon$  into its equivalent double inequality  $L - \varepsilon < x_n < L + \varepsilon$ ; this is useful in proofs where upper and lower bounds are handled separately. Condition (d) is the neighborhood formulation:  $x_n \in V_\varepsilon(L)$  is identical to conditions (b) and (c) but frames convergence as set membership, which is the natural language for the topological generalization. The index  $K$  plays the same role as  $N$  in the base definition [Definition 6.3.1](#); the letter changes here to signal that  $K$  is the tail offset in the sense of [Definition 6.4.1](#).

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Convergence — Neighborhood Formulation](#)

### 6.3.1 Limits

#### Theorem (Uniqueness of Limits)

**Theorem 6.3.4** (Uniqueness of Limits). *Let  $(x_n) \subseteq \mathbb{R}$ . If  $x_n \rightarrow L$  and  $x_n \rightarrow K$ , then  $L = K$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \, \forall K \in \mathbb{R} \\ \left( (x_n \rightarrow L \wedge x_n \rightarrow K) \implies L = K \right).$$

*Remark* (Theorem predicate reading).

$$\text{UniqueSequentialLimit}(x_n, L, K) := (\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K)) \implies L = K.$$

*Remark* (Negated quantified statement).

$$\exists (x_n) \subseteq \mathbb{R} \, \exists L \in \mathbb{R} \, \exists K \in \mathbb{R} \\ \left( (x_n \rightarrow L \wedge x_n \rightarrow K) \wedge L \neq K \right).$$

*Remark* (Negation predicate reading).

$$\neg \text{UniqueSequentialLimit}(x_n, L, K) := \text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K) \wedge L \neq K.$$

*Remark* (Failure modes). The only failure mode is the existence of one real sequence with two distinct real limits. In  $\mathbb{R}$  this is impossible because distinct real numbers can be separated by disjoint neighborhoods.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists(x_n) \subseteq \mathbb{R}}_{\text{some sequence}} \underbrace{x_n \rightarrow L \wedge x_n \rightarrow K}_{\text{converges to two values}} \underbrace{L \neq K}_{\text{distinct limits}} .$$

The theorem asserts this configuration is impossible in  $\mathbb{R}$ . The failure mode is not merely exotic: it occurs in non-Hausdorff topological spaces where distinct points cannot be separated by disjoint open sets. The proof that it cannot occur in  $\mathbb{R}$  is precisely an argument that  $\mathbb{R}$  has enough room to separate any two distinct points by disjoint  $\varepsilon$ -neighborhoods.

*Remark* (Interpretation). The theorem licenses the notation  $\lim_{n \rightarrow \infty} x_n = L$ : the definite article is justified precisely because the limit, when it exists, is unique. Without uniqueness the limit would be a relation rather than a function, and the notation would be ambiguous. The proof strategy is a separation argument: if  $L \neq K$  then  $|L - K| > 0$ , and choosing  $\varepsilon = |L - K|/2$  produces two disjoint neighborhoods that the tail of  $(x_n)$  cannot simultaneously occupy. The Hausdorff remark in the failure mode block is the forward pointer: when sequences are studied in general topological spaces, uniqueness of limits is equivalent to the Hausdorff separation axiom.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Triangle inequality](#)

#### Theorem (Limit Preserves Eventual Order)

**Theorem 6.3.5** (Limit Preserves Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . Suppose that  $x_n \rightarrow L$  and  $y_n \rightarrow M$ . If there exists  $N_0 \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N_0$ , then  $L \leq M$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall(x_n), (y_n) \subseteq \mathbb{R} \forall L, M \in \mathbb{R} \\ & \left( x_n \rightarrow L \wedge y_n \rightarrow M \wedge \exists N_0 \in \mathbb{N} \forall n \geq N_0 (x_n \leq y_n) \right) \\ & \implies L \leq M. \end{aligned}$$

*Remark* (Theorem predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M) \wedge \text{Eventually}(x_n, x_n \leq y_n)) \implies L \leq M. \blacksquare$$

*Remark* (Interpretation). An inequality that holds on a tail survives passage to the limit, but only as a weak inequality. Strict inequalities on the terms may collapse at the limit, as in  $0 < x_n$  with  $x_n \rightarrow 0$ .

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Uniqueness of Limits](#)

#### Theorem (Strict Limit Separation Gives Eventual Order)

**Theorem 6.3.6** (Strict Limit Separation Gives Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$  and  $y_n \rightarrow B$ . If  $A < B$ , then there exists  $N \in \mathbb{N}$  such that  $x_n < y_n$  for every  $n \geq N$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n), (y_n) \subseteq \mathbb{R} \forall A, B \in \mathbb{R} \\ \left( x_n \rightarrow A \wedge y_n \rightarrow B \wedge A < B \right) \\ \implies \exists N \in \mathbb{N} \forall n \geq N (x_n < y_n). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge A < B) \implies \text{Eventually}(x_n, x_n < y_n). \blacksquare$$

*Remark* (Interpretation). If the limiting values have a positive gap, then sufficiently late terms inherit the same order. The proof uses the open interval between  $A$  and  $B$ : eventually  $(x_n)$  lies near  $A$  and  $(y_n)$  lies near  $B$ , with room left between the two tails.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Uniqueness of Limits](#)

#### Theorem (Eventual Strict Comparison Preserves Weak Limit Order)

**Theorem 6.3.7** (Eventual Strict Comparison Preserves Weak Limit Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$  and  $y_n \rightarrow B$ .*

(a) *If there exists  $N \in \mathbb{N}$  such that  $x_n < y_n$  for every  $n \geq N$ , then  $A \leq B$ .*

(b) *If there exists  $N \in \mathbb{N}$  such that  $x_n > y_n$  for every  $n \geq N$ , then  $A \geq B$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n), (y_n) \subseteq \mathbb{R} \forall A, B \in \mathbb{R} \\ \left( x_n \rightarrow A \wedge y_n \rightarrow B \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n < y_n) \right) \implies A \leq B, \\ \left( x_n \rightarrow A \wedge y_n \rightarrow B \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n > y_n) \right) \implies A \geq B. \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned} &(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n, x_n < y_n)) \implies A \leq B, \\ &(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n, x_n > y_n)) \implies A \geq B. \end{aligned}$$

*Remark* (Interpretation). Strict comparison of terms does not force strict comparison of limits. The strict inequality can collapse in the limit, so the conclusion is weak: for example  $1/n > 0$  for every  $n \in \mathbb{N}$ , but both sides have limit 0.

*Remark* (Dependencies).

- [Limit Preserves Eventual Order](#)

#### Theorem (Constant Comparison for Sequence Limits)

**Theorem 6.3.8** (Constant Comparison for Sequence Limits). *Let  $(x_n)$  be a real sequence, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$ .*

- (a) *If there exists  $N \in \mathbb{N}$  such that  $x_n \leq B$  for every  $n \geq N$ , then  $A \leq B$ .*
- (b) *If there exists  $N \in \mathbb{N}$  such that  $x_n < B$  for every  $n \geq N$ , then  $A \leq B$ .*
- (c) *If there exists  $N \in \mathbb{N}$  such that  $x_n \geq B$  for every  $n \geq N$ , then  $A \geq B$ .*
- (d) *If there exists  $N \in \mathbb{N}$  such that  $x_n > B$  for every  $n \geq N$ , then  $A \geq B$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R} \forall A, B \in \mathbb{R} \\ &\quad \left( x_n \rightarrow A \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n \leq B) \right) \implies A \leq B, \\ &\quad \left( x_n \rightarrow A \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n < B) \right) \implies A \leq B, \\ &\quad \left( x_n \rightarrow A \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n \geq B) \right) \implies A \geq B, \\ &\quad \left( x_n \rightarrow A \wedge \exists N \in \mathbb{N} \forall n \geq N (x_n > B) \right) \implies A \geq B. \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned} &\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n, x_n \leq B) \implies A \leq B, \\ &\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n, x_n < B) \implies A \leq B, \\ &\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n, x_n \geq B) \implies A \geq B, \\ &\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n, x_n > B) \implies A \geq B. \end{aligned}$$

*Remark* (Interpretation). This is the constant version of the comparison theorem. A fixed real number  $B$  may be treated as the constant sequence with value  $B$ , so each conclusion is a direct application of eventual order preservation. Again, strict term inequalities produce only weak inequalities between limits.



Remark (Dependencies).

- Constant Sequence
- Limit Preserves Eventual Order
- Eventual Strict Comparison Preserves Weak Limit Order

### Theorem (Constant Squeeze Theorem)

**Theorem 6.3.9** (Constant Squeeze Theorem). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If there exists  $N_0 \in \mathbb{N}$  such that  $L \leq x_n \leq L$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \\ \left( \exists N_0 \in \mathbb{N} \forall n \geq N_0 (L \leq x_n \leq L) \right) \\ \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon). \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n, L \leq x_n \leq L) \implies \text{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). This is the degenerate squeeze theorem: the lower and upper barriers are the same constant sequence. It says that eventual equality to a real number is enough for convergence to that real number.

Remark (Dependencies).

- Convergence of a Sequence
- Constant Sequence

### Theorem (Sequence Squeeze Theorem)

**Theorem 6.3.10** (Sequence Squeeze Theorem). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be real sequences, and let  $L \in \mathbb{R}$ . Suppose that  $a_n \rightarrow L$  and  $b_n \rightarrow L$ . If there exists  $N_0 \in \mathbb{N}$  such that  $a_n \leq x_n \leq b_n$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \forall (a_n), (x_n), (b_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \\ \left( a_n \rightarrow L \wedge b_n \rightarrow L \wedge \exists N_0 \in \mathbb{N} \forall n \geq N_0 (a_n \leq x_n \leq b_n) \right) \\ \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon). \end{aligned}$$

*Remark* (Theorem predicate reading).

$(\text{ConvergentSequence}(a_n, L) \wedge \text{ConvergentSequence}(b_n, L) \wedge \text{Eventually}(x_n, a_n \leq x_n \leq b_n)) \implies \text{ConvergentSequence}(x_n, L)$

*Remark* (Interpretation). The middle sequence is trapped between two tails that approach the same number. Once both barriers lie inside an  $\varepsilon$ -neighborhood of  $L$ , the trapped sequence must also lie inside that neighborhood.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Limit Preserves Eventual Order](#)

#### Theorem (Absolute-Value Squeeze Theorem)

**Theorem 6.3.11** (Absolute-Value Squeeze Theorem). *Let  $(x_n)$  and  $(u_n)$  be real sequences, and let  $L \in \mathbb{R}$ . Suppose that  $u_n \rightarrow 0$ . If there exists  $N_0 \in \mathbb{N}$  such that  $|x_n - L| \leq u_n$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n), (u_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \\ & \left( u_n \rightarrow 0 \wedge \exists N_0 \in \mathbb{N} \, \forall n \geq N_0 \, (|x_n - L| \leq u_n) \right) \\ & \implies \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, (|x_n - L| < \varepsilon). \end{aligned}$$

*Remark* (Theorem predicate reading).

$(\text{ConvergentSequence}(u_n, 0) \wedge \text{Eventually}(x_n, |x_n - L| \leq u_n)) \implies \text{ConvergentSequence}(x_n, L).$  ■

*Remark* (Interpretation). This is the estimate form of the squeeze theorem. Instead of constructing two explicit bounding sequences, it bounds the error  $|x_n - L|$  by a null sequence.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Null Sequence](#)
- [Sequence Squeeze Theorem](#)

## 6.3.2 Algebra of Limits

### Algebra of Convergent Sequences

The algebra of convergent sequences records that limits are compatible with the ordinary algebraic operations on real numbers. The pointwise operations below turn real sequences

into new real sequences, and the theorem blocks that follow state how convergence passes through those operations. The reciprocal and quotient laws require nonzero denominator hypotheses, while the square-root law requires nonnegative terms.

### Definition (Pointwise Sum of Sequences)

**Definition 6.3.12** (Pointwise Sum of Sequences). Let  $(x_n)$  and  $(y_n)$  be real sequences. The **pointwise sum** of  $(x_n)$  and  $(y_n)$  is the real sequence  $(x_n + y_n)$  defined by

$$(x + y)_n := x_n + y_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  and  $(y_n)$  be real sequences.

$$\forall n \in \mathbb{N} \ ((x + y)_n = x_n + y_n).$$

*Remark* (Interpretation). The sum sequence is formed term by term. No limiting process is involved in the definition; convergence enters only when the behavior of the resulting sequence is studied.

*Remark* (Dependencies).

- [Real Sequence](#)

### Definition (Pointwise Difference of Sequences)

**Definition 6.3.13** (Pointwise Difference of Sequences). Let  $(x_n)$  and  $(y_n)$  be real sequences. The **pointwise difference** of  $(x_n)$  and  $(y_n)$  is the real sequence  $(x_n - y_n)$  defined by

$$(x - y)_n := x_n - y_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  and  $(y_n)$  be real sequences.

$$\forall n \in \mathbb{N} \ ((x - y)_n = x_n - y_n).$$

*Remark* (Interpretation). The difference sequence subtracts corresponding terms. It is equivalently the sum of  $(x_n)$  with the pointwise negation of  $(y_n)$ .

*Remark* (Dependencies).

- [Real Sequence](#)
- [Pointwise Sum of Sequences](#)

### Definition (Scalar Multiple of a Sequence)

**Definition 6.3.14** (Scalar Multiple of a Sequence). Let  $(x_n)$  be a real sequence, and let  $\alpha \in \mathbb{R}$ . The **scalar multiple**  $\alpha(x_n)$  is the real sequence  $(\alpha x_n)$  defined by

$$(\alpha x)_n := \alpha x_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and  $\alpha \in \mathbb{R}$ .  
 $\forall n \in \mathbb{N} \ ((\alpha x)_n = \alpha x_n)$ .

*Remark* (Interpretation). Scalar multiplication rescales every term by the same real number. It changes the vertical scale of the sequence but not its indexing.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Linear Combination of Real Sequences)

**Definition 6.3.15** (Linear Combination of Real Sequences). Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $\alpha, \beta \in \mathbb{R}$ . The real linear combination  $\alpha(x_n) + \beta(y_n)$  is the real sequence  $(\alpha x_n + \beta y_n)$  defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n), (y_n)$  be real sequences and  $\alpha, \beta \in \mathbb{R}$ .  
 $\forall n \in \mathbb{N} \ ((\alpha x + \beta y)_n = \alpha x_n + \beta y_n)$ .

*Remark* (Interpretation). A linear combination of sequences is built term by term from scalar multiples and sums. It is the common algebraic shape behind sequence limit rules and Cauchy closure under addition and scalar multiplication.

*Remark* (Dependencies).

- [Scalar Multiple of a Sequence](#)
- [Pointwise Sum of Sequences](#)

#### Definition (Pointwise Negation of a Sequence)

**Definition 6.3.16** (Pointwise Negation of a Sequence). Let  $(x_n)$  be a real sequence. The **pointwise negation** of  $(x_n)$  is the real sequence  $(-x_n)$  defined by

$$(-x)_n := -x_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.  
 $\forall n \in \mathbb{N} \ ((-x)_n = -x_n)$ .

*Remark* (Interpretation). Pointwise negation reflects the sequence across zero. It is the special scalar multiple corresponding to the scalar  $-1$ .

*Remark* (Dependencies).

- [Scalar Multiple of a Sequence](#)

#### Definition (Pointwise Product of Sequences)

**Definition 6.3.17** (Pointwise Product of Sequences). Let  $(x_n)$  and  $(y_n)$  be real sequences. The **pointwise product** of  $(x_n)$  and  $(y_n)$  is the real sequence  $(x_n y_n)$  defined by

$$(xy)_n := x_n y_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  and  $(y_n)$  be real sequences.  
 $\forall n \in \mathbb{N} \left( (xy)_n = x_n y_n \right).$

*Remark* (Interpretation). The product sequence multiplies corresponding terms. The product law for limits is subtler than the sum law because one factor must be controlled while the other is approximated.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Reciprocal Sequence)

**Definition 6.3.18** (Reciprocal Sequence). Let  $(x_n)$  be a real sequence such that  $x_n \neq 0$  for every  $n \in \mathbb{N}$ . The **reciprocal sequence** of  $(x_n)$  is the real sequence  $(1/x_n)$  defined by

$$\left( \frac{1}{x} \right)_n := \frac{1}{x_n} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence with  $\forall n \in \mathbb{N} (x_n \neq 0)$ .  
 $\forall n \in \mathbb{N} \left( \left( \frac{1}{x} \right)_n = \frac{1}{x_n} \right).$

*Remark* (Interpretation). The reciprocal sequence is defined only when division by each term is legal. For limit theorems, a nonzero limit guarantees eventual nonzero behavior, but the sequence-level construction itself requires nonzero terms wherever it is defined.

*Remark* (Dependencies).

- [Real Sequence](#)

### Definition (Pointwise Quotient of Sequences)

**Definition 6.3.19** (Pointwise Quotient of Sequences). Let  $(x_n)$  and  $(y_n)$  be real sequences such that  $y_n \neq 0$  for every  $n \in \mathbb{N}$ . The **pointwise quotient** of  $(x_n)$  by  $(y_n)$  is the real sequence  $(x_n/y_n)$  defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N} (y_n \neq 0). \\ &\forall n \in \mathbb{N} \left( \left(\frac{x}{y}\right)_n = \frac{x_n}{y_n} \right). \end{aligned}$$

*Remark* (Interpretation). The quotient sequence is the product of  $(x_n)$  with the reciprocal of  $(y_n)$ . Its convergence law is therefore a product law plus a reciprocal law.

*Remark* (Dependencies).

- [Pointwise Product of Sequences](#)
- [Reciprocal Sequence](#)

### Definition (Square Sequence)

**Definition 6.3.20** (Square Sequence). Let  $(x_n)$  be a real sequence. The **square sequence** of  $(x_n)$  is the real sequence  $(x_n^2)$  defined by

$$(x^2)_n := x_n^2 \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n) \text{ be a real sequence.} \\ &\forall n \in \mathbb{N} ((x^2)_n = x_n^2). \end{aligned}$$

*Remark* (Interpretation). The square sequence is the product of a sequence with itself. Its limit law is therefore a direct specialization of the product law.

*Remark* (Dependencies).

- [Pointwise Product of Sequences](#)

### Definition (Absolute Value Sequence)

**Definition 6.3.21** (Absolute Value Sequence). Let  $(x_n)$  be a real sequence. The **absolute value sequence** of  $(x_n)$  is the real sequence  $(|x_n|)$  defined by

$$(|x|)_n := |x_n| \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence.  
 $\forall n \in \mathbb{N} \left( (|x|)_n = |x_n| \right).$

*Remark* (Interpretation). The absolute value sequence records the magnitude of each term. It removes sign information while preserving distance from zero.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Square Root Sequence)

**Definition 6.3.22** (Square Root Sequence). Let  $(x_n)$  be a real sequence such that  $0 \leq x_n$  for every  $n \in \mathbb{N}$ . The **square root sequence** of  $(x_n)$  is the real sequence  $(\sqrt{x_n})$  defined by

$$(\sqrt{x})_n := \sqrt{x_n} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence with  $\forall n \in \mathbb{N} (0 \leq x_n)$ .  
 $\forall n \in \mathbb{N} \left( (\sqrt{x})_n = \sqrt{x_n} \right).$

*Remark* (Interpretation). The square root sequence is defined term by term on the non-negative real terms. The nonnegativity hypothesis is part of the construction, not a proof artifact.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Limit of a Scalar Multiple)

**Theorem 6.3.23** (Limit of a Scalar Multiple). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $\alpha x_n \rightarrow \alpha L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and let  $L, \alpha \in \mathbb{R}$ .  

$$\left[ \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon) \right]$$

$$\implies \left[ \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|\alpha x_n - \alpha L| < \varepsilon) \right].$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\alpha x_n, \alpha L).$$

*Remark* (Interpretation). Multiplying every term by a fixed scalar multiplies the limiting value by the same scalar. The case  $\alpha = 0$  collapses the sequence to the constant zero sequence.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Scalar Multiple of a Sequence](#)

#### Theorem (Limit of a Sum)

**Theorem 6.3.24** (Limit of a Sum). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n + y_n \rightarrow L + M$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}. \\ [x_n \rightarrow L \wedge y_n \rightarrow M] \implies [x_n + y_n \rightarrow L + M].$$

*Remark* (Predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)) \implies \text{ConvergentSequence}(x_n + y_n, L + M). \blacksquare$$

*Remark* (Interpretation). The sum law says that convergent sequences may be added before or after taking limits. The proof splits a target tolerance into two smaller tolerances, one for each summand.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Pointwise Sum of Sequences](#)
- [Triangle Inequality](#)

#### Theorem (Limit of a Negation)

**Theorem 6.3.25** (Limit of a Negation). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $-x_n \rightarrow -L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}. \\ [x_n \rightarrow L] \implies [-x_n \rightarrow -L].$$



*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(-x_n, -L).$$

*Remark* (Interpretation). Negation is scalar multiplication by  $-1$ , so this theorem is the scalar multiple law in its most common special case.

*Remark* (Dependencies).

- [Limit of a Scalar Multiple](#)
- [Pointwise Negation of a Sequence](#)

#### Theorem (Limit of a Difference)

**Theorem 6.3.26** (Limit of a Difference). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n - y_n \rightarrow L - M$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}. \\ &[x_n \rightarrow L \wedge y_n \rightarrow M] \implies [x_n - y_n \rightarrow L - M]. \end{aligned}$$

*Remark* (Predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)) \implies \text{ConvergentSequence}(x_n - y_n, L - M). \blacksquare$$

*Remark* (Interpretation). The difference law follows by adding  $(x_n)$  to the negation of  $(y_n)$ . It is therefore not a new analytic mechanism, but a useful named rule.

*Remark* (Dependencies).

- [Limit of a Sum](#)
- [Limit of a Negation](#)
- [Pointwise Difference of Sequences](#)

#### Theorem (Limit of a Product)

**Theorem 6.3.27** (Limit of a Product). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n y_n \rightarrow LM$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}. \\ &[x_n \rightarrow L \wedge y_n \rightarrow M] \implies [x_n y_n \rightarrow LM]. \end{aligned}$$

*Remark* (Predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)) \implies \text{ConvergentSequence}(x_n y_n, LM). \blacksquare$$

*Remark* (Interpretation). The product law uses both approximation and boundedness. A convergent sequence is eventually bounded, so one factor can be controlled while the other is made small.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Pointwise Product of Sequences](#)
- [Bounded Sequence](#)

#### Theorem (Nonzero Limit is Eventually Nonzero)

**Theorem 6.3.28** (Nonzero Limit is Eventually Nonzero). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$  with  $L \neq 0$ . If  $x_n \rightarrow L$ , then there exists  $N \in \mathbb{N}$  such that  $x_n \neq 0$  for every  $n \geq N$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ .

$$[L \neq 0 \wedge x_n \rightarrow L] \implies \exists N \in \mathbb{N} \forall n \geq N (x_n \neq 0).$$

*Remark* (Interpretation). A sequence converging to a nonzero number eventually stays away from zero. This is the denominator-control result needed for reciprocal and quotient rules.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)

#### Theorem (Limit of a Reciprocal)

**Theorem 6.3.29** (Limit of a Reciprocal). *Let  $(x_n)$  be a real sequence such that  $x_n \neq 0$  for every  $n \in \mathbb{N}$ , and let  $L \in \mathbb{R}$  with  $L \neq 0$ . If  $x_n \rightarrow L$ , then  $1/x_n \rightarrow 1/L$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence with  $\forall n \in \mathbb{N} (x_n \neq 0)$ , and let  $L \neq 0$ .

$$[x_n \rightarrow L] \implies \left[ \frac{1}{x_n} \rightarrow \frac{1}{L} \right].$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}\left(\frac{1}{x_n}, \frac{1}{L}\right).$$

*Remark* (Interpretation). Reciprocation is continuous away from zero. The nonzero limit supplies a tail on which the denominators stay uniformly separated from zero.

*Remark* (Dependencies).

- [Reciprocal Sequence](#)
- [Nonzero Limit is Eventually Nonzero](#)

### Theorem (Limit of a Quotient)

**Theorem 6.3.30** (Limit of a Quotient). *Let  $(x_n)$  and  $(y_n)$  be real sequences such that  $y_n \neq 0$  for every  $n \in \mathbb{N}$ , and let  $L, M \in \mathbb{R}$  with  $M \neq 0$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n/y_n \rightarrow L/M$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

Let  $(x_n), (y_n)$  be real sequences with  $\forall n \in \mathbb{N} (y_n \neq 0)$ , and let  $M \neq 0$ .

$$[x_n \rightarrow L \wedge y_n \rightarrow M] \implies \left[\frac{x_n}{y_n} \rightarrow \frac{L}{M}\right].$$

*Remark* (Predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)) \implies \text{ConvergentSequence}\left(\frac{x_n}{y_n}, \frac{L}{M}\right).$$

*Remark* (Interpretation). The quotient law is the product law applied to  $(x_n)$  and the reciprocal of  $(y_n)$ . The condition  $M \neq 0$  prevents the limiting denominator from collapsing.

*Remark* (Dependencies).

- [Pointwise Quotient of Sequences](#)
- [Limit of a Product](#)
- [Limit of a Reciprocal](#)

### Theorem (Limit of a Square)

**Theorem 6.3.31** (Limit of a Square). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $x_n^2 \rightarrow L^2$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ .

$$[x_n \rightarrow L] \implies [x_n^2 \rightarrow L^2].$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(x_n^2, L^2).$$

*Remark* (Interpretation). Squaring preserves limits because it is multiplication of a sequence by itself.

*Remark* (Dependencies).

- [Square Sequence](#)
- [Limit of a Product](#)

#### Theorem (Limit of an Absolute Value)

**Theorem 6.3.32** (Limit of an Absolute Value). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $|x_n| \rightarrow |L|$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ .

$$[x_n \rightarrow L] \implies [|x_n| \rightarrow |L|].$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(|x_n|, |L|).$$

*Remark* (Interpretation). Taking absolute values cannot increase distance by more than the original distance: the magnitude function is continuous on  $\mathbb{R}$ .

*Remark* (Dependencies).

- [Absolute Value Sequence](#)
- [Triangle Inequality](#)

#### Theorem (Positive Limit is Eventually Positive)

**Theorem 6.3.33** (Positive Limit is Eventually Positive). *Let  $(x_n)$  be a real sequence, and let  $L > 0$ . If  $x_n \rightarrow L$ , then there exists  $N \in \mathbb{N}$  such that  $0 < x_n$  for every  $n \geq N$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $(x_n)$  be a real sequence and let  $L > 0$ .

$$[x_n \rightarrow L] \implies \exists N \in \mathbb{N} \forall n \geq N (0 < x_n).$$

*Remark* (Interpretation). Convergence to a positive limit eventually traps the sequence inside the positive half-line. This result is the sign-control counterpart of the eventual nonzero theorem.

Remark (Dependencies).

- [Convergence of a Sequence](#)

#### Theorem (Limit of a Square Root)

**Theorem 6.3.34** (Limit of a Square Root). *Let  $(x_n)$  be a real sequence such that  $0 \leq x_n$  for every  $n \in \mathbb{N}$ , and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $0 \leq L$  and  $\sqrt{x_n} \rightarrow \sqrt{L}$ .*

*Go to proof.*

Remark (Standard quantified statement).

Let  $(x_n)$  be a real sequence with  $\forall n \in \mathbb{N} (0 \leq x_n)$ , and let  $L \in \mathbb{R}$ .

$$[x_n \rightarrow L] \implies [0 \leq L \wedge \sqrt{x_n} \rightarrow \sqrt{L}].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\sqrt{x_n}, \sqrt{L}).$$

Remark (Interpretation). The square-root function is continuous on the nonnegative real line. Near zero the proof uses the order relation; away from zero it can be rationalized by multiplying by the conjugate.

Remark (Dependencies).

- [Square Root Sequence](#)
- [Convergence of a Sequence](#)

#### Theorem (Polynomial Sequence Limit)

**Theorem 6.3.35** (Polynomial Sequence Limit). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $p \in \mathbb{R}[t]$  be a polynomial. If  $x_n \rightarrow L$ , then  $p(x_n) \rightarrow p(L)$ .*

*Go to proof.*

Remark (Standard quantified statement).

Let  $(x_n)$  be a real sequence,  $L \in \mathbb{R}$ , and  $p \in \mathbb{R}[t]$ .

$$[x_n \rightarrow L] \implies [p(x_n) \rightarrow p(L)].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(p(x_n), p(L)).$$

Remark (Interpretation). Polynomial limits are built from finitely many additions, scalar multiples, and products. The theorem packages those algebra laws into one reusable composition principle.

Remark (Dependencies).

- [Limit of a Sum](#)
- [Limit of a Scalar Multiple](#)
- [Limit of a Product](#)
- [Limit of a Square](#)

#### Theorem (Rational Sequence Limit)

**Theorem 6.3.36** (Rational Sequence Limit). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $p, q \in \mathbb{R}[t]$  be polynomials. Suppose  $q(L) \neq 0$  and  $q(x_n) \neq 0$  for every  $n \in \mathbb{N}$ . If  $x_n \rightarrow L$ , then*

$$\frac{p(x_n)}{q(x_n)} \rightarrow \frac{p(L)}{q(L)}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p, q \in \mathbb{R}[t]. \\ & [q(L) \neq 0 \wedge \forall n \in \mathbb{N} (q(x_n) \neq 0) \wedge x_n \rightarrow L] \\ & \implies \left[ \frac{p(x_n)}{q(x_n)} \rightarrow \frac{p(L)}{q(L)} \right]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}\left(\frac{p(x_n)}{q(x_n)}, \frac{p(L)}{q(L)}\right).$$

*Remark* (Interpretation). A rational expression is a quotient of two polynomial expressions. The condition  $q(L) \neq 0$  keeps the limiting denominator legitimate, and the pointwise condition  $q(x_n) \neq 0$  keeps every term of the quotient sequence defined.

*Remark* (Dependencies).

- [Polynomial Sequence Limit](#)
- [Limit of a Quotient](#)
- [Nonzero Limit is Eventually Nonzero](#)

## 6.4 Tails

#### Definition ( $M$ -Tail of a Sequence)

**Definition 6.4.1** ( $M$ -Tail of a Sequence). Let  $(x_n) \subseteq \mathbb{R}$  and let  $M \in \mathbb{N}$ . The  $M$ -**tail** of  $(x_n)$  is the sequence

$$(x_n)_{n \geq M} := (x_M, x_{M+1}, x_{M+2}, \dots).$$

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq \mathbb{R} \forall M \in \mathbb{N} \left( (x_n)_{n \geq M} = (x_M, x_{M+1}, x_{M+2}, \dots) \right).$$

*Remark* (Definition predicate reading).

$$\text{TailSequence}(x_n, M, y_k) := \forall k \in \mathbb{N} (y_k = x_{M+k-1}).$$

*Remark* (Interpretation). The  $M$ -tail discards the first  $M - 1$  terms and retains everything from index  $M$  onward. It is the formal object underlying the phrase *all but finitely many terms*: to say that all but finitely many terms of  $(x_n)$  satisfy a property  $P$  is precisely to say that some  $M$ -tail of  $(x_n)$  satisfies  $P$  termwise. The index  $M$  is an offset into the sequence, not a bound on the terms themselves; two sequences with identical  $M$ -tails share all asymptotic properties, differing only in finitely many initial values which convergence arguments discard entirely.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Convergence of a Tail)

**Theorem 6.4.2** (Convergence of a Tail). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $m \in \mathbb{N}$ . The  $m$ -tail  $(x_{m+n})_{n \in \mathbb{N}}$  converges if and only if  $(x_n)$  converges. Moreover, in this case,*

$$\lim_{n \rightarrow \infty} x_{m+n} = \lim_{n \rightarrow \infty} x_n.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \forall m \in \mathbb{N} \\ \left( (x_{m+n})_{n \in \mathbb{N}} \text{ converges} \iff (x_n) \text{ converges} \right) \\ \wedge \left( (x_n) \text{ converges} \implies \lim_{n \rightarrow \infty} x_{m+n} = \lim_{n \rightarrow \infty} x_n \right). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\text{TailConvergenceInvariant}(x_n, m) := \left( \exists L \in \mathbb{R} \text{ ConvergentSequence}(x_n, L) \right) \iff \left( \exists L \in \mathbb{R} \text{ ConvergentSequence}(x_{m+n}, L) \right)$$

*Remark* (Contrapositive quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \forall m \in \mathbb{N} \\ \left( (x_{m+n})_{n \in \mathbb{N}} \text{ diverges} \iff (x_n) \text{ diverges} \right). \end{aligned}$$

*Remark* (Contrapositive predicate reading).

$$\neg \text{TailConvergenceInvariant}(x_n, m) := \left( (x_n) \text{ diverges} \iff (x_{m+n})_{n \in \mathbb{N}} \text{ diverges} \right).$$

*Remark* (Interpretation). Convergence is a *tail property*: it depends only on the eventual behavior of the sequence, not on any finite initial segment. Discarding the first  $m$  terms neither creates nor destroys convergence, and when the limit exists it is preserved exactly. This theorem formalizes the informal principle that finitely many terms are irrelevant to asymptotic behavior. It is the rigorous justification for the common proof move of assuming without loss of generality that  $n$  is large enough for some auxiliary condition to hold: one simply passes to a suitable  $m$ -tail, applies the argument there, and concludes for the full sequence by this theorem. The contrapositive is equally useful: to show  $(x_n)$  diverges it suffices to show that some  $m$ -tail diverges.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [M-Tail of a Sequence](#)
- [Uniqueness of Limits](#)

#### Theorem (Convergence by Domination)

**Theorem 6.4.3** (Convergence by Domination). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $L \in \mathbb{R}$ . Let  $(a_n)$  be a sequence of positive real numbers with  $\lim_{n \rightarrow \infty} a_n = 0$ . If there exist a constant  $c > 0$  and an index  $m \in \mathbb{N}$  such that*

$$|x_n - L| \leq c a_n \quad \forall n \geq m,$$

*then  $\lim_{n \rightarrow \infty} x_n = L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \forall (a_n) \subseteq (0, \infty) \left( \left( \lim_{n \rightarrow \infty} a_n = 0 \wedge \exists c > 0 \exists m \in \mathbb{N} \forall n \geq m (|x_n - L| \leq c a_n) \right) \implies \lim_{n \rightarrow \infty} x_n = L \right).$$

*Remark* (Contrapositive quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \forall (a_n) \subseteq (0, \infty) \left( \lim_{n \rightarrow \infty} x_n \neq L \implies \lim_{n \rightarrow \infty} a_n \neq 0 \vee \forall c > 0 \forall m \in \mathbb{N} \exists n \geq m (|x_n - L| > c a_n) \right).$$

*Remark* (Contrapositive predicate reading).

$$\text{NotDominatedByNullError}(x_n, L, a_n) := \lim_{n \rightarrow \infty} a_n \neq 0 \vee \forall c > 0 \forall m \in \mathbb{N} \exists n \geq m (|x_n - L| > c a_n).$$

*Remark* (Interpretation). The theorem is a *domination principle*: it is not necessary to verify the  $\varepsilon$ - $N$  condition directly for  $(x_n)$ . It suffices to find a null sequence  $(a_n)$  – one converging to zero – that dominates the error  $|x_n - L|$  on some tail, up to a fixed scalar  $c$ . The scalar  $c$



is irrelevant to the conclusion: since  $a_n \rightarrow 0$ , the product  $c a_n \rightarrow 0$  regardless of the size of  $c$ , and any  $\varepsilon > 0$  is eventually beaten by  $c a_n$ . This theorem is the rigorous engine behind the common proof pattern of bounding  $|x_n - L|$  by a simpler expression and observing that the simpler expression tends to zero. The canonical instance is  $a_n = 1/n$ , where domination by  $c/n$  is sufficient to conclude convergence to  $L$ .

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [M-Tail of a Sequence](#)

#### Theorem (Ratio Limit Less Than One Implies Null)

**Theorem 6.4.4** (Ratio Limit Less Than One Implies Null). *Let  $(x_n)$  be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$$

*exists. If  $L < 1$ , then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = 0.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq (0, \infty) \forall L \in \mathbb{R} \\ &\left( \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = L \wedge L < 1 \right) \implies \lim_{n \rightarrow \infty} x_n = 0. \end{aligned}$$

*Remark* (Theorem predicate reading).

$$(\text{ConvergentSequence}(x_{n+1}/x_n, L) \wedge L < 1) \implies \text{NullSequence}(x_n).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\exists (x_n) \subseteq (0, \infty) \exists L \in \mathbb{R} \\ &\left( \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = L \wedge L < 1 \wedge \lim_{n \rightarrow \infty} x_n \neq 0 \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{ConvergentSequence}(x_{n+1}/x_n, L) \wedge L < 1 \wedge \text{NotConvergentTo}(x_n, 0).$$

*Remark* (Failure modes). A failure would be a positive sequence whose consecutive ratios settle below one in the limit but whose terms do not tend to zero. The theorem says this cannot happen: once the ratio limit is less than one, some tail has ratios bounded by a fixed number  $r < 1$ , forcing geometric decay.

*Remark* (Failure mode decomposition).

$$\underbrace{(x_n) \subseteq (0, \infty)}_{\text{positive terms}} \quad \underbrace{\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = L < 1}_{\text{eventual ratio gap}} \quad \underbrace{\lim_{n \rightarrow \infty} x_n \neq 0}_{\text{fails to be null}}.$$

The positive-term hypothesis makes each ratio meaningful. The strict gap  $L < 1$  gives a tail on which  $x_{n+1} \leq rx_n$  for some  $r < 1$ , and repeated iteration then gives domination by a null geometric sequence.

*Remark* (Interpretation). This is the sequence form of the ratio test. The theorem does not merely say that the terms decrease eventually; it says they decrease at an eventually geometric rate. The strict inequality  $L < 1$  is essential. When the ratio limit equals 1, the conclusion may hold, as with  $1/n$ , or fail, as with the constant sequence 1.

*Remark* (Dependencies).

- [Convergence of a Sequence](#)
- [Null Sequence](#)
- [M-Tail of a Sequence](#)
- [Convergence by Domination](#)

## 6.5 Monotonicity

### Monotonicity of Sequences

Monotonicity is an order condition on successive terms of a sequence. It is independent of convergence by itself, but when combined with boundedness it becomes one of the fundamental convergence mechanisms in real analysis.

#### Definition (Increasing Sequence)

**Definition 6.5.1** (Increasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **increasing** if

$$x_n \leq x_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is increasing} \iff \forall n \in \mathbb{N} (x_n \leq x_{n+1}).$$

*Remark* (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) := \forall n \in \mathbb{N} (x_n \leq x_{n+1}).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not increasing} \iff \exists n \in \mathbb{N} (x_{n+1} < x_n).$$

*Remark* (Negation predicate reading).

$$\neg \text{MonotoneIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N} (x_{n+1} < x_n).$$

*Remark* (Interpretation). An increasing sequence may stay flat. The term “increasing” is therefore used here in the non-strict sense: each term is at least as large as the one before it.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Decreasing Sequence)

**Definition 6.5.2** (Decreasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **decreasing** if

$$x_{n+1} \leq x_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is decreasing} \iff \forall n \in \mathbb{N} (x_{n+1} \leq x_n).$$

*Remark* (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(x_n) := \forall n \in \mathbb{N} (x_{n+1} \leq x_n).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not decreasing} \iff \exists n \in \mathbb{N} (x_n < x_{n+1}).$$

*Remark* (Negation predicate reading).

$$\neg \text{MonotoneDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N} (x_n < x_{n+1}).$$

*Remark* (Interpretation). A decreasing sequence may stay flat. The order condition says the sequence never rises as the index advances.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Strictly Increasing Sequence)

**Definition 6.5.3** (Strictly Increasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **strictly increasing** if

$$x_n < x_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is strictly increasing} \iff \forall n \in \mathbb{N} (x_n < x_{n+1}).$$

*Remark* (Definition predicate reading).

$$\text{StrictlyIncreasingSequence}(x_n) := \forall n \in \mathbb{N} (x_n < x_{n+1}).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not strictly increasing} \iff \exists n \in \mathbb{N} (x_{n+1} \leq x_n).$$

*Remark* (Negation predicate reading).

$$\neg \text{StrictlyIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N} (x_{n+1} \leq x_n).$$

*Remark* (Interpretation). Strict increase excludes equality between consecutive terms. Every strictly increasing sequence is increasing, but the converse need not hold.

*Remark* (Dependencies).

- [Real Sequence](#)
- [Increasing Sequence](#)

#### Definition (Strictly Decreasing Sequence)

**Definition 6.5.4** (Strictly Decreasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **strictly decreasing** if

$$x_{n+1} < x_n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is strictly decreasing} \iff \forall n \in \mathbb{N} (x_{n+1} < x_n).$$

*Remark* (Definition predicate reading).

$$\text{StrictlyDecreasingSequence}(x_n) := \forall n \in \mathbb{N} (x_{n+1} < x_n).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not strictly decreasing} \iff \exists n \in \mathbb{N} (x_n \leq x_{n+1}).$$

*Remark* (Negation predicate reading).

$$\neg \text{StrictlyDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N} (x_n \leq x_{n+1}).$$

*Remark* (Interpretation). Strict decrease excludes equality between consecutive terms. Every strictly decreasing sequence is decreasing, but the converse need not hold.

*Remark* (Dependencies).

- [Real Sequence](#)

- [Decreasing Sequence](#)

#### Definition (Monotone Sequence)

**Definition 6.5.5** (Monotone Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **monotone** if it is increasing or decreasing.

*Remark* (Standard quantified statement).

$$(x_n) \text{ is monotone} \\ \iff [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] \vee [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)].$$

*Remark* (Definition predicate reading).

$\text{MonotoneSequence}(x_n) := \text{MonotoneIncreasingSequence}(x_n) \vee \text{MonotoneDecreasingSequence}(x_n).$  ■

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not monotone} \\ \iff [\exists n \in \mathbb{N} (x_{n+1} < x_n)] \wedge [\exists m \in \mathbb{N} (x_m < x_{m+1})].$$

*Remark* (Negation predicate reading).

$\neg \text{MonotoneSequence}(x_n) \iff \neg \text{MonotoneIncreasingSequence}(x_n) \wedge \neg \text{MonotoneDecreasingSequence}(x_n).$  ■

*Remark* (Interpretation). Monotonicity means the sequence has one global direction. It may move upward, move downward, or stay constant, but it cannot reverse direction.

*Remark* (Dependencies).

- [Increasing Sequence](#)
- [Decreasing Sequence](#)

#### Definition (Eventually Increasing Sequence)

**Definition 6.5.6** (Eventually Increasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **eventually increasing** if there exists  $N \in \mathbb{N}$  such that

$$x_n \leq x_{n+1} \quad \text{for every } n \geq N.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is eventually increasing} \iff \exists N \in \mathbb{N} \forall n \geq N (x_n \leq x_{n+1}).$$

*Remark* (Definition predicate reading).

$\text{EventuallyIncreasingSequence}(x_n) := \exists N \in \mathbb{N} \forall n \geq N (x_n \leq x_{n+1}).$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not eventually increasing} \iff \forall N \in \mathbb{N} \exists n \geq N (x_{n+1} < x_n).$$

*Remark* (Negation predicate reading).

$$\neg \text{EventuallyIncreasingSequence}(x_n) \iff \forall N \in \mathbb{N} \exists n \geq N (x_{n+1} < x_n).$$

*Remark* (Interpretation). Eventual increase allows finitely many early reversals. Only the tail of the sequence must move in a nondecreasing direction.

*Remark* (Dependencies).

- [M-Tail of a Sequence](#)
- [Increasing Sequence](#)

#### Definition (Eventually Decreasing Sequence)

**Definition 6.5.7** (Eventually Decreasing Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **eventually decreasing** if there exists  $N \in \mathbb{N}$  such that

$$x_{n+1} \leq x_n \quad \text{for every } n \geq N.$$

*Remark* (Standard quantified statement).

$$(x_n) \text{ is eventually decreasing} \iff \exists N \in \mathbb{N} \forall n \geq N (x_{n+1} \leq x_n).$$

*Remark* (Definition predicate reading).

$$\text{EventuallyDecreasingSequence}(x_n) := \exists N \in \mathbb{N} \forall n \geq N (x_{n+1} \leq x_n).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not eventually decreasing} \iff \forall N \in \mathbb{N} \exists n \geq N (x_n < x_{n+1}).$$

*Remark* (Negation predicate reading).

$$\neg \text{EventuallyDecreasingSequence}(x_n) \iff \forall N \in \mathbb{N} \exists n \geq N (x_n < x_{n+1}).$$

*Remark* (Interpretation). Eventual decrease allows the initial terms to behave irregularly. The order condition begins only after some finite threshold.

*Remark* (Dependencies).

- [M-Tail of a Sequence](#)
- [Decreasing Sequence](#)

### Definition (Eventually Monotone Sequence)

**Definition 6.5.8** (Eventually Monotone Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **eventually monotone** if it is eventually increasing or eventually decreasing.

*Remark* (Standard quantified statement).

$$(x_n) \text{ is eventually monotone} \\ \iff [\exists N \in \mathbb{N} \forall n \geq N (x_n \leq x_{n+1})] \vee [\exists N \in \mathbb{N} \forall n \geq N (x_{n+1} \leq x_n)].$$

*Remark* (Definition predicate reading).

$\text{EventuallyMonotoneSequence}(x_n) := \text{EventuallyIncreasingSequence}(x_n) \vee \text{EventuallyDecreasingSequence}(x_n)$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not eventually monotone} \\ \iff [\forall N \in \mathbb{N} \exists n \geq N (x_{n+1} < x_n)] \wedge [\forall N \in \mathbb{N} \exists n \geq N (x_n < x_{n+1})].$$

*Remark* (Negation predicate reading).

$\neg \text{EventuallyMonotoneSequence}(x_n) \iff \neg \text{EventuallyIncreasingSequence}(x_n) \wedge \neg \text{EventuallyDecreasingSequence}(x_n)$

*Remark* (Interpretation). Eventual monotonicity is the tail version of monotonicity. It is often the right hypothesis in applications, because finitely many early terms do not affect convergence.

*Remark* (Dependencies).

- [Eventually Increasing Sequence](#)
- [Eventually Decreasing Sequence](#)
- [Convergence of a Tail](#)

### Theorem (Monotone Convergence Theorem)

**Theorem 6.5.9** (Monotone Convergence Theorem). Let  $(x_n)$  be a real sequence.

(a) If  $(x_n)$  is increasing and bounded above, then  $(x_n)$  converges and

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If  $(x_n)$  is decreasing and bounded below, then  $(x_n)$  converges and

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
& \forall(x_n) \subseteq \mathbb{R} \\
& \left( [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] \wedge [\exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)] \right) \\
& \implies \exists L \in \mathbb{R} \left( \lim_{n \rightarrow \infty} x_n = L \wedge L = \sup\{x_n : n \in \mathbb{N}\} \right), \\
& \left( [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)] \wedge [\exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n)] \right) \\
& \implies \exists L \in \mathbb{R} \left( \lim_{n \rightarrow \infty} x_n = L \wedge L = \inf\{x_n : n \in \mathbb{N}\} \right).
\end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned}
& \text{MonotoneIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \implies \text{ConvergentSequence}(x_n, \sup\{x_n : n \in \mathbb{N}\}) \\
& \text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \implies \text{ConvergentSequence}(x_n, \inf\{x_n : n \in \mathbb{N}\})
\end{aligned}$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
& \exists(x_n) \subseteq \mathbb{R} \left( [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] \wedge [\exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)] \right. \\
& \quad \left. \wedge (x_n) \text{ does not converge} \right), \\
& \exists(x_n) \subseteq \mathbb{R} \left( [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)] \wedge [\exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n)] \right. \\
& \quad \left. \wedge (x_n) \text{ does not converge} \right).
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned}
& \text{MonotoneIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \wedge \neg \exists L \in \mathbb{R} \text{ ConvergentSequence}(x_n, L), \\
& \text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \wedge \neg \exists L \in \mathbb{R} \text{ ConvergentSequence}(x_n, L).
\end{aligned}$$

*Remark* (Failure modes). The theorem would fail only if a one-directional bounded sequence could avoid settling to a real limit. Completeness of  $\mathbb{R}$  rules this out: an increasing bounded-above sequence is forced toward the least upper bound of its range, while a decreasing bounded-below sequence is forced toward the greatest lower bound of its range.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall n \in \mathbb{N} (x_n \leq x_{n+1})}_{\text{one direction}} \quad \underbrace{\exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M)}_{\text{ceiling}} \quad \underbrace{(x_n) \text{ does not converge}}_{\text{forbidden by completeness}}.$$

The decreasing case is dual: reverse the inequalities and replace the ceiling by a floor.

*Remark* (Interpretation). Boundedness prevents escape, and monotonicity prevents oscillation. Together they force convergence. The theorem is one of the standard forms in which the completeness of the real numbers becomes a sequence theorem.

*Remark* (Dependencies).



- [Convergence of a Sequence](#)
- [Increasing Sequence](#)
- [Decreasing Sequence](#)
- [Sequence Bounded Above](#)
- [Sequence Bounded Below](#)
- [Supremum](#)
- [Infimum](#)

#### Theorem (Strict Monotonicity Implies Monotonicity)

**Theorem 6.5.10** (Strict Monotonicity Implies Monotonicity). *Let  $(x_n)$  be a real sequence.*

- (a) *If  $(x_n)$  is strictly increasing, then  $(x_n)$  is increasing.*
- (b) *If  $(x_n)$  is strictly decreasing, then  $(x_n)$  is decreasing.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \\ \left( [\forall n \in \mathbb{N} (x_n < x_{n+1})] \implies [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] \right), \\ \left( [\forall n \in \mathbb{N} (x_{n+1} < x_n)] \implies [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)] \right). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned} \text{StrictlyIncreasingSequence}(x_n) &\implies \text{MonotoneIncreasingSequence}(x_n), \\ \text{StrictlyDecreasingSequence}(x_n) &\implies \text{MonotoneDecreasingSequence}(x_n). \end{aligned}$$

*Remark* (Interpretation). Strict monotonicity is stronger than non-strict monotonicity because every strict inequality is also a weak inequality.

*Remark* (Dependencies).

- [Strictly Increasing Sequence](#)
- [Strictly Decreasing Sequence](#)
- [Increasing Sequence](#)
- [Decreasing Sequence](#)

### Theorem (Bounded Monotone Sequence Equivalences)

**Theorem 6.5.11** (Bounded Monotone Sequence Equivalences). *Let  $(x_n)$  be a real sequence.*

- (a) *If  $(x_n)$  is increasing, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded above.*
- (b) *If  $(x_n)$  is decreasing, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded below.*
- (c) *If  $(x_n)$  is monotone, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned}\forall(x_n) \subseteq \mathbb{R} \\ [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] &\implies \left( \exists L \in \mathbb{R} (x_n \rightarrow L) \iff \exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M) \right), \\ [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)] &\implies \left( \exists L \in \mathbb{R} (x_n \rightarrow L) \iff \exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n) \right).\end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned}\text{MonotoneIncreasingSequence}(x_n) &\implies \left( \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L) \iff \text{BoundedAboveSequence}(x_n) \right), \\ \text{MonotoneDecreasingSequence}(x_n) &\implies \left( \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L) \iff \text{BoundedBelowSequence}(x_n) \right).\end{aligned}$$

*Remark* (Interpretation). For monotone sequences, the only obstruction to real convergence is escape in the monotone direction. Boundedness removes that obstruction.

*Remark* (Dependencies).

- [Monotone Convergence Theorem](#)
- [Bounded Sequence](#)
- [Convergence of a Sequence](#)

### Theorem (Eventually Monotone Convergence Theorem)

**Theorem 6.5.12** (Eventually Monotone Convergence Theorem). *Let  $(x_n)$  be a real sequence.*

- (a) *If  $(x_n)$  is eventually increasing and bounded above, then  $(x_n)$  converges.*
- (b) *If  $(x_n)$  is eventually decreasing and bounded below, then  $(x_n)$  converges.*
- (c) *If  $(x_n)$  is eventually monotone and bounded, then  $(x_n)$  converges.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned}
& \forall (x_n) \subseteq \mathbb{R} \\
& \left( \exists N \in \mathbb{N} \forall n \geq N (x_n \leq x_{n+1}) \wedge \exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M) \right) \\
& \implies \exists L \in \mathbb{R} (x_n \rightarrow L), \\
& \left( \exists N \in \mathbb{N} \forall n \geq N (x_{n+1} \leq x_n) \wedge \exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n) \right) \\
& \implies \exists L \in \mathbb{R} (x_n \rightarrow L).
\end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned}
& \text{EventuallyIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \implies \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L) \\
& \text{EventuallyDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \implies \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L)
\end{aligned}$$

*Remark* (Interpretation). This is the tail version of the Monotone Convergence Theorem. Once the tail is monotone and bounded in the right direction, finite initial behavior is irrelevant.

*Remark* (Dependencies).

- [Eventually Increasing Sequence](#)
- [Eventually Decreasing Sequence](#)
- [Convergence of a Tail](#)
- [Monotone Convergence Theorem](#)

### Theorem (Unbounded Monotone Divergence)

**Theorem 6.5.13** (Unbounded Monotone Divergence). *Let  $(x_n)$  be a real sequence.*

(a) *If  $(x_n)$  is increasing and is not bounded above, then  $x_n \rightarrow +\infty$ .*

(b) *If  $(x_n)$  is decreasing and is not bounded below, then  $x_n \rightarrow -\infty$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
& \forall (x_n) \subseteq \mathbb{R} \\
& \left( [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] \wedge [\forall M \in \mathbb{R} \exists n \in \mathbb{N} (M < x_n)] \right) \\
& \implies \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n), \\
& \left( [\forall n \in \mathbb{N} (x_{n+1} \leq x_n)] \wedge [\forall M \in \mathbb{R} \exists n \in \mathbb{N} (x_n < M)] \right) \\
& \implies \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M).
\end{aligned}$$

*Remark* (Theorem predicate reading).

$\text{MonotoneIncreasingSequence}(x_n) \wedge \neg \text{BoundedAboveSequence}(x_n) \implies \text{DivergesToPositiveInfinity}(x_n),$   
 $\text{MonotoneDecreasingSequence}(x_n) \wedge \neg \text{BoundedBelowSequence}(x_n) \implies \text{DivergesToNegativeInfinity}(x_n).$

*Remark* (Interpretation). For a monotone sequence, unboundedness has a direction. An increasing sequence that has no upper bound eventually exceeds every real number; the decreasing case is the order-dual statement.

*Remark* (Dependencies).

- Divergence to Positive Infinity
- Divergence to Negative Infinity
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below

#### Theorem (Algebraic Transformations Preserve Monotonicity)

**Theorem 6.5.14** (Algebraic Transformations Preserve Monotonicity). *Let  $(x_n)$  be a real sequence and let  $c, \alpha \in \mathbb{R}$ .*

- (a) *If  $(x_n)$  is increasing, then  $(x_n + c)$  is increasing.*
- (b) *If  $(x_n)$  is decreasing, then  $(x_n + c)$  is decreasing.*
- (c) *If  $(x_n)$  is increasing and  $\alpha > 0$ , then  $(\alpha x_n)$  is increasing.*
- (d) *If  $(x_n)$  is increasing and  $\alpha < 0$ , then  $(\alpha x_n)$  is decreasing.*
- (e) *If  $(x_n)$  is increasing, then  $(-x_n)$  is decreasing.*
- (f) *If  $(x_n)$  is decreasing, then  $(-x_n)$  is increasing.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \forall c, \alpha \in \mathbb{R} \\ [\forall n \in \mathbb{N} (x_n \leq x_{n+1})] &\implies [\forall n \in \mathbb{N} (x_n + c \leq x_{n+1} + c)], \\ (\alpha > 0 \wedge \forall n \in \mathbb{N} (x_n \leq x_{n+1})) &\implies [\forall n \in \mathbb{N} (\alpha x_n \leq \alpha x_{n+1})], \\ (\alpha < 0 \wedge \forall n \in \mathbb{N} (x_n \leq x_{n+1})) &\implies [\forall n \in \mathbb{N} (\alpha x_{n+1} \leq \alpha x_n)]. \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\begin{aligned} \text{MonotoneIncreasingSequence}(x_n) &\implies \text{MonotoneIncreasingSequence}(x_n + c), \\ \text{MonotoneIncreasingSequence}(x_n) \wedge \alpha > 0 &\implies \text{MonotoneIncreasingSequence}(\alpha x_n), \\ \text{MonotoneIncreasingSequence}(x_n) \wedge \alpha < 0 &\implies \text{MonotoneDecreasingSequence}(\alpha x_n). \end{aligned}$$

*Remark* (Interpretation). Translation preserves order, multiplication by a positive scalar preserves order, and multiplication by a negative scalar reverses order. Negation is the special case  $\alpha = -1$ .

*Remark* (Dependencies).

- [Pointwise Sum of Sequences](#)
- [Scalar Multiple of a Sequence](#)
- [Pointwise Negation of a Sequence](#)
- [Increasing Sequence](#)
- [Decreasing Sequence](#)

## 6.6 Subsequences

### Subsequences

Subsequences record what remains after passing to an infinite ordered selection of indices. They are the basic tool for extracting structure from a sequence: limits, oscillation, boundedness, compactness, and failure of convergence are all detected by suitable subsequences.

#### Definition (Strictly Increasing Index Map)

**Definition 6.6.1** (Strictly Increasing Index Map). Let  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  be a function. The function  $\sigma$  is a **strictly increasing index map** if

$$k < \ell \implies \sigma(k) < \sigma(\ell) \quad \text{for all } k, \ell \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\sigma : \mathbb{N} \rightarrow \mathbb{N} \text{ is strictly increasing} \iff \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)).$$

*Remark* (Definition predicate reading).

$$\text{StrictlyIncreasing}(\sigma) := \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)).$$

*Remark* (Negated quantified statement).

$$\sigma \text{ is not strictly increasing} \iff \exists k, \ell \in \mathbb{N} (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

*Remark* (Negation predicate reading).

$$\neg \text{StrictlyIncreasing}(\sigma) \iff \exists k, \ell \in \mathbb{N} (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

*Remark* (Interpretation). The index map selects terms without repetition and without changing their order. This is the structural difference between a subsequence and an arbitrary rearrangement.

*Remark* (Dependencies).

- [Function](#)
- [Natural Numbers](#)
- [Strict Order Successor Characterization](#)

#### Definition (Subsequence)

**Definition 6.6.2** (Subsequence). Let  $(x_n)$  be a real sequence. A **subsequence** of  $(x_n)$  is a sequence  $(x_{\sigma(k)})_{k \in \mathbb{N}}$ , where  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  is a strictly increasing index map.

*Remark* (Standard quantified statement).

$$\begin{aligned} (y_k) \text{ is a subsequence of } (x_n) \\ \iff \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N} (y_k = x_{\sigma(k)}) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{Subsequence}(y_k, x_n) := \exists \sigma \left( \text{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N} (y_k = x_{\sigma(k)}) \right).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} (y_k) \text{ is not a subsequence of } (x_n) \\ \iff \forall \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \exists k, \ell \in \mathbb{N} (k < \ell \wedge \sigma(\ell) \leq \sigma(k)) \vee \exists k \in \mathbb{N} (y_k \neq x_{\sigma(k)}) \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{Subsequence}(y_k, x_n).$$

*Remark* (Interpretation). A subsequence keeps the original chronological order but may skip terms. The new index  $k$  counts the selected terms, while  $\sigma(k)$  records where each selected term came from in the original sequence.

*Remark* (Dependencies).

- [Real Sequence](#)
- [Strictly Increasing Index Map](#)

### Definition (Subsequential Limit)

**Definition 6.6.3** (Subsequential Limit). Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ . The real number  $L$  is a **subsequential limit** of  $(x_n)$  if there exists a subsequence  $(x_{\sigma(k)})$  such that

$$\lim_{k \rightarrow \infty} x_{\sigma(k)} = L.$$

*Remark* (Standard quantified statement).

$L$  is a subsequential limit of  $(x_n)$

$$\iff \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \rightarrow \infty} x_{\sigma(k)} = L \right).$$

*Remark* (Definition predicate reading).

$$\text{SubsequentialLimit}(x_n, L) := \exists \sigma \left( \text{StrictlyIncreasing}(\sigma) \wedge \text{ConvergentSequence}(x_{\sigma(k)}, L) \right).$$

*Remark* (Interpretation). Subsequential limits detect limiting behavior along an infinite selection of terms. A sequence may have no ordinary limit but still have one or more subsequential limits.

*Remark* (Dependencies).

- [Subsequence](#)
- [Convergence of a Sequence](#)

### Definition (Has Convergent Subsequence)

**Definition 6.6.4** (Has Convergent Subsequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  **has a convergent subsequence** if there exist a strictly increasing index map  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  and a real number  $L$  such that

$$\lim_{k \rightarrow \infty} x_{\sigma(k)} = L.$$

*Remark* (Standard quantified statement).

$(x_n)$  has a convergent subsequence

$$\iff \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \exists L \in \mathbb{R} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \rightarrow \infty} x_{\sigma(k)} = L \right).$$

*Remark* (Definition predicate reading).

$$\text{HasConvergentSubsequence}(x_n) := \exists L \in \mathbb{R} \text{ SubsequentialLimit}(x_n, L).$$

*Remark* (Interpretation). This property asks for at least one convergent infinite selection. It does not assert convergence of the whole sequence and does not require uniqueness of the subsequential limit.

Remark (Dependencies).

- [Subsequential Limit](#)

#### Theorem (Subsequence Indices Dominate the Identity)

**Theorem 6.6.5** (Subsequence Indices Dominate the Identity). *Let  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  be a strictly increasing index map. Then*

$$k \leq \sigma(k) \quad \text{for every } k \in \mathbb{N}.$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\forall \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \left[ \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right] \implies \left[ \forall k \in \mathbb{N} (k \leq \sigma(k)) \right] \right).$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \implies \forall k \in \mathbb{N} (k \leq \sigma(k)).$$

Remark (Interpretation). A subsequence cannot outrun the original indexing backwards. By the time the subsequence has selected its  $k$ th term, it must have reached at least the  $k$ th original position.

Remark (Dependencies).

- [Strictly Increasing Index Map](#)
- [Well-Ordering Principle](#)

#### Theorem (Subsequences Preserve Limits)

**Theorem 6.6.6** (Subsequences Preserve Limits). *Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then every subsequence of  $(x_n)$  converges to  $L$ .*

[Go to proof.](#)

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \forall \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \left( x_n \rightarrow L \wedge \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right) \implies \lim_{k \rightarrow \infty} x_{\sigma(k)} = L \right).$$

Remark (Theorem predicate reading).

$$\text{ConvergentSequence}(x_n, L) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{ConvergentSequence}(x_{\sigma(k)}, L).$$

Remark (Interpretation). Subsequences cannot escape a limit already possessed by the whole sequence. The domination of subsequence indices by the identity transfers every tail estimate for  $(x_n)$  to a tail estimate for  $(x_{\sigma(k)})$ .



Remark (Dependencies).

- Subsequence
- Convergence of a Sequence
- Subsequence Indices Dominate the Identity

#### Theorem (Subsequential Limit of a Convergent Sequence)

**Theorem 6.6.7** (Subsequential Limit of a Convergent Sequence). *Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $L$  is the only possible subsequential limit of  $(x_n)$ .*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \forall L, K \in \mathbb{R} \\ \left( x_n \rightarrow L \wedge K \text{ is a subsequential limit of } (x_n) \right) \implies K = L. \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{ConvergentSequence}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K) \implies K = L.$$

Remark (Interpretation). A convergent sequence has no hidden alternative limiting behavior. Every convergent subsequence inherits the same limit, and uniqueness of limits then forces equality.

Remark (Dependencies).

- Subsequences Preserve Limits
- Uniqueness of Limits

#### Theorem (Divergence by Two Subsequential Limits)

**Theorem 6.6.8** (Divergence by Two Subsequential Limits). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  has two distinct subsequential limits, then  $(x_n)$  does not converge.*

*Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \forall L, K \in \mathbb{R} \\ \left( L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n) \wedge L \neq K \right) \\ \implies (x_n) \text{ does not converge.} \end{aligned}$$

Remark (Theorem predicate reading).

$$\text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K) \wedge L \neq K \implies \neg \exists A \in \mathbb{R} \text{ ConvergentSequence}(x_n, A).$$

*Remark* (Interpretation). Two different subsequential limits reveal persistent oscillation or splitting. If the whole sequence had a limit, every subsequence would have to inherit that same limit.

*Remark* (Dependencies).

- [Subsequential Limit](#)
- [Subsequential Limit of a Convergent Sequence](#)

#### Theorem (Boundedness Passes to Subsequences)

**Theorem 6.6.9** (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \ \forall \sigma : \mathbb{N} \rightarrow \mathbb{N} \\ \left( \exists M > 0 \ \forall n \in \mathbb{N} (|x_n| \leq M) \wedge \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right) \\ \implies \exists M > 0 \ \forall k \in \mathbb{N} (|x_{\sigma(k)}| \leq M). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{BoundedSequence}(x_{\sigma(k)}).$$

*Remark* (Interpretation). A subsequence cannot create new values; it only selects from the original sequence. Any global bound for the original sequence is automatically a bound for the selection.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Subsequence](#)

#### Theorem (Monotonicity Passes to Subsequences)

**Theorem 6.6.10** (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \ \forall \sigma : \mathbb{N} \rightarrow \mathbb{N} \\ \left( \left[ \forall n \in \mathbb{N} (x_n \leq x_{n+1}) \right] \wedge \left[ \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right] \right) \\ \implies \forall k \in \mathbb{N} (x_{\sigma(k)} \leq x_{\sigma(k+1)}). \end{aligned}$$

*Remark* (Theorem predicate reading).

$\text{MonotoneIncreasingSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{MonotoneIncreasingSequence}(x_{\sigma(k)})$ . ■

*Remark* (Interpretation). Selecting terms in the original order preserves the order pattern already present in the sequence. This is why the index map must be strictly increasing.

*Remark* (Dependencies).

- [Increasing Sequence](#)
- [Decreasing Sequence](#)
- [Subsequence](#)

#### Theorem (Subsequence of a Subsequence)

**Theorem 6.6.11** (Subsequence of a Subsequence). *Every subsequence of a subsequence of  $(x_n)$  is a subsequence of  $(x_n)$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \, \forall \sigma, \tau : \mathbb{N} \rightarrow \mathbb{N} \\ & \left( \left[ \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right] \wedge \left[ \forall k, \ell \in \mathbb{N} (k < \ell \implies \tau(k) < \tau(\ell)) \right] \right) \\ & \implies (x_{\sigma(\tau(k))})_{k \in \mathbb{N}} \text{ is a subsequence of } (x_n). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \wedge \text{StrictlyIncreasing}(\tau) \implies \text{Subsequence}(x_{\sigma(\tau(k))}, x_n).$$

*Remark* (Interpretation). Passing to subsequences is closed under iteration. The composed index map  $\sigma \circ \tau$  is still strictly increasing.

*Remark* (Dependencies).

- [Subsequence](#)
- [Composition](#)

#### Theorem (Eventual Properties Pass to Subsequences)

**Theorem 6.6.12** (Eventual Properties Pass to Subsequences). *Let  $P(n)$  be a property of indices. If there exists  $N \in \mathbb{N}$  such that  $P(n)$  holds for every  $n \geq N$ , then for every strictly increasing index map  $\sigma$  there exists  $K \in \mathbb{N}$  such that  $P(\sigma(k))$  holds for every  $k \geq K$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left( \exists N \in \mathbb{N} \forall n \geq N P(n) \wedge \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right) \\ & \implies \exists K \in \mathbb{N} \forall k \geq K P(\sigma(k)). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$\text{Eventually}(x_n, P) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{Eventually}(x_{\sigma(k)}, P).$$

*Remark* (Interpretation). Subsequences eventually enter every tail of the original sequence. Any property that holds on a tail of the original therefore holds on a tail of the subsequence.

*Remark* (Dependencies).

- [Subsequence Indices Dominate the Identity](#)
- [Subsequence](#)

#### Theorem (Frequent Properties Yield Subsequences)

**Theorem 6.6.13** (Frequent Properties Yield Subsequences). *Let  $P(n)$  be a property of indices. If for every  $N \in \mathbb{N}$  there exists  $n \geq N$  such that  $P(n)$  holds, then there exists a strictly increasing index map  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  such that  $P(\sigma(k))$  holds for every  $k \in \mathbb{N}$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[ \forall N \in \mathbb{N} \exists n \geq N P(n) \right] \implies \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N} P(\sigma(k)) \right).$$

*Remark* (Theorem predicate reading).

$$\text{Frequently}(x_n, P) \implies \exists \sigma (\text{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N} P(\sigma(k))).$$

*Remark* (Interpretation). An infinitely recurring property can be threaded into a subsequence. The construction chooses one later witness at each step.

*Remark* (Dependencies).

- [Well-Ordering Principle](#)
- [Subsequence](#)

#### Theorem (Subsequential Limits Respect Bounds)

**Theorem 6.6.14** (Subsequential Limits Respect Bounds). *Let  $(x_n)$  be a real sequence and let  $L$  be a subsequential limit of  $(x_n)$ . If  $m \leq x_n \leq M$  for every  $n \in \mathbb{N}$ , then  $m \leq L \leq M$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall L, m, M \in \mathbb{R} \\ \left( L \text{ is a subsequential limit of } (x_n) \wedge \forall n \in \mathbb{N} (m \leq x_n \leq M) \right) \implies m \leq L \leq M.$$

*Remark* (Theorem predicate reading).

$$\text{SubsequentialLimit}(x_n, L) \wedge \forall n \in \mathbb{N} (m \leq x_n \leq M) \implies m \leq L \leq M.$$

*Remark* (Interpretation). Bounds pass to subsequences, and limits preserve eventual inequalities. Hence subsequential limits remain inside any closed interval that contains all terms.

*Remark* (Dependencies).

- [Boundedness Passes to Subsequences](#)
- [Constant Comparison for Sequence Limits](#)

#### Theorem (Squeeze Passes to Subsequences)

**Theorem 6.6.15** (Squeeze Passes to Subsequences). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be real sequences. If  $a_n \leq x_n \leq b_n$  eventually, then for every subsequence  $(x_{\sigma(k)})$  there exists  $K \in \mathbb{N}$  such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \quad \text{for every } k \geq K.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\left( \exists N \in \mathbb{N} \forall n \geq N (a_n \leq x_n \leq b_n) \wedge \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \right) \\ \implies \exists K \in \mathbb{N} \forall k \geq K (a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)}).$$

*Remark* (Theorem predicate reading).

$$\text{Eventually}(x_n, a_n \leq x_n \leq b_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{Eventually}(x_{\sigma(k)}, a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)}). \blacksquare$$

*Remark* (Interpretation). Squeeze hypotheses are tail hypotheses, so they survive passage to subsequences.

*Remark* (Dependencies).

- [Eventual Properties Pass to Subsequences](#)
- [Sequence Squeeze Theorem](#)

#### Theorem (Monotone Subsequence Theorem)

**Theorem 6.6.16** (Monotone Subsequence Theorem). *Every real sequence has a monotone subsequence.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq \mathbb{R} \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge [\forall k \in \mathbb{N} (x_{\sigma(k)} \leq x_{\sigma(k+1)}) \vee \forall k \in \mathbb{N} (x_{\sigma(k+1)} \leq x_{\sigma(k)})] \right)$$

*Remark* (Theorem predicate reading).

$$\exists \sigma \left( \text{StrictlyIncreasing}(\sigma) \wedge \text{MonotoneSequence}(x_{\sigma(k)}) \right).$$

*Remark* (Interpretation). Every sequence contains an infinite ordered pattern. The proof uses the peak dichotomy: either there are infinitely many terms larger than all later terms, or there is an infinite rising chain.

*Remark* (Dependencies).

- [Monotone Sequence](#)
- [Subsequence](#)
- [Well-Ordering Principle](#)

#### Theorem (Bolzano-Weierstrass Theorem for Sequences)

**Theorem 6.6.17** (Bolzano-Weierstrass Theorem for Sequences). *Every bounded real sequence has a convergent subsequence.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq \mathbb{R} \left( \exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M) \implies \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \exists L \in \mathbb{R} [\forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \rightarrow \infty} x_{\sigma(k)} = L] \right)$$

*Remark* (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \text{HasConvergentSubsequence}(x_n).$$

*Remark* (Interpretation). Bolzano-Weierstrass is the compactness principle for bounded real sequences. Boundedness prevents escape, and the monotone subsequence theorem supplies an ordered subsequence to which the Monotone Convergence Theorem applies.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Monotone Subsequence Theorem](#)
- [Boundedness Passes to Subsequences](#)
- [Bounded Monotone Sequence Equivalences](#)

### Theorem (Sequential Compactness of Closed Bounded Intervals)

**Theorem 6.6.18** (Sequential Compactness of Closed Bounded Intervals). *Let  $a, b \in \mathbb{R}$  with  $a \leq b$ . Every sequence in  $[a, b]$  has a convergent subsequence whose limit belongs to  $[a, b]$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall a, b \in \mathbb{R} \, \forall (x_n) \subseteq \mathbb{R} \\ & \left( a \leq b \wedge \forall n \in \mathbb{N} \, (a \leq x_n \leq b) \right) \\ & \implies \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \, \exists L \in \mathbb{R} \left( \forall k, \ell \in \mathbb{N} \, (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \rightarrow \infty} x_{\sigma(k)} = L \wedge a \leq L \leq b \right). \end{aligned}$$

*Remark* (Theorem predicate reading).

$$[\forall n \in \mathbb{N} \, (a \leq x_n \leq b)] \implies \exists L \in [a, b] \, \text{SubsequentialLimit}(x_n, L).$$

*Remark* (Interpretation). Closed bounded intervals are sequentially compact: a sequence trapped in the interval has a convergent subsequence, and closedness keeps the subsequential limit inside the interval.

*Remark* (Dependencies).

- [Bolzano-Weierstrass Theorem for Sequences](#)
- [Subsequential Limits Respect Bounds](#)

## 6.7 Divergence

### Divergence — Quick Reference

| Concept                                          | Meaning                                                 |
|--------------------------------------------------|---------------------------------------------------------|
| <a href="#">Divergent sequence</a>               | Sequence that does not converge in $\mathbb{R}$         |
| <a href="#">Diverges to <math>+\infty</math></a> | Eventually exceeds every real bound                     |
| <a href="#">Diverges to <math>-\infty</math></a> | Eventually lies below every real bound                  |
| <a href="#">Oscillatory sequence</a>             | Divergence by persistent oscillation                    |
| <a href="#">Infinite divergence</a>              | Divergence to infinity implies real divergence          |
| <a href="#">Two subsequential limits</a>         | Distinct subsequential limits force divergence          |
| <a href="#">Bounded divergence</a>               | Bounded divergence yields multiple subsequential limits |

## Divergence

Divergence is not a single behavior. A sequence can fail to converge because it escapes upward, escapes downward, oscillates between incompatible limiting patterns, or remains

bounded while refusing to settle. The purpose of this section is to separate these failure modes and connect them to subsequences.

### Definition (Divergent Sequence)

**Definition 6.7.1** (Divergent Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **divergent** if there is no  $L \in \mathbb{R}$  such that  $x_n \rightarrow L$ .

*Remark* (Standard quantified statement).

$$(x_n) \text{ is divergent} \\ \iff \forall L \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| \geq \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{DivergentSequence}(x_n) := \forall L \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| \geq \varepsilon).$$

*Remark* (Negated quantified statement).

$$(x_n) \text{ is not divergent} \iff \exists L \in \mathbb{R} \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon).$$

*Remark* (Negation predicate reading).

$$\neg \text{DivergentSequence}(x_n) \iff \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L).$$

*Remark* (Interpretation). Divergence is the formal negation of real convergence. It says that every candidate limit fails at some fixed tolerance on every tail.

*Remark* (Dependencies).

- [Real Sequence](#)
- [Convergence of a Sequence](#)

### Definition (Divergence to Positive Infinity)

**Definition 6.7.2** (Divergence to Positive Infinity). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  **diverges to**  $+\infty$  if

$$\forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n).$$

*Remark* (Standard quantified statement).

$$x_n \rightarrow +\infty \iff \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n).$$

*Remark* (Definition predicate reading).

$$\text{DivergesToPositiveInfinity}(x_n) := \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n).$$



*Remark* (Negated quantified statement).

$$x_n \not\rightarrow +\infty \iff \exists M \in \mathbb{R} \forall N \in \mathbb{N} \exists n \geq N (x_n \leq M).$$

*Remark* (Interpretation). Divergence to  $+\infty$  means every horizontal threshold is eventually left below the sequence. It is not convergence to a real number; it is escape from the real line in the positive direction.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Divergence to Negative Infinity)

**Definition 6.7.3** (Divergence to Negative Infinity). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  **diverges to**  $-\infty$  if

$$\forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M).$$

*Remark* (Standard quantified statement).

$$x_n \rightarrow -\infty \iff \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M).$$

*Remark* (Definition predicate reading).

$$\text{DivergesToNegativeInfinity}(x_n) := \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M).$$

*Remark* (Negated quantified statement).

$$x_n \not\rightarrow -\infty \iff \exists M \in \mathbb{R} \forall N \in \mathbb{N} \exists n \geq N (M \leq x_n).$$

*Remark* (Interpretation). Divergence to  $-\infty$  is escape past every lower threshold. It is the order-dual of divergence to  $+\infty$ .

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Oscillatory Sequence)

**Definition 6.7.4** (Oscillatory Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is **oscillatory** if it is divergent, does not diverge to  $+\infty$ , and does not diverge to  $-\infty$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} (x_n) \text{ is oscillatory} \\ \iff & \left[ \forall L \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| \geq \varepsilon) \right] \\ & \wedge \left[ \exists M \in \mathbb{R} \forall N \in \mathbb{N} \exists n \geq N (x_n \leq M) \right] \\ & \wedge \left[ \exists M \in \mathbb{R} \forall N \in \mathbb{N} \exists n \geq N (M \leq x_n) \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\begin{aligned}\text{OscillatorySequence}(x_n) &:= \text{DivergentSequence}(x_n) \\ &\quad \wedge \neg \text{DivergesToPositiveInfinity}(x_n) \\ &\quad \wedge \neg \text{DivergesToNegativeInfinity}(x_n).\end{aligned}$$

*Remark* (Interpretation). Oscillation means failure to settle without one-directional escape. The sequence may remain bounded, as  $(-1)^n$  does, or it may be unbounded while still repeatedly returning in incompatible directions.

*Remark* (Dependencies).

- [Divergent Sequence](#)
- [Divergence to Positive Infinity](#)
- [Divergence to Negative Infinity](#)

#### Theorem (Divergence to Infinity Implies Real Divergence)

**Theorem 6.7.5** (Divergence to Infinity Implies Real Divergence). *Let  $(x_n)$  be a real sequence. If  $x_n \rightarrow +\infty$  or  $x_n \rightarrow -\infty$ , then  $(x_n)$  is divergent.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}&\left[ \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (M < x_n) \right] \\ &\quad \vee \left[ \forall M \in \mathbb{R} \exists N \in \mathbb{N} \forall n \geq N (x_n < M) \right] \\ &\implies \forall L \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| \geq \varepsilon).\end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{DivergesToPositiveInfinity}(x_n) \vee \text{DivergesToNegativeInfinity}(x_n) \implies \text{DivergentSequence}(x_n). \blacksquare$$

*Remark* (Interpretation). Infinity divergence is a stronger, directional failure of real convergence. A sequence escaping past every real threshold cannot approach a finite real limit.

*Remark* (Dependencies).

- [Divergent Sequence](#)
- [Divergence to Positive Infinity](#)
- [Divergence to Negative Infinity](#)

#### Theorem (Two Subsequential Limits Force Divergence)

**Theorem 6.7.6** (Two Subsequential Limits Force Divergence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  has two subsequential limits  $L, K \in \mathbb{R}$  with  $L \neq K$ , then  $(x_n)$  is divergent.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \exists L, K \in \mathbb{R} \left( L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n) \right) \\ \implies \forall A \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - A| \geq \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\begin{aligned} \exists L, K \in \mathbb{R} (L \neq K \wedge \text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K)) \\ \implies \text{DivergentSequence}(x_n). \end{aligned}$$

*Remark* (Interpretation). Convergence forces all subsequences to inherit the same limit. Two different subsequential limits therefore witness an irreparable conflict in the tail behavior of the original sequence.

*Remark* (Dependencies).

- [Subsequential Limit](#)
- [Subsequences Preserve Limits](#)
- [Uniqueness of Limits](#)
- [Divergence by Two Subsequential Limits](#)

#### Theorem (Unbounded Above Sequence Has a Positive-Infinity Subsequence)

**Theorem 6.7.7** (Unbounded Above Sequence Has a Positive-Infinity Subsequence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  is not bounded above, then there exists a subsequence  $(x_{\sigma(k)})$  such that  $x_{\sigma(k)} \rightarrow +\infty$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \left[ \forall M \in \mathbb{R} \exists n \in \mathbb{N} (M < x_n) \right] \\ \implies \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R} \exists K \in \mathbb{N} \forall k \geq K (M < x_{\sigma(k)}) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\neg \text{BoundedAboveSequence}(x_n) \implies \exists \sigma (\text{StrictlyIncreasing}(\sigma) \wedge \text{DivergesToPositiveInfinity}(x_{\sigma(k)})).$$

*Remark* (Interpretation). Unboundedness above may be sparse. This theorem says the high values can still be selected in increasing index order to form a subsequence that escapes to  $+\infty$ .

*Remark* (Dependencies).

- [Sequence Bounded Above](#)

- Subsequence
- Divergence to Positive Infinity

#### Theorem (Unbounded Below Sequence Has a Negative-Infinity Subsequence)

**Theorem 6.7.8** (Unbounded Below Sequence Has a Negative-Infinity Subsequence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  is not bounded below, then there exists a subsequence  $(x_{\sigma(k)})$  such that  $x_{\sigma(k)} \rightarrow -\infty$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[ \forall M \in \mathbb{R} \exists n \in \mathbb{N} (x_n < M) \right] \\ \implies \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} \left( \forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall M \in \mathbb{R} \exists K \in \mathbb{N} \forall k \geq K (x_{\sigma(k)} < M) \right).$$

*Remark* (Definition predicate reading).

$$\neg \text{BoundedBelowSequence}(x_n) \implies \exists \sigma (\text{StrictlyIncreasing}(\sigma) \wedge \text{DivergesToNegativeInfinity}(x_{\sigma(k)})).$$

*Remark* (Interpretation). Unboundedness below can be converted into a selected tail that escapes past every lower threshold. The proof is the order-reversed version of the positive-infinity subsequence construction.

*Remark* (Dependencies).

- Sequence Bounded Below
- Subsequence
- Divergence to Negative Infinity

#### Theorem (Bounded Divergence Produces Two Subsequential Limits)

**Theorem 6.7.9** (Bounded Divergence Produces Two Subsequential Limits). *Let  $(x_n)$  be a bounded real sequence. If  $(x_n)$  is divergent, then there exist  $L, K \in \mathbb{R}$  with  $L \neq K$  such that  $L$  and  $K$  are subsequential limits of  $(x_n)$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[ \exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M) \right] \wedge \left[ \forall A \in \mathbb{R} \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - A| \geq \varepsilon) \right] \\ \implies \exists L, K \in \mathbb{R} (L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)).$$

*Remark* (Definition predicate reading).

$$\text{BoundedSequence}(x_n) \wedge \text{DivergentSequence}(x_n) \implies \exists L, K \in \mathbb{R} (L \neq K \wedge \text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K)).$$

*Remark* (Interpretation). Bounded divergence cannot be explained by escape to infinity. In the real line, compactness forces convergent subsequences, and divergence forces at least two incompatible subsequential limits.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Divergent Sequence](#)
- [Subsequential Limit](#)
- [Bolzano-Weierstrass Theorem for Sequences](#)
- [Subsequences Preserve Limits](#)

## 6.8 Liminf and Limsup

### Liminf–Limsup — Quick Reference

| Concept                                      | Meaning                                         |
|----------------------------------------------|-------------------------------------------------|
| <a href="#">Tail supremum sequence</a>       | Supremum of each tail                           |
| <a href="#">Tail infimum sequence</a>        | Infimum of each tail                            |
| <a href="#">Limit superior</a>               | Limit of tail suprema                           |
| <a href="#">Limit inferior</a>               | Limit of tail infima                            |
| <a href="#">Liminf below limsup</a>          | Lower envelope never exceeds upper envelope     |
| <a href="#">Convergence criterion</a>        | Convergence iff liminf equals limsup            |
| <a href="#">Largest subsequential limit</a>  | Limsup is the top cluster value                 |
| <a href="#">Smallest subsequential limit</a> | Liminf is the bottom cluster value              |
| <a href="#">Oscillation criterion</a>        | Gap detects nonconvergence                      |
| <a href="#">Eventual-order squeeze</a>       | Squeeze estimates transfer to limsup and liminf |

## Liminf and Limsup

The limit superior and limit inferior measure the eventual upper and lower envelopes of a bounded sequence. Instead of asking whether all late terms settle to one number, they ask how high and how low the tails continue to reach. This makes them natural tools for detecting convergence, oscillation, and subsequential limits.

### Definition (Tail Supremum Sequence)

**Definition 6.8.1** (Tail Supremum Sequence). Let  $(x_n)$  be a bounded real sequence. The **tail supremum sequence** of  $(x_n)$  is the real sequence  $(s_n)$  defined by

$$s_n := \sup\{x_k : k \geq n\} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \left( s_n = \sup\{x_k : k \geq n\} \right).$$

*Remark* (Interpretation). The term  $s_n$  records the least ceiling of the  $n$ -th tail. As  $n$  grows, earlier terms are discarded, so the tail has fewer opportunities to reach large values.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Supremum](#)
- [M-Tail of a Sequence](#)

#### Definition (Tail Infimum Sequence)

**Definition 6.8.2** (Tail Infimum Sequence). Let  $(x_n)$  be a bounded real sequence. The **tail infimum sequence** of  $(x_n)$  is the real sequence  $(i_n)$  defined by

$$i_n := \inf\{x_k : k \geq n\} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \left( i_n = \inf\{x_k : k \geq n\} \right).$$

*Remark* (Interpretation). The term  $i_n$  records the greatest floor of the  $n$ -th tail. As earlier terms are discarded, the tail loses opportunities to reach small values.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Infimum](#)
- [M-Tail of a Sequence](#)

#### Definition (Limit Superior of a Sequence)

**Definition 6.8.3** (Limit Superior of a Sequence). Let  $(x_n)$  be a bounded real sequence, and let  $(s_n)$  be its tail supremum sequence. The **limit superior** of  $(x_n)$  is

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} s_n.$$

*Remark* (Standard quantified statement).

$$\limsup_{n \rightarrow \infty} x_n = L \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N \left( |s_n - L| < \varepsilon \right).$$

*Remark* (Definition predicate reading).

$$\text{LimsupSequence}(x_n, L) := \text{ConvergentSequence}(s_n, L).$$

*Remark* (Interpretation). The limit superior is the eventual upper envelope of the sequence. It is the number to which the ceilings of the tails descend.

*Remark* (Dependencies).

- [Tail Supremum Sequence](#)
- [Convergence of a Sequence](#)

#### Definition (Limit Inferior of a Sequence)

**Definition 6.8.4** (Limit Inferior of a Sequence). Let  $(x_n)$  be a bounded real sequence, and let  $(i_n)$  be its tail infimum sequence. The **limit inferior** of  $(x_n)$  is

$$\liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} i_n.$$

*Remark* (Standard quantified statement).

$$\liminf_{n \rightarrow \infty} x_n = L \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|i_n - L| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{LiminfSequence}(x_n, L) := \text{ConvergentSequence}(i_n, L).$$

*Remark* (Interpretation). The limit inferior is the eventual lower envelope of the sequence. It is the number to which the floors of the tails ascend.

*Remark* (Dependencies).

- [Tail Infimum Sequence](#)
- [Convergence of a Sequence](#)

#### Theorem (Tail Suprema Are Decreasing)

**Theorem 6.8.5** (Tail Suprema Are Decreasing). Let  $(x_n)$  be a bounded real sequence, and let  $(s_n)$  be its tail supremum sequence. Then  $(s_n)$  is decreasing.

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} (s_{n+1} \leq s_n).$$

*Remark* (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(s_n).$$

*Remark* (Interpretation). The  $(n+1)$ -st tail is contained in the  $n$ -th tail. Passing from a set to a smaller subset cannot raise its supremum.

*Remark* (Dependencies).

- [Tail Supremum Sequence](#)
- [Supremum Is Monotone Under Inclusion](#)

#### Theorem (Tail Infima Are Increasing)

**Theorem 6.8.6** (Tail Infima Are Increasing). *Let  $(x_n)$  be a bounded real sequence, and let  $(i_n)$  be its tail infimum sequence. Then  $(i_n)$  is increasing.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \ (i_n \leq i_{n+1}).$$

*Remark* (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(i_n).$$

*Remark* (Interpretation). The  $(n+1)$ -st tail is contained in the  $n$ -th tail. Passing from a set to a smaller subset cannot lower its infimum.

*Remark* (Dependencies).

- [Tail Infimum Sequence](#)
- [Infimum](#)

#### Theorem (Liminf Is Below Limsup)

**Theorem 6.8.7** (Liminf Is Below Limsup). *Let  $(x_n)$  be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

*Remark* (Interpretation). Every tail infimum lies below every corresponding tail supremum. Passing to the limits preserves this order.

*Remark* (Dependencies).

- [Limit Inferior of a Sequence](#)
- [Limit Superior of a Sequence](#)



- [Constant Comparison for Sequence Limits](#)

### Theorem (Convergence iff Liminf Equals Limsup)

**Theorem 6.8.8** (Convergence iff Liminf Equals Limsup). *Let  $(x_n)$  be a bounded real sequence, and let  $L \in \mathbb{R}$ . Then  $x_n \rightarrow L$  if and only if*

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$x_n \rightarrow L \iff \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L.$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) \iff (\text{LiminfSequence}(x_n, L) \wedge \text{LimsupSequence}(x_n, L)).$$

*Remark* (Interpretation). Convergence occurs exactly when the eventual lower envelope and eventual upper envelope collapse to the same number.

*Remark* (Dependencies).

- [Limit Inferior of a Sequence](#)
- [Limit Superior of a Sequence](#)
- [Convergence of a Sequence](#)
- [Sequence Squeeze Theorem](#)

### Theorem (Limsup Is the Largest Subsequential Limit)

**Theorem 6.8.9** (Limsup Is the Largest Subsequential Limit). *Let  $(x_n)$  be a bounded real sequence, and let  $S = \limsup_{n \rightarrow \infty} x_n$ . Then  $S$  is a subsequential limit of  $(x_n)$ , and every subsequential limit of  $(x_n)$  is less than or equal to  $S$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$S = \limsup_{n \rightarrow \infty} x_n \implies \left[ S \text{ is a subsequential limit of } (x_n) \wedge \forall L \in \mathbb{R} (L \text{ is a subsequential limit of } (x_n) \implies L \leq S) \right].$$

*Remark* (Definition predicate reading).

$$\text{LimsupSequence}(x_n, S) \implies (\text{SubsequentialLimit}(x_n, S) \wedge \forall L \in \mathbb{R} (\text{SubsequentialLimit}(x_n, L) \implies L \leq S)).$$

*Remark* (Interpretation). The limit superior is not merely an upper bound on subsequential behavior. It is actually attained as a subsequential limit, and no subsequential limit lies above it.

*Remark* (Dependencies).

- [Limit Superior of a Sequence](#)
- [Subsequential Limit](#)
- [Bolzano-Weierstrass Theorem for Sequences](#)

#### Theorem (Liminf Is the Smallest Subsequential Limit)

**Theorem 6.8.10** (Liminf Is the Smallest Subsequential Limit). *Let  $(x_n)$  be a bounded real sequence, and let  $I = \liminf_{n \rightarrow \infty} x_n$ . Then  $I$  is a subsequential limit of  $(x_n)$ , and every subsequential limit of  $(x_n)$  is greater than or equal to  $I$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$I = \liminf_{n \rightarrow \infty} x_n \implies \left[ I \text{ is a subsequential limit of } (x_n) \wedge \forall L \in \mathbb{R} (L \text{ is a subsequential limit of } (x_n) \implies I \leq L) \right].$$

*Remark* (Definition predicate reading).

$$\text{LiminfSequence}(x_n, I) \implies (\text{SubsequentialLimit}(x_n, I) \wedge \forall L \in \mathbb{R} (\text{SubsequentialLimit}(x_n, L) \implies I \leq L)).$$

*Remark* (Interpretation). The limit inferior is attained as a subsequential limit and no subsequential limit lies below it.

*Remark* (Dependencies).

- [Limit Inferior of a Sequence](#)
- [Subsequential Limit](#)
- [Bolzano-Weierstrass Theorem for Sequences](#)

#### Theorem (Oscillation Criterion via Liminf and Limsup)

**Theorem 6.8.11** (Oscillation Criterion via Liminf and Limsup). *Let  $(x_n)$  be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n < \limsup_{n \rightarrow \infty} x_n$$

*if and only if  $(x_n)$  has two distinct subsequential limits.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\liminf_{n \rightarrow \infty} x_n < \limsup_{n \rightarrow \infty} x_n$$

$$\iff \exists L, K \in \mathbb{R} (L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n)).$$

*Remark* (Definition predicate reading).

$$\exists I, S \in \mathbb{R} (\text{LiminfSequence}(x_n, I) \wedge \text{LimsupSequence}(x_n, S) \wedge I < S)$$

$$\iff \exists L, K \in \mathbb{R} (L \neq K \wedge \text{SubsequentialLimit}(x_n, L) \wedge \text{SubsequentialLimit}(x_n, K)).$$

*Remark* (Interpretation). A strict gap between the lower and upper envelopes is exactly persistent oscillation in bounded sequences. The sequence keeps returning near at least two incompatible limiting levels.

*Remark* (Dependencies).

- [Limsup Is the Largest Subsequential Limit](#)
- [Liminf Is the Smallest Subsequential Limit](#)
- [Two Subsequential Limits Force Divergence](#)

#### Theorem (Limsup Comparison Under Eventual Order)

**Theorem 6.8.12** (Limsup Comparison Under Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N$ , then*

$$\limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\exists N \in \mathbb{N} \forall n \geq N (x_n \leq y_n)$$

$$\implies \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n.$$

*Remark* (Interpretation). Eventual order of sequences passes to eventual order of their upper tail envelopes. Taking the limiting ceiling preserves that comparison.

*Remark* (Dependencies).

- [Limit Superior of a Sequence](#)
- [Limit Preserves Eventual Order](#)

### Theorem (Liminf Comparison Under Eventual Order)

**Theorem 6.8.13** (Liminf Comparison Under Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N$ , then*

$$\liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \exists N \in \mathbb{N} \forall n \geq N (x_n \leq y_n) \\ \implies \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n. \end{aligned}$$

*Remark* (Interpretation). Eventual order also passes to the lower tail envelopes. If one sequence is eventually below another, its eventual floor cannot sit above the other's eventual floor.

*Remark* (Dependencies).

- [Limit Inferior of a Sequence](#)
- [Limit Preserves Eventual Order](#)

### Theorem (Limsup Squeeze Under Eventual Order)

**Theorem 6.8.14** (Limsup Squeeze Under Eventual Order). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

*then*

$$\limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} b_n.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \exists N \in \mathbb{N} \forall n \geq N (a_n \leq x_n \leq b_n) \\ \implies \limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} b_n. \end{aligned}$$

*Remark* (Interpretation). If one sequence is eventually trapped between two others, then its eventual upper envelope is trapped between their eventual upper envelopes.

*Remark* (Dependencies).

- [Limsup Comparison Under Eventual Order](#)

### Theorem (Liminf Squeeze Under Eventual Order)

**Theorem 6.8.15** (Liminf Squeeze Under Eventual Order). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

*then*

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} b_n.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \exists N \in \mathbb{N} \forall n \geq N (a_n \leq x_n \leq b_n) \\ \implies \liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} b_n. \end{aligned}$$

*Remark* (Interpretation). The lower-envelope version of the squeeze principle says that eventual two-sided order traps the eventual floors as well as the eventual ceilings.

*Remark* (Dependencies).

- [Liminf Comparison Under Eventual Order](#)

*Remark* (Illustrative cases). For the alternating sequence  $x_n = (-1)^n$ , the tail suprema are constantly 1 and the tail infima are constantly  $-1$ , so

$$\limsup_{n \rightarrow \infty} (-1)^n = 1, \quad \liminf_{n \rightarrow \infty} (-1)^n = -1.$$

For any convergent sequence  $x_n \rightarrow L$ , both envelopes collapse to  $L$ .

## 6.9 Order-Theoretic Limit Constructions

### Order-Theoretic Limits — Quick Reference

| Statement                                      | Role                                        |
|------------------------------------------------|---------------------------------------------|
| <a href="#">Increasing sequence limit</a>      | Limit realized as a supremum                |
| <a href="#">Decreasing sequence limit</a>      | Limit realized as an infimum                |
| <a href="#">Tail extremals converge</a>        | Tail suprema and infima are monotone        |
| <a href="#">Bounded sequence limsup/liminf</a> | Bounded sequences have asymptotic envelopes |

### Order-Theoretic Limit Constructions

Completeness lets order data produce limits. An increasing sequence bounded above converges to the least ceiling of its range; a decreasing sequence bounded below converges to

the greatest floor of its range. Applying these same ideas to tail suprema and tail infima produces limsup and liminf for every bounded sequence.

### Theorem (Increasing Sequence Limit as Supremum)

**Theorem 6.9.1** (Increasing Sequence Limit as Supremum). *Let  $(x_n)$  be an increasing real sequence that is bounded above. Then*

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \left( \left[ \forall n \in \mathbb{N} (x_n \leq x_{n+1}) \right] \wedge \left[ \exists M \in \mathbb{R} \forall n \in \mathbb{N} (x_n \leq M) \right] \right) \\ \implies \lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}. \end{aligned}$$

*Remark* (Theorem predicate reading).

$\text{MonotoneIncreasingSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n) \implies \text{ConvergentSequence}(x_n, \sup\{x_n : n \in \mathbb{N}\})$

*Remark* (Interpretation). The supremum is the only possible destination: all terms lie below it, and the defining approximation property of the supremum forces the sequence eventually above every lower tolerance.

*Remark* (Dependencies).

- [Increasing Sequence](#)
- [Sequence Bounded Above](#)
- [Supremum](#)
- [Monotone Convergence Theorem](#)

### Theorem (Decreasing Sequence Limit as Infimum)

**Theorem 6.9.2** (Decreasing Sequence Limit as Infimum). *Let  $(x_n)$  be a decreasing real sequence that is bounded below. Then*

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \left( \left[ \forall n \in \mathbb{N} (x_{n+1} \leq x_n) \right] \wedge \left[ \exists m \in \mathbb{R} \forall n \in \mathbb{N} (m \leq x_n) \right] \right) \\ \implies \lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}. \end{aligned}$$

*Remark* (Theorem predicate reading).

$\text{MonotoneDecreasingSequence}(x_n) \wedge \text{BoundedBelowSequence}(x_n) \implies \text{ConvergentSequence}(x_n, \inf\{x_n : n \in \mathbb{N}\})$

*Remark* (Interpretation). The infimum is the decreasing analogue of the supremum target: the sequence falls toward its greatest lower barrier.

*Remark* (Dependencies).

- [Decreasing Sequence](#)
- [Sequence Bounded Below](#)
- [Infimum](#)
- [Monotone Convergence Theorem](#)

#### Theorem (Tail Suprema and Infima Converge)

**Theorem 6.9.3** (Tail Suprema and Infima Converge). *Let  $(x_n)$  be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \quad i_n = \inf\{x_k : k \geq n\}.$$

*Then  $(s_n)$  and  $(i_n)$  converge.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left( (x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left[ \lim_{n \rightarrow \infty} \sup\{x_k : k \geq n\} = S \wedge \lim_{n \rightarrow \infty} \inf\{x_k : k \geq n\} = I \right] \right).$$

*Remark* (Theorem predicate reading).

$\text{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \left( \text{ConvergentSequence}(\sup\{x_k : k \geq n\}, S) \wedge \text{ConvergentSequence}(\inf\{x_k : k \geq n\}, I) \right)$

*Remark* (Interpretation). Tail suprema decrease because later tails contain fewer terms; tail infima increase for the same reason. Boundedness supplies the opposite bound needed for monotone convergence.

*Remark* (Dependencies).

- [Tail Supremum Sequence](#)
- [Tail Infimum Sequence](#)
- [Tail Suprema Are Decreasing](#)
- [Tail Infima Are Increasing](#)
- [Monotone Convergence Theorem](#)

### Theorem (Bounded Sequence Has Limsup and Liminf)

**Theorem 6.9.4** (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left( (x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left( S = \limsup_{n \rightarrow \infty} x_n \wedge I = \liminf_{n \rightarrow \infty} x_n \right) \right).$$

*Remark* (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \left( \text{LimsupSequence}(x_n, S) \wedge \text{LiminfSequence}(x_n, I) \right).$$

*Remark* (Interpretation). For bounded sequences, limsup and liminf are not extra assumptions. They are guaranteed order-theoretic limits of the tail envelopes.

*Remark* (Dependencies).

- [Limit Superior of a Sequence](#)
- [Limit Inferior of a Sequence](#)
- [Tail Suprema and Infima Converge](#)

## 6.10 Cluster Values of Sequences

### Cluster Values — Quick Reference

| Concept                                               | Meaning                                        |
|-------------------------------------------------------|------------------------------------------------|
| <a href="#">Cluster value</a>                         | Value approached along a subsequence           |
| <a href="#">Cluster equals subsequential limit</a>    | Converts frequent approach to subsequences     |
| <a href="#">Bounded sequences have cluster values</a> | Compactness produces accumulation              |
| <a href="#">Extremal cluster values</a>               | Limsup and liminf are the outer cluster values |

### Cluster Values of Sequences

A convergent sequence has only one limiting value. A bounded divergent sequence may still have values that it approaches along selected subsequences. Cluster values name this repeated limiting behavior without requiring the entire sequence to converge.

#### Definition (Cluster Value of a Sequence)

**Definition 6.10.1** (Cluster Value of a Sequence). Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . The number  $L$  is a **cluster value** of  $(x_n)$  if for every  $\varepsilon > 0$  and every  $N \in \mathbb{N}$  there exists  $n \geq N$  such that

$$|x_n - L| < \varepsilon.$$



*Remark* (Standard quantified statement).

$L$  is a cluster value of  $(x_n) \iff \forall \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| < \varepsilon)$ .

*Remark* (Definition predicate reading).

$\text{SequenceClusterValue}(x_n, L) := \forall \varepsilon > 0 \forall N \in \mathbb{N} \exists n \geq N (|x_n - L| < \varepsilon)$ .

*Remark* (Negated quantified statement).

$L$  is not a cluster value of  $(x_n) \iff \exists \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| \geq \varepsilon)$ .

*Remark* (Interpretation). No matter how far out in the sequence one looks, some later term returns inside every neighborhood of  $L$ .

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Cluster Values Are Subsequential Limits)

**Theorem 6.10.2** (Cluster Values Are Subsequential Limits). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . Then  $L$  is a cluster value of  $(x_n)$  if and only if  $L$  is a subsequential limit of  $(x_n)$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \left( L \text{ is a cluster value of } (x_n) \iff L \text{ is a subsequential limit of } (x_n) \right)$ .

*Remark* (Theorem predicate reading).

$\text{SequenceClusterValue}(x_n, L) \iff \text{SubsequentialLimit}(x_n, L)$ .

*Remark* (Interpretation). The neighborhood formulation and the subsequence formulation describe the same phenomenon: infinitely many opportunities to get arbitrarily close to  $L$ .

*Remark* (Dependencies).

- [Cluster Value of a Sequence](#)
- [Subsequential Limit](#)
- [Frequent Properties Yield Subsequences](#)

#### Theorem (Bounded Sequences Have Cluster Values)

**Theorem 6.10.3** (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq \mathbb{R} \left( [\exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M)] \implies \exists L \in \mathbb{R} (L \text{ is a cluster value of } (x_n)) \right).$$

*Remark* (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \exists L \in \mathbb{R} \text{ SequenceClusterValue}(x_n, L).$$

*Remark* (Interpretation). Boundedness traps the sequence in a finite interval, and Bolzano-Weierstrass extracts a convergent subsequence from that trap.

*Remark* (Dependencies).

- [Bounded Sequence](#)
- [Bolzano-Weierstrass Theorem for Sequences](#)
- [Cluster Values Are Subsequential Limits](#)

#### Theorem (Limsup and Liminf Are Extremal Cluster Values)

**Theorem 6.10.4** (Limsup and Liminf Are Extremal Cluster Values). *Let  $(x_n)$  be a bounded real sequence. Then  $\limsup_{n \rightarrow \infty} x_n$  is the largest cluster value of  $(x_n)$ , and  $\liminf_{n \rightarrow \infty} x_n$  is the smallest cluster value of  $(x_n)$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq \mathbb{R} \left( (x_n) \text{ is bounded} \implies \left[ \limsup_{n \rightarrow \infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} (L \text{ is a cluster value of } (x_n) \implies L \leq \limsup_{n \rightarrow \infty} x_n) \right] \wedge \left[ \liminf_{n \rightarrow \infty} x_n \text{ is a cluster value of } (x_n) \wedge \forall L \in \mathbb{R} (L \text{ is a cluster value of } (x_n) \implies \liminf_{n \rightarrow \infty} x_n \leq L) \right] \right).$$

*Remark* (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies$$

$$\text{SequenceClusterValue}(x_n, \limsup_{n \rightarrow \infty} x_n) \wedge \forall L \in \mathbb{R} (\text{SequenceClusterValue}(x_n, L) \implies L \leq \limsup_{n \rightarrow \infty} x_n),$$

$$\text{BoundedSequence}(x_n) \implies$$

$$\text{SequenceClusterValue}(x_n, \liminf_{n \rightarrow \infty} x_n) \wedge \forall L \in \mathbb{R} (\text{SequenceClusterValue}(x_n, L) \implies \liminf_{n \rightarrow \infty} x_n \leq L).$$

*Remark* (Interpretation). Limsup and liminf describe the top and bottom edges of the sequence's subsequential behavior.

*Remark* (Dependencies).

- [Limit Superior of a Sequence](#)

- [Limit Inferior of a Sequence](#)
- [Limsup Is the Largest Subsequential Limit](#)
- [Liminf Is the Smallest Subsequential Limit](#)
- [Cluster Values Are Subsequential Limits](#)

## 6.11 Cauchy

### Cauchy Sequences

Cauchy sequences measure internal stabilization. Instead of comparing the terms to an already named limit, the Cauchy condition compares sufficiently late terms to each other. Over the real numbers this internal criterion is equivalent to convergence, and that equivalence is one of the central sequence forms of completeness.

#### Definition (Cauchy Sequence)

**Definition 6.11.1** (Cauchy Sequence). Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon).$$

*Remark* (Standard quantified statement).

$$\begin{aligned} (x_n) \text{ is Cauchy} \\ \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) := \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} (x_n) \text{ is not Cauchy} \\ \iff \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists m, n \geq N (|x_m - x_n| \geq \varepsilon). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{CauchySequence}(x_n) \iff \exists \varepsilon > 0 \forall N \in \mathbb{N} \exists m, n \geq N (|x_m - x_n| \geq \varepsilon).$$

*Remark* (Interpretation). The Cauchy condition says that the tails of the sequence eventually have arbitrarily small internal spread. It does not name a candidate limit; it only asserts that late terms become mutually close.

*Remark* (Dependencies).

- Real Sequence

### Theorem (Convergent Sequences Are Cauchy)

**Theorem 6.11.2** (Convergent Sequences Are Cauchy). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  converges, then  $(x_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \\ \left( (\exists L \in \mathbb{R} \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon)) \right. \\ \left. \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\forall(x_n) \subseteq \mathbb{R} \left( \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L) \implies \text{CauchySequence}(x_n) \right).$$

*Remark* (Interpretation). Once every sufficiently late term is close to the same limit, any two sufficiently late terms are close to each other. This theorem is the easy half of the Cauchy criterion.

*Remark* (Dependencies).

- Convergence of a Sequence
- Cauchy Sequence

### Theorem (Cauchy Sequences Are Bounded)

**Theorem 6.11.3** (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall(x_n) \subseteq \mathbb{R} \\ \left( (\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon)) \right. \\ \left. \implies \exists M > 0 \forall n \in \mathbb{N} (|x_n| \leq M) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\forall(x_n) \subseteq \mathbb{R} \left( \text{CauchySequence}(x_n) \implies \text{BoundedSequence}(x_n) \right).$$

*Remark* (Interpretation). A Cauchy sequence has one tightly controlled tail. The finitely many terms before that tail can be absorbed into a single larger bound.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Bounded Sequence](#)

#### Theorem (Cauchy Sequence with a Convergent Subsequence)

**Theorem 6.11.4** (Cauchy Sequence with a Convergent Subsequence). *Let  $(x_n)$  be a Cauchy real sequence. If a subsequence of  $(x_n)$  converges to  $L \in \mathbb{R}$ , then  $(x_n)$  converges to  $L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \\ \left( \left[ \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon) \right] \right. \\ \left. \wedge \left[ \exists \sigma : \mathbb{N} \rightarrow \mathbb{N} (\forall k, \ell \in \mathbb{N} (k < \ell \implies \sigma(k) < \sigma(\ell))) \wedge \forall \varepsilon > 0 \exists K \in \mathbb{N} \forall k \geq K (|x_{\sigma(k)} - L| < \varepsilon) \right] \right) \\ \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R} \forall L \in \mathbb{R} \left( \text{CauchySequence}(x_n) \wedge \text{SubsequentialLimit}(x_n, L) \implies \text{ConvergentSequence}(x_n, L) \right).$$

*Remark* (Interpretation). The Cauchy condition prevents the rest of the tail from wandering away from any convergent subsequence. A single subsequential limit therefore determines the limit of the whole sequence.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Subsequence](#)
- [Subsequential Limit](#)
- [Convergence of a Sequence](#)

#### Theorem (Cauchy Criterion for Real Sequences)

**Theorem 6.11.5** (Cauchy Criterion for Real Sequences). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  converges if and only if  $(x_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \\ \left( \exists L \in \mathbb{R} \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - L| < \varepsilon) \right. \\ \left. \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R} \left( \exists L \in \mathbb{R} \text{ConvergentSequence}(x_n, L) \iff \text{CauchySequence}(x_n) \right).$$

*Remark* (Interpretation). This theorem is the sequence form of completeness of  $\mathbb{R}$ . In an incomplete ambient space, a sequence may be internally stable without converging inside that space.

*Remark* (Dependencies).

- [Convergent Sequences Are Cauchy](#)
- [Cauchy Sequences Are Bounded](#)
- [Bolzano-Weierstrass Theorem for Sequences](#)
- [Cauchy Sequence with a Convergent Subsequence](#)

#### Theorem (Cauchy Criterion via Tails)

**Theorem 6.11.6** (Cauchy Criterion via Tails). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  is Cauchy if and only if for every  $\varepsilon > 0$  there exists  $N \in \mathbb{N}$  such that the  $N$ -tail has pairwise term distances less than  $\varepsilon$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} (x_n) \text{ is Cauchy} \\ \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall p, q \in \mathbb{N} (|x_{N+p} - x_{N+q}| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall p, q \in \mathbb{N} (|x_{N+p} - x_{N+q}| < \varepsilon).$$

*Remark* (Interpretation). The Cauchy condition is a tail property. Reindexing a sufficiently late tail does not change the substance of the condition; it only changes the names of the indices.

*Remark* (Dependencies).

- [Cauchy Sequence](#)

- [M-Tail of a Sequence](#)

#### Theorem (Cauchy Tail Diameter Criterion)

**Theorem 6.11.7** (Cauchy Tail Diameter Criterion). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} (x_n) \text{ is Cauchy} \\ \iff \forall \varepsilon > 0 \exists N_0 \in \mathbb{N} \forall N \geq N_0 \\ \forall m, n \geq N (|x_m - x_n| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0 \exists N_0 \in \mathbb{N} \forall N \geq N_0 \forall m, n \geq N (|x_m - x_n| < \varepsilon).$$

*Remark* (Interpretation). The tail-diameter form says that once one tail is small, every later tail is small as well. It records the monotone nature of tail control without introducing additional notation for diameters.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [M-Tail of a Sequence](#)

#### Theorem (Cauchy Sequences Have Vanishing Successive Differences)

**Theorem 6.11.8** (Cauchy Sequences Have Vanishing Successive Differences). *If  $(x_n)$  is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \rightarrow 0.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall (x_n) \subseteq \mathbb{R} \\ \left( \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_m - x_n| < \varepsilon) \right) \\ \implies \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_{n+1} - x_n| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{NullSequence}(|x_{n+1} - x_n|).$$

*Remark* (Interpretation). Cauchy control applies to every pair of late indices, so it applies in particular to consecutive late indices. The converse is false in general: small successive steps do not prevent accumulated drift.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Null Sequence](#)

#### Theorem (Scalar Multiples of Cauchy Sequences)

**Theorem 6.11.9** (Scalar Multiples of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence, and let  $\alpha \in \mathbb{R}$ . Then  $(\alpha x_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall \alpha \in \mathbb{R} \left( (x_n) \text{ is Cauchy} \implies (\alpha x_n) \text{ is Cauchy} \right).$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(\alpha x_n).$$

*Remark* (Interpretation). Multiplying all terms by one fixed real number rescales all pairwise tail distances by the same fixed factor.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Scalar Multiple of a Sequence](#)

#### Theorem (Sums of Cauchy Sequences)

**Theorem 6.11.10** (Sums of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n + y_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall (y_n) \subseteq \mathbb{R} \left( (x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n + y_n) \text{ is Cauchy} \right).$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n + y_n).$$



*Remark* (Interpretation). The triangle inequality splits the tail distance of the sum into the two tail distances already controlled by the original sequences.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Pointwise Sum of Sequences](#)

#### Theorem (Differences of Cauchy Sequences)

**Theorem 6.11.11** (Differences of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n - y_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall (y_n) \subseteq \mathbb{R} \\ \left( (x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n - y_n) \text{ is Cauchy} \right).$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n - y_n).$$

*Remark* (Interpretation). The difference theorem is the sum theorem applied after negating one sequence. It is often the form used to compare two approximating sequences.

*Remark* (Dependencies).

- [Sums of Cauchy Sequences](#)
- [Scalar Multiples of Cauchy Sequences](#)
- [Pointwise Difference of Sequences](#)

#### Theorem (Linear Combinations of Cauchy Sequences)

**Theorem 6.11.12** (Linear Combinations of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences, and let  $\alpha, \beta \in \mathbb{R}$ . Then  $(\alpha x_n + \beta y_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \forall (y_n) \subseteq \mathbb{R} \forall \alpha \in \mathbb{R} \forall \beta \in \mathbb{R} \\ \left( (x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (\alpha x_n + \beta y_n) \text{ is Cauchy} \right).$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \implies \text{CauchySequence}(\alpha x_n + \beta y_n).$$

*Remark* (Interpretation). The Cauchy real sequences are closed under finite real linear combinations. This is the additive vector-space part of Cauchy algebra.

*Remark* (Dependencies).

- [Linear Combination of Real Sequences](#)
- [Scalar Multiples of Cauchy Sequences](#)
- [Sums of Cauchy Sequences](#)

#### Theorem (Products of Cauchy Sequences)

**Theorem 6.11.13** (Products of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n y_n)$  is Cauchy.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \, \forall (y_n) \subseteq \mathbb{R} \\ & \left( (x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \implies (x_n y_n) \text{ is Cauchy} \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n y_n).$$

*Remark* (Interpretation). Products require boundedness in addition to tail closeness. The boundedness is supplied automatically because every Cauchy real sequence is bounded.

*Remark* (Dependencies).

- [Cauchy Sequences Are Bounded](#)
- [Pointwise Product of Sequences](#)

#### Theorem (Reciprocals of Cauchy Sequences)

**Theorem 6.11.14** (Reciprocals of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence. If there exist  $c > 0$  and  $N_0 \in \mathbb{N}$  such that  $|x_n| \geq c$  for every  $n \geq N_0$ , then  $(1/x_n)$  is Cauchy.*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \\ & \left( (x_n) \text{ is Cauchy} \wedge \exists c > 0 \, \exists N_0 \in \mathbb{N} \, \forall n \geq N_0 \, (|x_n| \geq c) \right. \\ & \quad \left. \implies (1/x_n) \text{ is Cauchy} \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \exists c > 0 \exists N_0 \in \mathbb{N} \forall n \geq N_0 (|x_n| \geq c) \implies \text{CauchySequence}(1/x_n).$$

*Remark* (Interpretation). The lower bound away from zero prevents division from amplifying small tail differences without control. Without it, reciprocals need not preserve Cauchy behavior.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Reciprocal Sequence](#)

### Theorem (Quotients of Cauchy Sequences)

**Theorem 6.11.15** (Quotients of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. If there exist  $c > 0$  and  $N_0 \in \mathbb{N}$  such that  $|y_n| \geq c$  for every  $n \geq N_0$ , then  $(x_n/y_n)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \forall (x_n) \subseteq \mathbb{R} \forall (y_n) \subseteq \mathbb{R} \\ & \left( (x_n) \text{ is Cauchy} \wedge (y_n) \text{ is Cauchy} \wedge \exists c > 0 \exists N_0 \in \mathbb{N} \forall n \geq N_0 (|y_n| \geq c) \right. \\ & \quad \left. \implies (x_n/y_n) \text{ is Cauchy} \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \wedge \exists c > 0 \exists N_0 \in \mathbb{N} \forall n \geq N_0 (|y_n| \geq c) \implies \text{CauchySequence}(x_n/y_n).$$

*Remark* (Interpretation). The quotient theorem is the product theorem combined with the reciprocal theorem for a denominator eventually bounded away from zero.

*Remark* (Dependencies).

- [Products of Cauchy Sequences](#)
- [Reciprocals of Cauchy Sequences](#)
- [Pointwise Quotient of Sequences](#)

### Theorem (Absolute Values of Cauchy Sequences)

**Theorem 6.11.16** (Absolute Values of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence. Then  $(|x_n|)$  is Cauchy.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left( (x_n) \text{ is Cauchy} \implies (|x_n|) \text{ is Cauchy} \right).$$

*Remark* (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(|x_n|).$$

*Remark* (Interpretation). The reverse triangle inequality bounds the distance between absolute values by the original distance between terms.

*Remark* (Dependencies).

- [Cauchy Sequence](#)
- [Absolute Value of a Sequence](#)

## 6.12 Applications of Sequences

### Sequence Applications — Quick Reference

| Construction                                              | Role                                    |
|-----------------------------------------------------------|-----------------------------------------|
| <a href="#">Newton sequence for <math>\sqrt{2}</math></a> | Iterative approximation model           |
| <a href="#">Newton approximation</a>                      | Converges to $\sqrt{2}$                 |
| <a href="#">Factorial partial sums</a>                    | Series approximants for $e$             |
| <a href="#">Approximation of <math>e</math></a>           | Factorial sums converge to $e$          |
| <a href="#">Compound-interest sequence</a>                | Classical exponential approximants      |
| <a href="#">Compound-interest approximation</a>           | Compound interest converges to $e$      |
| <a href="#">Decimal truncation sequence</a>               | Decimal approximants from below         |
| <a href="#">Decimal truncations</a>                       | Truncations converge to the target real |

### Applications of Sequences

The convergence theorems for sequences become numerical tools once the sequence is computable. Newton iteration turns an equation into a convergent recursive approximation; factorial partial sums and compound-interest terms approximate the Euler number; decimal truncations approximate arbitrary real numbers by finite decimals.

#### Definition (Newton Sequence for $\sqrt{2}$ )

**Definition 6.12.1** (Newton Sequence for  $\sqrt{2}$ ). The **Newton sequence for  $\sqrt{2}$**  is the real sequence  $(x_n)$  defined by

$$x_1 := \frac{3}{2}, \quad x_{n+1} := \frac{1}{2} \left( x_n + \frac{2}{x_n} \right) \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$x_1 = \frac{3}{2} \\ \wedge \quad \forall n \in \mathbb{N} \left( x_{n+1} = \frac{1}{2} \left( x_n + \frac{2}{x_n} \right) \right).$$

*Remark* (Interpretation). The recursion is Newton's method applied to the equation  $x^2 - 2 = 0$ . Each new term averages the current estimate with the correction  $2/x_n$ .

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Newton Approximation of $\sqrt{2}$ )

**Theorem 6.12.2** (Newton Approximation of  $\sqrt{2}$ ). *Let  $(x_n)$  be the Newton sequence for  $\sqrt{2}$ . Then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = \sqrt{2}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|x_n - \sqrt{2}| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, \sqrt{2}).$$

*Remark* (Interpretation). The Newton sequence is decreasing after the first term and bounded below by  $\sqrt{2}$ , so monotone convergence supplies a limit. Passing the recursion to the limit identifies the limit as the positive solution of  $L^2 = 2$ .

*Remark* (Dependencies).

- [Newton Sequence for  \$\sqrt{2}\$](#)
- [Monotone Convergence Theorem](#)
- [Rational Sequence Limit](#)
- [Square Root Sequence](#)

#### Definition (Factorial Partial Sums)

**Definition 6.12.3** (Factorial Partial Sums). The **factorial partial-sum sequence** is the real sequence  $(s_n)$  defined by

$$s_n := \sum_{k=0}^n \frac{1}{k!} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \left( s_n = \sum_{k=0}^n \frac{1}{k!} \right).$$

*Remark* (Interpretation). The sequence adds the reciprocal factorial terms one at a time. The terms decrease very rapidly, so the partial sums stabilize quickly.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Factorial Partial Sums Approximate $e$ )

**Theorem 6.12.4** (Factorial Partial Sums Approximate  $e$ ). *Let  $(s_n)$  be the factorial partial-sum sequence. Then  $(s_n)$  converges to the Euler number  $e$ :*

$$\lim_{n \rightarrow \infty} s_n = e.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|s_n - e| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(s_n, e).$$

*Remark* (Interpretation). This is the sequence form of the factorial-series definition of  $e$ . In practice the approximation is efficient because the error after  $n$  terms is controlled by a rapidly shrinking factorial tail.

*Remark* (Dependencies).

- [Factorial Partial Sums](#)
- [The Number  \$e\$](#)
- [Cauchy Criterion for Real Sequences](#)

#### Definition (Compound-Interest Sequence)

**Definition 6.12.5** (Compound-Interest Sequence). The **compound-interest sequence** is the real sequence  $(c_n)$  defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \left( c_n = \left( 1 + \frac{1}{n} \right)^n \right).$$

*Remark* (Interpretation). The term  $c_n$  is the value of one unit of principal after one unit of time when interest is compounded  $n$  times at nominal rate 1.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Compound-Interest Approximation of $e$ )

**Theorem 6.12.6** (Compound-Interest Approximation of  $e$ ). *Let  $(c_n)$  be the compound-interest sequence. Then*

$$\lim_{n \rightarrow \infty} c_n = e.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|c_n - e| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(c_n, e).$$

*Remark* (Interpretation). The compound-interest approximation gives a second natural sequence converging to the same constant as the factorial partial sums. It is slower numerically, but it explains why  $e$  appears in continuous growth.

*Remark* (Dependencies).

- [Compound-Interest Sequence](#)
- [The Number  \$e\$](#)
- [Sequence Squeeze Theorem](#)

#### Definition (Decimal Truncation Sequence)

**Definition 6.12.7** (Decimal Truncation Sequence). Let  $\alpha \in \mathbb{R}$ . The **decimal truncation sequence** of  $\alpha$  is the real sequence  $(d_n)$  defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \quad \text{for every } n \in \mathbb{N}.$$

*Remark* (Standard quantified statement).

$$\forall n \in \mathbb{N} \left( d_n = \frac{\lfloor 10^n \alpha \rfloor}{10^n} \right).$$

*Remark* (Interpretation). The number  $d_n$  is obtained by cutting  $\alpha$  off after  $n$  decimal places. Truncation is one-sided: it does not round to the nearest decimal.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Theorem (Decimal Truncations Converge)

**Theorem 6.12.8** (Decimal Truncations Converge). *Let  $\alpha \in \mathbb{R}$ , and let  $(d_n)$  be the decimal truncation sequence of  $\alpha$ . Then*

$$\lim_{n \rightarrow \infty} d_n = \alpha.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N (|d_n - \alpha| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{ConvergentSequence}(d_n, \alpha).$$

*Remark* (Interpretation). Decimal truncations converge because the truncation error is always less than  $10^{-n}$ . This is one of the most concrete examples of convergence: finite decimal data can approximate a real number to arbitrary accuracy.

*Remark* (Dependencies).

- [Decimal Truncation Sequence](#)
- [Sequence Squeeze Theorem](#)

## Proofs



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Sequence Convergence). *Let  $(x_n)$  be a real sequence, and let  $c \in \mathbb{R}$ . If  $x_n = c$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  converges to  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for constant-sequence-convergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-sequence-convergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Zero Sequence is Null). *Let  $(x_n)$  be a real sequence. If  $x_n = 0$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  is a null sequence.*

*Professional Standard Proof.*

TODO: professional standard proof for zero-sequence-is-null.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for zero-sequence-is-null.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Null Sequence). *Let  $(x_n)$  be a real sequence, and let  $c \in \mathbb{R}$ . If  $x_n = c$  for every  $n \in \mathbb{N}$ , then  $(x_n)$  is a null sequence if and only if  $c = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for constant-null-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-null-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Difference from the Limit is Null). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . The sequence  $(x_n)$  converges to  $L$  if and only if the sequence  $(x_n - L)$  is a null sequence.*

*Professional Standard Proof.*

TODO: professional standard proof for difference-from-limit-is-null. □

*Detailed Learning Proof.*

TODO: detailed learning proof for difference-from-limit-is-null. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Ultimately Constant Sequence Convergence). *Let  $(x_n)$  be a real sequence. If there exist  $N_0 \in \mathbb{N}$  and  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \geq N_0$ , then  $(x_n)$  converges to  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for ultimately-constant-sequence-convergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ultimately-constant-sequence-convergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Implies Ultimately Constant). *Every constant real sequence is ultimately constant.*

*Professional Standard Proof.*

TODO: professional standard proof for constant-implies-ultimately-constant. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-implies-ultimately-constant. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Ultimately Zero Sequence is Null). *Let  $(x_n)$  be a real sequence. If there exists  $N_0 \in \mathbb{N}$  such that  $x_n = 0$  for every  $n \geq N_0$ , then  $(x_n)$  is a null sequence.*

*Professional Standard Proof.*

TODO: professional standard proof for ultimately-zero-sequence-is-null. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ultimately-zero-sequence-is-null. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Ultimately Constant Null Sequence). *Let  $(x_n)$  be a real sequence. Suppose there exist  $N_0 \in \mathbb{N}$  and  $c \in \mathbb{R}$  such that  $x_n = c$  for every  $n \geq N_0$ . Then  $(x_n)$  is a null sequence if and only if  $c = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for ultimately-constant-null-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ultimately-constant-null-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Tail Equality Preserves Convergence). *Let  $(x_n)$  and  $(y_n)$  be real sequences. If there exists  $N_0 \in \mathbb{N}$  such that  $x_n = y_n$  for every  $n \geq N_0$ , then for every  $L \in \mathbb{R}$ ,  $(x_n)$  converges to  $L$  if and only if  $(y_n)$  converges to  $L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for tail-equality-preserves-convergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for tail-equality-preserves-convergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventually Bounded Above Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded above if and only if there exist  $N_0 \in \mathbb{N}$  and  $M \in \mathbb{R}$  such that  $x_n \leq M$  for every  $n \geq N_0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eventually-bounded-above-tail-formulation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eventually-bounded-above-tail-formulation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventually Bounded Below Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded below if and only if there exist  $N_0 \in \mathbb{N}$  and  $m \in \mathbb{R}$  such that  $m \leq x_n$  for every  $n \geq N_0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eventually-bounded-below-tail-formulation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eventually-bounded-below-tail-formulation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventually Bounded Tail Formulation). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded if and only if there exist  $N_0 \in \mathbb{N}$  and  $M > 0$  such that  $|x_n| \leq M$  for every  $n \geq N_0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eventually-bounded-tail-formulation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eventually-bounded-tail-formulation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Sequence If and Only If Bounded Above and Below). *Let  $(x_n)$  be a real sequence. The sequence  $(x_n)$  is bounded if and only if it is both bounded above and bounded below.*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-sequence-iff-bounded-above-and-below.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-sequence-iff-bounded-above-and-below.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Bound Gives Upper and Lower Bounds). *Let  $(x_n)$  be a real sequence, and let  $K > 0$ . If  $|x_n| \leq K$  for every  $n \in \mathbb{N}$ , then  $-K \leq x_n \leq K$  for every  $n \in \mathbb{N}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for absolute-bound-gives-upper-and-lower-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-bound-gives-upper-and-lower-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Upper and Lower Bounds Give an Absolute Bound). *Let  $(x_n)$  be a real sequence. If there exist  $m, M \in \mathbb{R}$  such that  $m \leq x_n \leq M$  for every  $n \in \mathbb{N}$ , then there exists  $K > 0$  such that  $|x_n| \leq K$  for every  $n \in \mathbb{N}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for upper-and-lower-bounds-give-absolute-bound. □

*Detailed Learning Proof.*

TODO: detailed learning proof for upper-and-lower-bounds-give-absolute-bound. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Scalar Multiple). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $\alpha x_n \rightarrow \alpha L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-scalar-multiple.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-scalar-multiple.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Sum). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n + y_n \rightarrow L + M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-sum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-sum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Negation). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $-x_n \rightarrow -L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-negation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-negation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Difference). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n - y_n \rightarrow L - M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-difference. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-difference. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Product). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n y_n \rightarrow LM$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-product. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-product. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Nonzero Limit is Eventually Nonzero). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$  with  $L \neq 0$ . If  $x_n \rightarrow L$ , then there exists  $N \in \mathbb{N}$  such that  $x_n \neq 0$  for every  $n \geq N$ .*

*Professional Standard Proof.*

TODO: professional standard proof for nonzero-limit-eventually-nonzero. □

*Detailed Learning Proof.*

TODO: detailed learning proof for nonzero-limit-eventually-nonzero. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Reciprocal). *Let  $(x_n)$  be a real sequence such that  $x_n \neq 0$  for every  $n \in \mathbb{N}$ , and let  $L \in \mathbb{R}$  with  $L \neq 0$ . If  $x_n \rightarrow L$ , then  $1/x_n \rightarrow 1/L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-reciprocal. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-reciprocal. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Quotient). *Let  $(x_n)$  and  $(y_n)$  be real sequences such that  $y_n \neq 0$  for every  $n \in \mathbb{N}$ , and let  $L, M \in \mathbb{R}$  with  $M \neq 0$ . If  $x_n \rightarrow L$  and  $y_n \rightarrow M$ , then  $x_n/y_n \rightarrow L/M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-quotient. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-quotient. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Square). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $x_n^2 \rightarrow L^2$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-square. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-square. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of an Absolute Value). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $|x_n| \rightarrow |L|$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-an-absolute-value. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-an-absolute-value. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Positive Limit is Eventually Positive). *Let  $(x_n)$  be a real sequence, and let  $L > 0$ . If  $x_n \rightarrow L$ , then there exists  $N \in \mathbb{N}$  such that  $0 < x_n$  for every  $n \geq N$ .*

*Professional Standard Proof.*

TODO: professional standard proof for positive-limit-eventually-positive. □

*Detailed Learning Proof.*

TODO: detailed learning proof for positive-limit-eventually-positive. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit of a Square Root). *Let  $(x_n)$  be a real sequence such that  $0 \leq x_n$  for every  $n \in \mathbb{N}$ , and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $0 \leq L$  and  $\sqrt{x_n} \rightarrow \sqrt{L}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-of-a-square-root.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-of-a-square-root.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Polynomial Sequence Limit). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $p \in \mathbb{R}[t]$  be a polynomial. If  $x_n \rightarrow L$ , then  $p(x_n) \rightarrow p(L)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for polynomial-sequence-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for polynomial-sequence-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Rational Sequence Limit). *Let  $(x_n)$  be a real sequence, let  $L \in \mathbb{R}$ , and let  $p, q \in \mathbb{R}[t]$  be polynomials. Suppose  $q(L) \neq 0$  and  $q(x_n) \neq 0$  for every  $n \in \mathbb{N}$ . If  $x_n \rightarrow L$ , then*

$$\frac{p(x_n)}{q(x_n)} \rightarrow \frac{p(L)}{q(L)}.$$

*Professional Standard Proof.*

TODO: professional standard proof for rational-sequence-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for rational-sequence-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Equivalence of Convergence Formulations). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $L \in \mathbb{R}$ . The following statements are equivalent.*

- (a)  $x_n \rightarrow L$ .
- (b)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (|x_n - L| < \varepsilon)$ .
- (c)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (L - \varepsilon < x_n < L + \varepsilon)$ .
- (d)  $\forall \varepsilon > 0 \exists K \in \mathbb{N} \forall n \geq K (x_n \in V_\varepsilon(L))$ .

*Professional Standard Proof.*

TODO: professional standard proof for equivalence-of-convergence-formulations. □

*Detailed Learning Proof.*

TODO: detailed learning proof for equivalence-of-convergence-formulations. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Uniqueness of Limits). *Let  $(x_n) \subseteq \mathbb{R}$ . If  $x_n \rightarrow L$  and  $x_n \rightarrow K$ , then  $L = K$ .*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit Preserves Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $L, M \in \mathbb{R}$ . Suppose that  $x_n \rightarrow L$  and  $y_n \rightarrow M$ . If there exists  $N_0 \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N_0$ , then  $L \leq M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-preserves-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-preserves-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Strict Limit Separation Gives Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$  and  $y_n \rightarrow B$ . If  $A < B$ , then there exists  $N \in \mathbb{N}$  such that  $x_n < y_n$  for every  $n \geq N$ .*

*Professional Standard Proof.*

TODO: professional standard proof for strict-limit-separation-gives-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for strict-limit-separation-gives-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventual Strict Comparison Preserves Weak Limit Order). *Let  $(x_n)$  and  $(y_n)$  be real sequences, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$  and  $y_n \rightarrow B$ .*

(a) *If there exists  $N \in \mathbb{N}$  such that  $x_n < y_n$  for every  $n \geq N$ , then  $A \leq B$ .*

(b) *If there exists  $N \in \mathbb{N}$  such that  $x_n > y_n$  for every  $n \geq N$ , then  $A \geq B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eventual-strict-comparison-preserves-weak-limit-order

□

*Detailed Learning Proof.*

TODO: detailed learning proof for eventual-strict-comparison-preserves-weak-limit-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Comparison for Sequence Limits). *Let  $(x_n)$  be a real sequence, and let  $A, B \in \mathbb{R}$ . Suppose that  $x_n \rightarrow A$ .*

- (a) *If there exists  $N \in \mathbb{N}$  such that  $x_n \leq B$  for every  $n \geq N$ , then  $A \leq B$ .*
- (b) *If there exists  $N \in \mathbb{N}$  such that  $x_n < B$  for every  $n \geq N$ , then  $A \leq B$ .*
- (c) *If there exists  $N \in \mathbb{N}$  such that  $x_n \geq B$  for every  $n \geq N$ , then  $A \geq B$ .*
- (d) *If there exists  $N \in \mathbb{N}$  such that  $x_n > B$  for every  $n \geq N$ , then  $A \geq B$ .*

*Professional Standard Proof.*

TODO: professional standard proof for constant-comparison-for-sequence-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-comparison-for-sequence-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Squeeze Theorem). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . If there exists  $N_0 \in \mathbb{N}$  such that  $L \leq x_n \leq L$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for constant-squeeze-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-squeeze-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sequence Squeeze Theorem). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be real sequences, and let  $L \in \mathbb{R}$ . Suppose that  $a_n \rightarrow L$  and  $b_n \rightarrow L$ . If there exists  $N_0 \in \mathbb{N}$  such that  $a_n \leq x_n \leq b_n$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequence-squeeze-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sequence-squeeze-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute-Value Squeeze Theorem). *Let  $(x_n)$  and  $(u_n)$  be real sequences, and let  $L \in \mathbb{R}$ . Suppose that  $u_n \rightarrow 0$ . If there exists  $N_0 \in \mathbb{N}$  such that  $|x_n - L| \leq u_n$  for every  $n \geq N_0$ , then  $x_n \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-squeeze-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-squeeze-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Convergence of a Tail). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $m \in \mathbb{N}$ . The  $m$ -tail  $(x_{m+n})_{n \in \mathbb{N}}$  converges if and only if  $(x_n)$  converges. Moreover, in this case,*

$$\lim_{n \rightarrow \infty} x_{m+n} = \lim_{n \rightarrow \infty} x_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for convergence-of-tail. □

*Detailed Learning Proof.*

TODO: detailed learning proof for convergence-of-tail. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Convergence by Domination). *Let  $(x_n) \subseteq \mathbb{R}$  and let  $L \in \mathbb{R}$ . Let  $(a_n)$  be a sequence of positive real numbers with  $\lim_{n \rightarrow \infty} a_n = 0$ . If there exist a constant  $c > 0$  and an index  $m \in \mathbb{N}$  such that*

$$|x_n - L| \leq c a_n \quad \forall n \geq m,$$

*then  $\lim_{n \rightarrow \infty} x_n = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for convergence-by-domination. □

*Detailed Learning Proof.*

TODO: detailed learning proof for convergence-by-domination. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Ratio Limit Less Than One Implies Null). *Let  $(x_n)$  be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$$

*exists. If  $L < 1$ , then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for ratio-limit-less-than-one-implies-null. □

*Detailed Learning Proof.*

TODO: detailed learning proof for ratio-limit-less-than-one-implies-null. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotone Convergence Theorem). *Let  $(x_n)$  be a real sequence.*

(a) *If  $(x_n)$  is increasing and bounded above, then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) *If  $(x_n)$  is decreasing and bounded below, then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for monotone-convergence-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-convergence-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Strict Monotonicity Implies Monotonicity). *Let  $(x_n)$  be a real sequence.*

(a) *If  $(x_n)$  is strictly increasing, then  $(x_n)$  is increasing.*

(b) *If  $(x_n)$  is strictly decreasing, then  $(x_n)$  is decreasing.*

*Professional Standard Proof.*

TODO: professional standard proof for strict-monotonicity-implies-monotonicity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for strict-monotonicity-implies-monotonicity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Monotone Sequence Equivalences). *Let  $(x_n)$  be a real sequence.*

- (a) If  $(x_n)$  is increasing, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded above.*
- (b) If  $(x_n)$  is decreasing, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded below.*
- (c) If  $(x_n)$  is monotone, then  $(x_n)$  converges if and only if  $(x_n)$  is bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-monotone-sequence-equivalences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-monotone-sequence-equivalences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventually Monotone Convergence Theorem). *Let  $(x_n)$  be a real sequence.*

*(a) If  $(x_n)$  is eventually increasing and bounded above, then  $(x_n)$  converges.*

*(b) If  $(x_n)$  is eventually decreasing and bounded below, then  $(x_n)$  converges.*

*(c) If  $(x_n)$  is eventually monotone and bounded, then  $(x_n)$  converges.*

*Professional Standard Proof.*

TODO: professional standard proof for eventually-monotone-convergence-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eventually-monotone-convergence-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Unbounded Monotone Divergence). *Let  $(x_n)$  be a real sequence.*

*(a) If  $(x_n)$  is increasing and is not bounded above, then  $x_n \rightarrow +\infty$ .*

*(b) If  $(x_n)$  is decreasing and is not bounded below, then  $x_n \rightarrow -\infty$ .*

*Professional Standard Proof.*

TODO: professional standard proof for unbounded-monotone-divergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for unbounded-monotone-divergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Algebraic Transformations Preserve Monotonicity). *Let  $(x_n)$  be a real sequence and let  $c, \alpha \in \mathbb{R}$ .*

- (a) *If  $(x_n)$  is increasing, then  $(x_n + c)$  is increasing.*
- (b) *If  $(x_n)$  is decreasing, then  $(x_n + c)$  is decreasing.*
- (c) *If  $(x_n)$  is increasing and  $\alpha > 0$ , then  $(\alpha x_n)$  is increasing.*
- (d) *If  $(x_n)$  is increasing and  $\alpha < 0$ , then  $(\alpha x_n)$  is decreasing.*
- (e) *If  $(x_n)$  is increasing, then  $(-x_n)$  is decreasing.*
- (f) *If  $(x_n)$  is decreasing, then  $(-x_n)$  is increasing.*

*Professional Standard Proof.*

TODO: professional standard proof for algebraic-transformations-preserve-monotonicity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for algebraic-transformations-preserve-monotonicity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Subsequence Indices Dominate the Identity). *Let  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  be a strictly increasing index map. Then*

$$k \leq \sigma(k) \quad \text{for every } k \in \mathbb{N}.$$

*Professional Standard Proof.*

TODO: professional standard proof for subsequence-indices-dominate-identity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsequence-indices-dominate-identity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Subsequences Preserve Limits). *Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then every subsequence of  $(x_n)$  converges to  $L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsequences-preserve-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsequences-preserve-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Subsequential Limit of a Convergent Sequence). *Let  $(x_n)$  be a real sequence and let  $L \in \mathbb{R}$ . If  $x_n \rightarrow L$ , then  $L$  is the only possible subsequential limit of  $(x_n)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsequential-limit-of-convergent-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsequential-limit-of-convergent-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Divergence by Two Subsequential Limits). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  has two distinct subsequential limits, then  $(x_n)$  does not converge.*

*Professional Standard Proof.*

TODO: professional standard proof for divergence-by-two-subsequential-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for divergence-by-two-subsequential-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for boundedness-passes-to-subsequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for boundedness-passes-to-subsequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

*Professional Standard Proof.*

TODO: professional standard proof for monotonicity-passes-to-subsequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotonicity-passes-to-subsequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Subsequence of a Subsequence). *Every subsequence of a subsequence of  $(x_n)$  is a subsequence of  $(x_n)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsequence-of-subsequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsequence-of-subsequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Eventual Properties Pass to Subsequences). *Let  $P(n)$  be a property of indices. If there exists  $N \in \mathbb{N}$  such that  $P(n)$  holds for every  $n \geq N$ , then for every strictly increasing index map  $\sigma$  there exists  $K \in \mathbb{N}$  such that  $P(\sigma(k))$  holds for every  $k \geq K$ .*

*Professional Standard Proof.*

TODO: professional standard proof for eventual-properties-pass-to-subsequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for eventual-properties-pass-to-subsequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Frequent Properties Yield Subsequences). *Let  $P(n)$  be a property of indices. If for every  $N \in \mathbb{N}$  there exists  $n \geq N$  such that  $P(n)$  holds, then there exists a strictly increasing index map  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$  such that  $P(\sigma(k))$  holds for every  $k \in \mathbb{N}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for frequent-properties-*yield*-subsequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for frequent-properties-*yield*-subsequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Subsequential Limits Respect Bounds). *Let  $(x_n)$  be a real sequence and let  $L$  be a subsequential limit of  $(x_n)$ . If  $m \leq x_n \leq M$  for every  $n \in \mathbb{N}$ , then  $m \leq L \leq M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for subsequential-limits-respect-bounds. □

*Detailed Learning Proof.*

TODO: detailed learning proof for subsequential-limits-respect-bounds. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Squeeze Passes to Subsequences). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be real sequences. If  $a_n \leq x_n \leq b_n$  eventually, then for every subsequence  $(x_{\sigma(k)})$  there exists  $K \in \mathbb{N}$  such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \quad \text{for every } k \geq K.$$

*Professional Standard Proof.*

TODO: professional standard proof for squeeze-passes-to-subsequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for squeeze-passes-to-subsequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotone Subsequence Theorem). *Every real sequence has a monotone subsequence.*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-subsequence-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-subsequence-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bolzano-Weierstrass Theorem for Sequences). *Every bounded real sequence has a convergent subsequence.*

*Professional Standard Proof.*

TODO: professional standard proof for bolzano-weierstrass-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bolzano-weierstrass-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sequential Compactness of Closed Bounded Intervals). *Let  $a, b \in \mathbb{R}$  with  $a \leq b$ . Every sequence in  $[a, b]$  has a convergent subsequence whose limit belongs to  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequential-compactness-closed-bounded-interval.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for sequential-compactness-closed-bounded-interval.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Divergence to Infinity Implies Real Divergence). *Let  $(x_n)$  be a real sequence. If  $x_n \rightarrow +\infty$  or  $x_n \rightarrow -\infty$ , then  $(x_n)$  is divergent.*

*Professional Standard Proof.*

TODO: professional standard proof for divergence-to-infinity-implies-real-divergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for divergence-to-infinity-implies-real-divergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Two Subsequential Limits Force Divergence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  has two subsequential limits  $L, K \in \mathbb{R}$  with  $L \neq K$ , then  $(x_n)$  is divergent.*

*Professional Standard Proof.*

TODO: professional standard proof for two-subsequential-limits-force-divergence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for two-subsequential-limits-force-divergence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Unbounded Above Sequence Has a Positive-Infinity Subsequence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  is not bounded above, then there exists a subsequence  $(x_{\sigma(k)})$  such that  $x_{\sigma(k)} \rightarrow +\infty$ .*

*Professional Standard Proof.*

TODO: professional standard proof for unbounded-above-has-positive-infinity-subsequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for unbounded-above-has-positive-infinity-subsequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Unbounded Below Sequence Has a Negative-Infinity Subsequence). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  is not bounded below, then there exists a subsequence  $(x_{\sigma(k)})$  such that  $x_{\sigma(k)} \rightarrow -\infty$ .*

*Professional Standard Proof.*

TODO: professional standard proof for unbounded-below-has-negative-infinity-subsequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for unbounded-below-has-negative-infinity-subsequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Divergence Produces Two Subsequential Limits). *Let  $(x_n)$  be a bounded real sequence. If  $(x_n)$  is divergent, then there exist  $L, K \in \mathbb{R}$  with  $L \neq K$  such that  $L$  and  $K$  are subsequential limits of  $(x_n)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-divergence-produces-two-subsequential-limits ■

□

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-divergence-produces-two-subsequential-limits.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Tail Suprema Are Decreasing). *Let  $(x_n)$  be a bounded real sequence, and let  $(s_n)$  be its tail supremum sequence. Then  $(s_n)$  is decreasing.*

*Professional Standard Proof.*

TODO: professional standard proof for tail-suprema-are-decreasing. □

*Detailed Learning Proof.*

TODO: detailed learning proof for tail-suprema-are-decreasing. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Tail Infima Are Increasing). *Let  $(x_n)$  be a bounded real sequence, and let  $(i_n)$  be its tail infimum sequence. Then  $(i_n)$  is increasing.*

*Professional Standard Proof.*

TODO: professional standard proof for tail-infima-are-increasing. □

*Detailed Learning Proof.*

TODO: detailed learning proof for tail-infima-are-increasing. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Liminf Is Below Limsup). *Let  $(x_n)$  be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for liminf-below-limsup. □

*Detailed Learning Proof.*

TODO: detailed learning proof for liminf-below-limsup. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Convergence iff Liminf Equals Limsup). *Let  $(x_n)$  be a bounded real sequence, and let  $L \in \mathbb{R}$ . Then  $x_n \rightarrow L$  if and only if*

$$\liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L.$$

*Professional Standard Proof.*

TODO: professional standard proof for convergence-iff-liminf-equals-limsup. □

*Detailed Learning Proof.*

TODO: detailed learning proof for convergence-iff-liminf-equals-limsup. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limsup Is the Largest Subsequential Limit). *Let  $(x_n)$  be a bounded real sequence, and let  $S = \limsup_{n \rightarrow \infty} x_n$ . Then  $S$  is a subsequential limit of  $(x_n)$ , and every subsequential limit of  $(x_n)$  is less than or equal to  $S$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limsup-largest-subsequential-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-largest-subsequential-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Liminf Is the Smallest Subsequential Limit). *Let  $(x_n)$  be a bounded real sequence, and let  $I = \liminf_{n \rightarrow \infty} x_n$ . Then  $I$  is a subsequential limit of  $(x_n)$ , and every subsequential limit of  $(x_n)$  is greater than or equal to  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for liminf-smallest-subsequential-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for liminf-smallest-subsequential-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Oscillation Criterion via Liminf and Limsup). *Let  $(x_n)$  be a bounded real sequence. Then*

$$\liminf_{n \rightarrow \infty} x_n < \limsup_{n \rightarrow \infty} x_n$$

*if and only if  $(x_n)$  has two distinct subsequential limits.*

*Professional Standard Proof.*

TODO: professional standard proof for oscillation-criterion-via-liminf-limsup. □

*Detailed Learning Proof.*

TODO: detailed learning proof for oscillation-criterion-via-liminf-limsup. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limsup Comparison Under Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N$ , then*

$$\limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for limsup-comparison-under-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-comparison-under-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Liminf Comparison Under Eventual Order). *Let  $(x_n)$  and  $(y_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that  $x_n \leq y_n$  for every  $n \geq N$ , then*

$$\liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for liminf-comparison-under-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for liminf-comparison-under-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limsup Squeeze Under Eventual Order). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

*then*

$$\limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} b_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for limsup-squeeze-under-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-squeeze-under-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Liminf Squeeze Under Eventual Order). *Let  $(a_n)$ ,  $(x_n)$ , and  $(b_n)$  be bounded real sequences. If there exists  $N \in \mathbb{N}$  such that*

$$a_n \leq x_n \leq b_n \quad \text{for every } n \geq N,$$

*then*

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} b_n.$$

*Professional Standard Proof.*

TODO: professional standard proof for liminf-squeeze-under-eventual-order. □

*Detailed Learning Proof.*

TODO: detailed learning proof for liminf-squeeze-under-eventual-order. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Increasing Sequence Limit as Supremum). *Let  $(x_n)$  be an increasing real sequence that is bounded above. Then*

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for increasing-sequence-limit-as-supremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for increasing-sequence-limit-as-supremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decreasing Sequence Limit as Infimum). *Let  $(x_n)$  be a decreasing real sequence that is bounded below. Then*

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for decreasing-sequence-limit-as-infimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decreasing-sequence-limit-as-infimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Tail Suprema and Infima Converge). *Let  $(x_n)$  be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \quad i_n = \inf\{x_k : k \geq n\}.$$

*Then  $(s_n)$  and  $(i_n)$  converge.*

*Professional Standard Proof.*

TODO: professional standard proof for tail-suprema-and-infima-converge. □

*Detailed Learning Proof.*

TODO: detailed learning proof for tail-suprema-and-infima-converge. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-sequence-has-limsup-and-liminf. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-sequence-has-limsup-and-liminf. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cluster Values Are Subsequential Limits). *Let  $(x_n)$  be a real sequence, and let  $L \in \mathbb{R}$ . Then  $L$  is a cluster value of  $(x_n)$  if and only if  $L$  is a subsequential limit of  $(x_n)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cluster-values-are-subsequential-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cluster-values-are-subsequential-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-sequences-have-cluster-values. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-sequences-have-cluster-values. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limsup and Liminf Are Extremal Cluster Values). *Let  $(x_n)$  be a bounded real sequence. Then  $\limsup_{n \rightarrow \infty} x_n$  is the largest cluster value of  $(x_n)$ , and  $\liminf_{n \rightarrow \infty} x_n$  is the smallest cluster value of  $(x_n)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limsup-liminf-extremal-cluster-values. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-liminf-extremal-cluster-values. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Convergent Sequences Are Cauchy). *Let  $(x_n)$  be a real sequence. If  $(x_n)$  converges, then  $(x_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for convergent-sequences-are-cauchy. □

*Detailed Learning Proof.*

TODO: detailed learning proof for convergent-sequences-are-cauchy. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-sequences-are-bounded.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-sequences-are-bounded.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Sequence with a Convergent Subsequence). *Let  $(x_n)$  be a Cauchy real sequence. If a subsequence of  $(x_n)$  converges to  $L \in \mathbb{R}$ , then  $(x_n)$  converges to  $L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-sequence-with-convergent-subsequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-sequence-with-convergent-subsequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Criterion for Real Sequences). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  converges if and only if  $(x_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-criterion-real-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-criterion-real-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Criterion via Tails). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  is Cauchy if and only if for every  $\varepsilon > 0$  there exists  $N \in \mathbb{N}$  such that the  $N$ -tail has pairwise term distances less than  $\varepsilon$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-criterion-via-tails. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-criterion-via-tails. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Tail Diameter Criterion). *Let  $(x_n)$  be a real sequence. Then  $(x_n)$  is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-tail-diameter-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-tail-diameter-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Sequences Have Vanishing Successive Differences). *If  $(x_n)$  is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \rightarrow 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-successive-differences-vanish. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-successive-differences-vanish. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Scalar Multiples of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence, and let  $\alpha \in \mathbb{R}$ . Then  $(\alpha x_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for scalar-multiple-cauchy-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for scalar-multiple-cauchy-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sums of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n + y_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for sum-cauchy-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sum-cauchy-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Differences of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n - y_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for difference-cauchy-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for difference-cauchy-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Linear Combinations of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences, and let  $\alpha, \beta \in \mathbb{R}$ . Then  $(\alpha x_n + \beta y_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for linear-combination-cauchy-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for linear-combination-cauchy-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Products of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. Then  $(x_n y_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for product-cauchy-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for product-cauchy-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Reciprocals of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence. If there exist  $c > 0$  and  $N_0 \in \mathbb{N}$  such that  $|x_n| \geq c$  for every  $n \geq N_0$ , then  $(1/x_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for reciprocal-cauchy-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for reciprocal-cauchy-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Quotients of Cauchy Sequences). *Let  $(x_n)$  and  $(y_n)$  be Cauchy real sequences. If there exist  $c > 0$  and  $N_0 \in \mathbb{N}$  such that  $|y_n| \geq c$  for every  $n \geq N_0$ , then  $(x_n/y_n)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for quotient-cauchy-sequences. □

*Detailed Learning Proof.*

TODO: detailed learning proof for quotient-cauchy-sequences. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Values of Cauchy Sequences). *Let  $(x_n)$  be a Cauchy real sequence. Then  $(|x_n|)$  is Cauchy.*

*Professional Standard Proof.*

TODO: professional standard proof for absolute-value-cauchy-sequence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute-value-cauchy-sequence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Newton Approximation of 2). *Let  $(x_n)$  be the Newton sequence for  $\sqrt{2}$ . Then  $(x_n)$  converges and*

$$\lim_{n \rightarrow \infty} x_n = \sqrt{2}.$$

*Professional Standard Proof.*

TODO: professional standard proof for newton-approximation-sqrt-two. □

*Detailed Learning Proof.*

TODO: detailed learning proof for newton-approximation-sqrt-two. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Factorial Partial Sums Approximate  $e$ ). *Let  $(s_n)$  be the factorial partial-sum sequence. Then  $(s_n)$  converges to the Euler number  $e$ :*

$$\lim_{n \rightarrow \infty} s_n = e.$$

*Professional Standard Proof.*

TODO: professional standard proof for factorial-partial-sums-approximate-e. □

*Detailed Learning Proof.*

TODO: detailed learning proof for factorial-partial-sums-approximate-e. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Compound-Interest Approximation of  $e$ ). *Let  $(c_n)$  be the compound-interest sequence. Then*

$$\lim_{n \rightarrow \infty} c_n = e.$$

*Professional Standard Proof.*

TODO: professional standard proof for compound-interest-approximation-e. □

*Detailed Learning Proof.*

TODO: detailed learning proof for compound-interest-approximation-e. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Decimal Truncations Converge). *Let  $\alpha \in \mathbb{R}$ , and let  $(d_n)$  be the decimal truncation sequence of  $\alpha$ . Then*

$$\lim_{n \rightarrow \infty} d_n = \alpha.$$

*Professional Standard Proof.*

TODO: professional standard proof for decimal-truncations-converge. □

*Detailed Learning Proof.*

TODO: detailed learning proof for decimal-truncations-converge. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

# Capstone

## Capstone Theorem: The Contraction Mapping Principle on $\mathbb{R}$

**Theorem.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ . Suppose there is a constant  $L$  with  $0 \leq L < 1$  such that

$$|f(x) - f(y)| \leq L|x - y| \quad \text{for all } x, y \in \mathbb{R}.$$

Fix  $x_0 \in \mathbb{R}$ , and define  $x_{n+1} = f(x_n)$ . Then  $(x_n)$  converges to a point  $x^* \in \mathbb{R}$ ,

$$f(x^*) = x^*,$$

this fixed point is unique, every starting point converges to it, and

$$|x_n - x^*| \leq L^n |x_0 - x^*|, \quad |x_n - x^*| \leq \frac{L^n}{1 - L} |x_1 - x_0|.$$

*Remark (Abstract).* This capstone is the chapter's completeness payoff. The hypothesis is only a distance-shrinking inequality; no continuity theorem is used. The orbit first proves itself Cauchy by geometric decay of its successive steps. Completeness then supplies a limit, and only afterward does the contraction inequality identify that limit as the fixed point.

*Remark (Orientation).* The orbit is the sequence

$$x_0, \quad f(x_0), \quad f(f(x_0)), \quad \dots$$

A fixed point is a point  $x^*$  satisfying  $f(x^*) = x^*$ . The contraction hypothesis says that applying  $f$  pulls every pair of points at least a factor  $L < 1$  closer. The proof cannot aim at  $x^*$ , because no fixed point has been produced yet. Instead it shows that the orbit crowds together internally, invokes the Cauchy criterion, and then checks what the limit must be.

*Remark (Step 1: one step is controlled by the previous step).* For every  $n \geq 1$ ,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| \leq L|x_n - x_{n-1}|.$$

This is the contraction hypothesis applied to the pair  $(x_n, x_{n-1})$ .

*Remark (Step 2: every step is controlled by the first step).* For every  $n \geq 0$ ,

$$|x_{n+1} - x_n| \leq L^n |x_1 - x_0|.$$

The case  $n = 0$  is equality. If the estimate holds at  $n - 1$ , then Step 1 gives

$$|x_{n+1} - x_n| \leq L|x_n - x_{n-1}| \leq L \cdot L^{n-1} |x_1 - x_0| = L^n |x_1 - x_0|.$$

Induction proves the estimate.

*Remark (Step 3: the orbit is Cauchy).* Let  $m > n$ . Telescoping and applying the triangle inequality gives

$$|x_m - x_n| = \left| \sum_{k=n}^{m-1} (x_{k+1} - x_k) \right| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq \sum_{k=n}^{m-1} L^k |x_1 - x_0|.$$



Since  $0 \leq L < 1$ ,

$$\sum_{k=n}^{m-1} L^k = L^n \frac{1 - L^{m-n}}{1 - L} \leq \frac{L^n}{1 - L}.$$

Hence

$$|x_m - x_n| \leq \frac{L^n}{1 - L} |x_1 - x_0|. \quad (*)$$

Because  $L^n \rightarrow 0$ , the right-hand side tends to 0. Thus for every  $\varepsilon > 0$  there is  $N$  such that  $m > n \geq N$  implies  $|x_m - x_n| < \varepsilon$ . The orbit is Cauchy.

*Remark* (Step 4: completeness produces the limit). By the Cauchy criterion for real sequences, every Cauchy sequence of real numbers converges. Therefore there is  $x^* \in \mathbb{R}$  such that

$$x_n \rightarrow x^*.$$

This is the only step in which a limit not given by a formula is brought into existence.

*Remark* (Step 5: the limit is a fixed point). For every  $n$ ,

$$\begin{aligned} |x^* - f(x^*)| &\leq |x^* - x_{n+1}| + |x_{n+1} - f(x^*)| \\ &= |x^* - x_{n+1}| + |f(x_n) - f(x^*)| \\ &\leq |x^* - x_{n+1}| + L|x_n - x^*|. \end{aligned}$$

The shifted sequence  $x_{n+1}$  also converges to  $x^*$ , and  $|x_n - x^*| \rightarrow 0$ . The right-hand side therefore tends to 0. Since  $|x^* - f(x^*)|$  is a fixed nonnegative number bounded above by a null sequence, it is 0. Hence  $f(x^*) = x^*$ .

*Remark* (Step 6: uniqueness and convergence from every start). If  $p$  and  $q$  are fixed points, then

$$|p - q| = |f(p) - f(q)| \leq L|p - q|.$$

Thus

$$0 \leq (1 - L)|p - q| \leq 0.$$

Since  $1 - L > 0$ , we have  $|p - q| = 0$ , so  $p = q$ . The construction above started from an arbitrary  $x_0$ , and uniqueness forces every such orbit to converge to the same fixed point.

*Remark* (Step 7: rates). Once  $x^*$  is known to be fixed,

$$|x_n - x^*| = |f(x_{n-1}) - f(x^*)| \leq L|x_{n-1} - x^*|.$$

Induction gives

$$|x_n - x^*| \leq L^n |x_0 - x^*|.$$

For the computable estimate, fix  $n$  in  $(*)$  and let  $m \rightarrow \infty$ . Using  $x_m \rightarrow x^*$  and preservation of eventual order under limits gives

$$|x_n - x^*| \leq \frac{L^n}{1 - L} |x_1 - x_0|.$$

*Remark* (Limits of the theorem). The strict inequality  $L < 1$  is essential. At  $L = 1$ , a map may preserve distances without having any fixed point; for example  $f(x) = x + 1$ . The general theorem replaces  $\mathbb{R}$  by any complete metric space and  $|x - y|$  by the metric  $d(x, y)$ . That version is the Banach fixed-point theorem and belongs to the metric-spaces layer.

*Remark* (Forward connection). Later, the continuity chapter names the hypothesis as a Lipschitz condition with constant  $L < 1$ . Here the name is deliberately avoided: the proof uses only the inequality, geometric decay, and Cauchy completeness. The same iteration idea becomes Picard iteration when differential equations are recast as fixed-point problems for integral operators.

*Remark* (Dependencies to state). • [Sequence](#)

- [Convergence of a Sequence](#)
- [Cauchy Sequence](#)

*Remark* (Dependencies to prove). • [Cauchy Criterion for Real Sequences](#)

- [Ratio Limit Less Than One Implies Null](#)
- [Subsequences Preserve Limits](#)
- [Constant Comparison for Sequence Limits](#)
- [Limit Preserves Eventual Order](#)

*Remark* (Dependency ceiling). This capstone may use only real-number arithmetic, absolute value and the triangle inequality, finite geometric sums, and sequence results routed at or before the Sequences chapter. It must not use continuity, Lipschitz maps as a named concept, metric-space completeness, or the Banach fixed-point theorem.

*Remark* (Dependencies).

- [Sequence](#)
- [Convergence of a Sequence](#)
- [Cauchy Sequence](#)
- [Cauchy Criterion for Real Sequences](#)
- [Ratio Limit Less Than One Implies Null](#)
- [Subsequences Preserve Limits](#)
- [Constant Comparison for Sequence Limits](#)
- [Limit Preserves Eventual Order](#)

# Chapter 7

## Series



### 7.1 Finite Series

*Remark* (Exposition). Tao introduces finite series by first fixing a finite real sequence indexed by consecutive integers and then defining its sum recursively [Tao \[2006\]](#). The source bundles these conventions into one definition; here they are split into atomic definitions.

#### Definition (Finite Real Sequence on an Integer Interval)

**Definition 7.1.1** (Finite Real Sequence on an Integer Interval). Let  $m, n \in \mathbb{Z}$  with  $m \leq n$ . A finite real sequence indexed from  $m$  to  $n$  is an assignment of a real number  $a_i$  to each integer  $i$  satisfying  $m \leq i \leq n$ . We denote such a sequence by  $(a_i)_{i=m}^n$ .

*Remark* (Interpretation). The indices form the integer interval  $\{m, m+1, \dots, n\}$ . The notation  $(a_i)_{i=m}^n$  records both the terms and the finite range of indices on which the terms are defined.

*Remark* (Dependencies).

- [Real Sequence](#)

#### Definition (Empty Finite Sum)

**Definition 7.1.2** (Empty Finite Sum). Let  $m, n \in \mathbb{Z}$ . If  $n < m$ , the finite sum from  $m$  to  $n$  is defined by

$$\sum_{i=m}^n a_i := 0.$$

*Remark* (Interpretation). When the upper index lies below the lower index, there are no terms to add. The sum is therefore assigned the additive identity 0.

*Remark* (Dependencies).

- [Finite Real Sequence on an Integer Interval](#)

#### Definition (Finite Sum)

**Definition 7.1.3** (Finite Sum). Let  $m, n \in \mathbb{Z}$ , and let real numbers  $a_i$  be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^n a_i$$

denotes the finite sum, or finite series, of the terms indexed from  $m$  to  $n$ .

*Remark* (Standard quantified statement).

$$\sum_{i=m}^n a_i \quad \text{denotes the finite sum of the terms } a_i \text{ indexed by } m \leq i \leq n.$$

*Remark* (Interpretation). This definition fixes the notation and name of the object being defined. The value of the object is then determined by the empty-sum convention and the recursive rule below.

*Remark* (Dependencies).

- [Finite Real Sequence on an Integer Interval](#)

#### Definition (Recursive Finite Sum)

**Definition 7.1.4** (Recursive Finite Sum). Let  $m \in \mathbb{Z}$ , and let real numbers  $a_i$  be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left( \sum_{i=m}^n a_i \right) + a_{n+1} \quad \text{whenever } n \geq m - 1.$$

*Remark* (Standard quantified statement).

$$\forall m, n \in \mathbb{Z}, \quad n \geq m - 1 \implies \sum_{i=m}^{n+1} a_i := \left( \sum_{i=m}^n a_i \right) + a_{n+1}.$$

*Remark* (Interpretation). The recursive rule says that extending the upper endpoint by one index extends the sum by adding exactly the new last term. Together with the empty-sum convention, this determines every finite sum.

*Remark* (Examples). For a sequence indexed from  $m$ , the first few cases are

$$\sum_{i=m}^{m-2} a_i = 0, \quad \sum_{i=m}^{m-1} a_i = 0, \quad \sum_{i=m}^m a_i = a_m,$$

and

$$\sum_{i=m}^{m+1} a_i = a_m + a_{m+1}, \quad \sum_{i=m}^{m+2} a_i = a_m + a_{m+1} + a_{m+2}.$$

*Remark* (Dependencies).

- [Empty Finite Sum](#)
- [Finite Sum](#)
- [Finite Real Sequence on an Integer Interval](#)

#### Lemma (Splitting a Finite Sum)

**Lemma 7.1.5** (Splitting a Finite Sum). *Let  $m \leq n < p$  be integers, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq p$ . Then*

$$\sum_{i=m}^n a_i + \sum_{i=n+1}^p a_i = \sum_{i=m}^p a_i.$$

*Go to proof.*

*Remark* (Interpretation). A finite sum over one integer interval may be split into two adjacent finite sums without changing its value.

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

#### Definition (Summation over a Finite Set)

**Definition 7.1.6** (Summation over a Finite Set). Let  $X$  be a finite set with  $n$  elements, where  $n \in \mathbb{N}$ , and let  $f : X \rightarrow \mathbb{R}$ . Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X.$$

The finite sum of  $f$  over  $X$  is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^n f(g(i)).$$

*Remark* (Standard quantified statement).

$$X = \{g(1), \dots, g(n)\} \implies \sum_{x \in X} f(x) := \sum_{i=1}^n f(g(i)).$$

*Remark (Interpretation).* The definition computes a sum over an unordered finite set by first listing its elements through a bijection from  $\{1, \dots, n\}$ . The notation  $\sum_{x \in X} f(x)$  records the set being summed over; the chosen listing is auxiliary.

*Remark (Dependencies).*

- [Finite Sum](#)

#### Proposition (Finite Summations Are Well-Defined)

**Proposition 7.1.7** (Finite Summations Are Well-Defined). *Let  $X$  be a finite set with  $n$  elements, where  $n \in \mathbb{N}$ , let  $f : X \rightarrow \mathbb{R}$ , and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X \quad \text{and} \quad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X$$

*be bijections. Then*

$$\sum_{i=1}^n f(g(i)) = \sum_{i=1}^n f(h(i)).$$

[Go to proof.](#)

*Remark (Interpretation).* The value of a finite summation over a finite set does not depend on which bijection is used to list the elements of that set. Tao notes that the corresponding issue for summations over infinite sets is more complicated.

*Remark (Dependencies).*

- [Summation over a Finite Set](#)

#### Proposition (Empty Finite Set Summation)

**Proposition 7.1.8** (Empty Finite Set Summation). *If  $X$  is empty, and  $f : X \rightarrow \mathbb{R}$  is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

[Go to proof.](#)

*Remark (Interpretation).* The finite summation over the empty set has value 0.

*Remark (Dependencies).*

- [Summation over a Finite Set](#)

#### Proposition (Singleton Finite Set Summation)

**Proposition 7.1.9** (Singleton Finite Set Summation). *If  $X$  consists of a single element,  $X = \{x_0\}$ , and  $f : X \rightarrow \mathbb{R}$  is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

[Go to proof.](#)

*Remark* (Interpretation). Summing a function over a one-element set gives the value of the function at that element.

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

#### Proposition (Substitution for Finite Set Summation)

**Proposition 7.1.10** (Substitution for Finite Set Summation). *If  $X$  is a finite set,  $f : X \rightarrow \mathbb{R}$  is a function, and  $g : Y \rightarrow X$  is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

*Go to proof.*

*Remark* (Interpretation). A finite summation over  $X$  may be computed by summing over any set bijective with  $X$ , after composing the summand with the bijection.

*Remark* (Dependencies).

- [Summation over a Finite Set](#)
- [Finite Summations Are Well-Defined](#)

#### Proposition (Integer Interval as a Finite Set)

**Proposition 7.1.11** (Integer Interval as a Finite Set). *Let  $n \leq m$  be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

*If  $a_i$  is a real number assigned to each integer  $i \in X$ , then*

$$\sum_{i=n}^m a_i = \sum_{i \in X} a_i.$$

*Go to proof.*

*Remark* (Interpretation). Summing over a consecutive integer interval agrees with summing over that same interval regarded as a finite set.

*Remark* (Dependencies).

- [Finite Sum](#)
- [Summation over a Finite Set](#)

### Proposition (Disjoint Union Finite Set Summation)

**Proposition 7.1.12** (Disjoint Union Finite Set Summation). *Let  $X$  and  $Y$  be disjoint finite sets, so  $X \cap Y = \emptyset$ , and let  $f : X \cup Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left( \sum_{x \in X} f(x) \right) + \left( \sum_{y \in Y} f(y) \right).$$

*Go to proof.*

*Remark (Interpretation).* A sum over a disjoint union splits into the sum over each component set.

*Remark (Dependencies).*

- [Summation over a Finite Set](#)

### Proposition (Addition Rule for Finite Set Summation)

**Proposition 7.1.13** (Addition Rule for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  and  $g : X \rightarrow \mathbb{R}$  be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

*Go to proof.*

*Remark (Interpretation).* Finite set summation distributes over pointwise addition.

*Remark (Dependencies).*

- [Summation over a Finite Set](#)

### Proposition (Scalar Multiplication Rule for Finite Set Summation)

**Proposition 7.1.14** (Scalar Multiplication Rule for Finite Set Summation). *Let  $X$  be a finite set, let  $f : X \rightarrow \mathbb{R}$  be a function, and let  $c$  be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

*Go to proof.*

*Remark (Interpretation).* A constant scalar may be pulled outside a finite set summation.

*Remark (Dependencies).*

- [Summation over a Finite Set](#)



### Proposition (Monotonicity for Finite Set Summation)

**Proposition 7.1.15** (Monotonicity for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  and  $g : X \rightarrow \mathbb{R}$  be functions such that  $f(x) \leq g(x)$  for all  $x \in X$ . Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$(\forall x \in X, f(x) \leq g(x)) \implies \sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

*Remark* (Interpretation). Pointwise order between two functions on a finite set gives the same order between their finite sums.

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

### Proposition (Triangle Inequality for Finite Set Summation)

**Proposition 7.1.16** (Triangle Inequality for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

*Go to proof.*

*Remark* (Interpretation). The absolute value of a finite set summation is bounded by the corresponding finite set summation of absolute values.

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

### Lemma (Iterated Summation over a Product)

**Lemma 7.1.17** (Iterated Summation over a Product). *Let  $X$  and  $Y$  be finite sets, and let  $f : X \times Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{x \in X} \left( \sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y).$$

*Go to proof.*

*Remark* (Interpretation). Summing first over  $Y$  for each  $x \in X$ , and then summing those values over  $X$ , gives the same value as summing over the product set  $X \times Y$ .

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

#### Corollary (Fubini's Theorem for Finite Series)

**Corollary 7.1.18** (Fubini's Theorem for Finite Series). *Let  $X$  and  $Y$  be finite sets, and let  $f : X \times Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{x \in X} \left( \sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y) = \sum_{(y, x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left( \sum_{x \in X} f(x, y) \right).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\sum_{x \in X} \left( \sum_{y \in Y} f(x, y) \right) = \sum_{y \in Y} \left( \sum_{x \in X} f(x, y) \right).$$

*Remark* (Interpretation). For a function on a finite product set, the order of summation over the two finite factors may be interchanged.

*Remark* (Dependencies).

- [Iterated Summation over a Product](#)
- [Summation over a Finite Set](#)

#### Lemma (Reindexing a Finite Sum)

**Lemma 7.1.19** (Reindexing a Finite Sum). *Let  $m \leq n$  be integers, let  $k$  be an integer, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

*Go to proof.*

*Remark* (Interpretation). Shifting every index by the same integer changes the labels of the summation but not the ordered list of terms being summed.

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

### Lemma (Finite Sum of Sums)

**Lemma 7.1.20** (Finite Sum of Sums). *Let  $m \leq n$  be integers, and let  $a_i$  and  $b_i$  be real numbers assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n (a_i + b_i) = \left( \sum_{i=m}^n a_i \right) + \left( \sum_{i=m}^n b_i \right).$$

*Go to proof.*

*Remark* (Interpretation). Finite summation distributes over termwise addition.

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

### Lemma (Finite Sum of Scalar Multiples)

**Lemma 7.1.21** (Finite Sum of Scalar Multiples). *Let  $m \leq n$  be integers, let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ , and let  $c \in \mathbb{R}$ . Then*

$$\sum_{i=m}^n (ca_i) = c \left( \sum_{i=m}^n a_i \right).$$

*Go to proof.*

*Remark* (Interpretation). A common scalar factor may be pulled outside a finite sum.

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

### Lemma (Triangle Inequality for Finite Series)

**Lemma 7.1.22** (Triangle Inequality for Finite Series). *Let  $m \leq n$  be integers, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i|.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall m, n \in \mathbb{Z}, \quad m \leq n \implies \left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i|.$$

*Remark* (Interpretation). The absolute value of a finite series is bounded by the corresponding finite series of absolute values.

Remark (Dependencies).

- [Recursive Finite Sum](#)

#### Lemma (Comparison Test for Finite Series)

**Lemma 7.1.23** (Comparison Test for Finite Series). *Let  $m \leq n$  be integers, and let  $a_i$  and  $b_i$  be real numbers assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Suppose that  $a_i \leq b_i$  for all  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n a_i \leq \sum_{i=m}^n b_i.$$

*Go to proof.*

Remark (Standard quantified statement).

$$\forall i (m \leq i \leq n \implies a_i \leq b_i) \implies \sum_{i=m}^n a_i \leq \sum_{i=m}^n b_i.$$

Remark (Interpretation). Termwise comparison of two finite series gives comparison of their finite sums.

Remark (Dependencies).

- [Recursive Finite Sum](#)

## 7.2 Infinite Series

#### Definition (Formal Infinite Series)

**Definition 7.2.1** (Formal Infinite Series). A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where  $m$  is an integer, and  $a_n$  is a real number for every integer  $n \geq m$ . This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots .$$

Remark (Standard quantified statement).

$\sum_{n=m}^{\infty} a_n$  denotes a formal infinite series with  $m \in \mathbb{Z}$  and  $a_n \in \mathbb{R}$  for every  $n \geq m$ .

Remark (Interpretation). The notation records an infinite list of real terms beginning at the integer index  $m$ . At this stage it is a formal expression rather than a claim that a value has already been assigned to the series.

*Remark* (Dependencies).

- [Real Sequence](#)

## Proofs

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Splitting a Finite Sum). *Let  $m \leq n < p$  be integers, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq p$ . Then*

$$\sum_{i=m}^n a_i + \sum_{i=n+1}^p a_i = \sum_{i=m}^p a_i.$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-splitting. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-splitting. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Finite Summations Are Well-Defined). *Let  $X$  be a finite set with  $n$  elements, where  $n \in \mathbb{N}$ , let  $f : X \rightarrow \mathbb{R}$ , and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X \quad \text{and} \quad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \rightarrow X$$

*be bijections. Then*

$$\sum_{i=1}^n f(g(i)) = \sum_{i=1}^n f(h(i)).$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-summations-well-defined. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-summations-well-defined. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Empty Finite Set Summation). *If  $X$  is empty, and  $f : X \rightarrow \mathbb{R}$  is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for empty-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for empty-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)



*Remark* (Return). [Return to Proposition](#)

**Proposition** (Singleton Finite Set Summation). *If  $X$  consists of a single element,  $X = \{x_0\}$ , and  $f : X \rightarrow \mathbb{R}$  is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

*Professional Standard Proof.*

TODO: professional standard proof for singleton-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for singleton-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Substitution for Finite Set Summation). *If  $X$  is a finite set,  $f : X \rightarrow \mathbb{R}$  is a function, and  $g : Y \rightarrow X$  is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

*Professional Standard Proof.*

TODO: professional standard proof for substitution-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for substitution-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)
- [Finite Summations Are Well-Defined](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Integer Interval as a Finite Set). *Let  $n \leq m$  be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

*If  $a_i$  is a real number assigned to each integer  $i \in X$ , then*

$$\sum_{i=n}^m a_i = \sum_{i \in X} a_i.$$

*Professional Standard Proof.*

TODO: professional standard proof for integer-interval-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for integer-interval-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Finite Sum](#)
- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Disjoint Union Finite Set Summation). *Let  $X$  and  $Y$  be disjoint finite sets, so  $X \cap Y = \emptyset$ , and let  $f : X \cup Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left( \sum_{x \in X} f(x) \right) + \left( \sum_{y \in Y} f(y) \right).$$

*Professional Standard Proof.*

TODO: professional standard proof for disjoint-union-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for disjoint-union-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Addition Rule for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  and  $g : X \rightarrow \mathbb{R}$  be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

*Professional Standard Proof.*

TODO: professional standard proof for addition-rule-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for addition-rule-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Scalar Multiplication Rule for Finite Set Summation). *Let  $X$  be a finite set, let  $f : X \rightarrow \mathbb{R}$  be a function, and let  $c$  be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

*Professional Standard Proof.*

TODO: professional standard proof for scalar-multiplication-rule-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for scalar-multiplication-rule-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Monotonicity for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  and  $g : X \rightarrow \mathbb{R}$  be functions such that  $f(x) \leq g(x)$  for all  $x \in X$ . Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

*Professional Standard Proof.*

TODO: professional standard proof for monotonicity-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotonicity-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Triangle Inequality for Finite Set Summation). *Let  $X$  be a finite set, and let  $f : X \rightarrow \mathbb{R}$  be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

*Professional Standard Proof.*

TODO: professional standard proof for triangle-inequality-finite-set-summation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for triangle-inequality-finite-set-summation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)



*Remark* (Return). [Return to Lemma](#)

**Lemma** (Iterated Summation over a Product). *Let  $X$  and  $Y$  be finite sets, and let  $f : X \times Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{x \in X} \left( \sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y).$$

*Professional Standard Proof.*

TODO: professional standard proof for iterated-summation-over-product. □

*Detailed Learning Proof.*

TODO: detailed learning proof for iterated-summation-over-product. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Fubini's Theorem for Finite Series). *Let  $X$  and  $Y$  be finite sets, and let  $f : X \times Y \rightarrow \mathbb{R}$  be a function. Then*

$$\sum_{x \in X} \left( \sum_{y \in Y} f(x, y) \right) = \sum_{(x, y) \in X \times Y} f(x, y) = \sum_{(y, x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left( \sum_{x \in X} f(x, y) \right).$$

*Professional Standard Proof.*

TODO: professional standard proof for fubini-theorem-finite-series. □

*Detailed Learning Proof.*

TODO: detailed learning proof for fubini-theorem-finite-series. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Iterated Summation over a Product](#)
- [Summation over a Finite Set](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Reindexing a Finite Sum). *Let  $m \leq n$  be integers, let  $k$  be an integer, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-reindexing. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-reindexing. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Finite Sum of Sums). *Let  $m \leq n$  be integers, and let  $a_i$  and  $b_i$  be real numbers assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n (a_i + b_i) = \left( \sum_{i=m}^n a_i \right) + \left( \sum_{i=m}^n b_i \right).$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-of-sums. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-of-sums. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Finite Sum of Scalar Multiples). *Let  $m \leq n$  be integers, let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ , and let  $c \in \mathbb{R}$ . Then*

$$\sum_{i=m}^n (ca_i) = c \left( \sum_{i=m}^n a_i \right).$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-scalar-multiple. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-scalar-multiple. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Triangle Inequality for Finite Series). *Let  $m \leq n$  be integers, and let  $a_i$  be a real number assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i|.$$

*Professional Standard Proof.*

TODO: professional standard proof for triangle-inequality-finite-series. □

*Detailed Learning Proof.*

TODO: detailed learning proof for triangle-inequality-finite-series. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Comparison Test for Finite Series). *Let  $m \leq n$  be integers, and let  $a_i$  and  $b_i$  be real numbers assigned to each integer  $i$  satisfying  $m \leq i \leq n$ . Suppose that  $a_i \leq b_i$  for all  $i$  satisfying  $m \leq i \leq n$ . Then*

$$\sum_{i=m}^n a_i \leq \sum_{i=m}^n b_i.$$

*Professional Standard Proof.*

TODO: professional standard proof for comparison-test-finite-series. □

*Detailed Learning Proof.*

TODO: detailed learning proof for comparison-test-finite-series. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Recursive Finite Sum](#)

# Chapter 8

## Function Sequences



### Proofs



# Chapter 9

## Elementary Functions



### 9.1 Hyperbolic Functions

| Hyperbolic Functions — Quick Reference |                                                          |
|----------------------------------------|----------------------------------------------------------|
| Concept                                | Meaning                                                  |
| Hyperbolic functions                   | $\sinh$ and $\cosh$ from even/odd exponential parts      |
| Hyperbolic Pythagorean identity        | $\cosh^2 x - \sinh^2 x = 1$                              |
| Addition formulas                      | Hyperbolic angle-addition identities                     |
| Tanh limits                            | $\tanh x \rightarrow \pm 1$ as $x \rightarrow \pm\infty$ |
| Inverse hyperbolic functions           | Logarithmic inverse formulas                             |

### Hyperbolic Functions

**Definition (Hyperbolic Functions)**

**Definition 9.1.1** (Hyperbolic Functions). For all  $x \in \mathbb{R}$ :

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional:  $\operatorname{sech} x := 1/\cosh x$ ,  $\operatorname{csch} x := 1/\sinh x$ ,  $\operatorname{coth} x := \cosh x / \sinh x$ .

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad e^x = \cosh x + \sinh x.$$

*Remark* (Definition predicate reading).

$$\text{HyperbolicCosh}(x, y) := y = \frac{e^x + e^{-x}}{2}. \quad \text{HyperbolicSinh}(x, y) := y = \frac{e^x - e^{-x}}{2}.$$

$$\text{HyperbolicTanh}(x, y) := \cosh x \neq 0 \wedge y = \frac{\sinh x}{\cosh x}.$$

*Remark* (Definition predicate reading).

$$\text{HyperbolicPythagorean}(x) := \text{HyperbolicCosh}(x, c) \wedge \text{HyperbolicSinh}(x, s) \wedge c^2 - s^2 = 1.$$

*Remark* (Negated quantified statement).

$$\neg \text{HyperbolicTanh}(x, y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note  $\cosh x > 0$  for all  $x$ , so  $\text{HyperbolicTanh}$  is defined on all of  $\mathbb{R}$ .

*Remark* (Contrast with trigonometric functions).

**Trigonometric**

$$\cos^2 x + \sin^2 x = 1$$

$$(\cos x)' = -\sin x$$

$$(\sin x)' = \cos x$$

$$e^{ix} = \cos x + i \sin x$$

Parametrises unit circle

**Hyperbolic**

$$\cosh^2 x - \sinh^2 x = 1$$

$$(\cosh x)' = \sinh x$$

$$(\sinh x)' = \cosh x$$

$$e^x = \cosh x + \sinh x$$

Parametrises unit hyperbola

The minus sign in  $\cosh^2 - \sinh^2 = 1$  versus  $\cos^2 + \sin^2 = 1$  reflects the substitution  $x \mapsto ix$  in Euler's formula.

### Proposition (Hyperbolic Pythagorean Identity)

**Proposition 9.1.2** (Hyperbolic Pythagorean Identity). *For all  $x \in \mathbb{R}$ :*

$$\cosh^2 x - \sinh^2 x = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \cosh^2 x - \sinh^2 x = 1.$$

*Remark* (Negated quantified statement).

$$\neg (\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

*Remark* (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

*Remark* (Dependencies).

- [Exponential Function](#)
- [ℝ Is a Field](#)

#### Proposition (Addition Formulas for Hyperbolic Functions)

**Proposition 9.1.3** (Addition Formulas for Hyperbolic Functions). *For all  $x, y \in \mathbb{R}$ :*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \quad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

*Remark* (Definition predicate reading).

$$\text{HyperbolicAddition}(x, y) :\equiv \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

*Remark* (Negated quantified statement).

$$\neg(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R} : \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y.$$

The negation has no witness.

*Remark* (Proof strategy). Substitute the definitions and expand, or note the formulas follow from  $e^{x+y} = e^x e^y$  by taking even and odd parts.

*Remark* (Comparison with trigonometric addition formulas). The  $\sinh$  formula matches  $\sin(x+y)$  exactly. The  $\cosh$  formula differs from  $\cos(x+y)$  only in the sign of the last term:  $+\sinh x \sinh y$  versus  $-\sin x \sin y$ . The sign difference is a direct consequence of  $\cosh^2 - \sinh^2 = 1$  versus  $\cos^2 + \sin^2 = 1$ .

*Remark* (Dependencies).

- [Hyperbolic Functions](#)
- [Exponential Function](#)
- [ℝ Is a Field](#)

#### Proposition (Limits of tanh)

**Proposition 9.1.4** (Limits of tanh).

$$\lim_{x \rightarrow +\infty} \tanh x = 1, \quad \lim_{x \rightarrow -\infty} \tanh x = -1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{LimitAtInfinity}(\tanh x, +\infty, 1) \wedge \text{LimitAtInfinity}(\tanh x, -\infty, -1).$$

*Remark* (Definition predicate reading).

$$\text{LimitAtInfinity}(\tanh x, +\infty, 1) \equiv \forall \varepsilon > 0, \exists M \in \mathbb{R}, \forall x : x > M \Rightarrow \text{OutputTolerance}(\tanh x, 1, \varepsilon). \blacksquare$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow +\infty} \tanh x = 1 \right) \iff \exists \varepsilon_0 > 0, \forall M > 0, \exists x > M : |\tanh x - 1| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Write  $\tanh x = (1 - e^{-2x})/(1 + e^{-2x})$ . As  $x \rightarrow +\infty$ ,  $e^{-2x} \rightarrow 0$ , giving  $\tanh x \rightarrow 1$ . The limit at  $-\infty$  follows by oddness:  $\tanh(-x) = -\tanh(x)$ .

*Remark* (Dependencies).

- [Hyperbolic Functions](#)
- [Exponential Function](#)
- [Addition on  \$\mathbb{R}\$](#)
- [Multiplication on  \$\mathbb{R}\$](#)

### Definition

**Definition 9.1.5** (Inverse Hyperbolic Functions). Since  $\sinh$  is strictly increasing and bijective  $\mathbb{R} \rightarrow \mathbb{R}$ , its inverse  $\text{arcsinh}$  is defined on all of  $\mathbb{R}$ . Since  $\cosh$  is strictly increasing on  $[0, \infty)$  with range  $[1, \infty)$ , its inverse  $\text{arccosh}$  is defined on  $[1, \infty)$ . Since  $\tanh$  is strictly increasing with range  $(-1, 1)$ ,  $\text{arctanh}$  is defined on  $(-1, 1)$ .

Explicit logarithmic formulas:

$$\text{arcsinh } x = \ln \left( x + \sqrt{x^2 + 1} \right), \quad x \in \mathbb{R}.$$

$$\text{arccosh } x = \ln \left( x + \sqrt{x^2 - 1} \right), \quad x \geq 1.$$

$$\text{arctanh } x = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right), \quad |x| < 1.$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \sinh(\text{arcsinh } x) = x \wedge \text{arcsinh}(\sinh x) = x.$$

$$\forall x \geq 1 : \cosh(\text{arccosh } x) = x. \quad \forall |x| < 1 : \tanh(\text{arctanh } x) = x.$$

*Remark* (Definition predicate reading).

$$\text{Arcsinh}(y, x) \equiv x \in \mathbb{R} \wedge \sinh x = y \equiv x = \ln \left( y + \sqrt{y^2 + 1} \right).$$

$$\text{Arctanh}(y, x) \equiv |y| < 1 \wedge \tanh x = y \equiv x = \frac{1}{2} \ln \left( \frac{1+y}{1-y} \right).$$

*Remark* (Negated quantified statement).

$$\neg \operatorname{Arctanh}(y, x) \iff |y| \geq 1 \vee \tanh x \neq y.$$

*Remark* (Derivation of arcsinh). Set  $y = \sinh x = (e^x - e^{-x})/2$  and let  $u = e^x > 0$ . Then  $y = (u - 1/u)/2$ , so  $u^2 - 2yu - 1 = 0$ . The positive root is  $u = y + \sqrt{y^2 + 1}$ , giving  $x = \ln(y + \sqrt{y^2 + 1})$ .

*Remark* (Derivatives).  $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2 + 1}$ ;  $(\operatorname{arccosh} x)' = 1/\sqrt{x^2 - 1}$  for  $x > 1$ ;  $(\operatorname{arctanh} x)' = 1/(1 - x^2)$  for  $|x| < 1$ . These follow from the inverse function theorem applied to  $\sinh$ ,  $\cosh$ , and  $\tanh$  respectively.

*Remark* (Dependencies).

- [Hyperbolic Functions](#)
- [Exponential Function](#)
- [Limit at Infinity](#)
- [Limit of a Quotient](#)

*Remark* (Dependencies).

- [Hyperbolic Functions](#)
- [Natural Logarithm](#)
- [Continuous Inverse Theorem](#)
- [Absolute Value on  \$\mathbb{R}\$](#)

## 9.2 The Logarithm

### Logarithmic Functions — Quick Reference

| Concept                                       | Meaning                                          |
|-----------------------------------------------|--------------------------------------------------|
| <a href="#">Natural logarithm</a>             | Inverse of the exponential on $(0, \infty)$      |
| <a href="#">Fundamental logarithmic limit</a> | $\ln(1 + x)/x \rightarrow 1$                     |
| <a href="#">Power dominates log</a>           | Logarithmic growth is below every positive power |

## The Logarithm

### Definition (Natural Logarithm)

**Definition 9.2.1** (Natural Logarithm). The **natural logarithm**  $\ln : (0, \infty) \rightarrow \mathbb{R}$  is the inverse of the exponential function: for  $x > 0$ ,

$$\ln x := \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently,  $\ln x$  is defined by the integral

$$\ln x := \int_1^x \frac{1}{t} dt.$$

*Remark* (Standard quantified statement).

$$\forall x > 0 : e^{\ln x} = x. \quad \forall y \in \mathbb{R} : \ln(e^y) = y.$$

*Remark* (Definition predicate reading).

$$\text{NaturalLog}(x, y) :\equiv x > 0 \wedge e^y = x :\equiv x > 0 \wedge y = \int_1^x \frac{1}{t} dt.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{\ln(1+x)}{x}, 0, 1\right) :\equiv \forall \varepsilon > 0, \exists \delta > 0, \forall x : \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{\ln(1+x)}{x}, 0, \varepsilon\right)$$

*Remark* (Negated quantified statement).

$$\neg \text{NaturalLog}(x, y) \iff x \leq 0 \vee e^y \neq x.$$

*Remark* (Two definitions, same function). The inverse-of-exponential definition requires  $e^x$  to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

*Remark* (Key properties). (i)  $\ln 1 = 0$ ;  $\ln e = 1$ .

(ii) **Functional equation:**  $\ln(xy) = \ln x + \ln y$  for all  $x, y > 0$ .

(iii)  $\ln(x/y) = \ln x - \ln y$ ;  $\ln(x^r) = r \ln x$  for  $r \in \mathbb{R}$ .

(iv)  $\ln$  is strictly increasing on  $(0, \infty)$ .

(v)  $\ln x \rightarrow +\infty$  as  $x \rightarrow \infty$ ;  $\ln x \rightarrow -\infty$  as  $x \rightarrow 0^+$ .

(vi)  $(\ln x)' = 1/x$  for all  $x > 0$ .

*Remark* (General logarithm). For  $a > 0$ ,  $a \neq 1$ , the **logarithm base**  $a$  is  $\log_a x := \ln x / \ln a$ . The common logarithm is  $\log_{10}$ ; the binary logarithm is  $\log_2$ .

### Proposition (Fundamental Logarithmic Limit)

**Proposition 9.2.2** (Fundamental Logarithmic Limit).

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x| < \delta \Rightarrow \left| \frac{\ln(1+x)}{x} - 1 \right| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \right) \iff \exists \varepsilon_0 > 0, \forall \delta > 0, \exists x : 0 < |x| < \delta \wedge \left| \frac{\ln(1+x)}{x} - 1 \right| \geq \varepsilon_0.$$

*Remark* (Equivalence with exponential limit). The fundamental exponential limit  $\lim(e^x - 1)/x = 1$  and the fundamental logarithmic limit  $\lim \ln(1+x)/x = 1$  are equivalent: each follows from the other via the substitution  $x \leftrightarrow e^x - 1$ . Both express the first-order Taylor approximation of the respective function at 0.

*Remark* (Proof strategy). From the series:  $\ln(1+x) = x - x^2/2 + x^3/3 - \dots$  (valid for  $|x| < 1$ ). Dividing by  $x$  gives  $1 - x/2 + x^2/3 - \dots \rightarrow 1$  as  $x \rightarrow 0$  by continuity of power series.

*Remark* (Dependencies).

- [Exponential Function](#)
- [Inverse Bijection](#)
- [Continuous Inverse Theorem](#)

### Proposition (Every Power Dominates the Logarithm)

**Proposition 9.2.3** (Every Power Dominates the Logarithm). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} x^r \ln x = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall r > 0 : \lim_{x \rightarrow \infty} x^{-r} \ln x = 0 \wedge \lim_{x \rightarrow 0^+} x^r \ln x = 0.$$

*Remark* (Definition predicate reading).

$\text{LimitAtInfinity}(x^{-r} \ln x, +\infty, 0) :\equiv \forall \varepsilon > 0, \exists M \in \mathbb{R}, \forall x : x > M \Rightarrow \text{OutputTolerance}(x^{-r} \ln x, 0, \varepsilon).$  ■

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow \infty} \frac{\ln x}{x^r} = 0 \right) \iff \exists \varepsilon_0 > 0, \forall M > 0, \exists x > M : x^{-r} |\ln x| \geq \varepsilon_0.$$

*Remark* (Proof strategy). For the first: substitute  $x = e^t$  to get  $t/e^{rt} \rightarrow 0$  by exponential dominance. For the second: substitute  $x = 1/t$  and reduce to the first:  $x^r \ln x = (-\ln t)/t^r \rightarrow 0$ .

*Remark* (Significance). These limits establish the position of  $\ln x$  in the growth hierarchy:  $\ln x$  is dominated by every positive power. Combined with the exponential dominance result, the full ordering is  $\ln x \ll x^r \ll e^x$  as  $x \rightarrow \infty$  for any  $r > 0$ . The second limit says that near  $0^+$ , the positive power  $x^r$  wins over the logarithmic blow-up  $|\ln x| \rightarrow \infty$ . Used in integrability arguments near 0.

*Remark* (Dependencies).

- [Natural Logarithm](#)
- [Fundamental Exponential Limit](#)
- [Function Limit](#)

*Remark* (Dependencies).

- [Natural Logarithm](#)
- [Power Function](#)
- [Exponential Dominates Every Power](#)
- [Limit at Infinity](#)

## 9.3 Power Functions and the Exponential

### Power and Exponential Functions — Quick Reference

| Concept                                       | Meaning                                        |
|-----------------------------------------------|------------------------------------------------|
| <a href="#">Power function</a>                | Real powers $x^r$ on positive inputs           |
| <a href="#">The number <math>e</math></a>     | Series and compound-interest limit definitions |
| <a href="#">Exponential function</a>          | Power series with $e^{x+y} = e^x e^y$          |
| <a href="#">Fundamental exponential limit</a> | $(e^x - 1)/x \rightarrow 1$                    |
| <a href="#">Compound-interest limit</a>       | $(1 + 1/n)^n \rightarrow e$                    |
| <a href="#">Growth hierarchy</a>              | $e^x$ dominates powers at infinity             |



## Power Functions and the Exponential

### Definition (Power Function)

**Definition 9.3.1** (Power Function). Let  $n \in \mathbb{N}$ . The **integer power function** is  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) := x^n$ . For  $r \in \mathbb{R}$  and  $x > 0$ , the **real power** is defined by

$$x^r := e^{r \ln x}.$$

*Remark* (Standard quantified statement).

$$\forall r \in \mathbb{R}, \forall x > 0 : x^r = e^{r \ln x}.$$

For  $r \in \mathbb{N}$  this reduces to the  $r$ -fold product  $x \cdot x \cdots x$ .

*Remark* (Definition predicate reading).

$$\text{RealPower}(x, r, y) :\equiv x > 0 \wedge y = e^{r \ln x}.$$

*Remark* (Laws of exponents). For  $x, y > 0$  and  $r, s \in \mathbb{R}$ :

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition  $x^r = e^{r \ln x}$ .

*Remark* (Interpretation). Defining  $x^r := e^{r \ln x}$  unifies all three cases — integer, rational, irrational exponents — into a single formula and makes differentiability of  $x^r$  immediate from that of  $e^x$  and  $\ln x$  via the chain rule:  $(x^r)' = r x^{r-1}$ .

### Definition (The Number $e$ )

**Definition 9.3.2** (The Number  $e$ ). The **Euler number** is

$$e := \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828 \dots$$

Both expressions define the same real number.

*Remark* (Standard quantified statement).

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \wedge e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

*Remark* (Interpretation). The series  $\sum 1/n!$  converges rapidly by comparison with a geometric series. The limit  $(1 + 1/n)^n$  arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

*Remark* (Dependencies).

- [Integer Powers in a Field](#)

- [Exponential Function](#)
- [Natural Logarithm](#)
- [Multiplication on  \$\mathbb{R}\$](#)

*Remark* (Dependencies).

- [Sequence](#)
- [Convergent Sequence](#)

### Definition (Exponential Function)

**Definition 9.3.3** (Exponential Function). The **exponential function**  $\exp : \mathbb{R} \rightarrow (0, \infty)$  is defined by the absolutely convergent power series

$$e^x := \exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{convergent absolutely since } \sum \frac{|x|^n}{n!} < \infty.$$

*Remark* (Definition predicate reading).

$$\text{Exponential}(x, y) \equiv y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

*Remark* (Key properties). (i)  $e^0 = 1$ ;  $e^1 = e$ .

(ii) **Functional equation:**  $e^{x+y} = e^x \cdot e^y$  for all  $x, y \in \mathbb{R}$ .

(iii)  $e^x > 0$  for all  $x$ ;  $e^{-x} = 1/e^x$ .

(iv)  $e^x$  is strictly increasing and bijective  $\mathbb{R} \rightarrow (0, \infty)$ .

(v)  $(e^x)' = e^x$  — the exponential is its own derivative.

*Remark* (General exponential). For  $a > 0$ ,  $a \neq 1$ :  $a^x := e^{x \ln a}$ . Strictly increasing if  $a > 1$ , strictly decreasing if  $0 < a < 1$ .  $(a^x)' = a^x \ln a$ . Only  $e^x$  is its own derivative.

### Proposition (Fundamental Exponential Limit)

**Proposition 9.3.4** (Fundamental Exponential Limit).

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{e^x - 1}{x}, 0, 1\right) := \forall \varepsilon > 0, \exists \delta > 0, \forall x : \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{e^x - 1}{x}, 0, \varepsilon\right)$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0, \forall \delta > 0, \exists x : 0 < |x| < \delta \wedge \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

*Remark* (Interpretation). This is the statement  $(e^x)'|_{x=0} = 1$ . The series gives  $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \rightarrow 1$  as  $x \rightarrow 0$  by continuity of power series at their center. Combined with the functional equation, this implies  $(e^x)' = e^x$  everywhere.

*Remark* (Dependencies).

- [The Number  \$e\$](#)
- [Integer Powers in a Field](#)
- [Sequence](#)
- [Convergent Sequence](#)

#### Proposition (Compound Interest Limit)

**Proposition 9.3.5** (Compound Interest Limit).

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0, \exists M > 0 : x > M \Rightarrow \left| \left(1 + \frac{1}{x}\right)^x - e \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{LimitAtInfinity}\left(\left(1 + \frac{1}{x}\right)^x, +\infty, e\right) := \forall \varepsilon > 0, \exists M \in \mathbb{R}, \forall x : x > M \Rightarrow \text{OutputTolerance}\left(\left(1 + \frac{1}{x}\right)^x, e, \varepsilon\right)$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \right) \iff \exists \varepsilon_0 > 0, \forall M > 0, \exists x > M : \left| \left(1 + \frac{1}{x}\right)^x - e \right| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Write  $(1 + 1/x)^x = e^{x \ln(1+1/x)}$ . Set  $t = 1/x \rightarrow 0^+$  and use the fundamental logarithmic limit  $\ln(1+t)/t \rightarrow 1$ . Then  $x \ln(1 + 1/x) = \ln(1+t)/t \rightarrow 1$ , so  $(1 + 1/x)^x = e^{x \ln(1+1/x)} \rightarrow e^1 = e$  by continuity of the exponential.

*Remark* (Dependencies).

- [Exponential Function](#)
- [Function Limit](#)
- [Derivative at a Point](#)

#### Proposition (Exponential Dominates Every Power)

**Proposition 9.3.6** (Exponential Dominates Every Power). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{x^r}{e^x} = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall r > 0 : \lim_{x \rightarrow \infty} x^r e^{-x} = 0.$$

*Remark* (Definition predicate reading).

$\text{LimitAtInfinity}(x^r e^{-x}, +\infty, 0) :\equiv \forall \varepsilon > 0, \exists M \in \mathbb{R}, \forall x : x > M \Rightarrow \text{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$  ■

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow \infty} \frac{x^r}{e^x} = 0 \right) \iff \exists \varepsilon_0 > 0, \forall M > 0, \exists x > M : x^r e^{-x} \geq \varepsilon_0.$$

*Remark* (Proof strategy). Pick integer  $n > r$ . The series bound  $e^x \geq x^n/n!$  gives  $x^r/e^x \leq n! x^{r-n} \rightarrow 0$  since  $r - n < 0$ .

*Remark* (Growth hierarchy). For any  $r > 0$ :

$$\ln x \ll x^r \ll e^x \quad \text{as } x \rightarrow \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

*Remark* (Dependencies).

- [The Number  \$e\$](#)
- [Exponential Function](#)

- [Natural Logarithm](#)
- [Fundamental Logarithmic Limit](#)
- [Limit at Infinity](#)

*Remark* (Dependencies).

- [Power Function](#)
- [Exponential Function](#)
- [Limit at Infinity](#)
- [Squeeze Theorem for Function Limits](#)

## 9.4 Trigonometric Functions

### Trigonometric Functions — Quick Reference

| Concept                                | Meaning                                       |
|----------------------------------------|-----------------------------------------------|
| <a href="#">Sine and cosine</a>        | Defined analytically by power series          |
| <a href="#">Pythagorean identity</a>   | $\sin^2 x + \cos^2 x = 1$                     |
| <a href="#">Addition formulas</a>      | Angle addition identities from exponentials   |
| <a href="#">Other trig functions</a>   | Tangent, cotangent, secant, cosecant          |
| <a href="#">Inverse trig functions</a> | Inverses on monotone restricted domains       |
| <a href="#">Fundamental trig limit</a> | $\sin x/x \rightarrow 1$ as $x \rightarrow 0$ |
| <a href="#">Second trig limit</a>      | $(1 - \cos x)/x^2 \rightarrow 1/2$            |

## Trigonometric Functions

### Definition (Sine and Cosine)

**Definition 9.4.1** (Sine and Cosine via Power Series). The **sine** and **cosine** functions are defined for all  $x \in \mathbb{R}$  by the absolutely convergent power series

$$\sin x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

*Remark* (Definition predicate reading).

$$\text{Sine}(x, y) := y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \quad \text{Cosine}(x, y) := y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

*Remark* (Key properties). (i)  $\sin 0 = 0$ ;  $\cos 0 = 1$ .

(ii)  $\sin(-x) = -\sin x$  (odd);  $\cos(-x) = \cos x$  (even).

(iii)  $(\sin x)' = \cos x$ ;  $(\cos x)' = -\sin x$ .

(iv)  $|\sin x| \leq 1$ ;  $|\cos x| \leq 1$  for all  $x$ .

(v)  $\sin^2 x + \cos^2 x = 1$  (Pythagorean identity).

(vi) Periodic:  $\sin(x + 2\pi) = \sin x$ ;  $\cos(x + 2\pi) = \cos x$ .

*Remark* (Relationship to  $e^{ix}$ ). Euler's formula:  $e^{ix} = \cos x + i \sin x$ , where  $e^{ix}$  is defined by the complex exponential series. This gives  $\cos x = \text{Re}(e^{ix})$  and  $\sin x = \text{Im}(e^{ix})$ . Euler's identity  $e^{i\pi} + 1 = 0$  follows immediately.

#### Proposition (Pythagorean Identity)

**Proposition 9.4.2** (Pythagorean Identity). *For all  $x \in \mathbb{R}$ :*

$$\sin^2 x + \cos^2 x = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R} : \sin^2 x + \cos^2 x = 1.$$

*Remark* (Definition predicate reading).

$$\text{PythagoreanIdentity}(x) := \text{Sine}(x, s) \wedge \text{Cosine}(x, c) \wedge s^2 + c^2 = 1.$$

*Remark* (Negated quantified statement).

$$\neg(\forall x : \sin^2 x + \cos^2 x = 1) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness — the identity holds universally.

*Remark* (Proof strategy). Define  $f(x) := \sin^2 x + \cos^2 x$ . Then  $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$ , so  $f$  is constant. Since  $f(0) = 0 + 1 = 1$ , the identity follows.

*Remark* (Dependencies).

- [Real Numbers](#)

- [Integer Powers in a Field](#)
- [Sequence](#)
- [Convergent Sequence](#)

#### Proposition (Addition Formulas)

**Proposition 9.4.3** (Addition Formulas). *For all  $x, y \in \mathbb{R}$ :*

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \quad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \sin(x + y) = \sin x \cos y + \cos x \sin y \wedge \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

*Remark* (Definition predicate reading).

$$\text{TrigAddition}(x, y) := \sin(x + y) = \sin x \cos y + \cos x \sin y \wedge \cos(x + y) = \cos x \cos y - \sin x \sin y. \blacksquare$$

*Remark* (Negated quantified statement).

$$\neg(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R} : \sin(x + y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

*Remark* (Proof strategy). From the Cauchy product:  $e^{i(x+y)} = e^{ix}e^{iy}$ . Taking real and imaginary parts yields both addition formulas simultaneously.

*Remark* (Derived identities).

$$\begin{aligned} \sin(2x) &= 2 \sin x \cos x \\ \cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1 \\ \sin^2 x &= \frac{1}{2}(1 - \cos 2x) \\ \cos^2 x &= \frac{1}{2}(1 + \cos 2x) \end{aligned}$$

*Remark* (Dependencies).

- [Sine and Cosine](#)
- [Derivative at a Point](#)
- [Zero Derivative Implies Constant](#)

#### Definition

**Definition 9.4.4** (Tangent, Cotangent, Secant, Cosecant).

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$  and  $\sec x$  are defined where  $\cos x \neq 0$ , i.e.,  $x \neq \pi/2 + k\pi$  for  $k \in \mathbb{Z}$ .

*Remark* (Standard quantified statement).

$$\forall x \in \mathbb{R}, \cos x \neq 0 : \tan x = \frac{\sin x}{\cos x}.$$

*Remark* (Definition predicate reading).

$$\text{Tangent}(x, y) := \cos x \neq 0 \wedge y = \frac{\sin x}{\cos x}.$$

*Remark* (Pythagorean identities for tan). Dividing  $\sin^2 x + \cos^2 x = 1$  by  $\cos^2 x$ :  $1 + \tan^2 x = \sec^2 x$ . Dividing by  $\sin^2 x$ :  $\cot^2 x + 1 = \csc^2 x$ .

*Remark* (Dependencies).

- [Sine and Cosine](#)
- [Addition on  \$\mathbb{R}\$](#)
- [Multiplication on  \$\mathbb{R}\$](#)

#### Definition

**Definition 9.4.5** (Inverse Trigonometric Functions). •  $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$  is the inverse of  $\sin|_{[-\pi/2, \pi/2]}$ .

•  $\arccos : [-1, 1] \rightarrow [0, \pi]$  is the inverse of  $\cos|_{[0, \pi]}$ .

•  $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$  is the inverse of  $\tan|_{(-\pi/2, \pi/2)}$ .

*Remark* (Standard quantified statement).

$$\forall y \in [-1, 1] : \sin(\arcsin y) = y. \quad \forall x \in [-\pi/2, \pi/2] : \arcsin(\sin x) = x.$$

*Remark* (Definition predicate reading).

$$\text{Arctan}(y, x) := x \in (-\pi/2, \pi/2) \wedge \tan x = y.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{\sin x}{x}, 0, 1\right) := \forall \varepsilon > 0, \exists \delta > 0, \forall x : \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{\sin x}{x}, 0, \varepsilon\right).$$

*Remark* (Negated quantified statement).

$$\neg \text{Arctan}(y, x) \iff x \notin (-\pi/2, \pi/2) \vee \tan x \neq y.$$

*Remark* (Key limits of arctan).

$$\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}, \quad \lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}.$$

$\arctan$  is bounded, strictly increasing, and continuous on  $\mathbb{R}$ ; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.



*Remark* (Dependencies).

- [Sine and Cosine](#)
- [ℝ Is a Field](#)

#### Theorem (Fundamental Trigonometric Limit)

**Theorem 9.4.6** (Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x| < \delta \Rightarrow \left| \frac{\sin x}{x} - 1 \right| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right) \iff \exists \varepsilon_0 > 0, \forall \delta > 0, \exists x : 0 < |x| < \delta \wedge \left| \frac{\sin x}{x} - 1 \right| \geq \varepsilon_0.$$

*Remark* (Proof strategy — series). From the series:  $\sin x/x = 1 - x^2/3! + x^4/5! - \cdots \rightarrow 1$  as  $x \rightarrow 0$  by continuity of power series at their center.

*Remark* (Proof strategy — geometric squeeze). For  $0 < x < \pi/2$ , area comparison of two triangles and a circular sector gives  $\cos x \leq \sin x/x \leq 1$ . Since  $\cos x \rightarrow 1$  as  $x \rightarrow 0^+$ , the Squeeze Theorem gives  $\sin x/x \rightarrow 1$ . Even symmetry extends this to  $x \rightarrow 0^-$ .

*Remark* (Significance). This is the statement  $(\sin x)'|_{x=0} = 1$ , which implies  $(\sin x)' = \cos x$  everywhere. It underlies all trigonometric differentiation.

*Remark* (Dependencies).

- [Sine and Cosine](#)
- [Tangent, Cotangent, Secant, Cosecant](#)
- [Continuous Inverse Theorem](#)
- [Closed Interval](#)

#### Proposition (Second Fundamental Trigonometric Limit)

**Proposition 9.4.7** (Second Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x| < \delta \Rightarrow \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}\left(\frac{1 - \cos x}{x^2}, 0, \frac{1}{2}\right) :\equiv \forall \varepsilon > 0, \exists \delta > 0, \forall x : \text{InputTolerance}(x, 0, \delta) \Rightarrow \text{OutputTolerance}\left(\frac{1 - \cos x}{x^2}, 0, \frac{1}{2}, \varepsilon\right)$$

*Remark* (Negated quantified statement).

$$\neg \left( \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) \iff \exists \varepsilon_0 > 0, \forall \delta > 0, \exists x : 0 < |x| < \delta \wedge \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

*Remark* (Proof strategy). Use the half-angle identity  $1 - \cos x = 2 \sin^2(x/2)$ :

$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left( \frac{\sin(x/2)}{x/2} \right)^2 \rightarrow \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

*Remark* (Dependencies).

- [Sine and Cosine](#)
- [Function Limit](#)
- [Squeeze Theorem for Function Limits](#)

*Remark* (Dependencies).

- [Sine and Cosine](#)
- [Trigonometric Addition Formulas](#)
- [Fundamental Trigonometric Limit](#)
- [Function Limit](#)

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Hyperbolic Pythagorean Identity). *For all  $x \in \mathbb{R}$ :*

$$\cosh^2 x - \sinh^2 x = 1.$$

*Professional Standard Proof.*

TODO: professional standard proof for hyperbolic-pythagorean.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for hyperbolic-pythagorean.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Addition Formulas for Hyperbolic Functions). *For all  $x, y \in \mathbb{R}$ :*

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y, \quad \cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

*Professional Standard Proof.*

TODO: professional standard proof for hyperbolic-addition. □

*Detailed Learning Proof.*

TODO: detailed learning proof for hyperbolic-addition. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Limits of ).

$$\lim_{x \rightarrow +\infty} \tanh x = 1, \quad \lim_{x \rightarrow -\infty} \tanh x = -1.$$

*Professional Standard Proof.*

TODO: professional standard proof for tanh-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for tanh-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Fundamental Logarithmic Limit).

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1.$$

*Professional Standard Proof.*

TODO: professional standard proof for limit-logarithm-fundamental. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-logarithm-fundamental. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Every Power Dominates the Logarithm). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} x^r \ln x = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for power-dominates-log. □

*Detailed Learning Proof.*

TODO: detailed learning proof for power-dominates-log. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Fundamental Exponential Limit).

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

*Professional Standard Proof.*

TODO: professional standard proof for limit-exponential-fundamental. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-exponential-fundamental. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Compound Interest Limit).

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

*Professional Standard Proof.*

TODO: professional standard proof for limit-compound-interest. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-compound-interest. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Exponential Dominates Every Power). *For every  $r > 0$ :*

$$\lim_{x \rightarrow \infty} \frac{x^r}{e^x} = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for exponential-dominates-power. □

*Detailed Learning Proof.*

TODO: detailed learning proof for exponential-dominates-power. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Pythagorean Identity). *For all  $x \in \mathbb{R}$ :*

$$\sin^2 x + \cos^2 x = 1.$$

*Professional Standard Proof.*

TODO: professional standard proof for pythagorean-identity.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for pythagorean-identity.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Addition Formulas). *For all  $x, y \in \mathbb{R}$ :*

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \quad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

*Professional Standard Proof.*

TODO: professional standard proof for trig-addition-formulas. □

*Detailed Learning Proof.*

TODO: detailed learning proof for trig-addition-formulas. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

*Professional Standard Proof.*

TODO: professional standard proof for fundamental-trig-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for fundamental-trig-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Second Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

*Professional Standard Proof.*

TODO: professional standard proof for limit-one-minus-cos. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-one-minus-cos. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

# Chapter 10

## Continuity



### 10.1 Limits

#### One-Sided Limits — Quick Reference

| Concept/Statement          | Meaning/Role                                           |
|----------------------------|--------------------------------------------------------|
| Right-hand limit           | Approach the point from inputs larger than $c$ .       |
| Left-hand limit            | Approach the point from inputs smaller than $c$ .      |
| Right sequential criterion | Test right limits by right-approaching sequences.      |
| Left sequential criterion  | Test left limits by left-approaching sequences.        |
| Matching criterion         | Two-sided limits exist exactly when side limits agree. |

#### Limits of Functions — Quick Reference

| Concept/Statement    | Meaning/Role                                          |
|----------------------|-------------------------------------------------------|
| Function limit       | Punctured input control near a cluster point.         |
| Sequential limit     | All approaching sequences have the same output limit. |
| Neighbourhood limit  | Neighbourhood formulation of the same condition.      |
| Uniqueness           | A function has at most one limit at a point.          |
| Sequential criterion | Sequence convergence is the main working test.        |
| Local boundedness    | Finite limits bound the function near the point.      |
| Squeeze theorem      | Ordered comparison transfers limits.                  |
| Locality             | Only behaviour near the approach point matters.       |

| Concept/Statement    | Meaning/Role                                             |
|----------------------|----------------------------------------------------------|
| Sum rule             | Limits add pointwise.                                    |
| Difference rule      | Limits subtract pointwise.                               |
| Scalar multiple rule | Scalars pass through limits.                             |
| Product rule         | Products of finite limits multiply.                      |
| Quotient rule        | Division is valid when the denominator limit is nonzero. |

## Limits of Functions

### Definition ( $\varepsilon$ -Closeness of a Function to a Value)

**Definition 10.1.1** ( $\varepsilon$ -Closeness of a Function to a Value). Let  $X \subseteq \mathbb{R}$ , let  $f : X \rightarrow \mathbb{R}$ , let  $L \in \mathbb{R}$ , and let  $\varepsilon > 0$ . The function  $f$  is  **$\varepsilon$ -close to  $L$**  if  $|f(x) - L| \leq \varepsilon$  for all  $x \in X$ .

*Remark* (Standard quantified statement).

$$f \text{ } \varepsilon\text{-close to } L \iff \forall x \in X \ (|f(x) - L| \leq \varepsilon).$$

*Remark* (Interpretation). A global condition: every output lies within  $\varepsilon$  of  $L$ , regardless of where in  $X$  the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of  $c$ .

### Definition (Local $\delta$ -Closeness Near a Point)

**Definition 10.1.2** (Local  $\delta$ -Closeness Near a Point). Let  $f : X \rightarrow \mathbb{R}$ , let  $L \in \mathbb{R}$ , let  $x_0$  be adherent to  $X$ , and let  $\varepsilon, \delta > 0$ . The function  $f$  is  **$\varepsilon$ -close to  $L$  near  $x_0$  within  $\delta$**  if

$$\forall x \in X \ (|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon).$$

*Remark* (Standard quantified statement).

$$\forall x \in X : |x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon.$$

*Remark* (Interpretation). A local condition: only outputs from the  $\delta$ -ball around  $x_0$  need lie within  $\varepsilon$  of  $L$ . The input condition here is Within — unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every  $\varepsilon$  such a  $\delta$  exists.

*Remark* (Dependencies).

- $\varepsilon$ -Closeness of a Function to a Value



### Definition (Limit of a Function)

**Definition 10.1.3** (Limit of a Function). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . A real number  $L$  is the **limit of  $f$  at  $c$** , written  $\lim_{x \rightarrow c} f(x) = L$ , if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Go to uniqueness proof.*

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading). Define

$$\begin{aligned} \text{OutputTol}(\varepsilon) &\iff \varepsilon > 0, \\ \text{InputTol}(\delta) &\iff \delta > 0, \\ \text{PuncturedNbhd}(x, c, \delta) &\iff 0 < |x - c| < \delta, \\ \text{OutputNbhd}(y, L, \varepsilon) &\iff |y - L| < \varepsilon. \end{aligned}$$

Then

$$\text{FunctionLimit}(f, A, c, L) \iff \forall \varepsilon \left( \text{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left( \text{InputTol}(\delta) \wedge \forall x \in A [\text{PuncturedNbhd}(x, c, \delta) \Rightarrow \text{OutputNbhd}(f(x), L, \varepsilon)] \right) \right)$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading).

$$\begin{aligned} \neg \text{FunctionLimit}(f, A, c, L) &\iff \exists \varepsilon_0 \left( \text{OutputTol}(\varepsilon_0) \wedge \forall \delta \left( \text{InputTol}(\delta) \Rightarrow \exists x \in A [\text{PuncturedNbhd}(x, c, \delta) \wedge \text{OutputNbhd}(f(x), L, \varepsilon_0)] \right) \right) \\ &\iff \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \left( 0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0 \right). \end{aligned}$$

*Remark* (Failure modes).

- The proposed value  $L$  is the wrong target: the function approaches some value  $M \neq L$  near  $c$ .
- The function does not settle near any single value as  $x$  approaches  $c$  through  $A$ .
- The outputs repeatedly stay at least a fixed distance from  $L$  no matter how tightly the inputs are restricted around  $c$ .

*Remark* (Failure mode decomposition).

$$\neg \text{FunctionLimit}(f, A, c, L) \iff \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \left( \underbrace{0 < |x - c| < \delta}_{\text{arbitrarily close to } c} \wedge \underbrace{|f(x) - L| \geq \varepsilon_0}_{\text{fixed output gap from } L} \right).$$

*Remark* (Interpretation). The quantifier structure is asymmetric by design. The output tolerance  $\varepsilon$  is a *demand*: the caller specifies how precise the output must be. The input tolerance  $\delta$  is a *response*: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because  $f$  may not be defined at  $c$ , and because continuity — not limits — is the concept that includes  $c$  itself. This is the deepest structural difference between the two definitions. Four critical features: (i)  $f$  need not be defined at  $c$ ; (ii)  $c$  must be a cluster point of  $A$  else the condition is vacuously satisfied by every  $L$ ; (iii)  $\delta$  may depend on  $\varepsilon$ ; (iv) the condition is about all  $x$  near  $c$  simultaneously, not just one.

*Remark* (Dependencies).

- [Cluster Point](#)

#### Definition (Sequential Definition of Limit)

**Definition 10.1.4** (Sequential Definition of Limit). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The limit  $\lim_{x \rightarrow c} f(x) = L$  in the **sequential sense** holds if for every sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$ , the sequence  $(f(x_n))$  converges to  $L$ .

*Remark* (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ (x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L).$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}_{\text{seq}}(f, c, L) := \forall (x_n) \subseteq A \setminus \{c\} : \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n), L).$$

*Remark* (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ (x_n \rightarrow c \wedge f(x_n) \not\rightarrow L).$$

*Remark* (Interpretation). The sequence  $(x_n)$  must lie in  $A \setminus \{c\}$ : the term  $x_n = c$  is excluded, mirroring the  $0 <$  in the  $\varepsilon$ - $\delta$  form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The  $\varepsilon$ - $\delta$  form is better for constructing explicit  $\delta$  functions.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Cluster Point](#)

#### Definition (Neighbourhood Form of Limit)

**Definition 10.1.5** (Neighbourhood Form of Limit). Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The limit  $\lim_{x \rightarrow c} f(x) = L$  in the **neighbourhood sense** holds if for every  $\varepsilon$ -neighbourhood  $V_\varepsilon(L)$  there exists a deleted  $\delta$ -neighbourhood  $V_\delta^*(c)$  such that

$$x \in V_\delta^*(c) \cap A \implies f(x) \in V_\varepsilon(L).$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in V_\delta^*(c) \cap A \ f(x) \in V_\varepsilon(L).$$

*Remark* (Interpretation). The neighbourhood form packages the same local limit condition in terms of sets around the input point and output value. The domain is still restricted to  $A$ , and the neighbourhood of  $c$  is deleted so the value of  $f$  at  $c$  itself is irrelevant.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [\$\varepsilon\$ -Neighbourhood](#)
- [Deleted  \$\varepsilon\$ -Neighbourhood](#)

### Proposition

**Proposition 10.1.6** (Neighbourhood Form of the Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every  $\varepsilon$ -neighbourhood  $V_\varepsilon(L)$  there exists a  $\delta$ -neighbourhood  $V_\delta(c)$  such that  $x \in V_\delta^*(c) \cap A$  implies  $f(x) \in V_\varepsilon(L)$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in V_\delta^*(c) \cap A \ f(x) \in V_\varepsilon(L) \right]. \end{aligned}$$

*Remark* (Interpretation). The equivalence is purely notational:  $V_\delta^*(c) \cap A$  is exactly the set of  $x \in A$  satisfying  $0 < |x - c| < \delta$ , and  $f(x) \in V_\varepsilon(L)$  is exactly  $\text{OutputTolerance}(f(x), L, \varepsilon)$ . The neighbourhood form makes the geometric content explicit: the image of every punctured  $\delta$ -slice of  $A$  must fit inside the  $\varepsilon$ -ball around  $L$ . This is the form of continuity used in metric space topology.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Neighbourhood Form of Limit](#)
- [\$\varepsilon\$ -Neighbourhood](#)
- [Deleted  \$\varepsilon\$ -Neighbourhood](#)

### Theorem

**Theorem 10.1.7** (Uniqueness of Limits of Functions). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $f$  has at most one limit at  $c$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow c} f(x) = L \wedge \lim_{x \rightarrow c} f(x) = L' \Rightarrow L = L'.$$

*Remark* (Failure modes). Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

*Remark* (Failure mode decomposition). Non-uniqueness would require two values  $L \neq L'$  each satisfying the limit definition. The  $\varepsilon = |L - L'|/2$  argument shows any  $x$  within  $\min(\delta_1, \delta_2)$  of  $c$  satisfies both  $|f(x) - L| < \varepsilon$  and  $|f(x) - L'| < \varepsilon$ , but then  $|L - L'| \leq |L - f(x)| + |f(x) - L'| < 2\varepsilon = |L - L'|$ , a contradiction. The cluster point condition is essential: it guarantees such an  $x$  exists.

*Remark* (Interpretation). The limit, when it exists, is uniquely determined by the local behavior of  $f$  near  $c$ . Uniqueness is what licenses writing  $\lim_{x \rightarrow c} f(x) = L$  as an equation rather than an assertion.

*Remark* (Dependencies).

- [Limit of a Function](#)

#### Proposition

**Proposition 10.1.8** (Sequential Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall (x_n) \subseteq A \setminus \{c\} \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L \right]. \end{aligned}$$

*Remark* (Negated quantified statement).

$$\neg \lim_{x \rightarrow c} f(x) = L \iff \exists (x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge f(x_n) \not\rightarrow L.$$

*Remark* (Interpretation). The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xrightarrow{\text{Sequential Criterion}} \text{algebra of function limits}.$$

Every sequence result — sum, product, quotient, squeeze — transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove  $\lim_{x \rightarrow c} f(x) = L$ , exhibit one sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$  and  $f(x_n) \not\rightarrow L$ .

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Sequential Definition of Limit](#)

### Corollary

**Corollary 10.1.9** (Three Equivalent Forms of the Function Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The  $\varepsilon$ - $\delta$  form of  $\lim_{x \rightarrow c} f(x) = L$ , the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in V_\delta^*(c) \cap A \ f(x) \in V_\varepsilon(L) \right] \\ & \iff \left[ \forall (x_n) \subseteq A \setminus \{c\} \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L \right]. \end{aligned}$$

*Remark* (Interpretation). All three encodings capture the same mathematical condition. The  $\varepsilon$ - $\delta$  form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Neighbourhood Form of Limit](#)
- [Sequential Definition of Limit](#)
- [Neighbourhood Form of the Limit](#)
- [Sequential Criterion for Function Limits](#)

### Theorem

**Theorem 10.1.10** (Limit Implies Local Boundedness). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  for some  $L \in \mathbb{R}$ , then there exist  $\delta, M > 0$  such that  $|f(x)| \leq M$  for all  $x \in A$  with  $0 < |x - c| < \delta$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow c} f(x) = L \Rightarrow \exists \delta > 0 \exists M > 0 \forall x \in A \ (0 < |x - c| < \delta \Rightarrow |f(x)| \leq M).$$

*Remark* (Proof sketch). Apply the definition with  $\varepsilon = 1$ : get  $\delta > 0$  such that  $|f(x) - L| < 1$  holds for all  $x$  satisfying  $0 < |x - c| < \delta$ . The triangle inequality then gives  $|f(x)| \leq |f(x) - L| + |L| < 1 + |L|$ . Set  $M = 1 + |L|$ .

*Remark* (Interpretation). A finite limit does more than control the output near  $L$ : it prevents the function from growing without bound near  $c$ . Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

*Remark* (Dependencies).

- [Limit of a Function](#)

#### Theorem

**Theorem 10.1.11** (Bounded Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $a, b \in \mathbb{R}$ . If  $a \leq f(x) \leq b$  for all  $x \in A$  with  $x \neq c$ , and  $\lim_{x \rightarrow c} f(x) = L$ , then  $a \leq L \leq b$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$(\forall x \in A \setminus \{c\} : a \leq f(x) \leq b) \wedge \lim_{x \rightarrow c} f(x) = L \Rightarrow a \leq L \leq b.$$

*Remark* (Interpretation). If every nearby value of  $f$  is trapped in  $[a, b]$ , the limit cannot escape outside  $[a, b]$ . This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

*Remark* (Dependencies).

- [Limit of a Function](#)

#### Theorem

**Theorem 10.1.12** (Squeeze Theorem for Function Limits). *Let  $f, g, h : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Suppose  $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} g(x) = L$ , and suppose there exists  $\delta_0 > 0$  such that*

$$\forall x \in A, 0 < |x - c| < \delta_0 \Rightarrow f(x) \leq h(x) \leq g(x).$$

*Then  $\lim_{x \rightarrow c} h(x) = L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} g(x) = L \wedge f \leq h \leq g \text{ near } c \Rightarrow \lim_{x \rightarrow c} h(x) = L.$$

*Remark* (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new  $\varepsilon$ - $\delta$  argument is needed. The condition  $f \leq h \leq g$  is only required near  $c$ , not globally.

*Remark* (Dependencies).

- [Limit of a Function](#)

- Bounded Limits

### Theorem

**Theorem 10.1.13** (Limits Are Local). *Let  $E \subseteq X \subseteq \mathbb{R}$ , let  $x_0$  be a cluster point of  $E$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $\delta > 0$ . Then*

$$\lim_{x \rightarrow x_0; x \in E} f(x) = L \iff \lim_{x \rightarrow x_0; x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow x_0; x \in E} f(x) = L \iff \lim_{x \rightarrow x_0; x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

*Remark* (Interpretation). The existence and value of  $\text{FunctionLimit}(f, x_0, L)$  depend only on the behavior of  $f$  within any fixed neighbourhood of  $x_0$ . Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around  $c$  without any loss of generality.

*Remark* (Dependencies).

- Limit of a Function

### Proposition

**Proposition 10.1.14** (Limit of Absolute Value). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then  $\lim_{x \rightarrow c} |f(x)| = |L|$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow c} f(x) = L \Rightarrow \lim_{x \rightarrow c} |f(x)| = |L|.$$

*Remark* (Negated quantified statement).

$$\lim_{x \rightarrow c} |f(x)| = |L| \not\Rightarrow \lim_{x \rightarrow c} f(x) = L.$$

*Remark* (Failure modes). The absolute values may converge while the signs of  $f(x)$  oscillate or approach the opposite sign from the proposed limiting value.

*Remark* (Proof strategy). The reverse triangle inequality  $||f(x)| - |L|| \leq |f(x) - L|$  feeds the  $\varepsilon$ - $\delta$  definition directly: the same  $\delta$  that delivers  $|f(x) - L| < \varepsilon$  also delivers  $||f(x)| - |L|| < \varepsilon$ .

*Remark* (Interpretation). Taking absolute values can collapse sign information, so convergence of  $f$  forces convergence of  $|f|$ , but convergence of  $|f|$  alone does not recover the original limiting sign.

*Remark* (Dependencies).

- [Limit of a Function](#)

### Theorem

**Theorem 10.1.15** (Composition of Limits). *Let  $f : A \rightarrow B$ , let  $g : B \rightarrow \mathbb{R}$ , let  $c_1$  be a cluster point of  $A$ , and let  $c_2 \in \mathbb{R}$  be a cluster point of  $B$ . Suppose  $\lim_{x \rightarrow c_1} f(x) = c_2$  and  $\lim_{y \rightarrow c_2} g(y) = L$ . If  $c_2 \in B$ , assume also that  $g(c_2) = L$ . Then  $\lim_{x \rightarrow c_1} g(f(x)) = L$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow c_1} f(x) = c_2 \wedge \lim_{y \rightarrow c_2} g(y) = L \wedge (c_2 \in B \Rightarrow g(c_2) = L) \Rightarrow \lim_{x \rightarrow c_1} g(f(x)) = L.$$

*Remark* (Failure modes). The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

*Remark* (Failure mode decomposition). The hypothesis  $g(c_2) = L$  is essential. Without it the composition can fail: take  $f \equiv c_2$  (constant) and  $g(c_2) \neq L$ . Then  $g(f(x)) = g(c_2) \neq L$  for all  $x$ , so  $\lim_{x \rightarrow c_1} g(f(x)) = L$  fails even though both component limits exist. The failure is that  $f(x) = c_2$  is excluded from the punctured condition on  $g$ , but  $f$  hits  $c_2$  constantly.

*Remark* (Interpretation). Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis  $g(c_2) = L$  bridges the gap: when  $f(x) = c_2$ , the output condition  $g(f(x)) = g(c_2) = L$  is satisfied directly. This hypothesis is equivalent to requiring  $g$  to be continuous at  $c_2$  (within  $B$ ).

*Remark* (Dependencies).

- [Limit of a Function](#)

### Theorem

**Theorem 10.1.16** (Monotone Limit Theorem). *Let  $a < b$  and let  $f : (a, b) \rightarrow \mathbb{R}$  be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing  $f$  and every  $x_0 \in (a, b)$ :*

$$f(x_0^-) = \sup\{f(x) : a < x < x_0\}, \quad f(x_0^+) = \inf\{f(x) : x_0 < x < b\}.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$f \text{ is monotone on } (a, b) \wedge \exists B > 0 \forall x \in (a, b) (|f(x)| \leq B) \Rightarrow \forall x_0 \in (a, b) : \lim_{x \rightarrow x_0^-} f(x) = f(x_0^-) \wedge \lim_{x \rightarrow x_0^+} f(x) = f(x_0^+)$$

*Remark* (Interpretation). Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing  $f$ , the values  $f(x)$  for  $x < x_0$  form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities — never oscillatory ones.



Remark (Dependencies).

- [Monotone Function](#)
- [Bounded Function](#)
- [Left-Hand Limit](#)
- [Right-Hand Limit](#)

### Proposition

**Proposition 10.1.17** (Cauchy Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  exists for some  $L \in \mathbb{R}$  if and only if*

$$\forall \text{OutputTolerance}(\varepsilon), \exists \text{InputTolerance}(\delta), \forall x, y \in A : 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R} \left( \lim_{x \rightarrow c} f(x) = L \right) \iff \forall \varepsilon > 0, \exists \delta > 0, \forall x, y \in A : 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

Remark (Interpretation). The Cauchy criterion tests limit existence without knowing  $L$ . Two outputs from any sufficiently small punctured ball must lie within  $\varepsilon$  of each other — the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of  $\mathbb{R}$  in exactly the same way.

Remark (Dependencies).

- [Limit of a Function](#)

## Algebra of Limits

### Theorem

**Theorem 10.1.18** (Sum Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (f + g)(x) = L + M.$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta_f > 0 \forall x \in A \ 0 < |x - c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \wedge \left[ \forall \varepsilon > 0 \exists \delta_g > 0 \forall x \in A \ 0 < |x - c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon \right] \\ & \Rightarrow \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |(f + g)(x) - (L + M)| < \varepsilon. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f + g, c, L + M).$$

*Remark* (Interpretation). Limits respect addition: sufficiently close values of  $f$  and  $g$  add to values of  $f + g$  close to the sum of the two limits.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Pointwise Sum of Functions](#)

#### Theorem

**Theorem 10.1.19** (Difference Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (f - g)(x) = L - M.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \lim_{x \rightarrow c} f(x) = L \right] \wedge \left[ \lim_{x \rightarrow c} g(x) = M \right] \\ & \implies \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \implies |(f - g)(x) - (L - M)| < \varepsilon. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f - g, c, L - M).$$

*Remark* (Interpretation). The difference rule is the sum rule applied to  $f + (-g)$ .

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Pointwise Difference of Functions](#)

#### Theorem

**Theorem 10.1.20** (Scalar Multiple Rule for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then*

$$\lim_{x \rightarrow c} (\alpha f)(x) = \alpha L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\left[ \lim_{x \rightarrow c} f(x) = L \right] \implies \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \implies |\alpha f(x) - \alpha L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \implies \text{FunctionLimit}(\alpha f, c, \alpha L).$$

*Remark* (Interpretation). Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Pointwise Scalar Multiple of a Function](#)

#### Theorem

**Theorem 10.1.21** (Product Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then*

$$\lim_{x \rightarrow c} (fg)(x) = LM.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \lim_{x \rightarrow c} f(x) = L \right] \wedge \left[ \lim_{x \rightarrow c} g(x) = M \right] \\ & \implies \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \implies |f(x)g(x) - LM| < \varepsilon. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(fg, c, LM).$$

*Remark* (Interpretation). The product rule combines closeness of  $f$  to  $L$  with local boundedness of  $g$  near  $c$ .

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Pointwise Product of Functions](#)

#### Theorem

**Theorem 10.1.22** (Quotient Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ ,  $\lim_{x \rightarrow c} g(x) = M$ , and  $M \neq 0$ , then there exists  $\delta_0 > 0$  such that  $g(x) \neq 0$  whenever  $x \in A$  and  $0 < |x - c| < \delta_0$ , and*

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \lim_{x \rightarrow c} f(x) = L \right] \wedge \left[ \lim_{x \rightarrow c} g(x) = M \right] \wedge M \neq 0 \\ & \Rightarrow \exists \delta_0 > 0 \forall x \in A \ 0 < |x - c| < \delta_0 \Rightarrow g(x) \neq 0, \\ & \text{and } \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow \left| \frac{f(x)}{g(x)} - \frac{L}{M} \right| < \varepsilon. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \wedge M \neq 0 \implies \text{FunctionLimit}(f/g, c, L/M).$$

*Remark* (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Pointwise Quotient of Functions](#)

## One-Sided Limits

### Definition (Right-Hand Limit)

**Definition 10.1.23** (Right-Hand Limit). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (c, \infty)$ . A real number  $L$  is the **right-hand limit** of  $f$  at  $c$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{RightLimit}(f, c, L) := \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < x - c < \delta \wedge |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading).

$$\neg \text{RightLimit}(f, c, L).$$

*Remark* (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to  $c$  from the right.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < x - c < \delta \wedge |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Interpretation). Only inputs to the right of  $c$  are tested. The value of  $f$  at  $c$ , if it is defined, is irrelevant to the right-hand limit.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Cluster Point](#)

#### Definition (Left-Hand Limit)

**Definition 10.1.24** (Left-Hand Limit). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (-\infty, c)$ . A real number  $L$  is the **left-hand limit** of  $f$  at  $c$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Definition predicate reading).

$$\text{LeftLimit}(f, c, L) := \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

*Remark* (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < c - x < \delta \wedge |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Negation predicate reading).

$$\neg \text{LeftLimit}(f, c, L).$$

*Remark* (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to  $c$  from the left.

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < c - x < \delta \wedge |f(x) - L| \geq \varepsilon_0.$$

*Remark* (Interpretation). Only inputs to the left of  $c$  are tested. This is the mirror image of the right-hand limit definition.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Cluster Point](#)

### Theorem

**Theorem 10.1.25** (Sequential Criterion for Right-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (c, \infty)$ . Then  $\lim_{x \rightarrow c^+} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (c, \infty)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall (x_n) \subseteq A \cap (c, \infty) \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty) \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L.$$

*Remark* (Interpretation). Right-hand limits can be tested by sequences approaching  $c$  from above. Every such sequence must force the corresponding output sequence toward  $L$ .

*Remark* (Dependencies).

- [Right-Hand Limit](#)
- [Sequential Definition of Limit](#)

### Corollary

**Corollary 10.1.26** (Sequential Criterion for Left-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (-\infty, c)$ . Then  $\lim_{x \rightarrow c^-} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (-\infty, c)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall (x_n) \subseteq A \cap (-\infty, c) \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c) \ x_n \rightarrow c \Rightarrow f(x_n) \rightarrow L.$$

*Remark* (Interpretation). Left-hand limits can be tested by sequences approaching  $c$  from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

*Remark* (Dependencies).

- [Left-Hand Limit](#)

- [Sequential Definition of Limit](#)

### Theorem

**Theorem 10.1.27** (Two-Sided Limit via One-Sided Limits). *Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . If  $c$  is a cluster point of both  $A \cap (c, \infty)$  and  $A \cap (-\infty, c)$ , then  $\lim_{x \rightarrow c} f(x) = L$  if and only if both  $\lim_{x \rightarrow c^+} f(x) = L$  and  $\lim_{x \rightarrow c^-} f(x) = L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \left[ \forall \varepsilon > 0 \exists \delta_+ > 0 \forall x \in A \ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon \right] \\ & \quad \wedge \left[ \forall \varepsilon > 0 \exists \delta_- > 0 \forall x \in A \ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon \right]. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \iff \text{RightLimit}(f, c, L) \wedge \text{LeftLimit}(f, c, L).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} & \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0 \\ & \iff \left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < x - c < \delta \wedge |f(x) - L| \geq \varepsilon_0 \right] \\ & \quad \vee \left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \ 0 < c - x < \delta \wedge |f(x) - L| \geq \varepsilon_0 \right]. \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{FunctionLimit}(f, c, L) \iff \neg \text{RightLimit}(f, c, L) \vee \neg \text{LeftLimit}(f, c, L).$$

*Remark* (Failure modes). A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

*Remark* (Failure mode decomposition).

$$\neg \text{FunctionLimit}(f, c, L) \iff \neg \text{RightLimit}(f, c, L) \vee \neg \text{LeftLimit}(f, c, L).$$

*Remark* (Interpretation). A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

*Remark* (Dependencies).

- [Limit of a Function](#)
- [Right-Hand Limit](#)
- [Left-Hand Limit](#)

## 10.2 Point Continuity

### Oscillation — Quick Reference

| Concept/Statement          | Meaning/Role                                          |
|----------------------------|-------------------------------------------------------|
| Oscillation on a set       | Diameter of the image over a set.                     |
| Oscillation at a point     | Limiting local oscillation near the point.            |
| Zero oscillation criterion | Continuity is exactly vanishing point oscillation.    |
| Discontinuity set          | Discontinuities are detected by positive oscillation. |

### Discontinuity — Quick Reference

| Concept/Statement           | Meaning/Role                             |
|-----------------------------|------------------------------------------|
| Point of discontinuity      | Negation of continuity at a point.       |
| Sequential discontinuity    | A bad sequence witnesses failure.        |
| Neighbourhood discontinuity | Neighbourhood form of the same failure.  |
| Equivalence lemma           | The discontinuity formulations agree.    |
| Types of discontinuity      | Removable, jump, and essential behavior. |

### Continuity — Quick Reference

| Concept/Statement        | Meaning/Role                                        |
|--------------------------|-----------------------------------------------------|
| Relative neighbourhood   | Neighbourhoods are read inside the domain.          |
| Point continuity         | The unpunctured $\varepsilon$ - $\delta$ condition. |
| Neighbourhood continuity | Images of nearby inputs stay near the value.        |
| Sequential continuity    | Convergent input sequences preserve limits.         |
| Oscillation criterion    | Continuity is equivalent to zero local oscillation. |

## Continuity

### Definition (Relative Neighborhood)

**Definition 10.2.1** (Relative Neighborhood). Let  $E \subseteq \mathbb{R}$  and  $a \in E$ . A set  $U \subseteq \mathbb{R}$  is a neighborhood of  $a$  in  $E$  if and only if there exists  $\delta > 0$  such that  $U = E \cap (a - \delta, a + \delta)$ . Equivalently, every neighborhood of  $a$  in  $E$  is a set of the form  $U_E(a) := E \cap (a - \delta, a + \delta)$  for some  $\delta > 0$ .

*Remark* (Standard quantified statement).

Let  $E \subseteq \mathbb{R}$ ,  $a \in E$ , and  $U \subseteq \mathbb{R}$ .

$U$  is a neighborhood of  $a$  in  $E \iff \exists \delta > 0 (U = E \cap (a - \delta, a + \delta))$ .

*Remark* (Definition predicate reading).

$\text{RelativeNeighborhood}(U, E, a) := \exists \delta > 0 (U = E \cap (a - \delta, a + \delta))$ .



*Remark* (Negated quantified statement).

Let  $E \subseteq \mathbb{R}$ ,  $a \in E$ , and  $U \subseteq \mathbb{R}$ .

$U$  fails to be a neighborhood of  $a$  in  $E \iff \forall \delta > 0 \ (U \neq E \cap (a - \delta, a + \delta))$ .

*Remark* (Negation predicate reading).

$\neg \text{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0 \ (U \neq E \cap (a - \delta, a + \delta))$ .

*Remark* (Failure modes). Failure occurs precisely when no radius  $\delta$  makes  $U$  coincide with the relative interval slice: for every  $\delta > 0$  either  $U$  contains a point outside  $E \cap (a - \delta, a + \delta)$  or the slice contains a point missing from  $U$ .

*Remark* (Failure mode decomposition).

$$\forall \delta > 0 \ \underbrace{(\exists x \in U : x \notin E \cap (a - \delta, a + \delta))}_{\text{extraneous point}} \vee \underbrace{(\exists y \in E \cap (a - \delta, a + \delta) : y \notin U)}_{\text{missing point}}.$$

*Remark* (Interpretation). Relative neighborhoods capture the local behavior of  $E$  around  $a$  by intersecting  $E$  with an ordinary open interval centered at  $a$ . The construction restricts the ambient notion of openness to the geometry of  $E$ , so it is suited to relative topologies. Failure simply means that  $U$  never matches the precise slice of  $E$  at any scale, typically because  $U$  omits some nearby points of  $E$  or includes points that fall outside  $E$  near  $a$ .

*Remark* (Dependencies).

- [Subset](#)
- [Epsilon Neighbourhood](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Continuity at a Point)

**Definition 10.2.2** (Continuity at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is **continuous at**  $c$  if and only if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A : (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon).$$

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ ,  $c \in A$ .

$f$  is continuous at  $c \iff \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)$ .

*Remark* (Definition predicate reading).

$\text{ContinuousAt}(f, A, c) := \forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)$ .

*Remark* (Negated quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ ,  $c \in A$ .

$f$  is not continuous at  $c \iff \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)$ .

*Remark* (Negation predicate reading).

$\neg \text{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0)$ .

*Remark* (Failure modes). Continuity at  $c$  fails precisely when there exists a persistent output gap  $\varepsilon_0 > 0$  such that no matter how small the chosen neighborhood of  $c$  is, a point of  $A$  within that neighborhood keeps the function values at least  $\varepsilon_0$  away from  $f(c)$ . The single failure branch records this inescapable oscillation near  $c$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{persistent gap}} \quad \underbrace{\forall \delta > 0}_{\text{every neighborhood}} \quad \underbrace{\exists x \in A}_{\text{nearby witness}} : |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0.$$

*Remark* (Interpretation). Continuity at  $c$  means that the output  $f(x)$  can be kept arbitrarily close to  $f(c)$  by taking  $x$  sufficiently close to  $c$  while staying in the domain  $A$ . The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on  $c$ , so nearby inputs cannot force the outputs to settle near  $f(c)$ . This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at  $c$ .

*Remark* (Dependencies).

- [Definition: Limit of a Function](#)

#### Definition (Continuity at a Point — Neighbourhood Form)

**Definition 10.2.3** (Continuity at a Point — Neighbourhood Form). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is continuous at  $c$  if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that

$$f(A \cap (c - \delta, c + \delta)) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ ,  $c \in A$ .

$f$  is continuous at  $c \iff \forall \varepsilon > 0 \exists \delta > 0 :$

$$f(A \cap (c - \delta, c + \delta)) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{ContinuousAt}(f, c) := \forall \varepsilon > 0 \exists \delta > 0 : f(A \cap (c - \delta, c + \delta)) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f : A \rightarrow \mathbb{R}, c \in A. \\ &\neg(f \text{ is continuous at } c) \iff \exists \varepsilon_0 > 0 \forall \delta > 0 : \\ &\quad f(A \cap (c - \delta, c + \delta)) \not\subseteq (f(c) - \varepsilon_0, f(c) + \varepsilon_0). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{DiscontinuousAt}(f, c) \iff \exists \varepsilon_0 > 0 \forall \delta > 0 : f(A \cap (c - \delta, c + \delta)) \not\subseteq (f(c) - \varepsilon_0, f(c) + \varepsilon_0).$$

*Remark* (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about  $c$ . This manifests by finding points of  $A$  arbitrarily close to  $c$  whose images either exceed the upper endpoint  $f(c) + \varepsilon_0$  or fall below the lower endpoint  $f(c) - \varepsilon_0$ .

*Remark* (Failure mode decomposition).

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \underbrace{\left( \exists x \in A \cap (c - \delta, c + \delta) : f(x) \geq f(c) + \varepsilon_0 \right)}_{\text{overshoot}} \vee \underbrace{\left( \exists x \in A \cap (c - \delta, c + \delta) : f(x) \leq f(c) - \varepsilon_0 \right)}_{\text{undershoot}}$$

*Remark* (Interpretation). Continuity at  $c$  demands that every output tolerance around  $f(c)$  has a matching neighborhood around  $c$  whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from  $A$  approaches  $c$  while their images either spike above or dip below  $f(c)$  by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above  $c$  must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs  $c$ .

#### Definition (Continuity at a Point — Sequential Form)

**Definition 10.2.4** (Continuity at a Point — Sequential Form). Let  $A \subseteq \mathbb{R}$ , let  $c \in A$ , and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is continuous at  $c$  if and only if for every sequence  $(x_n)_{n \in \mathbb{N}}$  in  $A$ , the convergence  $\lim_{n \rightarrow \infty} x_n = c$  implies  $\lim_{n \rightarrow \infty} f(x_n) = f(c)$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, c \in A, f : A \rightarrow \mathbb{R}. \\ &f \text{ is continuous at } c \iff \forall (x_n)_{n \in \mathbb{N}} \subseteq A \left( \lim_{n \rightarrow \infty} x_n = c \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(c) \right). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{ContinuousAt}(f, c) := \forall x_n \left( \text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n), f(c)) \right)$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, c \in A, f : A \rightarrow \mathbb{R}. \\ &f \text{ is not continuous at } c \iff \exists (x_n)_{n \in \mathbb{N}} \subseteq A \left( \lim_{n \rightarrow \infty} x_n = c \wedge \lim_{n \rightarrow \infty} f(x_n) \neq f(c) \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{ContinuousAt}(f, c) \iff \exists x_n \left( \text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \wedge \neg \text{ConvergentSequence}(f(x_n), f(c)) \right)$$

*Remark* (Failure modes). Continuity at  $c$  fails sequentially when there exists a sequence  $(x_n)_{n \in \mathbb{N}}$  in  $A$  with  $x_n \rightarrow c$  but  $f(x_n)$  does not converge to  $f(c)$ . This manifests in two distinct ways: either  $f(x_n)$  converges to some limit different from  $f(c)$ , or  $f(x_n)$  fails to converge entirely (oscillating without settling or diverging to infinity).

*Remark* (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n \in \mathbb{N}} \subseteq A : \underbrace{\lim_{n \rightarrow \infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n \rightarrow \infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

*Remark* (Interpretation). The sequential characterization asserts that information about  $f$  near  $c$  is completely encoded by its action on limits of sequences taken within the domain. Any approach to  $c$  through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that  $f$  cannot introduce jumps or oscillations that persist along every infinitesimal path toward  $c$ .

*Remark* (Dependencies).

[Definition 10.2.2.](#)

## Characterization of Discontinuity

### Definition (Point of Discontinuity)

**Definition 10.2.5** (Point of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , and let  $a \in A$ . The function  $f$  is discontinuous at  $a$  if and only if  $f$  is not continuous at  $a$ .

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f: A \rightarrow \mathbb{R}$ ,  $a \in A$ .

$$f \text{ is discontinuous at } a \iff \neg \left( \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A (|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon) \right).$$

*Remark* (Definition predicate reading).

$$\text{DiscontinuousAt}(f, a) := \neg \text{ContinuousAt}(f, a).$$

*Remark* (Negated quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f: A \rightarrow \mathbb{R}$ ,  $a \in A$ .

$$f \text{ is discontinuous at } a \iff \exists \varepsilon > 0 \forall \delta > 0 \exists x \in A (|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon).$$

*Remark* (Negation predicate reading).

$$\neg \text{DiscontinuousAt}(f, a) \iff \text{ContinuousAt}(f, a) \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A (|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon)$$

*Remark* (Failure modes). Discontinuity arises either because the nearby values of  $f$  do not approach any single real number as  $x \rightarrow a$ , or because a well-defined limit exists but disagrees with the actual value  $f(a)$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\neg \exists L \in \mathbb{R} \text{ FunctionLimit}(f, A, a, L)}_{\text{no limit exists}} \vee \underbrace{(\exists L \in \mathbb{R} \text{ FunctionLimit}(f, A, a, L) \wedge L \neq f(a))}_{\text{limit-value mismatch}}.$$

*Remark* (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

*Remark* (Dependencies).

[Definition 10.2.2.](#)

### Definition (Sequential Characterization of Discontinuity)

**Definition 10.2.6** (Sequential Characterization of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is discontinuous at  $c$  if and only if there exists a sequence  $(x_n)_{n \in \mathbb{N}} \subseteq A$  such that  $x_n \rightarrow c$  while  $f(x_n) \not\rightarrow f(c)$ .

*Remark* (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A [x_n \rightarrow c \wedge f(x_n) \not\rightarrow f(c)].$$

*Remark* (Definition predicate reading).

$$\text{DiscontinuousAt}(f, c) := \exists (x_n) \subseteq A : x_n \rightarrow c \wedge \neg(f(x_n) \rightarrow f(c)).$$

*Remark* (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A : x_n \rightarrow c \Rightarrow f(x_n) \rightarrow f(c).$$

*Remark* (Negation predicate reading).

$$\neg \text{DiscontinuousAt}(f, c) \iff \text{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A : x_n \rightarrow c \Rightarrow f(x_n) \rightarrow f(c). \blacksquare$$

*Remark* (Failure modes). The characterization fails — meaning  $f$  is NOT sequentially discontinuous — when every sequence  $(x_n)$  in  $A$  converging to  $c$  has  $f(x_n) \rightarrow f(c)$ . No sequential witness exists.

*Remark* (Failure mode decomposition).

$$\neg \text{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A : x_n \rightarrow c \Rightarrow f(x_n) \rightarrow f(c)}_{\text{every approach preserves the limit}}.$$

*Remark* (Interpretation). A discontinuity at  $c$  can be certified by a single sequence approaching  $c$  whose images refuse to settle at  $f(c)$ .

*Remark* (Dependencies).

[Definition 10.2.4](#); [Definition 10.2.5](#).

#### Definition (Neighbourhood Characterization of Discontinuity)

**Definition 10.2.7** (Neighbourhood Characterization of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The function  $f$  is discontinuous at  $c$  if and only if there exists a neighborhood  $V$  of  $f(c)$  such that for every neighborhood  $U$  of  $c$  relative to  $A$  there exists  $x \in U \cap A$  with  $f(x) \notin V$ .

*Remark* (Standard quantified statement).

$f$  is discontinuous at  $c \iff \exists V \text{ nbhd of } f(c) \forall U \text{ nbhd of } c \exists x \in U \cap A (f(x) \notin V)$ .

*Remark* (Definition predicate reading).

$\text{DiscontinuousAt}(f, c) := \exists V \text{ nbhd of } f(c) \forall U \text{ nbhd of } c \exists x \in U \cap A : f(x) \notin V$ .

*Remark* (Negated quantified statement).

$f$  is continuous at  $c \iff \forall V \text{ nbhd of } f(c) \exists U \text{ nbhd of } c \forall x \in U \cap A (f(x) \in V)$ .

*Remark* (Negation predicate reading).

$\neg \text{DiscontinuousAt}(f, c) \iff \text{ContinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c) \exists U \text{ nbhd of } c \forall x \in U \cap A (f(x) \in V)$

*Remark* (Failure modes). The neighbourhood discontinuity characterization fails — meaning  $f$  is actually continuous — when every output neighbourhood  $V$  of  $f(c)$  can be matched by some input neighbourhood  $U$  such that  $f$  maps all of  $U \cap A$  into  $V$ . No perpetually-missed output neighbourhood exists.

*Remark* (Failure mode decomposition).

$\neg \text{DiscontinuousAt}(f, c) \iff \forall V \text{ nbhd of } f(c) \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} : \underbrace{\forall x \in U \cap A : f(x) \in V}_{\text{all outputs inside tolerance}}.$

*Remark* (Interpretation). Discontinuity means a fixed output neighborhood  $V$  is perpetually missed no matter how tightly the input is restricted around  $c$ .

*Remark* (Dependencies).

[Definition 10.2.1](#).

#### Lemma (Equivalent Formulations of Discontinuity at a Point)

**Lemma 10.2.8** (Equivalent Formulations of Discontinuity at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , and let  $c \in A$ . The following are equivalent:

1.  $c$  is a point of discontinuity in the  $\varepsilon$ - $\delta$  sense.
2.  $c$  is a sequential point of discontinuity.
3.  $c$  is a neighbourhood point of discontinuity.

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
 & (1) \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A : 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
 \iff & (2) \exists (x_n) \subseteq A : x_n \rightarrow c \wedge \neg(f(x_n) \rightarrow f(c)) \\
 \iff & (3) \exists V \text{ nbhd of } f(c) \forall U \text{ nbhd of } c \exists x \in U \cap A : f(x) \notin V.
 \end{aligned}$$

*Remark* (Predicate reading).

$$\begin{aligned}
 \text{DiscontinuousAt}(f, c) & \iff \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A : 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\
 & \iff \exists (x_n) \subseteq A : x_n \rightarrow c \wedge \neg(f(x_n) \rightarrow f(c)) \\
 & \iff \exists V \text{ nbhd of } f(c) \forall U \text{ nbhd of } c \exists x \in U \cap A : f(x) \notin V.
 \end{aligned}$$

*Remark* (Failure modes). The equivalence cannot fail for functions at limit points of  $A$ : the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where  $c$  is not a limit point of  $A$ , the punctured-neighbourhood conditions are vacuously satisfied by every  $L$ , and the equivalence becomes trivial rather than informative.

*Remark* (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

*Remark* (Dependencies).

- [Point of Discontinuity](#)
- [Sequential Characterization of Discontinuity](#)
- [Neighbourhood Characterization of Discontinuity](#)

### Definition (Types of Discontinuity)

**Definition 10.2.9** (Types of Discontinuity). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $x_0 \in A$  with  $f$  discontinuous at  $x_0$ .

- (i) *Removable* iff  $\exists L \in \mathbb{R} : \lim_{x \rightarrow x_0} f(x) = L$  and  $L \neq f(x_0)$ .
- (ii) *Jump* iff  $\exists L_-, L_+ \in \mathbb{R} : \lim_{x \rightarrow x_0^-} f(x) = L_-, \lim_{x \rightarrow x_0^+} f(x) = L_+, \text{ and } L_- \neq L_+.$
- (iii) *Essential* iff it is not removable.

*Remark* (Standard quantified statement).

$$\begin{aligned}
 \text{Removable} & \iff \exists L \in \mathbb{R} \left( \lim_{x \rightarrow x_0} f(x) = L \wedge L \neq f(x_0) \right). \\
 \text{Jump} & \iff \exists L_-, L_+ \in \mathbb{R} \left( \lim_{x \rightarrow x_0^-} f(x) = L_- \wedge \lim_{x \rightarrow x_0^+} f(x) = L_+ \wedge L_- \neq L_+ \right). \\
 \text{Essential} & \iff \neg \exists L \in \mathbb{R} \left( \lim_{x \rightarrow x_0} f(x) = L \wedge L \neq f(x_0) \right).
 \end{aligned}$$

*Remark* (Definition predicate reading).

$\text{RemovableDiscontinuity}(f, x_0) := \exists L \in \mathbb{R} \left( \text{FunctionLimit}(f, A, x_0, L) \wedge L \neq f(x_0) \right).$

$\text{JumpDiscontinuity}(f, x_0) := \exists L_-, L_+ \in \mathbb{R} \left( \text{LeftLimit}(f, A, x_0, L_-) \wedge \text{RightLimit}(f, A, x_0, L_+) \wedge L_- \neq L_+ \right).$

$\text{EssentialDiscontinuity}(f, x_0) := \neg \text{RemovableDiscontinuity}(f, x_0).$

*Remark* (Negated quantified statement).

$$\begin{aligned}\neg \text{Removable} &\iff \forall L \in \mathbb{R} \left( \lim_{x \rightarrow x_0} f(x) \neq L \vee L = f(x_0) \right), \\ \neg \text{Jump} &\iff \forall L_-, L_+ \in \mathbb{R} \left( \lim_{x \rightarrow x_0^-} f(x) \neq L_- \vee \lim_{x \rightarrow x_0^+} f(x) \neq L_+ \vee L_- = L_+ \right), \\ \neg \text{Essential} &\iff \text{Removable}.\end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned}\neg \text{RemovableDiscontinuity}(f, x_0) &\iff \forall L \in \mathbb{R} \left( \neg \text{FunctionLimit}(f, A, x_0, L) \vee L = f(x_0) \right). \\ \neg \text{JumpDiscontinuity}(f, x_0) &\iff \forall L_-, L_+ \in \mathbb{R} \left( \neg \text{LeftLimit}(f, A, x_0, L_-) \vee \neg \text{RightLimit}(f, A, x_0, L_+) \right) \\ \neg \text{EssentialDiscontinuity}(f, x_0) &\iff \text{RemovableDiscontinuity}(f, x_0).\end{aligned}$$

*Remark* (Failure modes). • Removable discontinuity fails when no bilateral limit exists at  $x_0$ , or when the bilateral limit exists but equals  $f(x_0)$  (making the function actually continuous).

- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from  $f(x_0)$ ).

*Remark* (Failure mode decomposition).

$$\begin{aligned}\neg \text{RemovableDiscontinuity}(f, x_0) &\iff \underbrace{\neg \exists L \in \mathbb{R} \lim_{x \rightarrow x_0} f(x) = L}_{\text{no bilateral limit exists}} \vee \underbrace{\lim_{x \rightarrow x_0} f(x) = f(x_0)}_{\text{limit equals function value}}. \\ \neg \text{JumpDiscontinuity}(f, x_0) &\iff \underbrace{\neg \exists L_- \in \mathbb{R} \lim_{x \rightarrow x_0^-} f(x) = L_-}_{\text{no left limit}} \vee \underbrace{\neg \exists L_+ \in \mathbb{R} \lim_{x \rightarrow x_0^+} f(x) = L_+}_{\text{no right limit}} \\ &\quad \vee \underbrace{\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x)}_{\text{one-sided limits agree}}.\end{aligned}$$

*Remark* (Interpretation). Removable discontinuities can be healed by redefining  $f(x_0)$ . Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

*Remark* (Dependencies).

- [Point of Discontinuity](#)
- [Limit of a Function](#)
- [Left-Hand Limit](#)
- [Right-Hand Limit](#)



# Oscillation

## Definition (Oscillation on a Set)

**Definition 10.2.10** (Oscillation on a Set). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $E \subseteq A$ . The oscillation of  $f$  on  $E$  is the extended real number

$$\omega(f; E) := \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\} = \text{diam}(f(E)).$$

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f : A \rightarrow \mathbb{R}, E \subseteq A, \omega \in [0, \infty]. \\ &\omega(f; E) = \omega \iff \omega = \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\}. \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{OscillationOn}(f, E, \omega) := \omega = \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\}.$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f : A \rightarrow \mathbb{R}, E \subseteq A, \omega \in [0, \infty]. \\ &\omega(f; E) \neq \omega \iff \omega \neq \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\}. \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{OscillationOn}(f, E, \omega) \iff \omega \neq \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\}.$$

*Remark* (Failure modes). The proposed value  $\omega$  fails either because it is not an upper bound for pairwise differences of function values on  $E$ , or because it is an upper bound but not the least such upper bound.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists x_1, x_2 \in E : |f(x_1) - f(x_2)| > \omega}_{\text{not an upper bound}} \vee \underbrace{\exists \eta < \omega \forall x_1, x_2 \in E : |f(x_1) - f(x_2)| \leq \eta}_{\text{not least}}.$$

*Remark* (Interpretation). Oscillation on a set records the spread of the image — the diameter of  $f(E)$ . Small oscillation means all values of  $f$  on  $E$  lie close together. Zero oscillation means  $f$  is constant on  $E$ .

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Supremum](#)
- [Absolute Value on  \$\mathbb{R}\$](#)

### Definition (Oscillation at a Point)

**Definition 10.2.11** (Oscillation at a Point). Let  $E \subseteq \mathbb{R}$ , let  $f : E \rightarrow \mathbb{R}$ , and let  $x_0 \in \mathbb{R}$  satisfy  $U_\delta(x_0) \cap E \neq \emptyset$  for every  $\delta > 0$ , where  $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$ . The oscillation of  $f$  at  $x_0$  is the extended real number

$$\omega(f; x_0) = \lim_{\delta \rightarrow 0^+} \omega(f; U_\delta(x_0) \cap E),$$

with  $\omega(f; A)$  denoting the oscillation of  $f$  on the set  $A$ . Because the function  $\delta \mapsto \omega(f; U_\delta(x_0) \cap E)$  is nonincreasing, this right-hand limit exists (possibly  $+\infty$ ).

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } E \subseteq \mathbb{R}, f : E \rightarrow \mathbb{R}, x_0 \in \mathbb{R}, \omega \in [0, \infty]. \\ &[(\forall \delta > 0 : U_\delta(x_0) \cap E \neq \emptyset) \wedge \omega(f; x_0) = \omega] \\ &\iff \forall \varepsilon > 0 \exists \delta > 0 \forall r (0 < r < \delta \Rightarrow |\omega(f; U_r(x_0) \cap E) - \omega| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{OscillationAt}(f, x_0, \omega) := \lim_{\delta \rightarrow 0^+} \omega(f; U_\delta(x_0) \cap E) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } E \subseteq \mathbb{R}, f : E \rightarrow \mathbb{R}, x_0 \in \mathbb{R}, \omega \in [0, \infty]. \\ &\neg[(\forall \delta > 0 : U_\delta(x_0) \cap E \neq \emptyset) \wedge \omega(f; x_0) = \omega] \\ &\iff \left[ \exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \emptyset \right] \\ &\quad \vee \left( \exists \varepsilon_0 > 0 \forall \delta > 0 \exists r (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0) \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{OscillationAt}(f, x_0, \omega) \iff \left[ \exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \emptyset \right] \vee \left( \exists \varepsilon_0 > 0 \forall \delta > 0 \exists r (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0) \right).$$

*Remark* (Failure modes). The definition fails in two distinct ways: either the point  $x_0$  has a punctured neighborhood missing  $E$ , so there are no arbitrarily small samples to measure, or the local oscillation values near  $x_0$  refuse to stabilize, preventing the limit from existing.

*Remark* (Failure mode decomposition).

$$\neg \text{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \emptyset}_{\text{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0 \forall \delta > 0 \exists r (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)}_{\text{oscillation limit does not exist}}$$

*Remark* (Interpretation). Oscillation at  $x_0$  measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around  $x_0$ . The shrinking neighborhoods  $U_r(x_0) \cap E$  supply finer and finer samples, and the oscillation on each such set records the spread of  $f$  there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when  $f$  becomes arbitrarily close to a single value, i.e., when  $f$  is continuous at  $x_0$ . Failure occurs either because  $x_0$  has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

Remark (Dependencies).

Definition 10.2.10

**Theorem (Continuity Characterised by Oscillation)**

**Theorem 10.2.12** (Continuity Characterised by Oscillation). *Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , and let  $c \in A$ . Then  $f$  is continuous at  $c$  if and only if  $\omega(f; c) = 0$ . Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} & \text{Let } A \subseteq \mathbb{R}, f: A \rightarrow \mathbb{R}, c \in A. \\ & (\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)) \\ & \iff \\ & \left( \inf_{\delta > 0} \sup \{ |f(x) - f(y)| : x, y \in A, |x - c| < \delta, |y - c| < \delta \} = 0 \right). \end{aligned}$$

Remark (Predicate reading).

$$\text{ContinuousAt}(f, c) \iff \text{OscillationAt}(f, c, 0).$$

Remark (Negated quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f: A \rightarrow \mathbb{R}$ ,  $c \in A$ .

$$\begin{aligned} & \left( \forall \varepsilon > 0 \exists \delta > 0 \forall x \in A (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon) \right) \wedge \left( \inf_{\delta > 0} \sup \{ |f(x) - f(y)| : x, y \in A, |x - c| < \delta, |y - c| < \delta \} > 0 \right) \\ & \vee \\ & \left( \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0) \right) \wedge \left( \inf_{\delta > 0} \sup \{ |f(x) - f(y)| : x, y \in A, |x - c| < \delta, |y - c| < \delta \} = 0 \right) \end{aligned}$$

Remark (Negation predicate reading).

$$\neg(\text{ContinuousAt}(f, c) \iff \text{OscillationAt}(f, c, 0)).$$

Remark (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either  $f$  remains continuous at  $c$  while the oscillation retains a positive lower bound, or  $\omega(f; c)$  vanishes even though  $f$  does not satisfy the  $\varepsilon$ - $\delta$  continuity condition at  $c$ .

Remark (Failure mode decomposition).

$$\underbrace{\text{ContinuousAt}(f, c) \wedge \neg \text{OscillationAt}(f, c, 0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg \text{ContinuousAt}(f, c) \wedge \text{OscillationAt}(f, c, 0)}_{\text{Zero oscillation without continuity}}.$$

Remark (Interpretation). The theorem states that measuring the oscillation of  $f$  near  $c$  captures exactly the same local regularity as the classical  $\varepsilon$ - $\delta$  definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing  $f(x)$  to approach  $f(c)$ . Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of  $c$ .

*Remark* (Dependencies).

Definition 10.2.2, Definition 10.2.11.

### Proposition

**Proposition 10.2.13** (Discontinuity Set via Oscillation). *Let  $f : E \rightarrow \mathbb{R}$ . Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $f : E \rightarrow \mathbb{R}$ .

$$\forall x \in E : x \in D(f) \iff \omega(f; x) > 0 \iff \exists n \in \mathbb{N} : \omega(f; x) \geq \frac{1}{n}.$$

*Remark* (Predicate reading).

$$\begin{aligned} x \in D(f) &\iff \text{DiscontinuousAt}(f, x) \\ &\iff \exists \omega \text{ (OscillationAt}(f, x, \omega) \wedge \omega > 0) \\ &\iff \exists n \in \mathbb{N} \exists \omega \text{ (OscillationAt}(f, x, \omega) \wedge \omega \geq \tfrac{1}{n}). \end{aligned}$$

*Remark* (Negated quantified statement).

Let  $f : E \rightarrow \mathbb{R}$ .

$$\begin{aligned} &\neg \left[ \forall x \in E \left( x \in D(f) \iff \omega(f; x) > 0 \iff \exists n \in \mathbb{N} \omega(f; x) \geq \frac{1}{n} \right) \right] \\ &\iff \exists x \in E \left( (x \in D(f) \wedge \omega(f; x) = 0) \vee (x \notin D(f) \wedge \omega(f; x) > 0) \right. \\ &\quad \left. \vee (\omega(f; x) > 0 \wedge \forall n \in \mathbb{N} \omega(f; x) < \tfrac{1}{n}) \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned} \exists x \in E \left( \exists \omega \text{ (DiscontinuousAt}(f, x) \wedge \text{OscillationAt}(f, x, \omega) \wedge \omega = 0) \vee \right. \\ \left. \exists \omega \text{ (}\neg \text{DiscontinuousAt}(f, x) \wedge \text{OscillationAt}(f, x, \omega) \wedge \omega > 0) \vee \right. \\ \left. \exists \omega \text{ (OscillationAt}(f, x, \omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N} \omega < \tfrac{1}{n}) \right). \end{aligned}$$

*Remark* (Failure modes). The proposition fails precisely when some  $x \in E$  is marked as discontinuous although its oscillation vanishes, or when  $x$  is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts  $\omega(f; x) > 0$  into the countable union description.

*Remark* (Failure mode decomposition).

$$\exists x \in E \begin{cases} x \in D(f) \wedge \omega(f; x) = 0 & \text{(Type I),} \\ x \notin D(f) \wedge \omega(f; x) > 0 & \text{(Type II),} \\ \omega(f; x) > 0 \wedge \forall n \in \mathbb{N} \, \omega(f; x) < \frac{1}{n} & \text{(Type III).} \end{cases}$$

*Remark* (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces  $\omega(f; x)$  to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set  $\{x \in E : \omega(f; x) \geq 1/n\}$  is closed, the discontinuity set  $D(f)$  is an  $F_\sigma$ , a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of  $D(f)$  controls integrability.

*Remark* (Dependencies).

[Definition 10.2.5](#); [Definition 10.2.11](#).

## 10.3 Global Theorems

### Global Theorems — Quick Reference

| Concept/Statement                          | Meaning/Role                                     |
|--------------------------------------------|--------------------------------------------------|
| <a href="#">Bounded on a set</a>           | The output set is bounded in $\mathbb{R}$ .      |
| <a href="#">Boundedness theorem</a>        | Continuous functions on $[a, b]$ are bounded.    |
| <a href="#">Absolute extrema</a>           | Global maximum and minimum values on a set.      |
| <a href="#">Extreme value theorem</a>      | Continuous functions on $[a, b]$ attain extrema. |
| <a href="#">Intermediate value theorem</a> | Continuous functions do not skip values.         |
| <a href="#">Image interval theorem</a>     | $f([a, b])$ is a closed bounded interval.        |
| <a href="#">Darboux property</a>           | Interval values are preserved without gaps.      |

## Intermediate Value Theorem

### Definition (Bounded on a Set)

**Definition 10.3.1** (Bounded on a Set). Let  $A \subseteq \mathbb{R}$  and  $f: A \rightarrow \mathbb{R}$ . The function  $f$  is bounded on  $A$  if and only if there exists a constant  $M > 0$  such that  $|f(x)| \leq M$  for every  $x \in A$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ and } f: A \rightarrow \mathbb{R}. \\ &f \text{ is bounded on } A \iff \exists M > 0 \, \forall x \in A \, (|f(x)| \leq M). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{BoundedOn}(f, A) := \exists M > 0 \, \forall x \in A \, (|f(x)| \leq M).$$

*Remark* (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \forall x \in [a, b] (f(x) \neq c) \implies \left[ (\forall x \in [a, b] f(x) > c) \vee (\forall x \in [a, b] f(x) < c) \right].$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R} \text{ and } f: A \rightarrow \mathbb{R}. \\ &\neg(f \text{ bounded on } A) \iff \forall M > 0 \exists x \in A (|f(x)| > M). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{BoundedOn}(f, A) \iff \forall M > 0 \exists x \in A (|f(x)| > M).$$

*Remark* (Failure modes). Failure occurs precisely when every proposed positive bound  $M$  is exceeded by some value of  $|f(x)|$  with  $x \in A$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}} \quad \underbrace{\exists x \in A (|f(x)| > M)}_{\text{value exceeding } M}.$$

*Remark* (Interpretation). The definition packages the informal idea that the values of  $f$  stay within a fixed vertical strip over  $A$ . The witness  $M$  serves as a uniform ceiling for the magnitudes  $|f(x)|$ , so once such an  $M$  is known the graph of  $f$  never escapes the band  $[-M, M]$ . Failure means that the outputs of  $f$  grow without limit along some sequence of points inside  $A$ , so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Absolute Value on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

#### Theorem (Boundedness Theorem)

**Theorem 10.3.2** (Boundedness Theorem). *Let  $a < b$  and let  $f: [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then there exists  $M > 0$  such that  $|f(x)| \leq M$  for every  $x \in [a, b]$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } a < b \text{ and } f: [a, b] \rightarrow \mathbb{R}. \\ &\left( \forall c \in [a, b] \forall \varepsilon > 0 \exists \delta > 0 \forall x \in [a, b] (|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon) \right) \\ &\implies \exists M > 0 \forall x \in [a, b] (|f(x)| \leq M). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \implies \text{BoundedOn}(f, [a, b]).$$

*Remark* (Negated quantified statement).

Let  $a < b$  and  $f : [a, b] \rightarrow \mathbb{R}$ .

$$\left( \forall c \in [a, b] \forall \varepsilon > 0 \exists \delta > 0 \forall x \in [a, b] (|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon) \right) \\ \wedge \left( \forall M > 0 \exists x \in [a, b] (|f(x)| > M) \right).$$

*Remark* (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \neg \text{BoundedOn}(f, [a, b]).$$

*Remark* (Failure modes). Each component hypothesis is indispensable: without closedness (for instance  $f(x) = 1/x$  on  $(0, 1]$ ) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example  $f(x) = x$  on  $[0, \infty)$ ) linear growth forces unbounded values; without continuity (take  $f(0) = 0$  and  $f(x) = 1/x$  for  $x \in (0, 1]$ ) the discontinuity at 0 again creates arbitrarily large magnitudes.

*Remark* (Failure mode decomposition).

$$\neg [\text{ClosedInterval}(a, b) \wedge \text{BoundedSet}([a, b]) \wedge \text{ContinuousOn}(f, [a, b])] = \underbrace{\neg \text{ClosedInterval}(a, b)}_{\text{missing endpoint}} \vee \underbrace{\neg \text{BoundedSet}([a, b])}_{\text{unbounded}}$$

*Remark* (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval  $[a, b]$  can be covered by finitely many neighborhoods in which the oscillation of  $f$  is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

*Remark* (Dependencies).

[Definition 10.2.2](#); [Definition 10.3.1](#).

### Definition (Absolute Maximum and Absolute Minimum)

**Definition 10.3.3** (Absolute Maximum and Absolute Minimum). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . A point  $x^* \in A$  is an absolute maximum of  $f$  on  $A$  if and only if  $f(x^*) \geq f(x)$  for every  $x \in A$ . A point  $x_* \in A$  is an absolute minimum of  $f$  on  $A$  if and only if  $f(x_*) \leq f(x)$  for every  $x \in A$ .

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ ,  $x^*, x_* \in A$ .

$x^*$  is an absolute maximum of  $f$  on  $A \iff \forall x \in A (f(x) \leq f(x^*))$ .

$x_*$  is an absolute minimum of  $f$  on  $A \iff \forall x \in A (f(x_*) \leq f(x))$ .

*Remark* (Definition predicate reading).

$$\begin{aligned}(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A)) &\iff (x^* \in A) \wedge \forall x \in A (f(x) \leq f(x^*)), \\ (x_* \in A) \wedge \text{Minimum}(f(x_*), f(A)) &\iff (x_* \in A) \wedge \forall x \in A (f(x_*) \leq f(x)).\end{aligned}$$

*Remark* (Negated quantified statement).

$$\begin{aligned}\text{Let } A \subseteq \mathbb{R}, f: A \rightarrow \mathbb{R}, x^*, x_* \in \mathbb{R}. \\ x^* \text{ fails to be an absolute maximum of } f \text{ on } A \\ \iff (x^* \notin A) \vee (\exists x \in A (f(x) > f(x^*))). \\ x_* \text{ fails to be an absolute minimum of } f \text{ on } A \\ \iff (x_* \notin A) \vee (\exists x \in A (f(x_*) > f(x))).\end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned}\neg[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))] &\iff (x^* \notin A) \vee (\exists x \in A (f(x) > f(x^*))), \\ \neg[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))] &\iff (x_* \notin A) \vee (\exists x \in A (f(x_*) > f(x))).\end{aligned}$$

*Remark* (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain  $A$ , or because some  $x \in A$  produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in  $A$  with a strictly smaller value.

*Remark* (Failure mode decomposition).

$$\begin{aligned}\neg[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))] &= \underbrace{(x^* \notin A)}_{\text{external point}} \vee \underbrace{(\exists x \in A (f(x) > f(x^*)))}_{\text{better value}}, \\ \neg[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))] &= \underbrace{(x_* \notin A)}_{\text{external point}} \vee \underbrace{(\exists x \in A (f(x_*) > f(x)))}_{\text{better value}}.\end{aligned}$$

*Remark* (Interpretation). An absolute maximum identifies the highest value that  $f$  attains on its domain  $A$ , together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to  $A$ , so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above  $A$ , while the absolute minimum sits at the lowest point.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Order on  \$\mathbb{R}\$](#)



- [Maximum](#)
- [Minimum](#)

### Theorem (Extreme Value Theorem)

**Theorem 10.3.4** (Extreme Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. There exist points  $x_m, x_M \in [a, b]$  such that  $f(x_m) \leq f(x) \leq f(x_M)$  for every  $x \in [a, b]$ . Equivalently,  $f$  attains both an absolute maximum and an absolute minimum on  $[a, b]$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous.  
 $\exists x_m, x_M \in [a, b] \forall x \in [a, b] (f(x_m) \leq f(x) \leq f(x_M)).$

*Remark* (Predicate reading).

$\text{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b] (\text{Maximum}(f(x_M), f([a, b])) \wedge \text{Minimum}(f(x_m), f([a, b])))$ . ■

*Remark* (Failure modes). The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

*Remark* (Failure mode decomposition).

$\underbrace{\neg(a < b)}_{\text{invalid interval}} \vee \underbrace{\neg(f : [a, b] \rightarrow \mathbb{R})}_{\text{wrong function type}} \vee \underbrace{\exists c \in [a, b] \exists \varepsilon > 0 \forall \delta > 0 \exists x \in [a, b] (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon)}_{\text{loss of continuity}}.$

*Remark* (Interpretation). Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

*Remark* (Dependencies).

Uses [Definition 10.3.3](#) and [Theorem 10.3.2](#).

### Theorem (Location of Roots)

**Theorem 10.3.5** (Location of Roots). *Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. If either  $f(a) < 0 < f(b)$  or  $f(a) > 0 > f(b)$ , then there exists  $c \in (a, b)$  such that  $f(c) = 0$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$  and  $f : [a, b] \rightarrow \mathbb{R}$  continuous.  
 $(f(a) < 0 < f(b) \vee f(a) > 0 > f(b)) \implies \exists c \in (a, b) (f(c) = 0).$

*Remark* (Failure modes). The root conclusion can fail if  $f$  is not continuous on  $[a, b]$ , or if the endpoint values do not have opposite signs.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists c \in [a, b] \exists \varepsilon > 0 \forall \delta > 0 \exists x \in [a, b] (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon)}_{\text{loss of continuity}} \vee \underbrace{(f(a)f(b) \geq 0)}_{\text{no sign change}}.$$

*Remark* (Interpretation). Continuity forces the image of the interval  $[a, b]$  to contain every intermediate value between  $f(a)$  and  $f(b)$ . A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when  $f$  is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the  $x$ -axis must intersect the axis somewhere in between.

*Remark* (Dependencies).

[Bolzano's Intermediate Value Theorem.](#)

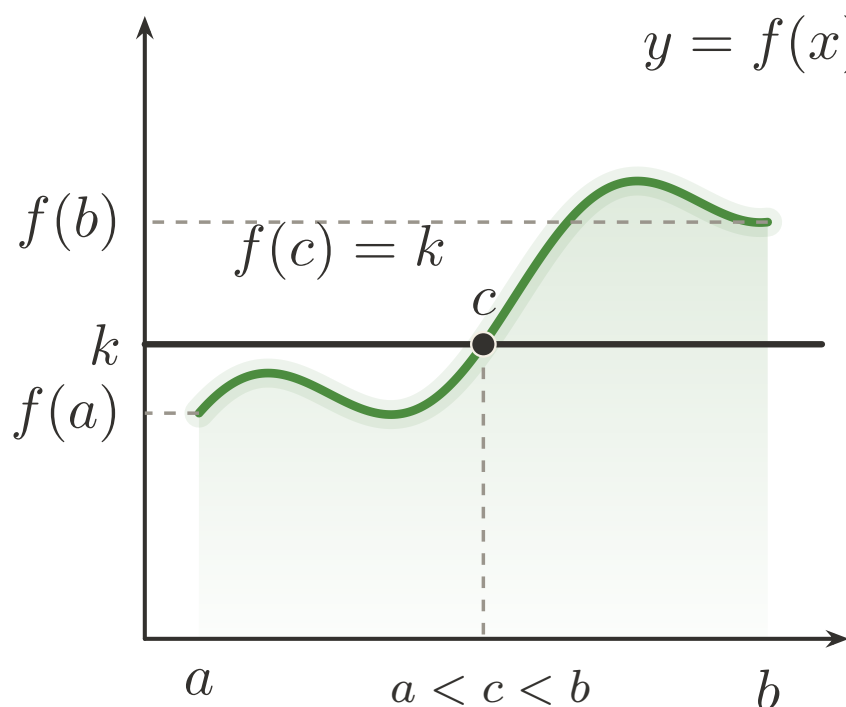


Figure 10.1: Intermediate Value Theorem: a continuous path hits every intermediate height.

#### Theorem (Bolzano's Intermediate Value Theorem)

**Theorem 10.3.6** (Bolzano's Intermediate Value Theorem). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f: I \rightarrow \mathbb{R}$  be continuous on  $I$ , and choose  $a, b \in I$  with  $f(a) < k < f(b)$ . Then there exists  $c \in I$  such that  $\min\{a, b\} < c < \max\{a, b\}$  and  $f(c) = k$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall I \subseteq \mathbb{R} \forall f : I \rightarrow \mathbb{R} \forall a, b \in I \forall k \in \mathbb{R} \left[ (\forall x, y \in I \forall z \in \mathbb{R} (x < z < y \Rightarrow z \in I)) \wedge (\forall c \in I \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)) \right]$$

*Remark* (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level  $k$  is not strictly between the two attained endpoint values.

*Remark* (Failure mode decomposition).

$$\begin{aligned} & \underbrace{\exists x, y \in I \exists z \in \mathbb{R} (x < z < y \wedge z \notin I)}_{\text{not an interval}} \\ & \vee \underbrace{\exists c \in I \exists \varepsilon > 0 \forall \delta > 0 \exists x \in I (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon)}_{\text{loss of continuity}} \\ & \vee \underbrace{\neg(f(a) < k < f(b))}_{\text{level not between endpoint values}}. \end{aligned}$$

*Remark* (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between  $f(a)$  and  $f(b)$ . The conclusion provides an interior point  $c$  whose image equals the prescribed level  $k$ , ensuring root-finding for  $f - k$ . If the hypothesis fails, either  $f$  is not continuous or the level  $k$  does not lie between the endpoint values, and no such interior point need exist.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Continuity at a Point](#)
- [Order on  \$\mathbb{R}\$](#)
- [Maximum](#)
- [Minimum](#)

#### Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)

**Theorem 10.3.7** (Image of Closed Bounded Interval Is Closed Bounded Interval). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. Then  $f([a, b]) = [\min_{[a, b]} f, \max_{[a, b]} f]$ . Go to proof.*

*Remark* (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \forall f : [a, b] \rightarrow \mathbb{R} :$$

$$\left( a < b \wedge \forall c \in [a, b] \forall \varepsilon > 0 \exists \delta > 0 \forall x \in [a, b] (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon) \right) \implies f([a, b]) = [\min_{[a, b]} f, \max_{[a, b]} f]$$

*Remark* (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \implies f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

*Remark* (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on  $[a, b]$ , with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image  $f([a, b])$  is a closed interval with endpoints given by the absolute extremal values of  $f$  on  $[a, b]$ , showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

*Remark* (Dependencies).

Extreme Value Theorem, Bolzano's Intermediate Value Theorem

#### Theorem (Preservation of Intervals)

**Theorem 10.3.8** (Preservation of Intervals). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be continuous. Then  $f(I)$  is an interval. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall I \subseteq \mathbb{R} \forall f : I \rightarrow \mathbb{R} :$$

$$\left[ \left( \forall x_1, x_2 \in I \forall z \in \mathbb{R} (x_1 < z < x_2 \implies z \in I) \right) \wedge \left( \forall c \in I \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I (|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon) \right) \right] \implies \forall y_1, y_2 \in f(I) \forall y \in \mathbb{R} (y_1 < y < y_2 \implies y \in f(I))$$

*Remark* (Predicate reading).

$$\text{ContinuousOn}(f, I) \implies \text{Interval}(f(I)).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} \exists I \subseteq \mathbb{R} \exists f : I \rightarrow \mathbb{R} \exists y_1, y_2 \in f(I) \exists y \in \mathbb{R} : \\ \left( \forall x_1, x_2 \in I \forall z \in \mathbb{R} (x_1 < z < x_2 \implies z \in I) \right) \\ \wedge \left( \forall c \in I \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I (|x - c| < \delta \implies |f(x) - f(c)| < \varepsilon) \right) \\ \wedge y_1 < y < y_2 \wedge y \notin f(I). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{Interval}(I) \wedge \text{ContinuousOn}(f, I) \wedge \neg \text{Interval}(f(I)).$$

*Remark* (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when  $I$  is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

Remark (Dependencies).

thm:bolzano-intermediate-value

### Theorem (Darboux Property)

**Theorem 10.3.9** (Darboux Property). *Let  $a, b \in \mathbb{R}$  satisfy  $a < b$ , let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and fix  $c \in \mathbb{R}$ . If  $f(x) \neq c$  for every  $x \in [a, b]$ , then either  $f(x) > c$  for all  $x \in [a, b]$  or  $f(x) < c$  for all  $x \in [a, b]$ . Go to proof.*

Remark (Standard quantified statement).

Let  $a < b$ ,  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and  $c \in \mathbb{R}$ .

$$\left( \forall x \in [a, b] f(x) \neq c \right) \Rightarrow \left[ \left( \forall x \in [a, b] f(x) > c \right) \vee \left( \forall x \in [a, b] f(x) < c \right) \right].$$

Remark (Negated quantified statement).

Let  $a < b$ ,  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and  $c \in \mathbb{R}$ .

$$\left( \forall x \in [a, b] f(x) \neq c \right) \wedge \left( \exists u \in [a, b] f(u) \leq c \right) \wedge \left( \exists v \in [a, b] f(v) \geq c \right).$$

Remark (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \left( \forall x \in [a, b] f(x) \neq c \right) \wedge \neg \left[ \left( \forall x \in [a, b] f(x) > c \right) \vee \left( \forall x \in [a, b] f(x) < c \right) \right]. \blacksquare$$

Remark (Contrapositive quantified statement).

Let  $a < b$ ,  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and  $c \in \mathbb{R}$ .

$$\left( \exists x \in [a, b] f(x) \leq c \right) \wedge \left( \exists y \in [a, b] f(y) \geq c \right) \Rightarrow \exists z \in [a, b] f(z) = c.$$

Remark (Contrapositive predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \text{MeetsBothSides}(f, [a, b], c) \implies \exists z \in [a, b] (f(z) = c).$$

Remark (Interpretation). A continuous function on a compact interval cannot oscillate across a level  $c$  without actually attaining that level. Stated contrapositive fashion, if  $c$  is never taken then the entire graph lies strictly above or strictly below the horizontal line  $y=c$ . The only way this guarantee can fail is when the function produces values on both sides of  $c$ , in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting  $c$  forces the image to stay in one of the open half-lines determined by  $c$ .

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

## 10.4 Uniform Continuity

### Uniform Continuity — Quick Reference

| Concept/Statement               | Meaning/Role                                                 |
|---------------------------------|--------------------------------------------------------------|
| Uniform continuity              | One $\delta(\varepsilon)$ works for all pairs in the domain. |
| Sequential test                 | Close inputs with separated outputs witness failure.         |
| Heine–Cantor                    | Continuity on $[a, b]$ upgrades to uniform continuity.       |
| Cauchy preservation             | Uniform continuity preserves Cauchy sequences.               |
| Lipschitz                       | A global slope bound gives a uniform modulus.                |
| Lipschitz $\Rightarrow$ uniform | Lipschitz control implies uniform continuity.                |
| Bi-Lipschitz maps               | Distances are controlled above and below.                    |

## Uniform Continuity

### Definition (Uniform Continuity)

**Definition 10.4.1** (Uniform Continuity). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The function  $f$  is uniformly continuous on  $A$  if and only if for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that for all  $x, u \in A$  the implication  $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$  holds.

*Remark* (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \rightarrow \mathbb{R}) \\ \implies \forall \varepsilon > 0 \exists \delta > 0 \forall x, u \in A (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

*Remark* (Definition predicate reading).

$$\text{UniformlyContinuous}(f, A) := \forall \varepsilon > 0 \exists \delta > 0 \forall x, u \in A (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

*Remark* (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \rightarrow \mathbb{R}) \\ \vee \exists \varepsilon > 0 \forall \delta > 0 \exists x, u \in A (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

*Remark* (Negation predicate reading).

$$\neg \text{UniformlyContinuous}(f, A) \iff \exists \varepsilon > 0 \forall \delta > 0 \exists x, u \in A (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

*Remark* (Failure modes). The definition fails exactly when a single positive tolerance  $\varepsilon_0$  exposes pairs of inputs arbitrarily close together whose function values remain at least  $\varepsilon_0$  apart, so no universal  $\delta$  works for that  $\varepsilon_0$ .

*Remark* (Failure mode decomposition).

$$\neg \text{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon_0)}_{\text{bad witness pair}}.$$

*Remark* (Interpretation). Uniform continuity demands that a single proximity scale  $\delta$  answers any prescribed accuracy  $\varepsilon$  simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of  $\delta$  because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length  $\delta$  horizontally and  $\varepsilon$  vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Absolute Value on  \$\mathbb{R}\$](#)
- [Order on  \$\mathbb{R}\$](#)

#### Theorem (Algebra of Uniform Continuity on a General Domain)

**Theorem 10.4.2** (Algebra of Uniform Continuity on a General Domain). (a) For every set  $A \subseteq \mathbb{R}$ , every pair of functions  $f, g : A \rightarrow \mathbb{R}$  that are uniformly continuous on  $A$ , and every scalar  $c \in \mathbb{R}$ , the functions  $f + g$ ,  $f - g$ , and  $cf$  are uniformly continuous on  $A$ .

(b) There exist a set  $A \subseteq \mathbb{R}$  and uniformly continuous functions  $f, g : A \rightarrow \mathbb{R}$  such that the product  $fg$  is not uniformly continuous on  $A$ .

*Go to proof.*

*Remark* (Standard quantified statement).

$\forall A \subseteq \mathbb{R} \forall f, g : A \rightarrow \mathbb{R} \forall c \in \mathbb{R} :$

$$\begin{aligned} & \left[ \left( \forall \varepsilon > 0 \exists \delta_f > 0 \forall x, y \in A (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon) \right) \right. \\ & \quad \wedge \left( \forall \varepsilon > 0 \exists \delta_g > 0 \forall x, y \in A (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon) \right) \Big] \Rightarrow \left[ \forall \varepsilon > 0 \exists \delta_1 > 0 \forall x, y \in A (|x - y| < \delta_1 \Rightarrow |f(x) + g(x) - f(y) - g(y)| < \varepsilon) \right. \\ & \quad \wedge \forall \varepsilon > 0 \exists \delta_2 > 0 \forall x, y \in A (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon) \\ & \quad \left. \wedge \forall \varepsilon > 0 \exists \delta_3 > 0 \forall x, y \in A (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon) \right], \end{aligned}$$

$$\begin{aligned} \exists A \subseteq \mathbb{R} \exists f, g : A \rightarrow \mathbb{R} : & \left[ \forall \varepsilon > 0 \exists \delta_f > 0 \forall x, y \in A (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon) \right] \\ & \wedge \left[ \forall \varepsilon > 0 \exists \delta_g > 0 \forall x, y \in A (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon) \right] \\ & \wedge \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x, y \in A (|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0). \end{aligned}$$

*Remark* (Predicate reading).

$\forall A \subseteq \mathbb{R} \forall f, g : A \rightarrow \mathbb{R} \forall c \in \mathbb{R} :$

$$\begin{aligned} & (\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A)) \implies (\text{UniformlyContinuous}(f + g, A) \wedge \text{UniformlyContinuous}(f - g, A) \wedge \text{UniformlyContinuous}(cf, A)) \\ \exists A \subseteq \mathbb{R} \exists f, g : A \rightarrow \mathbb{R} : & \text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A) \wedge \neg \text{UniformlyContinuous}(fg, A) \end{aligned}$$

*Remark* (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

*Remark* (Dependencies).

- Uniform Continuity
- Linear Combination of Real-Valued Functions
- Pointwise Difference of Functions

#### Theorem (Algebra of Uniform Continuity on a Bounded Domain)

**Theorem 10.4.3** (Algebra of Uniform Continuity on a Bounded Domain). *Let  $A \subseteq \mathbb{R}$  be bounded, and let  $f, g : A \rightarrow \mathbb{R}$  be uniformly continuous on  $A$ .*

- (a) *The product  $x \mapsto f(x)g(x)$  is uniformly continuous on  $A$ .*
- (b) *If there exists  $k > 0$  such that  $|g(x)| \geq k$  for all  $x \in A$ , then the quotient  $x \mapsto f(x)/g(x)$  is uniformly continuous on  $A$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$  be bounded and  $f, g : A \rightarrow \mathbb{R}$ .

$$\begin{aligned}
 & [\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)] \\
 & \wedge [\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)] \\
 & \Rightarrow \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon), \\
 & [\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)] \\
 & \wedge [\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)] \\
 & \wedge [\exists k > 0 \forall x \in A (|g(x)| \geq k)] \\
 & \Rightarrow \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
 \end{aligned}$$

*Remark* (Predicate reading).

Assume  $A \subseteq \mathbb{R}$  is bounded and  $f, g : A \rightarrow \mathbb{R}$ .

$\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A)$

$\implies \text{UniformlyContinuous}(fg, A),$

$\text{UniformlyContinuous}(f, A) \wedge \text{UniformlyContinuous}(g, A) \wedge \exists k > 0 \forall x \in A (|g(x)| \geq k)$

$\implies \text{UniformlyContinuous}(f/g, A).$



*Remark* (Failure modes). The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded,  $f$  is not uniformly continuous on  $A$ , or  $g$  is not uniformly continuous on  $A$ . The quotient conclusion can also fail if the denominator is not bounded away from zero on  $A$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\neg \text{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg \text{UniformlyContinuous}(f, A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg \text{UniformlyContinuous}(g, A)}_{\text{second factor not uniformly continuous}} \\ \vee \underbrace{\neg \exists k > 0 \forall x \in A : |g(x)| \geq k}_{\text{denominator not bounded away from zero}} .$$

*Remark* (Interpretation). Bounded subsets of  $\mathbb{R}$  prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for  $f$  and  $g$  can be combined to control  $fg$ . The quotient is uniformly continuous whenever  $g$  never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for  $|g|$  are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

*Remark* (Dependencies).

[Definition 10.4.1.](#)

#### Proposition (Sequential Characterization of Uniform Continuity)

**Proposition 10.4.4** (Sequential Characterization of Uniform Continuity). *Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . Then  $f$  is uniformly continuous on  $A$  if and only if for every pair of sequences  $(x_n)$  and  $(y_n)$  in  $A$  with  $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$  we have  $\lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon) \right] \iff \forall x_n, y_n : \mathbb{N} \rightarrow A (|x_n - y_n| \rightarrow 0 \Rightarrow |f(x_n) - f(y_n)| \rightarrow 0)$$

*Remark* (Predicate reading).

$$\text{UniformlyContinuous}(f, A) \iff \forall x_n, y_n : \mathbb{N} \rightarrow A (|x_n - y_n| \rightarrow 0 \Rightarrow |f(x_n) - f(y_n)| \rightarrow 0).$$

*Remark* (Negated quantified statement).

$$\exists A \subseteq \mathbb{R} \exists f : A \rightarrow \mathbb{R} : \\ \left( \left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon) \right] \right. \\ \left. \wedge \exists x_n, y_n : \mathbb{N} \rightarrow A (|x_n - y_n| \rightarrow 0 \wedge \neg(|f(x_n) - f(y_n)| \rightarrow 0)) \right) \\ \vee \left( \left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x, y \in A (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0) \right] \wedge \forall x_n, y_n : \mathbb{N} \rightarrow A (|x_n - y_n| \rightarrow 0 \Rightarrow |f(x_n) - f(y_n)| \rightarrow 0) \right)$$

*Remark* (Negation predicate reading).

$\exists A \subseteq \mathbb{R} \exists f : A \rightarrow \mathbb{R} :$

$$\left( \text{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \rightarrow A \left( |x_n - y_n| \rightarrow 0 \wedge \neg(|f(x_n) - f(y_n)| \rightarrow 0) \right) \right) \\ \vee \left( \neg \text{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \rightarrow A \left( |x_n - y_n| \rightarrow 0 \Rightarrow |f(x_n) - f(y_n)| \rightarrow 0 \right) \right).$$

*Remark* (Failure modes). The equivalence can fail in exactly two ways: either  $f$  is uniformly continuous while some pair of asymptotically coincident sequences in  $A$  yields outputs that do not converge together, or every such pair of sequences does converge together while  $f$  nonetheless fails to be uniformly continuous on  $A$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\text{UniformlyContinuous}(f, A)}_{\text{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \rightarrow A \left( |x_n - y_n| \rightarrow 0 \wedge \neg(|f(x_n) - f(y_n)| \rightarrow 0) \right)}_{\text{sequential test fails}} \\ \vee \underbrace{\neg \text{UniformlyContinuous}(f, A)}_{\text{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \rightarrow A \left( |x_n - y_n| \rightarrow 0 \Rightarrow |f(x_n) - f(y_n)| \rightarrow 0 \right)}_{\text{sequential test holds}}.$$

*Remark* (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of  $A$  are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in  $A$  but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

*Remark* (Dependencies).

[Definition 10.4.1](#)

#### Theorem (Heine–Cantor Theorem)

**Theorem 10.4.5** (Heine–Cantor Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then  $f$  is uniformly continuous on  $[a, b]$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $f : [a, b] \rightarrow \mathbb{R}$ .

$$\left[ \forall c \in [a, b] \forall \varepsilon > 0 \exists \delta > 0 \forall x \in [a, b] (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon) \right] \\ \implies \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in [a, b] (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

*Remark* (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Rightarrow \text{UniformlyContinuous}(f, [a, b]).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \rightarrow \mathbb{R}. \\ &\left[ \forall c \in [a, b] \forall \varepsilon > 0 \exists \delta > 0 \forall x \in [a, b] (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon) \right] \\ &\wedge \left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x, y \in [a, b] (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0) \right]. \end{aligned}$$

*Remark* (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \neg \text{UniformlyContinuous}(f, [a, b]).$$

*Remark* (Interpretation). Continuity on the compact interval  $[a, b]$  forces a single modulus of continuity that works simultaneously at every point, so oscillations of  $f$  cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

*Remark* (Dependencies).

[Definition 10.2.2](#); [Definition 10.4.1](#).

#### Proposition (Uniform Continuity Preserves Cauchy Sequences)

**Proposition 10.4.6** (Uniform Continuity Preserves Cauchy Sequences). *Let  $S \subseteq \mathbb{R}$ , let  $f : S \rightarrow \mathbb{R}$ , and let  $(x_n)$  be a sequence in  $S$ . If  $f$  is uniformly continuous on  $S$  and  $(x_n)$  is a Cauchy sequence in  $S$ , then  $(f(x_n))$  is a Cauchy sequence in  $\mathbb{R}$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } S \subseteq \mathbb{R}, f : S \rightarrow \mathbb{R}, (x_n)_{n \in \mathbb{N}} \subseteq S. \\ &\left[ \forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in S (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon) \right] \\ &\wedge \left[ \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|x_n - x_m| < \varepsilon) \right] \\ &\Rightarrow \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N (|f(x_n) - f(x_m)| < \varepsilon). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{UniformlyContinuous}(f, S) \wedge \text{CauchySequence}(x_n) \Rightarrow \text{CauchySequence}(f(x_n)).$$

*Remark* (Interpretation). Uniform continuity controls the change in  $f(x)$  solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when  $f$  is merely continuous and the sequence approaches a boundary point where

continuity loses uniform control, producing large oscillations in the image sequence. Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by  $(x_n)$ , ensuring that eventually all image terms lie in an arbitrarily small common interval.

*Remark* (Dependencies).

Definition 10.4.1; Definition 6.11.1.

### Definition (Lipschitz Condition)

**Definition 10.4.7** (Lipschitz Condition). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $K > 0$ . The function  $f$  satisfies the Lipschitz condition on  $A$  with constant  $K$  if and only if  $|f(x) - f(u)| \leq K|x - u|$  for every  $x, u \in A$ .

*Remark* (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \rightarrow \mathbb{R}) \wedge (K > 0) \\ \implies \forall x, u \in A (|f(x) - f(u)| \leq K|x - u|).$$

*Remark* (Definition predicate reading).

$$\text{Lipschitz}(f, A, K) := (K > 0) \wedge \forall x, u \in A (|f(x) - f(u)| \leq K|x - u|).$$

*Remark* (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \rightarrow \mathbb{R}) \vee (K \leq 0) \\ \vee \exists x, u \in A (|f(x) - f(u)| > K|x - u|).$$

*Remark* (Negation predicate reading).

$$\neg \text{Lipschitz}(f, A, K) \iff \neg(A \subseteq \mathbb{R}) \vee \neg(f : A \rightarrow \mathbb{R}) \vee (K \leq 0) \vee \exists x, u \in A (|f(x) - f(u)| > K|x - u|). \blacksquare$$

*Remark* (Failure modes). The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in  $A$  whose images separate faster than  $K$  times the separation of the points themselves.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg(A \subseteq \mathbb{R})}_{\text{bad domain}} \vee \underbrace{\neg(f : A \rightarrow \mathbb{R})}_{\text{bad function type}} \vee \underbrace{K \leq 0}_{\text{bad constant}} \vee \underbrace{\exists x, u \in A : |f(x) - f(u)| > K|x - u|}_{\text{excessive separation of outputs}}.$$

*Remark* (Interpretation). The Lipschitz condition requires a single constant  $K$  that simultaneously bounds the rate at which  $f$  can stretch distances across the entire set  $A$ . It encapsulates a global slope cap: every secant line through the graph has absolute slope at most  $K$ . Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by  $K$ , so nearby points cannot be blown apart faster than  $K$  times their original spacing.

Remark (Dependencies).

- [Function](#)

**Proposition (Lipschitz Implies Uniform Continuity)**

**Proposition 10.4.8** (Lipschitz Implies Uniform Continuity). *Let  $A \subseteq \mathbb{R}$  and  $f : A \rightarrow \mathbb{R}$ . If there exists  $K \geq 0$  such that  $|f(x) - f(y)| \leq K|x - y|$  for all  $x, y \in A$ , then  $f$  is uniformly continuous on  $A$ . [Go to proof](#).*

Remark (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ , and suppose  $\exists K \geq 0 \forall x, y \in A (|f(x) - f(y)| \leq K|x - y|)$ .  
 $\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in A (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)$ .

Remark (Predicate reading).

$$[\exists K \geq 0 \text{ Lipschitz}(f, A, K)] \Rightarrow \text{UniformlyContinuous}(f, A).$$

Remark (Contrapositive quantified statement). If  $f$  fails to be uniformly continuous on  $A$ , then  $f$  is not Lipschitz on  $A$  with any constant.

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f : A \rightarrow \mathbb{R}. \\ &\left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x, y \in A (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0) \right] \\ &\implies \forall K \geq 0 \exists x, y \in A (|f(x) - f(y)| > K|x - y|). \end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg \text{UniformlyContinuous}(f, A) \implies \neg \exists K \geq 0 : \text{Lipschitz}(f, A, K).$$

Remark (Interpretation). A Lipschitz bound controls the variation of  $f$  by the distance between inputs with a single constant  $K$ , so the same  $K$  yields a usable  $\delta = \varepsilon/K$  for every pair of points at once, proving uniform continuity. When  $K = 0$  the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

Remark (Strict implication note). The implication  $\text{Lipschitz} \Rightarrow \text{uniform continuity}$  is strict. The canonical counterexample is  $f(x) = \sqrt{x}$  on  $[0, 1]$ , which is uniformly continuous by the Heine–Cantor theorem but satisfies  $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \rightarrow \infty$  as  $x \rightarrow 0^+$ , so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable  $K$  for the Lipschitz condition; only an explicit Lipschitz bound does.

Remark (Dependencies).

[Definition 10.4.7](#), [Definition 10.4.1](#).

### Theorem (Algebra of Lipschitz Functions)

**Theorem 10.4.9** (Algebra of Lipschitz Functions). *Let  $A \subseteq \mathbb{R}$  and let  $f, g : A \rightarrow \mathbb{R}$  be Lipschitz with constants  $K_f, K_g \geq 0$  respectively.*

- (a) *For every  $c \in \mathbb{R}$ , the function  $cf$  is Lipschitz on  $A$  with constant  $|c|K_f$ .*
- (b) *The functions  $f + g$  and  $f - g$  are Lipschitz on  $A$  with constant  $K_f + K_g$ .*
- (c) *If  $B \subseteq \mathbb{R}$ ,  $h : B \rightarrow \mathbb{R}$  is Lipschitz with constant  $K_h \geq 0$ , and  $f(A) \subseteq B$ , then  $h \circ f$  is Lipschitz on  $A$  with constant  $K_h K_f$ .*
- (d) *If  $|f(x)| \leq M_f$  and  $|g(x)| \leq M_g$  for all  $x \in A$ , then  $fg$  is Lipschitz on  $A$  with constant  $M_g K_f + M_f K_g$ .*
- (e) *If  $|f(x)| \leq M_f$  for all  $x \in A$ , if  $|g(x)| \geq k > 0$  for all  $x \in A$ , and if  $g$  is Lipschitz on  $A$  with constant  $K_g$ , then  $f/g$  is Lipschitz on  $A$  with constant  $K_f/k + M_f K_g/k^2$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

- Let  $A \subseteq \mathbb{R}$ ,  $f, g : A \rightarrow \mathbb{R}$ ,  $K_f, K_g \geq 0$ ,
- $\forall x, y \in A : |f(x) - f(y)| \leq K_f |x - y|$  and  $|g(x) - g(y)| \leq K_g |x - y|$ .
- (a)  $\forall c \in \mathbb{R}, \forall x, y \in A : |cf(x) - cf(y)| \leq |c|K_f |x - y|$ .
  - (b)  $\forall x, y \in A : |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|$ .
  - (c) If  $B \subseteq \mathbb{R}$ ,  $f(A) \subseteq B$ ,  $h : B \rightarrow \mathbb{R}$ ,  $\forall u, v \in B : |h(u) - h(v)| \leq K_h |u - v|$ ,  
then  $\forall x, y \in A : |h(f(x)) - h(f(y))| \leq K_h K_f |x - y|$ .
  - (d) If  $|f(x)| \leq M_f$ ,  $|g(x)| \leq M_g \forall x \in A$ , then  $\forall x, y \in A :$   
 $|f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|$ .
  - (e) If  $|f(x)| \leq M_f$ ,  $|g(x)| \geq k > 0 \forall x \in A$ , then  $\forall x, y \in A :$   
 $\left| \frac{f(x)}{g(x)} - \frac{f(y)}{g(y)} \right| \leq \left( \frac{K_f}{k} + \frac{M_f K_g}{k^2} \right) |x - y|$ .

*Remark* (Predicate reading).

$$\begin{aligned}
 & \text{Lipschitz}(f, A, K_f) \wedge \text{Lipschitz}(g, A, K_g) \\
 \implies & \left[ \forall c \in \mathbb{R} : \text{Lipschitz}(cf, A, |c|K_f) \right] \\
 & \wedge \text{Lipschitz}(f + g, A, K_f + K_g) \wedge \text{Lipschitz}(f - g, A, K_f + K_g) \\
 & \wedge \left( \text{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \implies \text{Lipschitz}(h \circ f, A, K_h K_f) \right) \\
 & \wedge \left[ (|f| \leq M_f) \wedge (|g| \leq M_g) \implies \text{Lipschitz}(fg, A, M_g K_f + M_f K_g) \right] \\
 & \wedge \left[ (|f| \leq M_f) \wedge (|g| \geq k > 0) \implies \text{Lipschitz}\left(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\right) \right].
 \end{aligned}$$

*Remark* (Interpretation). Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

*Remark* (Dependencies).

Definition 10.4.7, Definition 10.3.1

#### Definition (Bi-Lipschitz Map)

**Definition 10.4.10** (Bi-Lipschitz Map). Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . The map  $f$  is bi-Lipschitz on  $A$  if and only if there exist constants  $0 < \alpha \leq K$  such that for every  $x, u \in A$ ,

$$\alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u|.$$

Any such pair  $(\alpha, K)$  is called a bi-Lipschitz constant pair for  $f$  on  $A$ .

*Remark* (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \rightarrow \mathbb{R}) \\ \implies \exists \alpha, K \in \mathbb{R} \left( 0 < \alpha \leq K \wedge \forall x, u \in A : \alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u| \right).$$

*Remark* (Definition predicate reading).

$$\text{BiLipschitz}(f, A, \alpha, K) := (0 < \alpha \leq K) \wedge \forall x, u \in A \left( \alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u| \right).$$

*Remark* (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \rightarrow \mathbb{R}) \\ \vee \forall \alpha, K \in \mathbb{R} \left( 0 < \alpha \leq K \implies \exists x, u \in A : |f(x) - f(u)| < \alpha|x - u| \vee |f(x) - f(u)| > K|x - u| \right).$$

*Remark* (Negation predicate reading).

$$\neg \text{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \leq K) \vee \exists x, u \in A \left( |f(x) - f(u)| < \alpha|x - u| \vee |f(x) - f(u)| > K|x - u| \right).$$

*Remark* (Failure modes). The property fails either because  $f$  collapses some pair of points faster than the allowed lower constant  $\alpha$ , or because it stretches some pair more than the upper constant  $K$  permits. Both distortions prevent a uniform two-sided control of distances.

*Remark* (Failure mode decomposition).

$$\exists x, u \in A : \underbrace{|f(x) - f(u)| < \alpha|x - u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x) - f(u)| > K|x - u|}_{\text{upper-distortion failure}}.$$

*Remark* (Interpretation). A bi-Lipschitz map distorts every distance in  $A$  by factors lying between  $\alpha$  and  $K$ , so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

*Remark* (Dependencies).

- [Function](#)

#### Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)

**Theorem 10.4.11** (Inverse of a Bi-Lipschitz Map Is Lipschitz). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose there exist constants  $\alpha, K > 0$  such that  $\alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y|$  for all  $x, y \in A$ . Then  $f$  is injective, the inverse mapping  $f^{-1} : f(A) \rightarrow A$  is well defined, and*

$$\frac{1}{K}|u - v| \leq |f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{\alpha}|u - v| \quad \text{for all } u, v \in f(A).$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $A \subseteq \mathbb{R}$ ,  $f : A \rightarrow \mathbb{R}$ ,  $\alpha, K > 0$ .

$$\begin{aligned} & \left[ \forall x, y \in A \left( \alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y| \right) \right] \\ \implies & \left[ \forall x, y \in A \left( f(x) = f(y) \Rightarrow x = y \right) \right. \\ & \wedge \exists g : f(A) \rightarrow A \left( \forall u \in f(A) \left( f(g(u)) = u \right) \right) \\ & \left. \wedge \forall u, v \in f(A) \left( \frac{1}{K}|u - v| \leq |g(u) - g(v)| \leq \frac{1}{\alpha}|u - v| \right) \right]. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{BiLipschitz}(f, A, \alpha, K) \Rightarrow (\text{Injective}(f) \wedge \text{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)).$$

*Remark* (Interpretation). A bi-Lipschitz bound simultaneously controls the expansion and contraction of  $f$ , so no two points of  $A$  can share the same image and the inverse is automatically a well-defined function on  $f(A)$ . The inequalities for  $f^{-1}$  follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

*Remark* (Dependencies).

[Definition 10.4.7](#); [Definition 10.4.10](#).



## 10.5 Monotone Functions

### Monotone Functions — Quick Reference

| Concept/Statement                    | Meaning/Role                                            |
|--------------------------------------|---------------------------------------------------------|
| <a href="#">One-sided limits</a>     | Monotone functions have finite or infinite side limits. |
| <a href="#">Continuity criterion</a> | Continuity means both side limits equal the value.      |
| <a href="#">Jump</a>                 | The size of the monotone discontinuity.                 |
| <a href="#">First-kind gaps</a>      | Monotone discontinuities are jumps.                     |
| <a href="#">Countability</a>         | Only countably many jumps can occur.                    |
| <a href="#">Continuous inverse</a>   | Strict monotonicity gives a continuous inverse.         |

### Limits Superior and Inferior — Quick Reference

| Concept/Statement                                                      | Meaning/Role                                         |
|------------------------------------------------------------------------|------------------------------------------------------|
| <a href="#">Function <math>\limsup</math> and <math>\liminf</math></a> | Tail suprema become punctured-neighbourhood suprema. |
| <a href="#">Order relation</a>                                         | Always $\limsup \geq \liminf$ .                      |
| <a href="#">Limit criterion</a>                                        | A finite limit exists exactly when they agree.       |

## Monotone Functions and Continuity

### Theorem (One-Sided Limits of Monotone Functions)

**Theorem 10.5.1** (One-Sided Limits of Monotone Functions). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $c$  be interior to  $I$ , and let  $f : I \rightarrow \mathbb{R}$  be increasing. Define*

$$L_- := \sup\{f(x) : x \in I, x < c\}, \quad L_+ := \inf\{f(x) : x \in I, c < x\}.$$

*Then both one-sided limits exist and satisfy*

$$\lim_{x \rightarrow c^-} f(x) = L_-, \quad \lim_{x \rightarrow c^+} f(x) = L_+.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned}
 & I \subseteq \mathbb{R} \wedge (\forall x, y \in I \forall z \in \mathbb{R} (x < z < y \Rightarrow z \in I)) \wedge f : I \rightarrow \mathbb{R} \\
 & \wedge (\forall x, y \in I (x < y \Rightarrow f(x) \leq f(y))) \wedge c \in I \wedge \exists a, b \in I (a < c < b) \\
 & \wedge L_- = \sup\{f(x) : x \in I, x < c\} \wedge L_+ = \inf\{f(x) : x \in I, c < x\} \\
 & \Rightarrow \\
 & \forall \varepsilon > 0 \exists \delta_- > 0 \forall x \in I ((0 < c - x < \delta_- \wedge x < c) \Rightarrow |f(x) - L_-| < \varepsilon) \\
 & \wedge \forall \varepsilon > 0 \exists \delta_+ > 0 \forall x \in I ((0 < x - c < \delta_+ \wedge c < x) \Rightarrow |f(x) - L_+| < \varepsilon).
 \end{aligned}$$

*Remark* (Predicate reading).

Let  $I \subseteq \mathbb{R}$  be an interval,  $f : I \rightarrow \mathbb{R}$  increasing, and  $c$  interior to  $I$ .

$$\text{LeftLimit}(f, c, \sup\{f(x) : x \in I, x < c\}) \wedge \text{RightLimit}(f, c, \inf\{f(x) : x \in I, x > c\}).$$

*Remark* (Interpretation). For an increasing function, the values just to the left of  $c$  form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when  $c$  is not an interior point or  $f$  is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

*Remark* (Dependencies).

- [Completeness of  \$\mathbb{R}\$](#)
- [Monotone Function](#)
- [Supremum](#)
- [Infimum](#)

#### Corollary (Continuity Criterion for Monotone Functions)

**Corollary 10.5.2** (Continuity Criterion for Monotone Functions). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be monotone increasing, and let  $c$  be an interior point of  $I$ . Then  $f$  is continuous at  $c$  if and only if its one-sided limits satisfy  $f(c^-) = f(c) = f(c^+)$ , where  $f(c^-) = \lim_{x \rightarrow c^-} f(x)$  and  $f(c^+) = \lim_{x \rightarrow c^+} f(x)$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $I \subseteq \mathbb{R}$  be an interval,  $f : I \rightarrow \mathbb{R}$  monotone increasing, and  $c$  an interior point of  $I$ .

$$(f \text{ is continuous at } c) \iff \left( \lim_{x \rightarrow c^-} f(x) = f(c) = \lim_{x \rightarrow c^+} f(x) \right).$$

*Remark* (Predicate reading).

$$\text{ContinuousAt}(f, c) \iff (\text{LeftLimit}(f, c, f(c)) \wedge \text{RightLimit}(f, c, f(c))).$$

*Remark* (Negated quantified statement).

Let  $I \subseteq \mathbb{R}$  be an interval,  $f : I \rightarrow \mathbb{R}$  monotone increasing, and  $c$  an interior point of  $I$ .

$$\neg(f \text{ is continuous at } c) \iff \left( \lim_{x \rightarrow c^-} f(x) < f(c) \right) \vee \left( f(c) < \lim_{x \rightarrow c^+} f(x) \right).$$

*Remark* (Negation predicate reading).

$$\neg \text{ContinuousAt}(f, c) \iff (\exists L^- [\text{LeftLimit}(f, c, L^-) \wedge L^- < f(c)]) \vee (\exists L^+ [\text{RightLimit}(f, c, L^+) \wedge f(c) < L^+])$$

*Remark* (Failure modes). Continuity can fail only through a jump: either the graph arrives at  $c$  from the left strictly below  $f(c)$ , or it departs to the right strictly above  $f(c)$ . Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

*Remark* (Failure mode decomposition).

$$\neg \text{ContinuousAt}(f, c) = \underbrace{(\exists L^- [\text{LeftLimit}(f, c, L^-) \wedge L^- < f(c)])}_{\text{left jump at } c} \vee \underbrace{(\exists L^+ [\text{RightLimit}(f, c, L^+) \wedge f(c) < L^+])}_{\text{right jump at } c}$$

*Remark* (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at  $c$  demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through  $c$ . Any failure manifests as a jump discontinuity: the graph either leaps upward at  $c$  or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

*Remark* (Dependencies).

Definition 10.2.2; Theorem 10.5.1.

#### Definition (Jump of a Function)

**Definition 10.5.3** (Jump of a Function). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be increasing, and let  $c$  be an interior point of  $I$ . If the one-sided limits  $f(c^-)$  and  $f(c^+)$  exist, the jump of  $f$  at  $c$  is

$$j_f(c) := f(c^+) - f(c^-).$$

*Remark* (Standard quantified statement).

$$\begin{aligned} \forall I \subseteq \mathbb{R} \forall f : I \rightarrow \mathbb{R} \forall c \in I \forall j \in \mathbb{R} \\ \left[ (\forall x, y \in I \forall z \in \mathbb{R} (x < z < y \Rightarrow z \in I)) \wedge \exists a, b \in I (a < c < b) \right. \\ \left. \wedge (\forall x, y \in I (x < y \Rightarrow f(x) \leq f(y))) \wedge \exists L_-, L_+ \in \mathbb{R} \left( \lim_{x \rightarrow c^-} f(x) = L_- \wedge \lim_{x \rightarrow c^+} f(x) = L_+ \right) \right] \\ \Rightarrow (j = j_f(c) \iff j = f(c^+) - f(c^-)). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{JumpOfFunction}(f, I, c, j) := \text{Interval}(I) \wedge \text{MonotoneIncreasing}(f, I) \wedge \text{InteriorPoint}(c, I) \wedge \exists L_-, L_+ \in \mathbb{R} \left( \lim_{x \rightarrow c^-} f(x) = L_- \wedge \lim_{x \rightarrow c^+} f(x) = L_+ \wedge j = f(c^+) - f(c^-) \right)$$

*Remark* (Negated quantified statement).

$$\begin{aligned} \forall I \subseteq \mathbb{R} \forall f : I \rightarrow \mathbb{R} \forall c \in I \forall j \in \mathbb{R} : \\ \neg(j = j_f(c)) \iff \neg(\forall x, y \in I \forall z \in \mathbb{R} (x < z < y \Rightarrow z \in I)) \\ \vee \neg(\forall x, y \in I (x < y \Rightarrow f(x) \leq f(y))) \vee \neg(\exists a, b \in I (a < c < b)) \\ \vee \forall L_-, L_+ \in \mathbb{R} \left( \lim_{x \rightarrow c^-} f(x) \neq L_- \vee \lim_{x \rightarrow c^+} f(x) \neq L_+ \vee j \neq L_+ - L_- \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$\neg \text{JumpOfFunction}(f, I, c, j) \iff \neg \text{Interval}(I) \vee \neg \text{MonotoneIncreasing}(f, I) \vee \neg \text{InteriorPoint}(c, I) \vee \forall L_-, L_+.$

*Remark* (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

*Remark* (Failure mode decomposition).

$$\underbrace{\neg(f(c^-) \text{ exists})}_{\text{missing left limit}} \vee \underbrace{\neg(f(c^+) \text{ exists})}_{\text{missing right limit}} \vee \underbrace{j \neq f(c^+) - f(c^-)}_{\text{incorrect jump magnitude}}.$$

*Remark* (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

*Remark* (Dependencies).

[Theorem 10.5.1.](#)

#### Proposition (Discontinuities of a Monotone Function Are of First Kind)

**Proposition 10.5.4** (Discontinuities of a Monotone Function Are of First Kind). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be monotone, and let  $a \in I$  be such that there exist points of  $I$  strictly less and strictly greater than  $a$ . If  $f$  is discontinuous at  $a$ , then both one-sided limits  $\lim_{x \uparrow a} f(x)$  and  $\lim_{x \downarrow a} f(x)$  exist in  $\mathbb{R}$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $I \subseteq \mathbb{R}$  be an interval,  $f : I \rightarrow \mathbb{R}$  monotone, and  $a \in I$  interior to  $I$ .

$$[f \text{ is discontinuous at } a] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim_{x \uparrow a} f(x) = L^-, \\ \lim_{x \downarrow a} f(x) = L^+. \end{cases}$$

*Remark* (Predicate reading).

$\text{MonotoneFunction}(f, I) \wedge \text{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R} \ (\text{LeftLimit}(f, a, L^-) \wedge \text{RightLimit}(f, a, L^+))$

*Remark* (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

*Remark* (Dependencies).

[Definition 10.2.5](#)

#### Corollary (Jump Intervals Are Disjoint)

**Corollary 10.5.5** (Jump Intervals Are Disjoint). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be nondecreasing. If  $a, b \in I$  are distinct points of discontinuity of  $f$ , then the open intervals  $(f(a^-), f(a^+))$  and  $(f(b^-), f(b^+))$  are disjoint. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall a, b \in I \left[ a \neq b \wedge (\exists \varepsilon_a > 0 \forall \delta > 0 \exists x \in I (|x-a| < \delta \wedge |f(x)-f(a)| \geq \varepsilon_a)) \wedge (\exists \varepsilon_b > 0 \forall \delta > 0 \exists y \in I (|y-b| < \delta \wedge |f(y)-f(b)| \geq \varepsilon_b)) \right]$$

*Remark* (Predicate reading).

$$\text{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \text{DiscontinuousAt}(f, a), \\ \text{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \emptyset.$$

*Remark* (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

*Remark* (Dependencies).

[Definition 10.5.3](#); [Proposition 10.5.4](#)

#### Theorem (Discontinuities of a Monotone Function Are Countable)

**Theorem 10.5.6** (Discontinuities of a Monotone Function Are Countable). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be monotone. Then the set  $D_f := \{c \in I : f \text{ is discontinuous at } c\}$  is at most countable. [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } I \subseteq \mathbb{R} \text{ be an interval and } f : I \rightarrow \mathbb{R} \text{ be monotone.} \\ &\#\{c \in I : f \text{ fails to be continuous at } c\} \leq \aleph_0. \end{aligned}$$

*Remark* (Predicate reading).

$$\text{MonotoneFunction}(f, I) \implies \#\{c \in I : \text{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

*Remark* (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap  $(f(c^-), f(c^+))$ . Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into  $\mathbb{Q}$ , forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

*Remark* (Dependencies).

[Theorem 10.5.1](#) and [Corollary 10.5.5](#).

### Corollary (Monotone Function Is Continuous on a Dense Set)

**Corollary 10.5.7** (Monotone Function Is Continuous on a Dense Set). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be monotone. Then the set  $\{x \in [a, b] : f \text{ is continuous at } x\}$  is dense in  $[a, b]$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $f : [a, b] \rightarrow \mathbb{R}$  be monotone.

$\forall u, v \in [a, b] (u < v \Rightarrow \exists c \in (u, v) \text{ such that } f \text{ is continuous at } c).$

*Remark* (Predicate reading).

$\text{MonotoneFunction}(f, [a, b]) \implies \text{DenseSubset}(\{x \in [a, b] : \text{ContinuousAt}(f, x)\}, [a, b]).$

*Remark* (Negated quantified statement).

$\exists u, v \in [a, b] \left( u < v \wedge \forall c \in (u, v) \exists \varepsilon_c > 0 \forall \delta > 0 \exists x \in [a, b] (|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_c) \right).$

*Remark* (Negation predicate reading).

$\exists u, v \in [a, b] \left( u < v \wedge \forall c \in (u, v) \neg \text{ContinuousAt}(f, c) \right).$

*Remark* (Interpretation). A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

*Remark* (Dependencies).

### Discontinuities of a Monotone Function Are Countable

### Proposition (Continuous Injective Is Strictly Monotone)

**Proposition 10.5.8** (Continuous Injective Is Strictly Monotone). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then  $f$  is injective if and only if  $f$  is strictly monotone on  $[a, b]$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ .

$$\left[ \forall x, y \in [a, b] (f(x) = f(y) \Rightarrow x = y) \right] \iff \left[ \forall x, y \in [a, b] (x < y \Rightarrow f(x) < f(y)) \right]$$
$$\vee \left[ \forall x, y \in [a, b] (x < y \Rightarrow f(y) < f(x)) \right].$$

*Remark* (Predicate reading).

$\text{ContinuousOn}(f, [a, b]) \implies (\text{Injective}(f) \iff \text{StrictlyMonotone}(f, [a, b])).$

*Remark* (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

*Remark* (Dependencies).

[Bolzano's Intermediate Value Theorem.](#)

#### Theorem (Continuous Inverse Theorem)

**Theorem 10.5.9** (Continuous Inverse Theorem). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be strictly monotone and continuous on  $I$ . Then the inverse function  $f^{-1} : f(I) \rightarrow I$  exists, is strictly monotone, and is continuous on  $f(I)$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $I \subseteq \mathbb{R}$  be an interval and  $f : I \rightarrow \mathbb{R}$ .

$$\begin{aligned} & \left[ (\forall x, y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon) \right] \\ & \implies \left[ \exists g : f(I) \rightarrow I \text{ such that } (\forall x \in I) g(f(x)) = x \wedge (\forall y \in f(I)) f(g(y)) = y \right. \\ & \quad \left. \wedge (\forall u, v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I) \forall \varepsilon > 0 \exists \eta > 0 \forall u \in f(I) (|u - y| < \eta \Rightarrow |g(u) - g(y)| < \varepsilon) \right] \end{aligned}$$

*Remark* (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of  $f^{-1}$  is the reflection of the graph of  $f$  across the diagonal line  $y = x$ , so the smoothness and strict increase or decrease are visibly preserved by symmetry.

*Remark* (Dependencies).

- [Bolzano's Intermediate Value Theorem](#)
- [Strictly Increasing Function](#)
- [Strictly Decreasing Function](#)
- [Continuity at a Point](#)

## Limits Superior and Inferior of Functions

### Definition (Limits Superior and Inferior of a Function)

**Definition 10.5.10** (Limits Superior and Inferior of a Function). Let  $f : E \rightarrow \mathbb{R}$  and let  $x_0$  be an accumulation point of  $E$ . For each  $\delta > 0$  set

$$U(\delta) = \{f(x) : x \in E, 0 < |x - x_0| < \delta\}.$$

The extended real numbers  $\limsup_{x \rightarrow x_0} f(x)$  and  $\liminf_{x \rightarrow x_0} f(x)$  are defined by

$$\limsup_{x \rightarrow x_0} f(x) = \inf_{\delta > 0} \sup U(\delta), \quad \liminf_{x \rightarrow x_0} f(x) = \sup_{\delta > 0} \inf U(\delta),$$

where each inner supremum or infimum is taken in  $\overline{\mathbb{R}}$  and the outer infimum or supremum ranges over all  $\delta > 0$ .

*Remark* (Standard quantified statement).

Let  $f : E \rightarrow \mathbb{R}$ ,  $x_0$  an accumulation point of  $E$ , and  $U(\delta) = \{f(x) : x \in E, 0 < |x - x_0| < \delta\}$ .

$$\limsup_{x \rightarrow x_0} f(x) = \inf_{\delta > 0} \sup U(\delta),$$

$$\liminf_{x \rightarrow x_0} f(x) = \sup_{\delta > 0} \inf U(\delta).$$

*Remark* (Definition predicate reading).

$$\text{FunctionLimsup}(L, f, E, x_0) := L = \inf_{\delta > 0} \sup U(\delta),$$

$$\text{FunctionLiminf}(l, f, E, x_0) := l = \sup_{\delta > 0} \inf U(\delta),$$

where  $U(\delta) = \{f(x) : x \in E, 0 < |x - x_0| < \delta\}$ .

*Remark* (Negated quantified statement).

Let  $f : E \rightarrow \mathbb{R}$ ,  $x_0$  an accumulation point of  $E$ , and  $U(\delta) = \{f(x) : x \in E, 0 < |x - x_0| < \delta\}$ .

$$\neg \left[ \limsup_{x \rightarrow x_0} f(x) = \inf_{\delta > 0} \sup U(\delta) \wedge \liminf_{x \rightarrow x_0} f(x) = \sup_{\delta > 0} \inf U(\delta) \right]$$

$$\iff \limsup_{x \rightarrow x_0} f(x) \neq \inf_{\delta > 0} \sup U(\delta) \vee \liminf_{x \rightarrow x_0} f(x) \neq \sup_{\delta > 0} \inf U(\delta).$$

*Remark* (Negation predicate reading).

$$\neg (\text{FunctionLimsup}(L, f, E, x_0) \wedge \text{FunctionLiminf}(l, f, E, x_0))$$

$$\iff L \neq \inf_{\delta > 0} \sup U(\delta) \vee l \neq \sup_{\delta > 0} \inf U(\delta),$$

where  $U(\delta) = \{f(x) : x \in E, 0 < |x - x_0| < \delta\}$ .

*Remark* (Failure modes). The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.



*Remark* (Failure mode decomposition).

$$\underbrace{\exists \delta > 0 : U(\delta) = \emptyset}_{\text{missing punctured samples}} \vee \underbrace{\left( \inf_{\delta > 0} \sup U(\delta) = +\infty \wedge L < +\infty \right)}_{\text{upper envelope blow-up}} \vee \underbrace{\left( \sup_{\delta > 0} \inf U(\delta) = -\infty \wedge l > -\infty \right)}_{\text{lower envelope collapse}}.$$

*Remark* (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to  $x_0$ , and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

*Remark* (Dependencies).

- [Function](#)
- [Supremum](#)
- [Infimum](#)
- [Epsilon Neighbourhood](#)

### Proposition

**Proposition 10.5.11** ( $\limsup \geq \liminf$ ). *Let  $A \subseteq \mathbb{R}$ , let  $x_0$  be a limit point of  $A$ , and let  $f : A \rightarrow \mathbb{R}$ . Then*

$$\limsup_{x \rightarrow x_0} f(x) \geq \liminf_{x \rightarrow x_0} f(x).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall A \subseteq \mathbb{R} \forall f : A \rightarrow \mathbb{R} \forall x_0 \in \mathbb{R} : \left( \forall \eta > 0 \exists x \in A \setminus \{x_0\} (|x - x_0| < \eta) \right) \implies \limsup_{x \rightarrow x_0} f(x) \geq \liminf_{x \rightarrow x_0} f(x).$$

*Remark* (Interpretation). The limsup records the largest cluster value that  $f$  attains near  $x_0$ , while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

*Remark* (Dependencies).

[Definition \(Limits Superior and Inferior of a Function\).](#)

## Proposition

**Proposition 10.5.12** (Limit Exists iff  $\limsup = \liminf$ ). *Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , let  $x_0$  be a limit point of  $A$  with  $x_0 \in \mathbb{R}$ , and let  $L \in \mathbb{R}$ . Then  $\lim_{x \rightarrow x_0} f(x)$  exists and equals  $L$  if and only if  $\limsup_{x \rightarrow x_0} f(x) = \liminf_{x \rightarrow x_0} f(x) = L$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f: A \rightarrow \mathbb{R}, x_0 \text{ a limit point of } A, L \in \mathbb{R}. \\ &(\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \setminus \{x_0\} (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon)) \\ &\iff \left( \limsup_{x \rightarrow x_0} f(x) = L = \liminf_{x \rightarrow x_0} f(x) \right). \end{aligned}$$

*Remark* (Predicate reading).

$$\text{FunctionLimit}(f, A, x_0, L) \iff \left( \limsup_{x \rightarrow x_0} f(x) = L = \liminf_{x \rightarrow x_0} f(x) \right).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R}, f: A \rightarrow \mathbb{R}, x_0 \text{ a limit point of } A, L \in \mathbb{R}. \\ &\left[ \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \setminus \{x_0\} (0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0) \right] \\ &\iff \left( \limsup_{x \rightarrow x_0} f(x) \neq L \vee \liminf_{x \rightarrow x_0} f(x) \neq L \right). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{FunctionLimit}(f, A, x_0, L) \iff \left( \limsup_{x \rightarrow x_0} f(x) \neq L \vee \liminf_{x \rightarrow x_0} f(x) \neq L \right).$$

*Remark* (Failure modes). The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach  $L$  in the epsilon-delta sense, or one of the two extremal envelopes near  $x_0$  does not equal  $L$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in A \setminus \{x_0\} (0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0)}_{\text{limit condition fails}} \vee \underbrace{\limsup_{x \rightarrow x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \vee \underbrace{\liminf_{x \rightarrow x_0} f(x) \neq L}_{\text{lower envelope mismatch}}$$

*Remark* (Interpretation). The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the  $\limsup$  exceeds the  $\liminf$ , as happens for  $f(x) = \sin(1/x)$  near  $x_0 = 0$ , the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

*Remark* (Dependencies).

[Definition 10.5.10](#)

## 10.6 Limits at Infinity

### Limits at Infinity — Quick Reference

| Concept/Statement        | Meaning/Role                                                  |
|--------------------------|---------------------------------------------------------------|
| Infinite adherent points | Unboundedness makes $\pm\infty$ available as approach points. |
| Limit at infinity        | Replace $0 <  x - c  < \delta$ by sufficiently large $ x $ .  |
| Sequential criterion     | Limits at infinity are tested along escaping sequences.       |
| Limit laws               | Algebraic limit rules carry over unchanged.                   |

## Limits at Infinity

### Definition (Infinite Adherent Points)

**Definition 10.6.1** (Infinite Adherent Points). Let  $X \subseteq \mathbb{R}$ . The point  $+\infty$  is adherent to  $X$  if and only if for every  $M \in \mathbb{R}$  there exists  $x \in X$  such that  $x > M$ . The point  $-\infty$  is adherent to  $X$  if and only if for every  $M \in \mathbb{R}$  there exists  $x \in X$  such that  $x < M$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} +\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R} \exists x \in X (x > M), \\ -\infty \text{ is adherent to } X &\iff \forall M \in \mathbb{R} \exists x \in X (x < M). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\begin{aligned} \text{PlusInfAdherent}(X) &:= \forall M \in \mathbb{R} \exists x \in X (x > M), \\ \text{MinusInfAdherent}(X) &:= \forall M \in \mathbb{R} \exists x \in X (x < M). \end{aligned}$$

*Remark* (Negated quantified statement).

$$\begin{aligned} \neg(+\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R} \forall x \in X (x \leq M), \\ \neg(-\infty \text{ is adherent to } X) &\iff \exists M \in \mathbb{R} \forall x \in X (x \geq M). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\begin{aligned} \neg \text{PlusInfAdherent}(X) &\iff \exists M \in \mathbb{R} \forall x \in X (x \leq M), \\ \neg \text{MinusInfAdherent}(X) &\iff \exists M \in \mathbb{R} \forall x \in X (x \geq M). \end{aligned}$$

*Remark* (Failure modes). Infinite adherence fails at  $+\infty$  exactly when  $X$  is bounded above, and fails at  $-\infty$  exactly when  $X$  is bounded below.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R} \forall x \in X (x \leq M)}_{\text{finite upper bound}} \quad \underbrace{\exists M \in \mathbb{R} \forall x \in X (x \geq M)}_{\text{finite lower bound}}.$$

*Remark* (Interpretation). Both finite and infinite adherence are instances of adherence in  $\overline{\mathbb{R}}$ : finite adherence uses balls; infinite adherence uses rays.

### Definition (Limit at Infinity)

**Definition 10.6.2** (Limit at Infinity). Let  $X \subseteq \mathbb{R}$  be such that for every  $M \in \mathbb{R}$  there exists  $x \in X$  with  $x > M$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $L \in \mathbb{R}$ . The limit of  $f$  at  $+\infty$  equals  $L$  if and only if for every  $\varepsilon > 0$  there exists  $M \in \mathbb{R}$  such that for every  $x \in X$ ,  $x > M$  implies  $|f(x) - L| < \varepsilon$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } X \subseteq \mathbb{R}, \forall M \in \mathbb{R} \exists x \in X (x > M), f : X \rightarrow \mathbb{R}, L \in \mathbb{R}. \\ &\lim_{x \rightarrow +\infty} f(x) = L \iff \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x \in X \\ &\quad (x > M \Rightarrow |f(x) - L| < \varepsilon). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{LimitAtInfinity}(f, +\infty, L) := \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x \in X (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } X \subseteq \mathbb{R}, \forall M \in \mathbb{R} \exists x \in X (x > M), f : X \rightarrow \mathbb{R}, L \in \mathbb{R}. \\ &\neg \left( \lim_{x \rightarrow +\infty} f(x) = L \right) \iff \exists \varepsilon_0 > 0 \forall M \in \mathbb{R} \exists x \in X \\ &\quad (x > M \wedge |f(x) - L| \geq \varepsilon_0). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0 \forall M \in \mathbb{R} \exists x \in X (x > M \wedge |f(x) - L| \geq \varepsilon_0).$$

*Remark* (Failure modes). The definition fails only when there is a fixed tolerance  $\varepsilon_0$  that the function cannot satisfy on any tail of  $X$ , meaning every attempt to choose  $M$  leaves some point  $x > M$  with  $|f(x) - L| \geq \varepsilon_0$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\text{fixed tolerance no tail succeeds}} \quad \underbrace{\forall M \in \mathbb{R}}_{\text{no tail succeeds}} \quad \underbrace{\exists x \in X (x > M \wedge |f(x) - L| \geq \varepsilon_0)}_{\text{far-point violation}}.$$

*Remark* (Interpretation). The limit at  $+\infty$  demands that eventually the function values stay within any prescribed accuracy of  $L$ . Intuitively, once  $x$  is large enough, the graph of  $f$  must lie in the horizontal band of height  $\varepsilon$  around  $L$ . Failure occurs when some positive tolerance is violated infinitely often by points escaping to  $+\infty$ . This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of  $f$  with respect to the ideal point at infinity.

*Remark* (Dependencies).

[Definition 10.6.1.](#)

## 10.7 Approximation

### Approximation — Quick Reference

| Concept/Statement         | Meaning/Role                                           |
|---------------------------|--------------------------------------------------------|
| Step approximation        | Uniform approximation by step functions.               |
| Piecewise linear function | Linear pieces joined across a partition.               |
| PL approximation          | Uniform approximation by polygonal functions.          |
| Weierstrass               | Uniform polynomial approximation on compact intervals. |
| Bernstein polynomial      | Constructive polynomial approximants on $[0, 1]$ .     |
| Bernstein convergence     | Bernstein polynomials converge uniformly.              |

## Approximation of Continuous Functions

### Theorem (Step Function Approximation)

**Theorem 10.7.1** (Step Function Approximation). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ . Then there exists a step function  $s_\varepsilon : [a, b] \rightarrow \mathbb{R}$  such that*

$$\sup_{x \in [a, b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

*Go to proof.*

*Remark* (Standard quantified statement).

Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ .

$\exists s_\varepsilon : [a, b] \rightarrow \mathbb{R}$  step function  $\forall x \in [a, b] (|f(x) - s_\varepsilon(x)| < \varepsilon)$ .

*Remark* (Predicate reading).

$\text{StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) := \text{StepFunction}(s_\varepsilon, [a, b]) \wedge \forall x \in [a, b] (|f(x) - s_\varepsilon(x)| < \varepsilon)$ .

*Remark* (Negated quantified statement).

Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ .

$\forall s : [a, b] \rightarrow \mathbb{R}$  step function  $\exists x \in [a, b] (|f(x) - s(x)| \geq \varepsilon)$ .

*Remark* (Negation predicate reading).

$\neg \exists s_\varepsilon \text{ StepApproximation}(f, s_\varepsilon, [a, b], \varepsilon) \iff \forall s : [a, b] \rightarrow \mathbb{R} (\text{StepFunction}(s, [a, b]) \Rightarrow \exists x \in [a, b] (|f(x) - s(x)| \geq \varepsilon))$

*Remark* (Failure modes). The statement fails precisely when a continuous  $f$  and tolerance  $\varepsilon$  are fixed but every step function leaves an error of at least  $\varepsilon$  at some point in  $[a, b]$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\forall s : [a, b] \rightarrow \mathbb{R} \text{ step function}}_{\text{all approximants}} \underbrace{\exists x \in [a, b] (|f(x) - s(x)| \geq \varepsilon)}_{\text{witnessed large deviation}}.$$

*Remark* (Interpretation). Uniform continuity on  $[a, b]$  guarantees that  $f$  does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track  $f$  within any prescribed uniform error. The typical construction samples  $f$  at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least  $\varepsilon$ , contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

*Remark* (Dependencies).

[Heine–Cantor Theorem](#).

#### Definition (Piecewise Linear Function)

**Definition 10.7.2** (Piecewise Linear Function). Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $g : [a, b] \rightarrow \mathbb{R}$ . The function  $g$  is piecewise linear if and only if there exist  $m \in \mathbb{N}$  with  $m \geq 1$  and points  $a = x_0 < x_1 < \cdots < x_m = b$  such that  $g|_{[x_{k-1}, x_k]}$  is affine for every  $k \in \{1, \dots, m\}$ .

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$  and  $g : [a, b] \rightarrow \mathbb{R}$ .

$g$  is piecewise linear

$$\iff \exists m \in \mathbb{N} \left( m \geq 1 \wedge \exists x_0, \dots, x_m \in [a, b] \left[ a = x_0 < x_1 < \cdots < x_m = b \right. \right. \\ \left. \left. \wedge \forall k \in \{1, \dots, m\} \, g|_{[x_{k-1}, x_k]} \text{ is affine} \right] \right).$$

*Remark* (Definition predicate reading).

$$\text{PiecewiseLinear}(g, [a, b]) := \exists m \in \mathbb{N} \left( m \geq 1 \wedge \exists x_0, \dots, x_m \in [a, b] \left[ a = x_0 < \cdots < x_m = b \wedge \forall k \in \{1, \dots, m\} \right. \right.$$

*Remark* (Negated quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$  and  $g : [a, b] \rightarrow \mathbb{R}$ .

$g$  is not piecewise linear

$$\iff \forall m \in \mathbb{N} \left( m \geq 1 \Rightarrow \forall x_0, \dots, x_m \in [a, b] \left[ a = x_0 < \cdots < x_m = b \right. \right. \\ \left. \left. \Rightarrow \exists k \in \{1, \dots, m\} \, g|_{[x_{k-1}, x_k]} \text{ is not affine} \right] \right).$$

*Remark* (Negation predicate reading).

$$\neg \text{PiecewiseLinear}(g, [a, b]) \iff \forall m \in \mathbb{N} \left( m \geq 1 \Rightarrow \forall x_0, \dots, x_m \in [a, b] \left[ a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in \{1, \dots, m\} \right. \right.$$

*Remark* (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of  $[a, b]$  is available, or because for every such partition at least one subinterval has a restriction of  $g$  that is not affine.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N} \ (m \geq 1 \Rightarrow \forall x_0, \dots, x_m \in [a, b] \ (a = x_0 < \dots < x_m = b \Rightarrow \dots))}_{\text{no valid finite affine subdivision}} \vee \underbrace{\exists k \in \{1, \dots, m\} \ g|_{[x_{k-1}, x_k]} \text{ is not affine}}_{\text{non-affine segment}}$$

*Remark* (Interpretation). A piecewise linear function on  $[a, b]$  has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

*Remark* (Dependencies).

- [Function](#)
- [Closed Interval](#)
- [Natural Numbers](#)
- [Order on  \$\mathbb{R}\$](#)

#### Theorem

**Theorem 10.7.3** (Piecewise Linear Approximation). *Let  $f: [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ . Then there exists a continuous piecewise linear function  $\ell_\varepsilon: [a, b] \rightarrow \mathbb{R}$  such that  $\sup_{x \in [a, b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

Let  $f: [a, b] \rightarrow \mathbb{R}$ ,  $f$  continuous,  $\varepsilon > 0$ .

$\exists$  continuous piecewise linear  $\ell_\varepsilon: [a, b] \rightarrow \mathbb{R} : \sup_{x \in [a, b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$ .

*Remark* (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding  $\varepsilon$  on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

*Remark* (Dependencies).

- [Heine–Cantor Theorem](#)
- [Piecewise Linear Function](#)
- [Continuity at a Point](#)

### Theorem (Weierstrass Approximation Theorem)

**Theorem 10.7.4** (Weierstrass Approximation Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. For every  $\varepsilon > 0$  there exists a polynomial  $p_\varepsilon \in \mathbb{R}[x]$  such that  $\|f - p_\varepsilon\|_{\infty, [a, b]} < \varepsilon$ . Go to proof.*

*Remark* (Standard quantified statement).

Let  $a < b$  in  $\mathbb{R}$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous.  
 $\forall \varepsilon > 0 \exists p \in \mathbb{R}[x] (\|f - p\|_{\infty, [a, b]} < \varepsilon).$

*Remark* (Predicate reading).

$\text{ContinuousOn}(f, [a, b]) \implies \forall \varepsilon > 0 \exists p \in \mathbb{R}[x] (\|f - p\|_{\infty, [a, b]} < \varepsilon).$

*Remark* (Interpretation). Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

*Remark* (Dependencies).

Depends on [thm:bernstein-approximation](#).

### Definition (Bernstein Polynomial)

**Definition 10.7.5** (Bernstein Polynomial). Let  $n \in \mathbb{N}$  and let  $f : [0, 1] \rightarrow \mathbb{R}$ . A function  $B_n : [0, 1] \rightarrow \mathbb{R}$  is called the  $n$ th Bernstein polynomial of  $f$  if and only if, for every  $x \in [0, 1]$ ,

$$B_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

*Remark* (Standard quantified statement).

Let  $n \in \mathbb{N}$ ,  $f : [0, 1] \rightarrow \mathbb{R}$ ,  $B_n : [0, 1] \rightarrow \mathbb{R}$ .

$B_n$  is the  $n$ th Bernstein polynomial of  $f$

$$\iff \forall x \in [0, 1] \left( B_n(x) = \sum_{k=0}^n f(k/n) \binom{n}{k} x^k (1-x)^{n-k} \right).$$

*Remark* (Definition predicate reading).

$\text{BernsteinPolynomial}(B_n, f, n) := \left( B_n : [0, 1] \rightarrow \mathbb{R} \right) \wedge \forall x \in [0, 1] \left( B_n(x) = \sum_{k=0}^n f(k/n) \binom{n}{k} x^k (1-x)^{n-k} \right).$



*Remark* (Interpretation). The  $n$ th Bernstein polynomial samples  $f$  at the uniformly spaced nodes  $k/n$ , weights those samples by the binomial probabilities of order  $n$ , and produces for each  $x$  an average that depends polynomially on  $x$ . This construction smooths  $f$  into a polynomial that agrees with  $f$  at the endpoints and approximates  $f$  uniformly as  $n$  grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

*Remark* (Dependencies).

- [Function](#)
- [Natural Numbers](#)
- [Multiplication on  \$\mathbb{R}\$](#)
- [Addition on  \$\mathbb{R}\$](#)

#### Theorem (Bernstein Approximation Theorem)

**Theorem 10.7.6** (Bernstein Approximation Theorem). *Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous, and for each  $n \in \mathbb{N}$  let  $B_n$  denote the  $n$ th Bernstein polynomial of  $f$ . Then for every  $\varepsilon > 0$  there exists  $n_\varepsilon \in \mathbb{N}$  such that  $\|f - B_n\|_\infty < \varepsilon$  for all  $n \geq n_\varepsilon$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous, and for each  $n \in \mathbb{N}$  let  $B_n$  be the  $n$ th Bernstein polynomial of  $f$ .  
 $\forall \varepsilon > 0 \exists n_\varepsilon \in \mathbb{N} \forall n \geq n_\varepsilon : \|f - B_n\|_\infty < \varepsilon.$

*Remark* (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on  $[0, 1]$  converge uniformly to the original function, so the entire graph of  $f$  can be approximated to any prescribed sup-norm accuracy by explicit degree- $n$  polynomials once  $n$  is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of  $f$ , which forces the uniform error to vanish as  $n \rightarrow \infty$ . This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when  $f$  is not continuous on  $[0, 1]$ , in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each  $B_n$  is a convex combination of the samples  $f(k/n)$ , so the polygons formed by these samples become finer and eventually hug the graph of  $f$  everywhere on the interval.

*Remark* (Dependencies).

[Definition \(Continuity at a Point\)](#), [Definition \(Bernstein Polynomial\)](#).

## 10.8 Gauge

### Gauges and Tagged Partitions — Quick Reference

| Concept/Statement        | Meaning/Role                                            |
|--------------------------|---------------------------------------------------------|
| Interval partition       | Finite subdivision of a compact interval.               |
| Tagged partition         | Each subinterval carries a selected tag.                |
| Gauge                    | A position-dependent positive tolerance.                |
| $\delta$ -fine partition | Each subinterval lies within its tag's gauge window.    |
| Cousin's theorem         | Every gauge on $[a, b]$ admits a fine tagged partition. |
| Mesh                     | The largest subinterval length.                         |
| Common refinement        | A partition refining two given partitions.              |

## Gauges and Tagged Partitions

### Definition (Partition)

**Definition 10.8.1** (Partition). Let  $a < b$  and set  $I = [a, b]$ . A finite ordered set  $P = \{x_0, \dots, x_n\}$  with  $n \in \mathbb{N}$  and  $n \geq 1$  is a partition of  $I$  if and only if  $x_0 = a$ ,  $x_n = b$ , and  $x_0 < x_1 < \dots < x_n$ . In this case the closed subintervals  $[x_{i-1}, x_i]$  for  $i = 1, \dots, n$  subdivide  $I$ .

*Remark* (Standard quantified statement).

Let  $a < b$ ,  $I = [a, b]$ ,  $n \in \mathbb{N}$ ,  $n \geq 1$ , and  $P = \{x_0, \dots, x_n\} \subseteq I$ .  
 $P$  is a partition of  $I$   
 $\iff (x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \dots, n\} (x_{i-1} < x_i))$ .

*Remark* (Definition predicate reading).

$\text{PartitionOfInterval}(P, I) := \exists n \in \mathbb{N} (n \geq 1 \wedge P = \{x_0, \dots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, \dots, n\} (x_{i-1} < x_i))$

*Remark* (Negated quantified statement).

Let  $a < b$ ,  $I = [a, b]$ ,  $n \in \mathbb{N}$ ,  $n \geq 1$ , and  $P = \{x_0, \dots, x_n\} \subseteq I$ .  
 $P$  is not a partition of  $I$   
 $\iff ((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \dots, n\} (x_{i-1} \geq x_i))$ .

*Remark* (Negation predicate reading).

$\neg \text{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N} (n \geq 1 \Rightarrow (P \neq \{x_0, \dots, x_n\} \vee P \not\subseteq I \vee x_0 \neq \min I \vee x_n \neq \max I \vee \exists i \in \{1, \dots, n\} (x_{i-1} \geq x_i)))$

*Remark* (Failure modes). The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

*Remark* (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\text{right endpoint missing}} \quad \vee \quad \underbrace{\exists i \in \{1, \dots, n\} (x_{i-1} \geq x_i)}_{\text{nodes not strictly increasing}}.$$

*Remark* (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals  $[x_{i-1}, x_i]$  tile  $[a, b]$  without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

*Remark* (Dependencies).

- [Closed Interval](#)
- [Natural Numbers](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition (Tagged Partition)

**Definition 10.8.2** (Tagged Partition). Let  $a, b \in \mathbb{R}$  with  $a < b$ . A list  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  is a tagged partition of  $[a, b]$  if and only if the subintervals determine a partition  $P = \{[x_{i-1}, x_i]\}_{i=1}^n$  of  $[a, b]$  with  $a = x_0 < x_1 < \dots < x_n = b$ , and each tag satisfies  $t_i \in [x_{i-1}, x_i]$  for  $i = 1, \dots, n$ . Equivalently, a tagged partition is exactly a partition  $P$  of  $[a, b]$  together with one tag chosen inside every subinterval of  $P$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\ &\dot{\mathcal{P}} \text{ is a tagged partition of } [a, b] \\ &\iff (a = x_0 < \dots < x_n = b \wedge \forall i \in \{1, \dots, n\} (t_i \in [x_{i-1}, x_i])). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) := \left( \text{Partition}(P, [a, b]) \wedge \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \wedge \forall i (t_i \in [x_{i-1}, x_i]) \right),$$

where  $\text{Partition}(P, [a, b])$  abbreviates  $a = x_0 < \dots < x_n = b$  for  $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ .

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\ &\dot{\mathcal{P}} \text{ is not a tagged partition of } [a, b] \\ &\iff (x_0 \neq a \vee x_n \neq b \vee \exists j \in \{1, \dots, n\} (x_{j-1} \geq x_j) \vee \exists i \in \{1, \dots, n\} (t_i \notin [x_{i-1}, x_i])). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \iff (x_0 \neq a) \vee (x_n \neq b) \vee \left( \exists j (x_{j-1} \geq x_j) \right) \vee \left( \exists i (t_i \notin [x_{i-1}, x_i]) \right).$$

*Remark* (Failure modes). A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

*Remark* (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\text{left endpoint defect}} \vee \underbrace{x_n \neq b}_{\text{right endpoint defect}} \vee \underbrace{\exists j (x_{j-1} \geq x_j)}_{\text{ordering defect}} \vee \underbrace{\exists i (t_i \notin [x_{i-1}, x_i])}_{\text{tag misplaced}}.$$

*Remark* (Interpretation). A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

*Remark* (Dependencies).

[Definition 10.8.1.](#)

#### Definition (Gauge)

**Definition 10.8.3** (Gauge). Let  $a, b \in \mathbb{R}$  with  $a < b$  and set  $I = [a, b]$ . A gauge on  $I$  is a function  $\delta : I \rightarrow (0, \infty)$ .

*Remark* (Standard quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ put } I = [a, b], \text{ and let } \delta : I \rightarrow \mathbb{R}. \\ &\delta \text{ is a gauge on } I \iff \forall x \in I (\delta(x) > 0). \end{aligned}$$

*Remark* (Definition predicate reading).

$$\text{Gauge}(\delta, I) := (\delta : I \rightarrow \mathbb{R}) \wedge \forall x \in I (\delta(x) > 0).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} &\text{Let } a, b \in \mathbb{R} \text{ with } a < b, \text{ put } I = [a, b], \text{ and let } \delta : I \rightarrow \mathbb{R}. \\ &\delta \text{ fails to be a gauge on } I \iff \exists x \in I (\delta(x) \leq 0). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{Gauge}(\delta, I) \iff \neg(\delta : I \rightarrow \mathbb{R}) \vee \exists x \in I (\delta(x) \leq 0).$$

*Remark* (Failure modes). A purported gauge fails either because it is not a function from  $I$  to  $\mathbb{R}$ , or because it takes on a nonpositive value somewhere in  $I$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\neg(\delta : I \rightarrow \mathbb{R})}_{\text{wrong function type}} \quad \vee \quad \underbrace{\exists x \in I (\delta(x) \leq 0)}_{\text{nonpositive gauge value}}$$

*Remark* (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

*Remark* (Dependencies).

- [Function](#)
- [Closed Interval](#)
- [Order on  \$\mathbb{R}\$](#)

#### Definition ( $\delta$ -Fine Partition)

**Definition 10.8.4** ( $\delta$ -Fine Partition). Let  $[a, b]$  be a closed interval, let  $\delta : [a, b] \rightarrow (0, \infty)$  be a gauge, and let  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be a tagged partition of  $[a, b]$ . The tagged partition  $\dot{\mathcal{P}}$  is  $\delta$ -fine if and only if for every  $i \in \{1, \dots, n\}$  and every  $x \in [x_{i-1}, x_i]$  the inequality  $|x - t_i| < \delta(t_i)$  holds.

*Remark* (Standard quantified statement).

Let  $[a, b]$  be a closed interval,  $\delta : [a, b] \rightarrow (0, \infty)$ , and  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  a tagged partition of  $[a, b]$ .  $\dot{\mathcal{P}}$  is  $\delta$ -fine  $\iff \forall i \in \{1, \dots, n\} \forall x \in [x_{i-1}, x_i] (|x - t_i| < \delta(t_i))$ .

*Remark* (Definition predicate reading).

$$\text{DeltaFine}(\dot{\mathcal{P}}, \delta) := \forall i \in \{1, \dots, n\} \forall x \in [x_{i-1}, x_i] (|x - t_i| < \delta(t_i)).$$

*Remark* (Negated quantified statement).

$$\exists i \in \{1, \dots, n\} \exists x \in [x_{i-1}, x_i] (|x - t_i| \geq \delta(t_i)).$$

*Remark* (Negation predicate reading).

$$\neg \text{DeltaFine}(\dot{\mathcal{P}}, \delta) \iff \exists i \in \{1, \dots, n\} \exists x \in [x_{i-1}, x_i] (|x - t_i| \geq \delta(t_i)).$$

*Remark* (Failure modes). A tagged partition fails to be  $\delta$ -fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius  $\delta(t_i)$ , so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

*Remark* (Failure mode decomposition).

$$\underbrace{\exists i \in \{1, \dots, n\} \exists x \in [x_{i-1}, x_i]}_{\text{choose offending subinterval and point}} \quad \underbrace{(|x - t_i| \geq \delta(t_i))}_{\text{distance not controlled by } \delta}.$$

*Remark* (Interpretation). A  $\delta$ -fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

*Remark* (Dependencies).

[Definition 10.8.3](#); [Definition 10.8.2](#)

#### Lemma

**Lemma 10.8.5** (Every Point Is Covered by a Tag). *Let  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be a tagged partition of  $[a, b]$  that is  $\delta$ -fine on  $[a, b]$ , and let  $x \in [a, b]$ . Then there exists  $i \in \{1, \dots, n\}$  such that  $|x - t_i| < \delta(t_i)$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

Let  $a < b$ ,  $\delta : [a, b] \rightarrow (0, \infty)$ , and  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be  $\delta$ -fine on  $[a, b]$ .  
 $\forall x \in [a, b] \exists i \in \{1, \dots, n\} (|x - t_i| < \delta(t_i))$ .

*Remark* (Interpretation). A  $\delta$ -fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

*Remark* (Dependencies).

[Definition 10.8.2](#), [Definition 10.8.4](#).

#### Theorem (Cousin's Theorem)

**Theorem 10.8.6** (Cousin's Theorem). *Every gauge  $\delta : [a, b] \rightarrow (0, \infty)$  on a closed interval  $[a, b]$  admits a  $\delta$ -fine tagged partition of  $[a, b]$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$  and let  $\delta : [a, b] \rightarrow (0, \infty)$ .  
 $\exists m \in \mathbb{N} \exists x_0, \dots, x_m \exists t_1, \dots, t_m$   
 $(a = x_0 < x_1 < \dots < x_m = b \wedge \forall j \in \{1, \dots, m\} (t_j \in [x_{j-1}, x_j])$   
 $\wedge \forall j \in \{1, \dots, m\} [x_{j-1}, x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j)))$ .

*Remark* (Predicate reading).

$$\forall \delta : [a, b] \rightarrow (0, \infty) \quad \exists \dot{\mathcal{P}} \left( \text{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \right. \\ \left. \wedge \text{DeltaFine}(\dot{\mathcal{P}}, \delta) \right).$$

*Remark* (Negated quantified statement).

$$\begin{aligned}
& \text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } \delta : [a, b] \rightarrow (0, \infty). \\
& \neg \left( \text{there exists a } \delta\text{-fine tagged partition of } [a, b] \right) \\
& \iff \forall m \in \mathbb{N} \forall x_0, \dots, x_m \forall t_1, \dots, t_m \\
& \quad \left[ (a = x_0 < x_1 < \dots < x_m = b \wedge \forall j (t_j \in [x_{j-1}, x_j])) \right. \\
& \quad \left. \Rightarrow \exists j \in \{1, \dots, m\} [x_{j-1}, x_j] \not\subseteq (t_j - \delta(t_j), t_j + \delta(t_j)) \right].
\end{aligned}$$

*Remark* (Negation predicate reading).

$$\exists \delta : [a, b] \rightarrow (0, \infty) \forall \dot{\mathcal{P}} \left( \text{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) \Rightarrow \neg \text{DeltaFine}(\dot{\mathcal{P}}, \delta) \right).$$

*Remark* (Failure modes). Failure would mean that a positive gauge  $\delta$  defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

*Remark* (Failure mode decomposition).

$$\underbrace{\forall \dot{\mathcal{P}} \text{ TaggedPartition}(\dot{\mathcal{P}}, [a, b])}_{\text{all tagged partitions}} \Rightarrow \underbrace{\neg \text{DeltaFine}(\dot{\mathcal{P}}, \delta)}_{\text{some interval escapes its gauge ball}}.$$

*Remark* (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

*Remark* (Dependencies).

[Definition 10.8.3](#), [Definition 10.8.2](#), [Definition 10.8.4](#).

*Remark* (Gauge proofs of the four classical theorems). Each proof follows the same pattern: define a gauge  $\delta$  from the continuity hypothesis, invoke Cousin's theorem for a  $\delta$ -fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

**Boundedness.** At each  $t$ , continuity gives  $\delta(t)$  with  $\text{Within}(x, t, \delta(t)) \Rightarrow |f(x) - f(t)| \leq 1$ . A  $\delta$ -fine partition with tags  $t_1, \dots, t_n$  gives  $K = \max_i |f(t_i)|$ ; the Covering Lemma yields  $|f(x)| \leq 1 + K$  for all  $x$ .

**EVT.** Let  $M = \sup f(I)$ ; suppose  $f(x) < M$  everywhere. At each  $t$ , take  $\delta(t)$  with  $\text{Within}(x, t, \delta(t)) \Rightarrow f(x) < \frac{1}{2}(M + f(t))$ . A  $\delta$ -fine partition produces  $\widetilde{M} = \frac{1}{2}(M + \max_i f(t_i)) < M$  bounding  $f$  everywhere — contradiction.

**Location of Roots.** Assume  $f(t) \neq 0$  for all  $t$ . At each  $t$ , take  $\delta(t)$  preserving  $\text{sgn}(f(t))$ . A  $\delta$ -fine partition forces consistent sign from  $f(a) < 0$  to  $f(b) > 0$  — contradiction.

**Heine–Cantor.** Given  $\varepsilon > 0$ , at each  $t$  let  $\delta(t)$  satisfy  $\text{Within}(x, t, 2\delta(t)) \Rightarrow |f(x) - f(t)| \leq \varepsilon/2$ . A  $\delta$ -fine partition yields  $\delta_\varepsilon = \min_i \delta(t_i) > 0$ . For  $|x - u| < \delta_\varepsilon$ : find  $t_i$  covering  $x$ ; then  $|u - t_i| \leq |u - x| + |x - t_i| < 2\delta(t_i)$ ; so  $|f(x) - f(u)| \leq \varepsilon$ .

## Tagged Partitions and Mesh

### Definition (Mesh of a Partition)

**Definition 10.8.7** (Mesh of a Partition). Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $P = \{x_0, \dots, x_n\}$  be a partition of  $[a, b]$ . A real number  $m$  is the *mesh* (or *norm*) of  $P$  if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of  $P$  is denoted by  $\|P\|$  and is the unique  $m$  satisfying this equivalence.

*Remark* (Standard quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $P = \{x_0, \dots, x_n\}$  be a partition of  $[a, b]$ .  
 $m$  is the mesh of  $P \iff m = \max_{1 \leq i \leq n} (x_i - x_{i-1})$ .

*Remark* (Definition predicate reading).

$$\text{NormOfPartition}(P, m) := m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

*Remark* (Negated quantified statement).

Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $P = \{x_0, \dots, x_n\}$  be a partition of  $[a, b]$ .  
 $\neg(m \text{ is the mesh of } P) \iff (m < \max_{1 \leq i \leq n} (x_i - x_{i-1})) \vee (m > \max_{1 \leq i \leq n} (x_i - x_{i-1})).$

*Remark* (Negation predicate reading).

$$\neg \text{NormOfPartition}(P, m) \iff (m < \max_{1 \leq i \leq n} (x_i - x_{i-1})) \vee (m > \max_{1 \leq i \leq n} (x_i - x_{i-1})).$$

*Remark* (Failure modes). Failure occurs either when  $m$  underestimates the largest subinterval of  $P$  or when  $m$  overshoots that largest subinterval.

*Remark* (Failure mode decomposition).

$$\neg \text{NormOfPartition}(P, m) = \underbrace{(m < \max_{1 \leq i \leq n} (x_i - x_{i-1}))}_{\text{Underestimate}} \vee \underbrace{(m > \max_{1 \leq i \leq n} (x_i - x_{i-1}))}_{\text{Overestimate}}.$$

*Remark* (Interpretation). The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

*Remark* (Dependencies).

[Definition 10.8.1](#).



### Definition (Refinement)

**Definition 10.8.8** (Refinement). Let  $[a, b] \subseteq \mathbb{R}$  and let  $P$  and  $Q$  be partitions of  $[a, b]$ . Then  $Q$  is a refinement of  $P$  if and only if every point of  $P$  is also a point of  $Q$ ; equivalently,  $P \subseteq Q$ .

*Remark* (Standard quantified statement).

$$\forall x (x \in P \Rightarrow x \in Q).$$

*Remark* (Definition predicate reading).

$$\text{Refinement}(Q, P) := P \subseteq Q.$$

*Remark* (Negated quantified statement).

$$\exists x (x \in P \wedge x \notin Q).$$

*Remark* (Negation predicate reading).

$$\neg \text{Refinement}(Q, P) \iff \exists x \in P (x \notin Q).$$

*Remark* (Failure modes). Failure occurs precisely when some point listed in  $P$  fails to appear in  $Q$ , meaning  $Q$  omits at least one of the partition points it was supposed to inherit from  $P$ .

*Remark* (Failure mode decomposition).

$$\underbrace{\exists x \in P (x \notin Q)}_{\text{point of } P \text{ omitted by } Q}.$$

*Remark* (Interpretation). Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition  $P \subseteq Q$  guarantees that every subinterval defined by  $P$  is still represented when one passes to  $Q$ , allowing comparisons of sums or integrals computed on different meshes. Failure means that  $Q$  discards at least one point of  $P$ , so information tied to that breakpoint would be lost when transitioning between partitions.

*Remark* (Dependencies).

[Definition 10.8.1.](#)

### Definition (Common Refinement)

**Definition 10.8.9** (Common Refinement). Let  $a < b$  and let  $P$  and  $Q$  be partitions of  $[a, b]$ . Their common refinement is the partition  $R$  obtained by taking the union of their node sets, so  $R = P \cup Q$  as sets of points in  $[a, b]$ .

*Remark* (Standard quantified statement).

Let  $a < b$  in  $\mathbb{R}$ , and let  $P, Q$  be partitions of  $[a, b]$ .

$\forall R$  partition of  $[a, b] : R$  is the common refinement of  $P$  and  $Q \iff R = P \cup Q$ .

*Remark* (Definition predicate reading).

$$\text{CommonRefinement}(R, P, Q) := (R = P \cup Q).$$

*Remark* (Negated quantified statement).

$$\begin{aligned} a < b \wedge P, Q \subseteq [a, b] \wedge R \subseteq [a, b] \\ \implies \neg(\forall x \in \mathbb{R} (x \in R \iff x \in P \cup Q)) \\ \iff (\exists x \in R (x \notin P \cup Q)) \vee (\exists x \in P \cup Q (x \notin R)). \end{aligned}$$

*Remark* (Negation predicate reading).

$$\neg \text{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

*Remark* (Failure modes). The claim that a partition  $R$  is the common refinement of  $P$  and  $Q$  fails exactly when  $R$  either contains a node not present in  $P \cup Q$  or omits a node that  $P \cup Q$  contributes. These are the only two structural ways in which  $R$  can differ from  $P \cup Q$ .

*Remark* (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\text{extra node}} \vee \underbrace{\exists x \in (P \cup Q) \setminus R}_{\text{missing node}}.$$

*Remark* (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in  $P$  or  $Q$ , and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

*Remark* (Dependencies).

[Definition 10.8.1](#)

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sum Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then  $\lim_{x \rightarrow c} (f + g)(x) = L + M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-sum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-sum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Difference Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then  $\lim_{x \rightarrow c} (f - g)(x) = L - M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-difference. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-difference. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Scalar Multiple Rule for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then  $\lim_{x \rightarrow c} (\alpha f)(x) = \alpha L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-scalar-multiple. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-scalar-multiple. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Product Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  and  $\lim_{x \rightarrow c} g(x) = M$ , then  $\lim_{x \rightarrow c} (fg)(x) = LM$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-product. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-product. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Quotient Rule for Function Limits). *Let  $f, g : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ ,  $\lim_{x \rightarrow c} g(x) = M$ , and  $M \neq 0$ , then there exists  $\delta_0 > 0$  such that  $g(x) \neq 0$  whenever  $x \in A$  and  $0 < |x - c| < \delta_0$ , and  $\lim_{x \rightarrow c} (f/g)(x) = L/M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-quotient. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-quotient. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Neighbourhood Form of the Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every  $\varepsilon$ -neighbourhood  $V_\varepsilon(L)$  there exists a  $\delta$ -neighbourhood  $V_\delta(c)$  such that  $x \in V_\delta^*(c) \cap A$  implies  $f(x) \in V_\varepsilon(L)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-neighbourhood-equiv. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-neighbourhood-equiv. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Uniqueness of Limits of Functions). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $f$  has at most one limit at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-unique. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-unique. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Sequential Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \setminus \{c\}$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequential-criterion-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sequential-criterion-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Three Equivalent Forms of the Function Limit). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . The  $\varepsilon$ - $\delta$  form of  $\lim_{x \rightarrow c} f(x) = L$ , the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

*Professional Standard Proof.*

TODO: professional standard proof for limit-three-equiv. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-three-equiv. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limit Implies Local Boundedness). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$  for some  $L \in \mathbb{R}$ , then there exist  $\delta, M > 0$  such that  $|f(x)| \leq M$  for all  $x \in A$  with  $0 < |x - c| < \delta$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-implies-local-bounding. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-implies-local-bounding. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Limits). *Let  $f : A \rightarrow \mathbb{R}$ , let  $c$  be a cluster point of  $A$ , and let  $a, b \in \mathbb{R}$ . If  $a \leq f(x) \leq b$  for all  $x \in A$  with  $x \neq c$ , and  $\lim_{x \rightarrow c} f(x) = L$ , then  $a \leq L \leq b$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Squeeze Theorem for Function Limits). *Let  $f, g, h : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Suppose  $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} g(x) = L$ , and suppose there exists  $\delta_0 > 0$  such that*

$$\forall x \in A, 0 < |x - c| < \delta_0 \Rightarrow f(x) \leq h(x) \leq g(x).$$

*Then  $\lim_{x \rightarrow c} h(x) = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for squeeze-function-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for squeeze-function-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Limits Are Local). *Let  $E \subseteq X \subseteq \mathbb{R}$ , let  $x_0$  be a cluster point of  $E$ , let  $f : X \rightarrow \mathbb{R}$ , and let  $\delta > 0$ . Then*

$$\lim_{x \rightarrow x_0; x \in E} f(x) = L \iff \lim_{x \rightarrow x_0; x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

*Professional Standard Proof.*

TODO: professional standard proof for limits-are-local. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limits-are-local. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Limit of Absolute Value). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . If  $\lim_{x \rightarrow c} f(x) = L$ , then  $\lim_{x \rightarrow c} |f(x)| = |L|$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limit-absolute-value. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limit-absolute-value. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Composition of Limits). *Let  $f : A \rightarrow B$ , let  $g : B \rightarrow \mathbb{R}$ , let  $c_1$  be a cluster point of  $A$ , and let  $c_2 \in \mathbb{R}$  be a cluster point of  $B$ . Suppose  $\lim_{x \rightarrow c_1} f(x) = c_2$  and  $\lim_{y \rightarrow c_2} g(y) = L$ . If  $c_2 \in B$ , assume also that  $g(c_2) = L$ . Then  $\lim_{x \rightarrow c_1} g(f(x)) = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for composition-of-limits.

□

*Detailed Learning Proof.*

TODO: detailed learning proof for composition-of-limits.

□

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotone Limit Theorem). *Let  $a < b$  and let  $f : (a, b) \rightarrow \mathbb{R}$  be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing  $f$  and every  $x_0 \in (a, b)$ :*

$$f(x_0^-) = \sup\{f(x) : a < x < x_0\}, \quad f(x_0^+) = \inf\{f(x) : x_0 < x < b\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for monotone-limit-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-limit-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Cauchy Criterion for Function Limits). *Let  $f : A \rightarrow \mathbb{R}$  and let  $c$  be a cluster point of  $A$ . Then  $\lim_{x \rightarrow c} f(x) = L$  exists for some  $L \in \mathbb{R}$  if and only if*

$$\forall \text{OutputTolerance}(\varepsilon), \exists \text{InputTolerance}(\delta), \forall x, y \in A : 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-criterion-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-criterion-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sequential Criterion for Right-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (c, \infty)$ . Then  $\lim_{x \rightarrow c^+} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (c, \infty)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequential-criterion-right-hand-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sequential-criterion-right-hand-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Sequential Criterion for Left-Hand Limits). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and suppose  $c$  is a cluster point of  $A \cap (-\infty, c)$ . Then  $\lim_{x \rightarrow c^-} f(x) = L$  if and only if for every sequence  $(x_n) \subseteq A \cap (-\infty, c)$  with  $x_n \rightarrow c$ , one has  $f(x_n) \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequential-criterion-left-hand-limit. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sequential-criterion-left-hand-limit. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Two-Sided Limit via One-Sided Limits). *Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . If  $c$  is a cluster point of both  $A \cap (c, \infty)$  and  $A \cap (-\infty, c)$ , then  $\lim_{x \rightarrow c} f(x) = L$  if and only if both  $\lim_{x \rightarrow c^+} f(x) = L$  and  $\lim_{x \rightarrow c^-} f(x) = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for two-sided-limit-iff-matching-one-sided-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for two-sided-limit-iff-matching-one-sided-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Lemma** (Equivalent Formulations of Discontinuity at a Point). *Let  $E \subseteq \mathbb{R}$ ,  $f : E \rightarrow \mathbb{R}$ ,  $x_0 \in E$ . The  $\varepsilon$ - $\delta$ , sequential, and neighbourhood formulations of discontinuity at  $x_0$  are mutually equivalent.*

*Professional Standard Proof.*

TODO: professional standard proof for equivalent-discontinuity-at-a-point. □

*Detailed Learning Proof.*

TODO: detailed learning proof for equivalent-discontinuity-at-a-point. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuity Characterised by Oscillation). *Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , and let  $c \in A$ . Then  $f$  is continuous at  $c$  if and only if  $\omega(f; c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for continuity-iff-zero-oscillation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for continuity-iff-zero-oscillation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Discontinuity Set via Oscillation). *Let  $f : E \rightarrow \mathbb{R}$ . Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

*Professional Standard Proof.*

TODO: professional standard proof for discontinuity-set-via-oscillation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for discontinuity-set-via-oscillation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Boundedness Theorem). *Every function continuous on a closed bounded interval  $[a, b]$  is bounded on  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for boundedness-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for boundedness-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Extreme Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ , and let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. There exist points  $x_m, x_M \in [a, b]$  such that  $f(x_m) \leq f(x) \leq f(x_M)$  for every  $x \in [a, b]$ . Equivalently,  $f$  attains both an absolute maximum and an absolute minimum on  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for extreme-value-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for extreme-value-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Location of Roots Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . If  $f(a) < 0 < f(b)$  or  $f(a) > 0 > f(b)$ , then there exists  $c \in (a, b)$  such that  $f(c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for location-of-roots. □

*Detailed Learning Proof.*

TODO: detailed learning proof for location-of-roots. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bolzano's Intermediate Value Theorem). *Let  $I$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , let  $a, b \in I$ , and let  $k \in \mathbb{R}$ . If  $f(a) < k < f(b)$  or  $f(a) > k > f(b)$ , then there exists  $c$  between  $a$  and  $b$  such that  $f(c) = k$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bolzano-intermediate-value. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bolzano-intermediate-value. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Image of a Closed Bounded Interval Is a Closed Bounded Interval). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on the closed bounded interval  $[a, b]$ . Then  $f([a, b]) = [m, M]$ , where  $m = \inf f([a, b])$  and  $M = \sup f([a, b])$ .*

*Professional Standard Proof.*

TODO: professional standard proof for image-of-closed-bounded-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for image-of-closed-bounded-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Preservation of Intervals). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ . Then  $f(I)$  is an interval.*

*Professional Standard Proof.*

TODO: professional standard proof for preservation-of-intervals. □

*Detailed Learning Proof.*

TODO: detailed learning proof for preservation-of-intervals. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Darboux Property). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on the closed interval  $[a, b]$ . For any  $c \in \mathbb{R}$ , if  $f(x) \neq c$  for all  $x \in [a, b]$ , then either  $f(x) > c$  for all  $x \in [a, b]$ , or  $f(x) < c$  for all  $x \in [a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for darboux-property. □

*Detailed Learning Proof.*

TODO: detailed learning proof for darboux-property. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Algebra of Uniform Continuity on a General Domain). *(a) For every set  $A \subseteq \mathbb{R}$ , every pair of functions  $f, g : A \rightarrow \mathbb{R}$  that are uniformly continuous on  $A$ , and every scalar  $c \in \mathbb{R}$ , the functions  $f + g$ ,  $f - g$ , and  $cf$  are uniformly continuous on  $A$ .*

*(b) There exist a set  $A \subseteq \mathbb{R}$  and uniformly continuous functions  $f, g : A \rightarrow \mathbb{R}$  such that the product  $fg$  is not uniformly continuous on  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for algebra-of-uniform-continuity-general. □

*Detailed Learning Proof.*

TODO: detailed learning proof for algebra-of-uniform-continuity-general. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Algebra of Uniform Continuity on a Bounded Domain). *Let  $A \subseteq \mathbb{R}$  be bounded, and let  $f, g : A \rightarrow \mathbb{R}$  be uniformly continuous on  $A$ .*

- (a) The product  $x \mapsto f(x)g(x)$  is uniformly continuous on  $A$ .*
- (b) If there exists  $k > 0$  such that  $|g(x)| \geq k$  for all  $x \in A$ , then the quotient  $x \mapsto f(x)/g(x)$  is uniformly continuous on  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for algebra-of-uniform-continuity-bounded. □

*Detailed Learning Proof.*

TODO: detailed learning proof for algebra-of-uniform-continuity-bounded. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Sequential Characterization of Uniform Continuity). *Let  $X \subseteq \mathbb{R}$  and  $f : X \rightarrow \mathbb{R}$ . Then  $f$  is uniformly continuous on  $X$  if and only if for every pair of sequences  $(x_n), (y_n)$  in  $X$  with  $|x_n - y_n| \rightarrow 0$ , we have  $|f(x_n) - f(y_n)| \rightarrow 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sequential-uniform-continuity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sequential-uniform-continuity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Heine-Cantor Theorem). *Every function continuous on a closed bounded interval  $[a, b]$  is uniformly continuous on  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for heine-cantor. □

*Detailed Learning Proof.*

TODO: detailed learning proof for heine-cantor. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Uniform Continuity Preserves Cauchy Sequences). *If  $f: S \rightarrow \mathbb{R}$  is uniformly continuous and  $(x_n)$  is a Cauchy sequence in  $S$ , then  $(f(x_n))$  is a Cauchy sequence in  $\mathbb{R}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for uniform-continuity-cauchy. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniform-continuity-cauchy. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Lipschitz Implies Uniform Continuity). *Let  $A \subseteq \mathbb{R}$  and  $f : A \rightarrow \mathbb{R}$ . If there exists  $K \geq 0$  such that  $|f(x) - f(y)| \leq K|x - y|$  for all  $x, y \in A$ , then  $f$  is uniformly continuous on  $A$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lipschitz-implies-uc. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lipschitz-implies-uc. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Algebra of Lipschitz Functions). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$  be Lipschitz on  $A$  with constants  $K_f$  and  $K_g$  respectively, and let  $c \in \mathbb{R}$ . Then:*

- (a)  $cf$  is Lipschitz with constant  $|c|K_f$
- (b)  $f + g$  and  $f - g$  are Lipschitz with constant  $K_f + K_g$
- (c) If  $h : B \rightarrow \mathbb{R}$  is Lipschitz on  $B$  with constant  $K_h$  and  $f(A) \subseteq B$ , then  $h \circ f$  is Lipschitz on  $A$  with constant  $K_h K_f$
- (d) If  $|f| \leq M_f$  and  $|g| \leq M_g$  on  $A$ , then  $fg$  is Lipschitz with constant  $M_g K_f + M_f K_g$
- (e) If  $|f| \leq M_f$  and  $|g(x)| \geq k > 0$  for all  $x \in A$ , then  $f/g$  is Lipschitz with constant  $K_f/k + M_f K_g/k^2$

*Professional Standard Proof.*

TODO: professional standard proof for algebra-of-lipschitz-functions. □

*Detailed Learning Proof.*

TODO: detailed learning proof for algebra-of-lipschitz-functions. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Inverse of a Bi-Lipschitz Map Is Lipschitz). *Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow f(A)$  be bi-Lipschitz with bounds  $\alpha$  and  $K$ , meaning*

$$\alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u| \quad \text{for all } x, u \in A.$$

*Then (a)  $f$  is injective, so  $f^{-1} : f(A) \rightarrow A$  exists; (b)  $f^{-1}$  is Lipschitz with constant  $1/\alpha$ ; (c)  $f^{-1}$  is bi-Lipschitz with bounds  $1/K$  and  $1/\alpha$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bilipschitz-inverse-is-lipschitz. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bilipschitz-inverse-is-lipschitz. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (). *Let  $A \subseteq \mathbb{R}$ , let  $x_0$  be a limit point of  $A$ , and let  $f : A \rightarrow \mathbb{R}$ . Then*

$$\limsup_{x \rightarrow x_0} f(x) \geq \liminf_{x \rightarrow x_0} f(x).$$

*Professional Standard Proof.*

TODO: professional standard proof for limsup-geq-liminf-function. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-geq-liminf-function. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Limit Exists iff  $=$ ). *Let  $A \subseteq \mathbb{R}$ , let  $f: A \rightarrow \mathbb{R}$ , let  $x_0$  be a limit point of  $A$  with  $x_0 \in \mathbb{R}$ , and let  $L \in \mathbb{R}$ . Then  $\lim_{x \rightarrow x_0} f(x)$  exists and equals  $L$  if and only if  $\limsup_{x \rightarrow x_0} f(x) = \liminf_{x \rightarrow x_0} f(x) = L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for limsup-liminf-limit-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for limsup-liminf-limit-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (One-Sided Limits of Monotone Functions). *Let  $f : I \rightarrow \mathbb{R}$  be monotone increasing on the interval  $I$ , and let  $c \in I$  be an interior point. Then both one-sided limits of  $f$  at  $c$  exist and are given by*

$$\lim_{x \rightarrow c^-} f(x) = \sup\{f(x) : x < c, x \in I\} \quad \text{and} \quad \lim_{x \rightarrow c^+} f(x) = \inf\{f(x) : x > c, x \in I\}.$$

*Moreover,  $f(c^-) \leq f(c) \leq f(c^+)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-one-sided-limits. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-one-sided-limits. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Continuity Criterion for Monotone Functions). *Let  $f : I \rightarrow \mathbb{R}$  be monotone increasing and let  $c \in I$  be an interior point. Then  $f$  is continuous at  $c$  if and only if  $f(c^-) = f(c^+)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-continuity-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-continuity-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Proposition** (Discontinuities of a Monotone Function Are of First Kind). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$  be monotone, and let  $a \in I$  be such that there exist points of  $I$  strictly less and strictly greater than  $a$ . If  $f$  is discontinuous at  $a$ , then both one-sided limits  $\lim_{x \uparrow a} f(x)$  and  $\lim_{x \downarrow a} f(x)$  exist in  $\mathbb{R}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-discontinuities-first-kind. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-discontinuities-first-kind. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Jump Intervals of a Monotone Function Are Disjoint). *Let  $f : I \rightarrow \mathbb{R}$  be a monotone function on an interval  $I$ , and let  $D \subseteq I$  be the set of all discontinuities of  $f$ . For each  $c \in D$ , the jump interval at  $c$  is  $J_c(f) := (f(c^-), f(c^+))$ . If  $c, d \in D$  with  $c \neq d$ , then  $J_c(f) \cap J_d(f) = \emptyset$ .*

*Professional Standard Proof.*

TODO: professional standard proof for jump-intervals-for-monotone-discontinuities. □

*Detailed Learning Proof.*

TODO: detailed learning proof for jump-intervals-for-monotone-discontinuities. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Discontinuities of a Monotone Function Are Countable). *Let  $f : I \rightarrow \mathbb{R}$  be a monotone function on an interval  $I$ . Then the set of discontinuities of  $f$  is at most countable.*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-discontinuities-countable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-discontinuities-countable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Monotone Function Is Continuous on a Dense Set). *Let  $f : I \rightarrow \mathbb{R}$  be a monotone function on an interval  $I$ . Then the set of points of continuity of  $f$  is dense in  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-continuous-on-dense-set. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-continuous-on-dense-set. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Proposition** (Continuous Injective Is Strictly Monotone). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then  $f$  is injective if and only if  $f$  is strictly monotone on  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for continuous-injective-iff-strictly-monotone. □

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous-injective-iff-strictly-monotone. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Inverse Theorem). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be strictly monotone and continuous on  $I$ . Then the inverse function  $f^{-1} : f(I) \rightarrow I$  exists, is strictly monotone, and is continuous on  $f(I)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for continuous-inverse-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous-inverse-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Step Function Approximation). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous and let  $\varepsilon > 0$ . Then there exists a step function  $s_\varepsilon : [a, b] \rightarrow \mathbb{R}$  such that*

$$\sup_{x \in [a, b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

*Professional Standard Proof.*

TODO: professional standard proof for step-function-approximation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for step-function-approximation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Piecewise Linear Approximation). *TODO: restate the theorem.*

*Professional Standard Proof.*

TODO: professional standard proof for piecewise-linear-approximation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for piecewise-linear-approximation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Weierstrass Approximation Theorem). *Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous. For every  $\varepsilon > 0$  there exists a polynomial  $p_\varepsilon \in \mathbb{R}[x]$  such that  $\|f - p_\varepsilon\|_{\infty, [a, b]} < \varepsilon$ .*

*Professional Standard Proof.*

TODO: professional standard proof for weierstrass-approximation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for weierstrass-approximation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bernstein Approximation Theorem). *TODO: restate the theorem.*

*Professional Standard Proof.*

TODO: professional standard proof for bernstein-approximation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bernstein-approximation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Lemma** (Every Point Is Covered by a Tag). *Let  $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$  be a tagged partition of  $[a, b]$  that is  $\delta$ -fine on  $[a, b]$ , and let  $x \in [a, b]$ . Then there exists  $i \in \{1, \dots, n\}$  such that  $|x - t_i| < \delta(t_i)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for every-point-covered-by-tag. □

*Detailed Learning Proof.*

TODO: detailed learning proof for every-point-covered-by-tag. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cousin's Theorem). *Every gauge  $\delta : [a, b] \rightarrow (0, \infty)$  on a closed interval  $[a, b]$  admits a  $\delta$ -fine tagged partition of  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cousins-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cousins-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



# Chapter 11

## Differentiation



### 11.1 The Tangent Problem

#### Secant and Tangent — Quick Reference

| Concept                      | Meaning                                          |
|------------------------------|--------------------------------------------------|
| <a href="#">Secant line</a>  | Line through two points of a graph               |
| <a href="#">Tangent line</a> | Limiting local line determined by the derivative |

*Remark* (Section guide). Before the derivative is a formula it is a question: what straight line best matches a curve at a single point? The route in is indirect: choose a second point, draw the computable secant line, and let that point slide toward the first. If the secants settle toward a limiting line, that line is the tangent and its slope is the derivative.

#### Definition (Secant Line)

**Definition 11.1.1** (Secant Line). Let  $f : A \rightarrow \mathbb{R}$ , and let  $x_1, x_2 \in A$  with  $x_1 \neq x_2$ . The *secant line* determined by the points  $(x_1, f(x_1))$  and  $(x_2, f(x_2))$  is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

*Remark* (Standard quantified statement).

$$\forall x_1, x_2 \in A (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \frac{f(x_2) - f(x_1)}{x_2 - x_1}).$$

*Remark* (Interpretation). With increments  $\Delta x := x_2 - x_1$  and  $\Delta y := f(x_2) - f(x_1)$ , the secant slope is  $\Delta y / \Delta x$  — the average rate of change of  $f$  over the interval  $[x_1, x_2]$ . This is an

exact algebraic identity; no limit is involved. As  $x_2 \rightarrow x_1$ , if the secant slope converges, its limit is the derivative  $f'(x_1)$  — the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point  $(x_1, f(x_1))$  as  $x_2$  approaches  $x_1$ , and the limit position is the tangent line.

### Definition (Difference Quotient)

**Definition 11.1.2** (Difference Quotient). Let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$ , and let  $x \in A$  with  $x \neq c$ . The *difference quotient* of  $f$  from  $c$  to  $x$  is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

*Remark* (Interpretation). The difference quotient is the slope of the secant line through  $(c, f(c))$  and  $(x, f(x))$ . It is the quantity whose limiting value, when it exists as  $x \rightarrow c$ , becomes the derivative.

*Remark* (Dependencies).

- [Secant Line](#)

### Definition (Tangent Line)

**Definition 11.1.3** (Tangent Line). Let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$ , and suppose that  $f$  is differentiable at  $c$ . The *tangent line* to the graph of  $f$  at the point  $(c, f(c))$  is the line through  $(c, f(c))$  with slope  $f'(c)$ . Its equation is

$$y = f(c) + f'(c)(x - c).$$

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

*Remark* (Interpretation). The tangent line is the limit of the family of secant lines through  $(c, f(c))$  and  $(x, f(x))$  as  $x \rightarrow c$ . More precisely: the slope  $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$  converges to  $f'(c)$ , so the tangent line is the line whose slope is that limit. It is also the unique linear function  $\ell(x) = f(c) + f'(c)(x - c)$  satisfying  $\ell(c) = f(c)$  and  $\ell'(c) = f'(c)$  — the best first-order approximation to  $f$  near  $c$ . The error  $f(x) - \ell(x) = o(x - c)$  as  $x \rightarrow c$ , meaning it vanishes faster than the displacement.

*Remark* (Dependencies).

- [Definition: Secant Line](#)
- [Definition: Derivative at a Point](#)

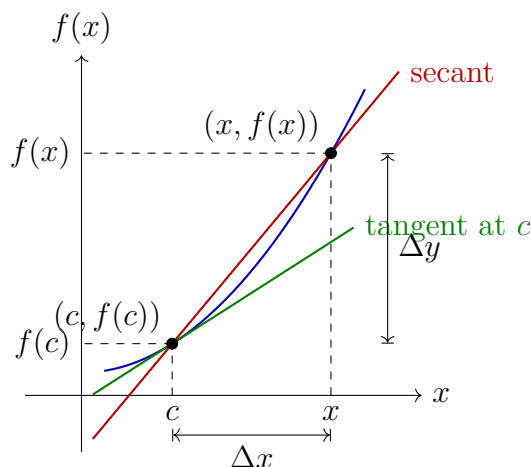


Figure 11.1: The secant line through  $(c, f(c))$  and  $(x, f(x))$  has slope  $\Delta y / \Delta x$ . As  $x \rightarrow c$ , the secant line rotates toward the tangent line at  $(c, f(c))$ , whose slope is  $f'(c)$ .

## 11.2 The Derivative

### Derivative Definition — Quick Reference

| Concept                              | Meaning                                                |
|--------------------------------------|--------------------------------------------------------|
| Derivative at a point                | Epsilon–delta limiting secant slope                    |
| Example: derivative of $x^2$         | The definition in a removable-singularity computation  |
| Topological derivative               | Neighbourhood formulation of the same limit            |
| Sequential derivative                | Sequence test for differentiability                    |
| Equivalent definitions               | Epsilon–delta, topological, and sequential forms agree |
| $h$ -form                            | Difference quotient written with $x = c + h$           |
| Differentiability implies continuity | Differentiability is stronger than continuity          |
| Example: one corner                  | Continuous but not differentiable at one point         |
| Example: sawtooth                    | Continuous with infinitely many isolated corners       |
| Example: Weierstrass function        | Continuous and nowhere differentiable                  |
| Uniqueness                           | The derivative value is single-valued                  |
| Identity derivative                  | Base case $D(x) = 1$                                   |
| Constant derivative                  | Base case $D(C) = 0$                                   |

*Remark* (Section guide). The derivative is the number the secant slopes approach, but the word “limit” has three useful faces. The epsilon–delta form is the contract; the topological form strips it to neighbourhoods; the sequential form is the disproving tool, since one bad approach sequence is enough.

### Definition (Derivative at a Point)

**Definition 11.2.1** (Derivative at a Point). Let  $f : D \rightarrow \mathbb{R}$  be a real-valued function and  $c \in \text{int}(D)$ . We say  $f$  is *differentiable at  $c$*  with *derivative*  $L = f'(c)$  if and only if:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left( 0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

*Remark* (Standard quantified statement).

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left( 0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

*Remark* (Definition predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{FunctionLimit} \left( \lambda x. \frac{f(x) - f(c)}{x - c}, D, c, L \right).$$

The predicate  $\text{DerivativeAt}(f, c, L)$  asserts that  $L$  is the value of the limit of the difference quotient of  $f$  at  $c$ .

*Remark* (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in D \left( 0 < |x - c| < \delta \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

*Remark* (Negation predicate reading).

$$\neg \text{DerivativeAt}(f, c, L) \iff \neg \text{FunctionLimit} \left( \lambda x. \frac{f(x) - f(c)}{x - c}, D, c, L \right).$$

*Remark* (Failure modes). • The difference quotient has no finite limit at  $c$  (e.g.  $|f(x) - f(c)|/|x - c| \rightarrow \infty$ ).

- The difference quotient oscillates without settling near any single value.
- The proposed value  $L$  is the wrong candidate.

*Remark* (Interpretation). The derivative is a limit of slopes of secant lines. The  $\varepsilon$ - $\delta$  form demands that the difference quotient can be made arbitrarily close to  $L$  by taking  $x$  sufficiently close to  $c$ . The input condition  $0 < |x - c|$  is punctured:  $x = c$  is excluded because  $(f(c) - f(c))/(c - c)$  is  $0/0$ .

*Remark* (Dependencies).

- [Limit of a Function](#)

*Example* (Derivative of  $x^2$  from the Definition). Let  $f(x) = x^2$  and fix  $c \in \mathbb{R}$ . For  $x \neq c$ ,

$$\frac{f(x) - f(c)}{x - c} = \frac{x^2 - c^2}{x - c} = x + c.$$

The singularity has been removed by exact algebra, so the limit as  $x \rightarrow c$  is  $2c$ . Hence  $f'(c) = 2c$ . Read through Carathéodory, this is the factor  $\varphi(x) = x + c$ , continuous at  $c$  with  $\varphi(c) = 2c$ .

### Definition (Topological Definition of Derivative at a Point)

**Definition 11.2.2** (Topological Definition of Derivative at a Point). Let  $f : D \rightarrow \mathbb{R}$  and  $c \in \text{int}(D)$ .  $f$  is differentiable at  $c$  with derivative  $L$  if for every neighbourhood  $V$  of  $L$ , there exists a neighbourhood  $U$  of  $c$  such that for all  $x \in (U \setminus \{c\}) \cap D$ :

$$\frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Standard quantified statement).

$$\forall V \text{ neighbourhood of } L \exists U \text{ neighbourhood of } c \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Definition predicate reading).

$$\text{DerivativeAt}_{\text{top}}(f, c, L) \iff \forall V \ni L \exists U \ni c \forall x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \in V.$$

*Remark* (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L \forall U \text{ neighbourhood of } c \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

*Remark* (Negation predicate reading).

$$\neg \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \exists V \ni L \forall U \ni c \exists x \in (U \setminus \{c\}) \cap D : \frac{f(x) - f(c)}{x - c} \notin V.$$

*Remark* (Failure modes). The topological derivative fails when a fixed output neighbourhood of  $L$  cannot be entered by the difference quotient no matter how tightly the input is restricted around  $c$ . Geometric reading: no punctured neighbourhood around  $c$  maps all secant slopes into any prescribed window around  $L$ .

*Remark* (Interpretation). The topological form replaces  $\varepsilon$ -balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

*Remark* (Dependencies).

- [Function](#)
- [Relative Neighborhood](#)
- [Subtraction on  \$\mathbb{R}\$](#)
- [Division on  \$\mathbb{R}\$](#)

### Definition (Sequential Definition of Derivative at a Point)

**Definition 11.2.3** (Sequential Definition of Derivative at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . We say that  $f$  is differentiable at  $c$  with derivative  $L$  if

$$\forall (x_n) \subseteq A \setminus \{c\} (x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L).$$

*Remark* (Standard quantified statement).

$$\forall(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L.$$

*Remark* (Definition predicate reading).

$$\text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \forall(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L.$$

*Remark* (Negated quantified statement).

$$\exists(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L.$$

*Remark* (Negation predicate reading).

$$\neg \text{DerivativeAt}_{\text{seq}}(f, c, L) \iff \exists(x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L.$$

*Remark* (Failure modes). Failure occurs when a single approach sequence  $(x_n) \rightarrow c$  exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than  $L$  (wrong slope candidate). A single such sequence certifies non-differentiability.

*Remark* (Failure mode decomposition).

$$\underbrace{x_n \rightarrow c}_{\text{approach condition}} \wedge \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g.,  $f(x) = |x|$  at  $c = 0$ ), or quotients growing without bound.

*Remark* (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence  $x_n \rightarrow c$  along which the difference quotient fails to converge to any single value.

*Remark* (Dependencies).

- [Sequence](#)
- [Convergent Sequence](#)
- [Derivative at a Point](#)
- [Limit of a Function](#)

### Theorem (Equivalence of Definitions of the Derivative at a Point)

**Theorem 11.2.4** (Equivalence of Definitions of the Derivative at a Point). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$  be a limit point of  $A$ . The following are equivalent:*

1.  *$f$  is differentiable at  $c$  in the  $\varepsilon$ - $\delta$  sense with derivative  $L$ .*
2.  *$f$  is differentiable at  $c$  in the topological sense with derivative  $L$ .*
3.  *$f$  is differentiable at  $c$  in the sequential sense with derivative  $L$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$(i) \iff (ii) \iff (iii).$$

*Remark* (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \text{DerivativeAt}_{\text{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of  $f$  at  $c$ .

*Remark* (Negated quantified statement).

$$\begin{aligned} \neg(i) &\iff \neg(ii) \iff \neg(iii) \\ &\iff \exists (x_n) \subseteq A \setminus \{c\} : x_n \rightarrow c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\rightarrow L. \end{aligned}$$

*Remark* (Failure modes). The equivalence cannot produce contradictory verdicts: if  $c$  is a limit point of  $A$ , all three forms agree. Degenerate case: if  $c$  is not a limit point of  $A$ , the punctured-ball and sequence conditions are vacuously satisfied by every candidate  $L$  simultaneously, making the equivalence trivially true for all  $L$  rather than characterizing a unique derivative.

*Remark* (Interpretation). The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

*Remark* (Dependencies).

- [Derivative at a Point](#)
- [Topological Definition](#)
- [Sequential Definition](#)

### Proposition (Derivative: $h$ -Form Equivalence)

**Proposition 11.2.5** (Derivative:  $h$ -Form Equivalence). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in \text{int}(A)$ . Then  $f$  is differentiable at  $c$  with derivative  $L$  if and only if*

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \iff \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

*Remark* (Negated quantified statement).

$$\neg \text{DerivativeAt}(f, c, L) \iff \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

*Remark* (Failure modes). The  $h$ -form fails in two branches: (i) The incremental ratio  $(f(c+h) - f(c))/h$  has no finite limit as  $h \rightarrow 0$  (corner, cusp, or vertical tangent at  $c$ ). (ii) The ratio converges to a finite value different from  $L$  (proposed  $L$  is the wrong slope).

*Remark* (Interpretation). The  $h$ -form substitutes  $x = c + h$  in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

*Remark* (Dependencies).

- [Derivative at a Point](#)

*Remark* (Discrete and continuous change). The forward difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of the derivative. The  $h$ -form above is its continuous descendant: replace the fixed step 1 by a shrinking step  $h$ , divide by  $h$ , and pass to the limit.

The kinship matters, but so does the divergence. The discrete product rule has an unavoidable shift:

$$\Delta(fg)(n) = (\Delta f)(n)g(n) + f(n+1)(\Delta g)(n).$$

The continuous product rule loses this shift because  $f(x+h) \rightarrow f(x)$  as  $h \rightarrow 0$ . Likewise, ordinary powers are not the clean basis for  $\Delta$ ; falling factorials are:

$$\Delta x^{\underline{k}} = k x^{\underline{k-1}}.$$

Passing to the limit is therefore not just a sharpening of a quotient. It restores symmetries that the discrete operator must carry as shifts.



## 11.2.1 First Consequences

### Theorem (Differentiable Implies Continuous)

**Theorem 11.2.6** (Differentiable Implies Continuous). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$ . If  $f$  is differentiable at  $c$ , then  $f$  is continuous at  $c$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DerivableAt}(f, c) \Rightarrow \text{ContinuousAt}(f, c).$$

*Remark* (Predicate reading).

$$\text{DifferentiableAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

*Remark* (Failure modes). The implication can only fail in the reverse direction: continuity at  $c$  does not force differentiability at  $c$ .

*Remark* (Failure mode decomposition). The converse fails:  $f(x) = |x|$  is continuous at 0 but not differentiable there.

*Remark* (Contrapositive quantified statement).

$$\neg \text{ContinuousAt}(f, c) \Rightarrow \neg \text{DerivableAt}(f, c).$$

*Remark* (Contrapositive predicate reading).

$$\neg \text{ContinuousAt}(f, c) \implies \neg \text{DifferentiableAt}(f, c).$$

*Remark* (Interpretation). Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at  $f(c)$  as  $x \rightarrow c$ .

*Remark* (Dependencies).

- [Derivative at a Point](#)
- [Continuity at a Point](#)

*Remark* (Converse failure). The theorem runs one way only. Differentiability forces continuity, but continuity does not force differentiability. The negation of the converse is

$$\exists f \exists c : \text{ContinuousAt}(f, c) \wedge \neg \text{DerivableAt}(f, c).$$

The examples below escalate the failure: one corner, infinitely many isolated corners, and no differentiability anywhere.

*Example* (One Corner:  $|x|$ ). The function  $f(x) = |x|$  is continuous on  $\mathbb{R}$ , but it is not differentiable at 0. Its one-sided derivatives are  $-1$  and  $+1$ , so the two one-sided slopes do not agree. Thus a single corner is enough to separate continuity from differentiability.

*Example* (Finitely Many Corners: A Sawtooth). On a compact interval, a triangular sawtooth made from finitely many line segments can be continuous while failing to be differentiable at each corner. For example,

$$s(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2 - x, & 1 \leq x \leq 2, \end{cases}$$

is continuous on  $[0, 2]$ , but the left derivative at 1 is 1 and the right derivative is  $-1$ . Adding more teeth gives any finite number of corner failures without breaking continuity.

*Example* (The Weierstrass Function). For  $0 < a < 1$  and an odd integer  $b$  satisfying  $ab > 1 + \frac{3}{2}\pi$ , the Weierstrass function

$$W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

is continuous on  $\mathbb{R}$ , because the series converges uniformly by the Weierstrass  $M$ -test. Classically, under the displayed condition, it is differentiable at no point. Thus continuity may hold everywhere while differentiability holds nowhere.

*Remark* (Proof status). The continuity of  $W$  follows from uniform convergence once the function-series machinery is available. The nowhere-differentiability proof is a classical result of Weierstrass and is deferred to the function-series development; it is used here only as a counterexample to the converse of Differentiable Implies Continuous.

*Remark* (Forward flag). Brownian motion has sample paths that are almost surely continuous everywhere and differentiable nowhere. This is the stochastic analogue of the Weierstrass phenomenon and one reason ordinary calculus must be replaced by the Itô integral in stochastic analysis.

### Theorem (Uniqueness of the Derivative)

**Theorem 11.2.7** (Uniqueness of the Derivative). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$ . If  $f$  is differentiable at  $c$ , its derivative is unique.*

[Go to proof.](#) [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L') \Rightarrow L = L'.$$

*Remark* (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L') \Longrightarrow L = L'.$$

*Remark* (Failure modes). The uniqueness proof assumes  $L \neq L'$  and takes  $\varepsilon = |L - L'|/2 > 0$ . The definition supplies  $\delta$  making the difference quotient within  $\varepsilon$  of both  $L$  and  $L'$  simultaneously, forcing  $|L - L'| < 2\varepsilon = |L - L'|$ , a contradiction. The cluster-point hypothesis on  $c$  is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

*Remark* (Contrapositive quantified statement).

$$L \neq L' \Rightarrow \neg(\text{DerivativeAt}(f, c, L) \wedge \text{DerivativeAt}(f, c, L')).$$

Two distinct candidates cannot both be the derivative of  $f$  at  $c$ .

*Remark* (Contrapositive predicate reading).

$$L \neq L' \implies \neg \text{DerivativeAt}(f, c, L) \vee \neg \text{DerivativeAt}(f, c, L').$$

*Remark* (Interpretation). Uniqueness licenses writing  $f'(c)$  as a definite value rather than a set of candidates.

*Remark* (Dependencies).

- [Derivative at a Point](#)

#### Proposition (Derivative of the Identity)

**Proposition 11.2.8** (Derivative of the Identity). *Let  $f(x) = x$  for all  $x \in \mathbb{R}$ . Then  $f'(c) = 1$  for every  $c \in \mathbb{R}$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall c \in \mathbb{R} : \frac{d}{dx}[x] \Big|_{x=c} = 1.$$

*Remark* (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

*Remark* (Dependencies).

- [Derivative at a Point](#)
- [Function](#)

#### Proposition (Derivative of a Constant)

**Proposition 11.2.9** (Derivative of a Constant). *Let  $k \in \mathbb{R}$  and let  $f(x) = k$  for all  $x \in \mathbb{R}$ . Then  $f'(c) = 0$  for every  $c \in \mathbb{R}$ .*

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall k \in \mathbb{R} \forall c \in \mathbb{R} : \frac{d}{dx}[k] \Big|_{x=c} = 0.$$

*Remark* (Interpretation). A constant function has zero change across every secant line, so its instantaneous rate of change is zero everywhere. Together with the derivative of the identity, this is one of the base cases for the algebraic computation rules.

*Remark* (Dependencies).

- [Derivative at a Point](#)

## 11.3 One-Sided Derivatives

### One-Sided Derivatives — Quick Reference

| Concept               | Meaning                                                |
|-----------------------|--------------------------------------------------------|
| Left-hand derivative  | Limiting slope approached from the left                |
| Right-hand derivative | Limiting slope approached from the right               |
| Two-sided criterion   | Differentiability iff both one-sided derivatives agree |
| Example: $ x $ at 0   | A continuous corner with unequal one-sided slopes      |

*Remark* (Section guide). A corner is a place where a curve has a left answer and a right answer and refuses to choose. One-sided derivatives interrogate each side alone, and their agreement criterion is the fastest way to certify many failures of differentiability.

### Definition (Left-Hand Derivative)

**Definition 11.3.1** (Left-Hand Derivative). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be such that  $c$  is a left-hand limit point of  $I$ . We say that  $f$  has a *left-hand derivative* at  $c$  if the limit

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted  $f'_-(c)$ :

$$f'_-(c) := \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}.$$

*Remark* (Standard quantified statement).

$$\exists L \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I \left( -(\delta) < x - c < 0 \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case  $f'_-(c) = L$ .

*Remark* (Definition predicate reading).

$$\text{LeftDerivativeAt}(f, c, L) \iff \text{LeftLimit} \left( \lambda x. \frac{f(x) - f(c)}{x - c}, c, L \right).$$

*Remark* (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in I \left( -\delta < x - c < 0 \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

*Remark* (Negation predicate reading).

$$\forall L \in \mathbb{R} \neg \text{LeftDerivativeAt}(f, c, L).$$

No finite  $L$  is the left-hand limit of the difference quotient.

*Remark* (Interpretation). The left-hand derivative  $f'_-(c)$  is the limit of the difference quotient as  $x$  approaches  $c$  *strictly from the left*. It measures the instantaneous rate of change of  $f$  at  $c$  as seen from the left side. The right-hand derivative  $f'_+(c)$  is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative  $f'(c)$  exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for  $f(x) = |x|$  at  $c = 0$ ,  $f'_-(0) = -1$  and  $f'_+(0) = +1$ ; since these disagree,  $f'(0)$  does not exist.

*Remark* (Dependencies).

- **Definition: Derivative at a Point**

#### Definition (Right-Hand Derivative)

**Definition 11.3.2** (Right-Hand Derivative). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be such that  $c$  is a right-hand limit point of  $I$ . We say that  $f$  has a *right-hand derivative* at  $c$  if the limit

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted  $f'_+(c)$ :

$$f'_+(c) := \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}.$$

*Remark* (Standard quantified statement).

$$\exists L \in \mathbb{R} \forall \varepsilon > 0 \exists \delta > 0 \forall x \in I \left( 0 < x - c < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

In this case  $f'_+(c) = L$ .

*Remark* (Definition predicate reading).

$$\text{RightDerivativeAt}(f, c, L) \iff \text{RightLimit} \left( \lambda x. \frac{f(x) - f(c)}{x - c}, c, L \right).$$

*Remark* (Negated quantified statement).

$$\forall L \in \mathbb{R} \exists \varepsilon > 0 \forall \delta > 0 \exists x \in I \left( 0 < x - c < \delta \wedge \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \right).$$

*Remark* (Negation predicate reading).

$$\forall L \in \mathbb{R} \neg \text{RightDerivativeAt}(f, c, L).$$

*Remark* (Interpretation). The right-hand derivative  $f'_+(c)$  is the instantaneous rate of change seen from inputs larger than  $c$ . It is the endpoint derivative available at the left endpoint of a closed interval, and it is one half of the two-sided derivative test at interior points.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

### Theorem

**Theorem 11.3.3** (Differentiability and One-Sided Derivatives). *Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f : I \rightarrow \mathbb{R}$ . Then  $f$  is differentiable at  $c \in I$  if and only if both one-sided derivatives  $f'_-(c)$  and  $f'_+(c)$  exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

*Go to proof.*

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \left( \text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L) \right) \text{ for some } L \in \mathbb{R}.$$

Remark (Negated quantified statement).

$$\neg \text{DifferentiableAt}(f, c) \iff \neg \left( \text{LeftDerivativeAt}(f, c, L) \wedge \text{RightDerivativeAt}(f, c, L) \right) \text{ for all } L \in \mathbb{R}.$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show  $f$  is not differentiable at  $c$ , compute  $f'_-(c)$  and  $f'_+(c)$  and observe they disagree. For  $f(x) = |x|$  at  $c = 0$ :  $f'_-(0) = -1 \neq +1 = f'_+(0)$ , so  $f$  is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval  $[a, b]$ : differentiable on the open interior  $(a, b)$  in the full sense, plus  $f'_+(a)$  and  $f'_-(b)$  both exist at the endpoints.

Remark (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

Example ( $|x|$  Has No Derivative at 0). For  $f(x) = |x|$  at  $c = 0$ , the one-sided difference quotients are

$$\frac{|x| - 0}{x - 0} = \frac{|x|}{x} = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Thus  $f'_-(0) = -1$  and  $f'_+(0) = 1$ . The two values disagree, so the two-sided derivative does not exist at 0, even though  $f$  is continuous there.

## 11.4 Carathéodory and the Chain Rule

### Carathéodory and Chain Rule — Quick Reference

| Concept                       | Role                                                        |
|-------------------------------|-------------------------------------------------------------|
| Carathéodory factorization    | Continuous slope factor for differentiability               |
| Carathéodory characterization | Differentiability iff the slope factor extends continuously |
| Chain Rule                    | Derivative of a composition                                 |
| Leibniz Rule                  | $n$ -th derivative of a product                             |
| Faa di Bruno                  | $n$ -th derivative of a composition                         |
| Faa di Bruno at $n = 2$       | Coefficient check for the second derivative                 |
| Faa di Bruno at $n = 3$       | Coefficient check for the third derivative                  |

### 11.4.1 Carathéodory Characterization

*Remark* (Section guide). Carathéodory replaces division by factorisation:  $f(x) - f(c) = (x - c)\varphi(x)$ , with  $\varphi$  continuous at  $c$ . Differentiability becomes continuity of the slope factor, and continuity composes. That is why the chain rule becomes clean rather than a fight with vanishing denominators.

#### Definition (Carathéodory Factorization)

**Definition 11.4.1** (Carathéodory Factorization). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . We say that  $f$  admits a *Carathéodory factorization* at  $c$  if there exists a function  $\phi : I \rightarrow \mathbb{R}$ , continuous at  $c$ , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case,  $f$  is differentiable at  $c$  and  $f'(c) = \phi(c)$ .

*Remark* (Standard quantified statement).

$$\exists \phi : I \rightarrow \mathbb{R} \left( \text{ContinuousAt}(\phi, c) \wedge \forall x \in I [f(x) = f(c) + (x - c)\phi(x)] \right).$$

*Remark* (Definition predicate reading). The existence of a Carathéodory factorization at  $c$  is equivalent to  $\text{DifferentiableAt}(f, c)$  by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \frac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of  $f$  at  $c$  is precisely the statement that this piecewise-defined  $\phi$  is continuous at  $c$ .

*Remark* (Interpretation). The Carathéodory factorization replaces the difference quotient — a function with a removable discontinuity at  $c$  — with a function  $\phi$  that is genuinely continuous at  $c$ . The factorization  $f(x) - f(c) = (x - c)\phi(x)$  holds everywhere on  $I$ , including at

$x = c$  where both sides equal zero. The derivative is recovered as  $\phi(c)$  without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

**Carathéodory vs. Lipschitz.** Both control the increment  $f(x) - f(c)$  by the factor  $(x - c)$ , but differently. The Carathéodory factorization requires the slope function  $\phi(x)$  to extend *continuously* to  $c$ ; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition  $|f(x) - f(c)| \leq L|x - c|$  only bounds the size of the increment. Carathéodory implies Lipschitz: if  $\phi$  is continuous at  $c$  it is bounded near  $c$ , so  $|f(x) - f(c)| = |\phi(x)||x - c| \leq K|x - c|$  for some  $K$  near  $c$ . The converse fails:  $f(x) = |x|$  is Lipschitz on  $\mathbb{R}$  with constant 1 but the slope function takes values  $\pm 1$  on either side of 0 with no continuous extension.

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Definition: Continuous at a Point](#)

#### Theorem

**Theorem 11.4.2** (Carathéodory Characterization of Differentiability). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . The following are equivalent:*

- (i)  $f$  is differentiable at  $c$ .
- (ii) There exists a function  $\phi : I \rightarrow \mathbb{R}$ , continuous at  $c$ , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

*In this case,  $\phi(c) = f'(c)$ . [Go to proof.](#) [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\begin{aligned} \text{DifferentiableAt}(f, c) &\iff \exists \phi : I \rightarrow \mathbb{R} \left( \text{ContinuousAt}(\phi, c) \right. \\ &\quad \left. \wedge \forall x \in I [f(x) = f(c) + (x - c)\phi(x)] \right). \end{aligned}$$

In this case  $\phi(c) = f'(c)$ .

*Remark* (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define  $\phi$  as the difference quotient extended by  $f'(c)$  at  $c$  and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from  $\phi$  and read off its limit as  $\phi(c)$ . The theorem's power is that it converts every future differentiability argument into a continuity argument: to show  $g \circ f$  is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of  $f$  and  $g$  yields one automatically.

*Remark* (Dependencies).



- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

#### Theorem

**Theorem 11.4.3** (Chain Rule). *Let  $I, J \subseteq \mathbb{R}$  be intervals, let  $f : I \rightarrow \mathbb{R}$  and  $g : J \rightarrow \mathbb{R}$ , and suppose that  $f(I) \subseteq J$ . Let  $c \in I$ . If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then the composition  $g \circ f$  is differentiable at  $c$ , and*

$$(g \circ f)'(c) = g'(f(c)) f'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, f(c)) \implies (g \circ f)'(c) = g'(f(c)) f'(c).$$

*Remark* (Interpretation). The derivative of a composition is the derivative of the outside function evaluated at the inside value, multiplied by the derivative of the inside function.

*Remark* (Interpretation). Let  $\phi_f$  and  $\phi_g$  be the Carathéodory slope factors for  $f$  at  $c$  and  $g$  at  $f(c)$ . Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x))(f(x) - f(c)) = \phi_g(f(x))\phi_f(x)(x - c).$$

The new slope factor is

$$\phi_{g \circ f}(x) = \phi_g(f(x))\phi_f(x),$$

which is continuous at  $c$  and has value  $g'(f(c))f'(c)$ . This proof never divides by  $f(x) - f(c)$ , so it avoids the failure in the informal  $\Delta g / \Delta f \cdot \Delta f / \Delta x$  argument when  $f(x) = f(c)$  near  $c$ .

*Remark* (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

#### Theorem (Leibniz Rule)

**Theorem 11.4.4** (Leibniz Rule). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $x \in I$ , and suppose that  $f, g : I \rightarrow \mathbb{R}$  are  $n$ -times differentiable on a neighbourhood of  $x$ . Then  $fg$  is  $n$ -times differentiable at  $x$ , and*

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{Diff}^n(f, x) \wedge \text{Diff}^n(g, x) \implies (fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

*Remark* (Interpretation). The binomial pattern is not an accident. Each of the  $n$  differentiations lands on either  $f$  or  $g$ , and  $\binom{n}{k}$  counts the ways to send exactly  $k$  differentiations to  $f$ . At  $n = 1$ , this is the ordinary product rule.

*Remark* (Dependencies).

- [Theorem: Product Rule](#)
- [Definition: Higher Derivatives](#)

#### Theorem (Faa di Bruno's Formula)

**Theorem 11.4.5** (Faa di Bruno's Formula). *Let  $I, J \subseteq \mathbb{R}$  be intervals, let  $x \in I$ , suppose  $g : I \rightarrow J$  is  $n$ -times differentiable on a neighbourhood of  $x$ , and suppose  $f : J \rightarrow \mathbb{R}$  is  $n$ -times differentiable on a neighbourhood of  $g(x)$ . Then  $f \circ g$  is  $n$ -times differentiable at  $x$ , and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^n m k_m = n} \frac{n!}{\prod_{m=1}^n k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^n (g^{(m)}(x))^{k_m},$$

where  $k_m \geq 0$  and  $k = \sum_{m=1}^n k_m$ . [Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{Diff}^n(g, x) \wedge \text{Diff}^n(f, g(x)) \implies (f \circ g)^{(n)}(x) = \sum_{\sum m k_m = n} \frac{n!}{\prod_m k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_m (g^{(m)}(x))^{k_m}.$$

*Remark* (Interpretation). The chain rule gives the first derivative of a composition. Repeated differentiation is deeper: product-rule terms and chain-rule terms interlace, and the book-keeping is governed by integer partitions. Faa di Bruno's formula is the all-orders chain rule.

*Remark* (Hypotheses). Smoothness is more than this formula needs. The  $n$ -th order statement requires  $n$  derivatives of  $g$  near  $x$  and  $n$  derivatives of  $f$  near  $g(x)$ . At  $n = 1$ , the formula is exactly the chain rule.

*Remark* (Structural reading). The tuple  $(k_1, \dots, k_n)$  records an integer partition of  $n$ :  $k_m$  is the number of parts of size  $m$ . The coefficient

$$\frac{n!}{\prod_m k_m! (m!)^{k_m}}$$

counts set partitions with that block-size profile. The outer derivative  $f^{(k)}$  is differentiated once per block, while each block of size  $m$  contributes one factor  $g^{(m)}$ .

*Remark* (Normalization). The formula uses the bare-product normalization: the coefficient contains  $(m!)^{k_m}$ , and the product contains  $(g^{(m)}(x))^{k_m}$ . Equivalently, one may put  $g^{(m)}(x)/m!$  inside the product and remove  $(m!)^{k_m}$  from the denominator. Mixing these two conventions gives the wrong coefficients.

*Remark* (Cross-volume callback). The constraint  $\sum_m mk_m = n$  is precisely the multiplicity form of an integer partition of  $n$ . Thus the number of summands is  $p(n)$ , the partition function from the discrete and number-system material, now appearing as bookkeeping for iterated differentiation.

*Remark* (Scope). This theorem is enrichment, not a critical dependency for the integration, measure, or manifold arcs. It is included because it is the complete all-orders chain rule and a useful callback to partitions.

*Remark* (Dependencies).

- [Theorem: Chain Rule](#)
- [Theorem: Leibniz Rule](#)
- [Definition: Higher Derivatives](#)
- [Definition: Class  \$C^k\$](#)

*Example* (Faa di Bruno at  $n = 2$ ). The partitions of 2 are represented by  $(k_1, k_2) = (2, 0)$  and  $(0, 1)$ . The first gives  $f''(g)(g')^2$ , and the second gives  $f'(g)g''$ . Hence

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g''.$$

The coefficient on  $f'(g)g''$  is 1, which is the quickest check that the normalization has not been mixed.

*Example* (Faa di Bruno at  $n = 3$ ). The partitions of 3 produce the three terms

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The middle coefficient 3 counts the three ways to choose which differentiation contributes the second-derivative block of  $g$ .

## 11.5 The Shape Questions

### Shape Questions — Quick Reference

| Concept               | Meaning                            |
|-----------------------|------------------------------------|
| Increasing at a point | Local right-up, left-down behavior |
| Decreasing at a point | Local right-down, left-up behavior |
| Relative minimum      | Locally smallest function value    |
| Relative maximum      | Locally largest function value     |
| Relative extremum     | Relative minimum or maximum        |
| Convex function       | Graph lies below its chords        |
| Concave function      | Graph lies above its chords        |
| Inflection point      | Local change in bending behavior   |

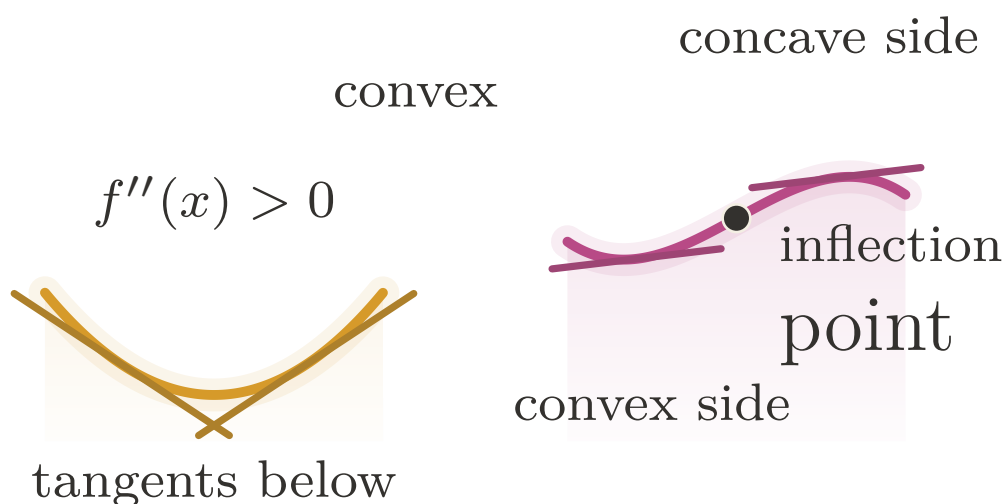


Figure 11.2: Convexity, concavity, and the change of bending at an inflection point.

### Definition (Convex Function)

**Definition 11.5.1** (Convex Function). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *convex* on  $I$  if for all  $x, y \in I$  and all  $\lambda \in [0, 1]$ ,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

*Remark* (Standard quantified statement).

$$\forall x, y \in I \quad \forall \lambda \in [0, 1] : \quad f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

*Remark* (Interpretation). Convexity is defined by chords, not derivatives. The graph lies below the line segment joining any two of its points. Derivative tests are later tools for detecting this geometric condition, not the definition of the condition.

Remark (Dependencies).

- [Closed Interval](#)

#### Definition (Concave Function)

**Definition 11.5.2** (Concave Function). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *concave* on  $I$  if  $-f$  is convex on  $I$ .

Remark (Standard quantified statement).

$$\text{ConcaveOn}(f, I) \iff \text{ConvexOn}(-f, I).$$

Remark (Interpretation). Concavity is the downward-bending counterpart of convexity. Equivalently, the graph of a concave function lies above each chord joining two of its points.

Remark (Dependencies).

- [Convex Function](#)

#### Definition (Inflection Point)

**Definition 11.5.3** (Inflection Point). Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$  be an interior point. The point  $c$  is an *inflection point* of  $f$  if  $f$  changes convexity at  $c$ : there is a neighbourhood  $(c - \delta, c + \delta) \subseteq I$  such that  $f$  is convex on one side of  $c$  and concave on the other side.

Remark (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \iff f \text{ changes convexity at } c.$$

Remark (Interpretation). An inflection point is a change in the direction of bending. It is not merely a point where  $f''(c) = 0$ ; that equation is only a necessary condition under additional hypotheses.

Remark (Dependencies).

- [Convex Function](#)
- [Concave Function](#)

## 11.5.1 Interior Extrema

### Interior Extrema — Quick Reference

| Statement                                 | Role                                                 |
|-------------------------------------------|------------------------------------------------------|
| <a href="#">Relative minimum</a>          | Local lower comparison point                         |
| <a href="#">Relative maximum</a>          | Local upper comparison point                         |
| <a href="#">Relative extremum</a>         | Umbrella term for local optima                       |
| <a href="#">Interior Extremum Theorem</a> | Interior differentiable extremum has zero derivative |
| <a href="#">Necessary condition</a>       | Extrema occur at critical behavior                   |
| <a href="#">Critical-point corollary</a>  | Derivative zero or derivative undefined              |

### Definition (Relative Minimum)

**Definition 11.5.4** (Relative Minimum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative minimum* at  $c$  if there exists  $\delta > 0$  such that

$$f(c) \leq f(x) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(c) < f(x) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then  $f$  has a *strict relative minimum* at  $c$ .

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \forall x \in A (|x - c| < \delta \implies f(c) \leq f(x)).$$

*Remark* (Definition predicate reading).

$$\text{LocalMinimum}(f, c) \iff \exists \delta > 0 \forall x \in A (\text{Within}(x, c, \delta) \implies f(c) \leq f(x)).$$

*Remark* (Interpretation).  $f$  has a relative minimum at  $c$  if  $f(c)$  is the smallest value of  $f$  in some open neighbourhood of  $c$  within the domain. The word *relative* (or *local*) signals that we are comparing  $f(c)$  only to nearby values, not to all values of  $f$  globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion — relative maximum — reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

*Remark* (Dependencies).

- [Function](#)
- [Subset](#)
- [Open Interval](#)
- [Order on  \$\mathbb{R}\$](#)

### Definition (Relative Maximum)

**Definition 11.5.5** (Relative Maximum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative maximum* at  $c$  if there exists  $\delta > 0$  such that

$$f(x) \leq f(c) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then  $f$  has a *strict relative maximum* at  $c$ .

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \forall x \in A (|x - c| < \delta \implies f(x) \leq f(c)).$$

*Remark* (Definition predicate reading).

$$\text{LocalMaximum}(f, c) \iff \exists \delta > 0 \forall x \in A (\text{Within}(x, c, \delta) \implies f(x) \leq f(c)).$$

*Remark* (Interpretation). A relative maximum is a point whose function value is at least as large as all nearby function values in the domain.

*Remark* (Dependencies).

- [Definition: Relative Minimum](#)

#### Definition (Relative Extremum)

**Definition 11.5.6** (Relative Extremum). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  has a *relative extremum* at  $c$  if  $f$  has either a relative minimum at  $c$  or a relative maximum at  $c$ .

*Remark* (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \vee \text{LocalMaximum}(f, c).$$

*Remark* (Definition predicate reading).

$$\text{LocalExtremum}(f, c) \iff \text{LocalMinimum}(f, c) \vee \text{LocalMaximum}(f, c).$$

*Remark* (Interpretation). Relative extrema are the local turning candidates of a graph: nearby values lie all on one side of  $f(c)$ .

*Remark* (Dependencies).

- [Definition: Relative Minimum](#)
- [Definition: Relative Maximum](#)

#### Theorem

**Theorem 11.5.7** (Interior Extremum Theorem). Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$  and  $f$  is differentiable at  $c$ , then

$$f'(c) = 0.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \wedge \text{DifferentiableAt}(f, c) \implies \text{DerivativeAt}(f, c, 0).$$

*Remark* (Contrapositive quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L \neq 0 \Rightarrow \neg \text{LocalExtremum}(f, c).$$

If  $f'(c)$  exists and is nonzero, then  $c$  is not a relative extremum.

*Remark* (Contrapositive predicate reading).

$$\text{DerivativeAt}(f, c, L) \wedge L \neq 0 \Rightarrow \neg \text{LocalMinimum}(f, c) \wedge \neg \text{LocalMaximum}(f, c).$$

*Remark* (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate — at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for  $x > c$  and non-negative for  $x < c$ ; since the derivative is a single limit, it is squeezed to zero. The converse fails:  $f'(c) = 0$  does not imply an extremum ( $f(x) = x^3$  at  $c = 0$ ). Critical points — where  $f'(c) = 0$  or  $f'(c)$  does not exist — are candidates for extrema, not guarantees.

*Remark* (Dependencies).

- [Definition: Relative Extremum](#)
- [Definition: Derivative at a Point](#)
- [Definition: Left-Hand Derivative](#)
- [Definition: Right-Hand Derivative](#)
- [Theorem: Differentiability and One-Sided Derivatives](#)

#### Theorem

**Theorem 11.5.8** (Necessary Condition for an Interior Extremum). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then either  $f'(c)$  does not exist, or*

$$f'(c) = 0.$$

*[Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \Rightarrow \neg \text{DifferentiableAt}(f, c) \vee \text{DerivativeAt}(f, c, 0).$$

*Remark* (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming  $f'(c)$  exists, the conclusion is a disjunction: either the derivative fails to exist at  $c$ , or it exists and equals zero. This is the correct form for identifying *critical points* — points that are candidates for extrema. The set of critical points is the union of points where  $f'$  is zero and points where  $f'$  does not exist.

*Remark* (Dependencies).



- [Theorem: Interior Extremum Theorem](#)

#### Corollary

**Corollary 11.5.9** (Necessary Condition for an Extremum). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \Rightarrow \text{DerivativeAt}(f, c, 0) \vee \neg \text{DifferentiableAt}(f, c).$$

*Remark* (Interpretation). An extremum can occur only at a critical point: either the derivative vanishes or the derivative fails to exist.

*Remark* (Dependencies).

- [Theorem: Necessary Condition for an Interior Extremum](#)

## 11.6 The Mean Value Theorems

#### Mean Value Theorems — Quick Reference

| Statement                                          | Role                                             |
|----------------------------------------------------|--------------------------------------------------|
| <a href="#">Rolle's Theorem</a>                    | Equal endpoint values force a horizontal tangent |
| <a href="#">Mean Value Theorem</a>                 | Some tangent slope equals the secant slope       |
| <a href="#">Cauchy Mean Value Theorem</a>          | Compares two functions' rates of change          |
| <a href="#">Derivative bound implies Lipschitz</a> | Turns a slope bound into a global modulus        |

*Remark* (Section guide). Rolle's theorem says that a differentiable curve returning to the same height must level off somewhere between. The Mean Value Theorem tilts that picture: between any two points, some interior tangent runs parallel to the chord. Cauchy's version compares two functions at once and later powers L'Hôpital's rule.

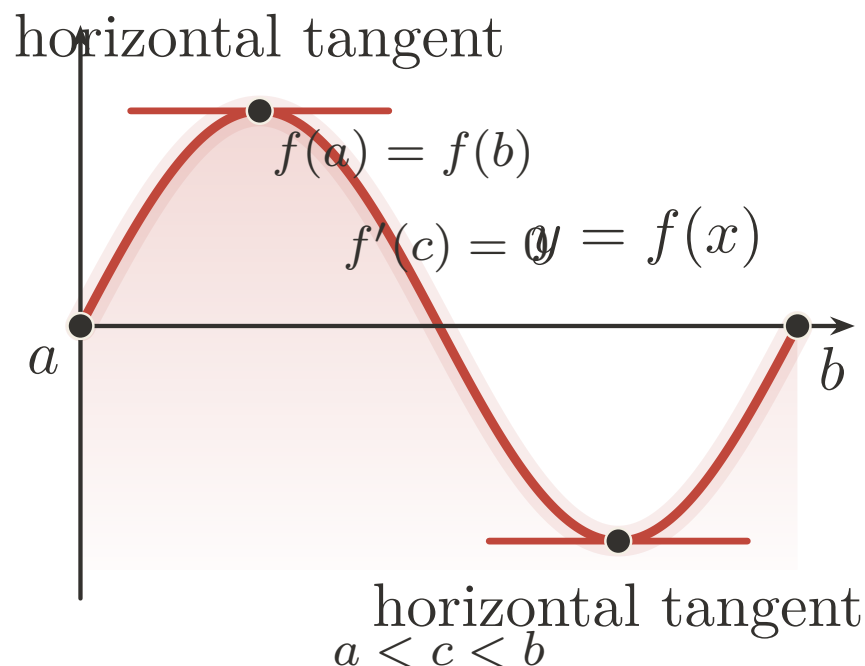


Figure 11.3: Rolle's Theorem: equal endpoint heights force a horizontal tangent.

#### Theorem (Rolle's Theorem)

**Theorem 11.6.1** (Rolle's Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and satisfies  $f(a) = f(b)$ . Then there exists  $c \in (a, b)$  such that*

$$f'(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \forall x \in (a, b) \text{ DifferentiableAt}(f, x) \wedge f(a) = f(b) \\ & \Rightarrow \exists c \in (a, b) \text{ DerivativeAt}(f, c, 0). \end{aligned}$$

*Remark* (Predicate reading).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \wedge f(a) = f(b) \\ & \Rightarrow \exists c \in (a, b) : \text{DerivativeAt}(f, c, 0). \end{aligned}$$

*Remark* (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let  $f(x) = \text{sgn}(x - 1/2)$  on  $[0, 1]$ . Then  $f(0) = f(1) = -1$ , but  $f$  has a jump discontinuity at  $x = 1/2$ . The derivative does not exist anywhere, so there is no horizontal tangent.
- **Differentiability fails:** Let  $f(x) = |x - 1/2|$  on  $[0, 1]$ . Then  $f(0) = f(1) = 1/2$ , and  $f$  is continuous, but  $f$  has a corner at  $x = 1/2$  where the derivative is undefined. For all  $c \neq 1/2$ , we have  $f'(c) = \pm 1 \neq 0$ .

- **Equal endpoints fail:** Let  $f(x) = x$  on  $[0, 1]$ . Then  $f$  is continuous and differentiable with  $f'(x) = 1 > 0$  everywhere, but  $f(0) = 0 \neq 1 = f(1)$ .

*Remark* (Contrapositive quantified statement).

$$(\forall c \in (a, b) : f'(c) \neq 0) \Rightarrow f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b) \vee f(a) \neq f(b).$$

*Remark* (Contrapositive predicate reading).

$$\begin{aligned} & (\forall c \in (a, b) : \neg \text{DerivativeAt}(f, c, 0)) \\ & \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)) \vee f(a) \neq f(b). \end{aligned}$$

*Remark* (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients — the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then  $f$  is constant and  $f' \equiv 0$  throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

*Remark* (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

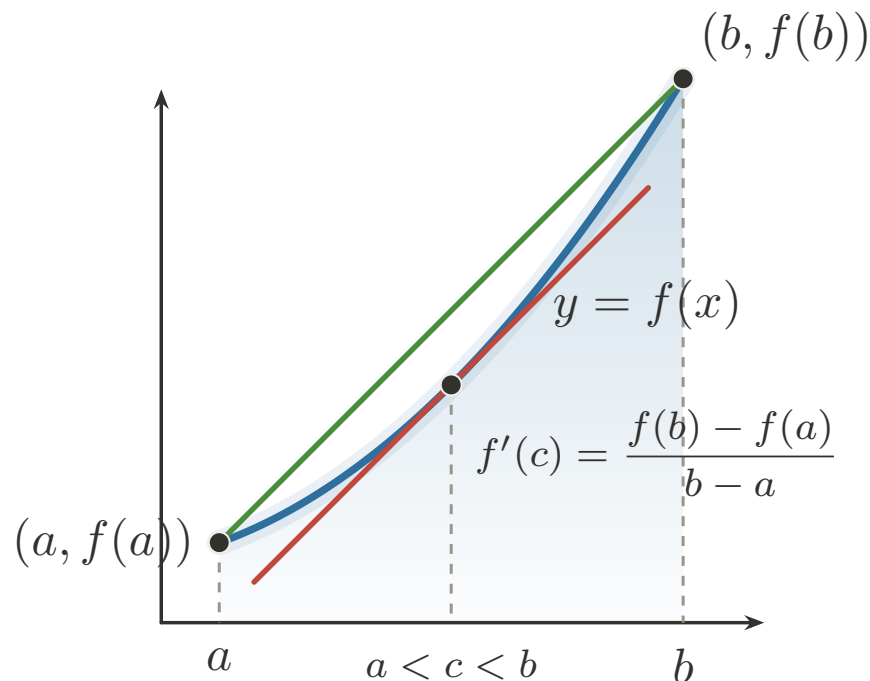


Figure 11.4: Mean Value Theorem: an interior tangent matches the secant slope.

### Theorem (Mean Value Theorem)

**Theorem 11.6.2** (Mean Value Theorem). Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \forall x \in (a, b) \text{ DifferentiableAt}(f, x) \\ & \Rightarrow \exists c \in (a, b) \text{ DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right). \end{aligned}$$

*Remark* (Predicate reading).

$$\begin{aligned} & \text{ContinuousOn}(f, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \\ & \implies \exists c \in (a, b) : \text{DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right). \end{aligned}$$

*Remark* (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let  $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$ . Then  $f$  has a jump discontinuity at  $x = 1/2$ , and the secant slope is  $\frac{f(1)-f(0)}{1-0} = 1$ . But  $f'(x) = 0$  wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let  $f(x) = |x|$  on  $[-1, 1]$ . Then  $f$  is continuous and the secant slope is  $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$ . But  $f'(x) = \pm 1$  everywhere  $f$  is differentiable (i.e., for  $x \neq 0$ ), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some  $c$  exists but gives no formula for it.

*Remark* (Contrapositive quantified statement).

$$\left( \forall c \in (a, b) : f'(c) \neq \frac{f(b) - f(a)}{b - a} \right) \Rightarrow f \text{ not continuous on } [a, b] \vee f \text{ not differentiable on } (a, b).$$

*Remark* (Contrapositive predicate reading).

$$\begin{aligned} & \left( \forall c \in (a, b) : \neg \text{DerivativeAt}\left(f, c, \frac{f(b)-f(a)}{b-a}\right) \right) \\ & \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)). \end{aligned}$$

*Remark* (Interpretation). The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point  $c$  is parallel to the secant line through  $(a, f(a))$  and  $(b, f(b))$ . The proof reduces to Rolle's Theorem by subtracting the secant: define  $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$ , which

satisfies  $g(a) = g(b) = f(a)$ , then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some  $c$  exists but gives no formula for it. Its immediate corollaries — the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary — each flow from the MVT by a one-line argument and are the theorem's primary applications.

*Remark* (Dependencies).

- [Theorem: Rolle's Theorem](#)
- [Definition: Derivative at a Point](#)

### Theorem (Cauchy Mean Value Theorem)

**Theorem 11.6.3** (Cauchy Mean Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f, g : [a, b] \rightarrow \mathbb{R}$  are continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

*If, in addition,  $g'(x) \neq 0$  for every  $x \in (a, b)$ , then  $g(b) \neq g(a)$  and*

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{ContinuousOn}(f, [a, b]) \wedge \text{ContinuousOn}(g, [a, b]) \wedge \text{DifferentiableOn}(f, (a, b)) \wedge \text{DifferentiableOn}(g, (a, b)) \\ \implies \exists c \in (a, b) [(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)].$$

*Remark* (Predicate reading).  $\text{CauchyMVT}(f, g, [a, b])$  asserts the existence of an interior point where the velocity vector  $(g'(c), f'(c))$  is parallel to the chord vector  $(g(b) - g(a), f(b) - f(a))$ . Taking  $g(x) = x$  recovers the ordinary Mean Value Theorem.

*Remark* (Negated quantified statement).

$$\forall c \in (a, b) : (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c).$$

*Remark* (Failure modes). The continuity and differentiability hypotheses fail for the same reasons they fail in the ordinary MVT: jumps break endpoint-to-interior control, and corners leave no derivative to match. The quotient form has the additional nonvanishing hypothesis  $g' \neq 0$  on  $(a, b)$ ; without it the cross-multiplied conclusion may still hold, but the displayed quotient can be undefined.

*Remark* (Contrapositive quantified statement).

$$(\forall c \in (a, b) : (f(b) - f(a))g'(c) \neq (g(b) - g(a))f'(c)) \\ \implies \neg \text{ContinuousOn}(f, [a, b]) \vee \neg \text{ContinuousOn}(g, [a, b]) \vee \neg \text{DifferentiableOn}(f, (a, b)) \vee \neg \text{DifferentiableOn}(g, (a, b))$$

*Remark* (Contrapositive predicate reading). If no interior point makes the derivative pair  $(g'(c), f'(c))$  parallel to the chord pair  $(g(b) - g(a), f(b) - f(a))$ , then at least one of the four regularity hypotheses fails: continuity of  $f$  or  $g$  on  $[a, b]$ , or differentiability of  $f$  or  $g$  on  $(a, b)$ .

*Remark* (Interpretation). This is the parametric Mean Value Theorem. For the planar curve  $t \mapsto (g(t), f(t))$ , some interior tangent is parallel to the chord joining its endpoints. It compares two rates of change rather than comparing one function to the horizontal axis. This is the form that later underlies L'Hopital-type ratio arguments and the Cauchy form of Taylor's remainder.

*Remark* (Dependencies).

- [Theorem: Rolle's Theorem](#)
- [Definition: Derivative at a Point](#)
- [Theorem: Sum Rule](#)
- [Theorem: Constant Multiple Rule](#)

#### Corollary (Derivative Bound Implies Lipschitz)

**Corollary 11.6.4** (Derivative Bound Implies Lipschitz). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If there exists  $M \geq 0$  such that*

$$|f'(x)| \leq M \quad \text{for every } x \in I,$$

*then  $f$  is Lipschitz on  $I$  with constant  $M$ :*

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every } x, y \in I.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$(\exists M \geq 0 \forall x \in I : |f'(x)| \leq M) \implies \forall x, y \in I : |f(x) - f(y)| \leq M|x - y|.$$

*Remark* (Predicate reading).

$$\text{BoundedDerivativeOn}(f, I, M) \implies \text{Lipschitz}(f, I, M).$$

*Remark* (Failure modes). The derivative bound must hold throughout the interval. A function such as  $f(x) = \sqrt{x}$  on  $[0, 1]$  is continuous and uniformly continuous, but its derivative is unbounded near 0, so no global Lipschitz constant follows from this test. The interval hypothesis is also essential: across a disconnected gap, the MVT cannot compare the two components.

*Remark* (Interpretation). This corollary converts a pointwise slope bound into a global modulus of continuity. It is the precise bridge from derivative estimates to the Lipschitz condition: controlling the size of  $f'$  controls every secant slope on the interval.

*Remark* (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Lipschitz Condition
- Definition: Derivative at a Point

## 11.7 Reading the Graph from $f'$ and $f''$

### 11.7.1 First-Order Shape

#### First-Order Shape — Quick Reference

| Statement                     | Role                                          |
|-------------------------------|-----------------------------------------------|
| Zero derivative               | Vanishing derivative forces constant behavior |
| Equal derivatives             | Functions differ by a constant                |
| Nonnegative derivative        | Detects nondecreasing behavior                |
| Nonpositive derivative        | Detects nonincreasing behavior                |
| Positive derivative           | Local increasing behavior                     |
| Negative derivative           | Local decreasing behavior                     |
| First derivative test: $\max$ | Sign change detects a local maximum           |
| First derivative test: $\min$ | Sign change detects a local minimum           |

*Remark* (Section guide). The Mean Value Theorem is now available, so the harvest can begin. Vanishing derivatives force constancy, derivative signs force monotonicity, and second derivatives detect bending when enough regularity is present. Darboux and the smoothness hierarchy show that derivatives carry hidden rigidity even when they are not continuous.

#### Definition (Increasing at a Point)

**Definition 11.7.1** (Increasing at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  is *increasing at  $c$*  if there exists  $\delta > 0$  such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have  $f(x) \leq f(c) \leq f(y)$ .

*Remark* (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \cap (c - \delta, c) \, \forall y \in A \cap (c, c + \delta) \, (f(x) \leq f(c) \leq f(y)).$$

*Remark* (Interpretation).  $f$  is increasing at  $c$  if  $c$  lies between the values of  $f$  on either side in some punctured neighbourhood:  $f$  is smaller to the left and larger to the right (weakly). This is a strictly local notion — it says nothing about  $f$  outside the  $\delta$ -neighbourhood of  $c$ . A function can be increasing at  $c$  without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of  $c$ , distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

**Definition (Decreasing at a Point)**

**Definition 11.7.2** (Decreasing at a Point). Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$ . We say that  $f$  is *decreasing at  $c$*  if there exists  $\delta > 0$  such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have  $f(x) \geq f(c) \geq f(y)$ .

Remark (Standard quantified statement).

$$\exists \delta > 0 \forall x \in A \cap (c - \delta, c) \forall y \in A \cap (c, c + \delta) (f(x) \geq f(c) \geq f(y)).$$

Remark (Interpretation). The function is decreasing at  $c$  when nearby values to the left sit above  $f(c)$  and nearby values to the right sit below it.

Remark (Dependencies).

- Definition: Increasing at a Point

**Theorem (Zero Derivative Implies Constant)**

**Theorem 11.7.3** (Zero Derivative Implies Constant). Let  $I := [a, b] \subseteq \mathbb{R}$  be a closed interval, and let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If

$$f'(x) = 0 \quad \text{for all } x \in (a, b),$$

then  $f$  is constant on  $I$ . *Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a, b) \text{ DerivativeAt}(f, x, 0) \Rightarrow \exists C \in \mathbb{R} \forall x \in I (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg(\exists C \in \mathbb{R} \forall x \in I (f(x) = C)) \Rightarrow \exists x \in (a, b) \neg \text{DerivativeAt}(f, x, 0).$$

If  $f$  is not constant on  $I$ , then  $f'$  is not identically zero on  $(a, b)$ .

Remark (Contrapositive predicate reading).

$$\neg \text{ConstantOn}(f, I) \Rightarrow \exists x \in (a, b) (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if  $f'(x) = 0$  everywhere on  $(a, b)$ , then for any two points  $x_1 < x_2$  in  $I$ , the MVT supplies a  $c$  with  $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$ , so  $f(x_1) = f(x_2)$ . Since this holds for all pairs,  $f$  is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments — if a solution is not identically the zero solution, its derivative is not identically zero somewhere.



Remark (Dependencies).

- Theorem: Mean Value Theorem

#### Corollary

**Corollary 11.7.4** (Equal Derivatives Imply Difference Is Constant). *Let  $I := [a, b] \subseteq \mathbb{R}$ , and let  $f, g : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If*

$$f'(x) = g'(x) \quad \text{for all } x \in (a, b),$$

*then there exists  $C \in \mathbb{R}$  such that  $f(x) = g(x) + C$  for all  $x \in I$ . [Go to proof.](#)*

Remark (Standard quantified statement).

$$\forall x \in (a, b) (f'(x) = g'(x)) \Rightarrow \exists C \in \mathbb{R} \forall x \in I (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R} \exists x \in I (f(x) \neq g(x) + C) \Rightarrow \exists x \in (a, b) (f'(x) \neq g'(x)).$$

If  $f - g$  is not a constant function, then  $f'$  and  $g'$  disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg \text{ConstantOn}(f - g, I) \Rightarrow \exists x \in (a, b) (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line — apply the previous theorem to  $h = f - g$ , whose derivative is  $(f - g)' = f' - g' = 0$ .

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

#### Theorem (Nondecreasing Iff Nonnegative Derivative)

**Theorem 11.7.5** (Nondecreasing Iff Nonnegative Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nondecreasing on  $I$  if and only if*

$$f'(x) \geq 0 \quad \text{for all } x \in I.$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\text{NondecreasingOn}(f, I) \iff \forall x \in I (f'(x) \geq 0).$$

*Remark* (Contrapositive quantified statement).

$$\exists x \in I (f'(x) < 0) \Rightarrow \neg \text{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

*Remark* (Contrapositive predicate reading).

$$\exists x \in I (f'(x) < 0) \Rightarrow \neg \text{NondecreasingOn}(f, I).$$

*Remark* (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction ( $f$  nondecreasing  $\Rightarrow f' \geq 0$ ) follows from the definition: the difference quotient is nonnegative when  $f$  is nondecreasing, so the limit is nonnegative. The reverse direction ( $f' \geq 0 \Rightarrow$  nondecreasing) is a consequence of the MVT: for  $x < y$ , the MVT gives  $f(y) - f(x) = f'(c)(y - x) \geq 0$ .

Note carefully:  $f' \geq 0$  characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires  $f' > 0$  at all but isolated points. The function  $f(x) = x^3$  satisfies  $f'(0) = 0$  but is strictly increasing;  $f' \geq 0$  with equality only at isolated points is sufficient for strict increase, but  $f' > 0$  everywhere is not necessary.

*Remark* (Dependencies).

- [Theorem: Mean Value Theorem](#)
- [Definition: Derivative at a Point](#)

#### Theorem

**Theorem 11.7.6** (Nonincreasing Iff Nonpositive Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nonincreasing on  $I$  if and only if*

$$f'(x) \leq 0 \quad \text{for all } x \in I.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I (f'(x) \leq 0).$$

*Remark* (Contrapositive quantified statement).

$$\exists x \in I (f'(x) > 0) \Rightarrow \neg \text{NonincreasingOn}(f, I).$$

*Remark* (Contrapositive predicate reading).

$$\exists x \in I (f'(x) > 0) \Rightarrow \neg \text{NonincreasingOn}(f, I).$$

*Remark* (Interpretation). Nonpositive derivative everywhere is exactly the infinitesimal signature of a function that never increases.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

#### Lemma

**Lemma 11.7.7** (Positive Derivative Implies Locally Increasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose that  $f$  is differentiable at  $c$ . If  $f'(c) > 0$ , then there exists  $\delta > 0$  such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

*Go to proof.*

Remark (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L > 0 \Rightarrow \exists \delta > 0 \forall x \in I \left( (c - \delta < x < c \Rightarrow f(x) < f(c)) \wedge (c < x < c + \delta \Rightarrow f(x) > f(c)) \right)$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0 \exists x \in I \left( (c - \delta < x < c \wedge f(x) \geq f(c)) \vee (c < x < c + \delta \wedge f(x) \leq f(c)) \right) \Rightarrow f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg \text{LocallyIncreasingAt}(f, c) \Rightarrow f'(c) \leq 0.$$

Remark (Interpretation). If the instantaneous rate of change is strictly positive at  $c$ , the function is genuinely climbing through  $c$ : it is strictly below  $f(c)$  immediately to the left and strictly above immediately to the right. The proof is a direct  $\varepsilon$ - $\delta$  argument: take  $\varepsilon = f'(c)/2 > 0$ ; the definition of the derivative supplies  $\delta$  such that the difference quotient is within  $\varepsilon$  of  $f'(c)$ , forcing it to be positive, which then forces the correct sign on  $f(x) - f(c)$ . This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

#### Lemma

**Lemma 11.7.8** (Negative Derivative Implies Locally Decreasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose that  $f$  is differentiable at  $c$ . If  $f'(c) < 0$ , then there exists  $\delta > 0$  such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DerivativeAt}(f, c, L) \wedge L < 0 \Rightarrow \exists \delta > 0 \forall x \in I \left( (c - \delta < x < c \Rightarrow f(x) > f(c)) \wedge (c < x < c + \delta \Rightarrow f(x) < f(c)) \right)$$

*Remark* (Contrapositive quantified statement).

$$\neg \text{LocallyDecreasingAt}(f, c) \Rightarrow f'(c) \geq 0.$$

*Remark* (Contrapositive predicate reading).

$$\neg \text{LocallyDecreasingAt}(f, c) \Rightarrow f'(c) \geq 0.$$

*Remark* (Interpretation). If the derivative at  $c$  is strictly negative, nearby values move downward as the input passes through  $c$  from left to right.

*Remark* (Dependencies).

- [Lemma: Positive Derivative Implies Locally Increasing](#)

#### Theorem (First Derivative Test: Relative Maximum)

**Theorem 11.7.9** (First Derivative Test: Relative Maximum). *Let  $I := [a, b] \subseteq \mathbb{R}$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$ . Suppose  $f$  is differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  such that  $(c - \delta, c + \delta) \subseteq I$  and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

*then  $f$  has a relative maximum at  $c$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$\begin{aligned} & \exists \delta > 0 \left[ (c - \delta, c + \delta) \subseteq I \wedge \forall x \in (c - \delta, c) (f'(x) \geq 0) \wedge \forall x \in (c, c + \delta) (f'(x) \leq 0) \right] \\ & \Rightarrow \text{LocalMaximum}(f, c). \end{aligned}$$

*Remark* (Interpretation). If the function is nondecreasing immediately to the left of  $c$  and nonincreasing immediately to the right, then  $c$  is a local maximum — the function climbs to  $c$  and then falls away. The proof integrates: by the nondecreasing characterisation applied on  $(c - \delta, c)$ ,  $f(x) \leq f(c)$  for  $x < c$ ; by the nonincreasing characterisation on  $(c, c + \delta)$ ,  $f(x) \leq f(c)$  for  $x > c$ . The sign condition on  $f'$  replaces the Interior Extremum Theorem's converse, which does not hold in general; this is a sufficient condition, not a necessary one. The test is applicable even when  $f'(c) = 0$  or  $f'(c)$  does not exist.

*Remark* (Dependencies).

- [Theorem: Nondecreasing Iff Nonnegative Derivative](#)
- [Theorem: Nonincreasing Iff Nonpositive Derivative](#)
- [Definition: Relative Maximum](#)

## Theorem

**Theorem 11.7.10** (First Derivative Test: Relative Minimum). *Let  $I := [a, b] \subseteq \mathbb{R}$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$ . Suppose  $f$  is differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  such that  $(c - \delta, c + \delta) \subseteq I$  and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

*then  $f$  has a relative minimum at  $c$ . Go to proof.*

*Remark* (Standard quantified statement).

$$\begin{aligned} \exists \delta > 0 \left[ (c - \delta, c + \delta) \subseteq I \wedge \forall x \in (c - \delta, c) (f'(x) \leq 0) \wedge \forall x \in (c, c + \delta) (f'(x) \geq 0) \right] \\ \Rightarrow \text{LocalMinimum}(f, c). \end{aligned}$$

*Remark* (Interpretation). If the derivative changes from nonpositive to nonnegative at  $c$ , the graph falls into  $c$  and then rises away from it, so  $c$  is a local minimum.

*Remark* (Dependencies).

- [Theorem: First Derivative Test: Relative Maximum](#)
- [Definition: Relative Minimum](#)

## 11.7.2 Second-Order Shape

### Second-Order Shape — Quick Reference

| Concept                                | Role                                       |
|----------------------------------------|--------------------------------------------|
| <a href="#">Second derivative</a>      | Derivative of the derivative               |
| <a href="#">Higher derivatives</a>     | Iterated derivatives $f^{(n)}$             |
| <a href="#">Convexity test</a>         | $f'' \geq 0$ detects convexity             |
| <a href="#">Concavity test</a>         | $f'' \leq 0$ detects concavity             |
| <a href="#">Second derivative test</a> | Classifies critical points                 |
| <a href="#">Inflection condition</a>   | Continuous $f''$ changes sign through zero |

## 11.7.3 Higher-Order Derivatives

### Definition (Second Derivative)

**Definition 11.7.11** (Second Derivative). Let  $f$  be differentiable on a neighbourhood of  $c$ . If  $f'$  is differentiable at  $c$ , the *second derivative* of  $f$  at  $c$  is

$$f''(c) := (f')'(c).$$

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f', c, f''(c)) \iff f''(c) = (f')'(c).$$

*Remark* (Interpretation). The first derivative measures instantaneous slope. The second derivative measures how that slope itself changes.

*Remark* (Dependencies).

- [Derivative at a Point](#)

#### Definition (Higher Derivatives)

**Definition 11.7.12** (Higher Derivatives). Define  $f^{(0)} := f$  and  $f^{(1)} := f'$ . For  $n \geq 2$ , if  $f^{(n-1)}$  is differentiable at  $c$ , define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

*Remark* (Standard quantified statement).

$$\forall n \geq 2 : f^{(n)}(c) = (f^{(n-1)})'(c)$$

whenever the derivative on the right exists.

*Remark* (Interpretation). Higher derivatives are obtained by iterating the derivative operator. Their existence is not automatic: each stage requires differentiability of the previous derivative.

*Remark* (Dependencies).

- [Second Derivative](#)

### 11.7.4 Second-Order Shape

#### Theorem (Second-Derivative Convexity Test)

**Theorem 11.7.13** (Second-Derivative Convexity Test). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If

$$f''(x) \geq 0 \quad \text{for all } x \in I,$$

then  $f$  is convex on  $I$ . *Go to proof.*

*Remark* (Standard quantified statement).

$$(\forall x \in I, f''(x) \geq 0) \implies \text{ConvexOn}(f, I).$$

*Remark* (Interpretation). Nonnegative second derivative means the slopes do not decrease. This is the derivative-based signature of convex bending.

*Remark* (Dependencies).

- [Second Derivative](#)
- [Convex Function](#)

### Theorem (Second-Derivative Concavity Test)

**Theorem 11.7.14** (Second-Derivative Concavity Test). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If

$$f''(x) \leq 0 \quad \text{for all } x \in I,$$

then  $f$  is concave on  $I$ . *Go to proof.*

*Remark* (Standard quantified statement).

$$(\forall x \in I, f''(x) \leq 0) \implies \text{ConcaveOn}(f, I).$$

*Remark* (Interpretation). Nonpositive second derivative means the slopes do not increase. This is the derivative-based signature of concave bending.

*Remark* (Interpretation). The second-derivative tests are narrower than the chord definitions of convexity and concavity. The function  $|x|$  is convex, but it is not twice differentiable at 0.

*Remark* (Dependencies).

- [Second Derivative](#)
- [Concave Function](#)

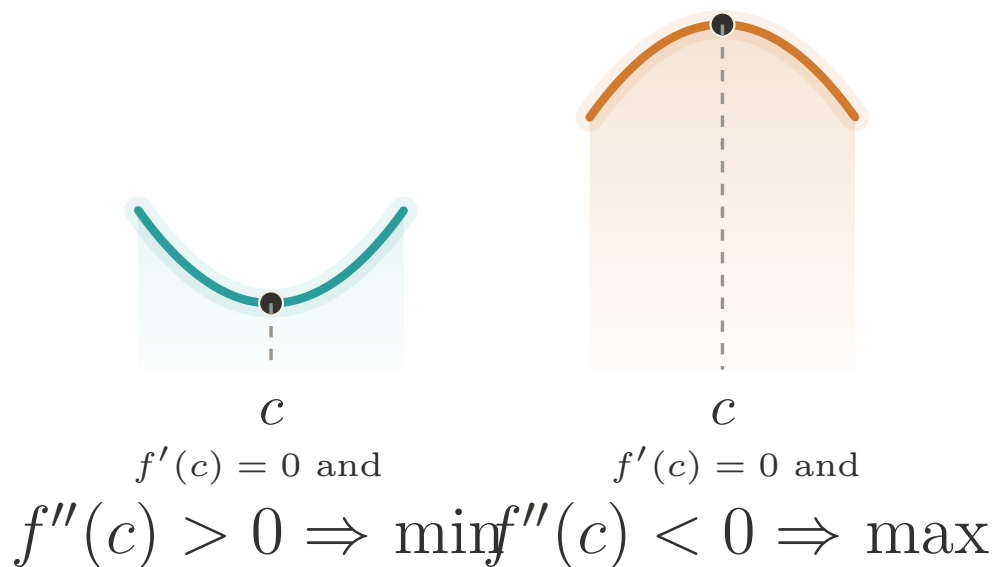


Figure 11.5: Second derivative test: the sign of  $f''(c)$  decides the local shape at a critical point.

### Theorem (Second Derivative Test)

**Theorem 11.7.15** (Second Derivative Test). *Let  $f$  be twice differentiable in a neighbourhood of  $c$ , and suppose  $f'(c) = 0$ .*

- *If  $f''(c) > 0$ , then  $f$  has a strict relative minimum at  $c$ .*
- *If  $f''(c) < 0$ , then  $f$  has a strict relative maximum at  $c$ .*

*Go to proof.*

*Remark* (Standard quantified statement).

$$f'(c) = 0 \wedge f''(c) > 0 \Rightarrow \text{StrictRelativeMinimum}(f, c),$$

and

$$f'(c) = 0 \wedge f''(c) < 0 \Rightarrow \text{StrictRelativeMaximum}(f, c).$$

*Remark* (Interpretation). At a critical point, positive second derivative means the graph bends upward; negative second derivative means it bends downward.

*Remark* (Dependencies).

- [Second Derivative](#)
- [Relative Minimum](#)
- [Relative Maximum](#)

### Proposition (Inflection Point Necessary Condition)

**Proposition 11.7.16** (Inflection Point Necessary Condition). *Let  $f$  have an inflection point at  $c$ . If  $f''$  exists on a neighbourhood of  $c$  and is continuous at  $c$ , then*

$$f''(c) = 0.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{InflectionPoint}(f, c) \wedge \text{ContinuousAt}(f'', c) \implies f''(c) = 0.$$

*Remark* (Interpretation). An inflection point is a change in bending. When the second derivative is continuous, such a change can pass through  $c$  only by passing through zero. The converse can fail:  $f''(c) = 0$  does not by itself guarantee an inflection point.

*Remark* (Dependencies).

- [Inflection Point](#)
- [Second Derivative](#)
- [Continuity at a Point](#)



## 11.7.5 Darboux's Theorem

### Darboux — Quick Reference

| Statement                         | Role                                             |
|-----------------------------------|--------------------------------------------------|
| <a href="#">Darboux's Theorem</a> | Derivatives have the intermediate value property |

### Theorem (Darboux's Theorem)

**Theorem 11.7.17** (Darboux's Theorem). *Let  $I := [a, b] \subseteq \mathbb{R}$  and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If  $k$  is a real number strictly between  $f'(a)$  and  $f'(b)$ , then there exists  $c \in (a, b)$  such that*

$$f'(c) = k.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall k \left( \min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \Rightarrow \exists c \in (a, b) (f'(c) = k) \right).$$

*Remark* (Interpretation). The Intermediate Value Theorem says continuous functions have the intermediate value property. Darboux's theorem says derivatives also have that property even when the derivative is not continuous. Thus a derivative cannot have a jump discontinuity. The proof uses differentiability of  $f$  to construct an auxiliary function  $g(x) = f(x) - kx$  and then applies the Extreme Value Theorem and the Interior Extremum Theorem.

*Remark* (Dependencies).

- [Theorem: Extreme Value Theorem](#)
- [Theorem: Interior Extremum Theorem](#)
- [Definition: Derivative at a Point](#)

## 11.7.6 Smoothness Classes

### Smoothness Classes — Quick Reference

| Concept                           | Meaning                                                          |
|-----------------------------------|------------------------------------------------------------------|
| Class $C^1$                       | Differentiable with continuous first derivative                  |
| Class $C^k$                       | $k$ derivatives with continuous top derivative                   |
| Class $C^\infty$                  | Continuous derivatives of every order                            |
| Class $C^\omega$                  | Equal to local Taylor series                                     |
| Smoothness tower                  | Strict hierarchy $C^\omega \subsetneq C^\infty \subsetneq \dots$ |
| Class $C^{1,1}$                   | $C^1$ with Lipschitz first derivative                            |
| Placement of $C^{1,1}$            | Refines the gap between $C^1$ and $C^2$                          |
| Bounded $f''$                     | Bounded second derivative gives $C^{1,1}$                        |
| Example: differentiable not $C^1$ | Oscillatory derivative discontinuity at 0                        |

### Definition (Continuously Differentiable; Class $C^1$ )

**Definition 11.7.18** (Continuously Differentiable; Class  $C^1$ ). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *continuously differentiable* on  $I$ , written  $f \in C^1(I)$ , if  $f$  is differentiable at every point of  $I$  and the derivative  $f' : I \rightarrow \mathbb{R}$  is continuous on  $I$ .

*Remark* (Standard quantified statement).

$$f \in C^1(I) \iff (\forall x \in I : \text{DifferentiableAt}(f, x)) \wedge \text{ContinuousOn}(f', I).$$

*Remark* (Interpretation). The condition  $f \in C^1(I)$  is strictly stronger than differentiability. A function may be differentiable everywhere while its derivative fails to be continuous; Darboux's Theorem still constrains that failure, because derivatives cannot jump. Thus the gap between differentiability and  $C^1$  is oscillatory rather than jump-like.

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Definition: Continuity at a Point](#)

*Example* (Differentiable Everywhere, but Not  $C^1$ ). Let

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At the origin,

$$f'(0) = \lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \rightarrow 0} x \sin(1/x) = 0$$

by squeezing. For  $x \neq 0$ ,

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

The first term tends to 0, but  $\cos(1/x)$  oscillates, so  $f'$  is not continuous at 0. Thus  $f$  is differentiable on all of  $\mathbb{R}$ , but  $f \notin C^1(\mathbb{R})$ . The failure is oscillatory, not a jump, as Darboux's theorem requires.

### Definition (Class $C^k$ )

**Definition 11.7.19** (Class  $C^k$ ). Let  $I \subseteq \mathbb{R}$  be an interval, let  $k \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is of *class  $C^k$*  on  $I$ , written  $f \in C^k(I)$ , if the derivatives  $f^{(0)}, f^{(1)}, \dots, f^{(k)}$  exist on  $I$  and  $f^{(k)}$  is continuous on  $I$ . By convention,  $C^0(I)$  is the class of continuous functions on  $I$ .

*Remark* (Standard quantified statement).

$$f \in C^k(I) \iff (\forall j \in \{0, 1, \dots, k\} : f^{(j)} \text{ exists on } I) \wedge \text{ContinuousOn}(f^{(k)}, I).$$

*Remark* (Predicate reading).

$$\text{ClassC}(f, I, k).$$

For a fixed interval  $I$ , the classes are nested:

$$f \in C^{k+1}(I) \implies f \in C^k(I).$$

*Remark* (Negated quantified statement).

$$f \notin C^k(I) \iff \exists x \in I : [f^{(k)} \text{ fails to exist at } x] \vee [f^{(k)} \text{ exists on } I \wedge \neg \text{ContinuousAt}(f^{(k)}, x)].$$

*Remark* (Negation predicate reading).  $\neg \text{ClassC}(f, I, k)$ : somewhere the  $k$ -th derivative either does not exist or exists but is not continuous. A single point where the top derivative is discontinuous certifies  $f \notin C^k(I)$ .

*Remark* (Interpretation). The notation  $C^k(I)$  records  $k$  layers of differentiability together with continuity at the top layer. This is the regularity scale used to state how many times a theorem may differentiate a function without leaving the controlled setting.

*Remark* (Dependencies).

- [Definition: Higher Derivatives](#)
- [Definition: Continuously Differentiable; Class  \$C^1\$](#)

#### Definition (Smooth Function; Class $C^\infty$ )

**Definition 11.7.20** (Smooth Function; Class  $C^\infty$ ). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *smooth*, or of *class  $C^\infty$* , written  $f \in C^\infty(I)$ , if  $f \in C^k(I)$  for every  $k \in \mathbb{N}$ . Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

*Remark* (Standard quantified statement).

$$f \in C^\infty(I) \iff \forall k \in \mathbb{N} : f \in C^k(I).$$

*Remark* (Negated quantified statement).

$$f \notin C^\infty(I) \iff \exists k \in \mathbb{N} : f \notin C^k(I).$$

A single order at which differentiability or continuity breaks demotes the function out of  $C^\infty$ .

*Remark* (Interpretation). Smoothness is the regularity language imported by differential geometry. Smooth manifolds are assembled from  $C^\infty$  transition maps, and the tangent-map construction seeded by the differential uses this vocabulary.

*Remark* (Dependencies).

- [Definition: Class  \$C^k\$](#)

### Definition (Real-Analytic Function; Class $C^\omega$ )

**Definition 11.7.21** (Real-Analytic Function; Class  $C^\omega$ ). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is *real-analytic*, or of *class  $C^\omega$* , written  $f \in C^\omega(I)$ , if for every  $a \in I$  there exists  $r > 0$  such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad \text{for every } x \in (a-r, a+r) \cap I.$$

*Remark* (Standard quantified statement).

$$f \in C^\omega(I) \iff \forall a \in I \exists r > 0 \forall x \in (a-r, a+r) \cap I : f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

*Remark* (Interpretation). Analytic regularity means that the Taylor series does more than approximate the function: near each point, the series equals the function. The infinite Taylor series and its convergence are developed in [Taylor's construction and error analysis](#), which supplies the series machinery behind this definition.

*Remark* (Dependencies).

- [Definition: Smooth Function; Class  \$C^\infty\$](#)
- [Definition: Taylor Polynomial at a Point](#)

### Theorem (The Smoothness Tower)

**Theorem 11.7.22** (The Smoothness Tower). *For every nondegenerate interval  $I$ , the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

*hold, and each displayed inclusion is strict. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\forall k \in \mathbb{N} : C^{k+1}(I) \subsetneq C^k(I), \quad C^\omega(I) \subsetneq C^\infty(I).$$

*Remark* (Failure modes). The inclusions do not fail; only their strictness needs witnesses.

- $C^k \setminus C^{k+1}$ :  $\varphi_k(x) = x^k|x|$ , whose  $k$ -th derivative is  $(k+1)!|x|$ , continuous but not differentiable at 0.
- $C^\infty \setminus C^\omega$ : the flat function  $e^{-1/x^2}$ , smooth with all derivatives zero at 0, so its Maclaurin series is zero while the function is not.

These separators prove every displayed containment is a genuine drop.

*Remark* (Interpretation). The inclusions are definitional; strictness is witnessed by examples. On any nondegenerate interval, translated and rescaled copies of  $x \mapsto x^k|x|$  separate  $C^k$  from  $C^{k+1}$ . Translated copies of the flat function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

separate  $C^\infty$  from  $C^\omega$ : all derivatives at the flat point vanish, so the Taylor series is identically zero, while the function is positive on one side of that point.

*Remark* (Dependencies).

- Definition: Class  $C^k$
- Definition: Smooth Function; Class  $C^\infty$
- Definition: Real-Analytic Function; Class  $C^\omega$
- Theorem: Darboux's Theorem

*Remark* (Section guide). Between  $C^1$  and  $C^2$  there is a regularity floor worth naming. A  $C^1$  function has a continuous derivative; a  $C^2$  function has a derivative that is itself differentiable. Many functions used in analysis live in the gap: their first derivative is continuous and controlled, but a classical second derivative may fail to exist at a few points. The control is a Lipschitz bound on  $f'$ , and the class is written  $C^{1,1}$ .

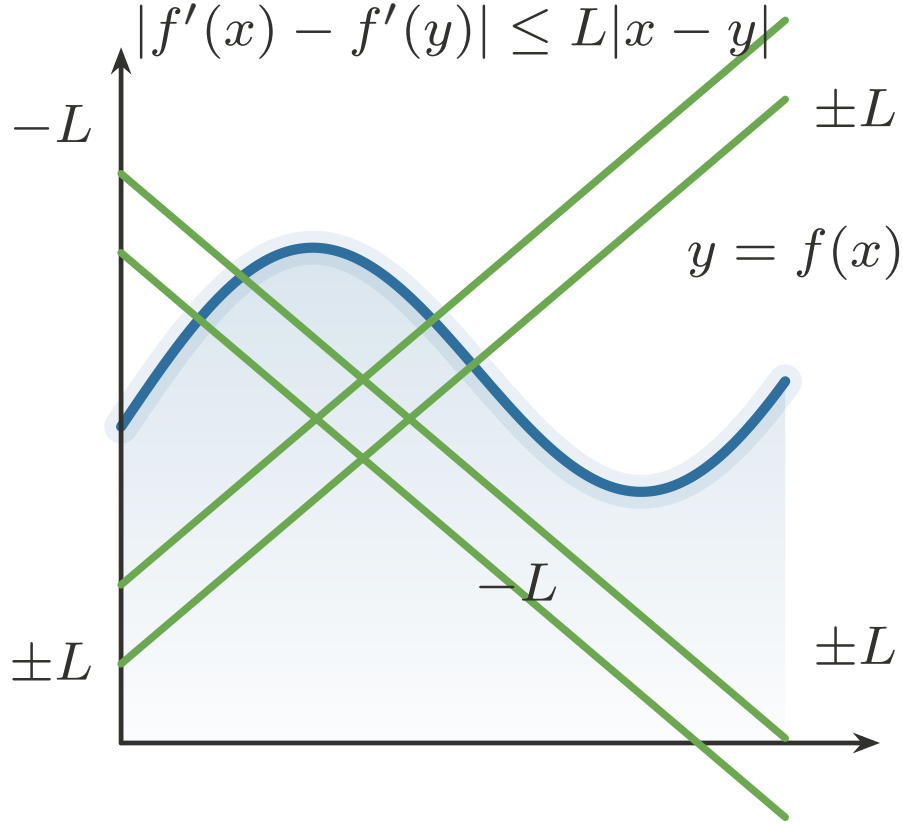


Figure 11.6: A Lipschitz derivative changes at a uniformly bounded rate.

#### Definition (Lipschitz-Derivative Class $C^{1,1}$ )

**Definition 11.7.23** (Lipschitz-Derivative Class  $C^{1,1}$ ). Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$ . The function  $f$  is of class  $C^{1,1}$  on  $I$ , written  $f \in C^{1,1}(I)$ , if  $f \in C^1(I)$  and  $f'$  is Lipschitz on  $I$ : there exists  $L \geq 0$  such that

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in I.$$

*Remark* (Standard quantified statement).

$$f \in C^{1,1}(I) \iff f \in C^1(I) \wedge \exists L \geq 0 \forall x, y \in I : |f'(x) - f'(y)| \leq L|x - y|.$$

*Remark* (Definition predicate reading).

$$\text{ClassC11}(f, I) \iff \text{ClassC}(f, I, 1) \wedge \exists L \geq 0 : \text{Lipschitz}(f', I, L).$$

The smallest admissible  $L$  is the Lipschitz constant of  $f'$ :

$$\sup_{x \neq y} \frac{|f'(x) - f'(y)|}{|x - y|}.$$

*Remark* (Negated quantified statement).

$$f \notin C^{1,1}(I) \iff f \notin C^1(I) \vee (\forall L \geq 0 \exists x, y \in I : |f'(x) - f'(y)| > L|x - y|).$$

*Remark* (Interpretation).  $C^{1,1}$  says that  $f'$  behaves as if  $|f''| \leq L$ , without requiring  $f''$  to exist everywhere. It is the exact regularity tracked by many first-order estimates: the derivative is not merely continuous, but changes at a bounded rate.

*Remark* (Dependencies).

- [Definition: Continuously Differentiable; Class  \$C^1\$](#)
- [Corollary: Derivative Bound Implies Lipschitz](#)

#### Theorem (Placement of $C^{1,1}$ )

**Theorem 11.7.24** (Placement of  $C^{1,1}$ ). *For every nondegenerate interval  $I$ ,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

*and  $C^{1,1}(I)$  is not contained in  $C^2(I)$ . If  $K$  is a compact interval, then*

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

*and the first inclusion is strict. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$(f \in C^{1,1}(I) \Rightarrow f \in C^1(I)), \quad (K \text{ compact} \wedge f \in C^2(K) \Rightarrow f \in C^{1,1}(K)),$$

with the displayed containments strict.

*Remark* (Failure modes). The global inclusion  $C^2(I) \subseteq C^{1,1}(I)$  is false on arbitrary intervals:  $f(x) = x^3$  is  $C^2(\mathbb{R})$ , but  $f'(x) = 3x^2$  is not Lipschitz on  $\mathbb{R}$ . Strictness is witnessed by two elementary functions.

- $C^{1,1} \setminus C^2$ :  $f(x) = x|x|$  has  $f'(x) = 2|x|$ , which is Lipschitz, but  $f''(0)$  does not exist.
- $C^1 \setminus C^{1,1}$ :  $f(x) = |x|^{4/3}$  has continuous first derivative, but  $f'$  has unbounded difference quotients at 0.

*Remark* (Interpretation).  $C^{1,1}$  is a genuine intermediate regularity condition, but it is not the same as  $C^2$ . A bounded second derivative forces the class; a Lipschitz first derivative may have corners; and a continuous first derivative may still vary too sharply to be Lipschitz.

*Remark* (Dependencies).

- [Definition: Lipschitz-Derivative Class  \$C^{1,1}\$](#)
- [Theorem: The Smoothness Tower](#)
- [Corollary: Derivative Bound Implies Lipschitz](#)

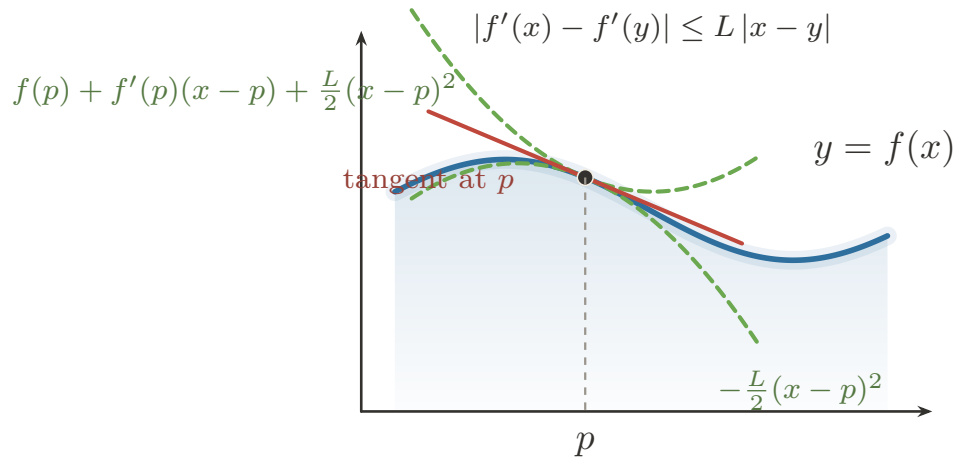


Figure 11.7:  $C^{1,1}$  control gives quadratic upper and lower bounds around a tangent line.

#### Corollary (Bounded Second Derivative Implies $C^{1,1}$ )

**Corollary 11.7.25** (Bounded Second Derivative Implies  $C^{1,1}$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f \in C^2(I)$ . If  $|f''(x)| \leq M$  for all  $x \in I$ , then  $f \in C^{1,1}(I)$ , and  $f'$  is Lipschitz with constant  $M$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$(f \in C^2(I) \wedge \forall x \in I : |f''(x)| \leq M) \Rightarrow f \in C^{1,1}(I) \wedge \text{Lipschitz}(f', I, M).$$

*Remark* (Interpretation). This is [Derivative Bound Implies Lipschitz](#) applied one level up, to  $f'$  in place of  $f$ . The converse requires almost-everywhere language and is therefore deferred.

*Remark* (Dependencies).

- [Corollary: Derivative Bound Implies Lipschitz](#)
- [Definition: Class  \$C^k\$](#)
- [Definition: Lipschitz-Derivative Class  \$C^{1,1}\$](#)

*Remark* (Forward flag). A Lipschitz derivative is differentiable at most points, but making that sentence precise requires measure zero. In one variable this passes through bounded variation; in several variables it becomes Rademacher's theorem. Those results belong to the integration and measure-theoretic development, not to this chapter.



## 11.8 The Algebra of Derivatives

### Algebra of Derivatives — Quick Reference

| Rule                      | Role                                                  |
|---------------------------|-------------------------------------------------------|
| Constant multiple         | Scalars pass through differentiation                  |
| Sum                       | Derivative distributes over addition                  |
| Product                   | Differentiates pointwise products                     |
| Quotient                  | Differentiates quotients where denominator is nonzero |
| Finite sum                | Iterates the sum rule over finitely many terms        |
| Extended product          | Iterates the product rule over finitely many factors  |
| Integer power             | Differentiates powers built from products             |
| Finite linear combination | Packages linearity in finite form                     |
| Inverse Function Theorem  | Produces a $C^1$ local inverse                        |
| Inverse derivative        | Gives the reciprocal derivative formula               |
| L'Hôpital 0/0             | Evaluates indeterminate zero ratios                   |
| L'Hôpital $\infty/\infty$ | Evaluates indeterminate divergent ratios              |

### 11.8.1 From Characterization to Computation

*Remark* (Section guide). With Carathéodory in hand, computing derivatives becomes book-keeping with continuous slope factors. The basic rules record one structural fact: differentiation is a linear operator that also obeys a product law.

*Remark* (Interpretation). Carathéodory's characterization turns differentiability into continuity of a slope factor. If

$$f(x) - f(c) = (x - c)\phi_f(x), \quad g(x) - g(c) = (x - c)\phi_g(x),$$

with  $\phi_f$  and  $\phi_g$  continuous at  $c$ , then the standard algebraic rules for derivatives follow from the algebra of continuous functions. The recurring move is no longer to manipulate a punctured quotient directly, but to construct a continuous factor whose value at  $c$  is the derivative.

*Remark* (Dependencies).

- Carathéodory Characterization of Differentiability
- Algebra of Limits

#### Theorem (Constant Multiple Rule)

**Theorem 11.8.1** (Constant Multiple Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable at  $c$ , then  $\alpha f$  is differentiable at  $c$ , and*

$$(\alpha f)'(c) = \alpha f'(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \text{DerivativeAt}(\alpha f, c, \alpha f'(c)).$$

*Remark* (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar  $\alpha$  factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Constant Multiple Rule for Limits](#)

#### Theorem (Sum Rule)

**Theorem 11.8.2** (Sum Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If  $f$  and  $g$  are differentiable at  $c$ , then  $f + g$  is differentiable at  $c$ , and*

$$(f + g)'(c) = f'(c) + g'(c).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(f + g, c, f'(c) + g'(c)).$$

*Remark* (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of  $f + g$  splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation:  $(\alpha f + \beta g)' = \alpha f' + \beta g'$ .

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Sum Rule for Limits](#)

#### Theorem (Product Rule)

**Theorem 11.8.3** (Product Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If  $f$  and  $g$  are differentiable at  $c$ , then  $fg$  is differentiable at  $c$ , and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \Rightarrow \text{DerivativeAt}(fg, c, f'(c)g(c) + f(c)g'(c)).$$

*Remark* (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of  $f$  weighted by the current value of  $g$ , plus the rate of change of  $g$  weighted by the current value of  $f$ . The proof inserts the auxiliary term  $f(x)g(c)$  to decouple the increments of  $f$  and  $g$ , isolating the individual difference quotients. Continuity of  $f$  at  $c$  (which follows from differentiability) is used to evaluate  $\lim_{x \rightarrow c} f(x) = f(c)$  in one of the factors.

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Theorem: Differentiable Implies Continuous](#)
- [Sum Rule for Limits](#)
- [Product Rule for Limits](#)

#### Theorem (Quotient Rule)

**Theorem 11.8.4** (Quotient Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f, g : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . Suppose that  $f$  and  $g$  are differentiable at  $c$  and that  $g(c) \neq 0$ . Then  $f/g$  is differentiable at  $c$ , and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge \text{DifferentiableAt}(g, c) \wedge g(c) \neq 0 \Rightarrow \text{DerivativeAt}\left(\frac{f}{g}, c, \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}\right).$$

*Remark* (Interpretation). The quotient rule is the product rule applied to  $f \cdot g^{-1}$ , but stated directly. The hypothesis  $g(c) \neq 0$  ensures the quotient is defined near  $c$  — continuity of  $g$  (from its differentiability) and the quotient rule for limits guarantee  $g(x) \neq 0$  in a deleted neighbourhood of  $c$ , so the difference quotient of  $f/g$  is well-defined throughout. The add-and-subtract technique with  $f(c)g(c)$  decouples the increments of  $f$  and  $g$  in the numerator, exactly as in the product rule proof.

*Remark* (Dependencies).

- [Definition: Derivative at a Point](#)
- [Theorem: Differentiable Implies Continuous](#)
- [Sum Rule for Limits](#)
- [Product Rule for Limits](#)
- [Quotient Rule for Limits](#)

## 11.8.2 Closure

### Corollary

**Corollary 11.8.5** (Finite Sum Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then  $\sum_{i=1}^n \alpha_i f_i$  is differentiable at  $c$ , and*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \in \{1, \dots, n\} \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\sum_i \alpha_i f_i, c, \sum_i \alpha_i f'_i(c)\right).$$

*Remark* (Interpretation). Finite sums and scalar weights do not introduce a new kind of differentiation rule; they package repeated use of the Sum Rule and Constant Multiple Rule.

*Remark* (Dependencies).

- [Theorem: Sum Rule](#)
- [Theorem: Constant Multiple Rule](#)

### Corollary

**Corollary 11.8.6** (Extended Product Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then  $\prod_{i=1}^n f_i$  is differentiable at  $c$ , and*

$$\left( \prod_{i=1}^n f_i \right)'(c) = \sum_{k=1}^n f'_k(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\prod_i f_i, c, \sum_k f'_k(c) \prod_{i \neq k} f_i(c)\right).$$

*Remark* (Interpretation). In a finite product, each term takes one turn being differentiated while all other factors are held at their values.

*Remark* (Dependencies).

- [Theorem: Product Rule](#)

### Corollary

**Corollary 11.8.7** (Power Rule for Integer Powers of a Function). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable at  $c$ , then  $f^n$  is differentiable at  $c$ , and*

$$(f^n)'(c) = n(f(c))^{n-1}f'(c).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \wedge n \in \mathbb{N} \Rightarrow \text{DerivativeAt}(f^n, c, n(f(c))^{n-1}f'(c)).$$

*Remark* (Interpretation). The integer power rule is the extended product rule applied to  $n$  identical factors.

*Remark* (Dependencies).

- [Corollary: Extended Product Rule](#)

### Theorem

**Theorem 11.8.8** (Finite Linear Combination Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall i \text{ DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\left(\sum_i \alpha_i f_i, c, \sum_i \alpha_i f'_i(c)\right).$$

*Remark* (Interpretation). The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules — it is their inductive closure. Stated in operator language: if  $\mathcal{D}(I)$  denotes the vector space of functions differentiable at  $c$ , then  $D_c : \mathcal{D}(I) \rightarrow \mathbb{R}$  defined by  $D_c(f) = f'(c)$  is a linear functional. The Finite Linear Combination Rule says  $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$ .

*Remark* (Dependencies).

- [Real-Linear Rule](#)
- [Linear Combination of Real-Valued Functions](#)
- [Theorem: Constant Multiple Rule](#)
- [Theorem: Sum Rule](#)

### 11.8.3 Global Forms on an Interval

#### Theorem

**Theorem 11.8.9** (Constant Multiple Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable on  $I$ , then  $\alpha f$  is differentiable on  $I$ , and*

$$(\alpha f)'(x) = \alpha f'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \text{ DifferentiableAt}(f, x) \Rightarrow \forall x \in I \text{ DerivativeAt}(\alpha f, x, \alpha f'(x)).$$

*Remark* (Interpretation). The pointwise constant-multiple rule becomes an interval statement by applying it at every point of the interval.

*Remark* (Dependencies).

- [Theorem: Constant Multiple Rule](#)

#### Theorem

**Theorem 11.8.10** (Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $f + g$  is differentiable on  $I$ , and*

$$(f + g)'(x) = f'(x) + g'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I (\text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x)) \Rightarrow \forall x \in I \text{ DerivativeAt}(f+g, x, f'(x)+g'(x)). \blacksquare$$

*Remark* (Interpretation). The derivative of a sum is computed point by point across the whole interval.

*Remark* (Dependencies).

- [Theorem: Sum Rule](#)

#### Theorem

**Theorem 11.8.11** (Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $fg$  is differentiable on  $I$ , and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \quad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall x \in I \left( \text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x) \right) \Rightarrow \forall x \in I \text{ DerivativeAt}(fg, x, f'(x)g(x) + f(x)g'(x))$$

*Remark* (Interpretation). The interval product rule records the product rule as an identity of derivative functions.

*Remark* (Dependencies).

- [Theorem: Product Rule](#)

#### Theorem

**Theorem 11.8.12** (Quotient Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . Suppose that  $f$  and  $g$  are differentiable on  $I$  and that  $g(x) \neq 0$  for all  $x \in I$ . Then  $f/g$  is differentiable on  $I$ , and*

$$\left( \frac{f}{g} \right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \text{for all } x \in I.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall x \in I \left( \text{DifferentiableAt}(f, x) \wedge \text{DifferentiableAt}(g, x) \wedge g(x) \neq 0 \right) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\frac{f}{g}, x, \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}\right)$$

*Remark* (Interpretation). The quotient rule on an interval is the pointwise quotient rule with the nonvanishing denominator hypothesis required at every point.

*Remark* (Dependencies).

- [Theorem: Quotient Rule](#)

#### Corollary

**Corollary 11.8.13** (Finite Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\sum_{i=1}^n \alpha_i f_i$  is differentiable on  $I$ , and*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\sum_i \alpha_i f_i, x, \sum_i \alpha_i f'_i(x)\right).$$

*Remark* (Interpretation). Finite linear combinations commute with differentiation as functions on the whole interval.

Remark (Dependencies).

- [Corollary: Finite Sum Rule](#)

#### Corollary

**Corollary 11.8.14** (Extended Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\prod_{i=1}^n f_i$  is differentiable on  $I$ , and*

$$\left( \prod_{i=1}^n f_i \right)'(x) = \sum_{k=1}^n f'_k(x) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(x) \quad \text{for all } x \in I.$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}\left(\prod_i f_i, x, \sum_k f'_k(x) \prod_{i \neq k} f_i(x)\right).$$

Remark (Interpretation). The product rule for many factors holds uniformly point by point on the interval.

Remark (Dependencies).

- [Corollary: Extended Product Rule](#)

#### Corollary

**Corollary 11.8.15** (Power Rule for Integer Powers of a Function on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable on  $I$ , then  $f^n$  is differentiable on  $I$ , and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \quad \text{for all } x \in I.$$

[Go to proof.](#)

Remark (Standard quantified statement).

$$\forall x \in I \text{ DifferentiableAt}(f, x) \wedge n \in \mathbb{N} \Rightarrow \forall x \in I \text{ DerivativeAt}(f^n, x, n(f(x))^{n-1} f'(x)).$$

Remark (Interpretation). The interval power rule says that the pointwise integer-power formula assembles into an equality of derivative functions.

Remark (Dependencies).

- [Corollary: Power Rule for Integer Powers of a Function](#)



### Theorem

**Theorem 11.8.16** (Finite Linear Combination Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\forall i \forall x \in I \text{ DifferentiableAt}(f_i, x) \Rightarrow \forall x \in I \text{ DerivativeAt}(\sum_i \alpha_i f_i, x, \sum_i \alpha_i f'_i(x)).$$

*Remark* (Interpretation). The pointwise statements above — each proved at a single point  $c$  — extend to global statements when the hypotheses hold at every point of  $I$ . If  $f$  is differentiable on  $I$ , then  $f' : I \rightarrow \mathbb{R}$  is itself a function and differentiation acts as a linear operator on the vector space  $\mathcal{D}(I)$  of functions differentiable on  $I$ :

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \quad (fg)' = f'g + fg', \quad \left( \frac{f}{g} \right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on  $I$ . The Finite Linear Combination Rule in its global form is precisely the statement that  $D : \mathcal{D}(I) \rightarrow \mathcal{F}(I)$  defined by  $D(f) = f'$  is a linear map between function spaces, where  $\mathcal{F}(I)$  denotes functions on  $I$ . The interval variants are not separate theorems — they are the pointwise results collected into function notation by applying the pointwise result at each  $x \in I$ .

*Remark* (Dependencies).

- [Real-Linear Rule](#)
- [Linear Combination of Real-Valued Functions](#)
- [Theorem: Finite Linear Combination Rule](#)

When  $f$  is strictly monotone and differentiable on  $I$  with  $f'(x) \neq 0$  for all  $x \in I$ , the inverse  $g : J \rightarrow I$  is differentiable on  $J = f(I)$  and

$$g' = \frac{1}{f' \circ g},$$

as functions on  $J$ . This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

## 11.8.4 Inverting the Operation

### Theorem (Inverse Function Theorem, One Variable)

**Theorem 11.8.17** (Inverse Function Theorem, One Variable). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$  be of class  $C^1$ , and let  $c \in I$  satisfy  $f'(c) \neq 0$ . Then there exist open intervals  $U \subseteq I$  and  $V \subseteq \mathbb{R}$ , with  $c \in U$  and  $f(c) \in V$ , such that:*

1.  $f|_U : U \rightarrow V$  is a bijection;
2. the inverse  $g = (f|_U)^{-1} : V \rightarrow U$  is of class  $C^1$ ;
3. for every  $y \in V$ ,

$$g'(y) = \frac{1}{f'(g(y))}.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$f \in C^1(I) \wedge c \in I \wedge f'(c) \neq 0$$

$$\implies \exists U \ni c \exists V \ni f(c) [\text{Bijective}(f|_U, U, V) \wedge (f|_U)^{-1} \in C^1(V) \wedge \forall y \in V : ((f|_U)^{-1})'(y) = 1/f'((f|_U)^{-1}(y))]$$

*Remark* (Predicate reading).  $\text{LocallyInvertible}(f, c)$  holds with a  $C^1$  local inverse. The condition  $f'(c) \neq 0$  upgrades local set-theoretic invertibility to local analytic invertibility: the inverse exists as a differentiable function and has controlled derivative.

*Remark* (Failure modes). The hypotheses fail in different ways.

- If  $f'(c) = 0$ , the inverse may exist without being differentiable. The map  $f(x) = x^3$  is globally bijective, but its inverse  $x \mapsto x^{1/3}$  is not differentiable at 0.
- If  $f'$  is not continuous at  $c$ , the sign of  $f'$  need not persist near  $c$ . Mere differentiability with  $f'(c) \neq 0$  is therefore not enough to force local monotonicity or local invertibility.

*Remark* (Contrapositive quantified statement).

$$\neg \text{LocallyInvertible}_{C^1}(f, c) \implies f \notin C^1(\text{near } c) \vee f'(c) = 0.$$

*Remark* (Contrapositive predicate reading). If  $f$  has no  $C^1$  local inverse at  $c$ , then either  $f$  fails to be continuously differentiable near  $c$ , or its derivative vanishes there. The witnesses  $x \mapsto x^3$  and  $x \mapsto x + 2x^2 \sin(1/x)$  isolate the two failure mechanisms.

*Remark* (Interpretation). This theorem is the existence result behind the inverse-derivative formula. A continuous nonzero derivative keeps its sign on a small interval; constant sign forces strict monotonicity; strict monotonicity gives a local inverse; and the Carathéodory characterization upgrades that inverse to a differentiable map with reciprocal slope. In several variables, the condition  $f'(c) \neq 0$  becomes invertibility of the differential.

*Remark* (Dependencies).

- Definition: Continuously Differentiable; Class  $C^1$
- Theorem: Monotonicity from the Sign of the Derivative
- Theorem: Continuous Inverse Theorem
- Theorem: Carathéodory Characterization of Differentiability

#### Corollary (Derivative of the Inverse Function)

**Corollary 11.8.18** (Derivative of the Inverse Function). *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse  $g$  satisfies*

$$g' = \frac{1}{f' \circ g}$$

*on  $V$ . More globally, if  $f : I \rightarrow J$  is a  $C^1$  bijection with  $f'(x) \neq 0$  for every  $x \in I$ , then  $f^{-1} : J \rightarrow I$  is  $C^1$  and*

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad \text{for every } y \in J.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$g = (f|_V)^{-1} \implies \forall y \in V : g'(y) = \frac{1}{f'(g(y))}.$$

*Remark* (Interpretation). The formula says that undoing a differentiable change of variable reverses the local stretch factor. If  $f$  multiplies small increments near  $x$  by  $f'(x)$ , then its inverse multiplies corresponding increments near  $f(x)$  by  $1/f'(x)$ .

*Remark* (Dependencies).

- Theorem: Inverse Function Theorem, One Variable
- Theorem: Chain Rule

## 11.8.5 Indeterminate Forms: L'Hôpital's Rule

### Theorem (L'Hôpital's Rule, 0/0 Form)

**Theorem 11.8.19** (L'Hôpital's Rule, 0/0 Form). *Let  $a < b$ , and let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable with  $g'(x) \neq 0$  for every  $x \in (a, b)$ . Suppose*

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

*for  $L \in \mathbb{R} \cup \{-\infty, +\infty\}$ . Then*

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

*The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\left( \lim_{x \rightarrow a^+} f(x) = 0 \wedge \lim_{x \rightarrow a^+} g(x) = 0 \wedge \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \right) \implies \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

*Remark* (Failure modes). The rule is a one-way implication and is routinely misapplied.

- *Form not indeterminate.* If  $f/g$  is not genuinely 0/0, differentiating numerator and denominator can answer a different problem.
- *$\lim f'/g'$  fails to exist.* With  $f(x) = x + \sin x$  and  $g(x) = x$  as  $x \rightarrow +\infty$ , the quotient  $f/g \rightarrow 1$ , but  $f'/g' = 1 + \cos x$  oscillates. Absence of  $\lim f'/g'$  says nothing by itself about  $\lim f/g$ .
- *$g'$  vanishes near the target.* The denominator derivative must stay nonzero so the Cauchy Mean Value Theorem applies and the derivative ratio is defined.

*Remark* (Interpretation). The 0/0 form is the Cauchy Mean Value Theorem read as a limit law. On a shrinking interval, a ratio of changes in  $f$  and  $g$  is represented by a ratio of derivatives at an interior point. If the derivative ratio settles, the original quotient is forced to settle to the same value.

*Remark* (Dependencies).

- [Theorem: Cauchy Mean Value Theorem](#)
- [Definition: Limit of a Function](#)
- [Theorem: Squeeze Theorem for Function Limits](#)

### Theorem (L'Hôpital's Rule, $\infty/\infty$ Form)

**Theorem 11.8.20** (L'Hôpital's Rule,  $\infty/\infty$  Form). *Let  $a < b$ , and let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable with  $g'(x) \neq 0$  for every  $x \in (a, b)$ . Suppose*

$$\lim_{x \rightarrow a^+} |g(x)| = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

*for  $L \in \mathbb{R} \cup \{-\infty, +\infty\}$ . Then*

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

*No separate hypothesis on  $\lim_{x \rightarrow a^+} f(x)$  is required. [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\left( \lim_{x \rightarrow a^+} |g(x)| = +\infty \wedge \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L \right) \implies \lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

*Remark* (Failure modes). L'Hôpital's Rule is a one-way implication and requires an indeterminate form. If the quotient is not of type  $0/0$  or  $\infty/\infty$ , differentiating numerator and denominator can change the problem. The derivative ratio must also have a limit: for  $f(x) = x + \sin x$  and  $g(x) = x$  as  $x \rightarrow +\infty$ , the quotient  $f/g$  tends to 1, but  $f'/g' = 1 + \cos x$  has no limit.

*Remark* (Interpretation). The  $\infty/\infty$  form compares two functions whose magnitudes escape. The Cauchy Mean Value Theorem controls the quotient after subtracting fixed endpoint values, and the divergent denominator makes those fixed values negligible.

*Remark* (Dependencies).

- [Theorem: Cauchy Mean Value Theorem](#)
- [Definition: Limit of a Function](#)
- [Theorem: Squeeze Theorem for Function Limits](#)

## 11.9 Taylor: Construction and Its Error

### Taylor Expansion — Quick Reference

| Concept                                                | Meaning                                                |
|--------------------------------------------------------|--------------------------------------------------------|
| <a href="#">Taylor polynomial</a>                      | Polynomial forced by derivative data at a point        |
| <a href="#">Taylor remainder</a>                       | Exact error after subtracting the Taylor polynomial    |
| <a href="#">Maclaurin polynomial</a>                   | Taylor polynomial centered at 0                        |
| <a href="#">Lagrange remainder</a>                     | Error controlled by an intermediate derivative value   |
| <a href="#">Peano remainder</a>                        | Local asymptotic error smaller than the retained order |
| <a href="#">Example: <math>(\sin x - x)/x^3</math></a> | Taylor and L'Hôpital compared on one limit             |
| <a href="#">Flat function</a>                          | Smooth function whose Taylor series misses it          |

Taylor's theorem is the derivative's most ambitious claim: local behaviour to finite order is encoded by derivatives at one point, with an error term that can be estimated. The first-order case is the seed of the closing reframe, where the derivative stops being only a number  $f'(c)$  and becomes the linear map  $h \mapsto f'(c)h$ .

### Definition (Taylor Polynomial at a Point)

**Definition 11.9.1** (Taylor Polynomial at a Point). Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have derivatives  $f^{(k)}(a)$  for  $0 \leq k \leq n$ . The **Taylor polynomial of order  $n$**  for  $f$  at  $a$  is

$$T_{n,a}f(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

*Remark* (Standard quantified statement).

Let  $I \subseteq \mathbb{R}$ ,  $a \in I$ ,  $n \in \mathbb{N}$ ,  $f : I \rightarrow \mathbb{R}$ .

If  $f^{(k)}(a)$  exists for every  $0 \leq k \leq n$ , then

$$\forall x \in I \left( T_{n,a}f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \right).$$

*Remark* (Interpretation). The Taylor polynomial is the unique polynomial of degree at most  $n$  whose first  $n$  derivatives at  $a$  agree with those of  $f$ . Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

*Remark* (Dependencies).

- [Derivative at a Point](#)
- [Power Rule for Integer Powers of a Function](#)
- [Finite Sum Rule](#)

### Definition (Taylor Remainder)

**Definition 11.9.2** (Taylor Remainder). Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have a Taylor polynomial  $T_{n,a}f$ . The **Taylor remainder of order  $n$**  at  $a$  is the function  $R_{n,a}f : I \rightarrow \mathbb{R}$  defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

*Remark* (Standard quantified statement).

$$\forall x \in I \left( R_{n,a}f(x) = f(x) - T_{n,a}f(x) \right).$$

*Remark* (Interpretation). The remainder is the exact error made by replacing  $f$  with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies).

- Taylor Polynomial at a Point

### Definition (Maclaurin Polynomial)

**Definition 11.9.3** (Maclaurin Polynomial). Let  $n \in \mathbb{N}$ , and let  $f$  be defined on an interval containing 0 with derivatives  $f^{(k)}(0)$  for  $0 \leq k \leq n$ . The **Maclaurin polynomial of order  $n$**  for  $f$  is

$$T_{n,0}f(x) := \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement).

$$\forall x \in I \left( T_{n,0}f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k \right).$$

Remark (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies).

- Taylor Polynomial at a Point

### Theorem (Taylor's Theorem with Lagrange Remainder)

**Theorem 11.9.4** (Taylor's Theorem with Lagrange Remainder). Let  $a, b \in \mathbb{R}$  with  $a < b$ , let  $n \in \mathbb{N}$ , and let  $f : [a, b] \rightarrow \mathbb{R}$ . Suppose that  $f, f', \dots, f^{(n)}$  are continuous on  $[a, b]$  and that  $f^{(n+1)}$  exists on  $(a, b)$ . Then, for every  $x \in (a, b)$ , there exists  $c$  strictly between  $a$  and  $x$  such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

*Go to proof.*

Remark (Standard quantified statement).

$$\forall x \in (a, b) \exists c \in (a, x) \left( f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree  $n$  has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point  $c$  is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

*Remark* (Dependencies).

- [Taylor Polynomial at a Point](#)
- [Taylor Remainder](#)
- [Mean Value Theorem](#)
- [Finite Sum Rule](#)
- [Power Rule for Integer Powers of a Function](#)

#### Corollary (Taylor Expansion with Peano Remainder)

**Corollary 11.9.5** (Taylor Expansion with Peano Remainder). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have derivatives up to order  $n$  in a neighbourhood of  $a$ . If  $f^{(n)}$  is continuous at  $a$ , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n) \quad \text{as } x \rightarrow a.$$

[Go to proof.](#)

*Remark* (Standard quantified statement).

$$\lim_{x \rightarrow a} \frac{f(x) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}{(x-a)^n} = 0.$$

*Remark* (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order  $n$ , and the remaining error is smaller than  $(x-a)^n$  near the center.

*Remark* (Dependencies).

- [Taylor Polynomial at a Point](#)
- [Taylor Remainder](#)
- [Limit of a Function](#)
- [Derivative at a Point](#)

#### Corollary (First-Order Peano Remainder)

**Corollary 11.9.6** (First-Order Peano Remainder). *Let  $f$  be differentiable at  $c$ . Then, as  $h \rightarrow 0$ ,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

[Go to proof.](#)



*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \Rightarrow \lim_{h \rightarrow 0} \frac{f(c+h) - f(c) - f'(c)h}{h} = 0.$$

*Remark* (Interpretation). The first-order Peano formula is the derivative rewritten as an approximation theorem. It says that the affine approximation  $f(c) + f'(c)h$  captures all first-order change, and the remaining error is smaller than  $h$  itself.

*Remark* (Dependencies).

- [Derivative at a Point](#)

*Example*  $((\sin x - x)/x^3 \text{ Two Ways})$ . Taylor expansion gives

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$

so

$$\frac{\sin x - x}{x^3} = \frac{-x^3/6 + o(x^3)}{x^3} \rightarrow -\frac{1}{6}.$$

L'Hôpital's rule reaches the same value by three differentiations:

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = \lim_{x \rightarrow 0} \frac{-\cos x}{6} = -\frac{1}{6}.$$

Taylor exposes the reason for the answer: the limit is the cubic coefficient.

#### Proposition (The Flat Function: Smooth but Not Analytic)

**Proposition 11.9.7** (The Flat Function: Smooth but Not Analytic). *Define  $f : \mathbb{R} \rightarrow \mathbb{R}$  by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

*Then:*

1.  $f \in C^\infty(\mathbb{R})$ ;
2.  $f^{(n)}(0) = 0$  for every  $n \geq 0$ ;
3. the Maclaurin series of  $f$  is identically 0, so  $f \notin C^\omega$  on any neighbourhood of 0.

*In particular,  $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \emptyset$ . [Go to proof](#).*

*Remark* (Standard quantified statement).

$$(\forall n \in \mathbb{N} : f^{(n)}(0) = 0) \wedge (\forall r > 0 \exists x : 0 < |x| < r \wedge f(x) \neq 0) \implies f \in C^\infty \setminus C^\omega.$$

*Remark* (Interpretation). The Taylor series sees only derivative data at the center, and all of that data is zero for the flat function at 0. The function is nevertheless positive away from 0. This separates smoothness from analyticity and supplies the strictness witness referenced by the smoothness tower.

*Remark* (Dependencies).

- Definition: Higher Derivatives
- Definition: Smooth Function; Class  $C^\infty$
- Definition: Real-Analytic Function; Class  $C^\omega$
- Definition: Maclaurin Polynomial
- Theorem: The Smoothness Tower

### 11.9.1 The Differential

*Remark* (Interpretation). The derivative value  $f'(c) \in \mathbb{R}$  is a number. The differential  $df_c$  is a linear map from increments to first-order changes. In one dimension these two objects are represented by the same scalar, but they play different roles.

#### Definition (Differentiability by a Differential)

**Definition 11.9.8** (Differentiability by a Differential). Let  $f$  be defined on a neighbourhood of  $c \in \mathbb{R}$ . The function  $f$  is *differentiable at  $c$  in the differential sense* if there exists a linear map  $L : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(c+h) - f(c) = L(h) + o(h) \quad \text{as } h \rightarrow 0.$$

In that case the *differential* of  $f$  at  $c$  is  $df_c := L$ .

*Remark* (Standard quantified statement).

$$\exists L : \mathbb{R} \rightarrow \mathbb{R} \text{ linear such that } f(c+h) - f(c) = L(h) + o(h).$$

*Remark* (Interpretation). The differential is the linear part of the increment. The function itself is approximated affinely by  $f(c) + df_c(x - c)$ , while  $df_c$  alone is the linear map acting on the displacement.

*Remark* (Dependencies).

- First-Order Peano Remainder

#### Theorem (Differential and Derivative Agree)

**Theorem 11.9.9** (Differential and Derivative Agree). *Let  $f$  be defined on a neighbourhood of  $c \in \mathbb{R}$ . Then  $f$  is differentiable at  $c$  in the ordinary derivative sense if and only if it is differentiable at  $c$  in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists df_c \text{ with } df_c(h) = f'(c)h.$$

*Remark* (Interpretation). This theorem is the bridge from the slope number to the linear map. The one-dimensional derivative is multiplication by the scalar  $f'(c)$ .

*Remark* (Dependencies).

- [Derivative at a Point](#)
- [Differentiability by a Differential](#)

#### Theorem (Uniqueness of the Differential)

**Theorem 11.9.10** (Uniqueness of the Differential). *If  $L_1, L_2 : \mathbb{R} \rightarrow \mathbb{R}$  are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \quad \text{as } h \rightarrow 0,$$

*then  $L_1 = L_2$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$(L_1(h) - L_2(h) = o(h)) \implies L_1 = L_2.$$

*Remark* (Interpretation). The differential is not a choice of linear approximation among many. If a first-order linear part exists, it is forced.

*Remark* (Dependencies).

- [Differentiability by a Differential](#)

#### Theorem (Differential Continuity Criterion)

**Theorem 11.9.11** (Differential Continuity Criterion). *If  $f$  is differentiable at  $c$  in the differential sense, then  $f$  is continuous at  $c$ . [Go to proof.](#)*

*Remark* (Standard quantified statement).

$$\text{DifferentiableByDifferentialAt}(f, c) \implies \text{ContinuousAt}(f, c).$$

*Remark* (Interpretation). From  $f(c+h) - f(c) = df_c(h) + o(h)$ , both terms on the right tend to zero as  $h \rightarrow 0$ . Continuity is therefore immediate in the differential language.

*Remark* (Dependencies).

- [Differentiability by a Differential](#)

#### Theorem (Chain Rule for Differentials)

**Theorem 11.9.12** (Chain Rule for Differentials). *If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

*[Go to proof.](#)*

*Remark* (Standard quantified statement).

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

*Remark* (Interpretation). In one dimension this says that composing two scalar multiplications multiplies their scalars. In several variables the same formula becomes matrix multiplication.

*Remark* (Dependencies).

- Chain Rule
- Differential and Derivative Agree

#### Theorem (Linearity of the Differential)

**Theorem 11.9.13** (Linearity of the Differential). *If  $f$  and  $g$  are differentiable at  $c$  and  $\alpha, \beta \in \mathbb{R}$ , then*

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

*Go to proof.*

*Remark* (Standard quantified statement).

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

*Remark* (Interpretation). The differential packages the linearity of differentiation as equality of linear maps.

*Remark* (Interpretation). On a manifold, the same construction becomes a tangent map

$$T_c f : T_c M \rightarrow T_{f(c)} N.$$

The one-dimensional differential is the prototype: replace real increments by tangent vectors and replace scalar multiplication by a linear map between tangent spaces.

*Remark* (Dependencies).

- Finite Linear Combination Rule
- Differential and Derivative Agree

## Proofs

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Equivalence of Definitions of the Derivative at a Point). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . Let  $L \in \mathbb{R}$ . The following are equivalent:*

(i) (**Epsilon-Delta**)  $\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \left( 0 < |x - c| < \delta \Rightarrow \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$

(ii) (**Neighbourhood**) *For every  $\varepsilon > 0$ , there exists a neighbourhood  $V$  of  $c$  such that*  
$$x \in (A \cap V) \setminus \{c\} \Rightarrow \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon.$$

(iii) (**Sequential**)  $\forall (x_n) \subseteq A \setminus \{c\} \left( x_n \rightarrow c \Rightarrow \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L \right).$

*Professional Standard Proof.*

TODO: professional standard proof for derivative-equivalence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for derivative-equivalence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Derivative:  $h$ -Form Equivalence). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . The following are equivalent:*

1.  *$f$  is differentiable at  $c$ .*
2. *The limit  $\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$  exists.*

*In this case,  $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for derivative- $h$ -form-equivalence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for derivative- $h$ -form-equivalence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Differentiability Implies Continuity). *Let  $f$  be defined on an open interval  $I$  containing  $c$ . If  $f$  is differentiable at  $c$ , then  $f$  is continuous at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for differentiable-implies-continuous. □

*Detailed Learning Proof.*

TODO: detailed learning proof for differentiable-implies-continuous. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Uniqueness of the Derivative). *Let  $f : A \rightarrow \mathbb{R}$  and  $c \in A$ . If  $f$  is differentiable at  $c$ , its derivative is unique.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-derivative. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-derivative. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Proposition](#)

**Proposition** (Derivative of the Identity). *Let  $I \subseteq \mathbb{R}$  be an open interval, and let  $\text{id}_I : I \rightarrow \mathbb{R}$  be the identity function defined by  $\text{id}_I(x) := x$ . Then  $\text{id}_I$  is differentiable at every  $c \in I$ , and  $\text{id}'_I(c) = 1$ .*

*Professional Standard Proof.*

TODO: professional standard proof for derivative-of-the-identity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for derivative-of-the-identity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Derivative of a Constant). *Let  $k \in \mathbb{R}$  and let  $f(x) = k$  for all  $x \in \mathbb{R}$ . Then  $f'(c) = 0$  for every  $c \in \mathbb{R}$ .*

*Professional Standard Proof.*

TODO: professional standard proof for derivative-of-a-constant. □

*Detailed Learning Proof.*

TODO: detailed learning proof for derivative-of-a-constant. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Differentiability and One-Sided Derivatives). *Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f : I \rightarrow \mathbb{R}$ . Fix  $c \in I$ . Then  $f$  is differentiable at  $c$  if and only if both one-sided derivatives  $f'_-(c)$  and  $f'_+(c)$  exist and are equal, in which case  $f'(c) = f'_-(c) = f'_+(c)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for differentiability-and-one-sided-derivatives. □

*Detailed Learning Proof.*

TODO: detailed learning proof for differentiability-and-one-sided-derivatives. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Carathéodory Characterization of Differentiability). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . The following are equivalent:*

(i)  *$f$  is differentiable at  $c$ .*

(ii) *There exists a function  $\phi : I \rightarrow \mathbb{R}$ , continuous at  $c$ , such that*

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

*In this case,  $\phi(c) = f'(c)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for caratheodory-characterization-of-differentiability. □

*Detailed Learning Proof.*

TODO: detailed learning proof for caratheodory-characterization-of-differentiability. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Chain Rule). *Let  $J, I \subseteq \mathbb{R}$  be open intervals, let  $f : J \rightarrow \mathbb{R}$  and  $g : I \rightarrow \mathbb{R}$  be functions with  $f(J) \subseteq I$ , and let  $c \in J$ . If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then  $g \circ f$  is differentiable at  $c$  and  $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for chain-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for chain-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Leibniz Rule). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $x \in I$ , and suppose that  $f, g : I \rightarrow \mathbb{R}$  are  $n$ -times differentiable on a neighbourhood of  $x$ . Then*

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

*Professional Standard Proof.*

TODO: professional standard proof for leibniz-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for leibniz-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Faa di Bruno's Formula). *Under the hypotheses of [Faa di Bruno's Formula](#),*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^n m k_m = n} \frac{n!}{\prod_{m=1}^n k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^n (g^{(m)}(x))^{k_m},$$

where  $k = \sum_{m=1}^n k_m$ .

*Professional Standard Proof.*

TODO: professional standard proof for faa-di-bruno. □

*Detailed Learning Proof.*

TODO: detailed learning proof for faa-di-bruno. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Interior Extremum Theorem). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$  and is differentiable at  $c$ , then  $f'(c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for interior-extremum-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for interior-extremum-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Necessary Condition for an Interior Extremum). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then either  $f'(c)$  does not exist, or*

$$f'(c) = 0.$$

*Professional Standard Proof.*

TODO: professional standard proof for necessary-condition-extremum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for necessary-condition-extremum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Necessary Condition for an Extremum). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $c \in I$ . If  $f$  has a relative extremum at  $c$ , then either  $f'(c)$  does not exist, or  $f'(c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for relative-extremum-necessary-condition. □

*Detailed Learning Proof.*

TODO: detailed learning proof for relative-extremum-necessary-condition. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Rolle's Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and satisfies  $f(a) = f(b)$ . Then there exists  $c \in (a, b)$  such that  $f'(c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for rolles-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for rolles-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Mean Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that*  
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Professional Standard Proof.*

TODO: professional standard proof for mean-value-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for mean-value-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Mean Value Theorem). *Let  $a, b \in \mathbb{R}$  with  $a < b$ . Suppose that  $f, g : [a, b] \rightarrow \mathbb{R}$  are continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists  $c \in (a, b)$  such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

*If  $g'(x) \neq 0$  for every  $x \in (a, b)$ , then  $g(b) \neq g(a)$  and the corresponding quotient identity holds.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-mean-value-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-mean-value-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Derivative Bound Implies Lipschitz). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If  $|f'(x)| \leq M$  for every  $x \in I$ , then*

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every } x, y \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for derivative-bound-implies-lipschitz. □

*Detailed Learning Proof.*

TODO: detailed learning proof for derivative-bound-implies-lipschitz. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Zero Derivative Implies Constant). *Let  $I := [a, b] \subseteq \mathbb{R}$  be a closed interval, and let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If  $f'(x) = 0$  for all  $x \in (a, b)$ , then  $f$  is constant on  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for zero-derivative-implies-constant. □

*Detailed Learning Proof.*

TODO: detailed learning proof for zero-derivative-implies-constant. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Equal Derivatives Imply Difference Is Constant). *Let  $I := [a, b] \subseteq \mathbb{R}$ , and let  $f, g : I \rightarrow \mathbb{R}$  be continuous on  $I$  and differentiable on  $(a, b)$ . If  $f'(x) = g'(x)$  for all  $x \in (a, b)$ , then there exists  $C \in \mathbb{R}$  such that  $f(x) = g(x) + C$  for all  $x \in I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for equal-derivatives-constant-difference. □

*Detailed Learning Proof.*

TODO: detailed learning proof for equal-derivatives-constant-difference. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Nondecreasing Iff Nonnegative Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nondecreasing on  $I$  if and only if  $f'(x) \geq 0$  for all  $x \in I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for nondecreasing-iff-nonneg-derivative. □

*Detailed Learning Proof.*

TODO: detailed learning proof for nondecreasing-iff-nonneg-derivative. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Nonincreasing Iff Nonpositive Derivative). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Then  $f$  is nonincreasing on  $I$  if and only if  $f'(x) \leq 0$  for all  $x \in I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for nonincreasing-iff-nonpos-derivative. □

*Detailed Learning Proof.*

TODO: detailed learning proof for nonincreasing-iff-nonpos-derivative. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Positive Derivative Implies Locally Increasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose  $f$  is differentiable at  $c$ . If  $f'(c) > 0$ , then there exists  $\delta > 0$  such that  $f(x) < f(c)$  for all  $x \in I$  with  $c - \delta < x < c$ , and  $f(x) > f(c)$  for all  $x \in I$  with  $c < x < c + \delta$ .*

*Professional Standard Proof.*

TODO: professional standard proof for positive-derivative-implies-locally-increasing. □

*Detailed Learning Proof.*

TODO: detailed learning proof for positive-derivative-implies-locally-increasing. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Negative Derivative Implies Locally Decreasing). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , let  $c \in I$ , and suppose  $f$  is differentiable at  $c$ . If  $f'(c) < 0$ , then there exists  $\delta > 0$  such that  $f(x) > f(c)$  for all  $x \in I$  with  $c - \delta < x < c$ , and  $f(x) < f(c)$  for all  $x \in I$  with  $c < x < c + \delta$ .*

*Professional Standard Proof.*

TODO: professional standard proof for negative-derivative-implies-locally-decreasing. □

*Detailed Learning Proof.*

TODO: detailed learning proof for negative-derivative-implies-locally-decreasing. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (First Derivative Test: Relative Maximum). *Let  $I := [a, b]$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$  with  $f$  differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  with  $(c - \delta, c + \delta) \subseteq I$  and  $f'(x) \geq 0$  for  $x \in (c - \delta, c)$  and  $f'(x) \leq 0$  for  $x \in (c, c + \delta)$ , then  $f$  has a relative maximum at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for first-derivative-test-maximum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for first-derivative-test-maximum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (First Derivative Test: Relative Minimum). *Let  $I := [a, b]$ , let  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ , and let  $c \in (a, b)$  with  $f$  differentiable on  $(a, c)$  and  $(c, b)$ . If there exists  $\delta > 0$  with  $(c - \delta, c + \delta) \subseteq I$  and  $f'(x) \leq 0$  for  $x \in (c - \delta, c)$  and  $f'(x) \geq 0$  for  $x \in (c, c + \delta)$ , then  $f$  has a relative minimum at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for first-derivative-test-minimum. □

*Detailed Learning Proof.*

TODO: detailed learning proof for first-derivative-test-minimum. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Second-Derivative Convexity Test). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If  $f''(x) \geq 0$  for all  $x \in I$ , then  $f$  is convex on  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for second-derivative-convexity-test. □

*Detailed Learning Proof.*

TODO: detailed learning proof for second-derivative-convexity-test. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Second-Derivative Concavity Test). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f : I \rightarrow \mathbb{R}$  be twice differentiable on  $I$ . If  $f''(x) \leq 0$  for all  $x \in I$ , then  $f$  is concave on  $I$ .*

*Professional Standard Proof.*

TODO: professional standard proof for second-derivative-concavity-test. □

*Detailed Learning Proof.*

TODO: detailed learning proof for second-derivative-concavity-test. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Second Derivative Test). *Let  $f$  be twice differentiable in a neighbourhood of  $c$ , and suppose  $f'(c) = 0$ . If  $f''(c) > 0$ , then  $f$  has a strict relative minimum at  $c$ . If  $f''(c) < 0$ , then  $f$  has a strict relative maximum at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for second-derivative-test. □

*Detailed Learning Proof.*

TODO: detailed learning proof for second-derivative-test. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Inflection Point Necessary Condition). *Let  $f$  have an inflection point at  $c$ . If  $f''$  exists on a neighbourhood of  $c$  and is continuous at  $c$ , then  $f''(c) = 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for inflection-point-necessary-condition. □

*Detailed Learning Proof.*

TODO: detailed learning proof for inflection-point-necessary-condition. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Darboux's Theorem). *Let  $I := [a, b] \subseteq \mathbb{R}$  and let  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . If  $k$  is a real number strictly between  $f'(a)$  and  $f'(b)$ , then there exists  $c \in (a, b)$  such that  $f'(c) = k$ .*

*Professional Standard Proof.*

TODO: professional standard proof for darbox. □

*Detailed Learning Proof.*

TODO: detailed learning proof for darbox. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (The Smoothness Tower). *For every nondegenerate interval  $I$ , the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

*hold, and each displayed inclusion is strict.*

*Professional Standard Proof.*

TODO: professional standard proof for smoothness-tower. □

*Detailed Learning Proof.*

TODO: detailed learning proof for smoothness-tower. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Placement of  $C^{1,1}$ ). *For every nondegenerate interval  $I$ ,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

*and  $C^{1,1}(I)$  is not contained in  $C^2(I)$ . If  $K$  is a compact interval, then*

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

*and the first inclusion is strict.*

*Professional Standard Proof.*

TODO: professional standard proof for c11-placement. □

*Detailed Learning Proof.*

TODO: detailed learning proof for c11-placement. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Bounded Second Derivative Implies  $C^{1,1}$ ). *Let  $I \subseteq \mathbb{R}$  be an interval and let  $f \in C^2(I)$ . If  $|f''(x)| \leq M$  for all  $x \in I$ , then  $f \in C^{1,1}(I)$ , and  $f'$  is Lipschitz with constant  $M$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bounded-second-derivative-implies-c11. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounded-second-derivative-implies-c11. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Multiple Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable at  $c$ , then  $\alpha f$  is differentiable at  $c$ , and*

$$(\alpha f)'(c) = \alpha f'(c).$$

*Professional Standard Proof.*

TODO: professional standard proof for constant-multiple-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-multiple-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sum Rule). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $c \in I$ , and let  $f, g : I \rightarrow \mathbb{R}$  be differentiable at  $c$ . Then  $f + g$  is differentiable at  $c$  and  $(f + g)'(c) = f'(c) + g'(c)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for sum-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sum-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Product Rule). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $c \in I$ , and let  $f, g : I \rightarrow \mathbb{R}$  be differentiable at  $c$ . Then  $fg$  is differentiable at  $c$  and  $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for product-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for product-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Quotient Rule). *Let  $I \subseteq \mathbb{R}$  be an open interval, let  $c \in I$ , and let  $f, g : I \rightarrow \mathbb{R}$  be differentiable at  $c$  with  $g(c) \neq 0$ . Then  $f/g$  is differentiable at  $c$  and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

*Professional Standard Proof.*

TODO: professional standard proof for quotient-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for quotient-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Finite Sum Rule). *If  $f_1, f_2, \dots, f_n$  are functions on an interval  $I$  to  $\mathbb{R}$  that are differentiable at  $c \in I$ , then:*

$$\left( \sum_{i=1}^n f_i \right)'(c) = \sum_{i=1}^n f'_i(c)$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Extended Product Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then  $\prod_{i=1}^n f_i$  is differentiable at  $c$ , and*

$$\left( \prod_{i=1}^n f_i \right)'(c) = \sum_{k=1}^n f'_k(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

*Professional Standard Proof.*

TODO: professional standard proof for extended-product-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for extended-product-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Power Rule for Integer Powers of a Function). *Let  $A \subseteq \mathbb{R}$ , let  $f : A \rightarrow \mathbb{R}$ , let  $c \in A$  be a limit point of  $A$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable at  $c$ , then  $f^n$  is differentiable at  $c$ , and*

$$(f^n)'(c) = n(f(c))^{n-1}f'(c).$$

*Professional Standard Proof.*

TODO: professional standard proof for power-rule-special-case. □

*Detailed Learning Proof.*

TODO: detailed learning proof for power-rule-special-case. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Finite Linear Combination Rule). *Let  $A \subseteq \mathbb{R}$ , let  $f_1, \dots, f_n : A \rightarrow \mathbb{R}$ , let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ , and let  $c \in A$  be a limit point of  $A$ . If each  $f_i$  is differentiable at  $c$ , then*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-linear-combination-rule. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-linear-combination-rule. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Constant Multiple Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $\alpha \in \mathbb{R}$ . If  $f$  is differentiable on  $I$ , then  $\alpha f$  is differentiable on  $I$ , and*

$$(\alpha f)'(x) = \alpha f'(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for constant-multiple-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for constant-multiple-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $f + g$  is differentiable on  $I$ , and*

$$(f + g)'(x) = f'(x) + g'(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for sum-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for sum-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . If  $f$  and  $g$  are differentiable on  $I$ , then  $fg$  is differentiable on  $I$ , and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for product-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for product-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Quotient Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f, g : I \rightarrow \mathbb{R}$ . Suppose that  $f$  and  $g$  are differentiable on  $I$  and that  $g(x) \neq 0$  for all  $x \in I$ . Then  $f/g$  is differentiable on  $I$ , and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for quotient-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for quotient-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Finite Sum Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\sum_{i=1}^n \alpha_i f_i$  is differentiable on  $I$ , and*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-sum-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-sum-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Extended Product Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, and let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then  $\prod_{i=1}^n f_i$  is differentiable on  $I$ , and*

$$\left( \prod_{i=1}^n f_i \right)'(x) = \sum_{k=1}^n f'_k(x) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for extended-product-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for extended-product-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Corollary** (Power Rule for Integer Powers of a Function on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f : I \rightarrow \mathbb{R}$ , and let  $n \in \mathbb{N}$ . If  $f$  is differentiable on  $I$ , then  $f^n$  is differentiable on  $I$ , and*

$$(f^n)'(x) = n(f(x))^{n-1}f'(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for power-rule-special-case-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for power-rule-special-case-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Finite Linear Combination Rule on an Interval). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $f_1, \dots, f_n : I \rightarrow \mathbb{R}$ , and let  $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ . If each  $f_i$  is differentiable on  $I$ , then*

$$\left( \sum_{i=1}^n \alpha_i f_i \right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

*Professional Standard Proof.*

TODO: professional standard proof for finite-linear-combination-rule-interval. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-linear-combination-rule-interval. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Inverse Function Theorem, One Variable). *Let  $I \subseteq \mathbb{R}$  be open, let  $f : I \rightarrow \mathbb{R}$  be  $C^1$ , and suppose  $f'(c) \neq 0$ . Then  $f$  has a  $C^1$  local inverse near  $c$ , and the inverse satisfies  $g'(y) = 1/f'(g(y))$ .*

*Professional Standard Proof.*

TODO: professional standard proof for inverse-function-theorem-one-variable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for inverse-function-theorem-one-variable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Derivative of the Inverse Function). *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse  $g$  satisfies  $g' = 1/(f' \circ g)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for inverse-function-derivative. □

*Detailed Learning Proof.*

TODO: detailed learning proof for inverse-function-derivative. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (L'Hôpital's Rule, 0/0 Form). *If  $f, g$  are differentiable near  $a^+$ ,  $g' \neq 0$ ,  $f(x) \rightarrow 0$ ,  $g(x) \rightarrow 0$ , and  $f'(x)/g'(x) \rightarrow L$ , then  $f(x)/g(x) \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lhopital-zero-over-zero. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lhopital-zero-over-zero. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (L'Hôpital's Rule,  $\infty/\infty$  Form). *If  $g(x)$  diverges in magnitude near  $a^+$ ,  $g' \neq 0$ , and  $f'(x)/g'(x) \rightarrow L$ , then  $f(x)/g(x) \rightarrow L$ .*

*Professional Standard Proof.*

TODO: professional standard proof for lhopital-infinity-over-infinity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lhopital-infinity-over-infinity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Differential and Derivative Agree). *The ordinary derivative at  $c$  exists if and only if the differential exists at  $c$ , and then  $df_c(h) = f'(c)h$ .*

*Professional Standard Proof.*

TODO: professional standard proof for differential-and-derivative-agree. □

*Detailed Learning Proof.*

TODO: detailed learning proof for differential-and-derivative-agree. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [First-Order Peano Remainder](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Uniqueness of the Differential). *If two linear maps give the same first-order expansion up to  $o(h)$ , then they are equal.*

*Professional Standard Proof.*

TODO: professional standard proof for uniqueness-of-the-differential. □

*Detailed Learning Proof.*

TODO: detailed learning proof for uniqueness-of-the-differential. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Differentiability by a Differential](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Differential Continuity Criterion). *If  $f$  is differentiable at  $c$  in the differential sense, then  $f$  is continuous at  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for differential-continuity-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for differential-continuity-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Differentiability by a Differential](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Chain Rule for Differentials). *If  $f$  is differentiable at  $c$  and  $g$  is differentiable at  $f(c)$ , then  $d(g \circ f)_c = dg_{f(c)} \circ df_c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for chain-rule-for-differentials. □

*Detailed Learning Proof.*

TODO: detailed learning proof for chain-rule-for-differentials. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Chain Rule](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Linearity of the Differential). *If  $f$  and  $g$  are differentiable at  $c$  and  $\alpha, \beta \in \mathbb{R}$ , then  $d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for linearity-of-the-differential. □

*Detailed Learning Proof.*

TODO: detailed learning proof for linearity-of-the-differential. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Finite Linear Combination Rule](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Taylor's Theorem with Lagrange Remainder). *Let  $a, b \in \mathbb{R}$  with  $a < b$ , let  $n \in \mathbb{N}$ , and let  $f : [a, b] \rightarrow \mathbb{R}$ . Suppose that  $f, f', \dots, f^{(n)}$  are continuous on  $[a, b]$  and that  $f^{(n+1)}$  exists on  $(a, b)$ . Then, for every  $x \in (a, b)$ , there exists  $c$  strictly between  $a$  and  $x$  such that*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

*Professional Standard Proof.*

TODO: professional standard proof for taylor-theorem-lagrange-remainder. □

*Detailed Learning Proof.*

TODO: detailed learning proof for taylor-theorem-lagrange-remainder. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.



*Remark* (Return). [Return to Theorem](#)

**Corollary** (Taylor Expansion with Peano Remainder). *Let  $I \subseteq \mathbb{R}$  be an interval, let  $a \in I$ , let  $n \in \mathbb{N}$ , and let  $f : I \rightarrow \mathbb{R}$  have derivatives up to order  $n$  in a neighbourhood of  $a$ . If  $f^{(n)}$  is continuous at  $a$ , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n) \quad \text{as } x \rightarrow a.$$

*Professional Standard Proof.*

TODO: professional standard proof for taylor-expansion-peano-remainder. □

*Detailed Learning Proof.*

TODO: detailed learning proof for taylor-expansion-peano-remainder. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: list mathematical dependencies.

*Remark* (Return). [Return to Corollary](#)

**Corollary** (First-Order Peano Remainder). *Let  $f$  be differentiable at  $c$ . Then  $f(c + h) = f(c) + f'(c)h + o(h)$  as  $h \rightarrow 0$ .*

*Professional Standard Proof.*

TODO: professional standard proof for first-order-peano-remainder. □

*Detailed Learning Proof.*

TODO: detailed learning proof for first-order-peano-remainder. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (The Flat Function: Smooth but Not Analytic). *The function  $f(x) = e^{-1/x^2}$  for  $x \neq 0$ , with  $f(0) = 0$ , is smooth on  $\mathbb{R}$ , has  $f^{(n)}(0) = 0$  for every  $n$ , and is not analytic at 0.*

*Professional Standard Proof.*

TODO: professional standard proof for flat-function. □

*Detailed Learning Proof.*

TODO: detailed learning proof for flat-function. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

# Chapter 12

## Integration



### 12.1 Partitions and Tags

*Remark* (Reframing note). This chapter is the narrative version of the reference list. The reference orders the definitions correctly, but the real subject is the sequence of defects and patches:

geometric problem  $\rightarrow$  Cauchy  $\rightarrow$  Riemann  
 $\rightarrow$  Darboux  $\rightarrow$  Stieltjes  
 $\rightarrow$  Henstock–Kurzweil and McShane  
 $\rightarrow$  Lebesgue.

Each integral answers a concrete complaint about the one before it. The point is not to memorize several integrals, but to understand why each one had to be invented.

#### The Geometric Problem

Forget integrals for a moment. The physical question is always the same:

Given a pointwise rate  $f$ , how much total quantity accumulates from  $a$  to  $b$ ?

A speedometer reads  $f(t)$ ; the accumulated quantity is distance. A wire has linear density  $f(x)$ ; the accumulated quantity is mass. A curve  $y = f(x)$  over  $[a, b]$  asks for signed area. The only available strategy is:

1. Chop  $[a, b]$  into thin slabs.
2. Pretend  $f$  is constant on each slab.
3. Add the resulting rectangles.

4. Refine the chopping and ask whether the sums settle.

Every construction below makes two choices: which height to use in a slab, and what it means for all these rectangle sums to settle to one number.

*Remark* (Construction comparison). The constructions differ by the height rule, the ruler, or the fineness rule:

| Construction      | Height rule                          | Ruler / fineness                                   |
|-------------------|--------------------------------------|----------------------------------------------------|
| Cauchy            | $f(x_{i-1})$ , left endpoint         | $\Delta x_i$                                       |
| Riemann           | $f(\xi_i)$ , arbitrary tag           | $\Delta x_i$                                       |
| Darboux           | $\inf_I f$ , $\sup_I f$ on each slab | $\Delta x_i$                                       |
| Riemann–Stieltjes | $f(\xi_i)$ , tagged height           | $\Delta \alpha_i$                                  |
| Henstock–Kurzweil | $f(\xi_i)$ , $\xi_i \in I_i$         | $\Delta x_i$ , $\delta(\xi_i)$ -fine, tag tethered |
| McShane           | $f(t_i)$ , $t_i$ may float           | $\Delta x_i$ , $\delta(t_i)$ -fine, tag untethered |

Henstock–Kurzweil does not change the sampled height or the ordinary ruler; it changes the admissible chopping by replacing a global mesh bound with a local gauge condition. McShane keeps the same gauge idea but frees the tag from the slab, which pulls the theory back toward Lebesgue’s absolute notion of size.

### Definition (Partition)

**Definition 12.1.1** (Partition of a Compact Interval). A *partition* of  $[a, b]$  is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned} \text{Partition}(P, a, b) \iff & \exists n \in \mathbb{N} \exists x_0, \dots, x_n \in \mathbb{R} \\ & (P = \{x_0, \dots, x_n\} \wedge x_0 = a \wedge x_n = b) \\ & \wedge \forall i (1 \leq i \leq n \Rightarrow x_{i-1} < x_i). \end{aligned}$$

### Definition (Subinterval Width)

**Definition 12.1.2** (Subinterval Width). The  $i$ -th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

### Definition (Mesh)

**Definition 12.1.3** (Mesh of a Partition). The *mesh* or *norm* of  $P$  is the width of its fattest slab:

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i.$$

*Remark* (Geometric reading). The mesh is the resolution dial. The phrase “as the partition is refined” will usually mean that  $\|P\| \rightarrow 0$ : no slab is allowed to remain fat.

### Definition (Tagged Partition)

**Definition 12.1.4** (Tagged Partition). A *tagged partition* is a partition  $P = \{x_0, \dots, x_n\}$  together with points  $\xi_i \in [x_{i-1}, x_i]$ . The tag is the point where the height  $f(\xi_i)$  is sampled.

### Definition (Refinement)

**Definition 12.1.5** (Refinement of a Partition). A partition  $P'$  *refines*  $P$  when  $P \subseteq P'$ . Refinement adds cut-points; it does not move or delete existing cut-points.

### Lemma (Common Refinement)

**Lemma 12.1.6** (Common Refinement of Two Partitions). *Any two partitions  $P$  and  $P'$  of  $[a, b]$  have a common refinement: order the finite set  $P \cup P'$  increasingly.*

*Remark* (Why the lemma is load-bearing). This small fact is the reason independently chosen partitions can be compared. Whenever a proof says “refine them together,” it is using this lemma. It is the structural engine behind Cauchy’s existence proof and Darboux’s squeeze.

## 12.2 The Cauchy Integral

### Geometric Intuition

Cauchy makes the simplest possible height choice: always sample the left edge of each slab. The result is a staircase of left-anchored rectangles. As the slabs thin out, the question is whether those staircase areas settle.

*Remark* (Engineering reading). Cauchy’s construction is the seed. It answers the geometric problem for the case where the curve is tame enough that left-edge staircases forget their initial sampling bias as the mesh shrinks. The construction is deliberately primitive: it fixes a height rule first and asks for convergence second.

### Definition (Cauchy Sum)

**Definition 12.2.1** (Left-Endpoint Cauchy Sum). For  $f : [a, b] \rightarrow \mathbb{R}$  and  $P = \{a = x_0 < \dots < x_n = b\}$ , define

$$C(f, P) = \sum_{i=1}^n f(x_{i-1}) \Delta x_i.$$

### Definition (Cauchy Integral)

**Definition 12.2.2** (Cauchy Integral). The function  $f$  is *Cauchy-integrable* with value  $L$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall P (\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon).$$

The value  $L$  is denoted  $\int_a^b f(x) dx$ .

### Proposition (Integral of a Constant)

**Proposition 12.2.3** (Cauchy Integral of a Constant). *If  $f(x) = c$  on  $[a, b]$ , then  $\int_a^b c \, dx = c(b - a)$ .*

### Theorem (Linearity)

**Theorem 12.2.4** (Linearity of the Cauchy Integral). *If  $f$  and  $g$  are Cauchy-integrable on  $[a, b]$ , then*

$$\int_a^b (\alpha f + \beta g) \, dx = \alpha \int_a^b f \, dx + \beta \int_a^b g \, dx.$$

### Theorem (Monotonicity)

**Theorem 12.2.5** (Monotonicity of the Cauchy Integral). *If  $f$  and  $g$  are Cauchy-integrable on  $[a, b]$  and  $f(x) \leq g(x)$  for every  $x \in [a, b]$ , then*

$$\int_a^b f \, dx \leq \int_a^b g \, dx.$$

### Corollary (Bounds)

**Corollary 12.2.6** (Bounds for the Cauchy Integral). *If  $f$  is continuous on  $[a, b]$ ,  $m = \min_{[a,b]} f$ , and  $M = \max_{[a,b]} f$ , then*

$$m(b - a) \leq \int_a^b f \, dx \leq M(b - a).$$

### Theorem (Triangle Inequality)

**Theorem 12.2.7** (Triangle Inequality for the Cauchy Integral). *If  $f$  and  $|f|$  are Cauchy-integrable on  $[a, b]$ , then*

$$\left| \int_a^b f \, dx \right| \leq \int_a^b |f| \, dx.$$

### Theorem (Interval Additivity)

**Theorem 12.2.8** (Interval Additivity for the Cauchy Integral). *If  $c \in [a, b]$  and  $f$  is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f \, dx = \int_a^c f \, dx + \int_c^b f \, dx.$$

### Definition (Oscillation on an Interval)

**Definition 12.2.9** (Oscillation on an Interval). For bounded  $f$  on an interval  $I$ , define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if  $\omega_f(I)$  is small, every sample height on  $I$  is close to every other sample height.

*Remark* (Oscillation as the visible error). Figure 12.1 shows the same quantity geometrically: the Darboux gap  $U(f, P) - L(f, P)$  is the sum of local oscillations  $\omega_f(I_i)$  multiplied by the slab widths  $\Delta x_i$ . The trapezoid and Simpson panels are numerical side branches; the load-bearing panel here is the oscillation identity.

## QUADRATURE & OSCILLATION

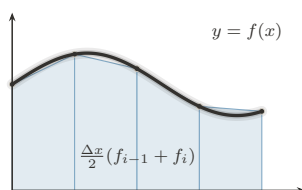


FIGURE 1. Trapezoid Rule:  
Chords: Exact for Affine  $f$

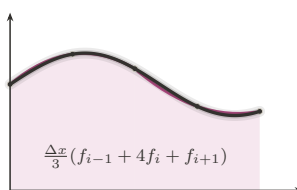


FIGURE 2. Simpson's Rule:  
Parabolas: Exact to Degree 3

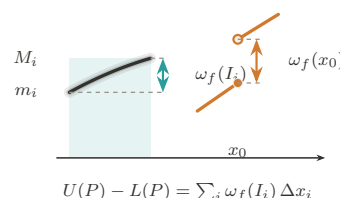


FIGURE 3. Oscillation:  
The Gap Generator  $\omega_f$

Figure 12.1: Quadrature and oscillation

### Theorem (Continuous Implies Cauchy-Integrable)

**Theorem 12.2.10** (Continuous Functions Are Cauchy-Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Cauchy-integrable.*

*Remark* (Proof skeleton). Continuity on the compact interval gives uniform continuity. Hence short slabs have small oscillation. Given two fine partitions, pass to their common refinement and compare sums slab by slab; each re-sampling error is bounded by  $\omega_f(I)$  times the slab width. The sums form a Cauchy net, and completeness of  $\mathbb{R}$  supplies the limit.

*Remark* (What continuity buys). Uniform continuity is the load-bearing ingredient. It converts pointwise control into one global scale: if the slabs are all shorter than that scale, then every slab has small oscillation. Without this global scale, Cauchy's left-endpoint definition has no mechanism for proving that unrelated partitions settle to the same number.

*Remark* (A first discontinuous edge case). A single jump should still have an area. If

$$f(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c, \end{cases}$$



on  $[a, b]$ , the only dangerous slab is the one crossing  $c$ . Its total possible error is bounded by its width, and that width tends to zero as the mesh tends to zero. This example is not hard; the point is that Cauchy's definition gives no internal test explaining why it is harmless. The proof above proves continuity is sufficient, not that controlled discontinuities are allowed.

### Theorem (Tag Independence)

**Theorem 12.2.11** (Tag Independence for Continuous Functions). *If  $f$  is continuous on  $[a, b]$ , then every choice of tags gives the same limit as the left-endpoint sums as  $\|P\| \rightarrow 0$ .*

*Remark* (Bug report). Cauchy has two defects. First, the left endpoint is privileged by convention; tag independence is a theorem only after continuity has already been assumed. Second, the existence proof gives no usable criterion for discontinuous functions. The patch request is to stop privileging the left endpoint and make tag independence part of the definition.

*Remark* (Patch request). The next definition should not say “sample on the left.” It should say: sample anywhere, even adversarially, and require all fine enough choices to agree. Riemann is exactly Cauchy's tag-independence theorem promoted into the definition of integrability.

## 12.3 The Riemann Integral

### Geometric Intuition

Riemann promotes Cauchy's tag-independence theorem into a definition. A function is integrable when every sufficiently fine staircase, even with adversarial tags, is forced near the same number.

*Remark* (Engineering reading). The sampling convention is no longer chosen by the author of the definition. It is chosen by a universal quantifier. A Riemann-integrable function is one whose accumulated area survives every possible choice of sample heights once the slabs are fine enough.

### Definition (Riemann Sum)

**Definition 12.3.1** (Riemann Sum). For a tagged partition  $(P, \xi)$ , define

$$S(f, P, \xi) = \sum_{i=1}^n f(\xi_i) \Delta x_i.$$

### Definition (Riemann Integral)

**Definition 12.3.2** (Riemann Integral). A function  $f : [a, b] \rightarrow \mathbb{R}$  is *Riemann-integrable* with value  $L$  if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall (P, \xi) (\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon).$$

*Remark* (The one symbol that changed everything). Cauchy quantifies over choppings. Riemann quantifies over choppings and every possible choice of heights:

$$\forall P \quad \rightsquigarrow \quad \forall (P, \xi).$$

That extra quantifier removes the arbitrary left endpoint and turns tag independence into the price of admission.

*Remark* (Payoff). The definition is no longer tied to continuity. It can admit bounded functions with discontinuities, because the question is not whether  $f$  is continuous but whether all fine tagged sums are forced into one narrow numerical corridor. The Cauchy criterion below makes that statement intrinsic: the value need not be known in advance.

*Remark* (The ruler function example). Thomae's function, also called the ruler function, is

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms,} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational, so its discontinuities are dense. Nevertheless its Riemann integral on any compact interval is 0. The reason is that the rational points where  $T$  can be large form only a finite set above any fixed height threshold. Fine partitions can isolate those finitely many tall spikes, while all remaining slabs have uniformly small height. This is exactly the kind of function Cauchy's continuity proof could not diagnose, but Riemann's tag-independent definition can admit.

#### Theorem (Continuous Implies Riemann-Integrable)

**Theorem 12.3.3** (Continuous Functions Are Riemann-Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann-integrable.*

#### Theorem (Linearity)

**Theorem 12.3.4** (Linearity of the Riemann Integral). *If  $f$  and  $g$  are Riemann-integrable on  $[a, b]$ , then*

$$\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx.$$

#### Theorem (Monotonicity)

**Theorem 12.3.5** (Monotonicity of the Riemann Integral). *If  $f$  and  $g$  are Riemann-integrable on  $[a, b]$  and  $f(x) \leq g(x)$  for every  $x \in [a, b]$ , then*

$$\int_a^b f dx \leq \int_a^b g dx.$$

### Theorem (Triangle Inequality)

**Theorem 12.3.6** (Triangle Inequality for the Riemann Integral). *If  $f$  is Riemann-integrable on  $[a, b]$ , then  $|f|$  is Riemann-integrable and*

$$\left| \int_a^b f \, dx \right| \leq \int_a^b |f| \, dx.$$

### Theorem (Interval Additivity)

**Theorem 12.3.7** (Interval Additivity for the Riemann Integral). *If  $c \in [a, b]$ , then  $f$  is Riemann-integrable on  $[a, b]$  if and only if it is Riemann-integrable on  $[a, c]$  and  $[c, b]$ , and in that case*

$$\int_a^b f \, dx = \int_a^c f \, dx + \int_c^b f \, dx.$$

### Theorem (Cauchy Criterion)

**Theorem 12.3.8** (Cauchy Criterion for Riemann Integrability). *A bounded function is Riemann-integrable if and only if for every  $\varepsilon > 0$  there is  $\delta > 0$  such that any two tagged partitions  $(P, \xi)$  and  $(Q, \eta)$  with  $\|P\|, \|Q\| < \delta$  have*

$$|S(f, P, \xi) - S(f, Q, \eta)| < \varepsilon.$$

*Remark* (Bug report). Riemann fixed the arbitrary tag, but the definition is operationally hostile. To verify integrability directly one must control all tagged partitions at once. The patch request is to remove tags from the verification problem: replace every possible sample height by the slab infimum and supremum, then squeeze.

*Remark* (Why this is hard to use). Riemann is logically clean but not mechanically friendly. Refining a partition does not make an arbitrary Riemann sum move monotonically, because every refinement lets the tags move again. There is no single upper or lower quantity improving under refinement. Darboux supplies exactly that missing monotone control.

## 12.4 The Darboux Integral

### Geometric Intuition

Darboux stops chasing tags. On each slab, the adversary can do no better than the supremum and no worse than the infimum. So build the upper staircase from  $\sup f$ , the lower staircase from  $\inf f$ , and ask whether the gap can be squeezed to zero.

*Remark* (Engineering reading). Darboux is the usable face of Riemann integration. It does not change the integral; it changes the test. Instead of quantifying over every possible tag, it brackets all tags at once.

### Definition (Darboux Sums)

**Definition 12.4.1** (Lower and Upper Darboux Sums). For bounded  $f$  on  $[a, b]$ , set

$$m_i = \inf_{[x_{i-1}, x_i]} f, \quad M_i = \sup_{[x_{i-1}, x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \quad U(f, P) = \sum_i M_i \Delta x_i.$$

### Lemma (Monotonicity Under Refinement)

**Lemma 12.4.2** (Darboux Sums Under Refinement). If  $P'$  refines  $P$ , then

$$L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P).$$

*Remark* (Geometric reading). Thinner slabs give lower rectangles room to rise and upper rectangles room to fall. This monotonicity is exactly what Riemann sums lack when tags move adversarially.

*Remark* (One-point refinement proof idea). It is enough to add one cut-point  $c$  inside one slab and iterate. The infimum over the whole slab is less than or equal to the infimum over each piece, so the lower contribution rises. The supremum over the whole slab is greater than or equal to the supremum over each piece, so the upper contribution falls.

### Definition (Darboux Integrability)

**Definition 12.4.3** (Darboux Integrability). The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \quad \bar{I} = \inf_P U(f, P).$$

The function  $f$  is *Darboux-integrable* if  $\underline{I} = \bar{I}$ .

### Theorem (Darboux Criterion)

**Theorem 12.4.4** (Darboux Criterion). A bounded function  $f$  is *Darboux-integrable* if and only if for every  $\varepsilon > 0$  there exists a partition  $P$  such that

$$U(f, P) - L(f, P) < \varepsilon.$$

*Remark* (The squeeze in one plate). Figure 12.2 places the Cauchy/Riemann tag choices, the lower–upper Darboux bracket, the refinement squeeze, and the gauge panel on the same page. The Darboux row is the chapter’s hinge: replacing moving tags by inf and sup turns integrability into the checkable  $U - L$  squeeze.

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## PROBES OF THE INTEGRAL

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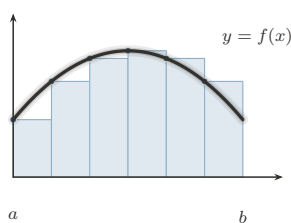


FIGURE 1. Left Riemann Sum:  
Tag at the Left Endpoint

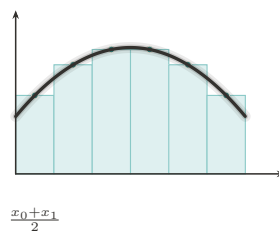


FIGURE 2. Midpoint Rule:  
The Centered Tag

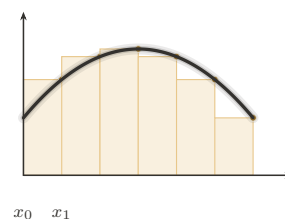


FIGURE 3. Right Riemann Sum:  
Tag at the Right Endpoint

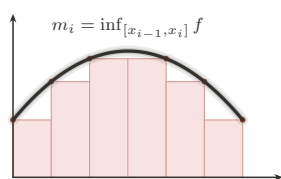


FIGURE 4. Lower Darboux Sum:  
The Infimum Probe

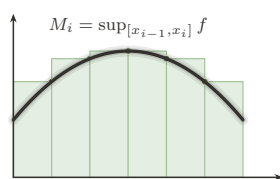


FIGURE 5. Upper Darboux Sum:  
The Supremum Probe

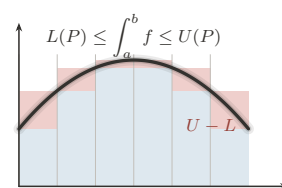


FIGURE 6. The Squeeze:  
Integrability as a Closing Gap

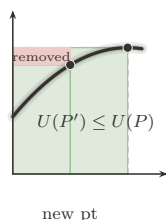


FIGURE 7. Partition Refinement:  
Adding Points Tightens the Sums

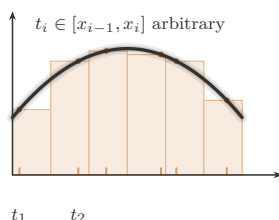


FIGURE 8. Tagged Partition:  
The General Riemann Sum

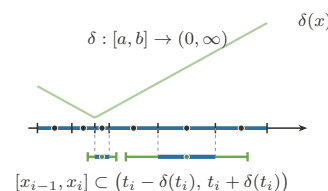


FIGURE 9. The Gauge:  
 $\delta$ -Fine Partitions (Henstock–Kurzweil)

Figure 12.2: Probes of the integral

### Theorem (Riemann Equals Darboux)

**Theorem 12.4.5** (Equivalence of Riemann and Darboux Integrability). *For bounded functions on  $[a, b]$ , Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

### Theorem (Continuous Functions Are Integrable)

**Theorem 12.4.6** (Continuous Functions Are Darboux-Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Darboux-integrable.*

### Theorem (Monotone Functions Are Integrable)

**Theorem 12.4.7** (Monotone Functions Are Darboux-Integrable). *Every monotone function  $f : [a, b] \rightarrow \mathbb{R}$  is Darboux-integrable.*

*Remark* (Why this is a harvest theorem). Monotone functions may have infinitely many discontinuities. Darboux still handles them because their total vertical movement is finite. On each slab the oscillation is just the endpoint jump across that slab, and these jumps telescope over the partition. Equal-width partitions therefore force  $U(f, P) - L(f, P)$  to zero.

### Theorem (Finitely Many Discontinuities)

**Theorem 12.4.8** (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable). *If  $f : [a, b] \rightarrow \mathbb{R}$  is bounded and has only finitely many discontinuities, then  $f$  is Darboux-integrable.*

*Remark* (Why finite bad sets are harmless). The Darboux gap is

$$U(f, P) - L(f, P) = \sum_i \omega_f([x_{i-1}, x_i]) \Delta x_i.$$

A finite set of discontinuities can be trapped in intervals of arbitrarily small total width. On the complement,  $f$  is continuous on compact pieces, so its oscillation is uniformly small on fine slabs. Thus the bad slabs are short and the good slabs are flat.

### Theorem (Algebra of Darboux-Integrable Functions)

**Theorem 12.4.9** (Sums and Scalar Multiples of Darboux-Integrable Functions). *If  $f$  and  $g$  are Darboux-integrable on  $[a, b]$ , then  $\alpha f + \beta g$  is Darboux-integrable for all  $\alpha, \beta \in \mathbb{R}$ .*

### Theorem (Products)

**Theorem 12.4.10** (Products of Darboux-Integrable Functions). *If  $f$  and  $g$  are Darboux-integrable on  $[a, b]$ , then  $fg$  is Darboux-integrable on  $[a, b]$ .*

### Theorem (Absolute Values)

**Theorem 12.4.11** (Absolute Values of Darboux-Integrable Functions). *If  $f$  is Darboux-integrable on  $[a, b]$ , then  $|f|$  is Darboux-integrable on  $[a, b]$ .*

### Theorem (Continuous Composition)

**Theorem 12.4.12** (Continuous Images of Darboux-Integrable Functions). *If  $f$  is Darboux-integrable on  $[a, b]$ ,  $\varphi$  is continuous on an interval containing  $f([a, b])$ , and  $f$  is bounded, then  $\varphi \circ f$  is Darboux-integrable on  $[a, b]$ .*

*Remark* (Why the harvest is cheap). The Darboux criterion reduces closure questions to controlling the upper-lower gap. Boundedness controls coefficients and products, while con-

tinuity of  $\varphi$  converts small oscillation of  $f$  into small oscillation of  $\varphi \circ f$ . This is the point of switching from tags to a squeeze: theorems become estimates on  $U(f, P) - L(f, P)$ .

*Remark* (Bug report). Darboux is a better tool for the same class. It does not enlarge Riemann integrability. The remaining defects now fork. One can change the ruler  $\Delta x$  to  $\Delta\alpha$ , which leads to Riemann–Stieltjes. Or one can diagnose the size of the discontinuity set, which leads to Lebesgue and also to the gauge-integral cure.

*Remark* (Residual failures). The Dirichlet function  $\mathbf{1}_{\mathbb{Q}}$  on  $[0, 1]$  has infimum 0 and supremum 1 on every slab, since rationals and irrationals are dense. Thus  $L(f, P) = 0$  and  $U(f, P) = 1$  for every partition, so the Darboux gap never closes. This is not a Darboux-specific failure; by equivalence, it is the failure of the whole Riemann family. The failure modes in Figure 12.3 separate the common pathologies: Dirichlet breaks the continuity-almost-everywhere condition, Thomae survives it, unbounded examples violate the boundedness hypothesis, and pointwise limits show why Riemann integration has weak limit behavior.

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## WHERE RIEMANN FAILS

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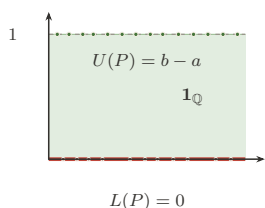


FIGURE 1. **Dirichlet  $\mathbf{1}_{\mathbb{Q}}$ :**  
**The Gap That Never Closes**

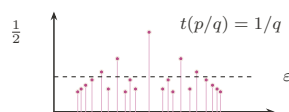


FIGURE 2. **Thomae's Function:**  
**Dense Jumps, Yet Integrable**

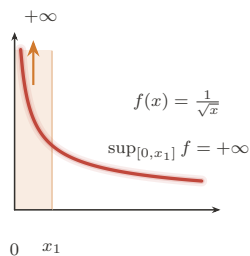


FIGURE 3. **Unbounded Integrand:**  
**Boundedness Is Not Optional**

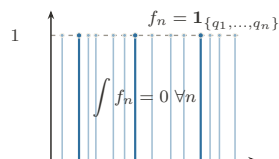


FIGURE 4. **Pointwise Limit:**  
 $\lim \int \neq \int \lim$  (Lebesgue's Cue)

Figure 12.3: Failure modes of the Riemann family

*Remark* (What Darboux makes visible). For Thomae's function, Darboux sees the opposite picture. On every slab the infimum is 0, because every slab contains irrationals. The supremum can be large only near rationals with small denominator; above any fixed height threshold, there are only finitely many such rationals. A partition can isolate those finitely

many tall rational spikes inside intervals of tiny total width, and the remaining upper rectangles are short. Thus the upper sum can be forced below any prescribed  $\varepsilon$ , while every lower sum is 0.

This is the practical gain: Riemann gives the correct definition, but Darboux gives the hand tool. Prove  $U - L$  can be squeezed, and the Riemann value follows from the equivalence theorem.

*Remark* (The fork). Darboux exposes two separate defects:

| Residual defect                | Successor         | Change                               |
|--------------------------------|-------------------|--------------------------------------|
| only ordinary length           | Riemann–Stieltjes | $\Delta x_i \mapsto \Delta \alpha_i$ |
| domain slabs see every bad set | Lebesgue / HK     | measure bad sets or adapt fineness   |

Stieltjes is therefore a sideways generalization, not the final repair of Riemann’s domain-partition limitation.

## 12.5 The Riemann–Stieltjes Integral

### Geometric Intuition

Riemann and Darboux weigh each slab by ordinary length,  $\Delta x_i$ . Riemann–Stieltjes changes the ruler. A slab is weighed by how much an integrator  $\alpha$  changes across it:

$$\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

If  $\alpha(x) = x$ , ordinary Riemann integration returns. If  $\alpha$  jumps, the integral sees a point mass.

*Remark* (Engineering reading). Stieltjes does not repair the discontinuity-set problem. It answers a different complaint: ordinary integration has only one ruler. By replacing  $\Delta x_i$  with  $\Delta \alpha_i$ , the integral can count regions with nonuniform weight and can encode discrete masses as jumps of  $\alpha$ .

#### Definition (Total Variation)

**Definition 12.5.1** (Total Variation). The total variation of  $\alpha$  on  $[a, b]$  is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

#### Definition (Bounded Variation)

**Definition 12.5.2** (Bounded Variation). The function  $\alpha$  has *bounded variation* on  $[a, b]$  if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.



### Proposition (Monotone Implies BV)

**Proposition 12.5.3** (Monotone Functions Have Bounded Variation). *If  $\alpha$  is monotone on  $[a, b]$ , then  $\alpha$  has bounded variation.*

*Remark* (Why bounded variation is the honest ruler condition). If  $\alpha$  is increasing, the increments telescope:

$$\sum_i (\alpha(x_i) - \alpha(x_{i-1})) = \alpha(b) - \alpha(a).$$

Bounded variation is the signed analogue of that finite ruler budget. It is what prevents the weighted sums from accumulating uncontrolled oscillatory ruler error.

### Definition (Riemann–Stieltjes Integral)

**Definition 12.5.4** (Riemann–Stieltjes Integral). The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)(\alpha(x_i) - \alpha(x_{i-1})).$$

When these sums have a tag-independent limit as  $\|P\| \rightarrow 0$ , the limit is denoted  $\int_a^b f d\alpha$ .

### Theorem (Continuous Integrand, BV Integrator)

**Theorem 12.5.5** (Continuous Integrand and BV Integrator). *If  $f$  is continuous on  $[a, b]$  and  $\alpha$  has bounded variation, then  $\int_a^b f d\alpha$  exists.*

### Theorem (Bilinearity)

**Theorem 12.5.6** (Bilinearity of the Riemann–Stieltjes Integral). *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g) d\alpha = \lambda \int_a^b f d\alpha + \mu \int_a^b g d\alpha$$

and

$$\int_a^b f d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f d\alpha + \mu \int_a^b f d\beta.$$

### Theorem (Interval Additivity)

**Theorem 12.5.7** (Interval Additivity for the Riemann–Stieltjes Integral). *If  $c \in [a, b]$  and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha.$$

### Theorem (Integration by Parts)

**Theorem 12.5.8** (Riemann–Stieltjes Integration by Parts). *If  $f$  and  $\alpha$  are such that one of  $\int_a^b f d\alpha$  and  $\int_a^b \alpha df$  exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a).$$

### Theorem (Smooth Integrator Reduction)

**Theorem 12.5.9** (Reduction to Riemann Integration for a  $C^1$  Integrator). *If  $\alpha \in C^1([a, b])$  and  $f$  is Riemann-integrable on  $[a, b]$ , then*

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x) dx.$$

### Theorem (Step Integrator)

**Theorem 12.5.10** (Step Integrators Give Finite Sums). *If  $\alpha$  is constant except for finitely many jumps at points  $c_k$  and  $f$  is continuous at each  $c_k$ , then*

$$\int_a^b f d\alpha = \sum_k f(c_k) \Delta\alpha(c_k).$$

*Remark* (Proof idea). The Cauchy common-refinement argument runs again. Uniform continuity makes  $\omega_f(I)$  small on short slabs, and bounded variation controls  $\sum_i |\Delta\alpha_i|$ . The error has the form

$$\sum_i \omega_f(I_i) |\Delta\alpha_i|,$$

so small oscillation multiplied by finite total variation is small.

*Remark* (Sums as integrals). If  $\alpha$  is a step function with jumps at points  $c_k$ , then the Stieltjes integral collapses to a weighted sum of the values  $f(c_k)$  times the jumps. This is the conceptual doorway to probability: a cumulative distribution function is an integrator, and expectation is an integral against that integrator.

*Remark* (Step integrator calculation). Suppose  $\alpha$  is constant except for finitely many jumps  $\Delta\alpha(c_k) = \alpha(c_k+) - \alpha(c_k-)$ . Any sufficiently fine partition has one tiny slab around each  $c_k$  carrying that jump, while all other slabs have  $\Delta\alpha_i = 0$ . If  $f$  is continuous at each  $c_k$ , then the tagged height on the  $k$ -th jump slab is forced toward  $f(c_k)$ , so

$$\int_a^b f d\alpha = \sum_k f(c_k) \Delta\alpha(c_k).$$

This is why a finite weighted sum is not a different species from an integral: it is a Stieltjes integral against a discrete ruler.

*Remark* (Shared-jump failure). The new pathology is real. On  $[0, 1]$ , let  $c = 1/2$ , let

$$\alpha(x) = \mathbf{1}_{[c,1]}(x), \quad f(x) = \mathbf{1}_{[c,1]}(x).$$

The only nonzero  $\Delta\alpha_i$  occurs on the slab containing  $c$ . If the tag for that slab is chosen left of  $c$ , its contribution is 0; if the tag is chosen at or right of  $c$ , its contribution is 1. This discrepancy does not shrink with the slab width, because the ruler has a whole unit of mass concentrated at the jump. Tag independence fails, so the Riemann–Stieltjes integral does not exist in this convention.

*Remark* (Bug report). Stieltjes is a sideways generalization, not the Lebesgue fix. Shared discontinuities of  $f$  and  $\alpha$  can destroy tag independence, and the standard existence theorem leans on the finite ruler budget  $V_a^b(\alpha) < \infty$ . Brownian paths have infinite variation almost surely, so stochastic integration cannot be ordinary pathwise Riemann–Stieltjes integration.

*Remark* (Patch request). The next stochastic patch is not “make Stieltjes a little better.” Brownian motion forces a different limiting mode: an  $L^2$  or probabilistic limit rather than a pathwise Riemann–Stieltjes limit. That is the structural reason Ito integration is not just Stieltjes integration with a rougher integrator.

## 12.6 The Henstock–Kurzweil Integral

### Where This Sits

Darboux exposed the Riemann disease before measure theory names it precisely: one uniform fineness parameter must serve the whole interval. There are two ways forward. Lebesgue will change what is measured. The gauge integrals keep the tagged-sum picture and change exactly one symbol: the constant fineness  $\delta > 0$  becomes a positive function  $\delta(x)$ .

*Remark* (Sibling cure). Henstock–Kurzweil belongs here, before McShane and the final measure-zero diagnosis, because it is still visibly a Riemann-style tagged-sum integral. It patches the uniform-mesh defect without abandoning slabs, tags, or sums. McShane will modify the tag rule one more time, producing a gauge-sum version of Lebesgue integration.

### Geometric Intuition

A gauge is adaptive mesh refinement written as analysis. It is tiny where the function needs tight local control and generous where the function is calm. A tagged slab is allowed only if it fits inside the local radius prescribed by its own tag.

*Remark* (The one-symbol upgrade). Riemann uses a positive number:

$$\exists \delta > 0.$$

Henstock–Kurzweil uses a positive function:

$$\exists \delta : [a, b] \rightarrow (0, \infty).$$

The mesh condition  $\|P\| < \delta$  becomes the local condition that each slab is  $\delta$ -fine at its tag.

### Definition (Gauge)

**Definition 12.6.1** (Gauge on an Interval). A *gauge* on  $[a, b]$  is a function

$$\delta : [a, b] \rightarrow (0, \infty).$$

### Definition (Delta-Fine Tagged Partition)

**Definition 12.6.2** ( $\delta$ -Fine Tagged Partition). A tagged partition  $(P, \xi)$  is  $\delta$ -fine if

$$[x_{i-1}, x_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)) \quad \text{for every } i.$$

### Definition (Henstock–Kurzweil Integral)

**Definition 12.6.3** (Henstock–Kurzweil Integral). A function  $f : [a, b] \rightarrow \mathbb{R}$  is *Henstock–Kurzweil integrable* with value  $A$  if

$$\forall \varepsilon > 0 \exists \delta : [a, b] \rightarrow (0, \infty) \forall (P, \xi) ((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon).$$

The value is denoted  $(\text{HK}) \int_a^b f$ .

### Lemma (Cousin)

**Lemma 12.6.4** (Cousin’s Lemma). *For every gauge  $\delta$  on  $[a, b]$ , there exists a  $\delta$ -fine tagged partition of  $[a, b]$ .*

*Remark* (Non-vacuity). Cousin’s Lemma prevents the definition from being empty. No matter how wildly the gauge varies, compactness of  $[a, b]$  guarantees that at least one tagged partition respects all the local step-size demands.

### Theorem (Riemann Sits Inside Henstock–Kurzweil)

**Theorem 12.6.5** (Riemann Integrable Implies Henstock–Kurzweil Integrable). *If  $f$  is Riemann-integrable on  $[a, b]$ , then  $f$  is Henstock–Kurzweil integrable on  $[a, b]$ , and the two integrals have the same value.*

*Remark* (Constant gauges). Riemann is the special case where the gauge is constant. If a Riemann proof hands over one mesh tolerance  $\delta_0$ , the gauge proof uses  $\delta(x) = \delta_0$  for every  $x$ .

### Lemma (Straddle)

**Lemma 12.6.6** (Straddle Lemma). *If  $F$  is differentiable at  $\xi$  with  $F'(\xi) = f(\xi)$ , then for every  $\varepsilon > 0$  there is  $\delta(\xi) > 0$  such that whenever  $u \leq \xi \leq v$ ,  $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$ ,*

$$|F(v) - F(u) - f(\xi)(v - u)| \leq \varepsilon(v - u).$$

### Theorem (FTC for the Gauge Integral)

**Theorem 12.6.7** (Fundamental Theorem for the Henstock–Kurzweil Integral). *If  $F : [a, b] \rightarrow \mathbb{R}$  is differentiable at every point and  $F' = f$ , then  $f$  is Henstock–Kurzweil integrable and*

$$(\text{HK}) \int_a^b f = F(b) - F(a).$$

### Theorem (Continuous Functions Are Henstock–Kurzweil Integrable)

**Theorem 12.6.8** (Continuous Functions Are Henstock–Kurzweil Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Henstock–Kurzweil integrable.*

*Remark* (What it buys). On a compact interval,

$$\mathcal{R}[a, b] \subsetneq \mathcal{M}[a, b] = \mathcal{L}[a, b] \subsetneq \mathcal{HK}[a, b].$$

Figure 12.4 records this inclusion lattice and the standard witness at each frontier. The first strict inclusion is witnessed by functions like  $\mathbf{1}_{\mathbb{Q}}$ , which are Lebesgue-integrable but not Riemann integrable. The second is witnessed by derivatives such as that of  $F(x) = x^2 \sin(1/x^2)$  with  $F(0) = 0$ : the derivative is HK-integrable by the gauge FTC even when its absolute value is not Lebesgue-integrable.

This inclusion statement anticipates the next volume’s Lebesgue theory. The present chapter has not built Lebesgue integration yet; it has only isolated the Riemann-side machinery, the gauge machinery, and the reason a measure-theoretic successor is needed.

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# THE INTEGRATION LATTICE

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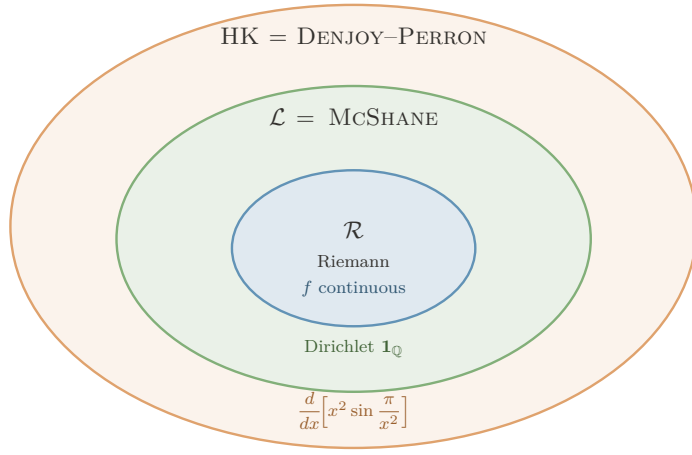


FIGURE. THE INTEGRATION LATTICE: EACH FRONTIER IS CROSSED BY EXACTLY ONE WITNESS.

$\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$ , with the strict gaps witnessed by:

- $\mathcal{L} \setminus \mathcal{R}$ :  $\mathbf{1}_{\mathbb{Q}}$  — Lebesgue (and McShane) integrable,  $\int = 0$ ; not Riemann.
- $\mathcal{HK} \setminus \mathcal{L}$ :  $F'$  for  $F(x) = x^2 \sin(\pi/x^2)$ ,  $F(0) = 0$  — HK-integrable ( $\int_0^1 F' = F(1) - F(0) = 0$ ) but  $\int_0^1 |F'| = \infty$ , so not Lebesgue.

**Engine:**  $\mathcal{L} = \text{McShane}$  uses *free* tags (absolute integration);  $\mathcal{HK}$  uses *tethered* tags, which preserve cancellation — the source of the outermost ring, and the reason HK integrates *every* everywhere-derivative.

Figure 12.4: The inclusion lattice for Riemann, Lebesgue–McShane, and Henstock–Kurzweil integration

*Remark* (Killer example). Let  $F(x) = x^2 \sin(1/x^2)$  for  $x \neq 0$ , and set  $F(0) = 0$ . Then  $F$  is differentiable at 0, with  $F'(0) = 0$ , while for  $x \neq 0$ ,

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2).$$

The oscillatory term has nonintegrable absolute size near 0, so this derivative is the standard witness that HK reaches beyond Lebesgue while the FTC remains exact.

*Remark* (HK’s own bug report). Henstock–Kurzweil is not simply “the best integral.” Lebesgue has stronger limit theorems, a complete  $L^1$  normed-space home, and a clean abstract measure-space setting. HK is the jewel for the one-dimensional FTC and conditional accumulation; Lebesgue and Ito remain the engines for probability and stochastic integration.

*Remark* (Computational bridge). A gauge  $\delta(x)$  is a spatially varying step-size field. Lipschitz control is the case where a constant gauge is enough; singular or non-Lipschitz regions demand a genuinely varying gauge. Cousin’s Lemma is the compactness guarantee that a valid adaptive partition exists.

*Remark* (One-page summary).

| Integral          | Patch                                     | Remaining defect            |
|-------------------|-------------------------------------------|-----------------------------|
| Cauchy            | left rectangles settle for continuous $f$ | privileged tag              |
| Riemann           | all tags must agree                       | hard to verify              |
| Darboux           | sup/inf squeeze                           | same class as Riemann       |
| Stieltjes         | change the ruler                          | needs BV; shared jumps      |
| Henstock–Kurzweil | local gauge fineness                      | weaker limit machinery      |
| McShane           | free gauge tags                           | same class as Lebesgue      |
| Lebesgue          | measure level sets                        | abstract machinery required |

The thread is the right-hand column: every integral is a direct response to a specific defect.

## 12.7 The McShane Integral

### Where This Sits

McShane is the gauge integral that looks almost indistinguishable from Henstock–Kurzweil until one notices where the tags are allowed to live. HK keeps each tag inside its own slab. McShane lets a tag control a slab from outside it, as long as the slab lies inside the tag’s gauge ball.

*Remark* (The tiny syntactic change). Henstock–Kurzweil says:

$$\xi_i \in [x_{i-1}, x_i] \quad \text{and} \quad [x_{i-1}, x_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)).$$

McShane drops only the first requirement. The interval still has to be locally small around its tag, but the tag need not lie in the interval it controls. That small freedom is decisive: it prevents the conditional cancellation that lets HK integrate every derivative.

*Remark* (HK versus McShane at a glance).

| Integral          | Gauge       | Tag position                                                 | Class on $[a, b]$ |
|-------------------|-------------|--------------------------------------------------------------|-------------------|
| Henstock–Kurzweil | $\delta(x)$ | $\xi_i \in I_i$                                              | $\mathcal{HK}$    |
| McShane           | $\delta(x)$ | $\xi_i \in [a, b]$ and $I_i \subset B(\xi_i, \delta(\xi_i))$ | $\mathcal{L}$     |

Both use a variable gauge. The difference is not the step-size field; it is whether the tag must sit inside the interval whose height it supplies. Figure 12.5 shows this tethered-versus-free distinction directly.

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## TWO WAYS TO TAG

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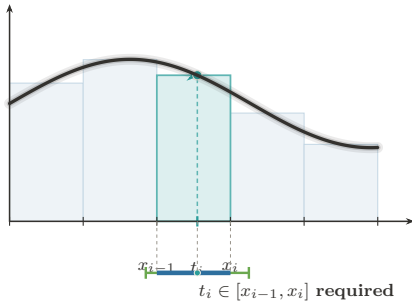


FIGURE 1. **Henstock–Kurzweil:**  
**Tag Tethered to Its Cell**

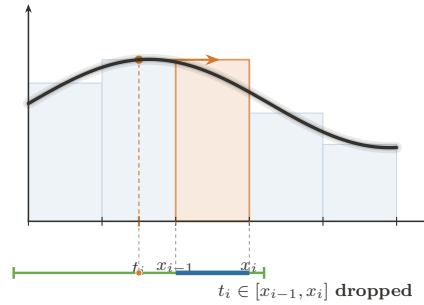


FIGURE 2. **McShane:**  
**The Tag Roams Free**

Same integrand, same gauge condition  $I_i \subseteq (t_i - \delta(t_i), t_i + \delta(t_i))$ . The *only* difference is whether the tag must sit in its own cell.

HK keeps the tether  $\Rightarrow$  conditional convergence survives  $\Rightarrow \mathcal{HK} = \text{Denjoy–Perron}$  (nonabsolute). McSHANE cuts it  $\Rightarrow$  only absolute integrals remain  $\Rightarrow \text{McShane} = \mathcal{L}^1$  (Lebesgue).

Figure 12.5: Henstock–Kurzweil versus McShane fineness

## Geometric Intuition

In HK, a tag is a sample point inside the slab. In McShane, a tag is more like a local address that authorizes nearby slabs. The gauge still says how far the tag’s influence reaches, but the rectangle height may be chosen from a point near the slab rather than inside it. This makes the integral less forgiving about sign-changing singular behavior and much closer to Lebesgue’s absolute notion of size.

### Definition (McShane-Fine Tagged Partition)

**Definition 12.7.1** (McShane-Fine Tagged Partition). Let  $\delta : [a, b] \rightarrow (0, \infty)$  be a gauge. A finite tagged partition  $\{([u_i, v_i], \xi_i)\}_{i=1}^n$  is *McShane  $\delta$ -fine* if the intervals  $[u_i, v_i]$  are non-overlapping, cover  $[a, b]$ , each  $\xi_i \in [a, b]$ , and

$$[u_i, v_i] \subset (\xi_i - \delta(\xi_i), \xi_i + \delta(\xi_i)) \quad \text{for every } i.$$

Unlike an HK tagged partition, the tag  $\xi_i$  is not required to lie in  $[u_i, v_i]$ .

*Remark* (Untethered tag). Figure 12.6 isolates the McShane move: a tag may sit outside the cell whose height it supplies, provided the cell lies inside the tag’s gauge ball. The ruler is still  $\Delta x_i$ ; the change is entirely in the admissible tag geometry.



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# THE MCSHANE INTEGRAL

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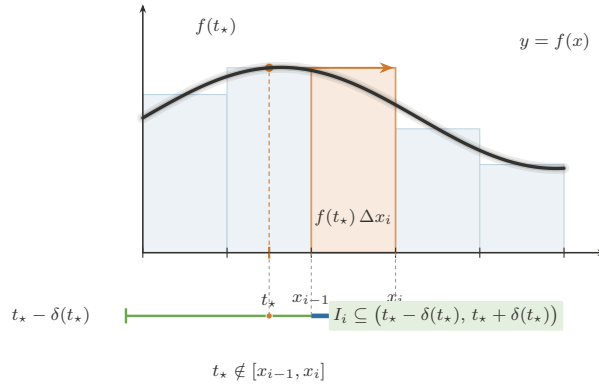


FIGURE. THE MCSHANE INTEGRAL — THE UNTETHERED TAG: A  $\delta$ -FINE PARTITION WHOSE TAGS ROAM FREE.

**Rule (the only change from Henstock–Kurzweil).** A tagged partition is *McShane  $\delta$ -fine* iff  $\forall i (I_i \subseteq (t_i - \delta(t_i), t_i + \delta(t_i)))$ , with tags  $t_i \in [a, b]$  arbitrary — the clause  $t_i \in I_i$  is dropped. The sum is still  $S = \sum_i f(t_i) \Delta x_i$ , and

$$f \text{ McShane integrable} \iff f \in L^1[a, b], \quad \text{with equal values.}$$

**Where it sits:**  $\mathcal{R} \subsetneq \mathcal{L} = \text{McShane} \subsetneq \mathcal{HK}$  (Riemann  $\subsetneq$  Lebesgue = McShane  $\subsetneq$  Henstock–Kurzweil).

Figure 12.6: The McShane integral and the untethered tag

## Definition (McShane Integral)

**Definition 12.7.2** (McShane Integral). A function  $f : [a, b] \rightarrow \mathbb{R}$  is *McShane integrable* with value  $A$  if

$$\forall \varepsilon > 0 \exists \delta : [a, b] \rightarrow (0, \infty) \forall (P, \xi) ((P, \xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon),$$

where

$$S(f, P, \xi) = \sum_i f(\xi_i)(v_i - u_i).$$

*Remark* (Relation to HK). Every McShane-fine partition is not necessarily HK-fine, because its tags may float outside their intervals. But every HK-fine partition is McShane-fine. Therefore McShane integrability is a stronger demand: the sums must survive all the usual gauge-fine tests and also the extra free-tag tests.

### Theorem (McShane Sits Between Riemann and HK)

**Theorem 12.7.3** (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable). *On a compact interval,*

$$\mathcal{R}[a, b] \subseteq \mathcal{M}[a, b] \subseteq \mathcal{HK}[a, b],$$

*and the integral values agree whenever a function belongs to the smaller class.*

### Theorem (McShane Equals Lebesgue)

**Theorem 12.7.4** (McShane Integrability Equals Lebesgue Integrability). *For real-valued functions on a compact interval,*

$$\mathcal{M}\mathcal{S}\langle \dashv \rangle[a, b] = \mathcal{L}[a, b],$$

*and the McShane and Lebesgue integral values agree.*

*Remark* (The Lebesgue connection). For real-valued functions on a compact interval, the McShane integral recovers the Lebesgue integral. In practical terms,

$$\mathcal{M}[a, b] = \mathcal{L}[a, b]$$

with equal values. Thus McShane is not a second route beyond Lebesgue in the way HK is. It is a gauge-sum presentation of Lebesgue integration.

*Remark* (Comparison chain). Combining the Riemann inclusion, the McShane–Lebesgue identification, and the HK derivative witness gives

$$\mathcal{R}[a, b] \subsetneq \mathcal{M}\mathcal{S}\langle \dashv \rangle[a, b] = \mathcal{L}[a, b] \subsetneq \mathcal{HK}[a, b].$$

The first strict inclusion is the old Riemann failure at dense null discontinuities; the second is the HK recovery of derivatives whose absolute size is not Lebesgue integrable.

*Remark* (Forward dependency). The proof that every Riemann-integrable function is McShane-integrable uses the measure-zero diagnosis in the next section: Riemann integrability means the discontinuity set is null. The exposition places McShane first because it belongs syntactically with the gauge integrals; the proof ledger records that it imports the Lebesgue criterion.

*Remark* (Why free tags force Lebesgue behavior). The HK derivative example

$$F'(x) = 2x \sin(1/x^2) - \frac{2}{x} \cos(1/x^2)$$

is conditionally integrable by HK because the tags remain inside the tiny slabs where the derivative is being locally linearized. McShane’s free tags can sample nearby oscillatory peaks in a way that destroys this conditional balance. What survives that stronger test is exactly the Lebesgue-integrable, absolutely controlled part of the gauge world.

*Remark* (What it buys). McShane gives a tagged-sum construction of the Lebesgue integral without starting from simple functions or sigma-algebras. Conceptually, that is its job in this chapter: it shows that adaptive Riemann-style sums can reach the Lebesgue class, while HK shows that keeping tags inside slabs reaches beyond Lebesgue to conditionally integrable derivatives.

*Remark* (Bug report). McShane is elegant but not the terminal fix for this chapter's story. If one wants a tagged-sum definition of the Lebesgue integral, it is excellent. If one wants the perfect one-dimensional Fundamental Theorem of Calculus, HK is strictly stronger. And if one wants probability, stochastic processes, and abstract spaces, the next volume still needs measure theory explicitly.

*Remark* (Handoff to measure zero). McShane reveals that gauge machinery and Lebesgue size are compatible. The next section names the exact size condition already controlling the older Riemann–Darboux family: the discontinuity set must have measure zero.

## 12.8 Measure Zero and the Lebesgue Criterion

### Geometric Intuition

Darboux integrability fails where upper and lower staircases refuse to meet. That refusal is local oscillation. Thus the exact Riemann question becomes: how large is the set where  $f$  is discontinuous?

*Remark* (Engineering reading). This final section is not another integral. It is the diagnostic scan of the entire Riemann–Darboux family, placed after Henstock–Kurzweil and McShane so the chapter ends by naming the exact obstruction that opens the Lebesgue volume. Once Darboux has made the gap visible, the remaining question is whether the bad part of the domain is small enough to cover by slabs of negligible total length.

#### Definition (Measure Zero)

**Definition 12.8.1** (Measure Zero). A set  $E \subseteq \mathbb{R}$  has *measure zero* if for every  $\varepsilon > 0$  there are open intervals  $I_k$  such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

#### Definition (Point Oscillation)

**Definition 12.8.2** (Point Oscillation). For bounded  $f$  on  $[a, b]$ , define

$$\omega_f(x) = \inf_{\delta > 0} \omega_f((x - \delta, x + \delta) \cap [a, b]).$$

Then  $f$  is continuous at  $x$  exactly when  $\omega_f(x) = 0$ .

*Remark* (Oscillation levels). The sets

$$D_\eta = \{x : \omega_f(x) \geq \eta\}$$

separate the bad set by severity. The full discontinuity set is

$$D = \bigcup_{k=1}^{\infty} D_{1/k}.$$

The proof of the criterion works by showing that positive-size high oscillation forces a permanent Darboux gap, while null high-oscillation sets can be covered by intervals of tiny total length.

### Theorem (Lebesgue Criterion)

**Theorem 12.8.3** (Lebesgue Criterion for Riemann Integrability). *A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

*has measure zero.*

*Remark* (Terminal diagnosis). Riemann integration works precisely when the set of bad points is invisible to length. The Dirichlet function  $\mathbf{1}_{\mathbb{Q}}$  is discontinuous everywhere, so  $D = [0, 1]$ , not a null set. The Lebesgue patch is to stop partitioning the domain into slabs and instead measure the sets where the function takes given ranges of values. Henstock–Kurzweil keeps tagged sums, but replaces uniform fineness by local fineness; McShane shows that a slightly stricter gauge rule recovers the Lebesgue class itself. The pathology plate in Figure 12.3 is the visual companion to this diagnosis.

*Remark* (Lebesgue handoff). The slogan is precise: Riemann partitions the domain, so every slab that meets a dense bad set still sees the bad behavior. Lebesgue partitions the range: it groups points by height and measures the set carrying each height. For  $\mathbf{1}_{\mathbb{Q}}$ , the height 1 lives on a null set and therefore contributes nothing.

## Appendix: Pathologies of the Riemann Integral

*Remark* (Source placement). This appendix is the converted support note from `integration-failures.tex`. Its standalone theorem machinery and document preamble have been removed; the material is kept as commentary so it does not duplicate the formal Lebesgue criterion in Section 12.8.3.

*Remark* (The one theorem behind the failures). The Riemann failures in Figure 12.3 are not a bag of examples. They are tests against the two hypotheses in the Lebesgue criterion: boundedness, and a discontinuity set of measure zero. Dirichlet fails because its discontinuity set is all of  $[0, 1]$ . Unbounded examples fail before the criterion even begins. Pointwise limits expose a different weakness: the Riemann class is not stable under the limit operations that Lebesgue integration was built to support.

*Remark* (Thomae as the diagnostic example). Thomae’s function is the useful warning. Its discontinuities are dense, but they are countable and therefore null. Darboux can isolate the finitely many large spikes above any fixed threshold inside intervals of tiny total length; outside those intervals, the upper sums are uniformly small. Thus density of the bad set is not the obstruction. Size is.

*Remark* (Through-line). The atlas failure note should be read as the bridge from Darboux to Measure Zero:  $U(f, P) - L(f, P)$  is local oscillation times length, so the only bad oscillation that matters is bad oscillation carried by a set of positive length. That observation closes the Riemann family and points directly to Lebesgue’s range-partition construction.

## Appendix: The Gauge Integrals

*Remark* (Source placement). This appendix is the converted support note from `gauge-integrals.tex`. The standalone preamble, local theorem declarations, and duplicate figure includes have been removed; the shared figures are the pinned figures already included in Sections 1.6 and 1.7.

*Remark* (The gauge framework). The gauge integrals replace Riemann’s single global mesh tolerance by a local step-size controller  $\delta : [a, b] \rightarrow (0, \infty)$ . Henstock–Kurzweil and McShane then differ by one geometric clause. HK requires the tag to sit inside the interval it controls. McShane drops that tether and allows the tag to range over  $[a, b]$ , provided the interval lies inside the tag’s gauge ball. The contrast is visualized in Figures 12.5 and 12.6.

*Remark* (Free tags and Lebesgue). McShane’s free tags force absolute, Lebesgue-style behavior. Since a tag can authorize nearby intervals without lying inside them, the sums can test whether the mass assigned to height bands is genuinely controlled. On compact intervals this recovers the Lebesgue integral:  $\mathcal{M}[a, b] = \mathcal{L}[a, b]$ , with equal values.

*Remark* (Tethered tags and the FTC). HK keeps the tag tethered to its interval, which preserves local cancellation for derivatives. That is why the HK integral has the exact one-dimensional FTC: if  $F$  is differentiable everywhere, then  $F'$  is HK-integrable and  $(\text{HK}) \int_a^b F' = F(b) - F(a)$ , even when  $F'$  is not Lebesgue integrable.

*Remark* (Lattice). The resulting compact-interval lattice is

$$\mathcal{R}[a, b] \subsetneq \mathcal{L}[a, b] = \mathcal{M}[a, b] \subsetneq \mathcal{HK}[a, b],$$

as recorded in Figure 12.4. This appendix is kept after the main chapter because the book has not yet built the measure theory needed to prove the McShane–Lebesgue identification.

## Proofs

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Common Refinement of Two Partitions). *If  $P$  and  $P'$  are partitions of  $[a, b]$ , then  $P \cup P'$ , ordered in increasing order, is a partition of  $[a, b]$  that refines both.*

*Professional Standard Proof.*

TODO: professional standard proof for common-refinement-partitions. □

*Detailed Learning Proof.*

TODO: detailed learning proof for common-refinement-partitions. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Cauchy Integral of a Constant). *If  $f(x) = c$  on  $[a, b]$ , then  $\int_a^b c \, dx = c(b - a)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-integral-constant. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-integral-constant. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Linearity of the Cauchy Integral). *If  $f$  and  $g$  are Cauchy-integrable, then the integral is linear.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-integral-linearity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-integral-linearity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotonicity of the Cauchy Integral). *If  $f \leq g$  pointwise and both functions are Cauchy-integrable, then the integral of  $f$  is no larger than the integral of  $g$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotonicity of the Cauchy integral. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotonicity of the Cauchy integral. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Cauchy Integral](#)

*Remark* (Return). [Return to Corollary](#)

**Corollary** (Bounds for the Cauchy Integral). *A continuous function whose values lie between  $m$  and  $M$  has integral between  $m(b - a)$  and  $M(b - a)$ .*

*Professional Standard Proof.*

TODO: professional standard proof for bounds for the Cauchy integral. □

*Detailed Learning Proof.*

TODO: detailed learning proof for bounds for the Cauchy integral. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Cauchy Integral of a Constant](#)
- [Monotonicity of the Cauchy Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Triangle Inequality for the Cauchy Integral). *If  $f$  and  $|f|$  are Cauchy-integrable, the absolute value of the integral is bounded by the integral of the absolute value.*

*Professional Standard Proof.*

TODO: professional standard proof for the Cauchy integral triangle inequality. □

*Detailed Learning Proof.*

TODO: detailed learning proof for the Cauchy integral triangle inequality. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Monotonicity of the Cauchy Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Interval Additivity for the Cauchy Integral). *When the relevant Cauchy integrals exist, the integral over  $[a, b]$  splits across an intermediate point  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for Cauchy integral interval additivity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Cauchy integral interval additivity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Cauchy Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Functions Are Cauchy-Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Cauchy-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for continuous-cauchy-integrable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous-cauchy-integrable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Tag Independence for Continuous Functions). *If  $f$  is continuous, all tagged sums have the same limit as the left sums.*

*Professional Standard Proof.*

TODO: professional standard proof for cauchy-tag-independence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cauchy-tag-independence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Functions Are Riemann-Integrable). *Every continuous function on a compact interval is Riemann-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for continuous functions being Riemann-integrable.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous functions being Riemann-integrable.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Tag Independence for Continuous Functions](#)
- [Riemann Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Linearity of the Riemann Integral). *The Riemann integral is linear on its domain of integrable functions.*

*Professional Standard Proof.*

TODO: professional standard proof for linearity of the Riemann integral. □

*Detailed Learning Proof.*

TODO: detailed learning proof for linearity of the Riemann integral. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann Integral](#)



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotonicity of the Riemann Integral). *If  $f \leq g$  pointwise and both functions are Riemann-integrable, then the integral of  $f$  is no larger than the integral of  $g$ .*

*Professional Standard Proof.*

TODO: professional standard proof for monotonicity of the Riemann integral. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotonicity of the Riemann integral. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Triangle Inequality for the Riemann Integral). *The Riemann integral satisfies the usual absolute-value bound.*

*Professional Standard Proof.*

TODO: professional standard proof for the Riemann integral triangle inequality. □

*Detailed Learning Proof.*

TODO: detailed learning proof for the Riemann integral triangle inequality. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Monotonicity of the Riemann Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Interval Additivity for the Riemann Integral). *Riemann integrability and the integral value split across an intermediate point.*

*Professional Standard Proof.*

TODO: professional standard proof for Riemann integral interval additivity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Riemann integral interval additivity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Cauchy Criterion for Riemann Integrability). *A bounded function is Riemann-integrable if and only if sufficiently fine tagged sums are mutually close.*

*Professional Standard Proof.*

TODO: professional standard proof for riemann-cauchy-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for riemann-cauchy-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Darboux Sums Under Refinement). *Refinement raises lower sums and lowers upper sums.*

*Professional Standard Proof.*

TODO: professional standard proof for darboux-refinement-squeeze. □

*Detailed Learning Proof.*

TODO: detailed learning proof for darboux-refinement-squeeze. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Darboux Criterion). *Darboux integrability is equivalent to making  $U(f, P) - L(f, P)$  arbitrarily small.*

*Professional Standard Proof.*

TODO: professional standard proof for darboux-criterion. □

*Detailed Learning Proof.*

TODO: detailed learning proof for darboux-criterion. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Equivalence of Riemann and Darboux Integrability). *For bounded functions, Riemann and Darboux integrability are equivalent.*

*Professional Standard Proof.*

TODO: professional standard proof for riemann-darboux-equivalence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for riemann-darboux-equivalence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Functions Are Darboux-Integrable). *Every continuous function on a compact interval is Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for continuous functions being Darboux-integrable.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous functions being Darboux-integrable.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Darboux Criterion](#)
- Heine–Cantor uniform continuity theorem.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Monotone Functions Are Darboux-Integrable). *Every monotone function  $f : [a, b] \rightarrow \mathbb{R}$  is Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-darboux-integrable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-darboux-integrable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable).  
*If  $f : [a, b] \rightarrow \mathbb{R}$  is bounded and has only finitely many discontinuities, then  $f$  is Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for finite-discontinuities-darboux-integrable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-discontinuities-darboux-integrable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Sums and Scalar Multiples of Darboux-Integrable Functions). *Linear combinations of Darboux-integrable functions are Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for Darboux-integrable linear combinations. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Darboux-integrable linear combinations. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Darboux Criterion](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Products of Darboux-Integrable Functions). *Products of Darboux-integrable functions are Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for products of Darboux-integrable functions. □

*Detailed Learning Proof.*

TODO: detailed learning proof for products of Darboux-integrable functions. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Darboux Criterion](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Absolute Values of Darboux-Integrable Functions). *The absolute value of a Darboux-integrable function is Darboux-integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for absolute values of Darboux-integrable functions.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for absolute values of Darboux-integrable functions.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Darboux Criterion](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Images of Darboux-Integrable Functions). *Continuous functions preserve Darboux integrability under composition with a bounded Darboux-integrable function.*

*Professional Standard Proof.*

TODO: professional standard proof for continuous composition preserving Darboux integrability.  $\square$

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous composition preserving Darboux integrability.  $\square$

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Darboux Criterion](#)

*Remark* (Return). [Return to Proposition](#)

**Proposition** (Monotone Functions Have Bounded Variation). *Every monotone function on  $[a, b]$  has bounded variation.*

*Professional Standard Proof.*

TODO: professional standard proof for monotone-bounded-variation. □

*Detailed Learning Proof.*

TODO: detailed learning proof for monotone-bounded-variation. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Integrand and BV Integrator). *If  $f$  is continuous and  $\alpha$  has bounded variation, then  $\int f d\alpha$  exists.*

*Professional Standard Proof.*

TODO: professional standard proof for rs-continuous-bv-existence. □

*Detailed Learning Proof.*

TODO: detailed learning proof for rs-continuous-bv-existence. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Bilinearity of the Riemann–Stieltjes Integral). *Whenever the displayed integrals exist, the Riemann–Stieltjes integral is linear in the integrand and in the integrator.*

*Professional Standard Proof.*

TODO: professional standard proof for Riemann–Stieltjes bilinearity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Riemann–Stieltjes bilinearity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann–Stieltjes Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Interval Additivity for the Riemann–Stieltjes Integral). *When the relevant integrals exist, the Riemann–Stieltjes integral over  $[a, b]$  splits across an intermediate point  $c$ .*

*Professional Standard Proof.*

TODO: professional standard proof for Riemann–Stieltjes interval additivity. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Riemann–Stieltjes interval additivity. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann–Stieltjes Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Riemann–Stieltjes Integration by Parts). *Under the Riemann–Stieltjes convention, existence of one integral implies existence of the paired integral and their sum is the endpoint product difference.*

*Professional Standard Proof.*

TODO: professional standard proof for Riemann–Stieltjes integration by parts. □

*Detailed Learning Proof.*

TODO: detailed learning proof for Riemann–Stieltjes integration by parts. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann–Stieltjes Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Reduction to Riemann Integration for a  $C^1$  Integrator). *If the integrator is  $C^1$ , then the Riemann–Stieltjes integral reduces to an ordinary Riemann integral against  $\alpha'$ .*

*Professional Standard Proof.*

TODO: professional standard proof for the  $C^1$  Riemann–Stieltjes reduction. □

*Detailed Learning Proof.*

TODO: detailed learning proof for the  $C^1$  Riemann–Stieltjes reduction. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann–Stieltjes Integral](#)
- [Riemann Integral](#)

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Step Integrators Give Finite Sums). *A Riemann–Stieltjes integral against a finite-step integrator is a finite weighted sum over the jump points, provided the integrand is continuous there.*

*Professional Standard Proof.*

TODO: professional standard proof for finite-step Riemann–Stieltjes integrators. □

*Detailed Learning Proof.*

TODO: detailed learning proof for finite-step Riemann–Stieltjes integrators. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- [Riemann–Stieltjes Integral](#)

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Cousin's Lemma). *For every gauge  $\delta$  on  $[a, b]$ , there exists a  $\delta$ -fine tagged partition of  $[a, b]$ .*

*Professional Standard Proof.*

TODO: professional standard proof for cousin. □

*Detailed Learning Proof.*

TODO: detailed learning proof for cousin. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Riemann Integrable Implies Henstock–Kurzweil Integrable). *If  $f$  is Riemann-integrable on  $[a, b]$ , then  $f$  is Henstock–Kurzweil integrable on  $[a, b]$ , with the same value.*

*Professional Standard Proof.*

TODO: professional standard proof for riemann-integrable-implies-hk-integrable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for riemann-integrable-implies-hk-integrable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Lemma](#)

**Lemma** (Straddle Lemma). *If  $F$  is differentiable at  $\xi$  with  $F'(\xi) = f(\xi)$ , then the one-slab linearization error across any sufficiently small interval straddling  $\xi$  is bounded by  $\varepsilon$  times its length.*

*Professional Standard Proof.*

TODO: professional standard proof for hk-straddle. □

*Detailed Learning Proof.*

TODO: detailed learning proof for hk-straddle. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.



*Remark* (Return). [Return to Theorem](#)

**Theorem** (Fundamental Theorem for the Henstock–Kurzweil Integral). *If  $F$  is differentiable on  $[a, b]$  and  $F' = f$ , then*

$$(HK) \int_a^b f = F(b) - F(a).$$

*Professional Standard Proof.*

TODO: professional standard proof for hk-fundamental-theorem. □

*Detailed Learning Proof.*

TODO: detailed learning proof for hk-fundamental-theorem. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Continuous Functions Are Henstock–Kurzweil Integrable). *Every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Henstock–Kurzweil integrable.*

*Professional Standard Proof.*

TODO: professional standard proof for continuous-hk-integrable. □

*Detailed Learning Proof.*

TODO: detailed learning proof for continuous-hk-integrable. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable). *On a compact interval,*

$$\mathcal{R}[a, b] \subseteq \mathcal{M}[a, b] \subseteq \mathcal{HK}[a, b],$$

*with agreement of integral values.*

*Professional Standard Proof.*

TODO: professional standard proof for riemann-mcshane-hk-inclusions. □

*Detailed Learning Proof.*

TODO: detailed learning proof for riemann-mcshane-hk-inclusions. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (McShane Integrability Equals Lebesgue Integrability). *For real-valued functions on a compact interval,*

$$\mathcal{M}\int \mathcal{S} \langle \cdot \rangle [a, b] = \mathcal{L}[a, b],$$

*with agreement of integral values.*

*Professional Standard Proof.*

TODO: professional standard proof for mcshane-equals-lebesgue. □

*Detailed Learning Proof.*

TODO: detailed learning proof for mcshane-equals-lebesgue. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

*Remark* (Return). [Return to Theorem](#)

**Theorem** (Lebesgue Criterion for Riemann Integrability). *A bounded function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann-integrable if and only if its discontinuity set has measure zero.*

*Professional Standard Proof.*

TODO: professional standard proof for lebesgue-criterion-riemann-integrability. □

*Detailed Learning Proof.*

TODO: detailed learning proof for lebesgue-criterion-riemann-integrability. □

*Remark* (Proof structure).

*Remark* (Dependencies).

- TODO: add dependencies for this proof.

# Bibliography

Terence Tao. *Analysis I*, volume 37 of *Texts and Readings in Mathematics*. Hindustan Book Agency, New Delhi, 2006. ISBN 978-8185931751.